

CHAPTER 8

INSTRUMENT SYSTEMS

CHAPTER OVERVIEW

SECTION	TITLE
8-1	Equipment Description and Data
8-2	Troubleshooting
8-3	On Aircraft Inspections
8-4	Maintenance Procedures
8-5	Maintenance Procedures (BENCH)

8-1/(8-2 Blank)

SECTION 8-1

EQUIPMENT DESCRIPTION AND DATA

SECTION OVERVIEW

PARAGRAPH	TITLE
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8-1-2	Miscellaneous Instruments Description and Data
8-1-3	Instrument Display System Description and Data
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8-1-1 FLIGHT INSTRUMENTS DESCRIPTION AND DATA.

8-1-1.1 Flight Instruments Description.

The flight instruments provide the pilots with information necessary for correct flight (Figure 8-1-1-1). The flight instruments include the barometric altimeters, airspeed indicators, vertical speed indicators, associated Pitot-static system, radar altimeters, horizontal situation indicators (HSIs), vertical situation indicators (VSIs), stabilator position indicators, and standby compass. For a complete description of the HSIs, VSIs, and radar altimeters, refer to TM 11-70-23.

8-1-1.2 Barometric Altimeters.

Two indicators, one on each side of the instrument panel, indicate altitude above or below sea level under standard conditions of temperature and atmospheric pressure (Figures 8-1-1-1 and 8-1-1-2). The range of the barometric altimeter is -1000 to 50,000 feet as indicated by three drum indicators and a pointer. The numeral on the 100-foot drum represents hundreds of feet. The pointer is a vernier indication of the hundreds drum as well as being an indication of trend information. Each digit on the 1,000-foot drum represents 1,000-foot intervals while each digit of the 10,000-foot drum represents 10,000-foot levels. In the space corresponding to zero, the 10,000-foot drum has striped lines. The combined readings of the three drums indicate the altitude of the helicopter. The barometric pressure zero set knob in the lower left corner is adjusted to compensate for varying barometric pressures. A small barometric scale, showing through a cutout in the dial between three and four markings, indicates the adjusted barometric pressure setting. The range of this scale is 28.1 to 31.0 inches of mercury. Each barometric altimeter has two connectors on the back side, one static and one electrical. The electrical connector provides a 28 vdc input to an internal vibrator that decreases the friction in the mechanism. Lighting for both the pilot's and copilot's barometric altimeters is provided by a lighted bezel. The pilot's barometric altimeter integral lighting is controlled by the INST LT PILOT FLT control, on the upper console. The copilot's barometric altimeter integral lighting is controlled by the CPLT FLT INST LTS control, on the upper console. The pilot's altimeter encoder

provides a digital output of pressure altitude to the transponder (IFF) (Figure 8-1-1-2).

8-1-1.3 Airspeed Indicators.

Two airspeed indicators, one on each side of the instrument panel, indicate helicopter speed in knots (Figure 8-1-1-1). The range is 0 to 250 knots, marked in five knot units. The indicators are differential pressure instruments, measuring the difference between impact pressure and static pressure. The two pressures are equal when the helicopter is stationary. As ram air pressure in the Pitot tube becomes greater than pressure in the static line, the diaphragm connected to the pressure line will expand, moving the airspeed needle upscale and indicating airspeed in knots. System installation error is noted on two placards (one each for the pilot and copilot), located on the sides of the lower console.

8-1-1.4 Vertical Speed Indicators.

Two vertical speed indicators, one on each side of the instrument panel, indicate helicopter rate of ascent or descent in feet per minute (fpm) (Figure 8-1-1-1). Range markings begin at level flight and are in units of 100 fpm up to 1000 fpm. From 1000, fpm units are in 500 fpm. Maximum indicated vertical speed is 6000 fpm. Indicator operation is controlled by pressure differential between two chambers. The vertical speed indicator is vented to cockpit atmosphere. The other chamber is the instrument case itself, connected by an air-restricting tube to the internal connection of the diaphragm supply line. The pointer may be zeroed externally by the adjusting screw in the lower left corner of the indicator.

8-1-1.5 Pitot-Static Pressure System.

The Pitot-static system provides pressure for operation of the differential pressure instruments. These are the barometric altimeters and airspeed and vertical speed indicators (Figure 8-1-1-3). Differential pressure used to actuate these instruments is created either by impact (Pitot) and static, or by static and trapped air pressures. The Pitot-static

system supplies both Pitot and static pressures to the instruments.

Pitot pressure is supplied through Pitot lines from two Pitot-static tubes to the airspeed indicators and airspeed and air data transducers (Figure 8-1-1-4). Static air pressure from the atmosphere is supplied through static lines from the Pitot-static tube to the altimeters and airspeed indicators. The Pitot-static tubes are mounted on top of each side of the cockpit and provide Pitot and/or static pressure to instrument lines. The lines from the Pitot-static tubes are routed down the sides of the cockpit and are connected to the applicable instruments mounted on the instrument panel. A filtered restrictor assembly and balance chamber installed in each Pitot line provides improved airspeed indicator damping. The restrictor filters (screens) provide protection from possible restrictor blockage caused by contaminants and airborne particles. The Pitot-static system has screw-capped drain ports, PITOT DRAIN and STATIC DRAIN, to release water vapor that has condensed to the lines. Two Pitot and two static drain ports are on the bottom of the helicopter under the front cabin.

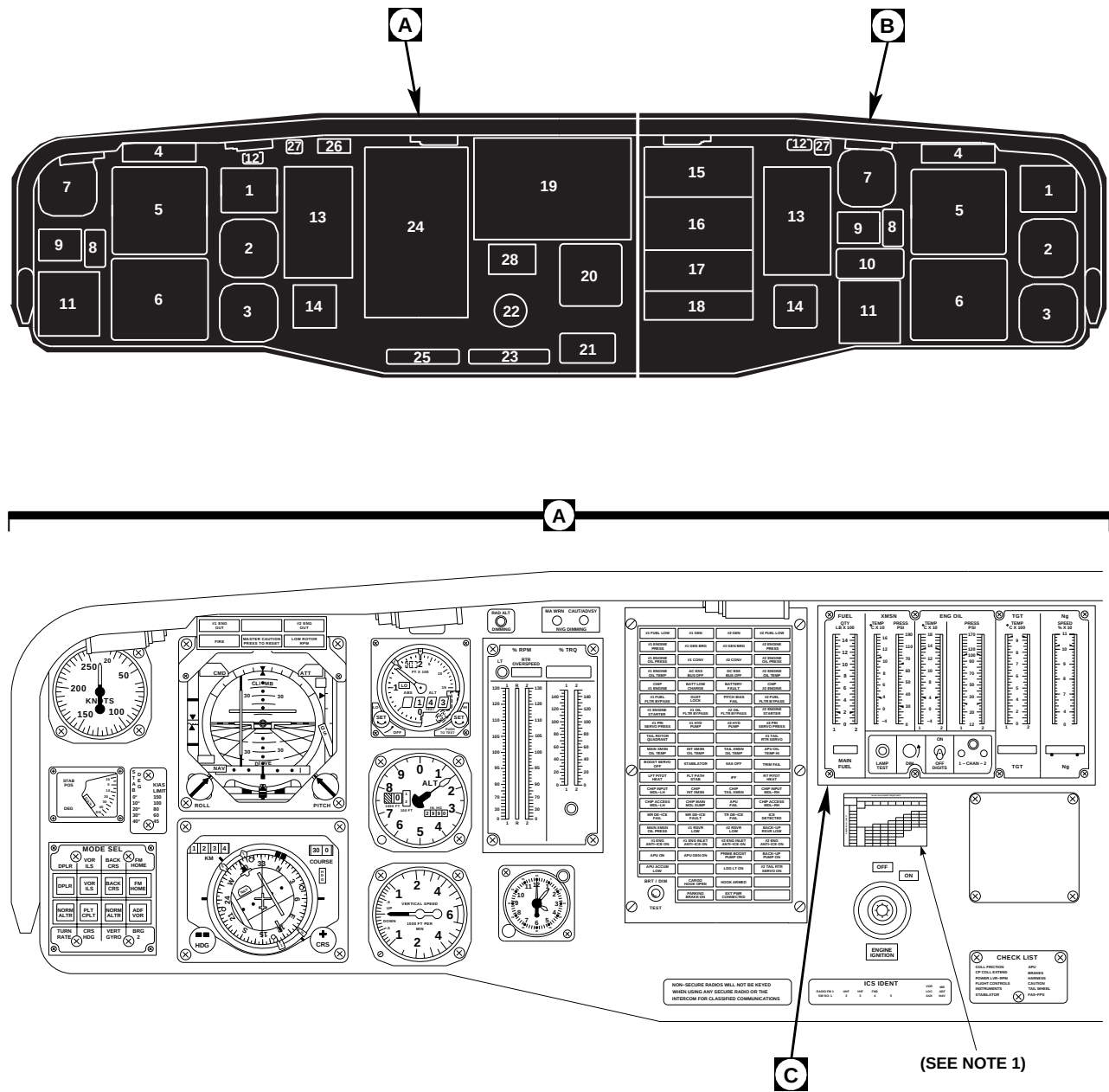
8-1-1.6 Pitot-Static Head Assembly.

The Pitot-static head assembly consists of a baseplate with a strut and probe tube. The base plate contains the Pitot tube fitting, two static tube fittings (S1 and S2) and an electrical connector wired to two de-icing heaters in the tube. The probe tube contains these pressure sensing ports: PITOT,

STATIC 1, and STATIC 2. PITOT pressure is sensed at the opening of the front end of the tube. STATIC 1 and STATIC 2 pressure is sensed at the contoured midsection of the tube. The Pitot-static head assembly is attached to a tapered mounting block assembly surrounded by an aerodynamic fairing that alleviates potential ice buildup at the head assembly/airframe interface.

8-1-1.7 Pitot-Static Heater System.

Pitot heaters, in the Pitot-static tubes, keep ice that could restrict air flow from forming on the tubes (Figures 8-1-1-5 and 8-1-1-6). Electrical power of 115 vac for the left Pitot heaters is supplied by the No. 1 ac primary bus through the LEFT PITOT HEAT circuit breaker on the copilot's circuit breaker panel. Electrical power of 115 vac for the right Pitot heaters is supplied by the No. 2 ac primary bus through the RT PITOT HEAT circuit breaker on the pilot's circuit breaker panel. DC power is supplied by the No. 1 dc primary bus through the NO. 1 ENG ANTI-ICE circuit breaker on the copilot's circuit breaker panel. When the PITOT HEAT switch is ON, power of 115 vac is fed through current sensors in the right and left relay panels to the right and left Pitot tube heaters, causing the de-icing heaters to go on. Also, 28 vdc is fed to a normally-open electronic switch within the current sensors. When a low heat or no heat condition is sensed by the current sensors, 28 vdc is fed through the electronic switch to the caution/advisory panel, lighting the RT or LFT PITOT HEAT caution capsule.

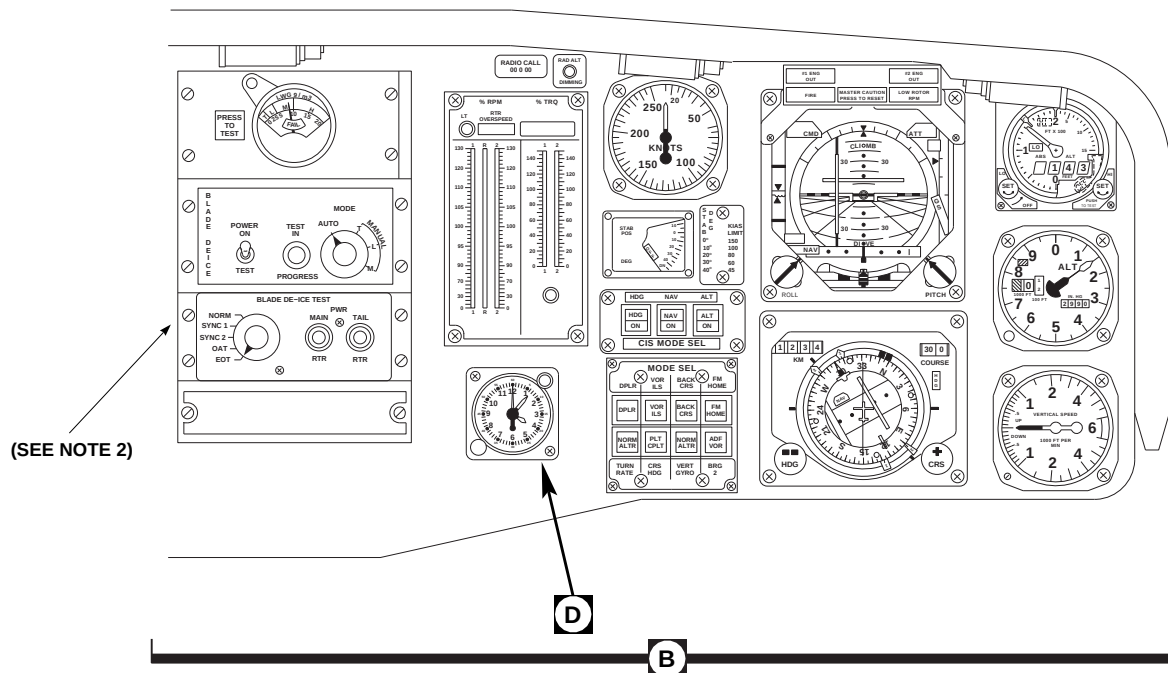


NOTES
S70-II 70984 - SUBQ THE COMPONENTS ARE INSTALLED AS FOLLOWS: BLADE DE-ICE CONTROL PANEL (TOP) BLADE DE-ICE TEST PANEL (CENTER) ICING RATE METER (BOTTOM)

EFFECTIVITY
NOT EQUIPPED WITH WEATHER RADAR.

FIGURE 8-1-1-1. INSTRUMENT PANEL FRONT VIEW (SHEET 1 OF 4)

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1. RADAR ALTIMETER
2. BAROMETRIC ALTIMETER
3. VERTICAL SPEED INDICATOR
4. MASTER WARNING PANEL
5. VERTICAL SITUATION INDICATOR
6. HORIZONTAL SITUATION INDICATOR
7. AIRSPEED INDICATOR
8. STABILATOR POSITION PLACARD
9. STABILATOR POSITION INDICATOR
10. CIS MODE SELECTOR
11. HSI / VSI MODE SELECT PANEL
12. RADIO CALL PLACARD
13. PILOT'S AND COPILOT'S DISPLAY UNIT
14. 8-DAY CLOCK OR DIGITAL CLOCK
15. ICING RATE METER (SEE NOTE 2)
16. BLADE DE-ICE CONTROL PANEL (SEE NOTE 2)
17. BLADE DE-ICE TEST PANEL (SEE NOTE 2)
18. BLANK PANEL
19. CENTRAL DISPLAY UNIT
20. BLANK PANEL
21. CHECKLIST
22. ENGINE IGNITION SWITCH
23. RADIO SELECT PLACARD
24. CAUTION / ADVISORY PANEL
25. SECURE RADIO WARNING PLACARD
26. SLS DIMMING CONTROL PANEL
27. RADAR ALTIMETER DIMMING CONTROL
28. S70-II DUAL ENGINE TORQUE LIMITS

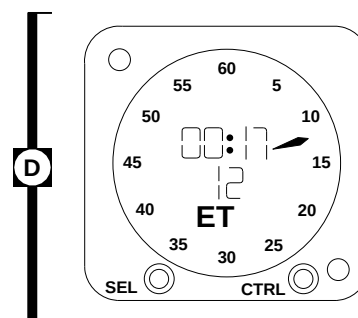
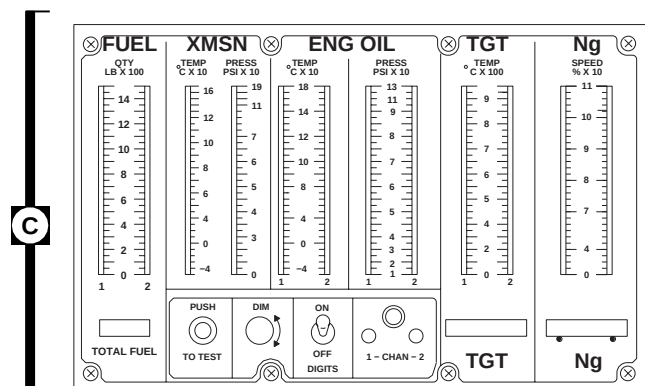
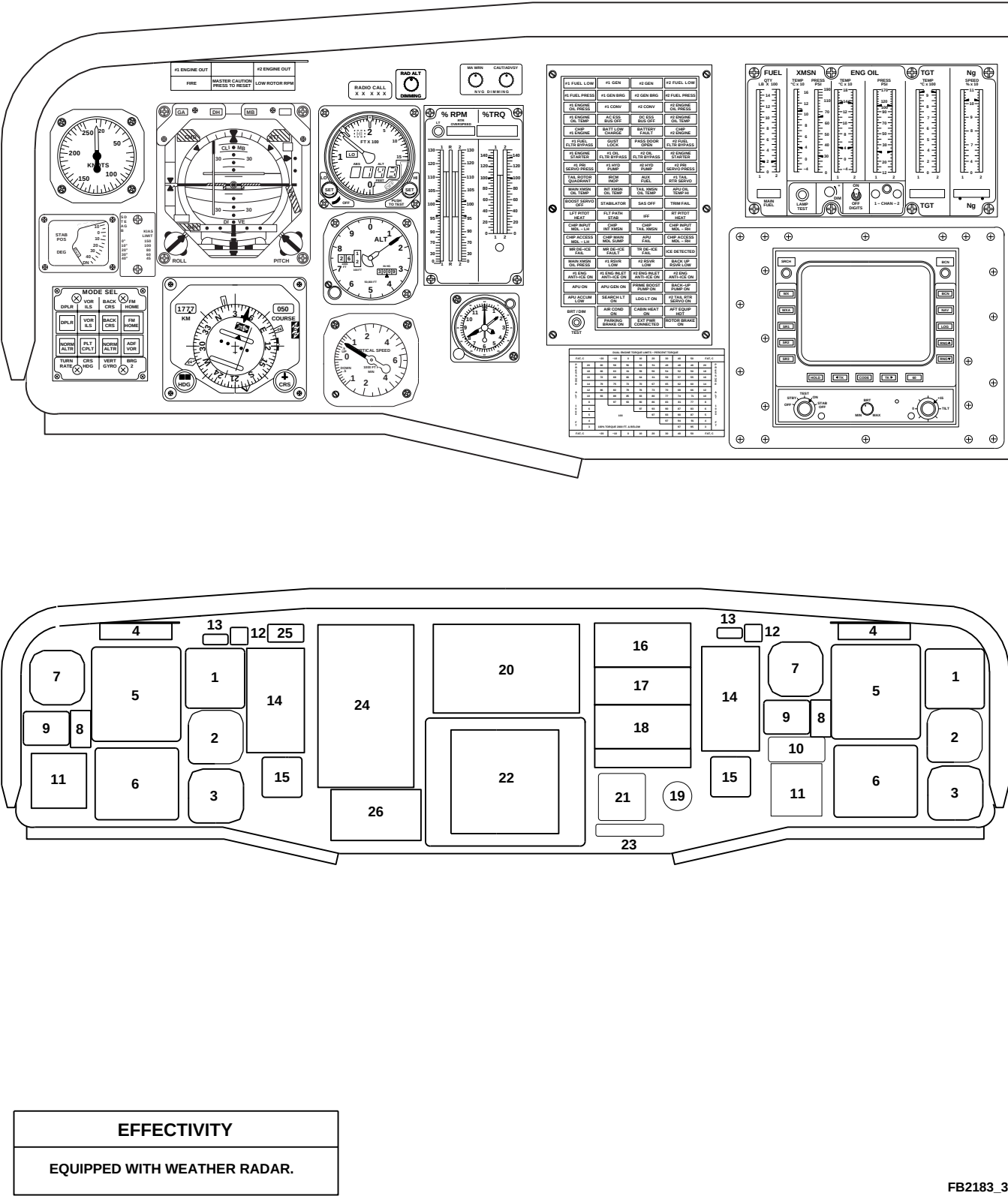
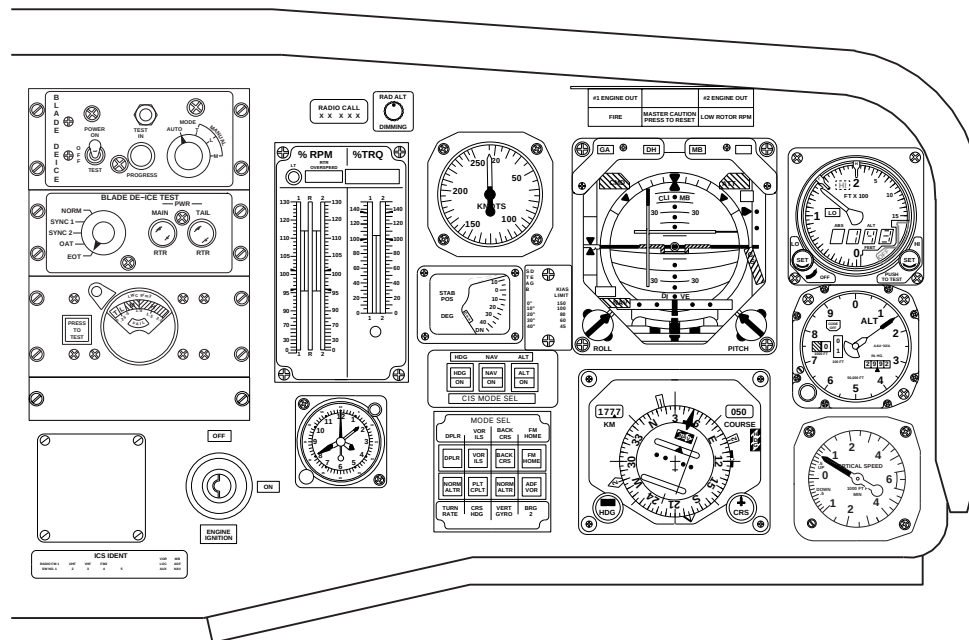


FIGURE 8-1-1-1. INSTRUMENT PANEL FRONT VIEW (SHEET 2 OF 4)

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- | | |
|-------------------------------------|--------------------------------------|
| 1. RADAR ALTIMETER | 16. BLADE DE-ICE CONTROL PANEL |
| 2. BAROMETRIC ALTIMETER | 17. BLADE DE-ICE TEST PANEL |
| 3. VERTICAL SPEED INDICATOR | 18. ICING RATE METER |
| 4. MASTER WARNING PANEL | 19. ENGINE IGNITION SWITCH |
| 5. VERTICAL SITUATION INDICATOR | 20. CENTRAL DISPLAY UNIT |
| 6. HORIZONTAL SITUATION INDICATOR | 21. BLANK PANEL |
| 7. AIRSPEED INDICATOR | 22. WEATHER RADAR |
| 8. STABILATOR POSITION PLACARD | 23. RADIO SELECT PLACARD |
| 9. STABILATOR POSITION INDICATOR | 24. CAUTION / ADVISORY PANEL |
| 10. CIS MODE INDICATOR | 25. SLS DIMMING CONTROL |
| 11. VSI / HSI MODE SELECTOR | 26. S70-II DUAL ENGINE TORQUE LIMITS |
| 12. RADAR ALTIMETER DIMMING CONTROL | |
| 13. RADIO CALL PLACARD | |
| 14. PILOT'S DISPLAY UNIT | |
| 15. 8-DAY CLOCK OR DIGITAL CLOCK | |

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FIGURE 8-1-1-1. INSTRUMENT PANEL FRONT VIEW (SHEET 4 OF 4)

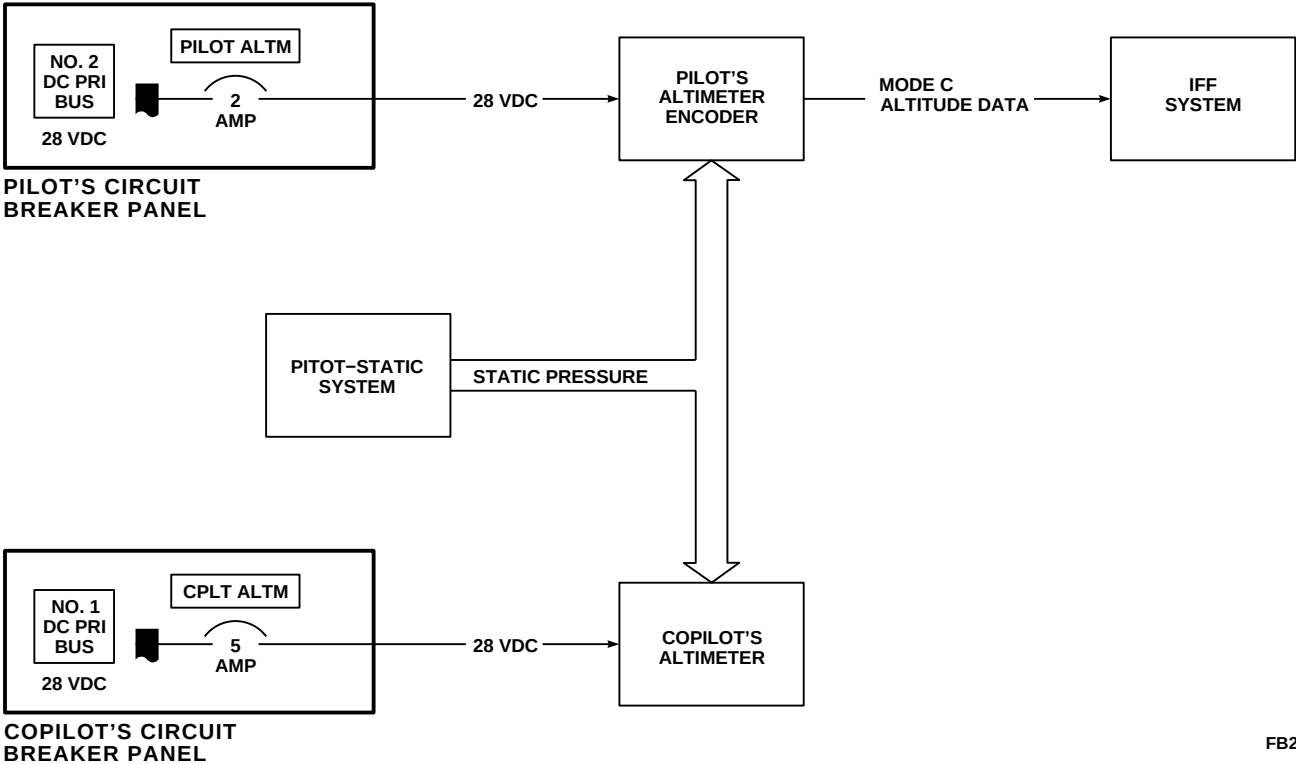


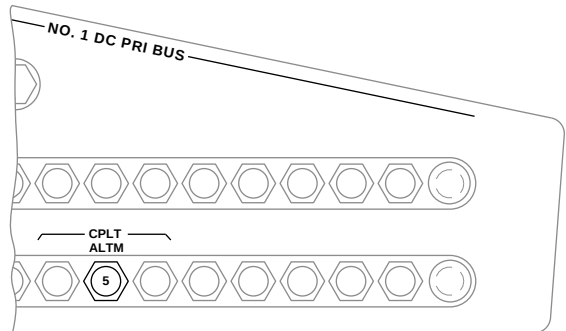
FIGURE 8-1-1-2. BAROMETRIC ALTIMETER BLOCK DIAGRAM

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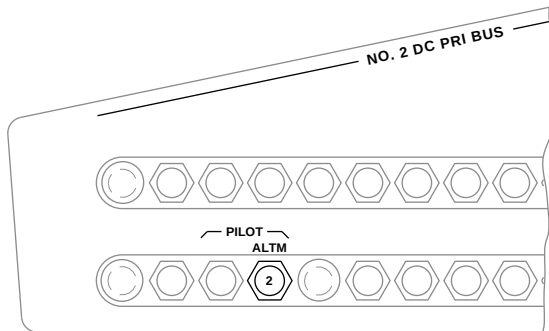
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P1	COPILOT'S BAROMETRIC ALTIMETER
P1R / J1R	COCKPIT BL 6 LH, STA 210
P20R / J20R	CABIN BL 40 RH, STA 248
P111 / J111	COCKPIT BL 7.5 LH, STA 197
P112 / J112	COCKPIT BL 6 RH, STA 197
P113 / J113	COCKPIT BL 6 RH, STA 197
P183	PILOT'S BAROMETRIC ALTIMETER LIGHTED, BEZEL ASSEMBLY
P193	COPILOT'S ENCODER ALTIMETER LIGHTED, BEZEL ASSEMBLY
P246 / J246	CABIN CEILING BL 5 RH, STA 247
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 25 RH
P280 / J280	CABIN CEILING BL 6 LH, STA 247
P306R	PILOT'S ALTIMETER ENCODER

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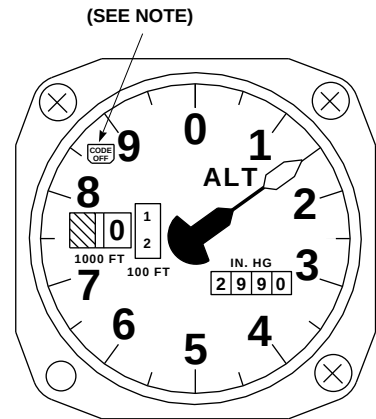
FIGURE 8-1-1-3. PITOT-STATIC PRESSURE SYSTEM LOCATION DIAGRAM (SHEET 2 OF 4)



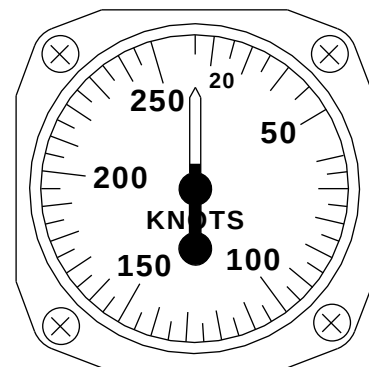
COPLOT'S CIRCUIT BREAKER PANEL



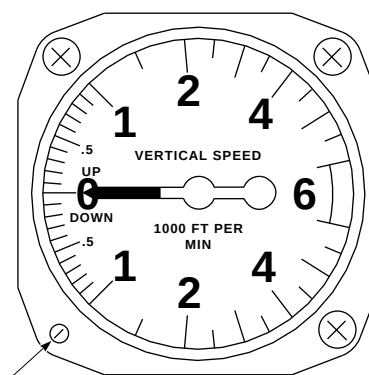
PILOT'S CIRCUIT BREAKER PANEL



**PILOT'S AND COPLOT'S
BAROMETRIC ALTIMETER**



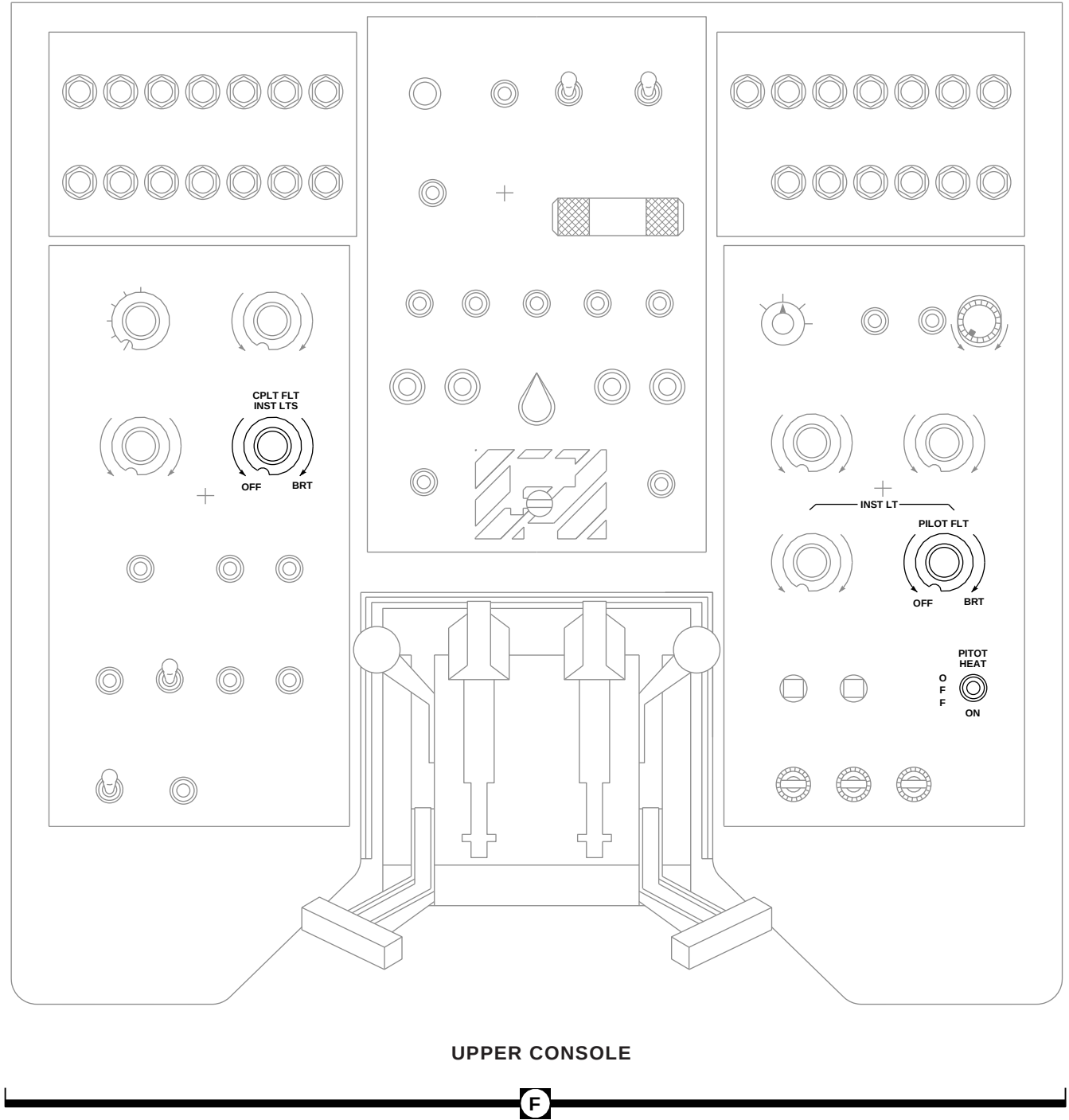
**PILOT'S AND COPLOT'S
AIRSPEED INDICATOR**



**PILOT'S AND COPLOT'S
VERTICAL SPEED INDICATOR**

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FIGURE 8-1-1-3. PITOT-STATIC PRESSURE SYSTEM LOCATION DIAGRAM (SHEET 3 OF 4)



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FIGURE 8-1-1-3. PITOT-STATIC PRESSURE SYSTEM LOCATION DIAGRAM (SHEET 4 OF 4)

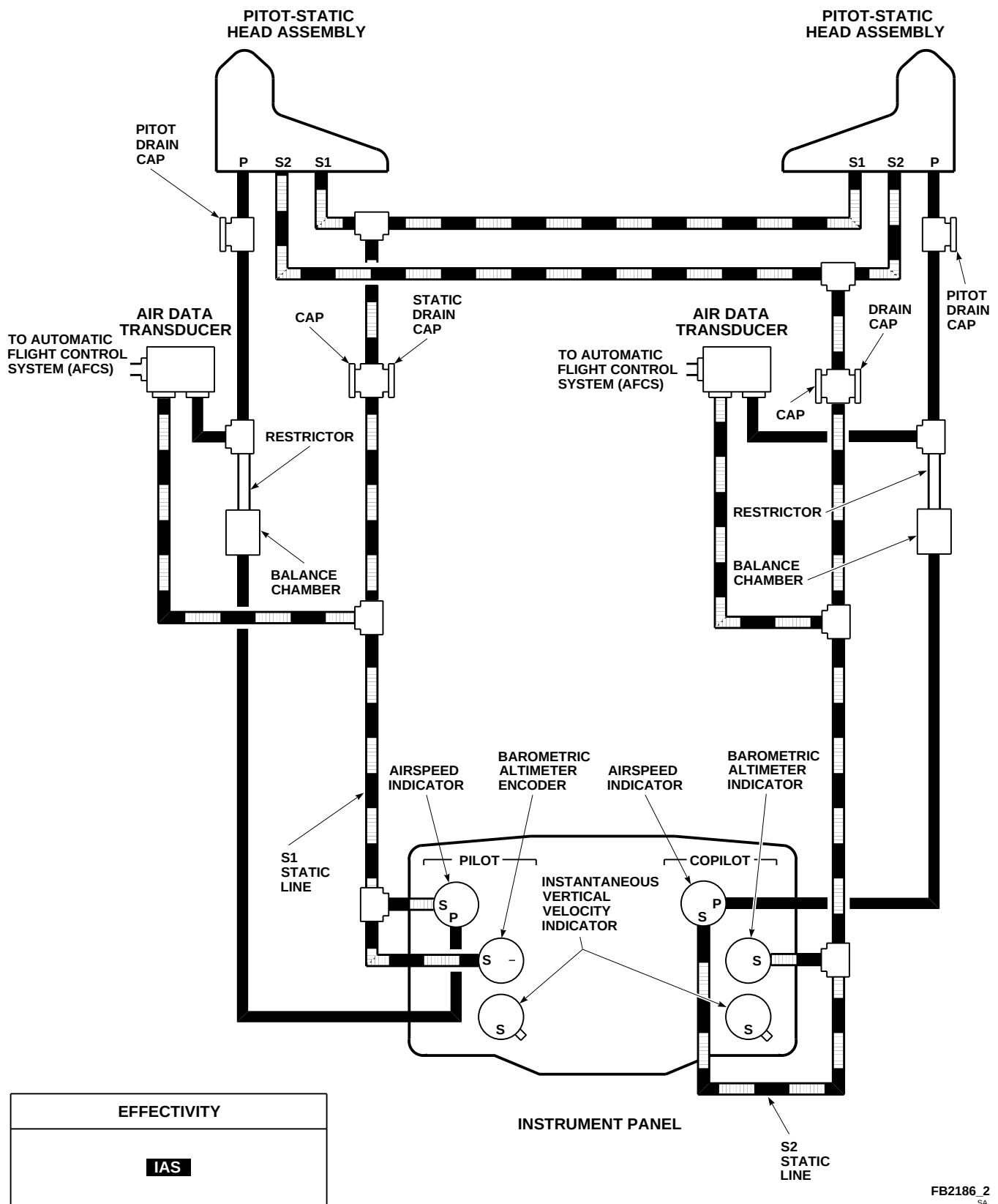
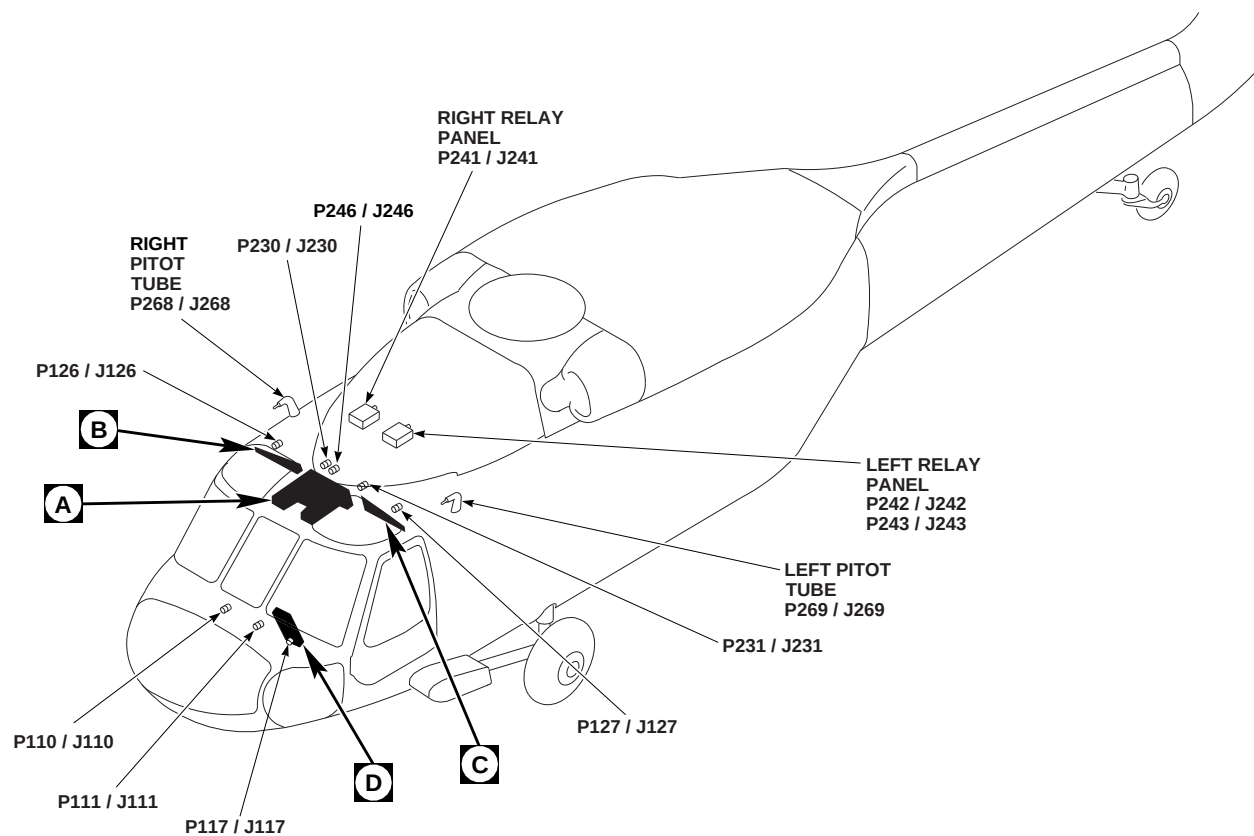
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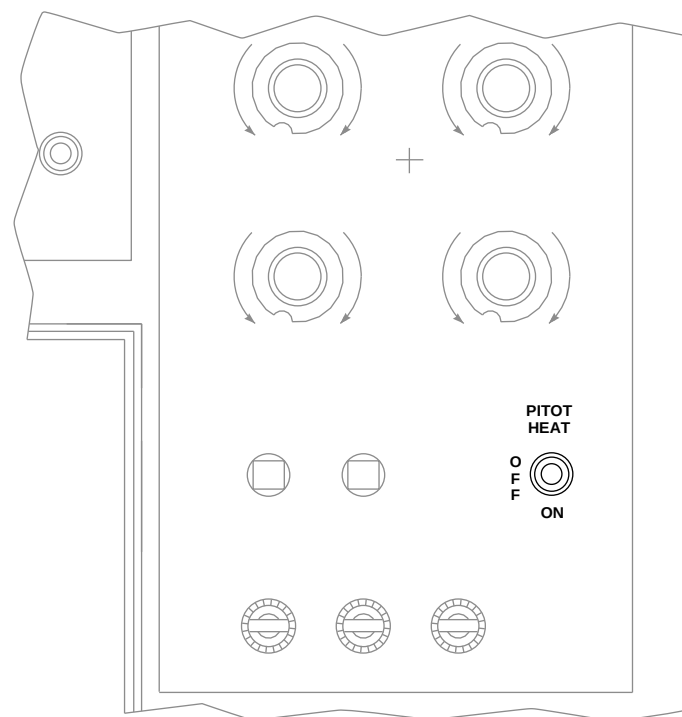
FIGURE 8-1-1-4. PITOT-STATIC PRESSURE SYSTEM BLOCK DIAGRAM



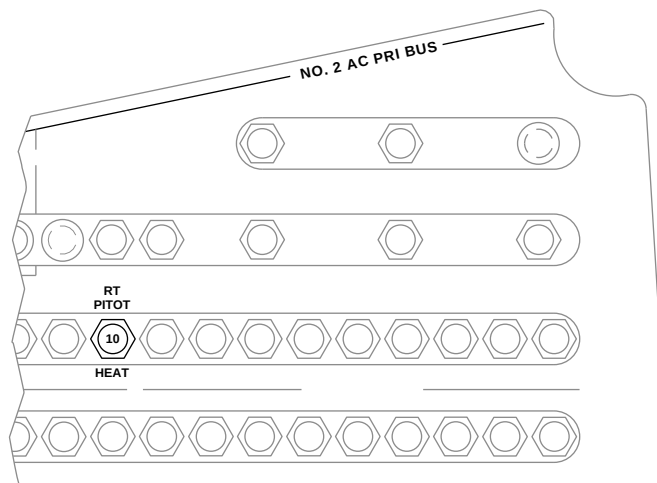
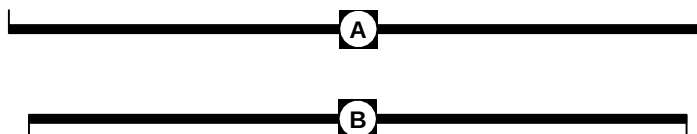
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P110 / J110	COCKPIT BL 7.5 RH, STA 197
P111 / J111	COCKPIT BL 7.5 LH, STA 197
P117 / J117	CAUTION / ADVISORY PANEL
P126 / J126	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 28 RH, STA 243
P127 / J127	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 28 LH, STA 243
P230 / J230	CABIN CEILING BL 8 RH, STA 247
P231 / J231	CABIN CEILING BL 9 LH, STA 247
P241 / J241	RIGHT RELAY PANEL
P242 / J242	LEFT RELAY PANEL
P243 / J243	LEFT RELAY PANEL
P246 / J246	CABIN CEILING BL 5 RH, STA 247
P268 / J268	RIGHT PITOT TUBE
P269 / J269	LEFT PITOT TUBE

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FIGURE 8-1-1-5. PITOT-STATIC HEATER LOCATION DIAGRAM (SHEET 1 OF 3)



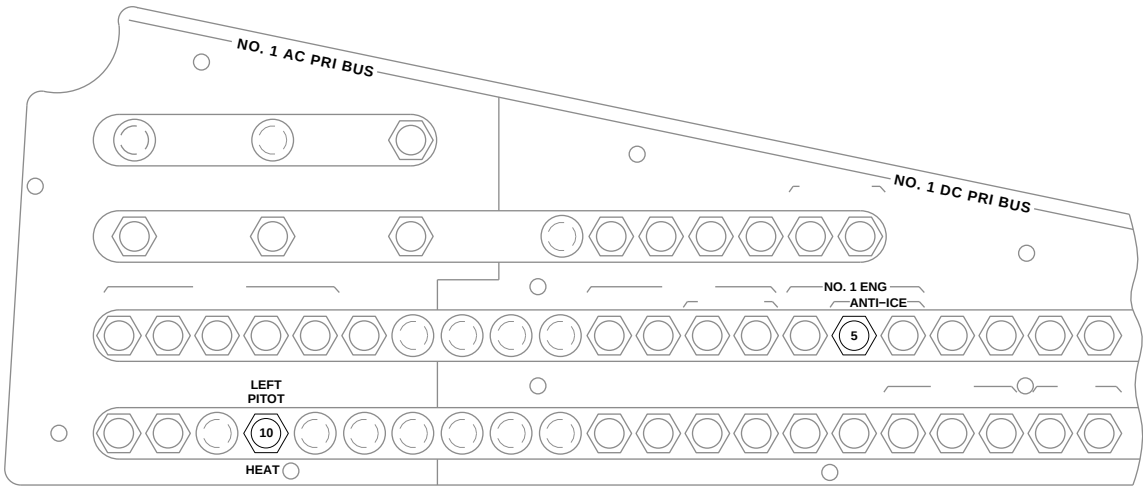
UPPER CONSOLE



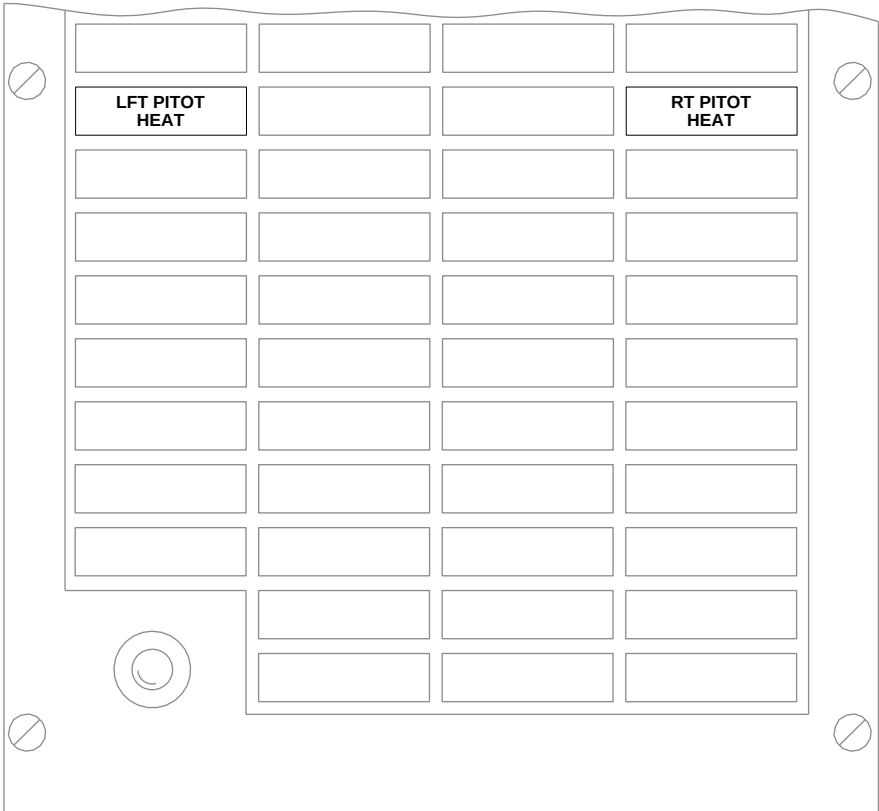
PILOT'S CIRCUIT BREAKER PANEL

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FIGURE 8-1-1-5. PITOT-STATIC HEATER LOCATION DIAGRAM (SHEET 2 OF 3)



COPILLOT'S CIRCUIT BREAKER PANEL



CAUTION/ADVISORY PANEL

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FIGURE 8-1-1-5. PITOT-STATIC HEATER LOCATION DIAGRAM (SHEET 3 OF 3)

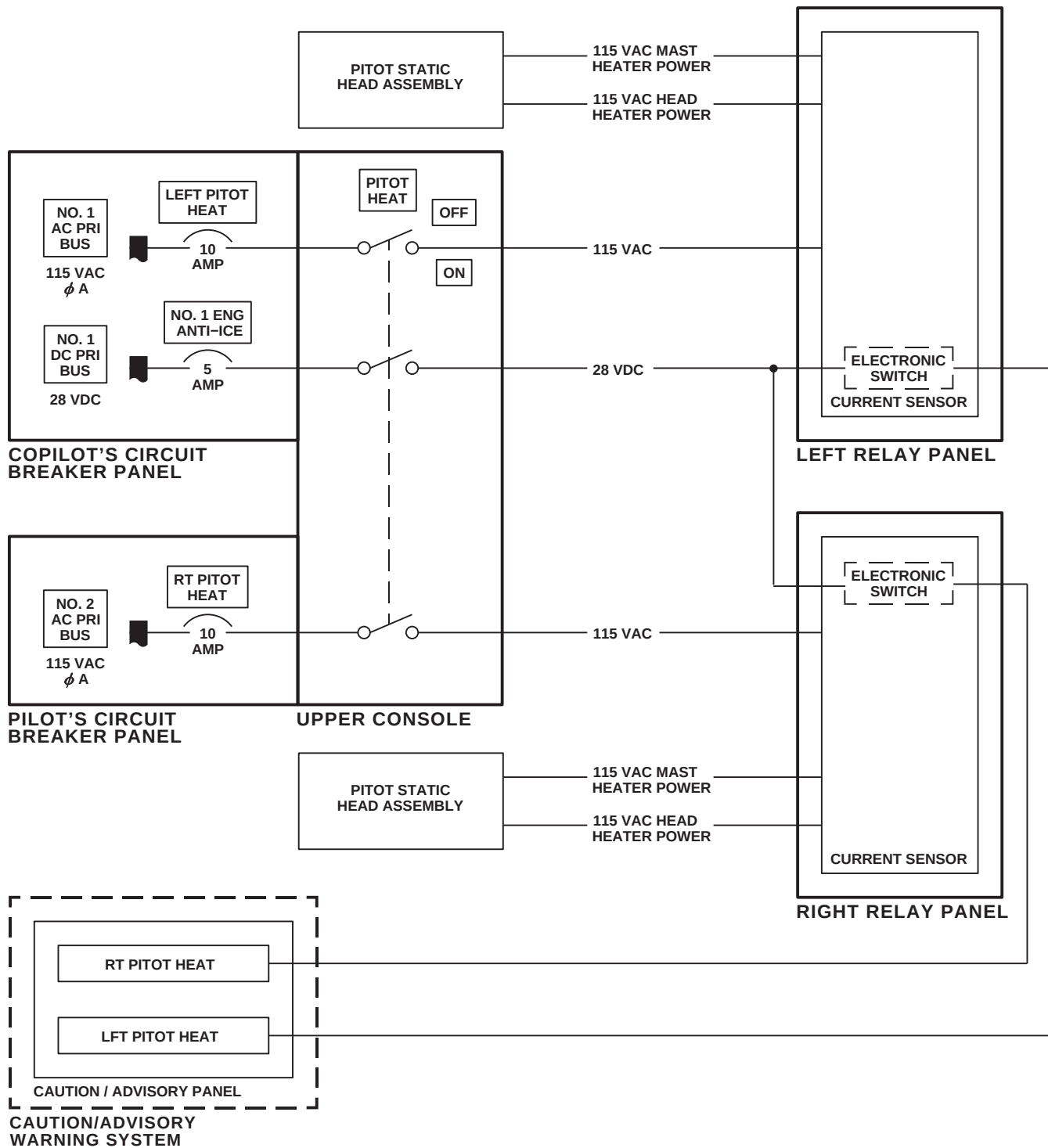


FIGURE 8-1-1-6. PITOT-STATIC HEATER BLOCK DIAGRAM

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8-1-2 MISCELLANEOUS INSTRUMENTS DESCRIPTION AND DATA.

8-1-2.1 Stabilator Position Indicator.

Two Stabilator Position Indicators, one on each side of the instrument panel, provide the pilot and copilot with an indication of stabilator position (Figure 8-1-2-1). The indicator range is marked from 45° DN to 10° UP. When power is lost or removed, the OFF flag will come into view, and the pointer will disappear behind the mask.

8-1-2.2 Stabilator Position Placard.

Two Stabilator Position Placards, one on each side of the instrument panel next to the stabilator position indicators (Figure 8-1-2-1), indicate the maximum allowable indicated airspeed in knots for a given stabilator position. This permits the pilot to keep forward airspeed within safe limits when flying in the manual mode or when the stabilator malfunctions and locks up in a position other than 0°.

8-1-2.3 Standby Magnetic Compass.

The Standby Magnetic Compass, located on the top of the instrument panel and housed in a filled sealed case, indicates heading relative to the magnetic north pole (Figure 8-1-2-2). It has a lubber line, a compass card, and a Permanent Magnet Compensating System. The lubber line is constructed so the parallax will be reduced to a minimum when reading the compass card. The compass card is nonmagnetic and is marked in 5° units. Cardinal headings are shown in enlarged letters: N for north at 0°, E for east at 90°, S for south at 180°, and W for west at 270°. The enlarged numerals 3, 6, 12, 15, 21, 24, 30, and 33 on the compass card indicate 30° units. The compensating system consists of permanent bar magnets that can be manually adjusted for removing compass deviations. The compensator screws are on the front behind a cover plate below the compass card and are marked N-S and E-W. The standby magnetic compass lighting receives power from the DC ESNTL BUS through the LIGHTS SEC PNL circuit breaker on the upper console.

8-1-2.4 Eight-Day Clock.

An Eight-Day, 24 Hour Clock is installed on each side of the instrument panel (Figure 8-1-2-3). The elapsed time knob is on the upper right corner of the clock. The clock is wound and set with a knob on the lower left corner.

8-1-2.5 Digital Clock.

Digital Clocks, replace the Eight-Day Clock on some helicopters. Digital clocks are installed on each side of the instrument panel (Figure 8-1-2-3). The digital clock has a six-digit liquid crystal display, 24 hour numerals, and a seconds indicator. The seconds indicator operates in clock or elapsed mode. The clock contains a replaceable battery that allows continuous timekeeping with helicopter power turned off.

8-1-2.6 Miscellaneous Switch Panel.

The Miscellaneous Switch Panel (MISC SW panel) (Figure 8-1-2-4) consists of:

- FUEL IND TEST Pushbutton Switch
- TAIL WHEEL Pushbutton Switch
- GYRO ERECT Pushbutton Switch
- TAIL SERVO/NORMAL/BACKUP Toggle Switch
- Lighted Information Panel
- Two Electrical Connectors, Rear of Panel

The FUEL IND TEST pushbutton switch, when pressed in and held, tests the Fuel Quantity Indicating System.

The TAIL WHEEL pushbutton switch locks and unlocks the tailwheel.

The GYRO ERECT pushbutton switch initiates a "fast erect" voltage to Attitude Indicating System's vertical gyros.

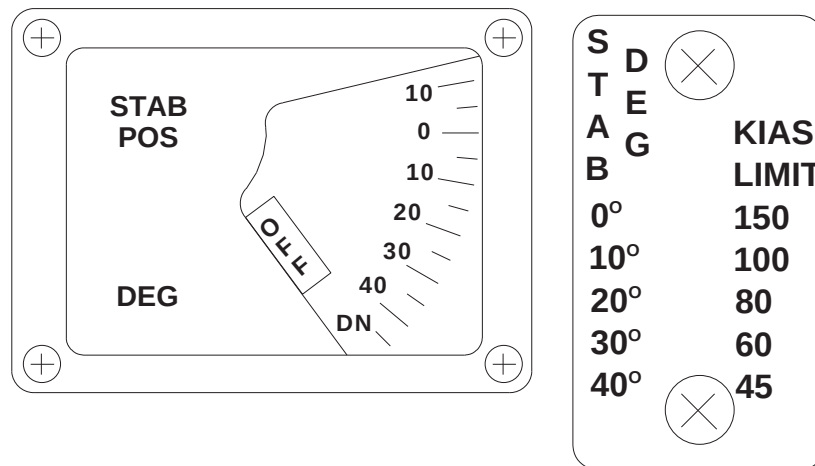
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FIGURE 8-1-2-1. STABILATOR POSITION INDICATOR AND PLACARD

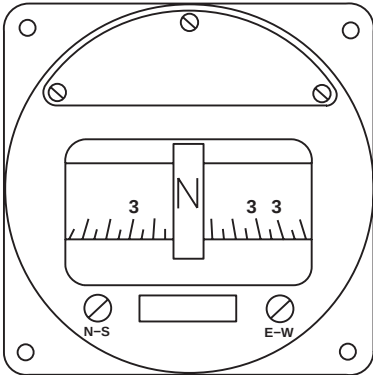
The TAIL SERVO/NORMAL/BACKUP toggle switch controls which Hydraulic System supplies pressure to the tail rotor servo.

8-1-2.7 Free-Air Thermometer.

Two free-air thermometers are installed, one in the left upper window and one in the right upper window of the cockpit (Figure 8-1-2-5). The thermometer is a self-indicating bimetallic instrument that displays the free-air temperature. The thermometer dial is marked from -70 - 40°C in 2° units. The 10° markings are indicated numerals. To avoid parallax, the pointer is mounted close to the dial.

8-1-2.8 Engine Instruments.

The engine instruments give the pilot and copilot indications of engine operating conditions. The engine instruments consist of a Central Display Unit, Copilot's Pilot's Display Unit (PDU), and Pilot's PDU. These three units are components of the Instrument Display System (IDS). Refer to PARA 8-1-3 for complete information on the Instrument Display System (IDS).



SHOWN WITH COVER PLATE REMOVED

FIGURE 8-1-2-2. STANDBY MAGNETIC COMPASS

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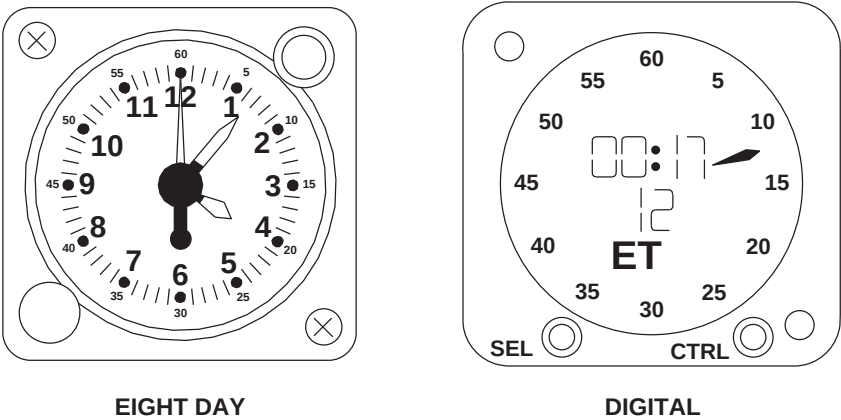
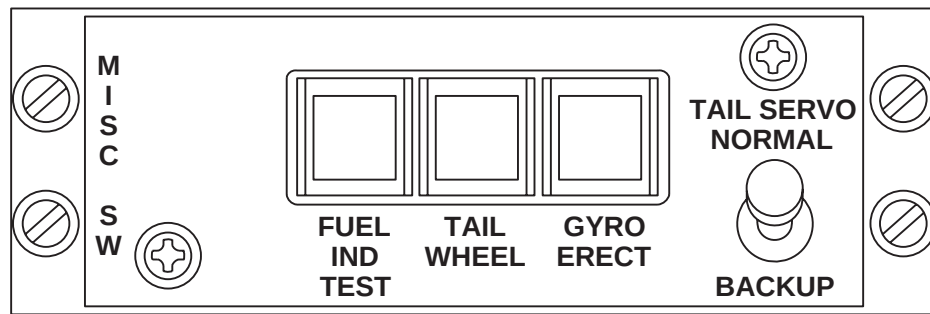


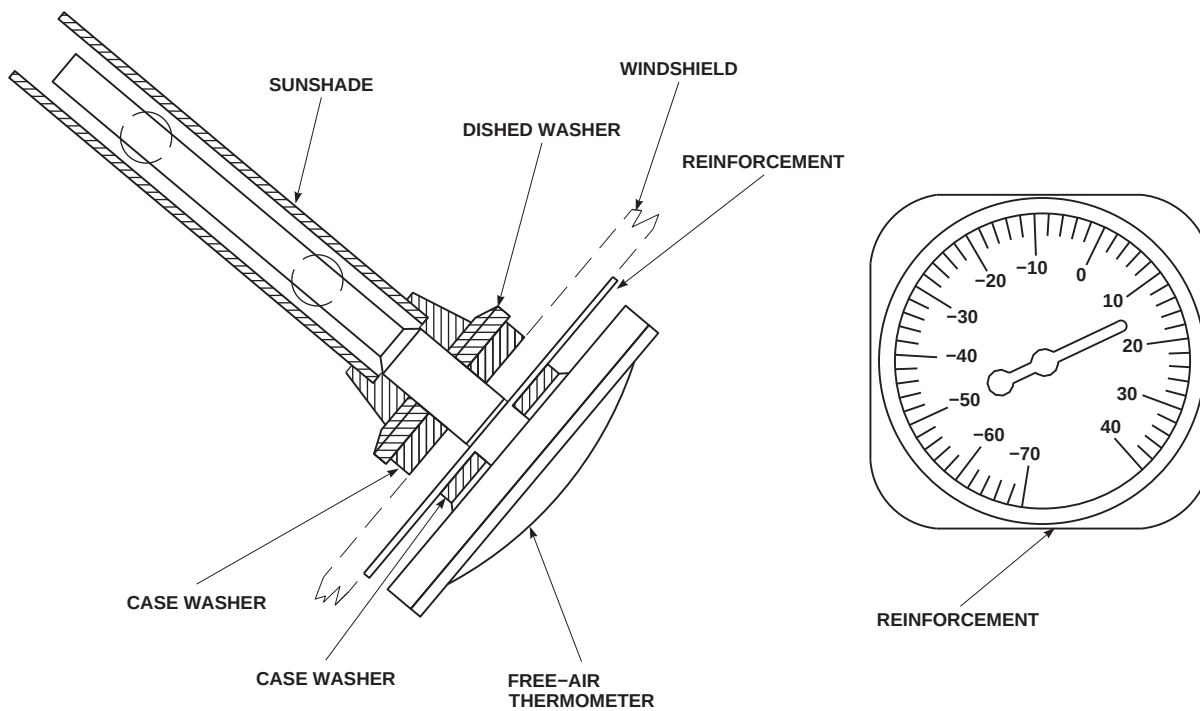
FIGURE 8-1-2-3. CLOCK

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FIGURE 8-1-2-4. MISCELLANEOUS SWITCH PANEL



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FIGURE 8-1-2-5. FREE-AIR THERMOMETER

8-1-3 INSTRUMENT DISPLAY SYSTEM DESCRIPTION AND DATA.

8-1-3.1 System Description.

The instrument display system (IDS), used in conjunction with engine and subsystem sensors (temperature, pressure, torque, fuel and RPM), provides the pilots with engine and subsystem monitoring (Figure 8-1-3-1). The IDS gives continuous indications of the parameters on vertical scales, digital readouts, and status lights. The system consists of a pilot's display unit (PDU), copilot's display unit (CPDU), and central display unit (CDU), on the instrument panel; No. 1 signal data converter (No. 1 SDC) and No. 2 signal data converter (No. 2 SDC), on the nose tunnel shelf; and a rotor overspeed reset switch, in the avionics compartment. Since the PDU and CPDU are identical and the No. 1 and No. 2 SDCs are identical, the IDS consists of three basic units: PDUs, SDCs, and the CDU.

8-1-3.2 Power Distribution.

System electrical power of 28 vdc is supplied by the No. 1 dc primary bus through the NO. 1 DC INST circuit breaker, and by the No. 2 dc primary bus through the NO. 2 DC INST circuit breaker (Figure 8-1-3-2). System electrical power of 115 vac is supplied by the No. 1 ac primary bus through the NO. 1 AC INST circuit breaker and by the No. 2 ac primary bus through the NO. 2 AC INST circuit breaker. The NO. 1 DC INST and NO. 1 AC INST circuit breakers, on the copilot's circuit breaker panel, provide power to the No. 1 SDC. The NO. 2 DC INST and NO. 2 AC INST circuit breakers, on the pilot's circuit breaker panel, provide power to the No. 2 SDC. Each SDC contains a logic power supply that feeds IDS digital and analog processing circuitry. Each SDC also contains a lamp supply that feeds IDS lamp display circuitry.

8-1-3.3 Faceplate Lighting.

Faceplate lighting for the CDU, CPDU, and PDU is controlled by the INST LT NON FLT, INST LT PILOT FLT, and CPLT FLT INST LTS controls on the upper console. CDU faceplate lighting voltage, between 0 and 115 vac, is applied from the INST LT NON FLT control through the No. 1 SDC

to the CDU. CPDU faceplate lighting voltage is applied from the CPLT FLT INST LTS control to the copilot's flight instrument lights 5V/115V transformer. The output voltage from the transformer is routed through the No. 1 SDC to the CPDU. PDU faceplate lighting voltage is applied from the INST LT PILOT FLT control to the pilot's flight instrument lights 5V/115V transformer. The output voltage from the transformer is routed through the No. 2 SDC to the PDU.

8-1-3.4 Sensor Signal Processing.

All engine and subsystem sensor signals monitored by the IDS are applied to the No. 1 or No. 2 SDC. The No. 1 SDC receives all No. 1 engine sensor signals (oil pressure, oil temperature, turbine gas temperature, gas generator tachometer, torque, power turbine tachometer), No. 1 fuel quantity sensor signal, main rotor speed sensor signal, No. 2 engine power turbine tachometer signal, and No. 2 engine torque sensor signal. The No. 2 SDC receives all No. 2 engine sensor signals (oil pressure, oil temperature, turbine gas temperature, gas generator tachometer, torque, power turbine tachometer), No. 2 fuel quantity sensor signal, main rotor speed sensor signal, No. 1 engine power turbine tachometer signal, No. 1 engine torque sensor signal, main transmission oil temperature sensor signal, and main transmission oil pressure sensor signal.

Within each SDC, the associated sensor signals, except for No. 1 and No. 2 fuel quantity, main transmission oil temperature, and main transmission oil pressure, are conditioned to a common digital format for multiplexing. The fuel quantity and main transmission sensor signals are conditioned and multiplexed within the CDU. After the sensor signals have been conditioned and multiplexed, the sensor data is routed to latching circuits in the CDU, PDU and CPDU. The latching circuits retain the last signal data until it is time to update. During update (twice per second), the latches activate lamp drivers that energize miniature lamps on the edge of the display modules. Light from the lamps is carried to the display panel face by fiber optic strips, giving visual

analog and digital displays corresponding to the level of the sensed parameter.

8-1-3.5 Display Lighting Control.

Each SDC contains a lamp power supply that limits the light intensity of the IDS displays. The No. 1 SDC lamp power supply provides voltage to all CPDU displays, alternate lamps on the CDU analog displays, and No. 1 engine and total fuel CDU digital displays. The No. 2 SDC lamp power supply provides voltage to all PDU displays, alternate lamps on the CDU analog display, and No. 2 engine CDU digital displays. The output voltages of the lamp power supplies are determined by three photocell outputs and the DIM control on the CDU. The three photocells, one on each display unit, sense the surrounding light level. The photocell sensing the highest level of light controls the input to the lamp power supply. The DIM control is a gain adjustment for the lamp power supply and sets the display lighting contrast level to be maintained by the photocells. Turning the DIM control clockwise, past the detent, sets the display lighting to a fixed preset level and disables the three photocells. The control loops of both lamp power supplies are tied together so both lamp supplies provide the same output voltage.

Main transmission oil pressure, No. 1 engine oil pressure, No. 2 engine oil pressure, No. 1 fuel quantity, No. 2 fuel quantity, and engine and rotor speed analog displays have low scale segments colored red and/or amber to indicate undesirable operating ranges. When the scale reading is above this low range, the IDS automatic bottom segment turn-off circuitry causes the bottom segment lamps to go off. If the display reading drops back into the undesirable operating range, the bottom segment lamps go on again.

8-1-3.6 System Interface.

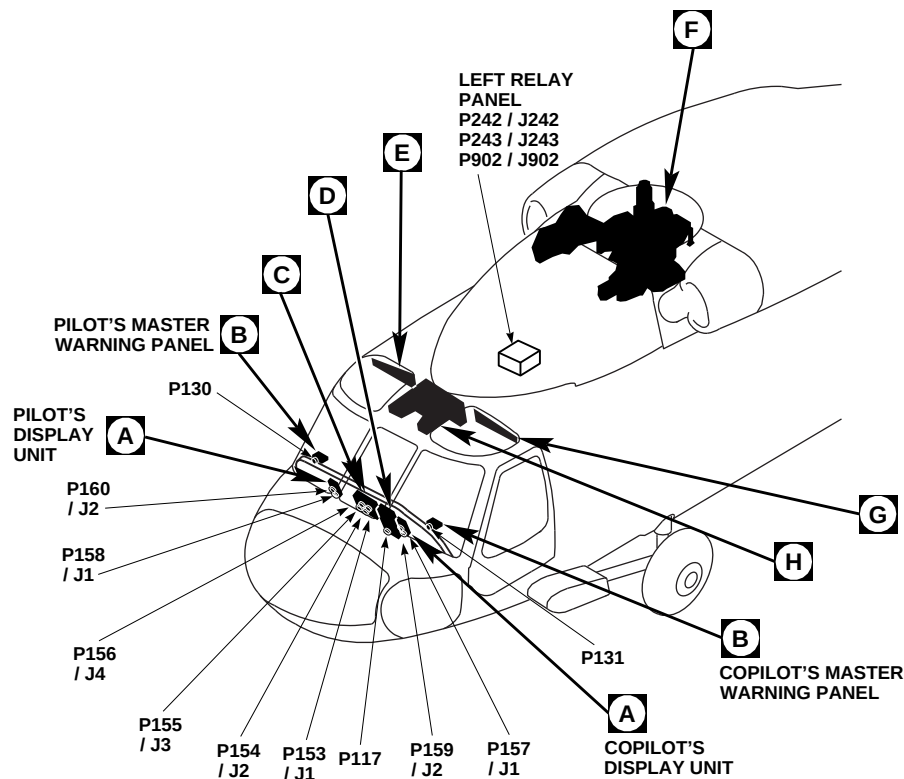
The IDS provides control voltages for these helicopter status capsules:

- (1) #1 ENG OUT

- (2) #2 ENG OUT
- (3) LOW ROTOR RPM
- (4) #1 ENGINE OIL PRESS
- (5) #1 ENGINE OIL TEMP
- (6) #2 ENGINE OIL PRESS
- (7) #2 ENGINE OIL TEMP

The #1 ENG OUT, #2 ENG OUT, and LOW ROTOR RPM capsules are on the pilot's and copilot's master warning panels, on the instrument panel. The remaining capsules are on the caution/advisory panel, also on the instrument panel. The #1 ENG OUT capsules will be on whenever the No. 1 engine gas generator tachometer (1 Ng SPEED % X 10) is less than 55%. The #2 ENG OUT capsules will be on whenever the No. 2 engine gas generator tachometer (2 Ng SPEED % X 10) is less than 55%. The LOW ROTOR RPM capsule will be on whenever the main rotor speed (RTR) is less than 95%. The #1 ENGINE OIL PRESS capsule will be on whenever the No. 1 engine oil pressure (1 ENG OIL PRESS PSI X 10) is less than 25 psi. The #1 ENGINE OIL TEMP capsule will be on whenever the No. 1 engine oil temperature (1 ENG OIL TEMP °C X 10) is more than 150°C. The #2 ENGINE OIL PRESS capsule will be on whenever the No. 2 engine oil pressure (2 ENG OIL PRESS PSI X 10) is less than 25 psi. The #2 ENGINE OIL TEMP capsule will be on whenever the No. 2 engine oil temperature (2 ENG OIL TEMP °C X 10) is more than 150°C.

The control voltages provided by the IDS to the LOW ROTOR RPM and #1 and #2 ENG OUT capsules also control low rotor RPM and engine-out audible warning signals. The IDS signals energize a relay in the left relay panel that controls an audible warning circuit. For a further description of the audible warning circuit, refer to the Caution/Advisory Warning System (PARA 8-1-7).

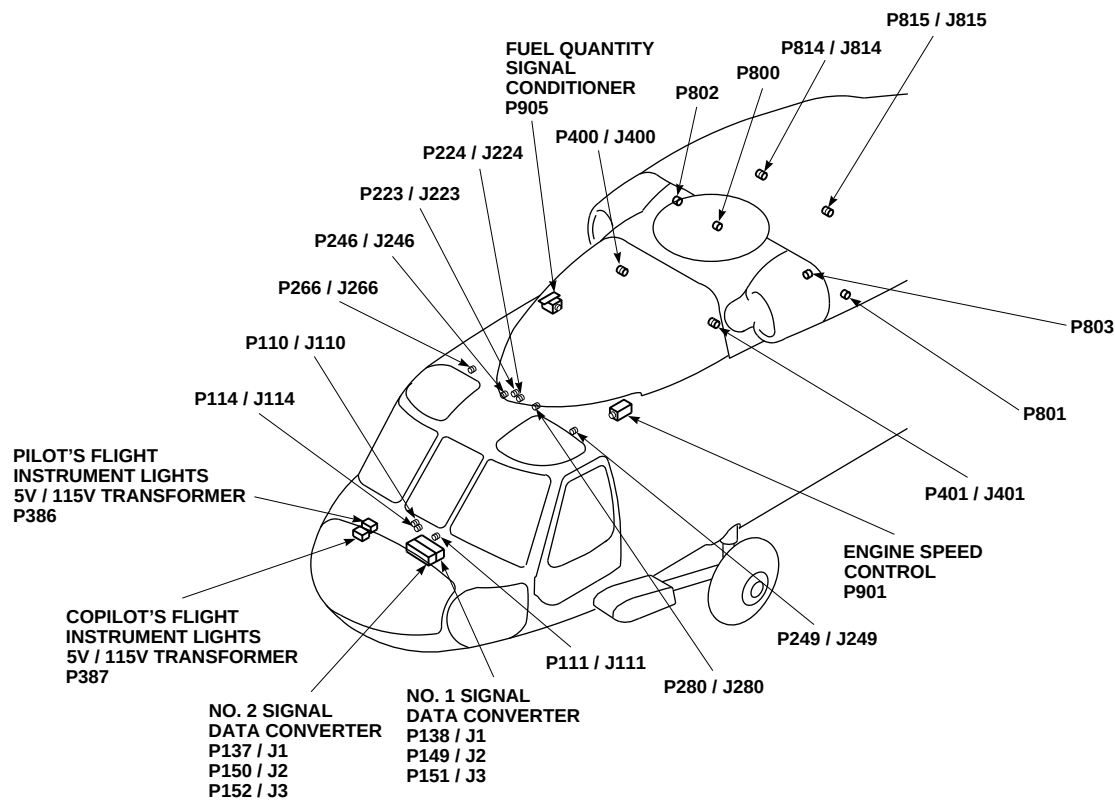


NOTES	
1.	S70-I
2.	S70-II

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P110 / J110	COCKPIT BL 7.5 RH, STA 197
P111 / J111	COCKPIT BL 7.5 LH, STA 197
P114 / J114	COCKPIT BL 8 RH, STA 200
P117	CAUTION / ADVISORY PANEL
P130	PILOT'S MASTER WARNING PANEL
P131	COPILOT'S MASTER WARNING PANEL
P137 / J1	NO. 2 SIGNAL DATA CONVERTER
P138 / J1	NO. 1 SIGNAL DATA CONVERTER
P149 / J2	NO. 1 SIGNAL DATA CONVERTER
P150 / J2	NO. 2 SIGNAL DATA CONVERTER
P151 / J3	NO. 1 SIGNAL DATA CONVERTER
P152 / J3	NO. 2 SIGNAL DATA CONVERTER
P153 / J1	CENTRAL DISPLAY UNIT
P154 / J2	CENTRAL DISPLAY UNIT

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FIGURE 8-1-3-1. INSTRUMENT DISPLAY SYSTEM LOCATION DIAGRAM (SHEET 1 OF 6)



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P155 / J3	CENTRAL DISPLAY UNIT
P156 / J4	CENTRAL DISPLAY UNIT
P157 / J1	COPILOT'S DISPLAY UNIT
P158 / J1	PILOT'S DISPLAY UNIT
P159 / J2	COPILOT'S DISPLAY UNIT
P160 / J2	PILOT'S DISPLAY UNIT
P223 / J223	COCKPIT CEILING BL 3 LH, STA 243
P224 / J224	COCKPIT CEILING BL 3 RH, STA 245
P242 / J242	LEFT RELAY PANEL
P243 / J243	LEFT RELAY PANEL
P246 / J246	CABIN CEILING BL 5 RH, STA 247
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH, STA 247
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 25 RH, STA 247
P280 / J280	CABIN CEILING BL 6 LH, STA 247
P386	PILOT'S FLIGHT INSTRUMENT LIGHTS 5V / 115V TRANSFORMER

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P387	COPILOT'S FLIGHT INSTRUMENT LIGHTS 5V / 115V TRANSFORMER
P400 / J400	CABIN CEILING BL 24 RH, STA 318
P401 / J401	CABIN CEILING BL 24 LH, STA 318
P412	MAIN XMSN OIL TEMPERATURE SENSOR
P443	MAIN ROTOR SPEED SENSOR
P444	MAIN XMSN OIL PRESSURE SENSOR
P800	NO. 2 ENGINE
P801	NO. 1 ENGINE
P802	NO. 2 ENGINE
P803	NO. 1 ENGINE
P814 / J814	ENGINE FIREWALL BL 14 RH, STA 373
P815 / J815	ENGINE FIREWALL BL 14 LH, STA 373
P901	ENGINE SPEED CONTROL
P902 / J902	LEFT RELAY PANEL
P905	FUEL QUANTITY SIGNAL CONDITIONER

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FIGURE 8-1-3-1. INSTRUMENT DISPLAY SYSTEM LOCATION DIAGRAM (SHEET 2 OF 6)

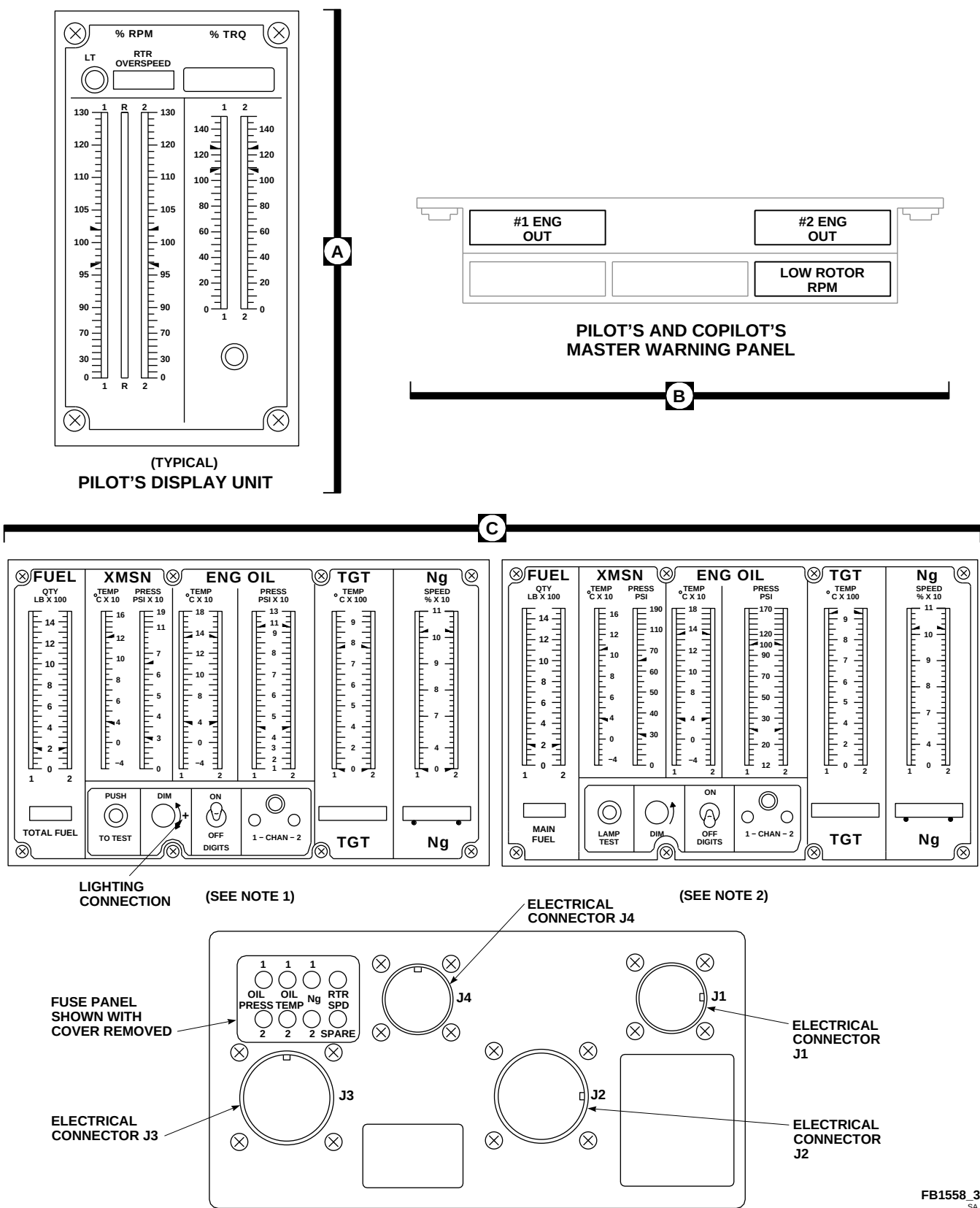
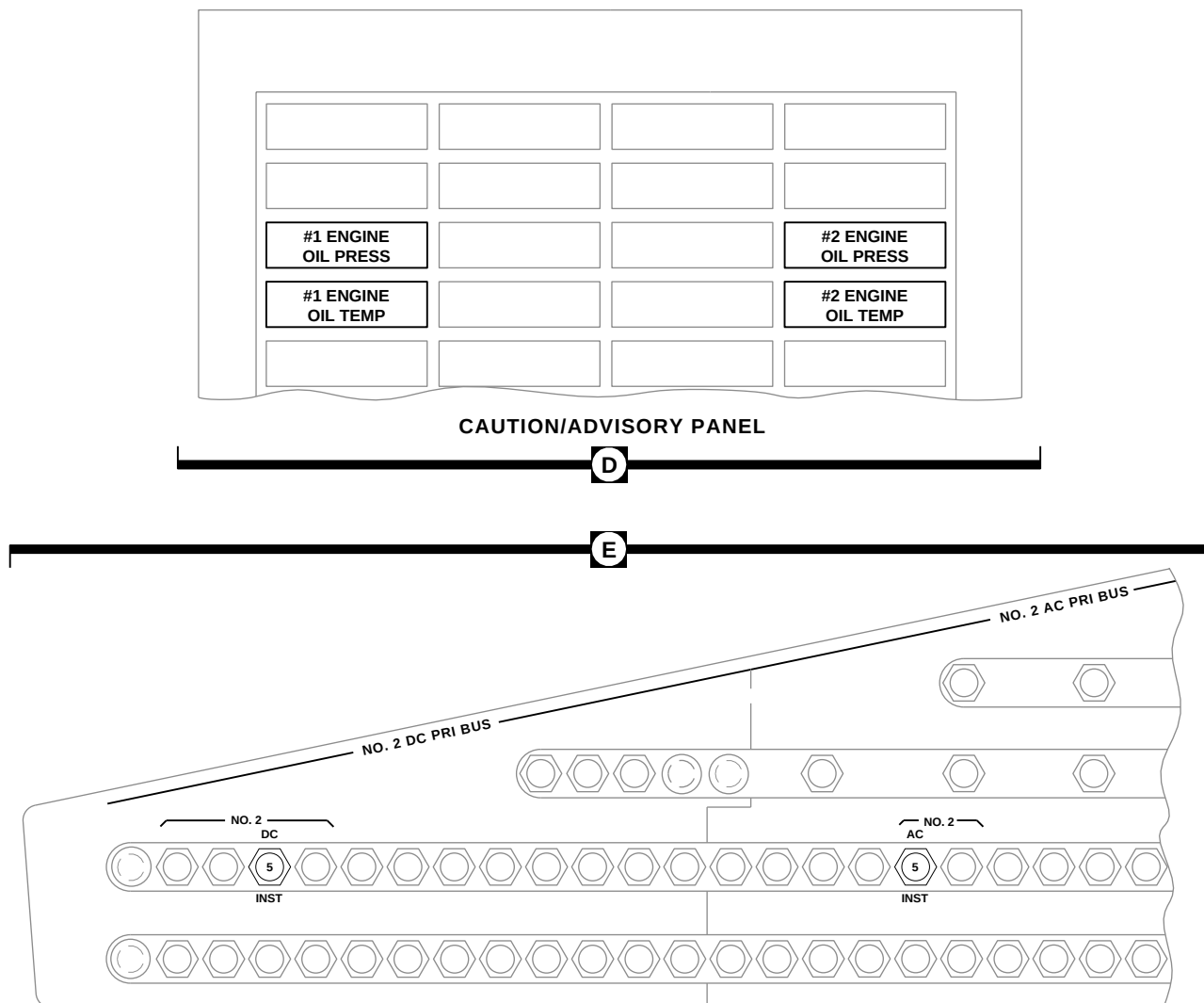


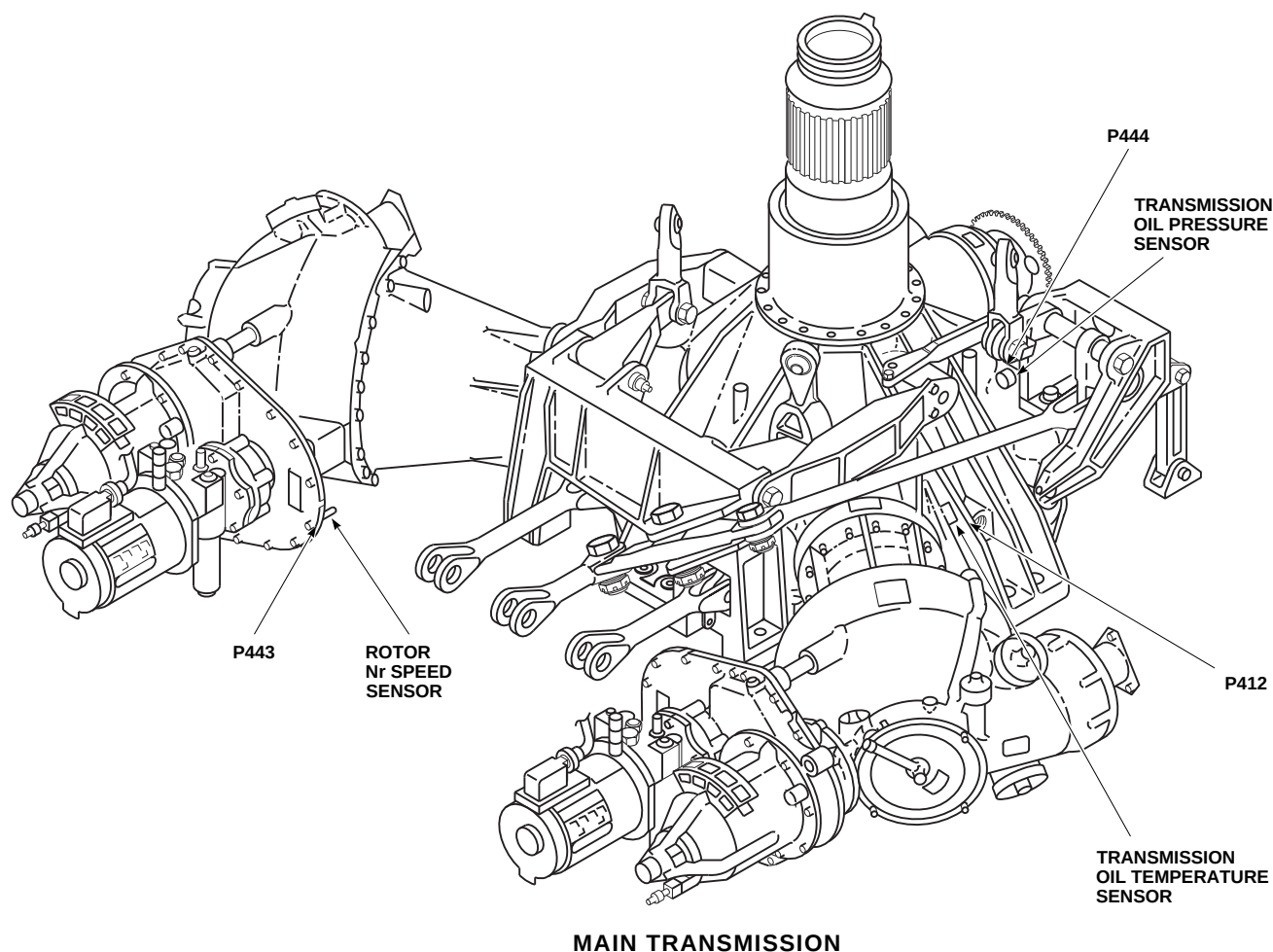
FIGURE 8-1-3-1. INSTRUMENT DISPLAY SYSTEM LOCATION DIAGRAM (SHEET 3 OF 6)



PILOT'S CIRCUIT BREAKER PANEL

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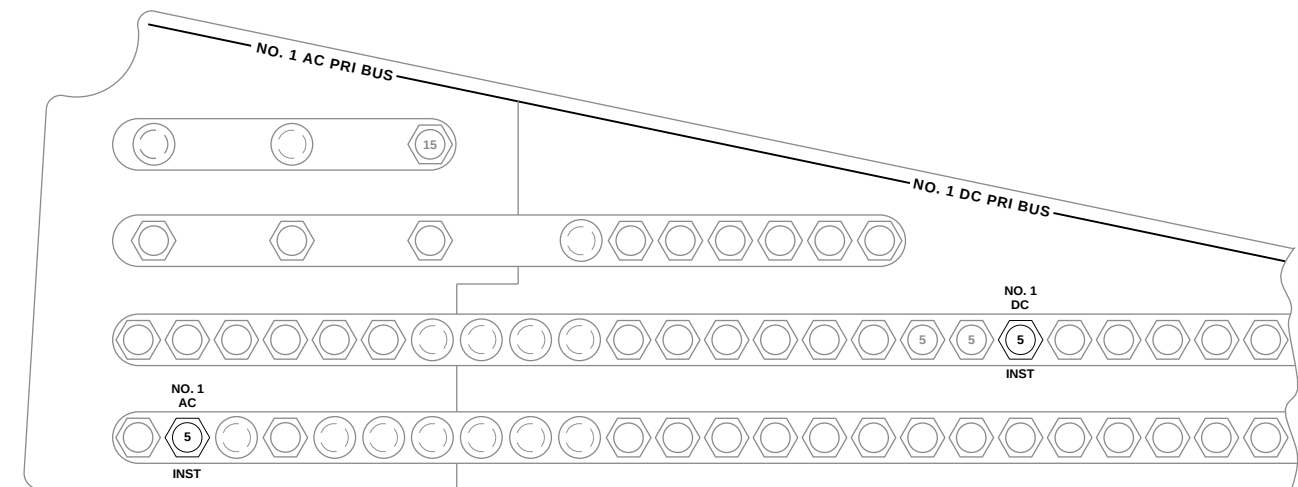
FIGURE 8-1-3-1. INSTRUMENT DISPLAY SYSTEM LOCATION DIAGRAM (SHEET 4 OF 6)



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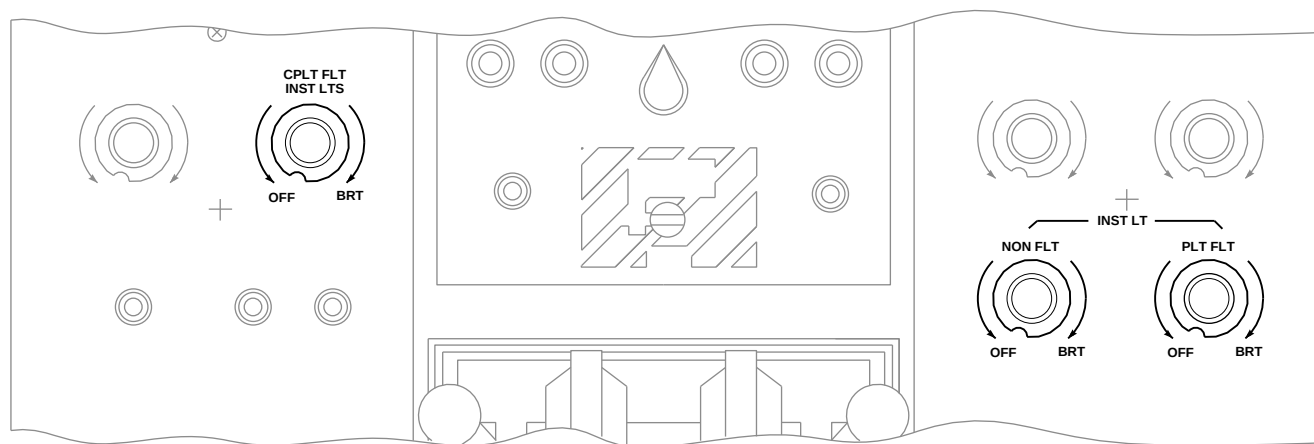
FIGURE 8-1-3-1. INSTRUMENT DISPLAY SYSTEM LOCATION DIAGRAM (SHEET 5 OF 6)



COPLOT'S CIRCUIT BREAKER PANEL

G

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UPPER CONSOLE

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FIGURE 8-1-3-1. INSTRUMENT DISPLAY SYSTEM LOCATION DIAGRAM (SHEET 6 OF 6)

8-1-3-8

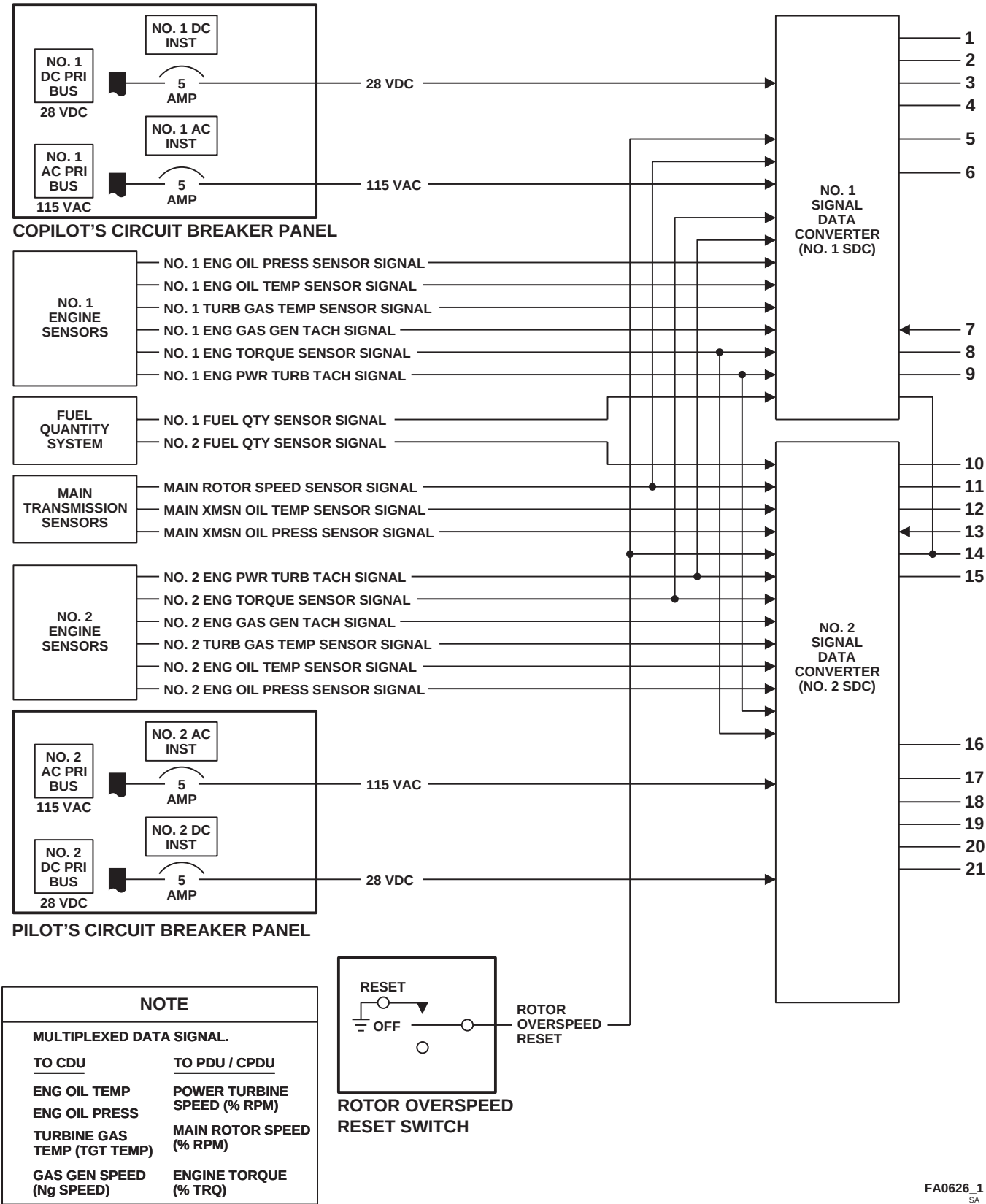


FIGURE 8-1-3-2. INSTRUMENT DISPLAY SYSTEM BLOCK DIAGRAM (SHEET 1 OF 2)

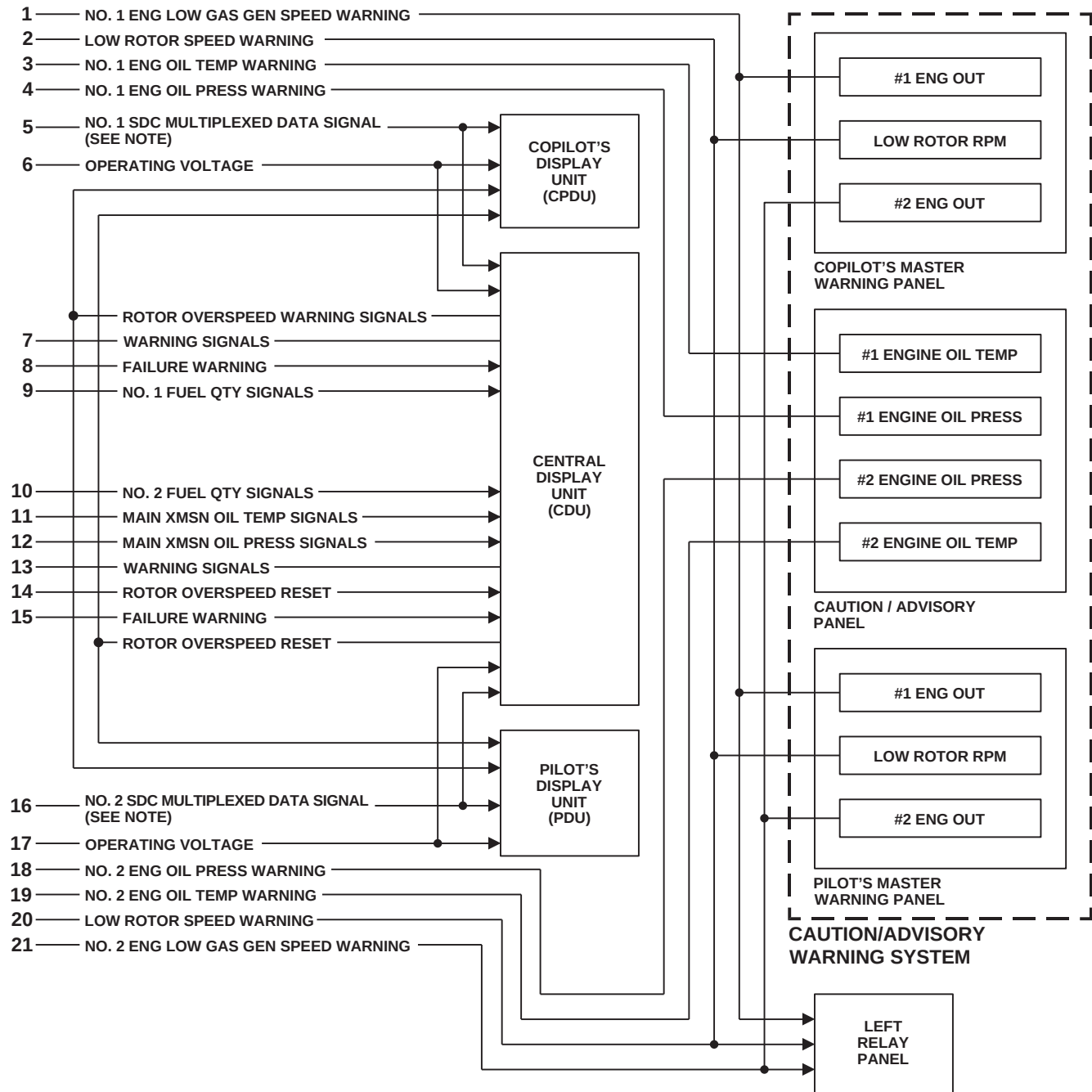
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FIGURE 8-1-3-2. INSTRUMENT DISPLAY SYSTEM BLOCK DIAGRAM (SHEET 2 OF 2)

8-1-4 SIGNAL DATA CONVERTER DESCRIPTION AND DATA.

8-1-4.1 Signal Data Converter Description.

The signal data converter (SDC) processes a variety of helicopter sensor signals and provides proportional digital signals in a multiplexed format to the central display unit (CDU) and the pilot's display unit (PDU) (Figure 8-1-4-1). The CDU and PDU provide visual indications of the processed sensor signals. The SDC operates on helicopter 115 vac, 400 Hz and 28 vdc power. The SDC contains the circuitry to process the following helicopter sensor signals: engine oil pressure, engine oil temperature, No. 1 and No. 2 engine torque, engine turbine gas temperature (TGT), No. 1 and No. 2 engine power turbine tachometer (% RPM), engine gas generator tachometer (Ng), and rotor RPM. The SDC updates this data twice per second. The fuel quantity, main transmission oil temperature, and main transmission oil pressure signals are routed through the SDC and processed in the CDU. The more important signals are processed through redundant circuits for reliability. Monitoring circuits are used to ensure the validity of the displayed data. The SDC also provides output voltages to the helicopter caution/advisory warning system when any of the following conditions exists: low engine oil pressure, high engine oil temperature, low engine Ng, and low rotor speed. There are three external electrical connectors on the SDC: J1 which provides input/output interface with helicopter power and sensors, J2 which provides signal and power input/output interface with the PDU, and J3 which provides signal and power input/output interface with the CDU. The SDC contains the following modules:

- (a) Lamp power supply (A2)
- (b) Logic power supply (A3)
- (c) Interface No. 4 module (A4)
- (d) Interface No. 3 module (A5)
- (e) Interface No. 2 module (A6)
- (f) Interface No. 1 module (A7)
- (g) Analog processor No. 2 module (A8)

- (h) Analog processor No. 1 module (A9)
- (i) Digital processor module (A10)
- (j) Voltage regulator module (A11)

8-1-4.2 Lamp Power Supply (A2).

The lamp power supply A2 is a high-current, low-voltage unit (Figure 8-1-4-2). External power of 28 vdc and unregulated 15.5 vdc and -15.5 vdc are provided to the lamp power supply. The lamp supply control voltage is provided from the voltage regulator module. The output of the lamp power supply can be varied between 200 millivolts and 6 vdc determined by the control voltage input.

8-1-4.3 Logic Power Supply (A3).

The logic power supply receives 115 vac, 400 Hz power from the helicopter power system (Figure 8-1-4-2). The power supply uses a step-down multi-secondary winding transformer and conventional diode rectifiers to generate various dc voltages and ac excitation for the engine and main transmission oil pressure sensors.

8-1-4.4 Interface No. 4 Module (A4).

The interface No. 4 module A4 contains engine oil pressure interface circuitry with its -5 vdc reference voltage regulator, a floating 5 vdc regulator for the temperature compensating unit, and turbine gas temperature interface circuitry (Figures 8-1-4-2 and 8-1-4-3). A 10 vac, 400 Hz reference voltage from the logic power supply is used with the engine oil pressure sensor signal (400 Hz) to generate a dc analog signal proportional to the pressure. The dc analog voltage is applied to the A/T converter which produces the engine oil pressure update pulse when enabled by the multiplexer signal from the analog processor. The analog reset signal generated by the analog processor resets the A/T converter. The update pulse is applied to the engine oil pressure lamp driver module in the CDU. A floating 8 vdc is supplied to the interface No. 4 module from the logic power supply. This input is applied to a voltage regulator that generates 5 vdc output to the temperature compensating unit. The

output of the thermocouple probes that sense the TGT is applied to the interface circuitry through a temperature compensating unit. A 5 vdc regulator supplies a floating output to the temperature compensating unit which compensates for ambient temperature changes. The conditioned thermocouple signal is applied to the analog and digital A/T converters which supply TGT analog and digital update pulses to the CDU. The TGT circuitry also receives analog reset and multiplex signals generated by the analog processor and digital reset and multiplex signals generated by the digital processor. The interface No. 4 module receives -10 vdc, 10 vdc, and 5 vdc operating voltages from the voltage regulator module.

8-1-4.5 Interface No. 3 Module (A5).

The interface No. 3 module A5 contains voltage regulators that provide 11 vdc, -11 vdc, and -5 vdc, and interface circuitry for No. 1 and No. 2 engine torque signals (Figures 8-1-4-2 and 8-1-4-3). The 11 vdc and -11 vdc regulators operate from the 14.5 vdc and -14.5 vdc floating input from the logic power supply. The ground isolated -5 vdc regulator operates from the -11 vdc regulator. The -5 vdc is used as a reference voltage for the A/T converters. Since the engine control unit (ECU) is floating, all reference voltages must also be isolated. The torque sensor signals are conditioned by the ECU and applied to buffer amplifiers. The outputs of the buffers are applied to differential amplifiers. The differential amplifiers provide signals to the A/T analog and digital converters which produce updated analog and digital No. 1 and No. 2 engine torque signals to both PDUs. The No. 1 and No. 2 engine torque circuits receive reset and multiplex signals generated by the analog and digital processors. The interface No. 3 module receives -10 vdc, 10 vdc, and 5 vdc from the voltage regulator module.

8-1-4.6 Interface No. 2 Module (A6).

The interface No. 2 module contains interface circuitry for the engine oil temperature, No. 1 engine power turbine tachometer signal (% RPM 1) and No. 2 engine power turbine tachometer signal (% RPM 2) (Figures 8-1-4-2 and 8-1-4-3). The oil

temperature sensor provides a variable resistance input. This resistance forms part of the feedback loop of an operational amplifier. The dc voltage output is proportional to the oil temperature and is applied to the A/T converter which produces an update engine oil temperature signal for the CDU. The engine oil temperature circuits receive analog reset and multiplex signals generated by the analog processor. The 1.28 MHz reference frequency is used to convert both No. 1 and No. 2 power turbine tachometer signals to a dc analog voltage proportional to the sensor input frequency. There are identical circuits for the engine No. 1 and engine No. 2 sensors. The dc analog signals are fed to the A/T converters which produce update % RPM 1 and % RPM 2 signals to the PDU. The engine No. 1 and No. 2 circuits receive reset and multiplex signals generated by the analog processor. The interface No. 2 module receives -10 vdc, and 10 vdc, and 5 vdc operating power from the voltage regulator module.

8-1-4.7 Interface No. 1 Module (A7).

The interface No. 1 module A7 contains interface circuitry for the main rotor speed and Ng signal inputs (Figures 8-1-4-2 and 8-1-4-3). The 1.28 MHz reference voltage from the digital processor module is used with the main rotor speed signal to generate a dc analog signal proportional to the input frequency. The dc analog signal is fed to the A/T converter which produces an update rotor speed signal to both the PDU and the CDU. The rotor speed circuits receive an analog reset pulse and rotor speed multiplex signal generated by the analog processor. The 1.28 MHz reference frequency is also used to convert the gas generator tachometer signal (Ng SPEED) to a dc analog voltage proportional to the sensor input frequency. The dc analog signal is fed to the A/T analog converter and the A/T digital converter. The A/T analog converter receives an analog reset signal and Ng SPEED analog multiplex signal generated by the analog processor and sends an Ng SPEED analog update signal to the CDU for the analog display. The A/T digital converter receives a digital reset signal and Ng SPEED multiplex signal generated by the digital processor and sends an Ng SPEED digital update pulse to the CDU for the digital display.

play. The interface No. 1 module receives -10 vdc, 10 vdc, and 5 vdc operating power from the voltage regulator module.

8-1-4.8 Analog Processor No. 2 Module (A8).

The analog processor No. 2 module A8 contains the last stages of the A and B shift registers, overflow counters, circuitry to generate reset pulses, decoding circuitry to generate multiplex frame pulses, and failure warning detectors (Figures 8-1-4-2 and 8-1-4-3). The No. 2 module receives the 12.8 kHz clock signal and the output of the last stage of the A and B registers from the analog processor No. 1 module. The A and B monitor signals from the analog processor No. 1 module are combined with the monitor signals in the analog processor No. 2 module to generate the analog A or analog B select signal. The select gates are inhibited when a failure warning signal is received from the digital processor. Shift register reset pulses are produced and sent to the No. 1 analog processor. In addition, the analog reset pulse is produced which resets the A/T converters after amplification by the voltage regulator. Decoding circuits decode the output of a counter to produce the analog multiplex signal that controls the sequence in which the various analog signals are enabled for updating in the interface circuitry.

8-1-4.9 Analog Processor No. 1 Module (A9).

The analog processor No. 1 module A9 contains two shift registers, an A register and a B register (Figures 8-1-4-2 and 8-1-4-3). It also contains the AND/OR select gates which connect either the A or the B register to the output for the analog data word to the PDU and the CDU. The output of the last stage of the shift registers is connected to the No. 2 analog processor module. The shift registers are clocked by a 128 kHz signal from the digital processor. The output of the last stage is also fed to a monitor circuit for each register. The output of the monitors is fed to the analog processor No. 2 module. The A or B shift register select signal is received from the analog processor No. 2 module. The analog processor No. 1 module receives 5 vdc operating power from the voltage regulator module.

8-1-4.10 Digital Processor Module (A10).

The digital processor module A10 generates clock signals for both digital and analog processors and digital multiplexing signals (Figures 8-1-4-2 and 8-1-4-3). In addition, it provides binary coded decimal (BCD) information for use on the digital displays in the CDU and PDU. Two crystal-controlled oscillators generate 1.28 MHz frequencies in a redundant configuration. The 1.28 MHz is used in the No. 1 and No. 2 interface modules. The 1.28 MHz signal is divided down to provide the analog and digital clock signals. Redundant decade counters A and B produce BCD data from the 128 kHz clock signal. The last stages of the counter are used to generate a reset pulse, multiplex sequencing pulses, digital test frequency, and a monitor input signal. The output of the monitor provides the digital select A or digital select B signal. The digital reset pulse is fed to the voltage regulator for amplification. Analog clock inhibit signals are received from the PDU and the CDU and are connected through circuitry to the analog processor No. 1 module. Digital clock inhibit signals are received from the CDU and PDU for use in the digital processor. These signals stop the clock pulses during update. This prevents any clock inputs to either the analog or digital processors from being processed during transfer of data into the displays.

8-1-4.11 Voltage Regulator Module (A11).

The voltage regulator module contains three regulators (Figures 8-1-4-2 and 8-1-4-3). These are -10 volt, 10 volt, and 5 volt. In addition, there are buffer amplifiers for the analog and digital reset signals and the error amplifier of the lamp power supply control. The 10 vdc regulator operates on 15.5 vdc input power. The 15.5 vdc is also fed out to the lamp power supply. The -10 vdc regulator operates on -15.5 vdc input power. The -15.5 vdc is also fed out to the lamp power supply. The 5 vdc regulator operates on 8 vdc power. The 8 vdc is also sent to the PDU. The analog reset is a narrow positive pulse generated on the analog processor No. 2 module. This pulse is used to reset all the A/T converters synchronously with the resetting of the analog processor. The voltage regulator module amplifies this pulse for use in the SDC and the CDU. A similar circuit does the same for the digital

reset signal from the digital processor. The lamp power supply error amplifier receives a control signal from the auto dim circuit in the CDU and the

lamp power supply output. Operational amplifiers compare the two signals and produce a lamp supply control signal for the lamp power supply.

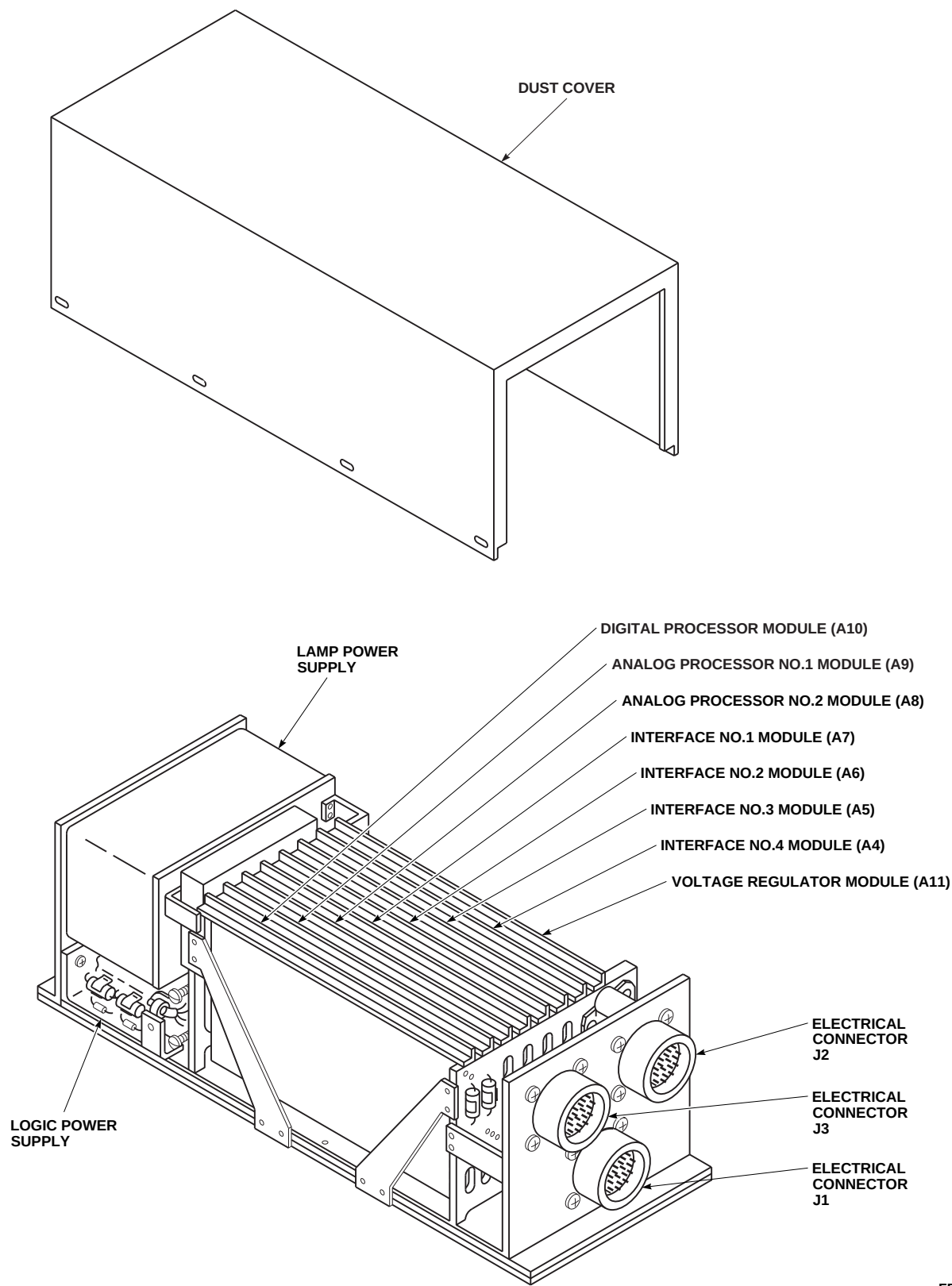


FIGURE 8-1-4-1. SIGNAL DATA CONVERTER PARTS LOCATION

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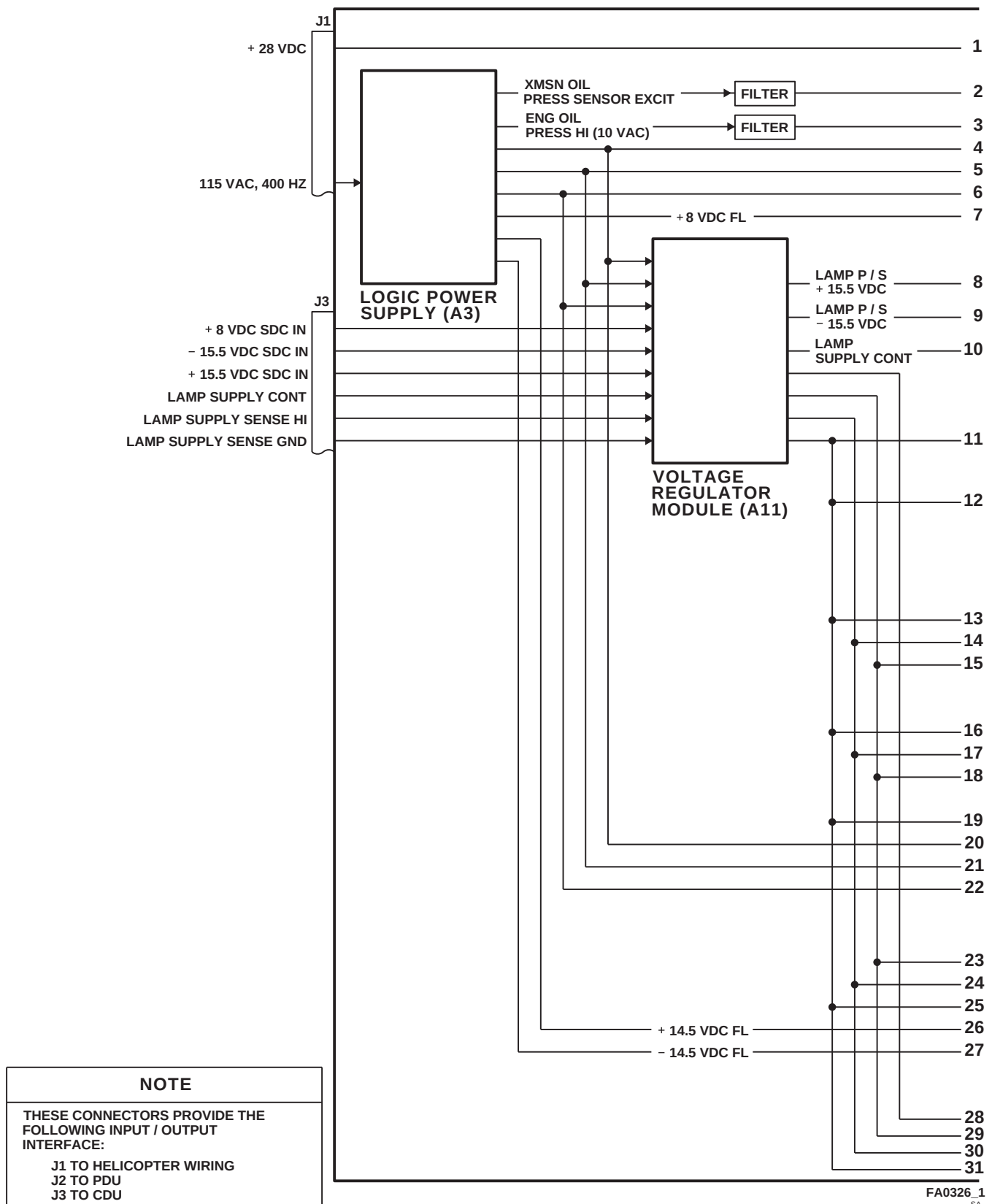
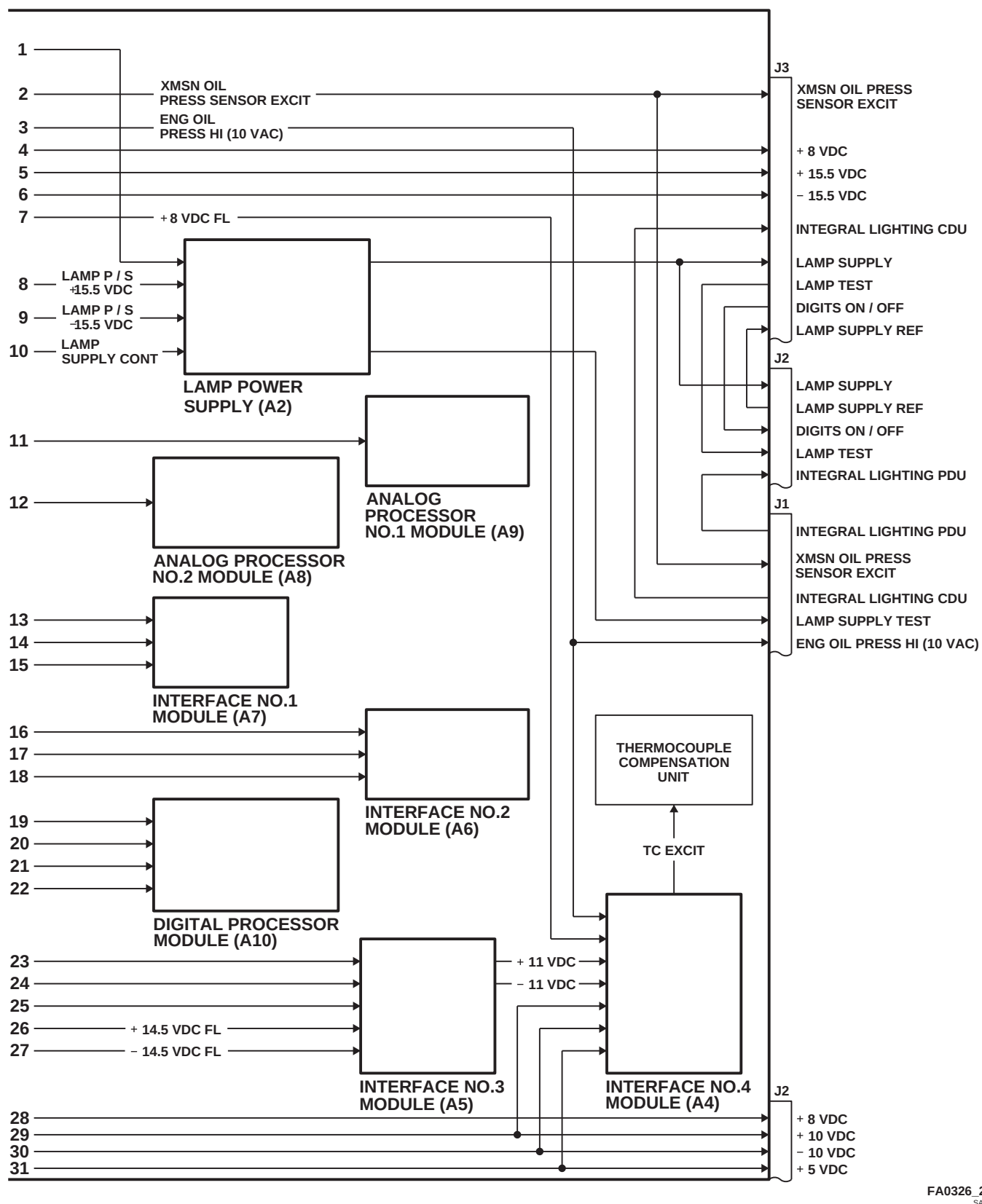


FIGURE 8-1-4-2. SIGNAL DATA CONVERTER POWER DISTRIBUTION (SHEET 1 OF 2)



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FIGURE 8-1-4.2. SIGNAL DATA CONVERTER POWER DISTRIBUTION (SHEET 2 OF 2)

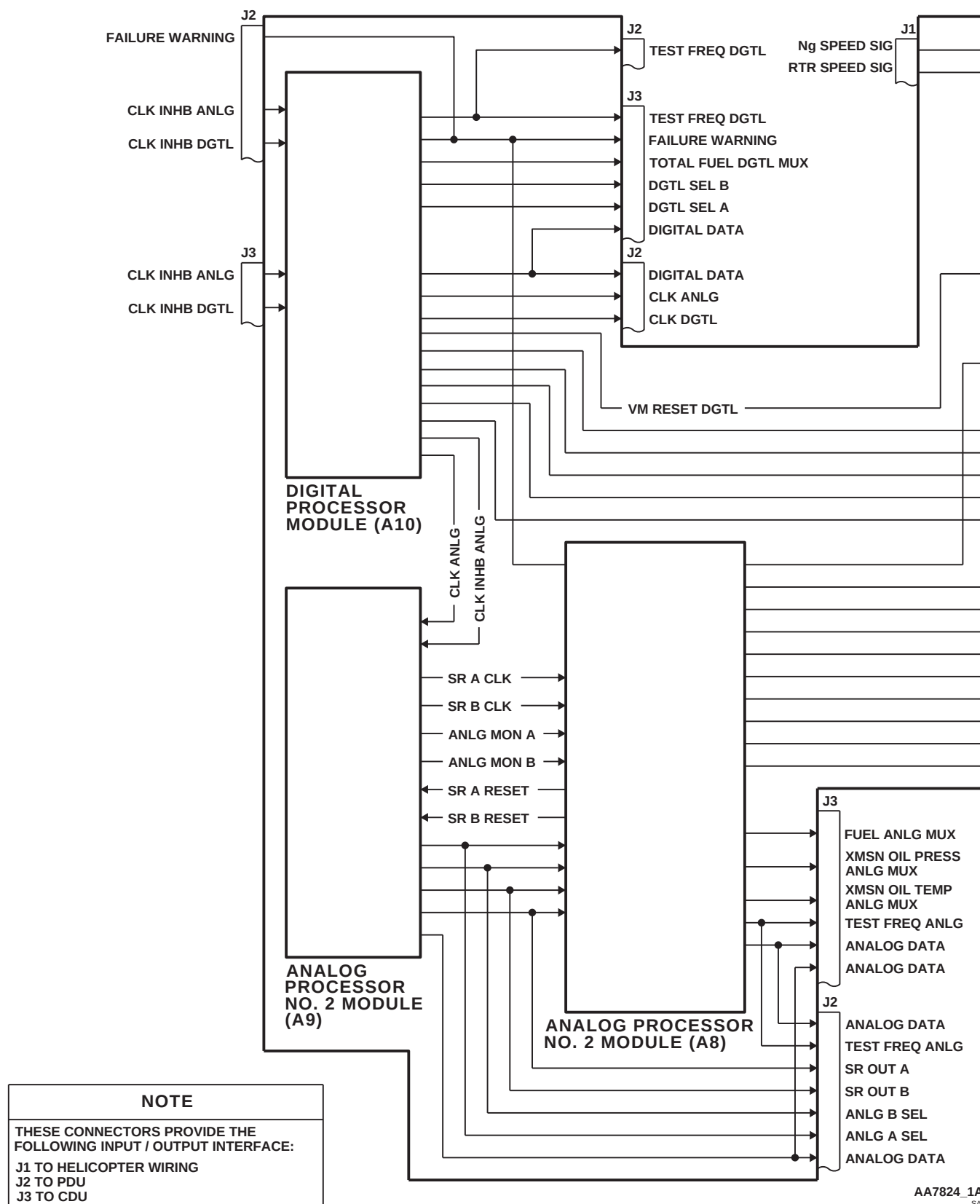


FIGURE 8-1-4-3. SIGNAL DATA CONVERTER SIGNAL FLOW DIAGRAM (SHEET 1 OF 2)

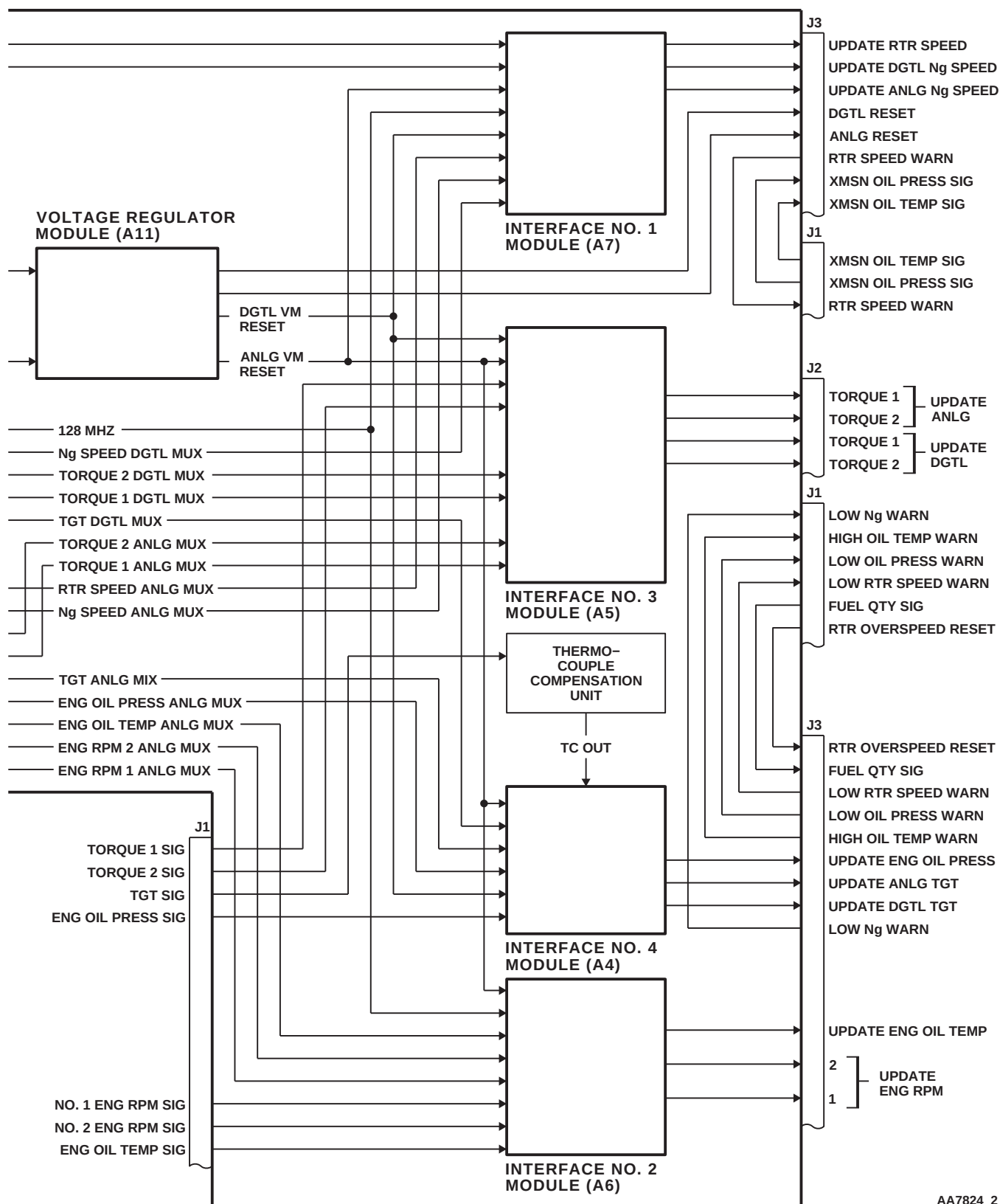


FIGURE 8-1-4-3. SIGNAL DATA CONVERTER SIGNAL FLOW DIAGRAM (SHEET 2 OF 2)

8-1-4-9/(8-1-4-10 Blank)

8-1-5 CENTRAL DISPLAY UNIT DESCRIPTION AND DATA.

8-1-5.1 Central Display Unit.

The central display unit (CDU), in the center of the instrument panel, receives signal and power inputs from both the No. 1 and No. 2 signal data converters (SDC) (Figure 8-1-5-1). The CDU contains 12 analog displays, five digital displays, and two failure lights. The CDU receives the following multiplexed data signals from the SDCs: No. 1 and No. 2 engine oil temperature, No. 1 and No. 2 engine oil pressure, No. 1 and No. 2 engine turbine gas temperature (TGT), and No. 1 and No. 2 engine gas generator tachometer (Ng). These parameters are displayed on the CDU analog scales. The No. 1 and No. 2 engine turbine gas temperature and No. 1 and No. 2 engine gas generator tachometer information are also displayed on CDU digital readouts. The No. 1 and No. 2 fuel quantity signals

from the No. 1 SDC and No. 2 SDC, respectively, and main transmission oil temperature and pressure signals from the No. 2 SDC are conditioned and multiplexed by the CDU for analog display. Total fuel quantity is displayed on a digital readout. Additional information on CDU analog displays is given in the following tabular data. The CDU failure lights, CHAN 1 and CHAN 2, are part of the IDS fault detection circuit. A failure of any SDC or CDU processing circuit, CDU or PDU display driver module, or SDC logic power supply will cause the associated display channel to turn off or switch to backup processor and will light the associated CHAN failure light. Failure of lamp power supply within an SDC will cause every second display light on the CDU to go off. The parameters displayed and the limits of safe, caution and warning conditions are listed in Table 8-1-5-1.

Table 8-1-5-1. CDU Ranges.

Parameter	Scale Range
No. 1 and No. 2 fuel quantity sensor signal (FUEL QTY 1 and 2)	0-1500 LBS GREEN SEGMENT: 200-1500 LBS AMBER SEGMENT: 0-200 LBS
Main transmission oil temperature sensor signal (XMSN TEMP)	-50°C TO 170°C GREEN SEGMENT: -50°C TO 120°C AMBER SEGMENT: 120°C TO 140°C RED SEGMENT: 140°C TO 170°C
Main transmission oil pressure sensor signal (XMSN PRESS)	0-190 PSI GREEN SEGMENT: 30-65 PSI AMBER SEGMENT: 20-30 PSI, AND 65-130 PSI RED SEGMENT: 0-20 PSI, AND 130-190 PSI
No. 1 and No. 2 engine oil temperature sensor signal (ENG OIL TEMP 1 and 2)	-50°C TO 180°C GREEN SEGMENT: -50°C TO 135°C AMBER SEGMENT: 135°C TO 150°C RED SEGMENT: 150°C TO 180°C

Table 8-1-5-1. CDU Ranges. (Cont)

Parameter	Scale Range
No. 1 and No. 2 engine oil pressure sensor signal (ENG OIL PRESS 1 and 2)	10-130 PSIOR 12-170 PSI GREEN SEGMENT: 45-100 PSI AMBER SEGMENT: 25-45 PSI (ON CDU NOT MODIFIED) AMBER SEGMENT: 20-45 PSI (ON MODIFIED CDU) RED SEGMENT: 10-25OR 12-25, AND 100-130 PSI (ON CDU NOT MODIFIED) RED SEGMENT: 10-20 OR 12-20 AND 100-130 PSI (ON MODIFIED CDU)
No. 1 and No. 2 engine turbine inlet gas temperature sensor signal (TGT TEMP 1 and 2)	0°C TO 950°C GREEN SEGMENT: 0°C TO 775°C AMBER SEGMENT: 775°C TO 850°C RED SEGMENT: 850°C TO 950°C
No. 1 and No. 2 engine gas generator tachometer signal (Ng SPEED 1 and 2)	0%-110% GREEN SEGMENT: 0%-98% AMBER SEGMENT: 98%-102% RED SEGMENT: 102%-110%

The CDU also contains a PUSH TO TEST or LAMP TEST switch that, when pressed, causes all CDU vertical scale lamps, digital readouts, and CHAN 1 and CHAN 2 failure lights and the PDU and CPDU RTR OVERSPEED lights to go on; a DIGITS switch that gives ON-OFF control for all digital displays; a photocell to sense ambient light for automatic level adjustment; and a DIM control that sets the level of display lighting to be determined by the photocells. On the back panel of the CDU, there are four electrical connectors (J1, J2, J3, and J4), and eight fuses (seven used, one spare). The CDU contains the following modules:

- (1) No. 1 fuel quantity lamp driver module (A3)
- (2) No. 2 fuel quantity lamp driver module (A4)
- (3) Transmission oil temperature lamp driver module (A5)
- (4) Transmission oil pressure lamp driver module (A6)

- (5) No. 1 engine oil temperature lamp driver module (A7)
- (6) No. 2 engine oil temperature lamp driver module (A8)
- (7) No. 1 engine oil pressure lamp driver module (A9)
- (8) No. 2 engine oil pressure lamp driver module (A10)
- (9) No. 1 TGT lamp driver module (A11)
- (10) No. 2 TGT lamp driver module (A12)
- (11) No. 1 Ng speed lamp driver module (A13)
- (12) No. 2 Ng speed lamp driver module (A14)
- (13) Interface No. 1 module (A15)
- (14) Interface No. 2 module (A16)
- (15) Voltage regulator module (A17)

8-1-5-2

- (16) Digital readout module (A18)

8-1-5.1.1 Lamp Driver Modules.

All the lamp driver modules operate in basically the same way (Figures 8-1-5-2 and 8-1-5-3). The only difference is some modules (No. 1 and No. 2 fuel quantity, No. 1 and No. 2 engine oil pressure, and main transmission oil pressure modules) have automatic bottom segment turnoff. This feature automatically turns off the low scale red and/or amber segments when the scale reading reaches the green segment. The low scale segments are automatically turned on when the scale reading drops below the green segment. Analog data is fed to the lamp drivers from the analog processor shift registers in the SDCs. The analog processors also supply the update pulses and test frequency. The analog update initiates the transfer of the data into the parallel-in/parallel-out shift registers. The output of the registers is applied to transistor lamp drivers to light the appropriate lamps. Light from the lamps is carried to the display panel faceplate by fiber optic strips, giving visual displays corresponding to the level of the sensed parameter. Power for the lamps is supplied by lamp supply voltages from SDC No. 1 and SDC No. 2. The No. 1 SDC supplies lamp voltage to all even-numbered lamps while the No. 2 SDC supplies lamp voltage to all the odd-numbered lamps on the vertical scale. A fault detection circuit monitors the test frequency and update pulse. In the absence of the test frequency or an update pulse, the fault detection circuit produces a failure warning signal and a reset pulse which resets the shift register, turning off all module display lamps on that module. When the CDU PUSH TO TEST or LAMP TEST switch is pressed, a lamp test voltage is applied to the driver module causing all of its lamps to go on.

8-1-5.1.2 Interface No. 1 Module (A15).

The interface No. 1 module contains the signal conditioning circuitry for the transmission oil pressure signal and the auto dim control circuitry for the CDU, pilot's PDU and copilot's PDU (Figures 8-1-5-2 and 8-1-5-3). A 10 vac, 400 Hz reference voltage from the SDC logic power supply is used with the transmission oil pressure sensor signal (400 Hz) to generate a dc analog signal

proportional to the pressure. The dc analog is applied to the A/T converter which produces the transmission oil pressure update pulse when enabled by the multiplexer signal from the SDC analog processor. The analog reset signal from the SDC resets the A/T converter. The update pulse is applied to the transmission oil pressure lamp driver module. The auto dim circuitry provides a reference voltage for the SDC lamp supply that is proportional to the ambient light level. The three photocells, one on the CDU, and one on both PDUs, sense the surrounding light level. The photocell sensing the highest level of light will provide the controlling input to the interface No. 1 module. The DIM control, on the CDU, is a gain adjustment which sets the display light level output. Turning the DIM control clockwise, past the detent, sets the display lighting to a fixed preset level and disables the three photocells.

8-1-5.1.3 Interface No. 2 Module (A16).

The interface No. 2 module contains the signal conditioning circuitry for transmission oil temperature and the No. 1 and No. 2 fuel quantity signals (Figures 8-1-5-2 and 8-1-5-3). It also contains the transmission oil temperature A/T converter for the analog scale, the No. 1 and No. 2 A/T converters for the fuel quantity analog scales and the digital A/T converter for total fuel. The output from the No. 1 and No. 2 fuel quantity sensors are fed to a fuel quantity conditioning circuit where they are converted to a dc voltage. The outputs are applied to the No. 1 and No. 2 fuel quantity A/T converters. They are summed by the A/T converter for the digital display for total fuel. The interface No. 2 module also receives analog reset and multiplex signals from the analog processor in the SDC and generates the FUEL 1 and FUEL 2 update pulses to the respective lamp driver modules. The digital reset and total fuel multiplex signal are provided by the digital processor in the SDC, and the total fuel update pulse is applied to the digital readout module. The transmission oil temperature sensor, a variable resistance, is fed to a conditioning circuit where it is converted to a dc voltage proportional to the oil temperature. The proportional dc voltage is then applied to the transmission oil temperature A/T converter. The analog reset signal and the transmission oil temperature multiplex signal from

the SDC are used by the A/T converter to generate the transmission oil temperature update pulse which is applied to the transmission oil temperature lamp driver module.

8-1-5.1.4 Voltage Regulator (A17).

The voltage regulator receives both power and selective multiplexed analog data/update pulses from both the No. 1 and No. 2 SDCs (Figures 8-1-5-2 and 8-1-5-3). The voltage regulator provides over current protection (in conjunction with Q1) and regulated -10 vdc, 10 vdc, and 5 vdc operating voltages for the CDU. In addition, the voltage regulator provides output signals when any specific helicopter parameter high/low warning condition is detected. The regulator provides three rotor overspeed warning voltages to both PDUs and also provides output voltages through the appropriate CDU fuse and the SDCs to the helicopter caution/advisory warning system when any of the following conditions exists: No. 1 or No. 2 low engine oil pressure, No. 1 or No. 2 high engine oil temperature, No. 1 or No. 2 low engine Ng, or low rotor speed. When any of the rotor overspeed conditions (127%, 137%, or 142%) are detected, a relay on the voltage regulator is energized (latched) and connects the lamp power supply voltage to the respective overspeed lamps on both PDUs, causing the lamps to go on. The PDU lamps will now remain on and the regulator relay will remain latched even when the overspeed condition no longer exists (safe RPM). An external reset signal from the helicopter system to the voltage regulator module is required to reset the latching relay and cause the PDU overspeed lamps to go off. The remaining warning detection circuits on the voltage regulator

will reset whenever the warning (unsafe) condition no longer exists.

8-1-5.1.5 Digital Readout Module (A18).

The digital readout module provides digital readouts for total fuel, No. 1 and No. 2 engine gas generator tachometer (Ng SPEED 1 and 2) and No. 1 and No. 2 engine turbine gas temperature sensor signals (TGT TEMP 1 and 2) (Figures 8-1-5-2 and 8-1-5-3). The digital readout module receives digital data from both SDCs. The digital data is fed to latching circuits on the digital module. When a parameter update pulse is received, the incoming digital data is coupled through the latching circuitry to the seven-segment decoder drivers which light the appropriate lamps. Digital data as well as the appropriate update, multiplexing, test frequency, and lamp power supply voltages for total fuel, No. 1 engine gas generator tachometer (Ng 1) and No. 1 engine turbine gas temperature (TGT TEMP 1) are provided by the No. 1 SDC. The equivalent voltages for No. 2 engine gas generator tachometer (Ng 2) and No. 2 engine turbine gas temperature (TGT TEMP 2) display are provided by the No. 2 SDC. Each update pulse and the test frequency are applied to a monitor circuit on the digital readout module. In the absence of an update pulse or test frequency, the monitor circuit produces a failure warning signal which lights the appropriate CHAN 1 or CHAN 2 warning light. In addition, the warning signal turns off the fault detected digital readout. All digital readouts can be blanked by placing the DIGITS switch to OFF. When the CDU PUSH TO TEST or LAMP TEST switch is engaged, all digital segments will light and will be displayed in the digital readout.

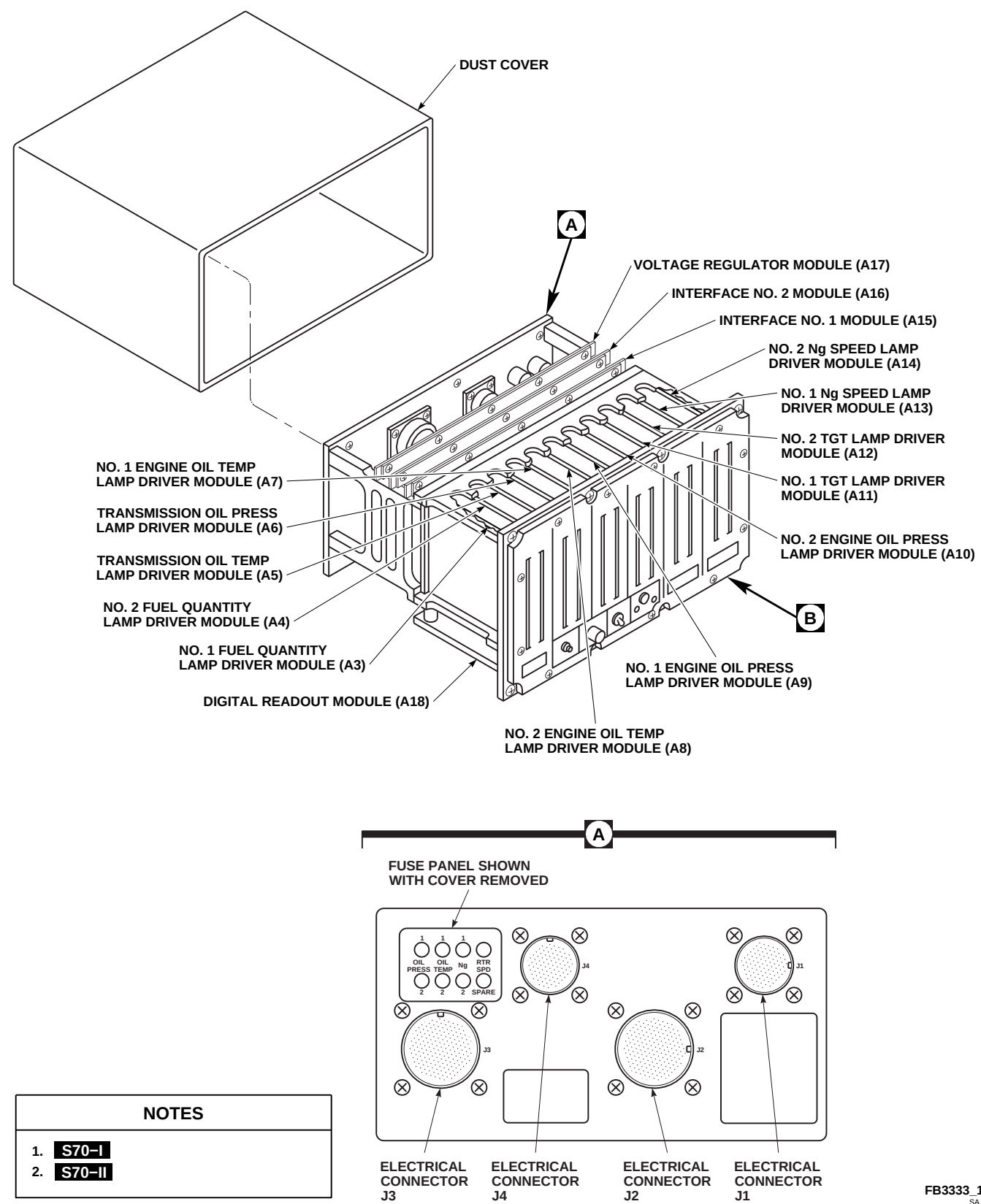


FIGURE 8-1-5-1. CENTRAL DISPLAY UNIT PARTS LOCATION DIAGRAM (SHEET 1 OF 2)

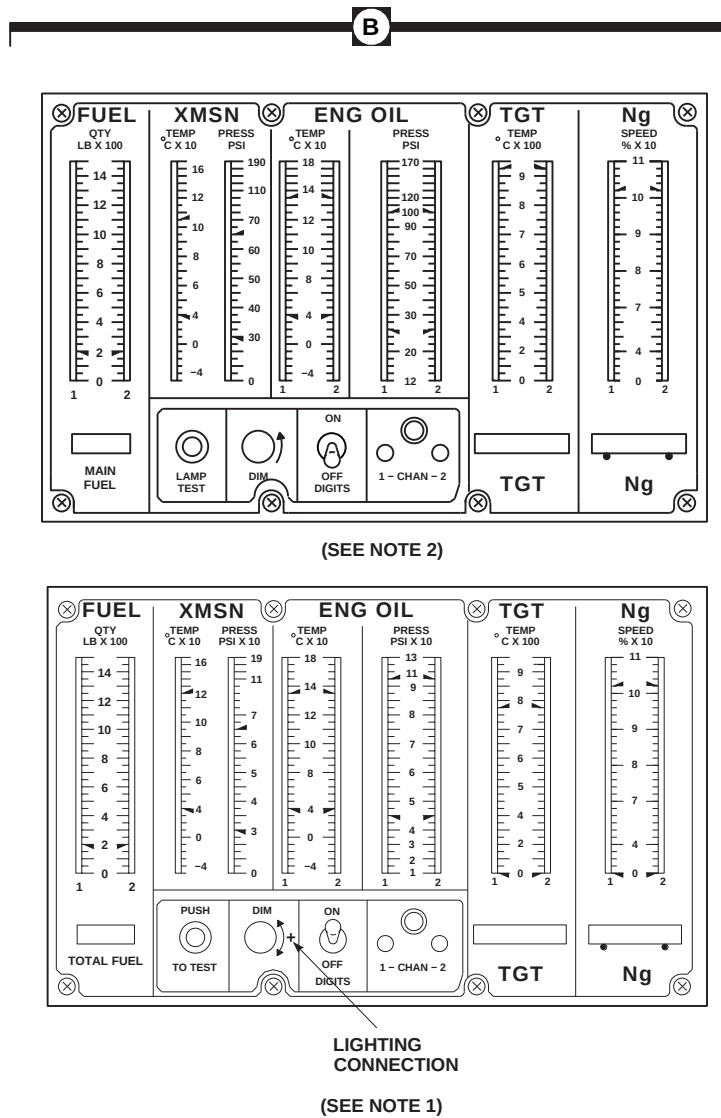
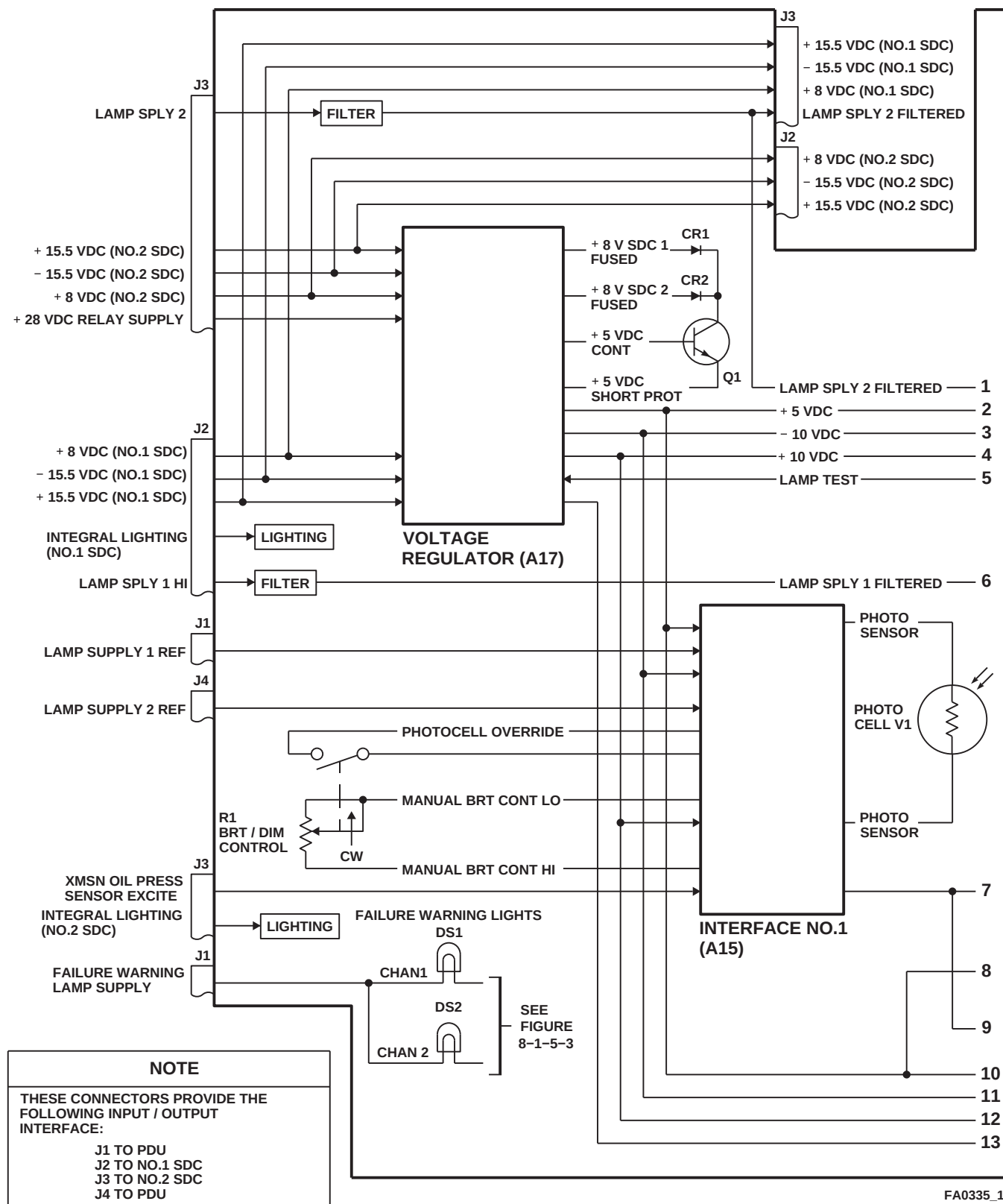
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FIGURE 8-1-5-1. CENTRAL DISPLAY UNIT PARTS LOCATION DIAGRAM (SHEET 2 OF 2)



FA0335_1
SA

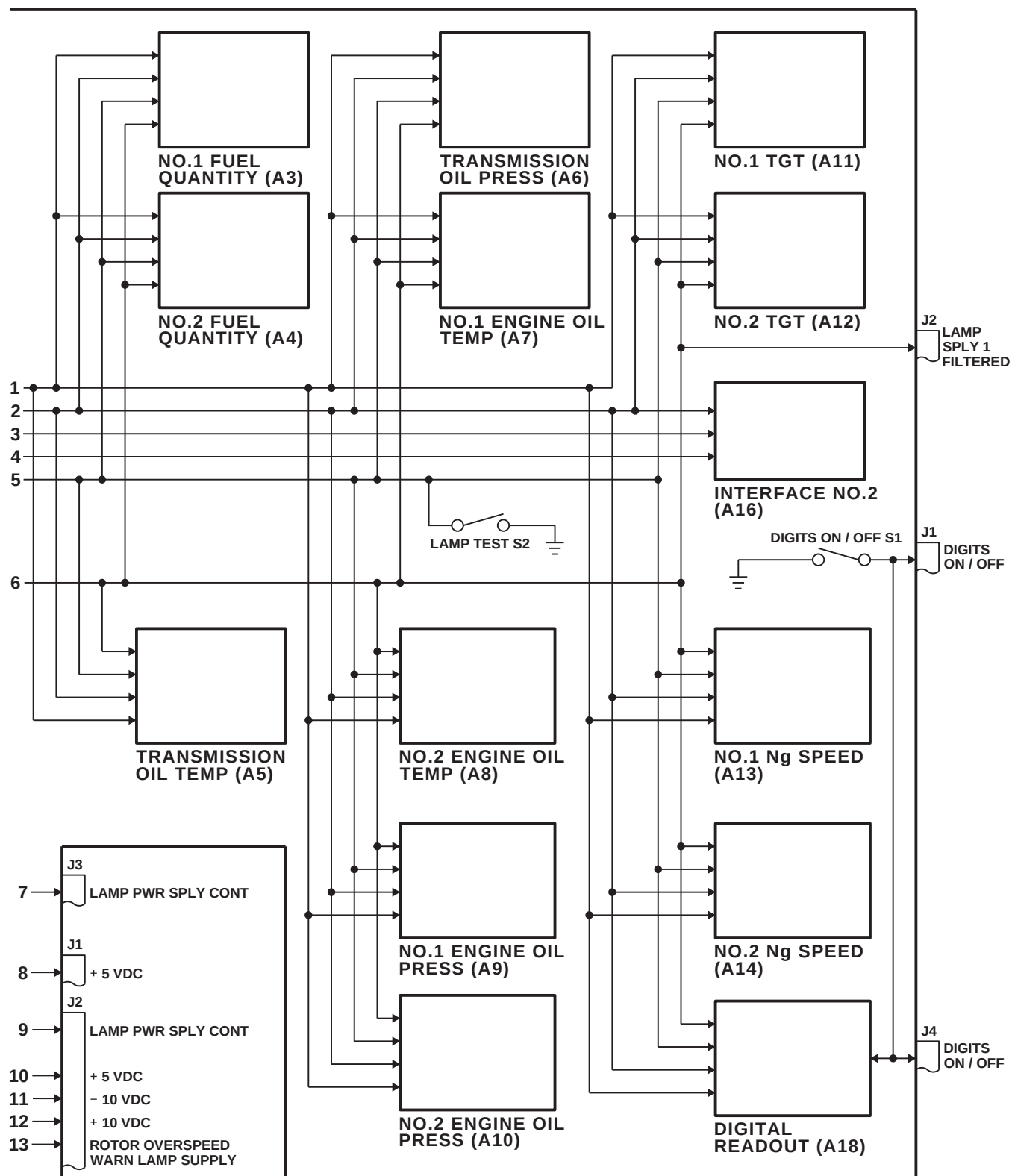
FA0335_2
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FIGURE 8-1-5-2. CENTRAL DISPLAY UNIT POWER DISTRIBUTION DIAGRAM
(SHEET 2 OF 2)

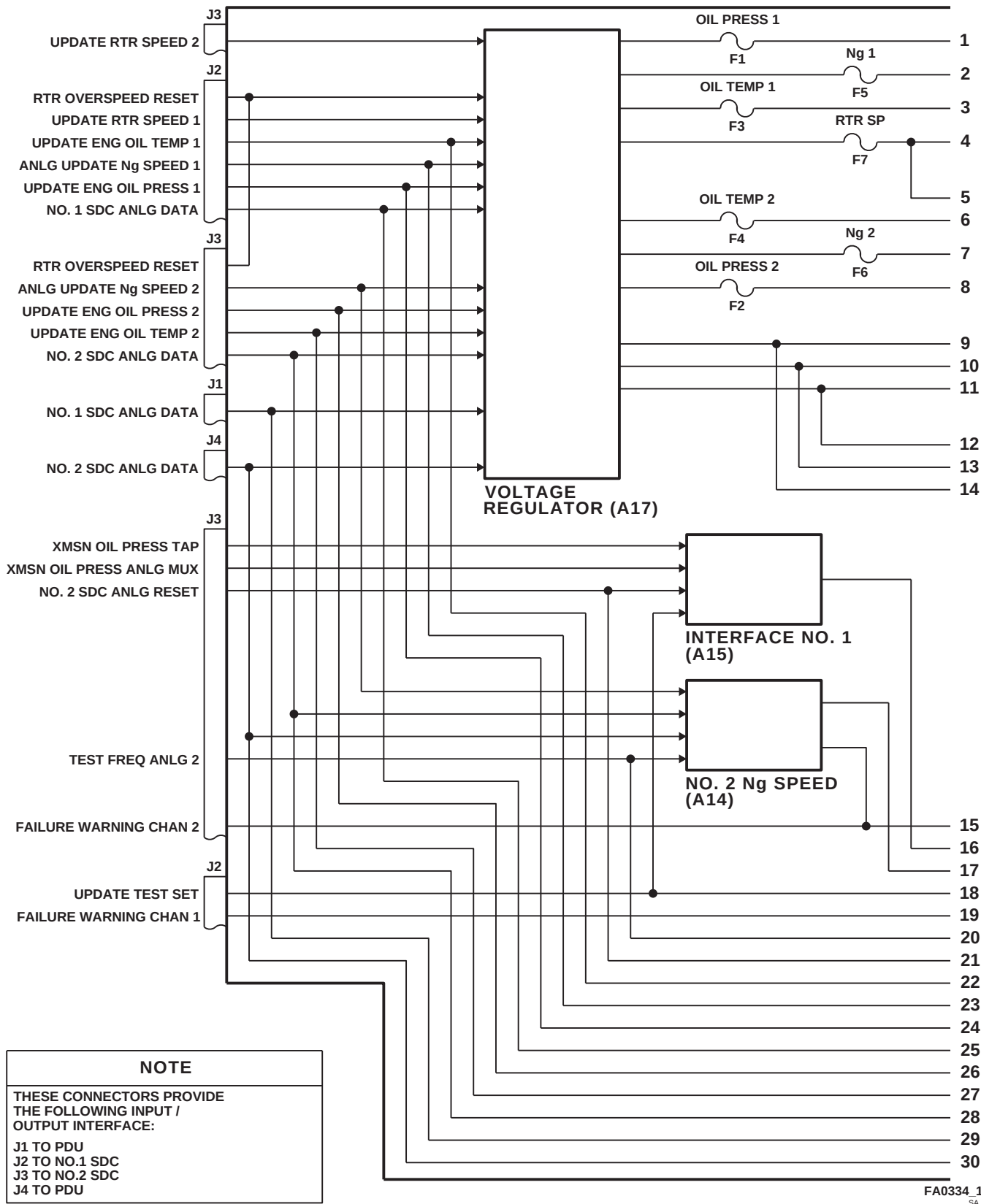


FIGURE 8-1-5-3. CENTRAL DISPLAY UNIT SIGNAL FLOW DIAGRAM (SHEET 1 OF 4)

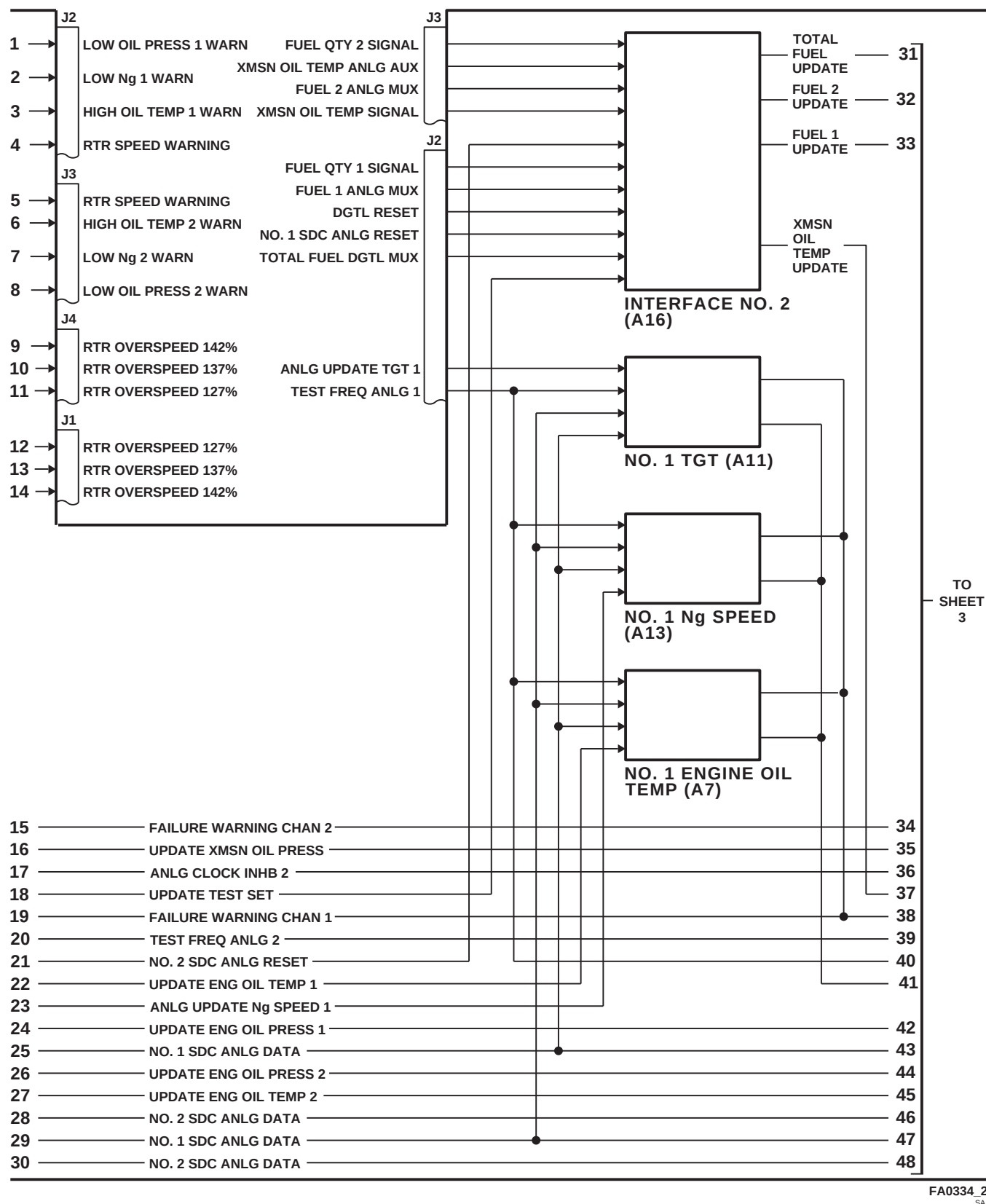
FA0334.2
SA

FIGURE 8-1-5-3. CENTRAL DISPLAY UNIT SIGNAL FLOW DIAGRAM (SHEET 2 OF 4)

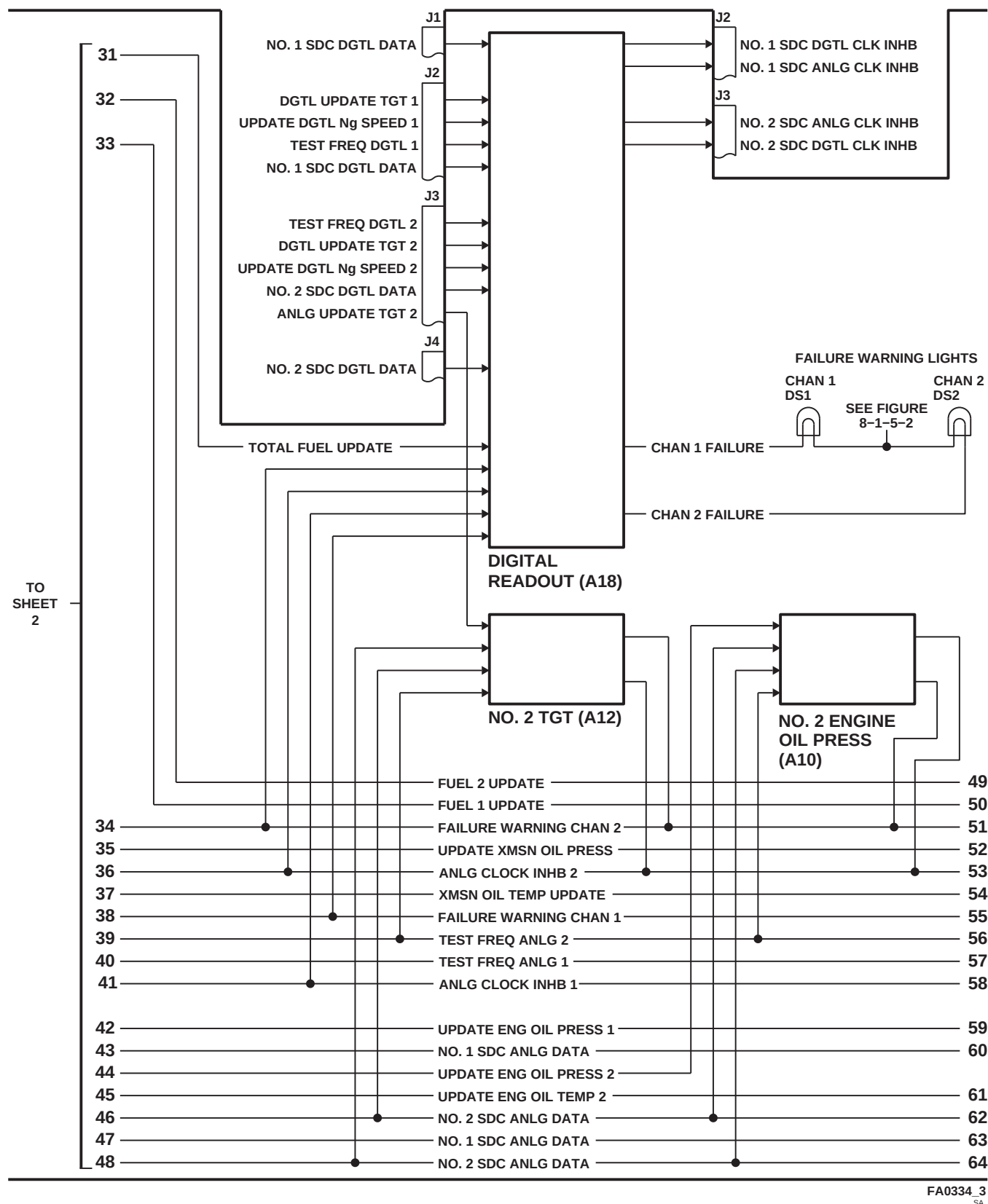


FIGURE 8-1-5-3. CENTRAL DISPLAY UNIT SIGNAL FLOW DIAGRAM (SHEET 3 OF 4)

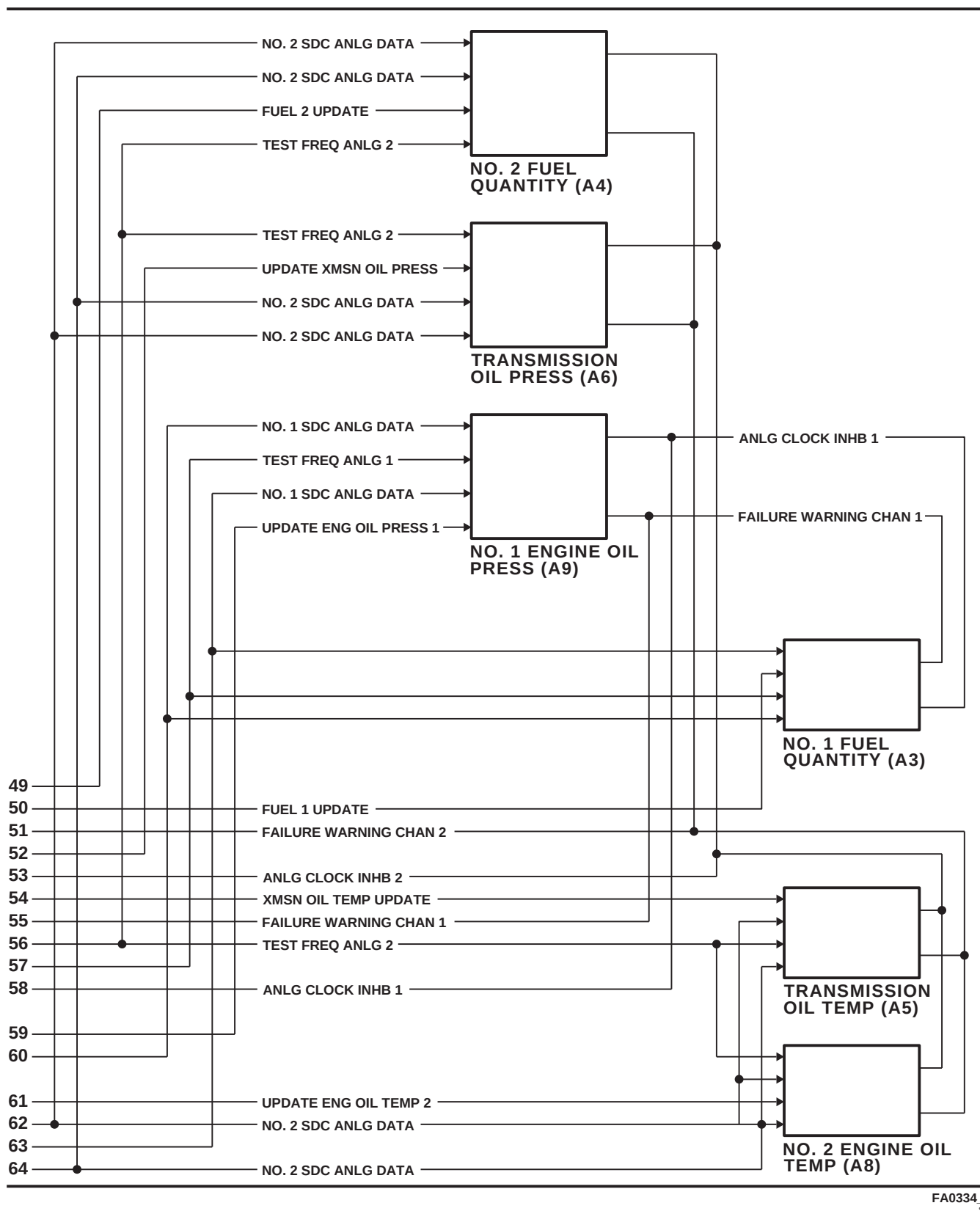


FIGURE 8-1-5-3. CENTRAL DISPLAY UNIT SIGNAL FLOW DIAGRAM (SHEET 4 OF 4)

8-1-6 PILOT’S DISPLAY UNIT DESCRIPTION AND DATA.

8-1-6.1 Pilot’s Display Unit.

The pilot’s display unit (PDU) receives signal and power inputs from an associated signal data converter (SDC) and central display unit (CDU) (Figure 8-1-6-1). The PDU contains five analog vertical scale displays, two digital displays, three indicator lights and a photocell. The PDU receives the following multiplexed data signals from the SDC: percentage of RPM speed for No. 1 and No. 2 engines and main rotor, and percentage of torque for the No. 1 and No. 2 engines. These parameters are displayed on the PDU analog scales. The percentage of torque for the No. 1 and No. 2 engines are also displayed on the digital readouts. The parameters displayed, and the limits of safe, caution, and warning conditions are listed in Table 8-1-6-1.

The PDU also contains a LT switch which, when pressed, lights the vertical displays and digital readouts. The RTR OVERSPEED lights go on individually to indicate overspeeds of 127%, 137%, and 142%. The CDU provides power to each of the lights. On the back panel of the PDU are two electrical connectors (J1 and J2). The PDU contains the following modules: No. 1 engine RPM lamp driver module (A3), rotor RPM lamp driver module (A4), No. 2 engine RPM lamp driver module (A5), No. 1 engine torque (analog) lamp driver module (A6), No. 2 engine torque (analog) lamp driver module (A7), torque digital readout module (A8), and autodim module (A9).

Table 8-1-6-1. PDU Ranges.

Parameter	Scale Range
S70-I No. 1 and No. 2 engine power turbine speed sensor signal (% RPM 1 and 2)	0% TO 130% GREEN SEGMENT: 95% TO 101% AMBER SEGMENT: 91% TO 95% AND 101% TO 105% RED SEGMENT: 0% TO 91% AND 105% TO 130%
S70-II No. 1 and No. 2 engine power turbine speed sensor signal (% RPM 1 and 2)	0% TO 130% GREEN SEGMENT: 95% TO 101% AMBER SEGMENT: 91% TO 95% AND 101% TO 110% RED SEGMENT: 0% TO 91% AND 105% TO 130%
S70-I Main rotor speed sensor signal (% RPM R)	0% TO 130% GREEN SEGMENT: 95% TO 101% AMBER SEGMENT: 91% TO 95% AND 101% TO 110% RED SEGMENT: 0% TO 91% AND 110% TO 130%

Table 8-1-6-1. PDU Ranges. (Cont)

Parameter	Scale Range
S70-II Main rotor speed sensor signal (% RPM R)	0% TO 130% GREEN SEGMENT: 96% TO 101% AMBER SEGMENT: 91% TO 96% AND 101% TO 110% RED SEGMENT: 0% TO 91% AND 110% TO 130%
S70-I No. 1 and No. 2 engine torque sensor signal (% TRQ 1 and 2)	0% TO 150% GREEN SEGMENT: 0% TO 100% AMBER SEGMENT: 100% TO 110% RED SEGMENT: 110% TO 150%
S70-II No. 1 and No. 2 engine torque sensor signal (% TRQ 1 and 2)	0% TO 150% GREEN SEGMENT: 0% TO 100% AMBER SEGMENT: 100% TO 135% RED SEGMENT: 135% TO 150%

8-1-6.1.1 Lamp Driver Modules.

All the lamp driver modules operate in basically the same way (Figures 8-1-6-2 and 8-1-6-3). The only difference is that some modules (No. 1 and No. 2 engine RPM and rotor RPM modules) have automatic bottom segment turnoff. This feature automatically turns off the low scale red and/or amber segments when the scale reading reaches the green segment. The low scale segments are automatically turned on when the scale reading drops below the green segment. Analog data is fed to the lamp drivers from the analog processor shift registers in the SDCs. The analog processors also supply the update pulses and test frequency. The analog update initiates the transfer of the data into the parallel-in/parallel-out shift registers. The output of the registers is applied to transistor lamp drivers to light the appropriate lamps. Light from the lamps is carried to the display panel faceplate by fiber optic strips, giving visual displays corresponding to the level of the sensed parameter. Power for the lamps is supplied by lamp supply voltages from the SDC. A monitor circuit checks for the presence of update pulses and a test frequency. Should either signal fail, the monitor circuit turns off the display of that module and applies a failure warning signal to the CDU.

8-1-6.1.2 Torque Digital Readout Module (A8).

The torque digital readout module provides digital readouts for percentage of No. 1 and No. 2 engine torque (Figures 8-1-6-2 and 8-1-6-3). The module receives digital data from the SDC and stores the data in a latching circuit when an update pulse is present. The data is fed to two decoder drivers which light the seven-segment units and tens displays, and to two transistors that drive separate lamps for the "hundreds" display (numeral 1 only). Each update pulse and a test frequency are applied to a monitor circuit that checks for the presence of update pulses. If the update pulses fail to correspond with the test frequency, the monitor circuit turns off the digital readout and applies a failure warning signal to the CDU. When the LT switch is pressed, all digital segments will go on.

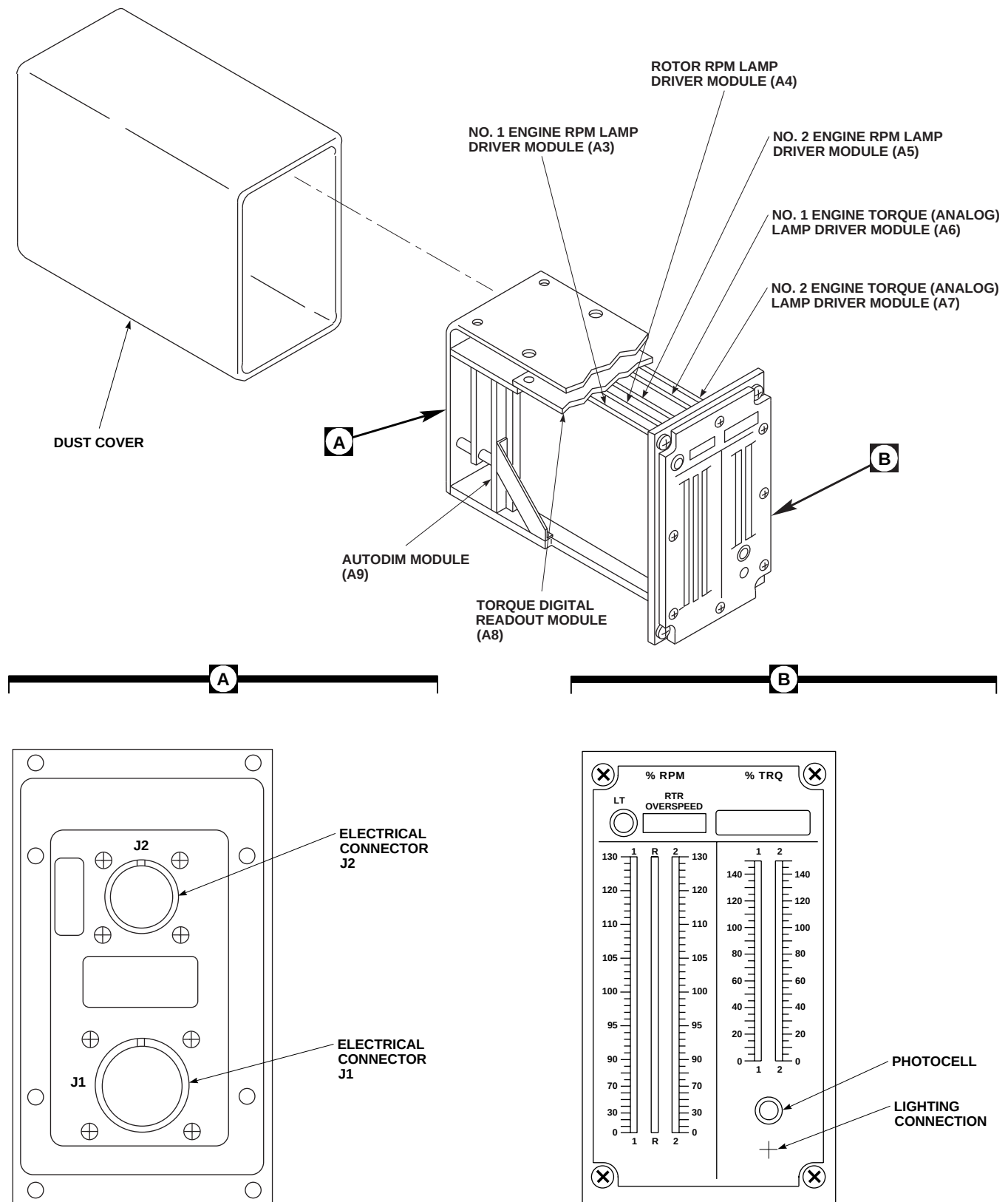
8-1-6.1.3 Autodim Module (A9).

The autodim module produces a dc lamp supply voltage which is proportional to ambient light detected by the photocell on the front panel (Figures 8-1-6-2 and 8-1-6-3). The lamp supply voltage is routed out to the CDU as a reference for the CDU dimming control circuit. The autodim

8-1-6-2

module contains a lamp test supply circuit for testing all analog and digital displays. When the LT switch is pressed, the circuit provides five separate lamp test voltages for the analog displays and a

logic signal for the digital displays. The module also contains clock inhibit drivers, which provide the combined analog and digital update pulses to the analog and digital clock inhibit circuits.



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FIGURE 8-1-6-1. PILOT'S DISPLAY UNIT PARTS LOCATION DIAGRAM

8-1-6-4

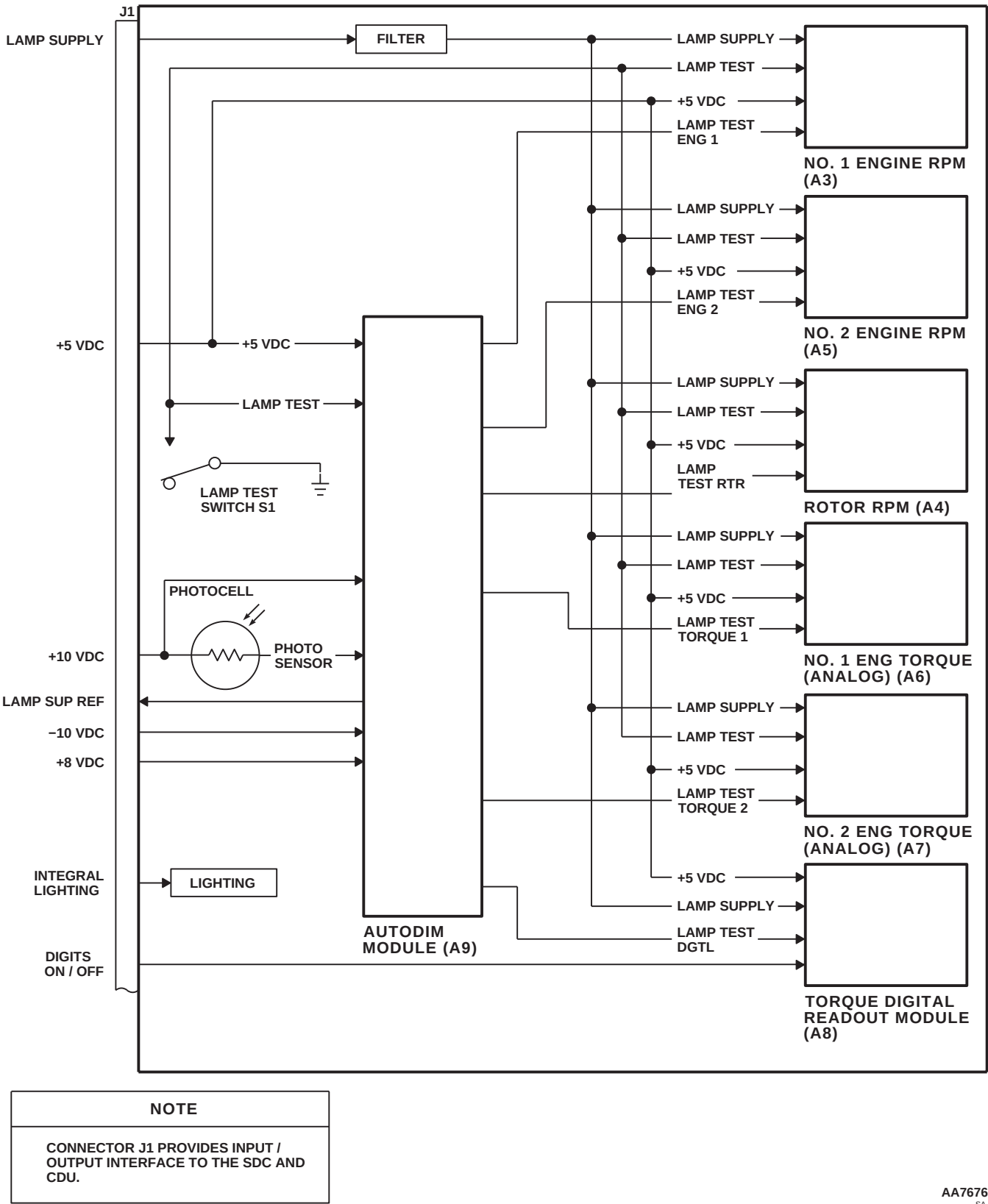
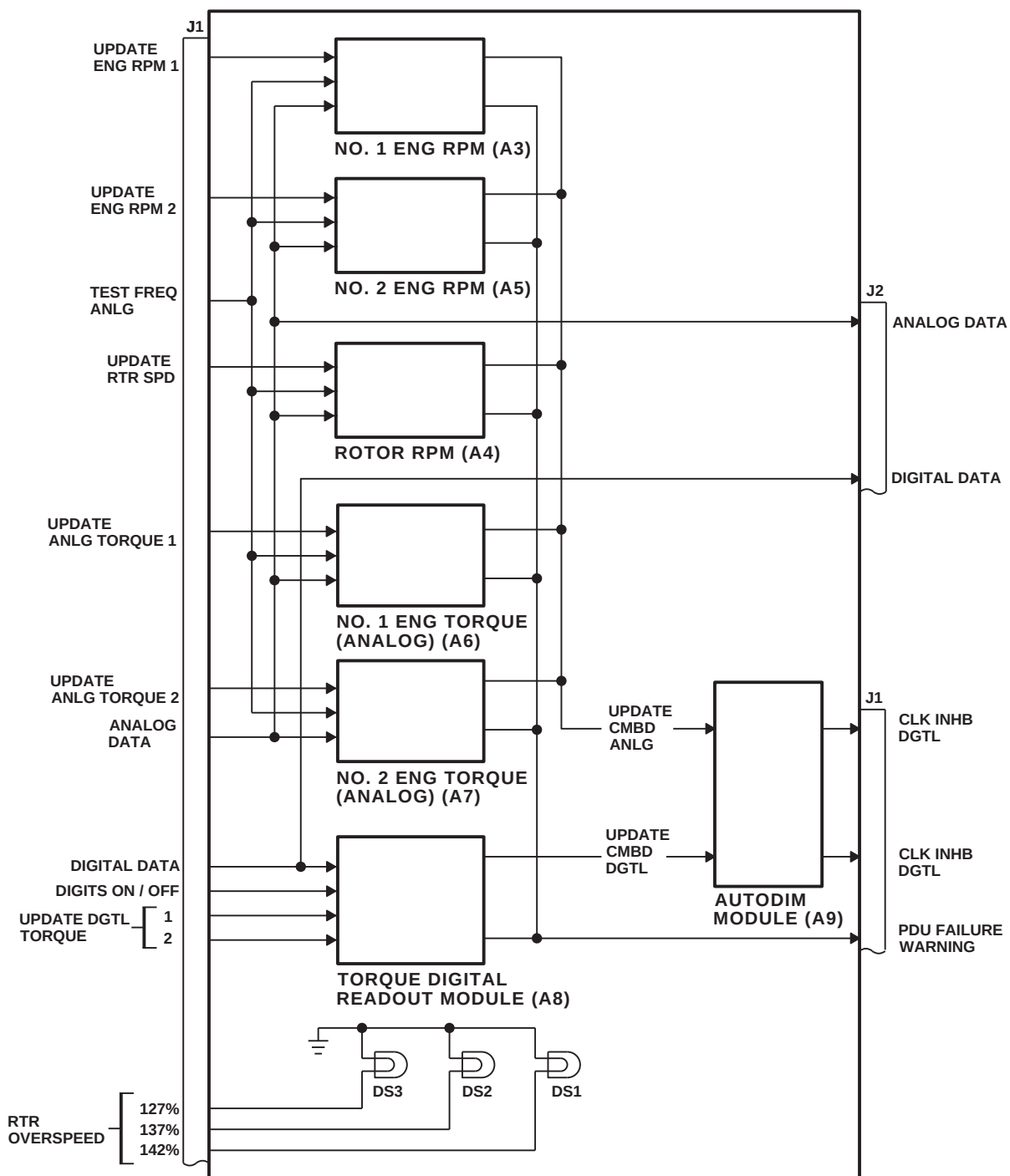


FIGURE 8-1-6-2. PILOT'S DISPLAY UNIT POWER DISTRIBUTION DIAGRAM

**NOTE**

THESE CONNECTORS PROVIDE THE FOLLOWING INPUT / OUTPUT INTERFACE:
 J1 TO SDC AND CDU
 J2 TO CDU

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FIGURE 8-1-6-3. PILOT'S DISPLAY UNIT SIGNAL FLOW DIAGRAM

8-1-7 CAUTION/ADVISORY WARNING SYSTEM DESCRIPTION AND DATA.

8-1-7.1 System Description.

The caution/advisory warning system gives visual capsule indications for up to 82 helicopter subsystem conditions and aural indications for three of these conditions (Figures 8-1-7-1 and 8-1-7-2). Each indicator capsule has a colored legend that is visible when the capsule lights to indicate the condition monitored. Green advisory capsules indicate actuation and normal operation of helicopter equipment. Yellow caution capsules indicate unsafe flight conditions and system failures. Red warning capsules indicate unsafe flight conditions requiring immediate action. When an advisory, caution, or warning condition exists, the related capsule goes on and remains on until the condition is removed or the trouble is corrected. On helicopters equipped with caution/advisory panel, 70550-01107-102, there are seven latching caution capsules that remain on once turned on and do not go off until reset. On helicopters with caution/advisory panel, 70550-01107-103, these capsules do not latch. When a caution capsule goes on, two MASTER CAUTION PRESS TO RESET capsules also go on to attract the attention of the pilot and copilot to the caution condition. The MASTER CAUTION PRESS TO RESET capsules will remain on until the trouble is corrected or until either capsule is pressed to reset both capsules for another caution condition. The system provides aural warnings to the pilot's and copilot's ICS stations in addition to the visual warnings for low rotor RPM, engine out, and stabilator shutdown conditions. A steady tone is provided for either low rotor RPM or engine out conditions, and a beeping tone is provided for stabilator shutdown. The system also gives indicator light test and dimming control signals to other helicopter systems. The caution/advisory warning system consists of a caution/advisory panel, on the center of the instrument panel; two master warning panels, one on each side of the instrument panel glare shield; an

audible warning unit, on the canted bulkhead on the left side of the avionics compartment; and two dimming controls, on the left side of the instrument panel.

Sixty-four of the caution/advisory capsules are yellow while 18 are green. The 64 yellow capsules are caution capsules that indicate unsafe flight condition system failures. The 18 green capsules are advisory capsules that indicate actuation or normal operation of various helicopter subsystems. With the exception of two caution capsules, 1 and 62, all of the caution and advisory capsules light steady. Capsules 1 and 62 flash when activated. Whenever a fault signal is applied to one of the 64 caution capsules, a 28 vdc output from the caution/advisory panel is also produced. In the helicopter installation, this output is used to power external lights on two master warning panels. The external master caution signal may be reset by unlatching a circuit in the caution/advisory panel with an external reset command voltage. The capsules are arranged on the front of the caution/advisory panel in 4 horizontal rows, with 19 capsules in the extreme left vertical row and 21 capsules in each of the remaining 3 rows. The upper portion displays 64 caution indications. The lower portion displays 18 advisory indications. On panel 70550-01107-102, there are seven latching capsules (13, 14, 21, 33, 42, 74, and 75). Once they are turned on, they remain on until a reset is applied. All the latching capsules are identical in operation except 33, which has a 30-second time delay before turn-on. The caution/advisory panel also consists of a 3-position momentary-type contact toggle switch with panel markings of BRT/DIM and TEST, a frame assembly, two side covers, four circuit channel cards and one control card which plug into mating connectors within the frame, two input/output signal and power connectors, and chassis-mounted electronic components. Descriptions of the caution and advisory capsules are listed in Table 8-1-7-1.

Table 8-1-7-1. Caution/Advisory Panel Capsules.

Legend	Indication
#1 FUEL LOW	Flashes between 3 to 5 flashes per second when left fuel tank level is about 172 pounds (20 minutes) remaining at normal cruise flight.
#1 FUEL PRESS	Left engine fuel pressure is below 8.0 to 9.0 psi between engine-driven low pressure fuel pump and high pressure vane fuel pump.
#1 ENGINE OIL PRESS	Left engine oil pressure is below 25 psi.
#1 ENGINE OIL TEMP	Left engine oil temperature is above 150°C.
CHIP #1 ENGINE	Left engine chip detector in scavenge oil system has metal chip or particles buildup.
#1 FUEL FLTR BYPASS	Left engine fuel filter has 7.5 psid across filter.
#1 ENGINE STARTER	Left engine start valve is open.
#1 PRI SERVO PRESS	First stage pressure is shut off, has dropped below 1950 to 2050 psi range, or servo pilot valve is jammed.
TAIL ROTOR QUADRANT	Goes on when a tail rotor cable is broken or disconnected.
MAIN XMSN OIL TEMP	Main transmission oil temperature is above 120°C.
BOOST SERVO OFF	Indicates loss of 2nd stage hydraulic pressure to the boost servo, or a boost servo jam.
LFT PITOT HEAT	Indicates left pitot heater element is not receiving power with PITOT HEAT switch in ON position.
CHIP INPUT MDL - LH	Indicates a metal particle has been detected by the chip detector.
CHIP ACCESS MDL - LH	Indicates a metal particle has been detected by the chip detector.
MR DE-ICE FAIL	Indicates a short or open in the main rotor de-ice system, which will disable the system.
MAIN XMSN OIL PRESS	Main transmission oil pressure is below 14 psi.
#1 GEN	Left generator is not supplying power to the buses.

8-1-7-2

Table 8-1-7-1. Caution/Advisory Panel Capsules. (Cont)

Legend	Indication
#1 GEN BRG	Left generator main bearing is worn or has failed.
#1 CONV	Left converter (ac to dc current) has no output.
AC ESS BUS OFF	Indicates that no power (115 vac phase B) is being supplied to the ac essential bus.
BATT LOW CHARGE	Indicates that the battery charge state is at or below 22.7 vdc or 40% charge.
GUST LOCK	Indicates the gust lock is engaged.
#1 OIL FLTR BYPASS	Left engine oil filter pressure differential is between 60 and 80 psi.
#1 HYD PUMP	Left hydraulic pump output pressure is below 1950 to 2050 psi range.
INT XMSN OIL TEMP	Intermediate gear box oil temperature is above 140°C.
STABILATOR	Stabilator system is turned on but is in the manual mode.
FLT PATH STAB	Indicates that FPS is inoperative.
CHIP INT XMSN	Indicates a metal particle has been detected by the chip detector.
CHIP MAIN MDL SUMP	Indicates a metal particle has been detected by the chip detector.
MR DE-ICE FAULT	Indicates partial failure of the blade de-ice system. Uneven shedding of ice can be expected.
#1 RSVR LOW	Hydraulic fluid level has dropped below 60% of full capacity.
#2 GEN	Right generator is not supplying power to the buses.
#2 GEN BRG	Right generator main bearing is worn or has failed.
#2 CONV	Right converter (ac to dc current) has no output.

Table 8-1-7-1. Caution/Advisory Panel Capsules. (Cont)

Legend	Indication
DC ESS BUS OFF	Indicates that no power (28 vdc) is being supplied to the dc essential bus.
BATTERY FAULT (NOTE 2)	Indicates that the battery has exceeded safe operating temperature (overtemperature) or a battery cell dissimilarity exists.
#2 OIL FLTR BYPASS	Right engine oil filter pressure differential is between 60 and 80 psi.
#2 HYD PUMP	Right hydraulic pump output pressure is below 1950 to 2050 psi range.
AUX FUEL (NOTE 1)	Indicates one or more auxiliary fuel tanks are empty, fuel flow is desired but not present, or the system is operating in a degraded mode.
TAIL XMSN OIL TEMP	Tail gear box oil temperature is above 140°C.
SAS OFF	Hydraulic pressure supplied to the SAS actuator is below 1950 to 2050 psi range.
IFF	Mode 4 is not capable of responding to interrogation.
CHIP TAIL XMSN	Indicates a metal particle has been detected by the chip detector.
APU FAIL	APU was automatically shut down by the electronic sequence unit.
TR DE-ICE FAIL	Indicates a short or open in a tail rotor blade de-ice element.
#2 RSVR LOW	Hydraulic fluid level has dropped below 60% of full capacity.
#2 FUEL LOW	Flashes between 3 to 5 flashes per second when right fuel level is about 172 pounds (20 minutes) remaining at normal cruise flight.
#2 FUEL PRESS	Right engine fuel pressure is below 9.0 psi between engine-driven low pressure fuel pump and high pressure fuel pump.

8-1-7-4

Table 8-1-7-1. Caution/Advisory Panel Capsules. (Cont)

Legend	Indication
#2 ENGINE OIL PRESS	Right engine oil pressure is below 25 psi.
#2 ENGINE OIL TEMP	Right engine oil temperature is above 150°C.
CHIP #2 ENGINE	Right engine chip detector in scavenge oil system has metal chips or particles buildup.
#2 FUEL FLTR BYPASS	Right fuel filter has 7.5 psid across filter.
#2 ENGINE STARTER	Right engine start valve is open.
#2 PRI SERVO PRESS	Second stage pressure is shut off, has dropped below 1950 to 2050 psi range, or servo pilot valve is jammed.
#1 TAIL RTR SERVO	Hydraulic pressure to the first stage tail rotor servo is below 1950 to 2050 psi range.
APU OIL TEMP HI	APU oil temperature is above 149°C.
TRIM FAIL	Indicates that yaw, roll or pitch trim actuators are not responding accurately to computer signals.
RT PITOT HEAT	Indicates right pitot heat element is not receiving power.
CHIP INPUT MDL - RH	Indicates a metal particle has been detected by the chip detector.
CHIP ACCESS MDL - RH	Indicates a metal particle has been detected by the chip detector.
ICE DETECTED	Indicates that ice is on the ice detector.
BACK-UP RSVR LOW	Hydraulic fluid level has dropped below 60% of full capacity.
#1 ENG ANTI-ICE ON	Indicates that No. 1 engine anti-ice/start bleed valve is open.
APU ON	APU speed is above 90% for 1.5 seconds.
APU ACCUM LOW	APU accumulator pressure is below 2600 psi.

Table 8-1-7-1. Caution/Advisory Panel Capsules. (Cont)

Legend	Indication
#1 ENG INLET ANTI-ICE ON	Indicates the No. 1 engine inlet temperature is 93°C or above.
APU GEN ON	APU generator output is accepted and being supplied to the helicopter.
SEARCH LT ON	Either pilot or copilot has selected SEARCH LT ON.
CARGO HOOK OPEN	Indicates that cargo hook load beam is not latched.
PARKING BRAKE ON	Indicates that PARKING BRAKE handle is pulled.
#2 ENG INLET ANTI-ICE ON	Indicates that No. 2 engine inlet temperature is 93°C or above.
PRIME BOOST PUMP ON	Prime boost pump switch is at PRIME or BOOST.
LDG LT ON	Either pilot or copilot has selected LDG LT ON.
HOOK ARMED	The cargo hook release system is armed.
EXT PWR CONNECTED	Indicates that external power plug is connected to helicopter's EXT POWER connector and BATT switch is ON, if EXT PWR switch is OFF.
#2 ENG ANTI-ICE ON	Indicates that No. 2 engine inlet anti-ice/start bleed valve is open.
BACK-UP PUMP ON	Backup pump pressure is being supplied at a pressure above 2350 psi.
#2 TAIL RTR SERVO ON	Pressure to second stage tail rotor servo is above 2350 psi.
AUX SRCHLT ON	Indicates the auxiliary searchlight is on.
ROTOR BRAKE ON	Indicates that rotor brake is on.

NOTES

1. **W/O ESSS** AUX FUEL capsule is not functional.

8-1-7.2 System Operation.

The caution/advisory warning system gets dc electrical power from the upper console circuit breaker panel and the copilot’s circuit breaker panel (Figure 8-1-7-3). Electrical power of 28 vdc is supplied by the dc essential bus and routed through the CAUT/ADVSY PNL circuit breaker to the caution/advisory panel, the master warning panels, and the left relay panel. The 28 vdc is also applied to the INST LT PILOT FLT control. When the control is turned from OFF, 28 vdc is applied to the caution/advisory panel for lamp dimming control. Electrical power of 28 vdc is supplied by the No. 1 dc primary bus and routed through the LIGHTS CAUT ADVSY circuit breaker for lamp test power. Electrical power of 28 vdc is applied to the PNL LTS switches on the pilot’s and copilot’s cyclic stick grips from the LIGHTS ADVSY circuit breaker. When the switches are pressed to turn off panel lights, the 28 vdc is applied to the caution/advisory panel for soft lighting dimming control.

The caution/advisory panel receives 28 vdc and ground signals from helicopter systems to light the caution and advisory capsules listed in Table 8-1-7-1. For most of the caution capsules and all of the advisory capsules, a 28 vdc signal from the monitored system is applied directly to the capsule circuit to light the capsule. These capsules will remain on until the input signal is removed. For chip detection and generator bearing failure monitoring, 28 vdc caution capsule power is applied to ground sensing circuits in the caution/advisory panel. When a chip is detected or a generator bearing fails, a ground is applied to the ground sensing circuit. The sensing circuit then applies the 28 vdc power to the caution capsule circuit to light the capsule. With the exception of the #1 and #2 FUEL LOW capsules, all caution and

advisory capsules light steady. The #1 and #2 FUEL LOW capsules flash at a rate of about four flashes per second when activated by a low fuel caution input.

When any caution capsule lights, a signal is applied to a master caution light circuit in the caution/advisory panel. This circuit applies a 28 vdc signal to the pilot’s and copilot’s master warning panels to light the MASTER CAUTION PRESS TO RESET capsules. These capsules remain on until the caution condition is removed or until manually reset by pressing either capsule. Pressing either capsule applies a 28 vdc reset signal to the master caution light circuit to reset it for another caution input. When the MASTER CAUTION PRESS TO RESET capsules are activated by a low fuel caution, they flash as do the #1 and #2 FUEL LOW caution capsules. A diode installed in the warning reset line to the caution/advisory panel eliminates audible warning reset relay transients from preventing reset of the master warning caution circuits when the MASTER CAUTION PRESS TO RESET is pressed and released.

The pilot’s and copilot’s master warning panels receive 28 vdc signals from the instrument display system (IDS) and the fire detection system to light the warning capsules listed in Table 8-1-7-2. The IDS provides warning signals for low engine gas turbine speed (engine-out) and low rotor RPM. The #1 and #2 ENG OUT capsules light steady when activated. The LOW ROTOR RPM capsules flash at three to five flashes per second when activated. The fire detection system generates a fire warning signal when a fire is detected in the No. 1 engine, No. 2 engine, or APU area. The 28 vdc warning signal is routed through a fire detection circuit in the left relay panel to the pilot’s and copilot’s master warning panels to light the FIRE capsules.

Table 8-1-7-2. Master Warning Panel Capsules.

Legend	Fault
#1 ENG OUT	No. 1 engine gas turbine speed (Ng) is below 55%.

Table 8-1-7-2. Master Warning Panel Capsules. (Cont)

Legend	Fault
FIRE	Indicates a fire detector has actuated a fire warning circuit.
MASTER CAUTION PRESS TO RESET	Indicates a caution light on the caution panel has been actuated by failed system.
#2 ENG OUT	No. 2 engine gas turbine speed (Ng) is below 55%.
LOW ROTOR RPM	Rotor speed is below 95% RPM R.

The audible warning function is controlled by the left relay panel. Warning signals from the IDS (low rotor RPM and No. 1 or No. 2 engine out) and the stabilator system are routed through the left relay panel to activate the audible warning unit, which generates both beeping and steady warning tones. The warning tones are applied to the left relay panel which selects either one for application to the intercommunication system. The stabilator system warning signal is given priority over both IDS warning signals. With a stabilator warning condition present, the 28 vdc stabilator warning signal is applied through the normally-closed contacts of relay K49 to the audible warning unit and to the solenoid of relay K44. Relay K44 energizes, selecting the beeping warning tone output from the audible warning unit. The warning tone is fed from the left relay panel through the audio junction box assembly to the pilot's and copilot's intercommunication system stations. With a No. 1 or No. 2 engine-out warning condition present, the 28 vdc signals are applied through the normally-closed contacts of relay K48 to the audible warning unit. The steady warning tone is selected by the de-energized contacts of relay K44 and routed to the pilot's and copilot's stations. With a low rotor RPM warning condition present during flight, the 28 vdc signal is applied through the energized contacts of relay K46 to the audible warning unit. The steady warning signal is applied to the pilot's and copilot's stations as described for engine-out warning condition. The low rotor RPM warning is disabled with weight on wheels. The audible warnings for stabilator shutdown and engine-out condi-

tions may be cleared by pressing one of the MASTER CAUTION PRESS TO RESET capsules. Pressing a capsule applies 28 vdc to the left relay panel audio reset circuit consisting of relays K48 and K49. Relay K48 energizes to disable an engine-out warning, and relay K49 energizes to disable a stabilator shutdown warning.

The caution/advisory panel contains a BRT/DIM-TEST switch that enables testing and changing the light intensity of all caution, advisory, and warning capsules. The switch is momentary in both the BRT/DIM and TEST positions. Placing the switch to TEST applies 28 vdc to all caution and advisory capsule circuits to light all capsules. A 28 vdc test signal is also applied to the master warning panels to light all warning capsules. As in normal operation, the #1 and #2 FUEL LOW caution capsules and the LOW ROTOR RPM warning capsules flash when activated by the test signal. With the switch placed to TEST, a 28 vdc test signal is also applied to the pilot's and copilot's HSI/VSI mode select panels to brighten them. The panels provide ground signals to light the mode select switch lights, the VSI advisory lights, and the CIS mode select panel switch light. The BRT/DIM position of the caution/advisory panel BRT/DIM-TEST switch permits changing the light intensities of all caution, advisory, and warning capsules. When the INST LT PILOT FLT control is turned from OFF, 28 vdc is applied to the caution/advisory panel to enable the light dimming circuit. When the BRT/DIM-TEST switch is placed to BRT/ DIM, 28 vdc is applied to a dimming logic circuit. The dimming

logic circuit applies a ground to the caution and advisory light dimming circuits to dim all caution/advisory panel capsules. A ground is also applied to the master caution and master warning light dimming circuits in the caution/advisory panel. These circuits apply ground outputs to dim the master warning panel capsules. A dim control ground is also applied from the caution/advisory panel to the indicator light dimming and the T-handle light dimming circuits in the left relay panel, and to the stabilator control/auto flight control panel to enable dimming of other helicopter lights. When the BRT/DIM-TEST switch is placed to BRT/DIM again or if the 28 vdc input to the switch is removed, the ground output from the dimming logic circuit is removed, and all caution/advisory panel, master warning panel, and all other helicopter lights that were dimmed now go on bright. Further dimming of the caution/advisory and master warning panel capsules is provided when the soft lighting system (SLS) is used. When the pilot's or copilot's PNL LTS switch is pressed, the following panel indicating lights are disabled: CARGO HOOK EMERG REL, rescue hoist control panel (when installed) SQUIB IND light, IRCM control panel ON and INOP lights, miscellaneous switch panel TAIL WHEEL LOCK and UNLK lights, pilot's and copilot's radar altimeter indicator LOW warning lights and ALT FEET digital readouts, blade de-ice test panel MAIN RTR and TAIL RTR PWR lights, blade de-ice control panel TEST IN PROGRESS light, all failure and advisory lights on the stabilator control/auto flight control panel, the keyboard and digital readout lights on the Doppler computer display, pushbutton switch lights on the CIS mode select panel, indicator lights on the ESSS and auxiliary searchlight control panels, lighted pushbutton switches on the pilot's and copilot's HSI/VSI mode select panels, and the DH, GA, and MB lights on the pilot's and copilot's VSIs. The No. 1 engine, No. 2 engine, and APU fire control T-handles may also go to the dim operating mode, depending on the lighting loads selected when the PNL LTS switch is pressed. Also, when the pilot's or copilot's PNL LTS switch is pressed, a 28 vdc signal is applied to the caution/advisory panel dimming logic circuit to enable the caution/advisory and master warning dimming circuits. The CAUT/ADVSY NVG DIMMING control on the instrument panel can now be used to dim the caution/

advisory capsules to the intensity required for SLS use. The MA WRN NVG DIMMING control on the instrument panel can be used to dim the master warning capsules to SLS intensity.

The panel light dimming system will enter an out of synchronization condition if the PNL LTS switch is pressed to establish SLS conditions and left in that position when the helicopter is shut down. When electrical power is restored to the helicopter, the caution/advisory panel lights will be bright while the lower console warning lights will be off. To restore synchronization, the PNL LTS switch must be pressed and released, followed by removing and then restoring helicopter electrical power.

8-1-7.3 Caution/Advisory Panel.

The control circuits for distributing input and output signals within the caution/advisory panel are on four channel cards and one control card (Figures 8-1-7-2 and 8-1-7-4). These are designated cards A1 through A5.

8-1-7.3.1 Channel Card A1.

Channel card A1 contains positive input seeking circuitry that controls 27 caution capsules. When a fault is detected by any of those associated helicopter systems, 28 vdc input is applied to channel card A1. Channel card A1 then supplies a voltage to the corresponding capsule, causing the capsule lamps to go on. Whenever a caution capsule goes on, an output from the channel card is also applied (through channel card A4) to light external master caution lights. Master caution lights remain on until either master caution capsule is reset (pressed) or until the fault that caused the capsule to light is no longer present. Pressing either master caution capsule provides a reset signal to channel card A4, turning off the master caution capsules while the caution capsule on the caution/advisory panel will remain on until the fault that caused the capsule to light is no longer present.

8-1-7.3.2 Channel Card A2.

Channel card A2 contains positive input seeking circuitry that controls 22 caution capsules and three advisory capsules. The operation of the caution

capsule circuitry is identical to channel card A1 caution capsule circuitry (PARA 8-1-7.3.1). The operation of all the advisory capsules (whether controlled by channel card A2, A3, or A4) is the same. The advisory capsule will be on whenever the corresponding positive input is present and will automatically go/be off when the positive input is removed/not present.

8-1-7.3.3 Channel Card A3.

Channel card A3 contains seven negative (50 ohms or less) input-seeking circuits that control seven caution capsules and 13 positive input-seeking circuits that control 13 advisory capsules. The operation of caution capsules (5 and 66) is identical to channel card A1 caution circuitry, except that these capsules require a negative (50 ohms or less) input. The operation of the five remaining caution capsules (21, 33, 42, 74, and 75) is identical to channel card A1 caution circuitry, except that these capsules require a negative (50 ohms or less) input. On panel, 70550-01107-102, these five caution capsules also have latching circuits that remain on once turned on and do not go off until reset. When the fault is removed, a latched capsule can be turned off by applying either a master or special reset signal. A master reset occurs when operational power is removed from the caution/advisory panel for longer than 200 msec. When a master reset is applied, all latched capsules will be off. A special reset signal is generated when power is removed from the particular system being monitored. With a special reset applied, only the associated capsule resets. One special reset input resets capsules 33, 74, and 75 from channel card A3 and capsules 13 and 14 from channel card A4. A second special reset signal only resets capsule 21, while another special reset signal only resets capsule 42. Caution capsule 33 control circuitry also requires that the negative input be present for at least 30 seconds before capsule 33 is turned on. The operation of the 13 advisory capsules is identical to channel card A2 advisory capsule circuitry (PARA 8-1-7.3.2).

8-1-7.3.4 Channel Card A4.

Channel card A4 contains the four negative input-seeking circuits that control four caution capsules

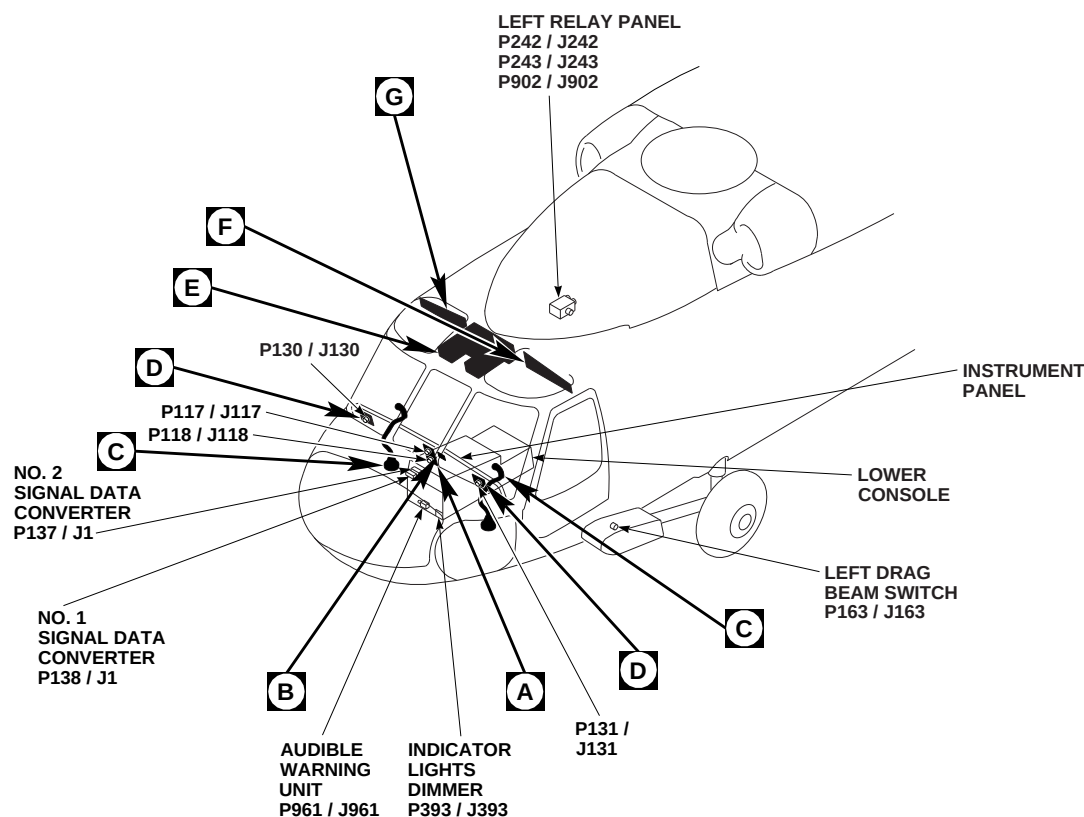
(29, 32, 50 and 53), two negative input-seeking circuits (latching type) that control two caution capsules (13 and 14), a flasher circuit and two positive input-seeking circuits that control two caution capsules (1 and 62), and two positive input-seeking circuits that control two advisory capsules (81 and 82). These capsules require a negative (50 ohms or less) input. The operation of caution capsules (13 and 14) is identical to channel card A1 caution circuitry, except that these capsules require a negative (50 ohms or less) input. On panel, 70550-01107-102, caution capsules 13 and 14 also have latching circuits that remain on once turned on and do not go off until reset. The two types of reset signals are described in PARA 8-1-7.3.3. The operation of caution capsules (1 and 62) is identical to channel A1 caution circuitry, except that these capsules are driven by a flashing circuit, causing the capsules and master warning capsules to flash. The advisory capsules (81 and 82) will be on whenever the corresponding positive input is present and will automatically go/be off when the positive input is removed/not present.

8-1-7.3.5 Bright/Dim/SLS Control Card A5.

Card A5 contains the circuits that permit changing the intensity of the capsules on the caution/advisory panel (and on the external master warning panels) from bright to dim. Two levels of dim intensity are provided. Card A5 contains three voltage regulators, with a Darlington-pair transistor circuit at the output of each regulator. One half of each Darlington circuit is on card A5; the other half is chassis mounted. Chassis-mounted Q1 is in the return to ground path of the lighting circuit for all capsules in the external master warning panel; Q2 is in the return path for all 18 advisory capsules on the caution/advisory panel; Q3 is in the return path for all 64 caution capsules. In the absence of a dimming enable signal, each of the three Darlington-pairs conduct full on and provide a low impedance return path to ground for all lamps in all panels. Therefore, all lamps light at full brightness. When a dimming enable signal is externally applied and the BRT/DIM-TEST switch S1 is placed to BRT/DIM, circuit conditions change. Switch S1 applies the enabling signal to channel card A4 to energize a logic circuit. A ground is then applied to the three voltage regulators on card A5. The output of the

regulations biases the Darlington-pair transistors so that current flow is reduced. With increased resistance in the lamp return circuits, all capsules light at reduced intensity. To restore their intensity to full brightness, press and release the BRT/DIM switch. This de-energizes the logic circuit on card A4, removing the ground input signal to the three voltage regulators. Pressing the BRT/DIM switch energizes the relays and reapplies the ground to the voltage regulations which control current flow in the Darlington-pair transistors, causing all lights to dim. Card A5 also contains two additional voltage regulators and three relays for SLS lighting. The SLS circuit logic will activate SLS if a 28 vdc enable is present upon initial power application and is removed. The SLS circuit logic will activate SLS if a 0 vdc enable is present upon initial power ap-

plication and 28 vdc is applied. One voltage regulator controls current flow in Darlington-pair Q1, which is in the ground return lighting circuit for all capsules in the external master warning panels. Current flow in Q1 is held to a low level; therefore, lamps on this external panel have a high impedance in their return path and light at lowest intensity. Variation of the input voltage to the lamps is provided by an external control. The second voltage regulator on card A5 is used to control current flow in Darlington-pairs Q2 and Q3. With this circuit arrangement, the lamps in both the advisory and caution capsules of the caution/advisory panel operate at the identical low intensity. A separate external control is provided to vary the intensity of all 82 lamps on this panel.



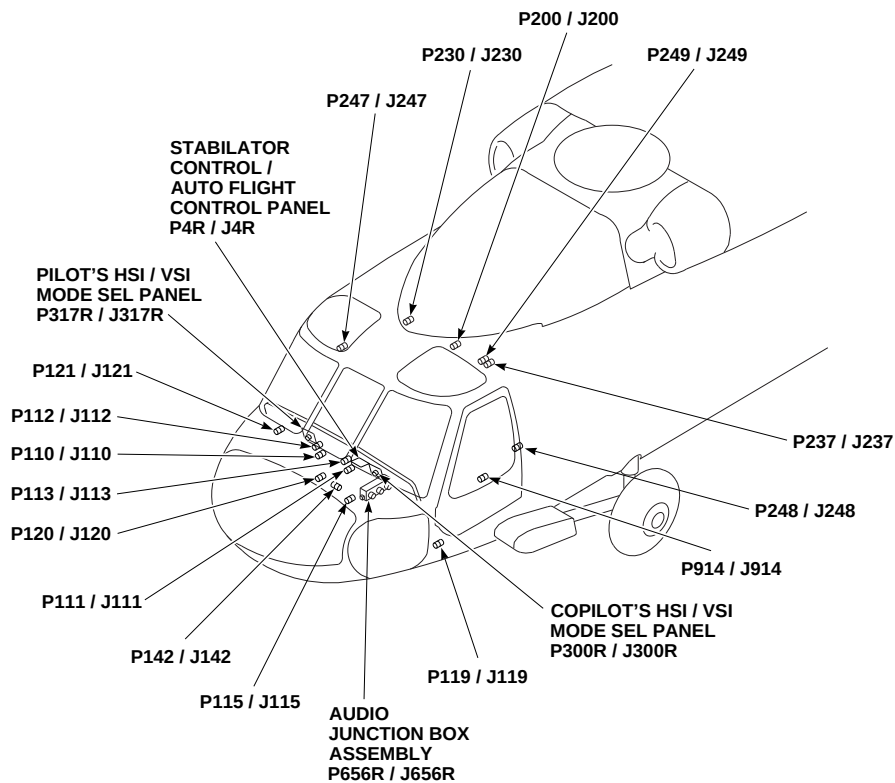
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P4R / J4R	STABILATOR CONTROL / AUTO FLIGHT CONTROL PANEL
P110 / J110	COCKPIT BL 7.5 RH, STA 197
P111 / J111	COCKPIT BL 7.5 LH, STA 197
P112 / J112	COCKPIT BL 6 RH, STA 197
P113 / J113	COCKPIT BL 6 LH, STA 197
P115 / J115	COCKPIT BL 6 LH, STA 210
P117 / J117	CAUTION / ADVISORY PANEL
P118 / J118	CAUTION / ADVISORY PANEL
P119 / J119	COCKPIT TUB BL 30 LH, STA 220
P120 / J120	COCKPIT TUB BL 30 RH, STA 220
P121 / J121	COCKPIT BL 10 RH, STA 195
P130 / J130	PILOT'S MASTER WARNING PANEL

NOTES

1. **ESSS**
2. NOT INSTALLED ON **702375-SUBQ**

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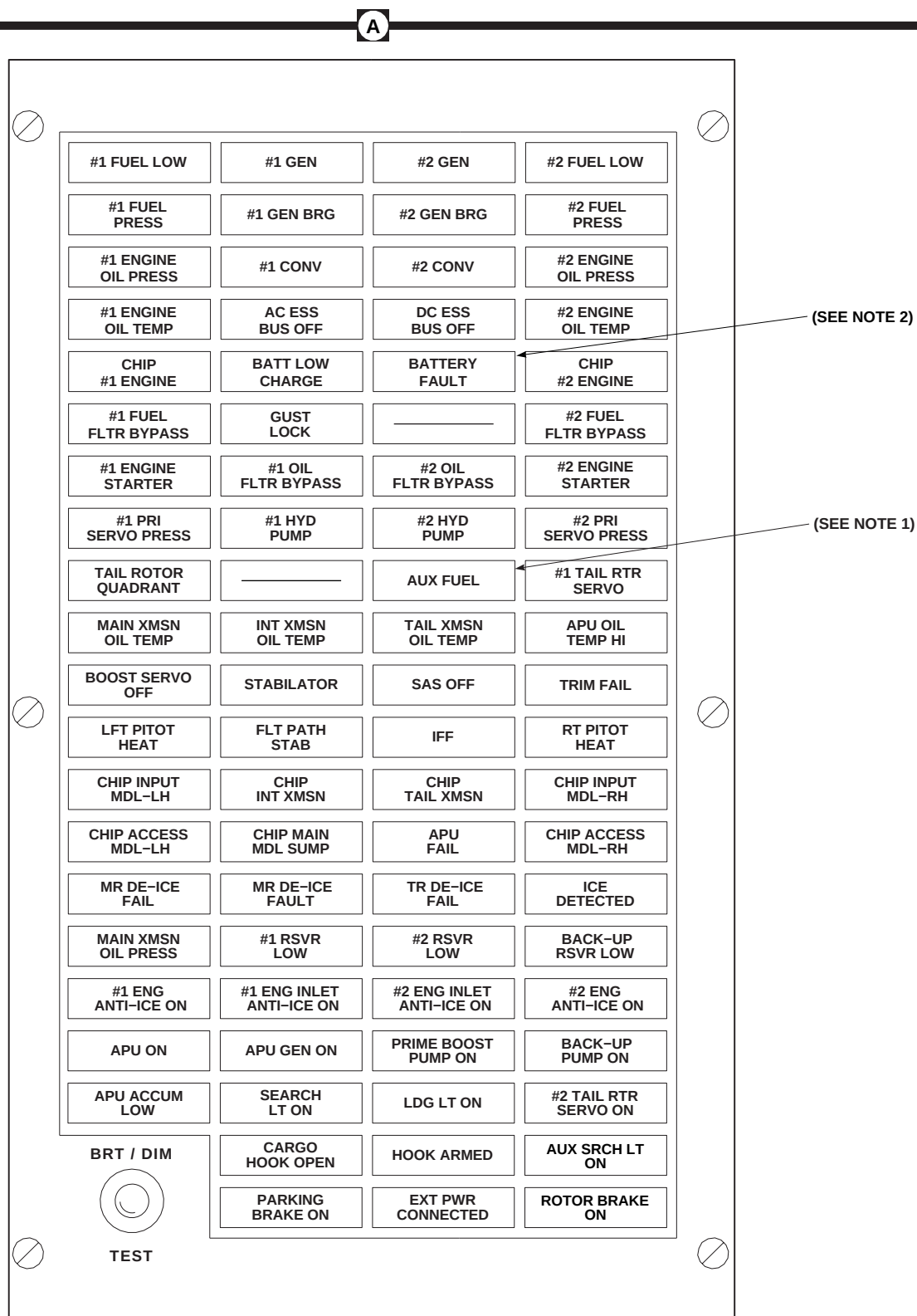
FIGURE 8-1-7-1. CAUTION/ADVISORY WARNING SYSTEM LOCATION DIAGRAM
(SHEET 1 OF 7)



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P131 / J131	COPILOT'S MASTER WARNING PANEL
P137 / J1	NO. 2 SIGNAL DATA CONVERTER
P138 / J1	NO. 1 SIGNAL DATA CONVERTER
P142 / J142	COCKPIT, BL 3 RH, STA 210
P163 / J163	LEFT DRAG BEAM SWITCH
P200 / J200	CABIN CEILING, BL 13 LH, STA 246
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P237 / J237	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 25 LH
P242 / J242	LEFT RELAY PANEL
P243 / J243	LEFT RELAY PANEL

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P247 / J247	CABIN, BL 40 RH, STA 248
P248 / J248	CABIN, BL 40 LH, STA 248
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P300R / J300R	COPILOT'S HSI / VSI MODE SELECT PANEL
P317R / J317R	PILOT'S HSI / VSI MODE SELECT PANEL
P393 / J393	INDICATOR LIGHTS DIMMER
P656R / J656R	JUNCTION BOX ASSEMBLY
P902 / J902	LEFT RELAY PANEL
P914 / J914	COCKPIT, BL 24 LH, STA 247
P961 / J961	AUDIBLE WARNING UNIT

**FIGURE 8-1-7-1. CAUTION/ADVISORY WARNING SYSTEM LOCATION DIAGRAM
(SHEET 2 OF 7)**



**FIGURE 8-1-7-1. CAUTION/ADVISORY WARNING SYSTEM LOCATION DIAGRAM
(SHEET 3 OF 7)**

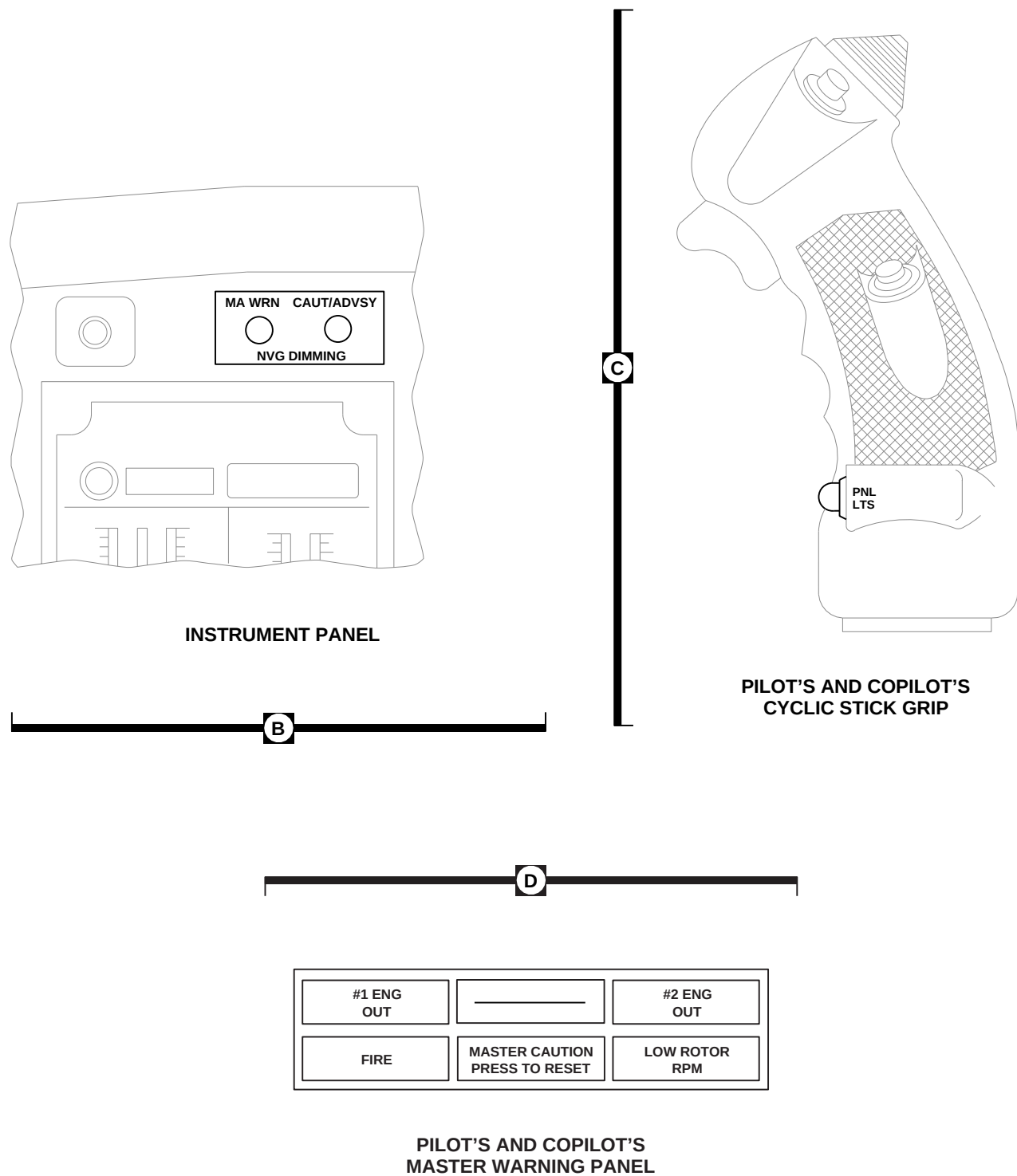
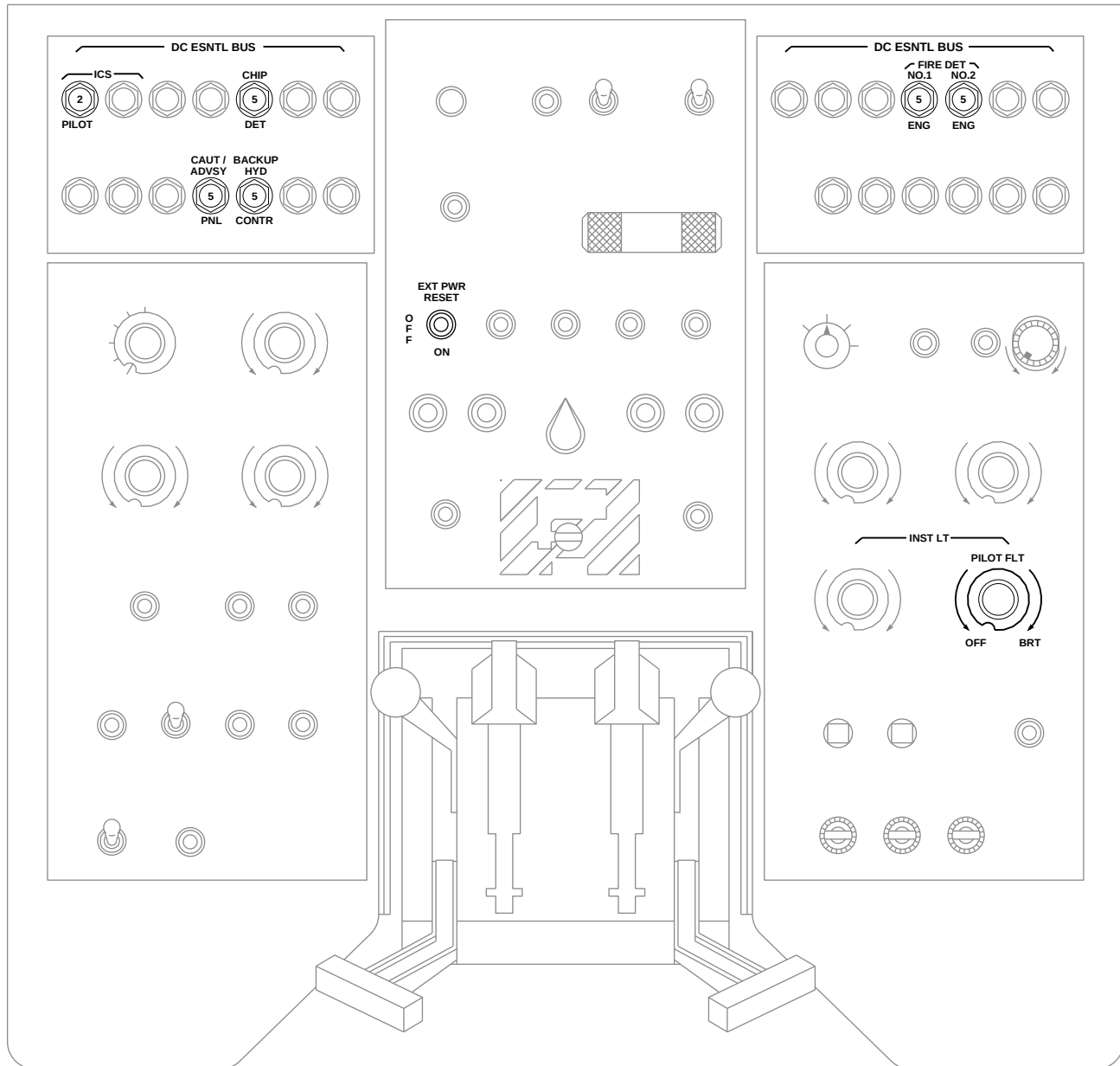


FIGURE 8-1-7-1. CAUTION/ADVISORY WARNING SYSTEM LOCATION DIAGRAM
(SHEET 4 OF 7)

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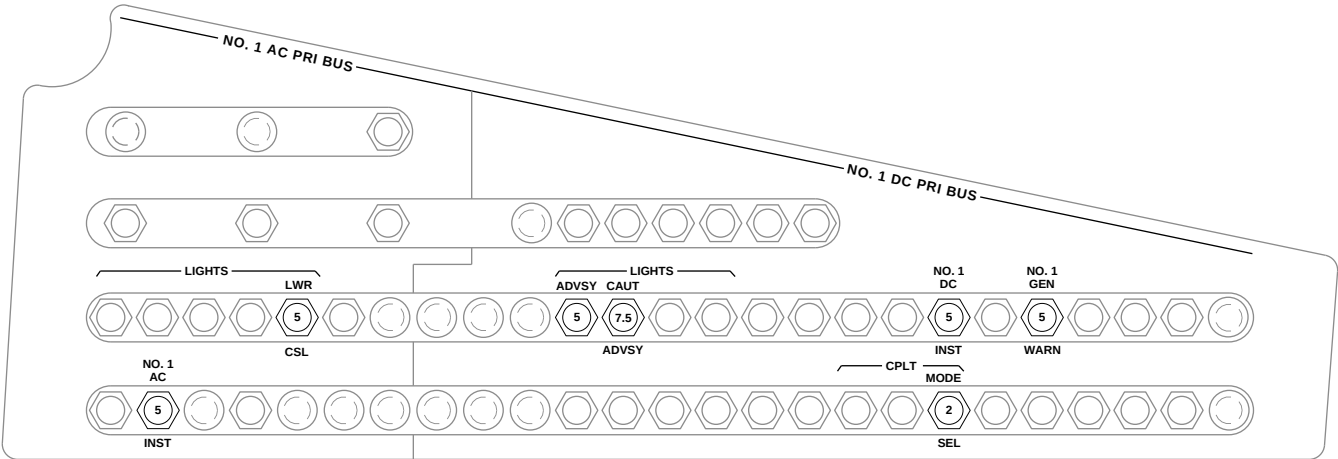
E



UPPER CONSOLE

FT1347_5
SAFIGURE 8-1-7-1. CAUTION/ADVISORY WARNING SYSTEM LOCATION DIAGRAM
(SHEET 5 OF 7)

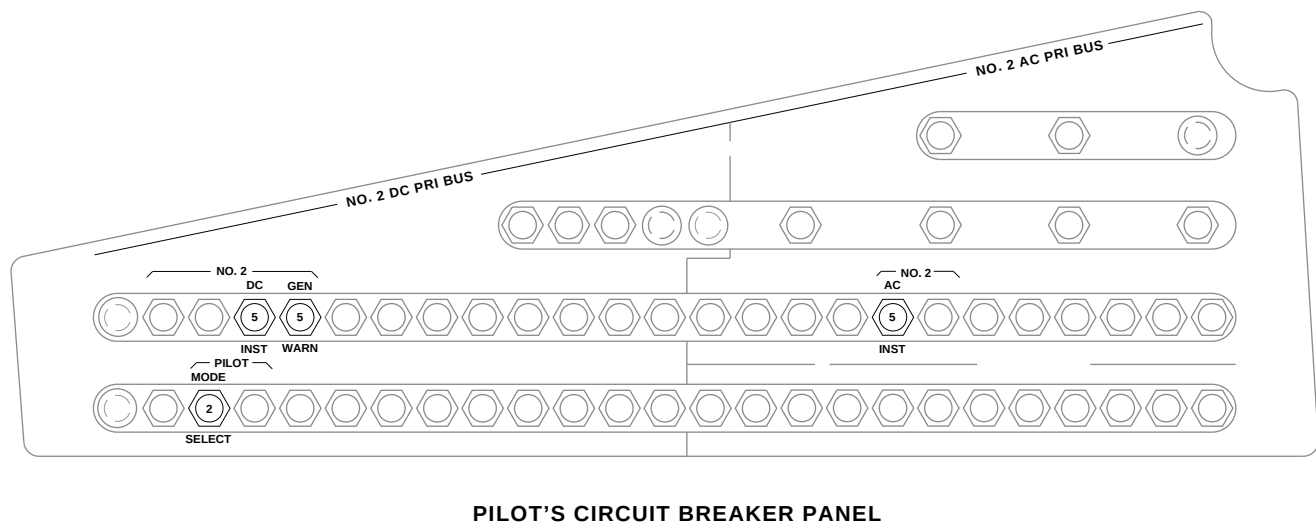
8-1-7-16



COPILOT'S CIRCUIT BREAKER PANEL

F

FIGURE 8-1-7-1. CAUTION/ADVISORY WARNING SYSTEM LOCATION DIAGRAM
(SHEET 6 OF 7)



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**FIGURE 8-1-7-1. CAUTION/ADVISORY WARNING SYSTEM LOCATION DIAGRAM
(SHEET 7 OF 7)**

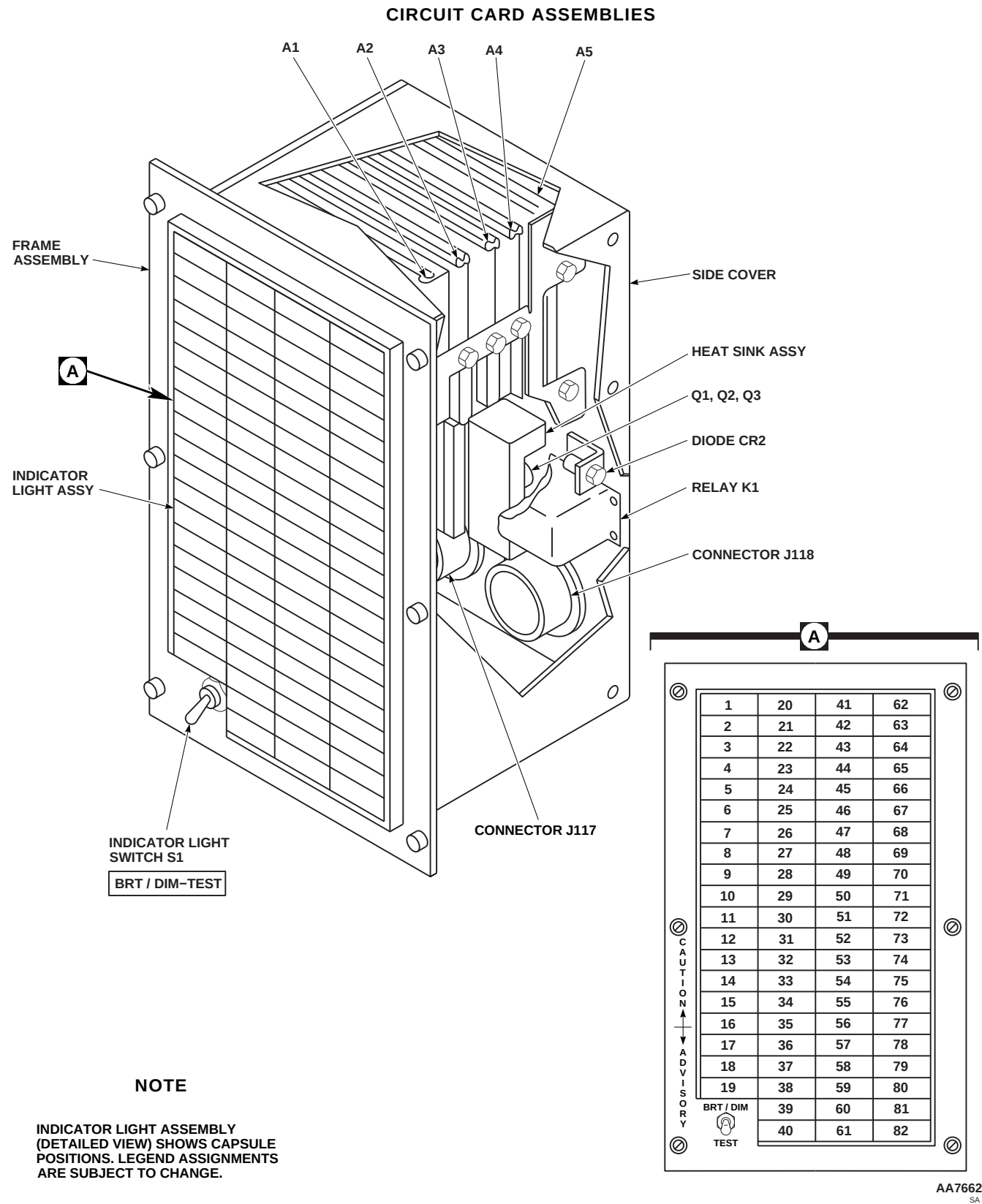


FIGURE 8-1-7-2. CAUTION/ADVISORY PANEL PARTS LOCATION DIAGRAM

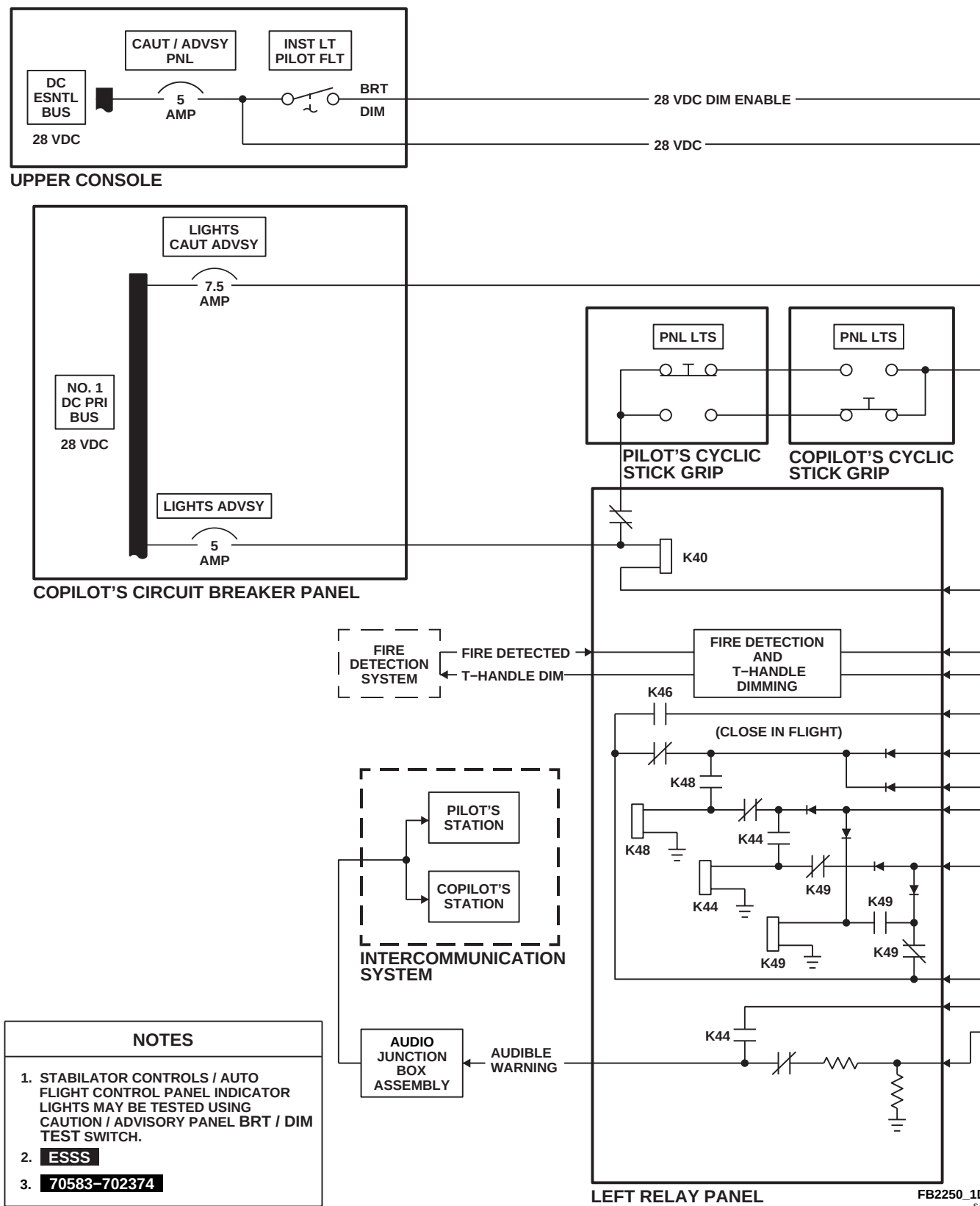


FIGURE 8-1-7-3. CAUTION/ADVISORY WARNING SYSTEM BLOCK DIAGRAM
(SHEET 1 OF 7)

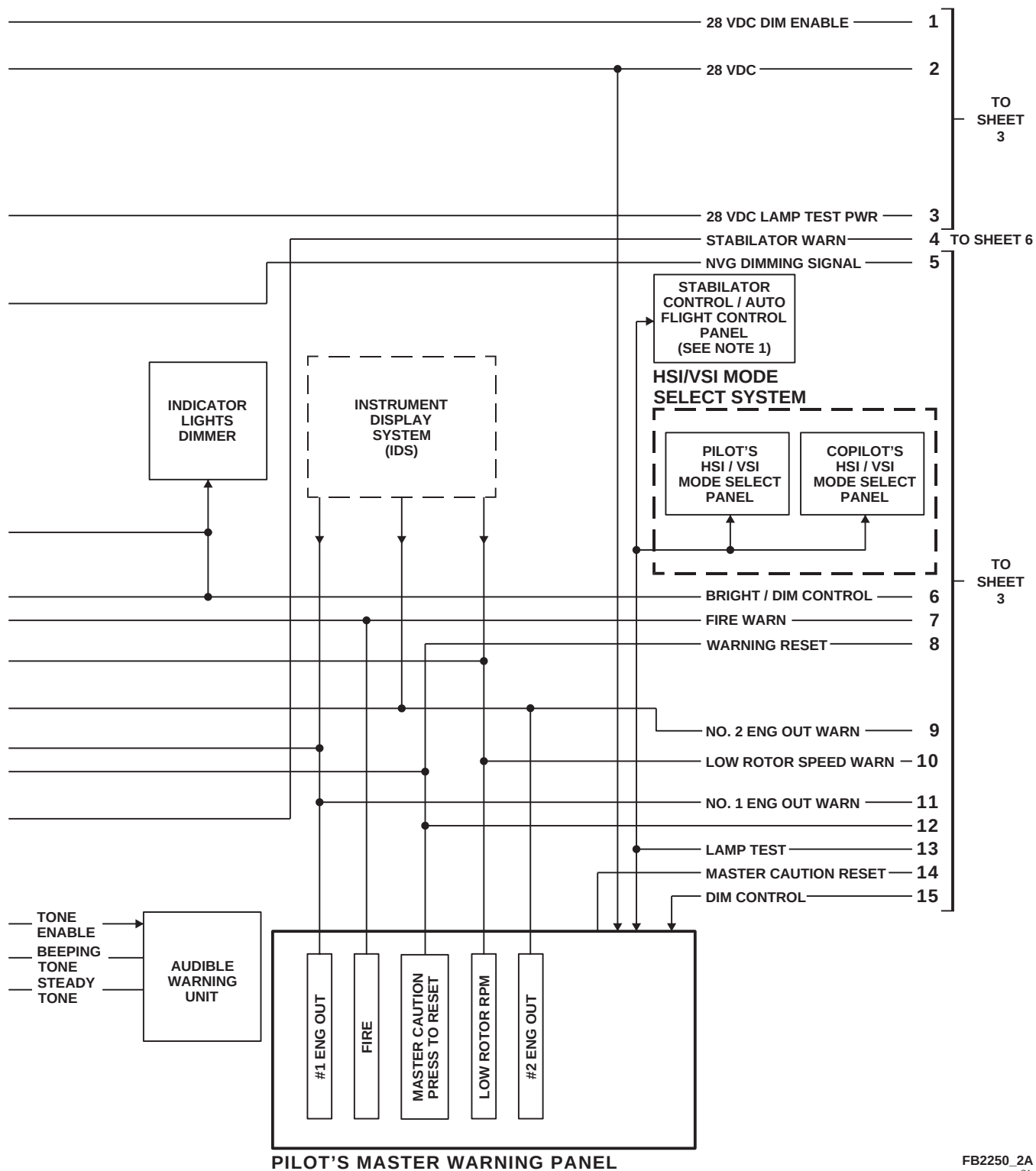


FIGURE 8-1-7-3. CAUTION/ADVISORY WARNING SYSTEM BLOCK DIAGRAM
(SHEET 2 OF 7)

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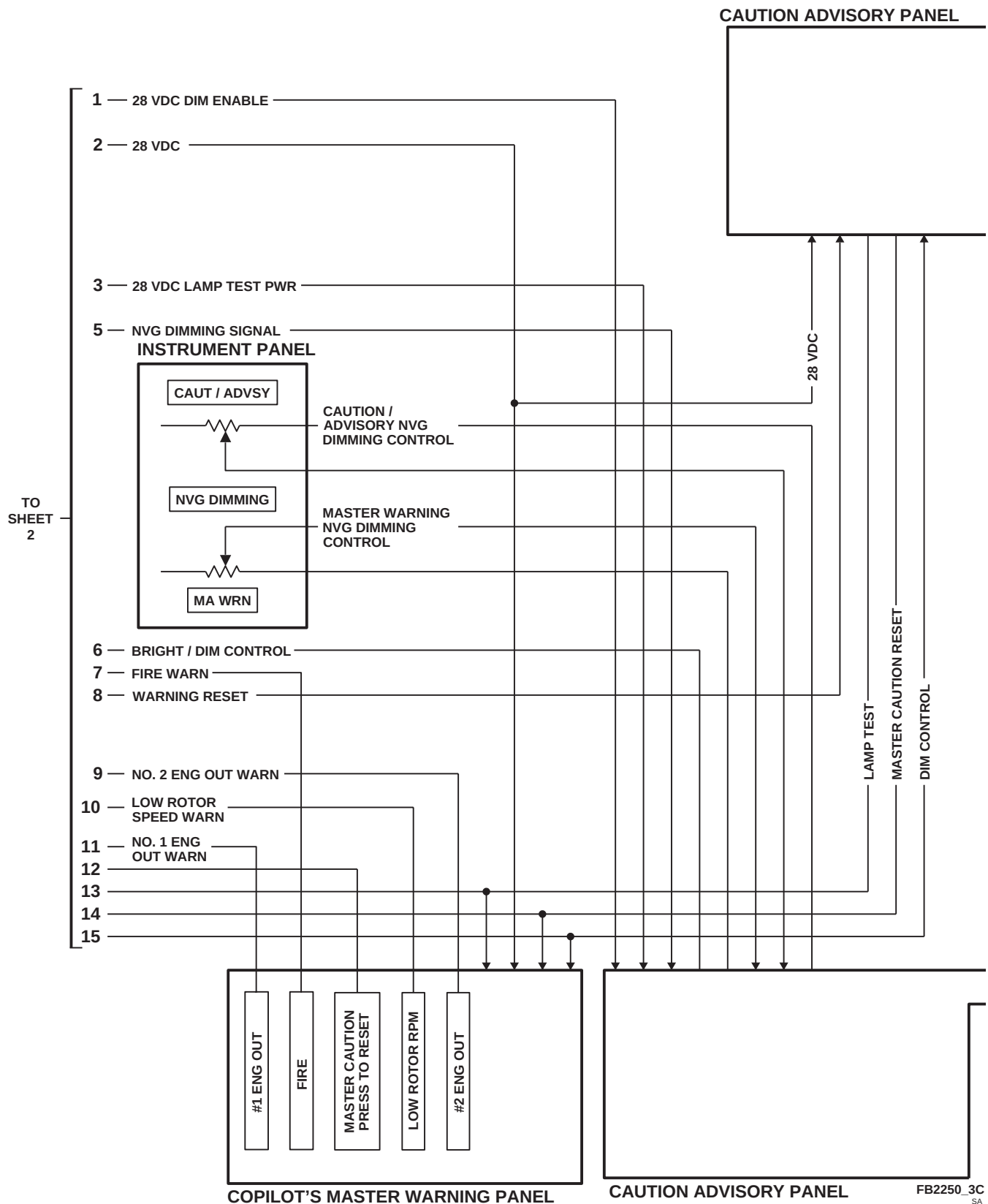
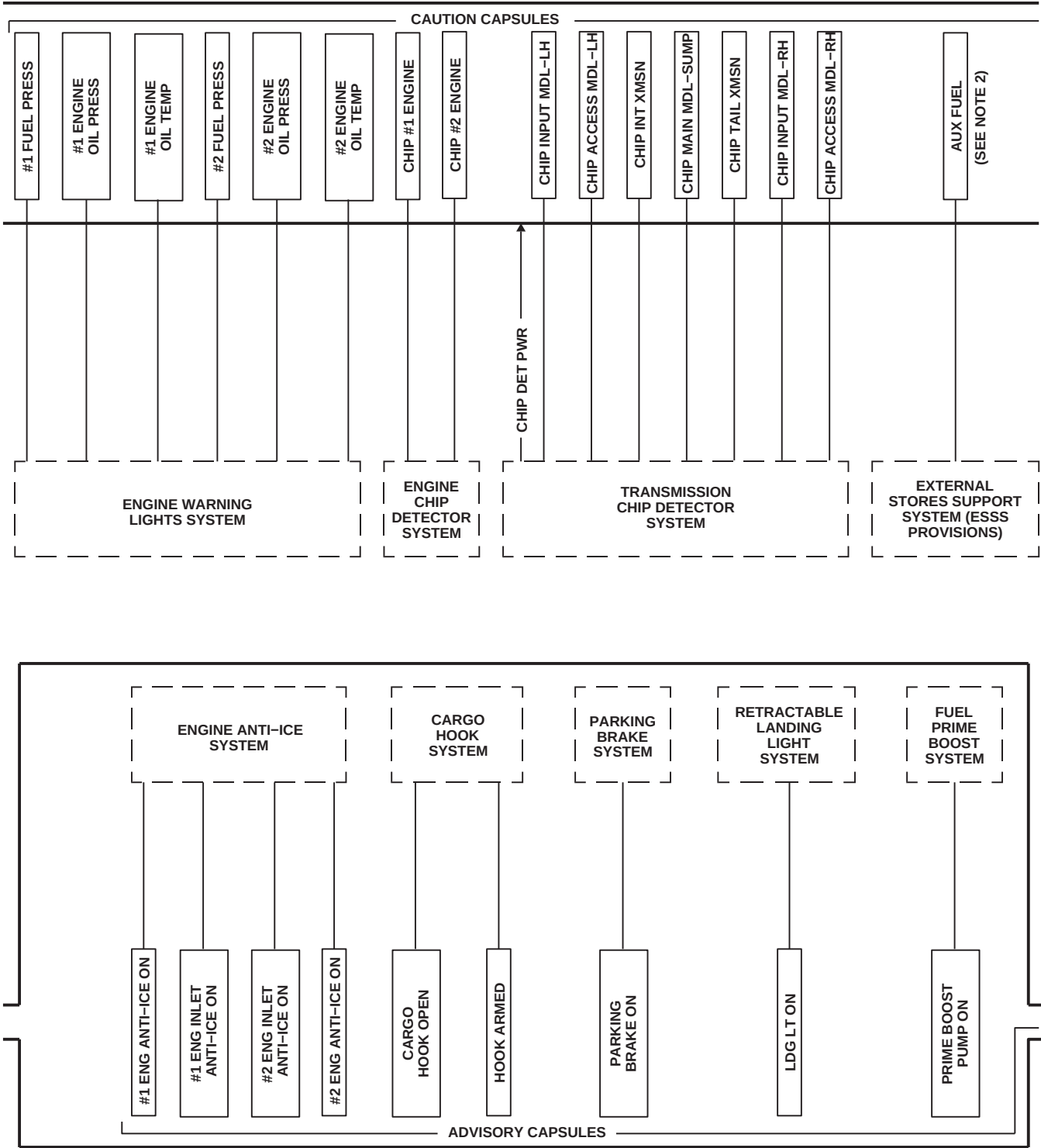


FIGURE 8-1-7-3. CAUTION/ADVISORY WARNING SYSTEM BLOCK DIAGRAM
(SHEET 3 OF 7)



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FIGURE 8-1-7-3. CAUTION/ADVISORY WARNING SYSTEM BLOCK DIAGRAM
(SHEET 4 OF 7)

CAUTION ADVISORY PANEL

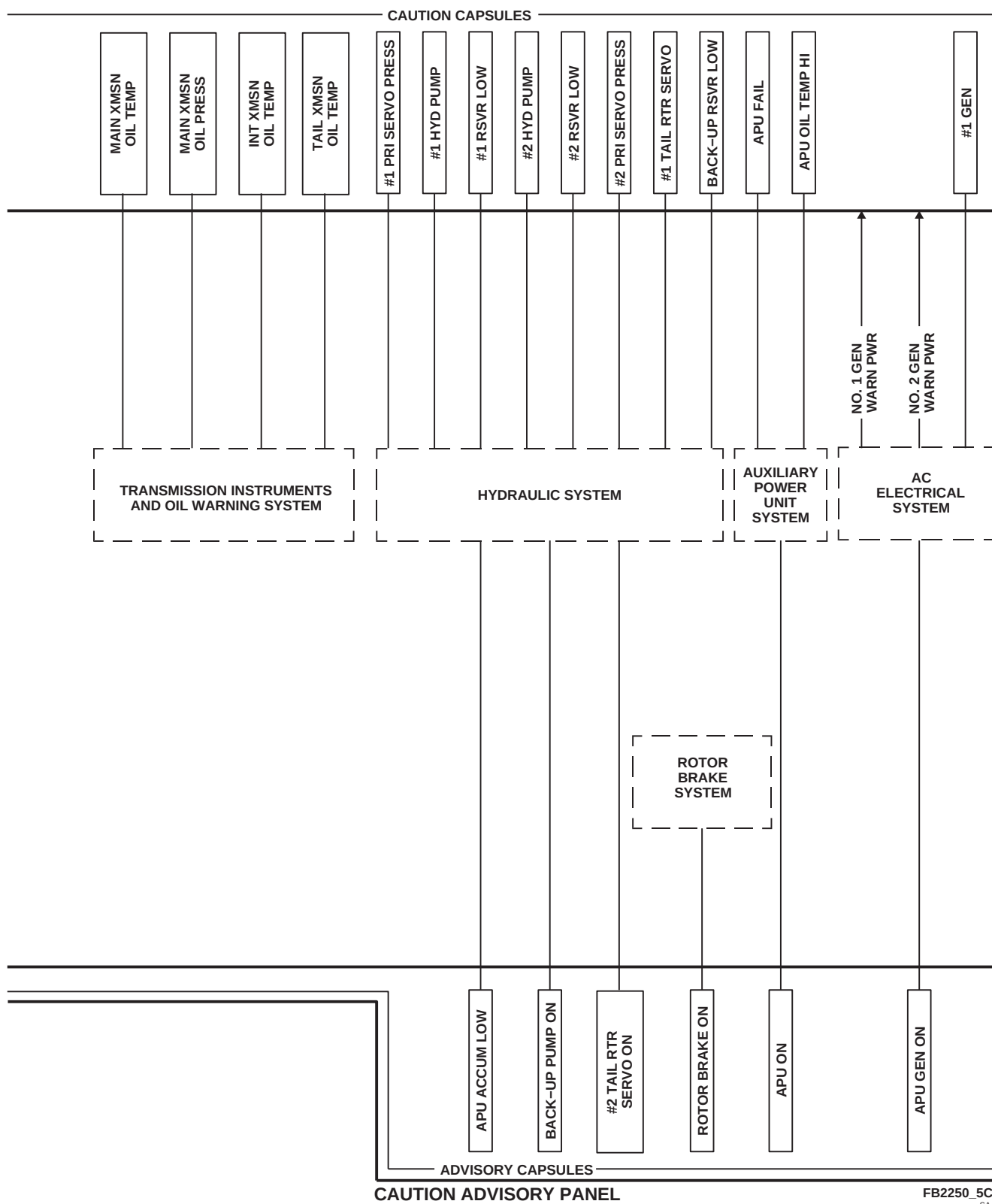
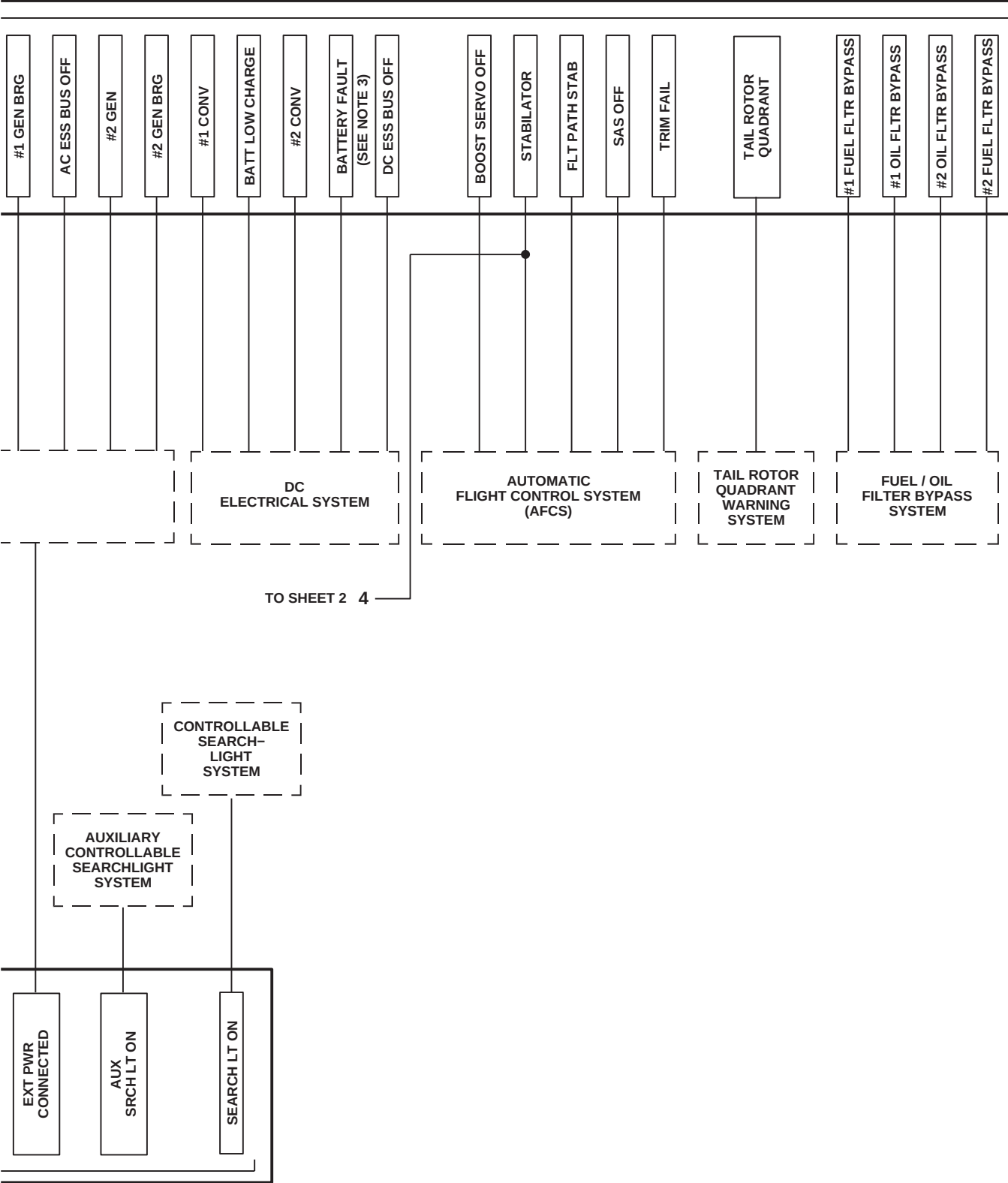
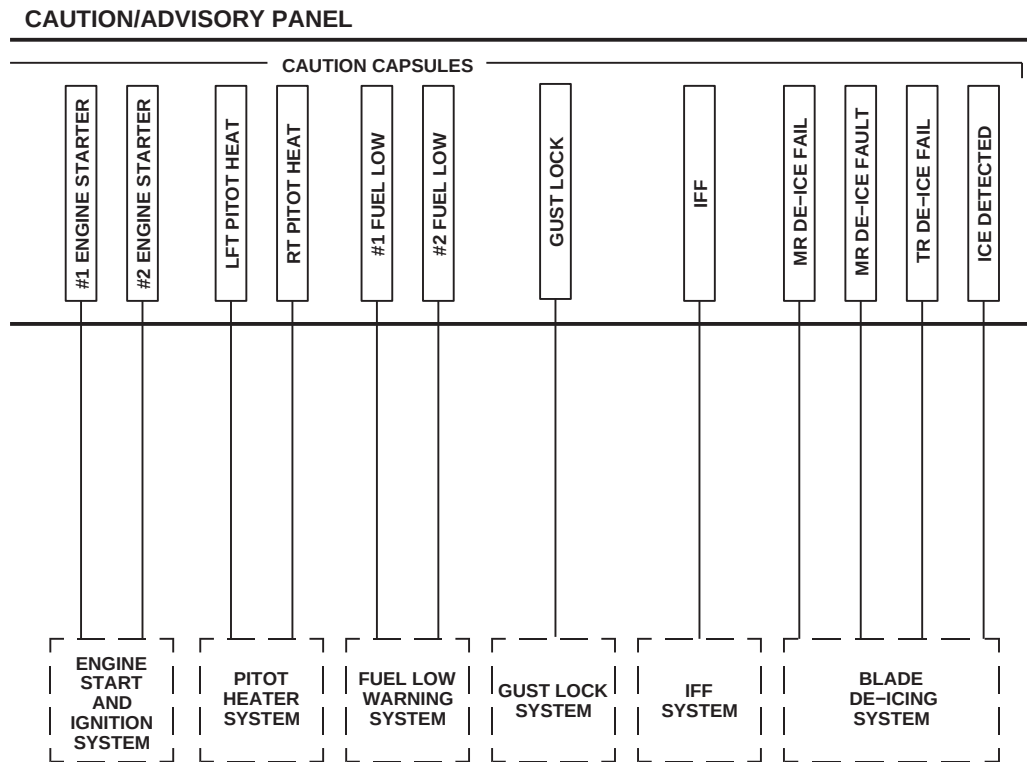
FB2250_5C
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FIGURE 8-1-7-3. CAUTION/ADVISORY WARNING SYSTEM BLOCK DIAGRAM
(SHEET 5 OF 7)



FC2250_6D
SA

FIGURE 8-1-7-3. CAUTION/ADVISORY WARNING SYSTEM BLOCK DIAGRAM
(SHEET 6 OF 7)



FB2250_7C
SA

**FIGURE 8-1-7-3. CAUTION/ADVISORY WARNING SYSTEM BLOCK DIAGRAM
(SHEET 7 OF 7)**

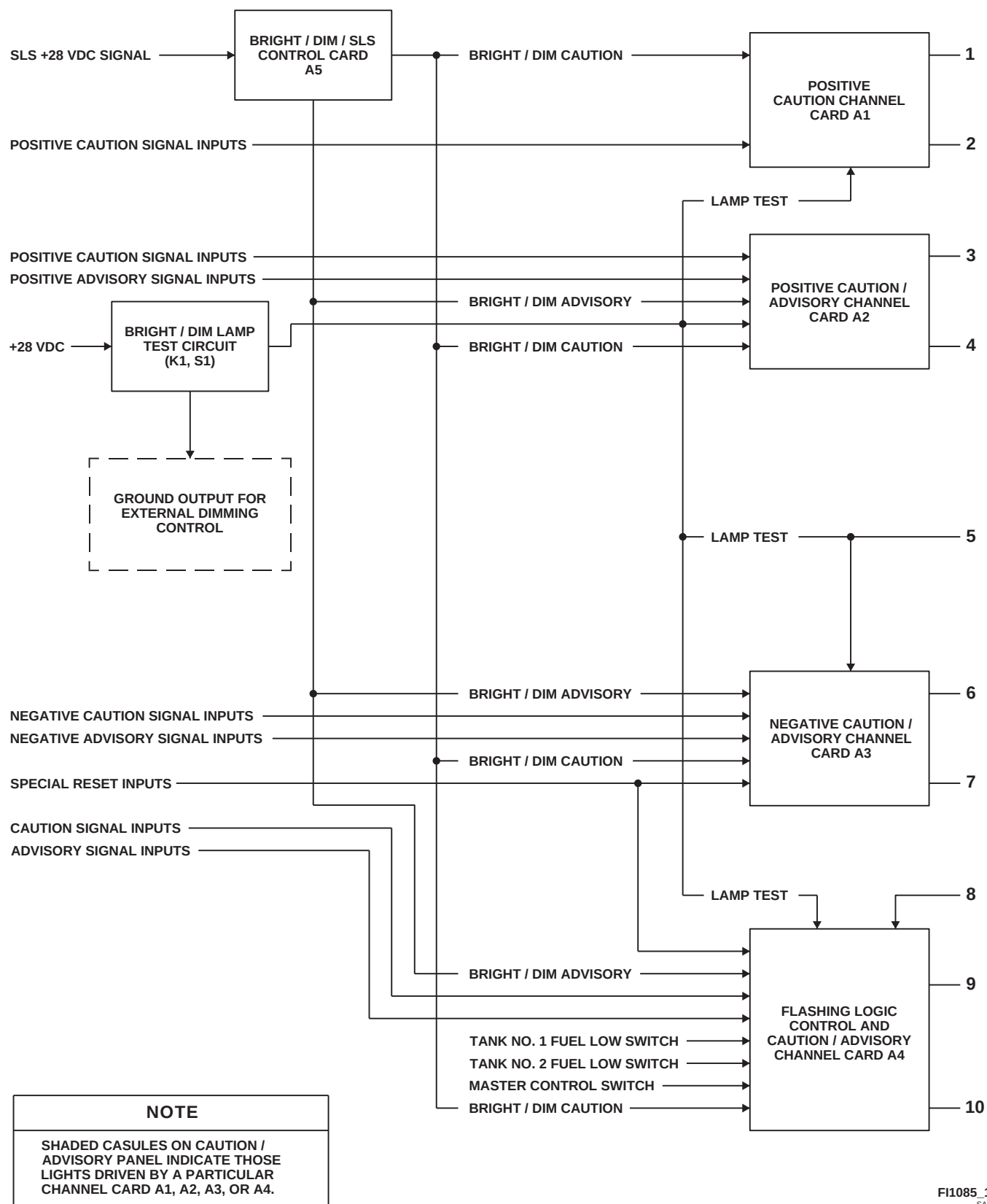


FIGURE 8-1-7-4. CAUTION/ADVISORY PANEL BLOCK DIAGRAM (SHEET 1 OF 2)

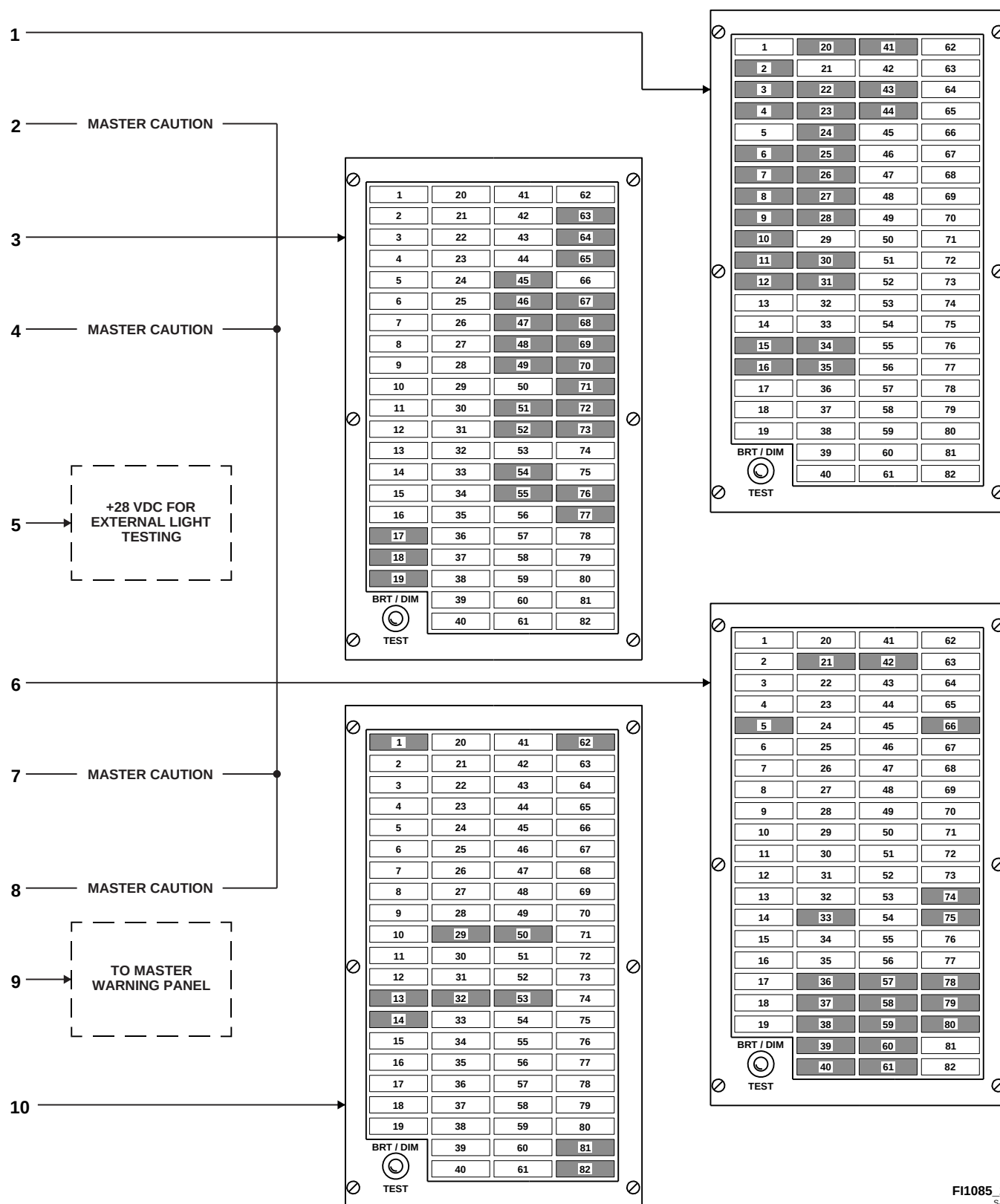
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FIGURE 8-1-7-4. CAUTION/ADVISORY PANEL BLOCK DIAGRAM (SHEET 2 OF 2)

SECTION 8-2

TROUBLESHOOTING

SECTION OVERVIEW

PARAGRAPH	TITLE
8-2-1	Pitot-Static System
8-2-2	Pitot Heater
8-2-3	Instrument Display System
8-2-4	Caution/Advisory Warning System
8-2-5	Digital Clock
8-2-6	Central Display Unit (70450-01043-122) (BENCH)
8-2-7	Central Display Unit (70450-01916-105) (BENCH)
8-2-8	Pilots Display Unit (BENCH)
8-2-9	Signal Data Converter (BENCH)
8-2-10	Caution/Advisory Panel (BENCH)
8-2-11	Miscellaneous Switch Panel (BENCH)

8-2-1 PITOT-STATIC SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45
- Pitot and Static System Test Set

Tools

- Electrical Repairer Toolkit
- Locally-Made Pitot-Static System Test Set Power Cable Assembly, Figure H-49, Appendix H
- Locally-Made Pitot-Static Test Set Tee Air Hose Assembly, Figure H-49, Appendix H
- Pitot-Static Head Adapters, 884DB (2 Required)

Materials/Parts

- Ethylene Glycol, Item 137, Appendix D

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available



Damage could occur to Pitot-static components if reverse pressure is ap-

plied to Pitot-static heads during operational/troubleshooting procedure. Make sure test set and connections are properly set up before applying power to the test set. Use extreme caution when removing the Pitot-static test set and adapters. Slowly return Pitot and static pressure to 0. Do not pull Pitot-static adapters off Pitot heads until all readings are at 0, since this could cause damage to components.

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

Special Instructions

The Pitot-static operational/troubleshooting procedure is subdivided as follows:

- (1) Setup
- (2) Barometric Altimeter/Vertical Speed Indicator
- (3) Airspeed Indicator
- (4) Barometric Altimeter Lighted Bezel Assembly
- (5) Shutdown

8-2-1.1 Setup.

- Connect external electrical power.
- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Make sure upper console PITOT HEAT switch is placed OFF.
- Remove covers from Pitot tubes.
- Drain Pitot-static lines.
- Make sure the following circuit breakers are pulled out:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
RT PITOT HEAT	Pilot's
LEFT PITOT HEAT	Copilot's

temperatures, use mixture of one-half water and one-half Ethylene glycol (Item 137, Appendix D).

WARNING

To prevent injury to personnel, make sure Pitot heaters are off (PITOT HEAT switch on upper console panel is OFF), and Pitot heads are cool.

- g. Attach locally-made tee air hoses to test set.
- h. Attach P2 of locally-made power cable to power connector at rear of test set.

NOTE

Qualified personnel may use external power or power test set using vac or vdc power.

- i. Attach P1 of power cable to 28 vdc cabin utility receptacle.
- j. Cap all tee air hoses.
- k. Do a leak check of test set and tee air hoses.
- l. Attach tee air hoses to Pitot-static head adapters, making sure Pitot and static lines are not crossed.
- m. Attach static tee air hoses to pilot's and copilot's vertical speed indicators.

CAUTION

To prevent damage to Pitot-static heads, use only pure Ethylene glycol and not antifreeze.

NOTE

Make sure that vent holes on Pitot-static head adapters are not plugged.

- n. Wet inside surface of Pitot-static head adapters with water to permit easier installation and prevent wear of sealing rings. In freezing

- o. Using twisting motion, slide Pitot-static head adapter connected to test set over pilot's (right) Pitot head until fully seated.
- p. Using twisting motion, slide other Pitot-static head adapter over copilot's (left) Pitot head until fully seated.
- q. Turn on electrical power (PARA 1-6-16).
- r. Set test set altimeter to 29.92 in. hg.

8-2-1.2 Barometric Altimeter/Vertical Speed Indicator.

NOTE

In the following steps, barometric altimeter/vertical speed indicator readings shall not exceed tolerance ratings specified on test set calibration sticker.

- a. Do Setup, if not already done.
- b. Set pilot's and copilot's vertical speed indicator pointers to indicate 0 fpm with zero adjust screw (lower left side of instrument).
- c. Set pilot's and copilot's barometric altimeters to scale reading of 29.92 in. hg.

RESULT: Pilot's and copilot's barometric altimeters shall indicate from -50 to 50 feet.

- If result is not as specified, check for leaky adapter sealing rings and loose tee air hose

8-2-1-2

connections to pilot's and copilot's vertical speed indicators and altimeters.

- (1) If leaky seals are present, replace seals.
- (2) If loose fittings are present, tighten fittings.
- (3) If trouble remains, replace pilot's or copilot's barometric altimeter or vertical speed indicator, as required (PARA 8-4-6 or PARA 8-4-4).

d. Listen to pilot's and copilot's barometric altimeters to make sure internal vibrator is working.

RESULT:

- (1) Pilot's barometric altimeter encoder vibrator shall be working.
 - (2) Copilot's barometric altimeter vibrator shall be working.
- If result (1) is not as specified, go to TABLE NO. 8-2-1-1.
 - If result (2) is not as specified, go to TABLE NO. 8-2-1-2.
- e. Set test set altitude indicator to 0 feet.
- f. While monitoring pilot's barometric altimeter footage needle, turn pilot's barometric altimeter pressure knob from fully clockwise to fully counterclockwise.

RESULT:

- (1) Pilot's altimeter footage indication shall change about 2500 feet.
 - (2) Pilot's barometric altimeter footage needle shall not bind or skip.
- If results are not as specified, replace barometric altimeter (PARA 8-4-6).

g. Set pilot's barometric altimeter scale to 29.92 in. hg.

RESULT: Pilot's barometric altimeter shall indicate from -50 to 50 feet.

- If result is not as specified, replace pilot's barometric altimeter encoder (PARA 8-4-6).
- h. Repeat steps f. and g. for copilot's barometric altimeter.
- i. Set test set altimeter indicator to 500 feet.

RESULT: Pilot's and copilot's barometric altimeters shall indicate from 440 to 560 feet.

- If result is not as specified, replace faulty barometric altimeter (PARA 8-4-6).



To prevent damage to equipment, do not exceed 5000 feet per minute climb or descent on Pitot-static test set.

- j. Set test set altitude indicator to 1000 feet and allow readings to stabilize.
- k. Set test set altimeter indicator to 2500 feet.

RESULT: Pilot's and copilot's barometric altimeters shall indicate from 2430 to 2570 feet.

- If result is not as specified, replace faulty barometric altimeter (PARA 8-4-6).

l. Set test set altimeter indicator to 5000 feet.

RESULT: Pilot's and copilot's barometric altimeters shall indicate from 4920 to 5080 feet.

- If result is not as specified, replace faulty barometric altimeter (PARA 8-4-6).

m. Set test set altimeter indicator to 10,000 feet.

RESULT: Pilot's and copilot's barometric altimeters shall indicate from 9,900 to 10,100 feet.

- If result is not as specified, replace faulty barometric altimeter (PARA 8-4-6).

n. Set test set altitude indicator to 15,500 feet.

o. Close off vacuum source from test set.

p. Using helicopter's clock or a watch, check pilot's and copilot's barometric altimeter indications for 2 minutes, at 1-minute periods.

RESULT: Pilot's and copilot's barometric altimeters indications shall not change by more than 150 ft/min.

- If result is not as specified, go to TABLE NO. 8-2-1-3.

CAUTION

Use extreme caution when removing the Pitot-static test set and adapters. Do not exceed the maximum allowable feet per minute descent on Pitot-static test set. Slowly return Pitot and static pressure to 0. Do not pull Pitot-static adapters off Pitot heads until all readings are at 0, since this could cause damage to components.

q. Open Pitot-static test set pressure release valves and return pressure to 0 psi.

r. If no other check is to be done, go to Shutdown.

8-2-1.3 Airspeed Indicator.

a. Do Setup, if not already done.

b. Set test set airspeed indicator to 200 knots and allow readings to stabilize.

RESULT: Pilot's and copilot's airspeed indicators shall indicate from 196.5 to 203.5 knots.

- If result is not as specified, check for leaky adapter sealing rings, and loose tee air hose connections to pilot's and copilot's airspeed indicators.

(a) If leaky seals are present, replace seals.

(b) If loose fittings are present, tighten fittings.

(c) If trouble remains, replace pilot's or copilot's airspeed indicator, as required (PARA 8-4-5).

c. Using helicopter's clock or a watch, check pilot's airspeed indicator indication for 2 minutes.

RESULT: Pilot's airspeed indicator shall indicate from 196.5 to 203.5 knots.

- If result is not as specified, visually check for damaged Pitot-static head.

(a) If Pitot-static head is damaged, replace Pitot-static head (PARA 8-4-1).

(b) If Pitot-static head is not damaged, replace airspeed indicator (PARA 8-4-5).

d. Slowly decrease test set pressure until test set airspeed indicator indicates 100 knots.

RESULT: Pilot's airspeed indicator shall indicate from 98 to 102 knots.

- If result is not as specified, replace airspeed indicator (PARA 8-4-5).

e. Set test set airspeed indicator to 0 knots.

f. If no other check is to be done, go to Shutdown.

8-2-1-4

8-2-1.4 Barometric Altimeter Lighted Bezel Assembly.

- a. Turn INST LT PILOT FLT control from OFF to BRT.

RESULT: Pilot’s barometric altimeter lighted bezel assembly shall brighten.

- If result is not as specified, go to TABLE NO. 8-2-1-4.

- b. Turn INST LT PILOT FLT control from BRT to OFF.

RESULT: Pilot’s barometric altimeter lighted bezel assembly shall dim and go off.

- If result is not as specified, troubleshoot instrument panel lights PARA 9-2-4).

- c. Turn CPLT FLT INST LTS control from OFF to BRT.

RESULT: Copilot’s barometric altimeter lighted bezel assembly shall brighten.

- If result is not as specified, go to TABLE NO. 8-2-1-5.

- d. Turn CPLT FLT INST LTS control from BRT to OFF.

RESULT: Copilot’s barometric altimeter lighted bezel assembly shall dim and go off.

- If result is not as specified, troubleshoot instrument panel lights PARA 9-2-4).

- e. If no other check is to be done, go to Shutdown.

8-2-1.5 Shutdown.

- a. Turn off and disconnect external electrical power (PARA 1-6-16).
- b. Disconnect test setup.
- c. If Ethylene glycol was used to install Pitot-static head adapters, wipe off Pitot heads with soft rag after removing adapters.
- d. Install covers on Pitot tubes.

TABLE NO. 8-2-1-1. Pilot’s Barometric Altimeter Encoder Vibrator Does Not Work.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 vdc between P306R-P and P306R-M.	Step 1. If voltage is as specified, replace barometric altimeter encoder (PARA 8-4-6). Go to 4. Step 2. If voltage is not as specified, go to 2.
2. Check continuity between P306R-M and ground.	Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
3. Check continuity between P306R-P and PILOT ALTM circuit breaker terminal 1 and P306R-c and PILOT ALTM circuit breaker terminal 1.	Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to 4.

TABLE NO. 8-2-1-1. Pilot's Barometric Altimeter Encoder Vibrator Does Not Work. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
4. Procedure completed.	

TABLE NO. 8-2-1-2. Copilot's Barometric Altimeter Vibrator Does Not Work.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 vdc between P1-P and P1-M.	Step 1. If voltage is as specified, replace altimeter (PARA 8-4-6). Go to 4. Step 2. If voltage is not as specified, go to 2.
2. Check continuity between P1-M, P1-a, and P1-X and ground.	Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
3. Check continuity between P1-P and CPLT ALTM circuit breaker terminal 1.	Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
4. Procedure completed.	

TABLE NO. 8-2-1-3. Pilot's And Copilot's Barometric Altimeters Change More Than 150 Ft/Min.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check that static line connections are tight.	Step 1. If connections are tight, go to 2. Step 2. If connections are not tight, tighten static line connections (PARA 8-4-3). Go to 4.
2. Visually check for damaged static lines.	

TABLE NO. 8-2-1-3. Pilot's And Copilot's Barometric Altimeters Change More Than 150 Ft/Min. (Cont)

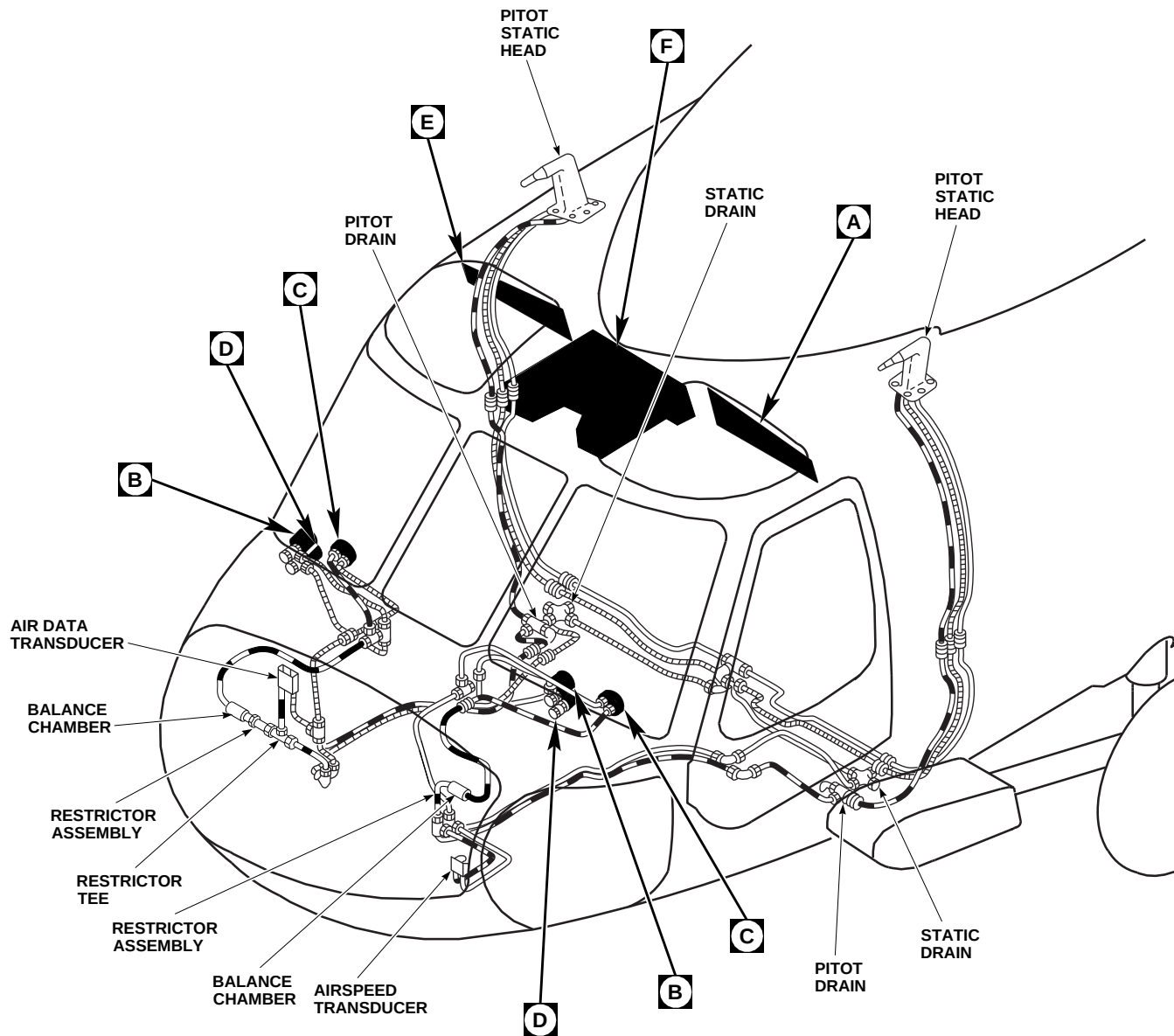
TEST OR INSPECTION	CORRECTIVE ACTION
Step 1. If damaged lines are present, replace static tubing (PARA 8-4-3). Go to 4. Step 2. If damaged lines are not present, go to 3.	
3. Visually check for damaged Pitot-static heads.	Step 1. If damaged heads are present, replace Pitot-static head assembly, as required (PARA 8-4-1). Go to 4. Step 2. If damaged heads are not present, replace pilot's or copilot's barometric altimeter, as required (PARA 8-4-6). Go to 4.
4. Procedure completed.	

TABLE NO. 8-2-1-4. Pilot's Barometric Altimeter Lighted Bezel Assembly Does Not Brighten.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Turn INST LT PILOT FLT control to BRT. Go to 2.	
2. Check for 28 vdc between P183-A and ground.	Step 1. If voltage is not as specified, troubleshoot instrument panel lights (PARA 9-2-4). Go to 4. Step 2. If voltage is as specified, go to 3.
3. Check continuity between P183-B and ground.	Step 1. If continuity is present, replace lighted bezel assembly (PARA 8-4-6). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
4. Procedure completed.	

TABLE NO. 8-2-1-5. Copilot's Barometric Altimeter Lighted Bezel Assembly Does Not Brighten.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Turn CPLT FLT INST LTS control to BRT. Go to 2.	
2. Check for 28 vdc between P193-A and ground.	Step 1. If voltage is not as specified, troubleshoot instrument panel lights (PARA 9-2-4). Go to 4. Step 2. If voltage is as specified, go to 3.
3. Check continuity between P193-B and ground.	Step 1. If continuity is present, replace lighted bezel assembly (PARA 8-4-6). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
4. Procedure completed.	



LEGEND	
	PITOT
	STATIC (1)
	STATIC (2)

NOTE
PILOT'S BAROMETRIC ALTIMETER ENCODER.

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FIGURE 8-2-1-1. PITOT-STATIC SYSTEM LOCATION DIAGRAM (SHEET 1 OF 4)

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P1	COPILOT'S BAROMETRIC ALTIMETER
P1R / J1R	COCKPIT BL 6 LH, STA 210
P20R / J20R	CABIN BL 40 RH, STA 248
P111 / J111	COCKPIT BL 7.5 LH, STA 197
P112 / J112	COCKPIT BL 6 RH, STA 197
P113 / J113	COCKPIT BL 6 RH, STA 197
P183	PILOT'S BAROMETRIC ALTIMETER LIGHTED, BEZEL ASSEMBLY
P193	COPILOT'S ENCODER ALTIMETER LIGHTED, BEZEL ASSEMBLY
P246 / J246	CABIN CEILING BL 5 RH, STA 247
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 25 RH
P280 / J280	CABIN CEILING BL 6 LH, STA 247
P306R	PILOT'S ALTIMETER ENCODER

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FIGURE 8-2-1-1. PITOT-STATIC SYSTEM LOCATION DIAGRAM (SHEET 2 OF 4)

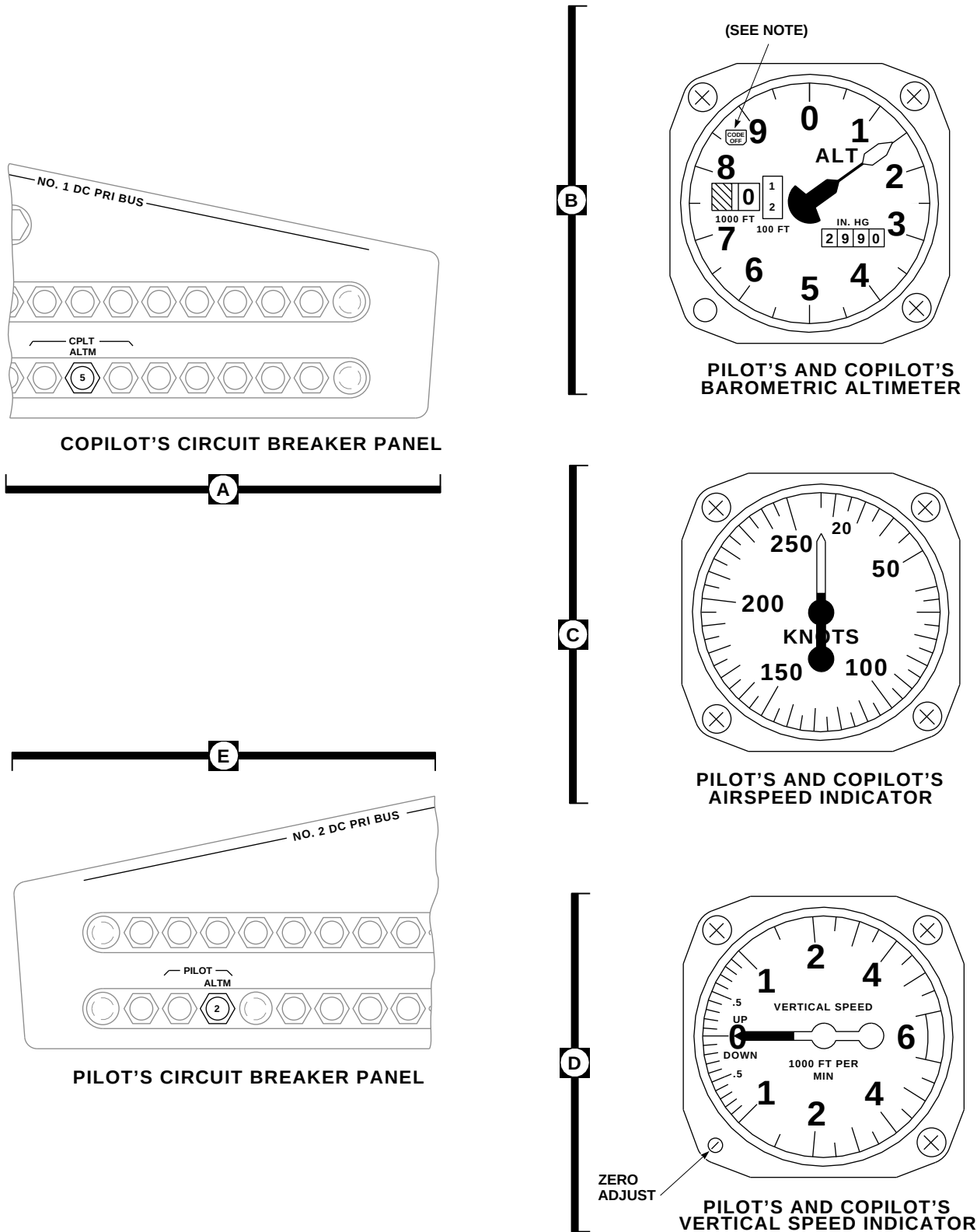
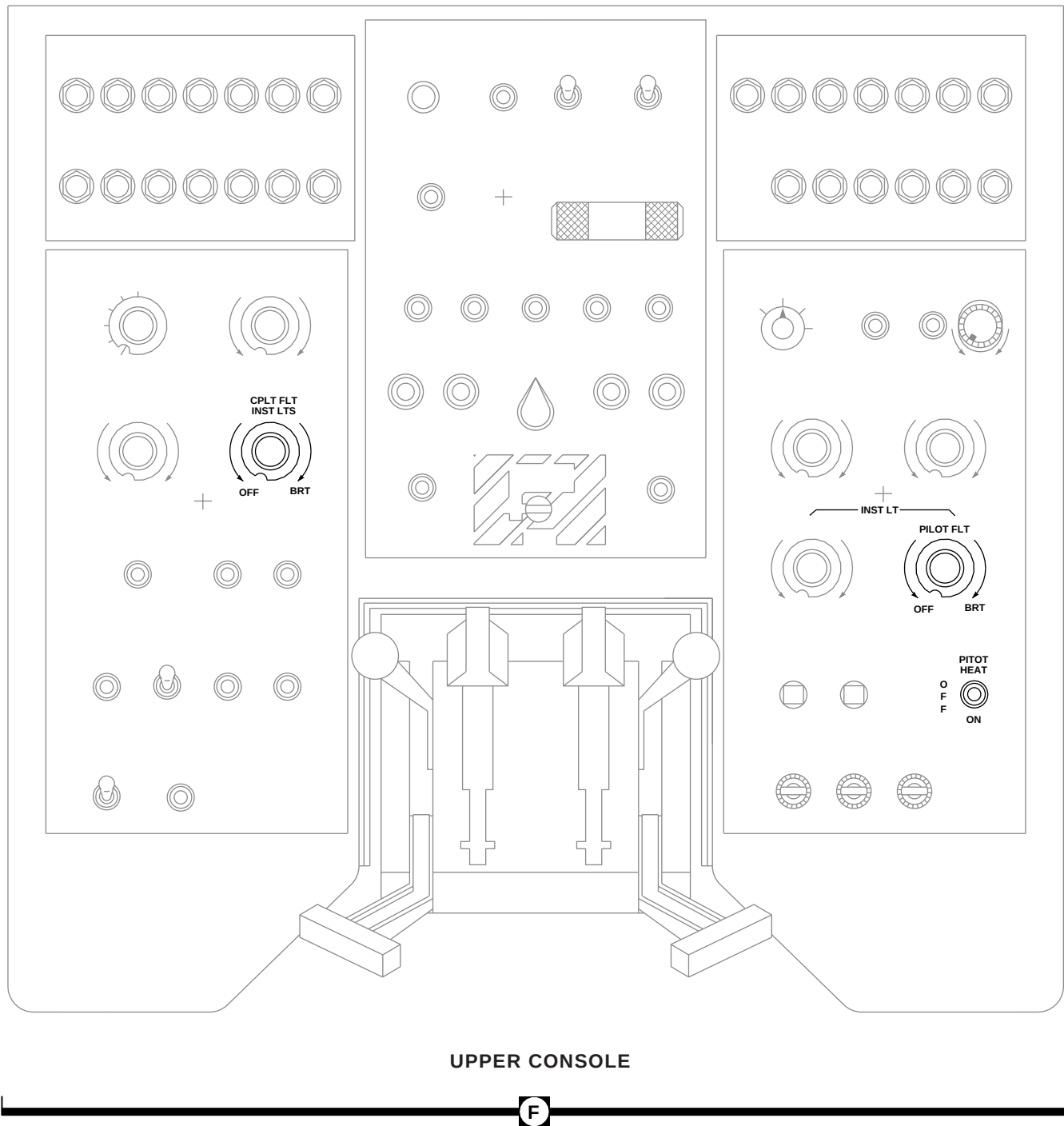


FIGURE 8-2-1-1. PITOT-STATIC SYSTEM LOCATION DIAGRAM (SHEET 3 OF 4)

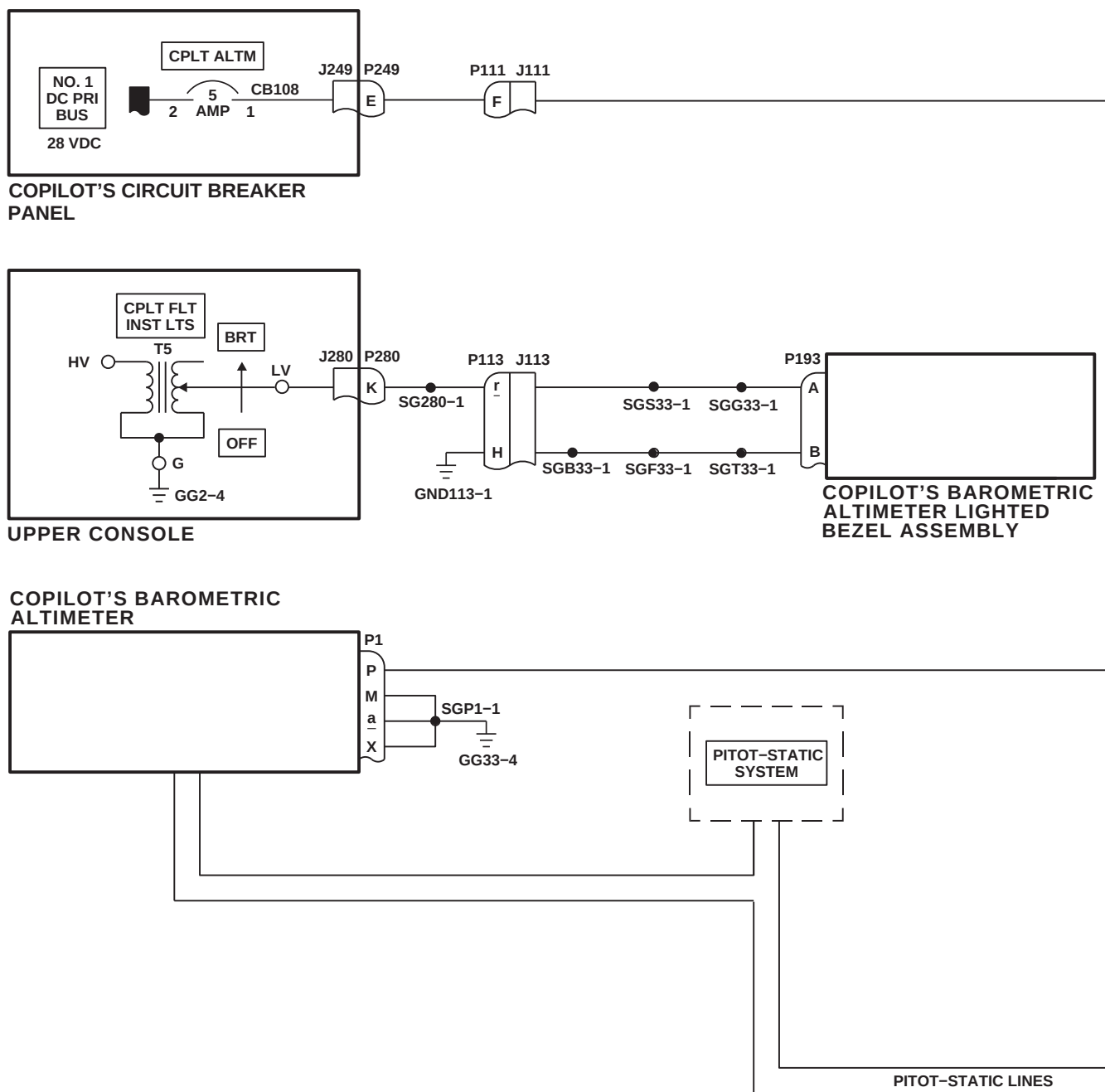
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FIGURE 8-2-1-1. PITOT-STATIC SYSTEM LOCATION DIAGRAM (SHEET 4 OF 4)

8-2-1-12



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FIGURE 8-2-1-2. BAROMETRIC ALTIMETER SCHEMATIC DIAGRAM (SHEET 1 OF 2)

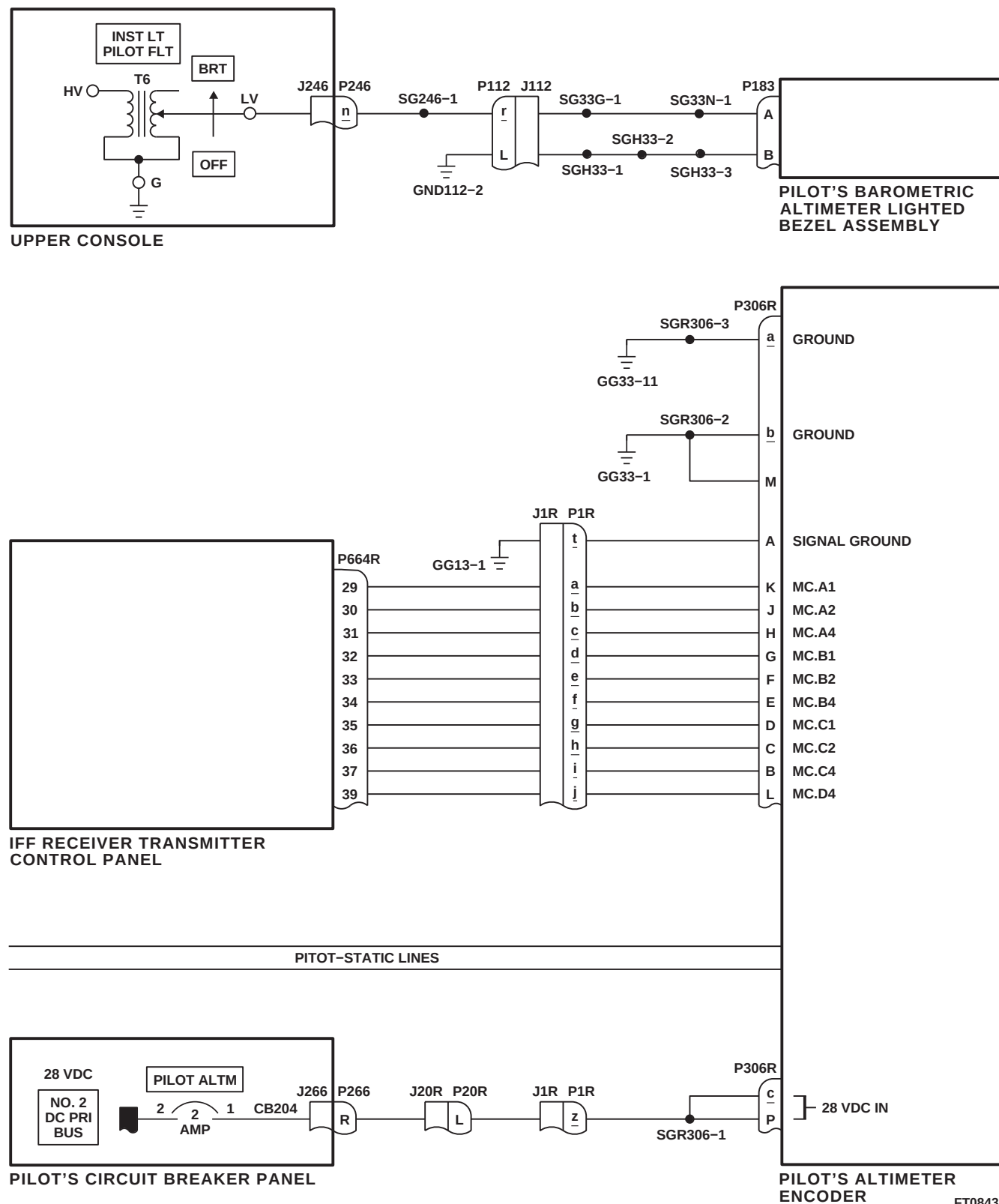


FIGURE 8-2-1-2. BAROMETRIC ALTIMETER SCHEMATIC DIAGRAM (SHEET 2 OF 2)

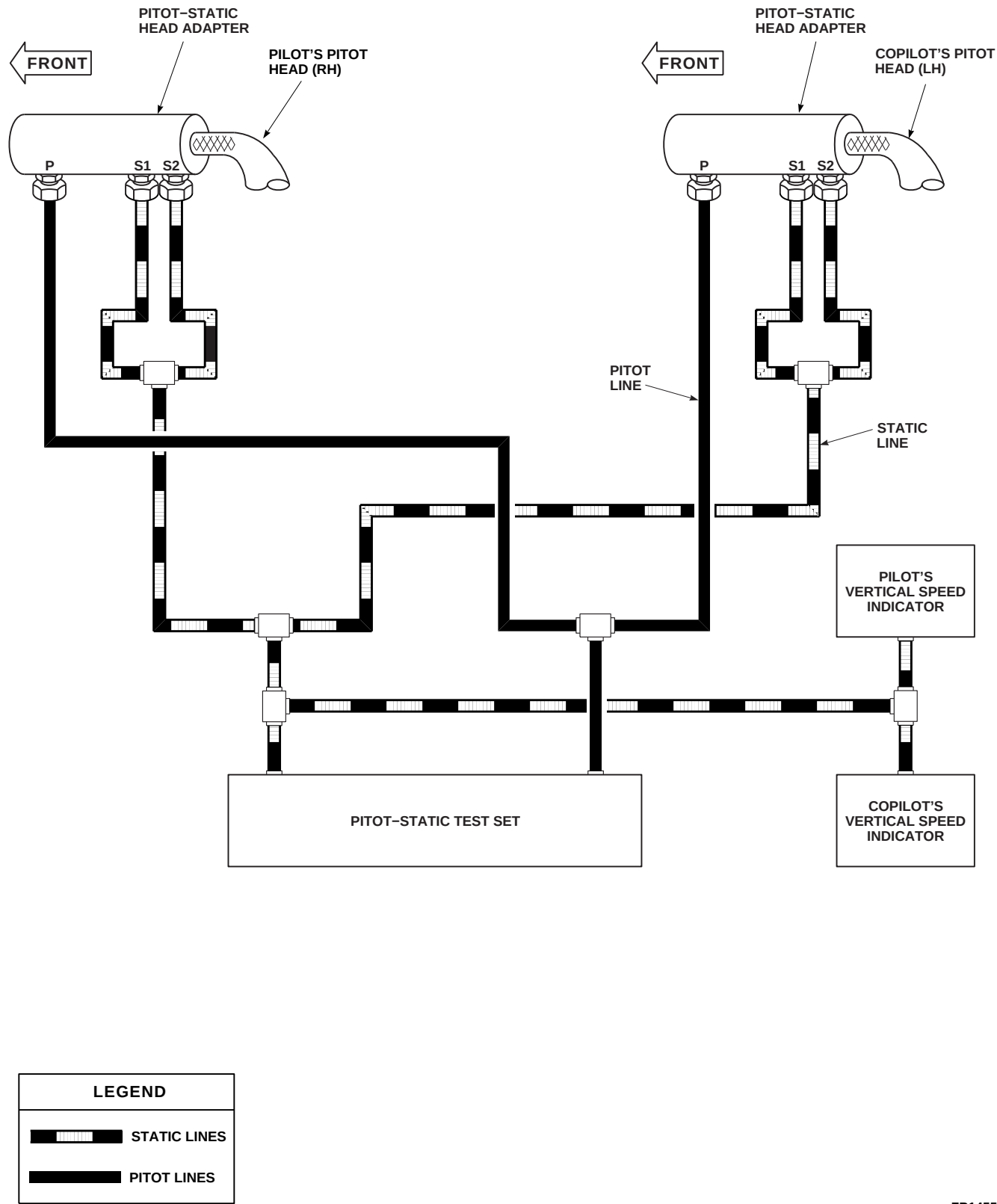


FIGURE 8-2-1-3. PITOT-STATIC SYSTEM TEST SETUP DIAGRAM

8-2-2 PITOT HEATER.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Connected or APU Operational

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

8-2-2.1 Operational/Troubleshooting Procedure.



- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- b. Make sure upper console PITOT HEAT switch is placed to the OFF position.
- c. Remove covers from Pitot tubes.
- d. Turn on electrical power (PARA 1-6-16).



To avoid skin burn, carefully feel for heat at Pitot tubes.

- e. Carefully feel for heat at left and right Pitot tubes.

RESULT: No heat shall be felt.

- If result is not as specified, replace PITOT HEAT switch S30 (PARA 9-4-5).

Do not leave PITOT HEAT switch to ON position any longer than necessary. Doing so may cause damage to equipment.

- f. Place upper console PITOT HEAT switch to ON position and carefully feel for heat at Pitot tubes.

RESULT:

- (1) Heat shall be felt at left Pitot tube.
- (2) LFT PITOT HEAT capsule on caution/advisory panel shall be off.
- (3) Heat shall be felt at right Pitot tube.
- (4) RT PITOT HEAT capsule on caution/advisory panel shall be off.
- If all results are not as specified, replace PITOT HEAT switch S30 (PARA 9-4-5).
- If results (1) and (2) are not as specified, go to TABLE NO. 8-2-2-1.

- If results (3) and (4) are not as specified, go to TABLE NO. 8-2-2-2.
 - g. Place upper console PITOT HEAT switch to OFF position.
 - h. Pull out RT PITOT HEAT circuit breaker (pilot's circuit breaker panel).
 - i. Place upper console PITOT HEAT switch to ON position.
- RESULT: RT PITOT HEAT capsule on caution/advisory panel shall be on.
- If result is not as specified, go to TABLE NO. 8-2-2-3.
- j. Place upper console PITOT HEAT switch to OFF position.
- k. Push in RT PITOT HEAT circuit breaker (pilot's circuit breaker panel) and pull out LEFT PITOT HEAT circuit breaker (copilot's circuit breaker panel).
- l. Place upper console PITOT HEAT switch to ON position.
- RESULT: LFT PITOT HEAT capsule on caution/advisory panel shall be on.
- If result is not as specified, go to TABLE NO. 8-2-2-4.
- m. Turn off electrical power (PARA 1-6-16).
- n. After Pitot tubes have cooled, install covers.

TABLE NO. 8-2-2-1. No Heat At Left Pitot Tube And LFT PITOT HEAT Caution Capsule Is On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check resistance between: J269-A and J269-D (9 to 29 ohms) J269-B and J269-C (45 to 55 ohms) Step 1. If resistance is as specified, go to 2. Step 2. If resistance is not as specified, replace left Pitot tube (PARA 8-4-3). Go to 9. 2. Check continuity between: P269-C and ground P269-D and ground Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 9. 3. Check for 9 to 29 ohms between P242-K and ground. Step 1. If resistance is as specified, go to 4. Step 2. If resistance is not as specified, repair/replace wiring between P242-K and P269-A (Appendix F). Go to 9.	

**TABLE NO. 8-2-2-1. No Heat At Left Pitot Tube And LFT PITOT HEAT Caution Capsule Is On.
(Cont)**

TEST OR INSPECTION CORRECTIVE ACTION
4. Check for 45 to 55 ohms between P242-<u>m</u> and ground. Step 1. If resistance is as specified, go to 5. Step 2. If resistance is not as specified, repair/replace wiring between P242-<u>m</u> and P269-B (Appendix F). Go to 9. 5. Check for 115 vac between P242-<u>r</u> and ground. Step 1. If voltage is as specified, replace left relay panel (PARA 9-4-2). Go to 9. Step 2. If voltage is not as specified, go to 6. 6. Check for 115 vac between PITOT HEAT switch terminal 6 and ground. Step 1. If voltage is as specified, repair/replace wiring between terminal 6 and P242-<u>r</u>(Appendix F). Go to 9. Step 2. If voltage is not as specified, go to 7. 7. Check for 115 vac between PITOT HEAT switch terminal 5 and ground. Step 1. If voltage is as specified, replace switch S30 (PARA 9-4-5). Go to 9. Step 2. If voltage is not as specified, go to 8. 8. Check for 115 vac between LEFT PITOT HEAT circuit breaker terminal 1 and ground. Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and switch S30 terminal 5 (Appendix F). Go to 9. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 9. 9. Procedure completed.

TABLE NO. 8-2-2-2. No Heat At Right Pitot Tube And RT PITOT HEAT Caution Capsule Is On.

TEST OR INSPECTION CORRECTIVE ACTION
1. Check resistance between:

**TABLE NO. 8-2-2-2. No Heat At Right Pitot Tube And RT PITOT HEAT Caution Capsule Is On.
(Cont)**

TEST OR INSPECTION	CORRECTIVE ACTION
J268-A and J268-D (9 to 29 ohms) J268-B and J268-C (45 to 55 ohms) Step 1. If resistance is as specified, go to 2. Step 2. If resistance is not as specified, replace right Pitot tube (PARA 8-4-3). Go to 9. 2. Check continuity between: P268-C and ground P268-D and ground Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 9. 3. Check for 9 to 29 ohms between P241-c and ground. Step 1. If resistance is as specified, go to 4. Step 2. If resistance is not as specified, repair/replace wiring between P241-c and P268-A (Appendix F). Go to 9. 4. Check for 45 to 55 ohms between P241-r and ground. Step 1. If resistance is as specified, go to 5. Step 2. If resistance is not as specified, repair/replace wiring between P241-r and P268-B (Appendix F). Go to 9. 5. Check for 115 vac between PITOT HEAT switch terminal 3 and ground. Step 1. If voltage is as specified, go to 6. Step 2. If voltage is not as specified, go to 7. 6. Check continuity between PITOT HEAT switch terminal 3 and P241-m. Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to 9. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 9. 7. Check for 115 vac between PITOT HEAT switch terminal 2 and ground.	

**TABLE NO. 8-2-2-2. No Heat At Right Pitot Tube And RT PITOT HEAT Caution Capsule Is On.
(Cont)**

TEST OR INSPECTION	CORRECTIVE ACTION
Step 1. If voltage is as specified, replace switch S30 (PARA 9-4-5). Go to 9. Step 2. If voltage is not as specified, go to 8.	8. Check continuity between PITOT HEAT switch terminal 2 and RT PITOT HEAT circuit breaker terminal 1.
Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to 9. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 9.	9. Procedure completed.

TABLE NO. 8-2-2-3. RT PITOT HEAT Caution/Advisory Capsule Is Not On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check helicopter configuration.	Step 1. EMEP , go to 2. Step 2. W/O EMEP , go to 4.
EMEP	2. Remove and check pin filtered adapter from P117 (PARA 9-4-98).
Step 1. If pin filtered adapter is good, go to 3. Step 2. If pin filtered adapter is not good, replace pin filtered adapter (PARA 9-4-98). Go to 9.	EMEP
3. Install pin filtered adapter (PARA 9-4-98). Go to 4.	4. Check for 28 vdc between P117-60 and ground.
Step 1. If voltage is as specified, troubleshoot caution/advisory warning system (PARA 8-2-4). Go to 9. Step 2. If voltage is not as specified, go to 5.	5. Check for 28 vdc between P241-d and ground.
Step 1. If voltage is as specified, go to 6.	

TABLE NO. 8-2-2-3. RT PITOT HEAT Caution/Advisory Capsule Is Not On. (Cont)

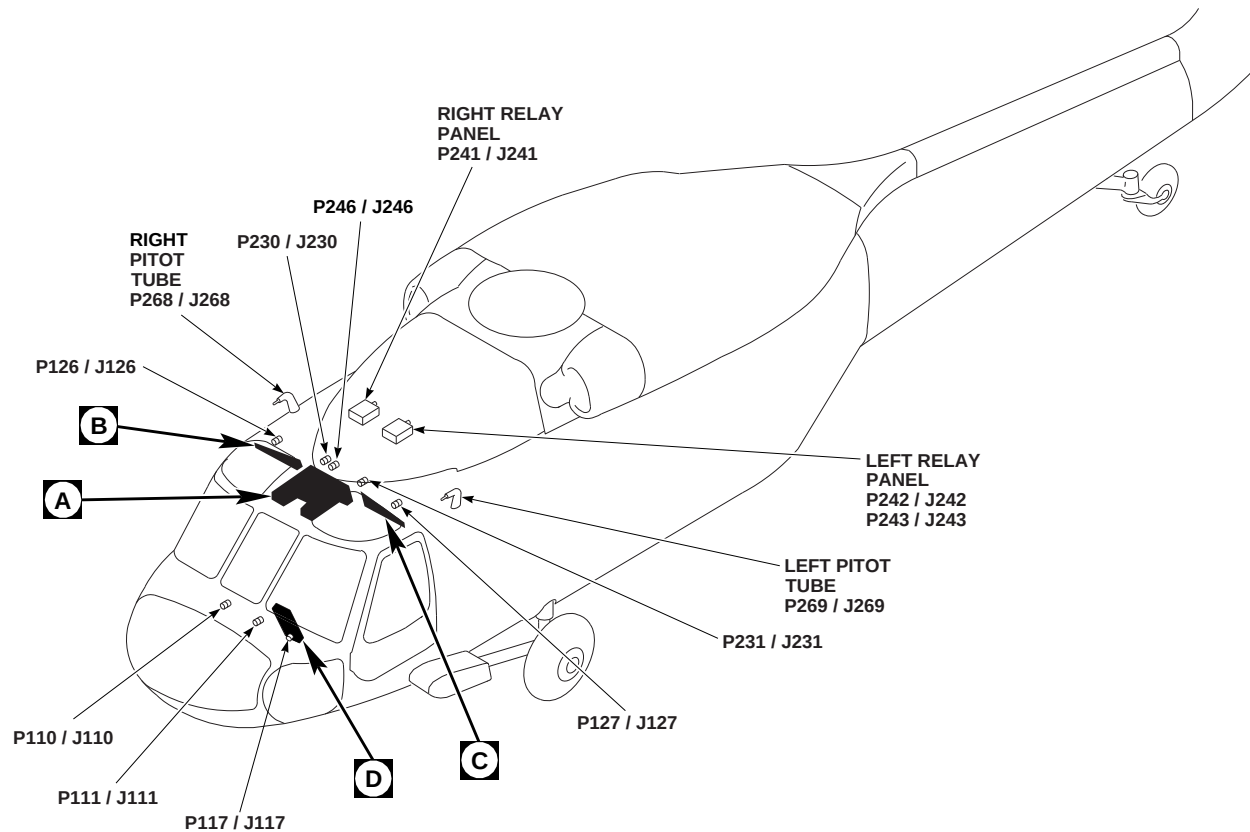
TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If voltage is not as specified, go to 7.
6. Check continuity between P241-f and P117-60.	Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to 9. Step 2. If continuity is not present, repair/replace wiring between P241-f and P117-60 (Appendix F). Go to 9.
7. Check for 28 vdc between P230-u and ground.	Step 1. If voltage is as specified, repair/replace wiring between P230-u and P241-d (Appendix F). Go to 9. Step 2. If voltage is not as specified, go to 8.
8. Check for 28 vdc between NO. 1 ENG ANTI-ICE circuit breaker terminal 1 and ground.	Step 1. If voltage is as specified, replace switch S30 (PARA 9-4-5). Go to 9. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 9.
9. Procedure completed.	

TABLE NO. 8-2-2-4. LFT PITOT HEAT Caution/Advisory Capsule Is Not On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check helicopter configuration.	Step 1. EMEP, go to 2. Step 2. W/O EMEP, go to 4.
2. Remove and check pin filtered adapter from P117 (PARA 9-4-98).	Step 1. If pin filtered adapter is good, go to 3. Step 2. If pin filtered adapter is not good, replace pin filtered adapter (PARA 9-4-98). Go to 9.
3. Install pin filtered adapter (PARA 9-4-98). Go to 4.	

TABLE NO. 8-2-2-4. LFT PITOT HEAT Caution/Advisory Capsule Is Not On. (Cont)

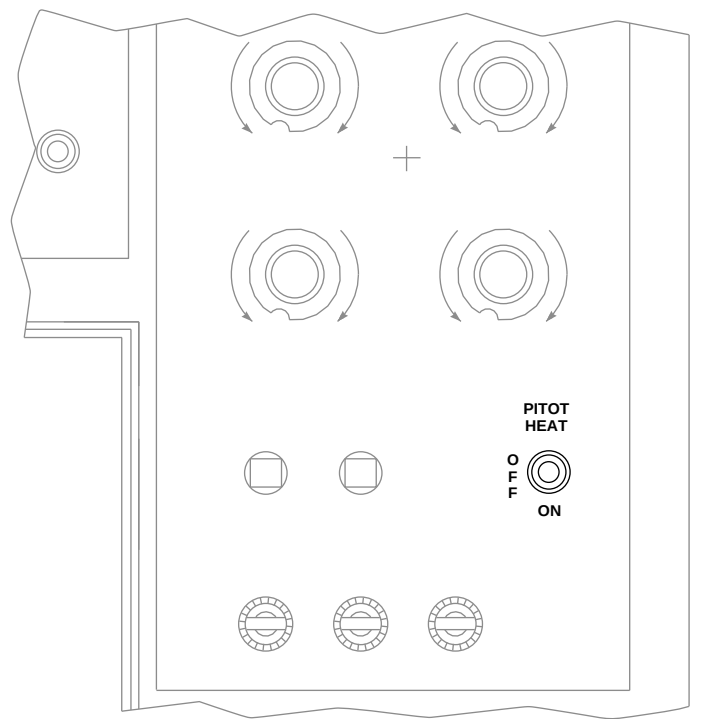
TEST OR INSPECTION CORRECTIVE ACTION
4. Check for 28 vdc between P117-12 and ground. Step 1. If voltage is as specified, troubleshoot caution/advisory warning system (PARA 8-2-4). Go to 9. Step 2. If voltage is not as specified, go to 5. 5. Check for 28 vdc between P243-X and ground. Step 1. If voltage is as specified, go to 6. Step 2. If voltage is not as specified, go to 7. 6. Check continuity between P242-<u>b</u> and P117-12. Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to 9. Step 2. If continuity is not present, repair/replace wiring between P242-<u>b</u> and P117-12 (Appendix F). Go to 9. 7. Check for 28 vdc between P230-<u>u</u> and ground. Step 1. If voltage is as specified, repair/replace wiring between P230-<u>u</u> and P243-X (Appendix F). Go to 9. Step 2. If voltage is not as specified, go to 8. 8. Check for 28 vdc between NO. 1 ENG ANTI-ICE circuit breaker terminal 1 and ground. Step 1. If voltage is as specified, replace switch S30 (PARA 9-4-5). Go to 9. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 9. 9. Procedure completed.



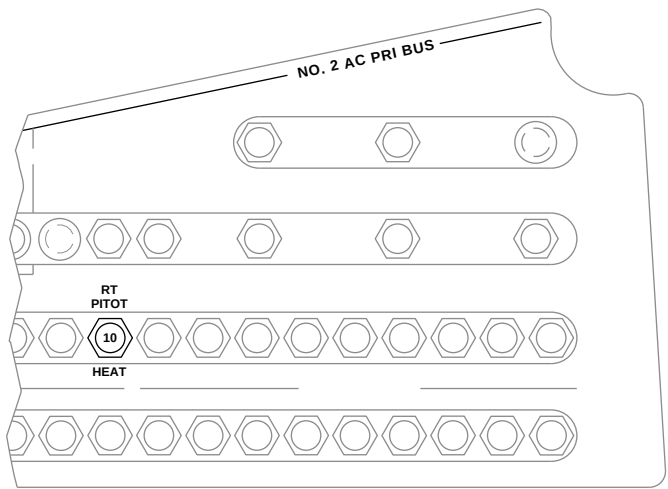
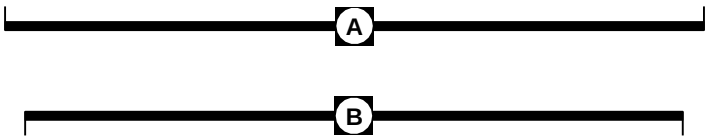
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P110 / J110	COCKPIT BL 7.5 RH, STA 197
P111 / J111	COCKPIT BL 7.5 LH, STA 197
P117 / J117	CAUTION / ADVISORY PANEL
P126 / J126	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 28 RH, STA 243
P127 / J127	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 28 LH, STA 243
P230 / J230	CABIN CEILING BL 8 RH, STA 247
P231 / J231	CABIN CEILING BL 9 LH, STA 247
P241 / J241	RIGHT RELAY PANEL
P242 / J242	LEFT RELAY PANEL
P243 / J243	LEFT RELAY PANEL
P246 / J246	CABIN CEILING BL 5 RH, STA 247
P268 / J268	RIGHT PITOT TUBE
P269 / J269	LEFT PITOT TUBE

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FIGURE 8-2-2-1. PITOT HEATER SYSTEM LOCATION DIAGRAM (SHEET 1 OF 3)



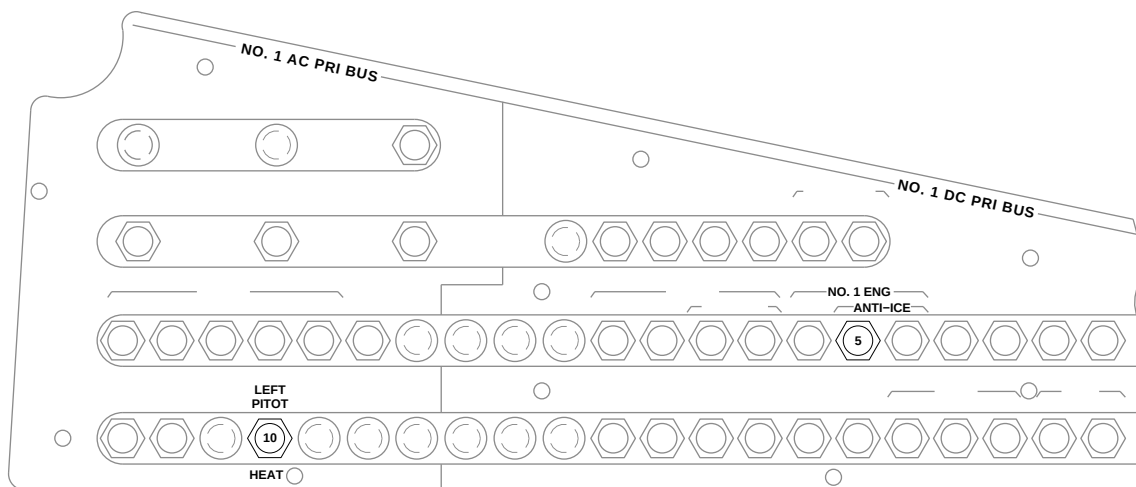
UPPER CONSOLE



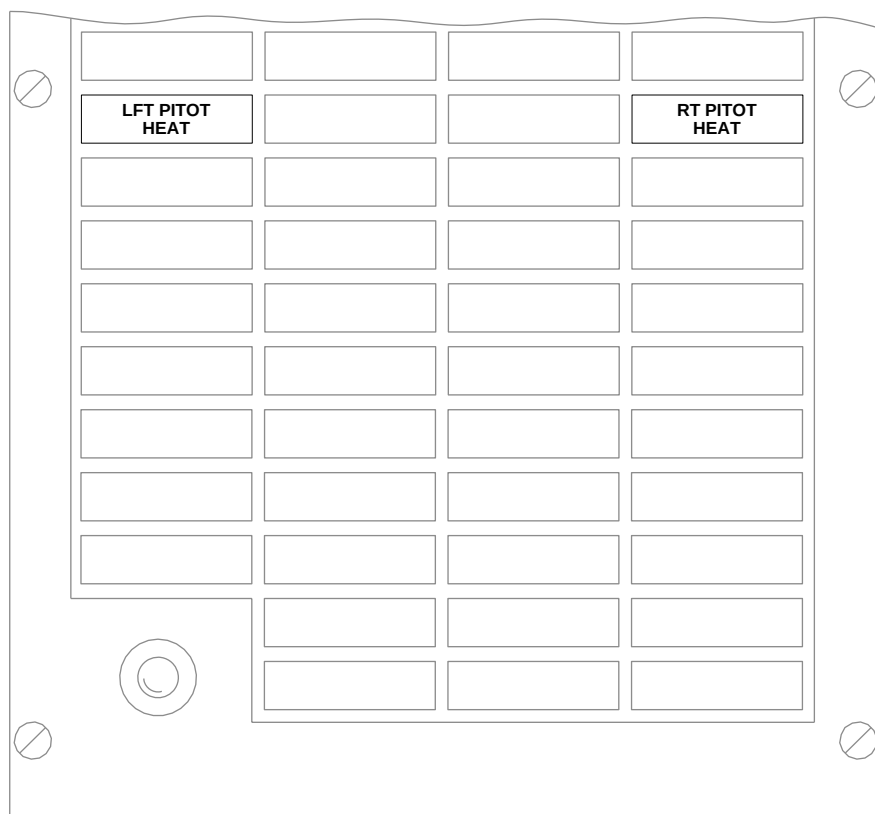
PILOT'S CIRCUIT BREAKER PANEL

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FIGURE 8-2-2-1. PITOT HEATER SYSTEM LOCATION DIAGRAM (SHEET 2 OF 3)



COPILOT'S CIRCUIT BREAKER PANEL

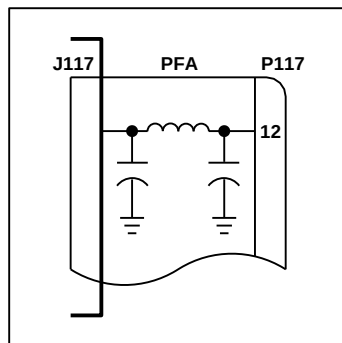
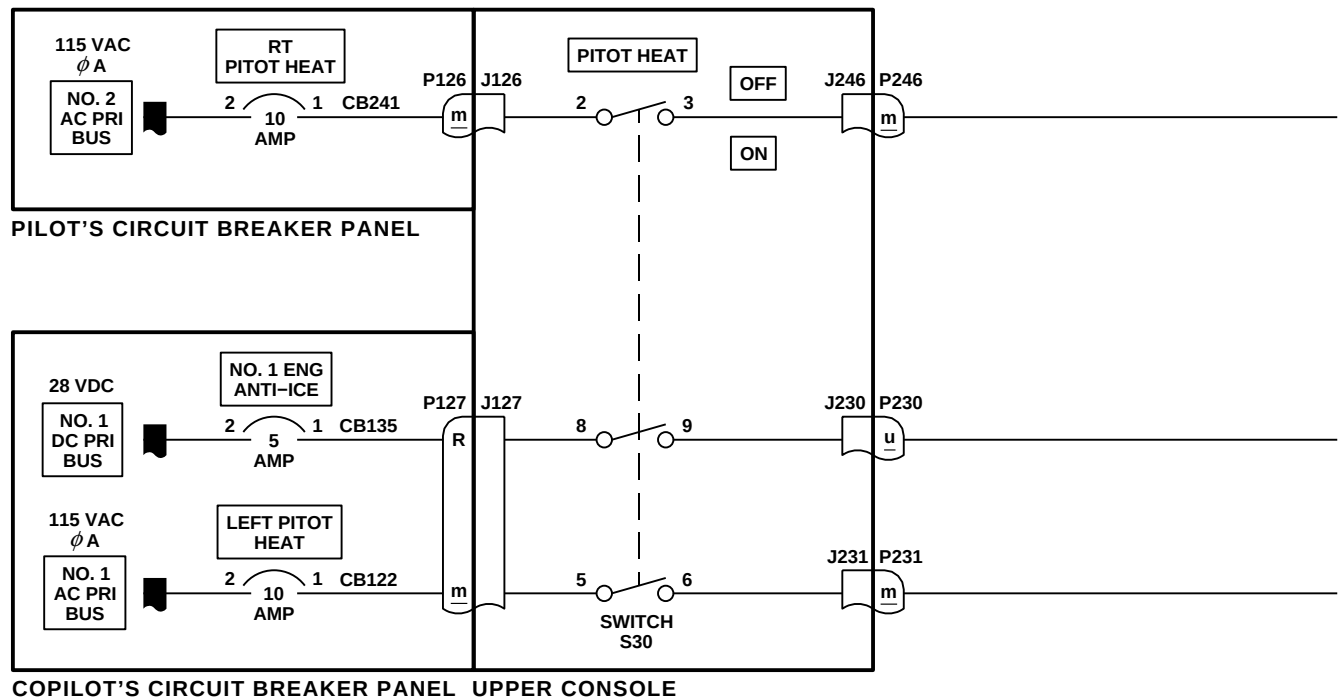


CAUTION/ADVISORY PANEL

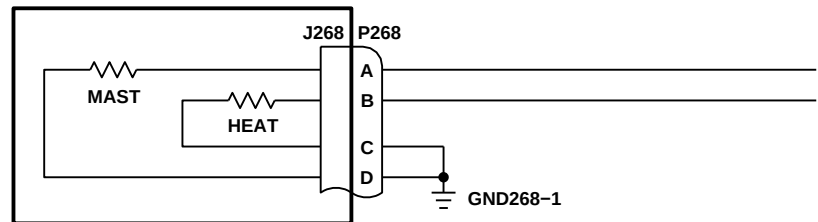
AA2680_3
SA

FIGURE 8-2-2-1. PITOT HEATER SYSTEM LOCATION DIAGRAM (SHEET 3 OF 3)

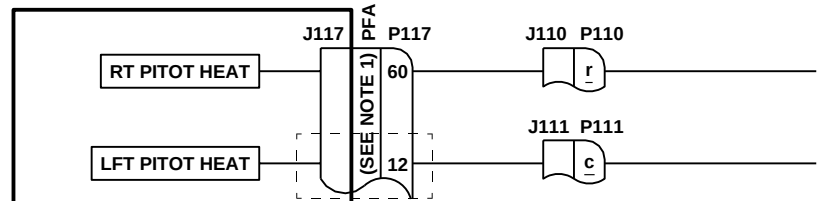
8-2-2-10



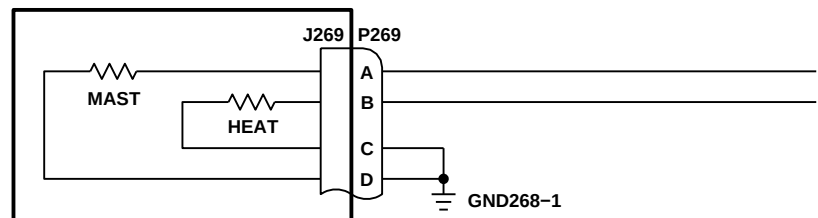
DETAIL A
(SEE NOTE 2)



RIGHT PITOT TUBE HEATER



CAUTION/ADVISORY PANEL (SEE DETAIL A)



LEFT PITOT TUBE HEATER

NOTES	
1.	W/O EMEP PIN FILTERED ADAPTER (PFA) IS NOT INSTALLED.
2.	P117-12 SHOWS TYPICAL CONFIGURATION FOR ALL PIN FILTERED ADAPTERS.

FT0841_1

FIGURE 8-2-2-2. PITOT HEATER SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 2)

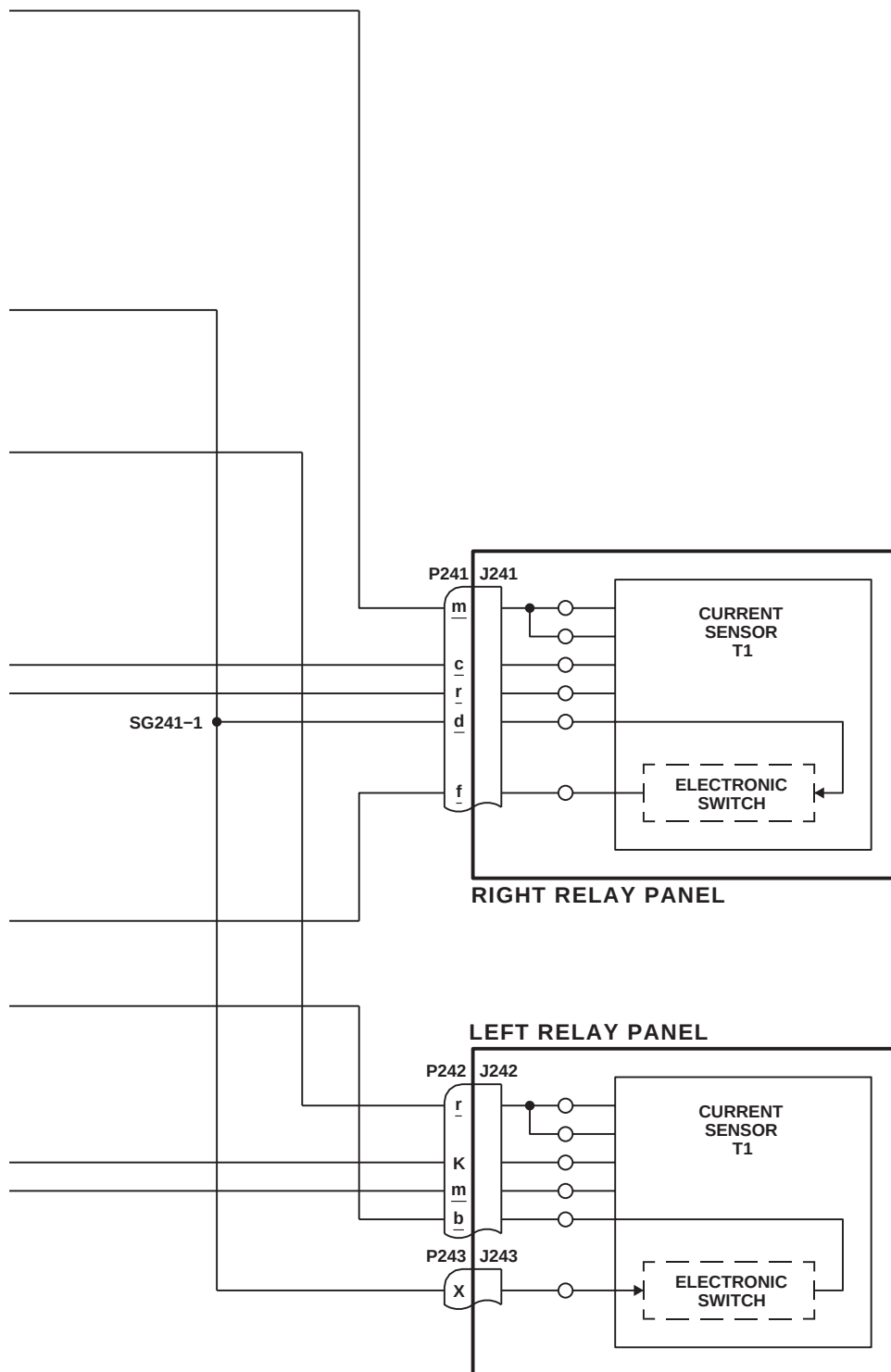
FB1452_2A
SA

FIGURE 8-2-2-2. PITOT HEATER SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 2)

8-2-3 INSTRUMENT DISPLAY SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit
- Line Test Set, Instrument Display, 100-601164-000
- Extended Cable LTS-28V
- Extended Cable LTS-115V

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or Auxiliary Power Unit (APU) Operational

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

Special Instructions

The Instrument Display System (IDS) operational/troubleshooting procedure is subdivided as follows:

- (1) Setup
- (2) % RPM R (Main Rotor Speed)
- (3) FUEL QTY (Fuel Quantity)
- (4) ENG OIL PRESS (Engine Oil Pressure)
- (5) ENG OIL TEMP (Engine Oil Temperature)
- (6) Ng SPEED (Engine Gas Generator Tachometer)
- (7) % TRQ (Engine Torque)
- (8) XMSN PRESS (Main Transmission Oil Pressure)
- (9) XMSN TEMP (Main Transmission Oil Temperature)
- (10) % RPM 1-2 (Engine Power Turbine Tachometer)
- (11) TGT TEMP (Turbine Gas Temperature)
- (12) Lamp Power Supply
- (13) Faceplate Lighting
- (14) Shutdown

8-2-3.1 Setup.

- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Place Central Display Unit (CDU) DIGITS switch to ON and DIM control to about mid-position.

- Connect Line Test Set (LTS) 28 VDC cable between utility receptacle J258 in upper cabin area behind pilot's seat and LTS.
- Connect LTS 115 VAC cable between utility receptacle J257 in upper cabin area behind pilot's seat and LTS.
- Disconnect cable assemblies from J1 on No. 1 Signal Data Converter (SDC) and J1 on No. 2 SDC.

f. Position LTS switches as follows:

INPUT POWER	OFF
All CAL-NOR switches	NOR
All HI-LO switches	HI
All NOR controls	Fully counterclockwise

g. Connect LTS cable assembly 419-741 between NO. 1 SIGNAL DATA CONVERTER on LTS and J1 on No. 1 SDC.

h. Connect second LTS cable assembly 419-741 between NO. 2 SIGNAL DATA CONVERTER on LTS and J1 on No. 2 SDC.

i. Turn on electrical power (PARA 1-6-16).

j. Place LTS - INPUT POWER switch to SIM ONLY.

RESULT:

- (1) All CDU and PDU (pilot's and copilot's) vertical displays shall be at zero, digital readouts shall indicate about zero, and the following LTS lights shall go on:

LOW ROTOR SPD
NO. 1 ENGINE OUT
NO. 2 ENGINE OUT
NO. 1 LOW OIL PRESS
NO. 2 LOW OIL PRESS

- (2) Ng SPEED 1 or 2 vertical display or digital readout shall indicate between 0 and 0.5%.

- If LOW ROTOR SPD light on LTS is off, check CDU RTR SPD fuse.

- (a) If fuse is good, replace CDU (PARA 8-4-10).

- (b) If fuse is not good, replace fuse (PARA 8-4-13).

- If NO. 1 or NO. 2 ENGINE OUT light on LTS is off, go to TABLE NO. 8-2-3-1.
- If NO. 1 or NO. 2 LOW OIL PRESS light on LTS is off, go to TABLE NO. 8-2-3-2.
- If Ng SPEED readout is not as specified, perform Ng SPEED check.

If trouble remains, troubleshoot No. 1 or No. 2 Engine Instruments And Warning Lights System, as applicable (PARA 4-2-2 or PARA 4-2-3).

- If ENG OIL PRESS 1 or 2 display does not indicate zero, perform ENG OIL PRESS check.

If trouble remains, troubleshoot No. 1 or No. 2 Engine Instruments And Warning Lights System, as applicable (PARA 4-2-2 or PARA 4-2-3).

- If XMSN PRESS display does not indicate zero, perform XMSN PRESS check.

If trouble remains, troubleshoot Transmission Chip Detector, Instruments, And Oil Warning System (PARA 6-2-2).

- If % RPM 1 or 2 display does not indicate zero, perform % RPM 1-2 check.

If trouble remains, troubleshoot No. 1 or No. 2 Engine Instruments And Warning Lights System, as applicable (PARA 4-2-2 or PARA 4-2-3).

- If % RPM R display does not indicate zero, perform % RPM R check.

If trouble remains, troubleshoot Transmission Chip Detector, Instruments, And Oil Warning System (PARA 6-2-2).

- If % TRQ vertical displays or digital readouts do not indicate zero, perform % TRQ check.

If trouble remains, troubleshoot No. 1 or No. 2 Engine Instruments And Warning Lights System, as applicable (PARA 4-2-2 or PARA 4-2-3).

- k. Momentarily press LTS-LOW ROTOR SPD reset button.

RESULT:

- (1) RTR OVERSPEED lights on PDUs (pilot's and copilot's) shall be off.
 - (2) CHAN 1 warning light on CDU shall be off.
 - (3) CHAN 2 warning light on CDU shall be off.
 - (4) ENG OIL PRESS, XMSN PRESS, and Ng SPEED vertical displays, plus the associated Ng digital displays on CDU shall all indicate zero.
 - (5) The % RPM and % TRQ vertical displays, and the associated digital readouts on PDUs (pilot's and copilot's) shall indicate zero.
- If result (1) is not as specified, replace CDU (PARA 8-4-10).
 - If result (2) is not as specified, go to TABLE NO. 8-2-3-3.
 - If result (3) is not as specified, go to TABLE NO. 8-2-3-4.
 - If result (4) or (5) is not as specified, note trouble and continue operational/troubleshooting procedure.
 - If results (4) and (5) are not as specified, replace CDU (PARA 8-4-10).
- l. Press and hold CDU PUSH TO TEST or LAMP TEST switch and turn DIM control

fully counterclockwise, then fully clockwise. Next, adjust DIM control for normal display.

RESULT:

- (1) All CDU vertical scale lamps shall go on.
 - (2) All CDU digital readouts shall indicate 888, except for MAIN FUEL or TOTAL FUEL, which shall indicate 8888.
 - (3) Three RTR OVERSPEED lights on PDUs (pilot's and copilot's) shall go on.
 - (4) CHAN 1 and CHAN 2 warning lights on CDU shall go on.
 - (5) CDU display lighting shall vary with CDU's DIM control setting, except CHAN 1 and CHAN 2 warning lights.
- If any two or more adjacent lamps on one vertical scale are off, or if one lamp between the green and amber, amber and red, or green and red ranges on a scale is off, go to TABLE NO. 8-2-3-5.
 - If alternate vertical scale lamps are off, note trouble and continue with steps m. and n.
- **EMEP** If result (2) is not as specified, replace Direct View Filament (DVF) module (PARA 8-4-12).■
- If trouble remains, replace CDU (PARA 8-4-10).
- **W/O EMEP** If result (2) is not as specified, replace CDU (PARA 8-4-10).■
- If result (3) is not as specified, go to TABLE NO. 8-2-3-6.
 - If result (4) or (5) is not as specified, replace CDU (PARA 8-4-10).
 - If DIM control varies alternate display lamps only, go to TABLE NO. 8-2-3-7.

- m. Release CDU PUSH TO TEST or LAMP TEST switch.
- n. Press and hold pilot's PDU LT switch.

RESULT:

- (1) All PDU (pilot's) vertical scale lamps except the top segments (130% RPM) shall go on.
- (2) All PDU (pilot's) digital readouts shall indicate 188.

- If result (1) is not as specified and alternate CDU scale lamps were off in step l., replace No. 2 SDC (PARA 8-4-15).

- **EMEP** If any two or more adjacent lamps on one vertical scale are off, or if one lamp between the green and amber, amber and red, or green and red ranges on a scale is off, replace Bar Graph Filament (BGF) lamp (PARA 8-4-18).◀

If trouble remains, replace pilot's PDU (PARA 8-4-16).

- **W/O EMEP** If any two or more adjacent lamps on one vertical scale are off, or if one lamp between the green and amber, amber and red, or green and red ranges on a scale is off, replace pilot's PDU (PARA 8-4-16).◀

- **EMEP** If result (2) is not as specified, replace DVF module (PARA 8-4-18).◀

If trouble remains, replace pilot's PDU (PARA 8-4-16).

- **W/O EMEP** If result (2) is not as specified, replace pilot's PDU (PARA 8-4-16).◀

- o. Release pilot's PDU LT switch.
- p. Press and hold copilot's PDU LT switch.

RESULT:

- (1) All PDU (copilot's) vertical scale lamps except the top segments (130% RPM) shall go on.

- (2) All PDU (copilot's) digital readouts shall indicate 188.

- If result (1) is not as specified and alternate CDU scale lamps were off in step l., replace No. 1 SDC (PARA 8-4-15).

- **EMEP** If any two or more adjacent lamps on one vertical scale are off, or if one lamp between the green and amber, amber and red, or green and red ranges on a scale is off, replace Bar Graph Filament (BGF) lamp (PARA 8-4-18).◀

If trouble remains, replace copilot's PDU (PARA 8-4-16).

- **W/O EMEP** If any two or more adjacent lamps on one vertical scale are off, or if one lamp between the green and amber, amber and red, or green and red ranges on a scale is off, replace copilot's PDU (PARA 8-4-16).◀

- **EMEP** If result (2) is not as specified, replace Direct View Filament (DVF) module (PARA 8-4-18).◀

If trouble remains, replace copilot's PDU (PARA 8-4-16).

- **W/O EMEP** If result (2) is not as specified, replace copilot's PDU (PARA 8-4-16).◀

- q. Release copilot's PDU LT switch.

NOTE

Make sure the CDU DIM control is at midposition.

- r. Using a flashlight, position light beam on PDUs (pilot's and copilot's) and CDU faceplates separately.

RESULT: All displays shall increase in brightness as light beam is positioned on each unit.

- If result is not as specified for PDUs (pilot's and copilot's), interchange PDUs.

- (a) If trouble remains on same PDU, replace display unit (PARA 8-4-16).
- (b) If trouble does not remain on same PDU, repair/replace wiring between P153-4 and P157-30, or between P156-4 and P158-30 as required (Appendix F).
 - If result is not as specified, replace CDU (PARA 8-4-10).
- s. Press CDU PUSH TO TEST or LAMP TEST switch and adjust DIM control to get a normal viewing brightness.
- (6) Right RTR OVERSPEED light (indicating about 142%) shall go on.
 - If result (1) is not as specified, go to TABLE NO. 8-2-3-8.
 - If result (2) is not as specified, replace malfunctioning PDU (PARA 8-4-16).
 - If result (3) is not as specified, replace CDU (PARA 8-4-10).
 - If results (4), (5), and (6) are not as specified, go to TABLE NO. 8-2-3-9.
- c. Slowly turn LTS - ROTOR SPEED variable control fully counterclockwise.

8-2-3.2 % RPM R (Main Rotor Speed).

- a. Perform Setup, if not already done.

NOTE

With line test set connected, RTR OVERSPEED lamps on PDUs (pilot's and copilot's) are reset by ROTOR OVERSPD RESET switch on line test set.

- b. Slowly turn LTS - ROTOR SPEED variable control fully clockwise.

RESULT:

- (1) % RPM R vertical scale indication on PDUs (pilot's and copilot's) shall advance smoothly to full scale.
- (2) At 96% indication, lower red and amber scale segments shall go off.
- (3) LOW ROTOR SPD light on LTS shall go off.
- (4) Left RTR OVERSPEED light (indicating about 127%) on PDUs (pilot's and copilot's) shall go on.
- (5) Center RTR OVERSPEED light (indicating about 137%) shall go on.
- (1) % RPM R vertical scale indications on PDUs (pilot's and copilot's) shall decrease smoothly.
- (2) At 95% indication, lower red and amber scale segments shall go on.
- (3) LOW ROTOR SPD light on LTS shall go on.
- (4) All three RTR OVERSPEED lights on PDUs (pilot's and copilot's) shall remain ON.
 - If result (1) or (2) is not as specified, replace malfunctioning PDU (PARA 8-4-16).
 - If result (3) is not as specified, check RTR SPD fuse.
 - (a) If fuse is good, replace CDU (PARA 8-4-10).
 - (b) If fuse is not good, replace fuse (PARA 8-4-13).
 - If result (4) is not as specified, replace CDU (PARA 8-4-10).
- d. Press and release LTS - ROTOR OVERSPD RESET switch.

RESULT: Three RTR OVERSPEED lights on PDUs (pilot's and copilot's) shall go off.

- If result is not as specified, replace CDU (PARA 8-4-10).

e. Place LTS - ROTOR SPEED CAL-NOR switch to CAL.

RESULT:

- (1) % RPM R (pilot's PDU) scale shall indicate 101% or 102%.
 - (2) % RPM R (copilot's PDU) scale shall indicate 101% or 102%.
- If result (1) is not as specified, replace No. 2 SDC (PARA 8-4-15).
 - If result (2) is not as specified, replace No. 1 SDC (PARA 8-4-15).

f. Place LTS - ROTOR SPEED HI-LO switch to LO.

RESULT:

- (1) % RPM R (pilot's PDU) scales shall be blank (0%).
 - (2) % RPM R (copilot's PDU) scales shall be blank (0%).
- If result (1) is not as specified, replace No. 2 SDC (PARA 8-4-15).
 - If result (2) is not as specified, replace No. 1 SDC (PARA 8-4-15).

g. Place LTS - ROTOR SPEED CAL-NOR switch to NOR.

h. If no other check is to be done, go to Shutdown.

8-2-3.3 FUEL QTY (Fuel Quantity).

a. Perform Setup, if not already done.

b. Slowly turn LTS - FUEL QUANTITY NO. 1 variable control fully clockwise.

RESULT:

- (1) FUEL QTY 1 vertical scale indication on CDU shall advance smoothly to full scale.
- (2) MAIN FUEL or TOTAL FUEL digital readout on CDU shall advance to about 1500.
- (3) At about 250 pounds indication, amber scale segment shall go off.

- If result (1) or (2) is not as specified, go to TABLE NO. 8-2-3-10.

- If result (3) is not as specified, replace CDU (PARA 8-4-10).

c. Slowly turn LTS - FUEL QUANTITY NO. 1 variable control fully counterclockwise.

RESULT:

- (1) FUEL QTY 1 vertical scale indication on CDU shall decrease smoothly to zero.
- (2) MAIN FUEL or TOTAL FUEL digital readout on CDU shall decrease to zero.
- (3) At about 200 pounds indication, amber scale segment shall go on.

- If results are not as specified, replace CDU (PARA 8-4-10).

d. Place LTS - FUEL QUANTITY NO. 1 CAL-NOR switch to CAL.

RESULT:

- (1) FUEL QTY 1 vertical scale on CDU shall indicate 1200 pounds.
- (2) MAIN FUEL or TOTAL FUEL digital readout on CDU shall indicate between 1210 and 1230 pounds.

8-2-3-6 Change 8

- If results are not as specified, replace CDU (PARA 8-4-10).
- e. Place LTS - FUEL QUANTITY NO. 1 CAL-NOR switch to NOR.
- f. Slowly turn LTS - FUEL QUANTITY NO. 2 variable control fully clockwise.

RESULT:

- (1) FUEL QTY 2 vertical scale indication on CDU shall advance smoothly to full scale.
 - (2) MAIN FUEL or TOTAL FUEL digital readout on CDU shall advance to about 1500.
 - (3) At about 250 pounds indication, amber scale segment shall go off.
- If results (1) and (2) are not as specified, go to TABLE NO. 8-2-3-10.
 - If result (3) is not as specified, replace CDU (PARA 8-4-10).
- g. Slowly turn LTS - FUEL QUANTITY NO. 2 variable control fully counterclockwise.

RESULT:

- (1) FUEL QTY 2 scale on CDU shall decrease smoothly to zero.
 - (2) MAIN FUEL or TOTAL FUEL digital readout on CDU shall decrease to zero.
 - (3) At about 200 pounds indication, amber scale segment shall go on.
- If results are not as specified, replace CDU (PARA 8-4-10).
- h. Place LTS - FUEL QUANTITY NO. 1 CAL-NOR switch to CAL, FUEL QUANTITY NO. 2 CAL-NOR switch to CAL, and NO. 2 HI-LO switch to LO.

RESULT:

- (1) FUEL QTY 2 vertical scale on CDU shall be blank.
 - (2) MAIN FUEL or TOTAL FUEL digital readout on CDU shall indicate between 1210 and 1230 pounds.
- If results are not as specified, replace CDU (PARA 8-4-10).
- i. Place LTS - FUEL QUANTITY NO. 1 HI-LO switch to LO.

RESULT:

- (1) FUEL QTY 1 vertical scale on CDU shall be blank.
 - (2) MAIN FUEL or TOTAL FUEL digital readout on CDU shall indicate between 0 and 10 pounds.
- If results are not as specified, replace CDU (PARA 8-4-10).
- j. Place LTS - FUEL QUANTITY NO. 2 HI-LO switch to HI.

RESULT:

- (1) FUEL QTY 2 vertical scale on CDU shall indicate 1200 pounds.
 - (2) MAIN FUEL or TOTAL FUEL digital readout on CDU shall indicate between 1210 and 1230 pounds.
- If results are not as specified, replace CDU (PARA 8-4-10).
- k. Place LTS - FUEL QUANTITY NO. 1 HI-LO switch to HI.

RESULT: MAIN FUEL or TOTAL FUEL digital readout on CDU shall indicate between 2420 and 2460 pounds.

- If result is not as specified, replace CDU (PARA 8-4-10).

- l. Place LTS - FUEL QUANTITY NO. 1 and NO. 2 CAL-NOR switches to NOR.

- m. If no other check is to be done, go to Shutdown.

8-2-3.4 ENG OIL PRESS (Engine Oil Pressure).

- a. Perform Setup, if not already done.
- b. Slowly turn LTS - ENGINE OIL PRESS NO. 1 variable control fully clockwise.

RESULT:

- (1) ENG OIL PRESS 1 vertical scale indication on CDU shall advance smoothly to full scale.
- (2) **S70-II** At 22 psi indication, NO. 1 LOW OIL PRESS light on LTS shall go off.■
- (3) **S70-I** At 25 psi indication, NO. 1 LOW OIL PRESS LIGHT on LTS shall go off.■
- (4) **S70-II** At 26 psi indication, lower red and amber scale segments shall go off.■
- (5) **S70-I** At 47.5 psi indication, lower red and amber scale segments shall go off.■

- If result (1) is not as specified, go to TABLE NO. 8-2-3-11.
- If result (2), (3), (4) or (5) is not as specified, replace CDU (PARA 8-4-10).

- c. Slowly turn LTS - ENGINE OIL PRESS NO. 1 variable control fully counterclockwise.

RESULT:

- (1) ENG OIL PRESS 1 vertical scale indication shall decrease smoothly.
- (2) **S70-II** At 26 psi indication, lower red and amber scale segments shall go on.■

- (3) **S70-I** At 45 psi indication, lower red and amber scale segments shall go on.■

- (4) **S70-II** At 22 psi indication, NO. 1 LOW OIL PRESS light on LTS shall go on.■

- (5) **S70-I** At 20 psi indication, NO. 1 LOW OIL PRESS light on LTS shall go on.■

- If result (1), (2), or (3) is not as specified, replace CDU (PARA 8-4-10).
- If result (4) or (5) is not as specified, go to TABLE NO. 8-2-3-2.

- d. Place LTS - ENGINE OIL PRESS NO. 1 CAL-NOR switch to CAL.

RESULT: ENG OIL PRESS 1 scale on CDU shall indicate 110 psi or 120 psi.

- If result is not as specified, replace No. 1 SDC (PARA 8-4-15).

- e. Place LTS - ENGINE OIL PRESSURE NO. 1 HI-LO switch to LO.

RESULT: ENG OIL PRESS 1 scale on CDU shall be blank.

- If result is not as specified, replace No. 1 SDC (PARA 8-4-15).

- f. Place LTS - ENGINE OIL PRESS NO. 1 CAL-NOR switch to NOR.

- g. Slowly turn LTS - ENGINE OIL PRESS NO. 2 variable control fully clockwise.

RESULT:

- (1) ENG OIL PRESS 2 vertical scale indications on CDU shall advance smoothly to full scale.
- (2) **S70-II** At 22 psi indication, NO. 2 LOW OIL PRESS light on LTS shall go off.■

8-2-3-8 Change 8

- (3) **S70-I** At 25 psi indication, NO. 2 LOW OIL PRESS LIGHT on LTS shall go off.■
- (4) **S70-II** At 26 psi indication, lower red and amber scale segments shall go off.■
- (5) **S70-I** At 47.5 psi indication, lower red and amber scale segments shall go off.■
- If result (1) is not as specified, go to TABLE NO. 8-2-3-11.
- If result (2), (3), (4), or (5) is not as specified, replace CDU (PARA 8-4-10).
- h. Slowly turn LTS - ENGINE OIL PRESS NO. 2 variable control fully counterclockwise.

RESULT:

- (1) ENG OIL PRESS 2 vertical scale indication on CDU shall decrease smoothly.
- (2) **S70-II** At 26 psi indication, lower red and amber scale segments shall go on.■
- (3) **S70-I** At 45 psi indication, lower red and amber scale segments shall go on.■
- (4) **S70-II** At 22 psi indication, NO. 2 LOW OIL PRESS light on LTS shall go on.■
- (5) **S70-I** At 20 psi indication, NO. 2 LOW OIL PRESS light on LTS shall go on.■
- If result (1), (2), or (3) is not as specified, replace CDU (PARA 8-4-10).
- If result (4) or (5) is not as specified, go to TABLE NO. 8-2-3-2.
- i. Place LTS - ENGINE OIL PRESS NO. 2 CAL-NOR switch to CAL.

RESULT: ENG OIL PRESS 2 scale on CDU shall indicate 110 or 120 psi.

- If result is not as specified, replace No. 2 SDC (PARA 8-4-15).

- j. Place LTS - ENG OIL PRESS NO. 2 HI-LO switch to LO.

RESULT: ENG OIL PRESS 2 scale on CDU shall be blank.

- If result is not as specified, replace No. 2 SDC (PARA 8-4-15).
- k. Place LTS - ENGINE OIL PRESS NO. 2 CAL-NOR switch to NOR.
- l. If no other check is to be done, go to Shutdown.

8-2-3.5 ENG OIL TEMP (Engine Oil Temperature).

- a. Perform Setup, if not already done.
- b. Slowly turn LTS - ENGINE OIL TEMP NO. 1 variable control fully clockwise.

RESULT:

- (1) ENG OIL TEMP 1 vertical scale indication on CDU shall advance smoothly to full scale.
- (2) At 302°F (150°C) indication, NO. 1 HIGH OIL TEMP light on LTS shall go on.
- If result (1) is not as specified, go to TABLE NO. 8-2-3-12.
- If result (2) is not as specified, go to TABLE NO. 8-2-3-13.
- c. Slowly turn LTS - ENGINE OIL TEMP NO. 1 variable control fully counterclockwise.

RESULT:

- (1) ENG OIL TEMP 1 vertical scale indication on CDU shall decrease smoothly.
- (2) At 284°F (140°C) indication, NO. 1 HIGH OIL TEMP light on LTS shall go off.
- If results are not as specified, replace CDU (PARA 8-4-10).

- d. Place LTS - ENGINE OIL TEMP NO. 1 CAL-NOR switch to CAL.

RESULT: ENG OIL TEMP 1 scale on CDU shall indicate 284°F (140°C).

- If result is not as specified, replace No. 1 SDC (PARA 8-4-15).

- e. Place LTS - NO. 1 ENGINE OIL TEMP HI-LO switch to LO.

RESULT:

- (1) **S70-II** ENG OIL TEMP 1 scale on CDU shall be blank.
- (2) **S70-I** ENG OIL TEMP 1 scale on CDU shall indicate -58°F (-50°C).

- If result (1) or (2) is not as specified, replace No. 1 SDC (PARA 8-4-15).

- f. Place LTS - NO. 1 ENGINE OIL TEMP CAL-NOR switch to NOR.

- g. Slowly turn LTS - ENGINE OIL TEMP NO. 2 variable control fully clockwise.

RESULT:

- (1) ENG OIL TEMP 2 vertical scale indication on CDU shall advance smoothly to full scale.
- (2) At 302°F (150°C) indication, NO. 2 HIGH OIL TEMP light on LTS shall go on.

- If result (1) is not as specified, go to TABLE NO. 8-2-3-12.
- If result (2) is not as specified, go to TABLE NO. 8-2-3-13.

- h. Slowly turn LTS - ENGINE OIL TEMP NO. 2 variable control fully counterclockwise.

RESULT:

- (1) ENG OIL TEMP 2 vertical scale indication on CDU shall decrease smoothly.

- (2) At 284°F (140°C) indication, NO. 2 HIGH OIL TEMP light on LTS shall go off.

- If results are not as specified, replace CDU (PARA 8-4-10).

- i. Place LTS - ENGINE OIL TEMP NO. 2 CAL-NOR switch to CAL.

RESULT: ENG OIL TEMP 2 scale on CDU shall indicate 284°F (140°C).

- If result is not as specified, replace No. 2 SDC (PARA 8-4-15).

- j. Place LTS - NO. 2 ENGINE OIL TEMP HI-LO switch to LO.

RESULT:

- (1) **S70-II** ENG OIL TEMP 2 scale on CDU shall be blank.
- (2) **S70-I** ENG OIL TEMP 2 scale on CDU shall indicate -58°F (-50°C).

- If result (1) or (2) is not as specified, replace No. 2 SDC (PARA 8-4-15).

- k. Place LTS - ENGINE OIL TEMP NO. 2 CAL-NOR switch to NOR.

- l. If no other check is to be done, go to Shutdown.

8-2-3.6 Ng SPEED (Engine Gas Generator Tachometer).

- a. Perform Setup, if not already done.
- b. Slowly turn LTS - GAS GEN SPEED NO. 1 variable control fully clockwise.

RESULT:

- (1) Ng SPEED 1 vertical scale on CDU shall advance smoothly to full scale.
- (2) Ng SPEED digital readout indications on CDU shall advance smoothly to full scale.

- (3) At about 55% indication, NO. 1 ENGINE OUT light on LTS shall go off.

- If result (1) is not as specified, go to TABLE NO. 8-2-3-14.
- If result (2) is not as specified, go to TABLE NO. 8-2-3-15.
- If result (3) is not as specified, replace CDU (PARA 8-4-10).

- c. Slowly turn LTS - GAS GEN SPEED NO. 1 variable control fully counterclockwise.

RESULT:

- (1) Ng SPEED 1 vertical scale and digital readout indications on CDU shall decrease smoothly.
 - (2) At about 55% indication, NO. 1 ENGINE OUT light on LTS shall go on.
- If result (1) is not as specified, replace CDU (PARA 8-4-10).
 - If result (2) is not as specified, go to TABLE NO. 8-2-3-1.

- d. Place LTS - GAS GEN SPEED NO. 1 CAL-NOR switch to CAL.

RESULT:

- (1) Ng SPEED 1 vertical scale shall indicate 98% or 100%.
 - (2) Ng SPEED 1 digital readout shall indicate between 99.5% and 100.5%.
- If results are not as specified, replace No. 1 SDC (PARA 8-4-15).

- e. Place LTS - GAS GEN SPEED NO. 1 HI-LO switch to LO.

RESULT:

- (1) Ng SPEED 1 vertical scale on CDU shall be blank.

- (2) Ng SPEED 1 digital readout on CDU shall indicate between 0 and 0.5%.

- If results are not as specified, replace No. 1 SDC (PARA 8-4-15).

- f. Place LTS - GAS GEN SPEED NO. 1 CAL-NOR switch to NOR.

- g. Slowly turn LTS - GAS GEN SPEED NO. 2 variable control fully clockwise.

RESULT:

- (1) Ng SPEED 2 vertical scale indication on CDU shall advance smoothly to full scale.
- (2) Ng SPEED 2 digital readout indication on CDU shall advance smoothly to full scale.
- (3) At about 55% indication, NO. 2 ENGINE OUT light on LTS shall go off.

- If result (1) is not as specified, go to TABLE NO. 8-2-3-14.
- If result (2) is not as specified, go to TABLE NO. 8-2-3-15.
- If result (3) is not as specified, replace CDU (PARA 8-4-10).

- h. Slowly turn LTS - GAS GEN SPEED NO. 2 variable control fully counterclockwise.

RESULT:

- (1) Ng SPEED 2 vertical scale and digital readout indications on CDU shall decrease smoothly.
- (2) At about 55% indication, NO. 2 ENGINE OUT light on LTS shall go on.

- If result (1) is not as specified, replace CDU (PARA 8-4-10).
- If result (2) is not as specified, go to TABLE NO. 8-2-3-1.

- i. Place LTS - GAS GEN SPEED NO. 2 CAL-NOR switch to CAL.

RESULT:

- (1) Ng SPEED 2 vertical scale on CDU shall indicate 98% or 100%.
- (2) Ng SPEED 2 digital readout on CDU shall indicate between 99.5% and 100.5%.
- If results are not as specified, replace No. 2 SDC (PARA 8-4-15).

- j. Place LTS - GAS GEN SPEED NO. 2 HI/LO switch to LO.

RESULT:

- (1) Ng SPEED 2 vertical scale on CDU shall be blank.
- (2) Ng SPEED 2 digital readout on CDU shall indicate between 0 and 0.5%.
- If results are not as specified, replace No. 2 SDC (PARA 8-4-15).

- k. Place LTS - GAS GEN SPEED NO. 2 CAL-NOR switch to NOR.

- l. If no other check is to be done, go to Shutdown.

8-2-3.7 % TRQ (Engine Torque).

- a. Perform Setup, if not already done.
- b. Slowly turn LTS - TORQUE NO. 1 variable control fully clockwise.

RESULT: % TRQ 1 (pilot's and copilot's PDU) vertical scale and digital readout indications shall advance smoothly to full scale.

- If result is not as specified, go to TABLE NO. 8-2-3-16.

- c. Slowly turn LTS - TORQUE NO. 1 variable control fully counterclockwise.

RESULT:

- (1) % TRQ 1 (pilot's PDU) vertical scale and digital readout indications shall decrease smoothly.
- (2) % TRQ 1 (copilot's PDU) vertical scale and digital readout indications shall decrease smoothly.

- If result (1) is not as specified, replace pilot's PDU (PARA 8-4-16).

- If result (2) is not as specified, replace copilot's PDU (PARA 8-4-16).

- d. Place LTS - TORQUE NO. 1 CAL-NOR switch to CAL.

RESULT:

- (1) % TRQ 1 (pilot's PDU) vertical scales shall indicate 100% or 105%.
- (2) % TRQ 1 (pilot's PDU) digital readouts shall indicate between 103% and 105%.
- (3) % TRQ 1 (copilot's PDU) vertical scales shall indicate 100% or 105%.
- (4) % TRQ 1 (copilot's PDU) vertical scales shall indicate 103% or 105%.

- If result (1) or (2) is not as specified, replace No. 2 SDC (PARA 8-4-15).

- If result (3) or (4) is not as specified, replace No. 1 SDC (PARA 8-4-15).

- e. Place LTS - TORQUE NO. 1 HI-LO switch to LO.

RESULT:

- (1) % TRQ 1 (pilot's PDU) digital readouts shall be 0 or 1%.
- (2) % TRQ 1 (pilot's PDU) vertical scales shall be blank.

- (3) % TRQ 1 (copilot's PDU) digital readouts shall be 0 or 1%.
- (4) % TRQ 1 (copilot's PDU) vertical scales shall be blank.
 - If result (1) or (2) is not as specified, replace No. 2 SDC (PARA 8-4-15).
 - If result (3) or (4) is not as specified, replace No. 1 SDC (PARA 8-4-15).
- f. Place LTS - TORQUE NO. 1 CAL-NOR switch to NOR.
- g. Slowly turn LTS - TORQUE NO. 2 variable control fully clockwise.

RESULT: % TRQ 2 (pilot's and copilot's PDU) vertical scale and digital readout indications shall advance smoothly to full scale.

- If result is not as specified, go to TABLE NO. 8-2-3-16.
- h. Slowly turn LTS - TORQUE NO. 2 variable control fully counterclockwise.

RESULT:

- (1) % TRQ 2 (pilot's PDU) vertical scale and digital readout indication shall decrease smoothly.
- (2) % TRQ 2 (copilot's PDU) vertical scale and digital readout indication shall decrease smoothly.
 - If result (1) is not as specified, replace pilot's PDU (PARA 8-4-16).
 - If result (2) is not as specified, replace copilot's PDU (PARA 8-4-16).
- i. Place LTS - TORQUE NO. 2 CAL-NOR switch to CAL.

RESULT:

- (1) % TRQ 2 (pilot's PDU) vertical scales shall indicate 100% or 105%.

- (2) % TRQ 2 (pilot's PDU) digital readouts shall indicate between 103% and 105%.
- (3) % TRQ 2 (copilot's PDU) vertical scales shall indicate 100% or 105%.
- (4) % TRQ 2 (copilot's PDU) digital readouts shall indicate between 103% and 105%.
 - If result (1) or (2) is not as specified, replace No. 2 SDC (PARA 8-4-15).
 - If result (3) or (4) is not as specified, replace No. 1 SDC (PARA 8-4-15).
- j. Place LTS - TORQUE NO. 2 HI-LO switch to LO.

RESULT:

- (1) % TRQ 2 (pilot's PDU) digital readout shall be 0 or 1%.
- (2) % TRQ 2 (pilot's PDU) vertical scales shall be blank.
- (3) % TRQ 2 (copilot's PDU) digital readout shall be 0 or 1%.
- (4) % TRQ 2 (copilot's PDU) vertical scales shall be blank.
 - If result (1) or (2) is not as specified, replace No. 2 SDC (PARA 8-4-15).
 - If result (3) or (4) is not as specified, replace No. 1 SDC (PARA 8-4-15).
- k. Place LTS - TORQUE NO. 2 CAL-NOR switch to NOR.
- l. If no other check is to be done, go to Shutdown.

8-2-3.8 XMSN PRESS (Transmission Oil Pressure).

- a. Perform Setup, if not already done.
- b. Slowly turn LTS - XMSN OIL PRESS variable control fully clockwise.

RESULT:

- (1) XMSN PRESS vertical scale indication on CDU shall advance smoothly to full scale.
- (2) At 32.5 psi indication, lower red and amber scale segments shall go off.
 - If result (1) is not as specified, go to TABLE NO. 8-2-3-17.
 - If result (2) is not as specified, replace CDU (PARA 8-4-10).
- c. Slowly turn LTS - XMSN OIL PRESS variable control fully counterclockwise.

RESULT:

- (1) XMSN PRESS vertical scale indication on CDU shall decrease smoothly.
- (2) At 30 psi indication, lower red and amber scale segments shall go on.
 - If results are not as specified, replace CDU (PARA 8-4-10).
- d. Place LTS - XMSN OIL PRESS CAL-NOR switch to CAL.

RESULT: XMSN PRESS vertical scale shall indicate 130 psi.

- If result is not as specified, replace CDU (PARA 8-4-10).
- e. Place LTS - XMSN OIL PRESS HI-LO switch to LO.

RESULT: XMSN PRESS vertical scale shall be blank.

- If result is not as specified, replace CDU (PARA 8-4-10).
- f. Place LTS - XMSN OIL PRESS CAL-NOR switch to NOR.
- g. If no other check is to be done, go to Shutdown.

8-2-3.9 XMSN TEMP (Transmission Oil Temperature).

- a. Perform Setup, if not already done.
- b. Slowly turn LTS - XMSN OIL TEMP variable control fully clockwise.

RESULT: XMSN TEMP vertical scale indication on CDU shall advance smoothly to full scale.

- If result is not as specified, go to TABLE NO. 8-2-3-18.
- c. Slowly turn LTS - XMSN OIL TEMP variable control fully counterclockwise.

RESULT: XMSN TEMP vertical scale indication on CDU shall decrease smoothly.

- If result is not as specified, replace CDU (PARA 8-4-10).
- d. Place LTS - XMSN OIL TEMP CAL-NOR switch to CAL.

RESULT: XMSN TEMP vertical scale on CDU shall indicate 266°F (130°C).

- If result is not as specified, replace CDU (PARA 8-4-10).
- e. Place LTS - XMSN OIL TEMP HI-LO switch to LO.

RESULT:

- (1) **S70-II** XMSN TEMP vertical scale on CDU shall be blank. 
- (2) **S70-I** XMSN TEMP vertical scale on CDU shall indicate -58°F (-50°C). 
- If result (1) or (2) is not as specified, replace CDU (PARA 8-4-10).
- f. Place LTS - XMSN OIL TEMP CAL-NOR switch to NOR.

- g. If no other check is to be done, go to Shutdown.

8-2-3.10 % RPM 1-2 (Engine Power Turbine Tachometer).

- a. Perform, if not already done.
- b. Slowly turn LTS - ENGINE % RPM NO. 1 variable control fully clockwise.

RESULT:

- (1) % RPM 1 (pilot's and copilot's PDU) vertical scale indications shall advance smoothly to full scale.
 - (2) **S70-II** At 95% indication, pilot's PDU lower red and red amber scale segments shall go off.■
 - (3) **S70-I** At 96% indication, pilot's PDU lower red and amber scale segments shall go off.■
 - (4) **S70-II** At 95% indication, copilot's PDU lower red and amber scale segments shall go off.■
 - (5) **S70-I** At 96% indication, copilot's PDU lower red and amber scale segments shall go off.■
- If result (1) is not as specified, go to TABLE NO. 8-2-3-19.
 - If result (2) or (3) is not as specified, replace pilot's PDU (PARA 8-4-16).
 - If result (4) or (5) is not as specified, replace copilot's PDU (PARA 8-4-16).
- c. Slowly turn LTS - ENGINE % RPM NO. 1 variable control fully counterclockwise.

RESULT:

- (1) % RPM 1 (pilot's PDU) vertical scale indications shall decrease smoothly.

- (2) **S70-II** At 96% indication, pilot's PDU lower red and amber scale segments shall go on.■

- (3) **S70-I** At 95% indication, pilot's PDU lower red and amber scale segments shall go on.■

- (4) % RPM 1 (copilot's PDU) vertical scale indications shall decrease smoothly.

- (5) **S70-II** At 96% indication, copilot's PDU lower red and amber scale segments shall go on.■

- (6) **S70-I** At 95% indication, copilot's PDU lower red and amber scale segments shall go on.■

- If result (1), (2), or (3) is not as specified, replace pilot's PDU (PARA 8-4-16).
- If result (4), (5), or (6) is not as specified, replace copilot's PDU (PARA 8-4-16).

- d. Place LTS - ENGINE % RPM NO. 1 CAL-NOR switch to CAL.

RESULT:

- (1) % RPM 1 (pilot's PDU) vertical scales shall indicate 95% or 96%.
 - (2) % RPM 1 (copilot's PDU) vertical scales shall indicate 95% or 96%.
- If result (1) is not as specified, replace No. 2 SDC (PARA 8-4-15).
 - If result (2) is not as specified, replace No. 1 SDC (PARA 8-4-15).

- e. Place LTS - ENGINE % RPM NO. 1 HI-LO switch to LO.

RESULT:

- (1) % RPM 1 (pilot's PDU) vertical scales shall be blank.

- (2) % RPM 1 (copilot's PDU) vertical scales shall be blank.

- If result (1) is not as specified, replace No. 2 SDC (PARA 8-4-15).
- If result (2) is not as specified, replace No. 1 SDC (PARA 8-4-15).

f. Place LTS - ENGINE % RPM NO. 1 CAL-NOR switch to NOR.

g. Slowly turn LTS - ENGINE % RPM NO. 2 variable control fully clockwise.

RESULT:

- (1) PDUs (pilot's and copilot's) % RPM 2 vertical scale indications shall advance smoothly to full scale.
- (2) **S70-II** At 95% indication, pilot's PDU lower red and red amber scale segments shall go off. 
- (3) **S70-I** At 96% indication, pilot's PDU lower red and amber scale segments shall go off. 
- (4) **S70-II** At 95% indication, copilot's PDU lower red and amber scale segments shall go off. 
- (5) **S70-I** At 96% indication, copilot's PDU lower red and amber scale segments shall go off. 

- If result (1) is not as specified, go to TABLE NO. 8-2-3-19.
- If result (2) or (3) is not as specified, replace pilot's PDU (PARA 8-4-16).
- If result (4) or (5) is not as specified, replace copilot's PDU (PARA 8-4-16).

h. Slowly turn LTS - ENGINE % RPM NO. 2 variable control fully counterclockwise.

RESULT:

- (1) % RPM 2 (pilot's PDU) vertical scale indications shall decrease smoothly.
- (2) **S70-II** At 96% indication, pilot's PDU lower red and amber scale segments shall go on. 
- (3) **S70-I** At 95% indication, pilot's PDU lower red and amber scale segments shall go on. 
- (4) % RPM 1 (copilot's PDU) vertical scale indications shall decrease smoothly.
- (5) **S70-II** At 96% indication, copilot's PDU lower red and amber scale segments shall go on. 
- (6) **S70-I** At 95% indication, copilot's PDU lower red and amber scale segments shall go on. 

- If result (1), (2), or (3) is not as specified, replace pilot's PDU (PARA 8-4-16).
- If result (4), (5), or (6) is not as specified, replace copilot's PDU (PARA 8-4-16).

i. Place LTS - ENGINE % RPM NO. 2 CAL-NOR switch to CAL, and HI-LO switch to HI.

RESULT:

- (1) % RPM 2 (pilot's PDU) vertical scales shall indicate 95% or 96%.
- (2) % RPM 2 (copilot's PDU) vertical scales shall indicate 95% or 96%.

- If result (1) is not as specified, replace No. 2 SDC (PARA 8-4-15).
- If result (2) is not as specified, replace No. 1 SDC (PARA 8-4-15).

j. Place LTS - ENGINE % RPM NO. 2 HI-LO switch to LO.

RESULT:

- (1) % RPM 2 (pilot's PDU) vertical scales shall be blank.
 - (2) % RPM 2 (copilot's PDU) vertical scales shall be blank.
 - If result (1) is not as specified, replace No. 2 SDC (PARA 8-4-15).
 - If result (2) is not as specified, replace No. 1 SDC (PARA 8-4-15).
- k. Place LTS - ENGINE % RPM NO. 2 CAL-NOR switch to NOR.
- l. If no other check is to be done, go to Shutdown.

8-2-3.11 TGT TEMP (Turbine Gas Temperature).

- a. Perform Setup, if not already done.
- b. Slowly turn LTS - TURBINE GAS TEMP NO. 1 variable control fully clockwise.

RESULT:

- (1) TGT TEMP 1 vertical scale indication on CDU shall advance smoothly to full scale.
 - (2) TGT TEMP 1 digital readout indication on CDU shall advance smoothly.
 - If result (1) is not as specified, go to TABLE NO. 8-2-3-20.
 - If result (2) is not as specified, go to TABLE NO. 8-2-3-21.
- c. Slowly turn LTS - TURBINE GAS TEMP NO. 1 variable control fully counterclockwise.

RESULT: TGT TEMP 1 vertical scale and digital readout indications on CDU shall decrease smoothly.

- If result is not as specified, replace CDU (PARA 8-4-10).

- d. Place LTS - TURBINE GAS TEMP NO. 1 CAL-NOR switch to CAL, and TURBINE GAS TEMP NO. 1 HI-LO switch to HI.

RESULT:

- (1) TGT TEMP 1 vertical scale on CDU shall indicate 1472°F (800°C).
- (2) TGT TEMP 1 digital readout on CDU shall indicate between 1482° and 1504°F (806° and 818°C).
- If results are not as specified, replace No. 1 SDC (PARA 8-4-15).

- e. Place LTS - TURBINE GAS TEMP NO. 1 HI-LO switch to LO.

RESULT:

- (1) TGT TEMP 1 vertical scale on CDU shall be blank.
- (2) TGT TEMP 1 digital readout on CDU shall indicate between 32° to 46°F (0° to 8°C).
- If results are not as specified, replace CDU (PARA 8-4-10).

- f. Place LTS - TURBINE GAS TEMP NO. 1 CAL-NOR switch to NOR.
- g. Slowly turn LTS - TURBINE GAS TEMP NO. 2 variable control fully clockwise.

RESULT:

- (1) TGT TEMP 2 vertical scale indication on CDU shall advance smoothly to full scale.
- (2) TGT TEMP 2 digital readout indication on CDU shall advance.
- If result (1) is not as specified, go to TABLE NO. 8-2-3-20.
- If result (2) is not as specified, go to TABLE NO. 8-2-3-21.

- h. Slowly turn LTS - TURBINE GAS TEMP NO. 2 variable control fully counterclockwise.

RESULT: TGT TEMP 2 vertical scale and digital readout indications on CDU shall decrease smoothly.

- If result is not as specified, replace CDU (PARA 8-4-10).
- i. Place LTS - TURBINE GAS TEMP NO. 2 CAL-NOR switch to CAL and HI-LO switch to HI.

RESULT:

- (1) TGT TEMP 2 vertical scale on CDU shall indicate 1472°F (800°C).
 - (2) TGT TEMP 2 digital readout on CDU shall indicate between 1482° to 1504°F (806° and 818°C).
- If results are not as specified, replace No. 2 SDC (PARA 8-4-15).

- j. Place LTS - TURBINE GAS TEMP NO. 2 HI-LO switch to LO.

RESULT:

- (1) TGT TEMP 2 vertical scale on CDU shall be blank.
 - (2) TGT TEMP 2 digital readout on CDU shall indicate between 32° to 46°F (0° to 8°C).
- If results are not as specified, replace CDU (PARA 8-4-10).

- k. Place LTS - TURBINE GAS TEMP NO. 2 CAL-NOR switch to NOR.

- l. If no other check is to be done, go to Shutdown.

8-2-3.12 Lamp Power Supply.

- a. Perform Setup, if not already done.

- b. Turn all LTS variable controls fully clockwise.

- c. Place and hold LTS - LAMP SUPPLY OVERLOAD switch to NO. 1.

RESULT:

- (1) All lamps on copilot's PDU shall be off, except RTR OVERSPEED lamps.
 - (2) Alternate lamps on CDU shall be off.
- If results (1) and (2) are not as specified, replace No. 1 SDC (PARA 8-4-15).
 - If result (1) is not as specified, replace copilot's PDU (PARA 8-4-16).
 - If result (2) is not as specified, replace CDU (PARA 8-4-10).

- d. Release LTS - LAMP SUPPLY OVERLOAD switch.

RESULT: All PDU (pilot's and copilot's) and CDU lamps shall go on.

- If result is not as specified, replace No. 1 SDC (PARA 8-4-15).

- e. Place and hold LTS - LAMP SUPPLY OVERLOAD switch to NO. 2.

RESULT:

- (1) All lamps on pilot's PDU shall be off, except RTR OVERSPEED lamps.
 - (2) Alternate lamps on CDU shall be off.
- If results (1) and (2) are not as specified, replace No. 2 SDC (PARA 8-4-15).
 - If result (1) is not as specified, replace pilot's PDU (PARA 8-4-16).
 - If result (2) is not as specified, replace CDU (PARA 8-4-10).

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- f. Release LTS - LAMP SUPPLY OVERLOAD switch.

RESULT: All PDU (pilot’s and copilot’s) and CDU lamps shall go on.

- If result is not as specified, replace No. 2 SDC (PARA 8-4-15).

- g. Press LTS ROTOR OVERSPD RESET switch to reset RTR OVERSPEED lights if lights remain lit.

- h. Position LTS switches as follows:

INPUT POWER	OFF
All CAL-NOR switches	NOR
All HI-LO switches	HI
All NOR controls	Fully counterclockwise

- i. Turn off electrical power (PARA 1-6-16).
- j. Connect J1 to No. 2 SDC and reconnect cable assemblies to No. 1 and No. 2 SDC.
- k. If no other check is to be done, go to Shutdown.

8-2-3.13 Faceplate Lighting.

NOTE

Make sure Shutdown has been performed before proceeding to step a.

- a. Turn on electrical power (PARA 1-6-16).
- b. Turn upper console INST LT NON FLT control to BRT.

RESULT: CDU faceplate lights shall go on bright.

- If result is not as specified, go to TABLE NO. 8-2-3-22.

- c. Turn upper console INST LT PILOT FLT control to BRT.

RESULT: PDU (pilot’s) faceplate lights shall go on bright.

- If result is not as specified, go to TABLE NO. 8-2-3-23.

- d. Turn upper console CPLT FLT INST LTS control to BRT.

RESULT: PDU (copilot’s) faceplate lights shall go on bright.

- If result is not as specified, go to TABLE NO. 8-2-3-24.

- e. Turn upper console INST LT NON FLT, INST LT PILOT FLT, and CPLT FLT INST LTS controls to comfortable viewing level.

RESULT:

- (1) TGT TEMP vertical displays and associated TGT digital readouts on CDU shall indicate approximate ambient temperature of helicopter (assuming cooldown has occurred).
- (2) ENG OIL TEMP vertical displays on CDU shall indicate approximate ambient temperature of helicopter (assuming cooldown has occurred).
- (3) XMSN TEMP vertical displays on CDU shall indicate approximate ambient temperature of helicopter (assuming cooldown has occurred).
- (4) FUEL QTY vertical displays on CDU shall indicate actual fuel on helicopter and associated MAIN FUEL or TOTAL FUEL digital readout shall indicate sum of two vertical scale readings.
- (5) LOW ROTOR RPM, #1 ENG OUT, and #2 ENG OUT capsules on pilot’s and

copilot's master warning panels shall go on (LOW ROTOR RPM capsule flashes, others are on steady).

- (6) #1 ENGINE OIL PRESS and #2 ENGINE OIL PRESS capsules on caution/advisory panel shall go on.

- If result (1) is not as specified, perform TGT TEMP check using line test set.

If trouble remains, troubleshoot No. 1 or No. 2 Engine Instruments And Warning Lights System, as required (PARA 4-2-2 or PARA 4-2-3).

- If result (2) is not as specified, perform ENG OIL TEMP check using line test set.

If trouble remains, troubleshoot No. 1 or No. 2 Engine Instruments And Warning Lights System, as required (PARA 4-2-2 or PARA 4-2-3).

- If result (3) is not as specified, perform XM-SN TEMP check using line test set.

- If result (4) is not as specified, perform FUEL QTY check using line test set.

If trouble remains, troubleshoot Fuel Quantity System (PARA 10-2-5 or PARA 10-2-6).

- If result (5) is not as specified, go to TABLE NO. 8-2-3-25.

- If result (6) is not as specified, go to TABLE NO. 8-2-3-26.

- f. If no other check is to be done, go to Shutdown.

8-2-3.14 Shutdown.

- Turn off electrical power (PARA 1-6-16).
- Disconnect LTS setup from helicopter.

TABLE NO. 8-2-3-1. NO. 1 ENGINE OUT And/Or NO. 2 ENGINE OUT Lights Are Off.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Are both lights off?	Step 1. If both lights are off, replace CDU (PARA 8-4-10). Go to 7. Step 2. If both lights are not off, go to 2.
2. Is NO. 1 ENGINE OUT light off?	Step 1. If light is off, go to 3. Step 2. If light is on, go to 5.
3. Check if Ng 1 fuse is good.	Step 1. If fuse is good, go to 4. Step 2. If fuse is not good, replace fuse (PARA 8-4-13). Go to 7.
4. Check continuity between P151-31 and P154-31.	

TABLE NO. 8-2-3-1. NO. 1 ENGINE OUT And/Or NO. 2 ENGINE OUT Lights Are Off. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
Step 1. If continuity is present, replace No. 1 SDC (PARA 8-4-15). Go to 7. Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 7. Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 7.
5. Check if Ng 2 fuse is good.
Step 1. If fuse is good, go to 6. Step 2. If fuse is not good, replace fuse (PARA 8-4-13). Go to 7.
6. Check continuity between P152-31 and P155-31.
Step 1. If continuity is present, replace No. 2 SDC (PARA 8-4-15). Go to 7. Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 7. Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 7.
7. Procedure completed.

TABLE NO. 8-2-3-2. NO. 1 LOW OIL PRESS And/Or NO. 2 LOW OIL PRESS Lights Are Off.

TEST OR INSPECTION CORRECTIVE ACTION
1. Are both lights off?
Step 1. If both lights are off, replace CDU (PARA 8-4-10). Go to 7. Step 2. If both lights are not off, go to 2.
2. Check if No. 1 LOW OIL PRESS light is off.
Step 1. If light is off, go to 3. Step 2. If light is on, go to 5.
3. Check if NO. 1 OIL PRESS fuse is good.
Step 1. If fuse is good, go to 4. Step 2. If fuse is not good, replace fuse (PARA 8-4-13). Go to 7.
4. Check continuity between P151-30 and P154-30.

TABLE NO. 8-2-3-2. NO. 1 LOW OIL PRESS And/Or NO. 2 LOW OIL PRESS Lights Are Off. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, replace No. 1 SDC (PARA 8-4-15). Go to 7.
 Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 7.
 Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 7.

5. Check if No. 2 OIL PRESS fuse is good.

Step 1. If fuse is good, go to 6.
 Step 2. If fuse is not good, replace fuse (PARA 8-4-13). Go to 7.

6. Check continuity between P152-30 and P155-30.

Step 1. If continuity is present, replace No. 2 SDC (PARA 8-4-15). Go to 7.
 Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 7.
 Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 7.

7. Procedure completed.

TABLE NO. 8-2-3-3. CHAN 1 Light Is On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Replace No. 1 SDC (PARA 8-4-15). Go to 2.

Step 1. If trouble remains, replace CDU (PARA 8-4-10). Go to 2.
 Step 2. If trouble still remains, replace PDU (PARA 8-4-16). Go to 2.

2. Procedure completed.

TABLE NO. 8-2-3-4. CHAN 2 Light Is On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Replace No. 2 SDC (PARA 8-4-15). Go to 2.

TABLE NO. 8-2-3-4. CHAN 2 Light Is On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If trouble remains, replace CDU (PARA 8-4-10). Go to 2. Step 2. If trouble still remains, replace PDU (PARA 8-4-16). Go to 2. 2. Procedure completed.

TABLE NO. 8-2-3-5. With CDU PUSH TO TEST or LAMP TEST Switch Pressed, Two Or More Adjacent Lamps On One Scale Are Off, Or One Lamp Between Green And Amber, Amber And Red, Or Green And Red Ranges On A Scale Is Off.

TEST OR INSPECTION	CORRECTIVE ACTION
	NOTE: These modules are interchangeable: Transmission Oil Temp (A5) No. 1 Engine Oil Temp (A7) No. 2 Engine Oil Temp (A8) No. 1 TGT (A11) No. 2 TGT (A12) No. 1 Ng Speed (A13) No. 2 Ng Speed (A14) No. 1 Fuel Quantity (A3) No. 2 Fuel Quantity (A4) No. 1 Engine Oil Press (A9) No. 2 Engine Oil Press (A10) 1. Interchange modules in accordance with above. Does problem follow module? Step 1. If problem follows module, go to 2. Step 2. If problem does not follow module, replace CDU (PARA 8-4-10). Go to 5. 2. Check helicopter configuration. Step 1. EMEP , go to 3. Step 2. W/O EMEP , go to 5. EMEP 3. Replace faulty Bar Graph Filament (BGF) lamp (PARA 8-4-12). Go to 4.

TABLE NO. 8-2-3-5. With CDU PUSH TO TEST or LAMP TEST Switch Pressed, Two Or More Adjacent Lamps On One Scale Are Off, Or One Lamp Between Green And Amber, Amber And Red, Or Green And Red Ranges On A Scale Is Off. (Cont)

	TEST OR INSPECTION CORRECTIVE ACTION
	4. If trouble remains, replace CDU (PARA 8-4-10). Go to 7.
W/O EMEP	5. Replace display lamp assembly (PARA 8-4-14). Go to 6.
	6. If trouble remains, replace CDU (PARA 8-4-10). Go to 7.
	7. Procedure completed.

TABLE NO. 8-2-3-6. With CDU PUSH TO TEST or LAMP TEST Switch Pressed, One Or More RTR OVERSPEED Lights Are Off.

	TEST OR INSPECTION CORRECTIVE ACTION
	1. Is same light off on both PDUs?
	Step 1. If the same light is off, replace CDU (PARA 8-4-10). Go to 4.
	Step 2. If different lights are off, go to 2.
	2. Interchange PDUs. Go to 3.
	3. Is trouble on same PDU?
	Step 1. If trouble is on same display unit, replace malfunctioning CDU or PDU (PARA 8-4-10 or PARA 8-4-16). Go to 4.
	Step 2. If trouble is on different display unit, repair/replace wiring, as required, between the following (Appendix F), then go to 4:
	P153-5 and P157-31
	P153-6 and P157-32
	P153-7 and P157-33
	P156-5 and P158-31
	P156-6 and P158-32
	P156-7 and P158-33
	4. Procedure completed.

TABLE NO. 8-2-3-7. CDU DIM Control Varies Alternate CDU Display Lamps Only.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check that CDU DIM control varies CDU display lamps.	Step 1. If CDU display lamps vary, go to 2. Step 2. If CDU display lamps do not vary, go to 3.
2. Check continuity between P151-19 and P154-19.	Step 1. If continuity is present, replace No 1. SDC (PARA 8-4-15). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
3. Check continuity between P152-19 and P155-19.	Step 1. If continuity is present, replace No. 2 SDC (PARA 8-4-15). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
4. Procedure completed.	

TABLE NO. 8-2-3-8. Pilot's Or Copilot's PDU % RPM R Display Is Inoperative.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Interchange PDUs. Go to 2.	
2. Is trouble on same PDU?	Step 1. If trouble is on same PDU, replace malfunctioning PDU (PARA 8-4-16). Go to 4. Step 2. If trouble is on different PDU, go to 3.
3. Check continuity between:	
P149-18 and P157-18 (copilot's PDU)	
P150-18 and P158-18 (pilot's PDU)	
	Step 1. If continuity is present, replace No. 1 SDC or No. 2 SDC as required (PARA 8-4-15). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.

TABLE NO. 8-2-3-8. Pilot's Or Copilot's PDU % RPM R Display Is Inoperative. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

4. Procedure completed.

TABLE NO. 8-2-3-9. One Or More RTR OVERSPEED Lights Are Off.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Is same light off on pilot's and copilot's PDU?

Step 1. If the same light is off, replace CDU (PARA 8-4-10). Go to **4**.

Step 2. If different light is off, go to **2**.

2. Interchange PDUs. Go to 3.

3. Is trouble on same PDU?

Step 1. If trouble is still on same display unit, replace malfunctioning PDU (PARA 8-4-16). Go to **4**.

Step 2. If trouble is on different display unit, repair/replace wiring, as required, between the following (Appendix F), then go to **4**:

P153-5 and P157-31

P153-6 and P157-32

P153-7 and P157-33

P156-5 and P158-31

P156-6 and P158-32

P156-7 and P158-33

4. Procedure completed.

TABLE NO. 8-2-3-10. FUEL QTY 1, FUEL QTY 2, And/Or MAIN FUEL or TOTAL FUEL Display Inoperative.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Are all three displays inoperative?

TABLE NO. 8-2-3-10. FUEL QTY 1, FUEL QTY 2, And/Or MAIN FUEL or TOTAL FUEL Display Inoperative. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
	Step 1. If all three displays are inoperative, replace CDU (PARA 8-4-10). Go to 7. Step 2. If only one or two displays are inoperative, go to 2.
2. Is FUEL QTY 1 inoperative?	Step 1. If FUEL QTY 1 is inoperative, go to 3. Step 2. If FUEL QTY 1 is operative, go to 4.
3. Check continuity between P151-41 and P154-41.	Step 1. If continuity is present, replace No. 1 SDC (PARA 8-4-15). Go to 7. Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 7. Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 7.
4. Is FUEL QTY 2 inoperative?	Step 1. If FUEL QTY 2 is inoperative, go to 5. Step 2. If FUEL QTY 2 is operative, go to 6.
5. Check continuity between P152-41 and P155-41.	Step 1. If continuity is present, replace No. 2 SDC (PARA 8-4-15). Go to 7. Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 7. Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 7.
6. Check continuity between P151-43 and P154-43.	Step 1. If continuity is present, replace No. 1 SDC (PARA 8-4-15). Go to 7. Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 7. Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 7.
7. Procedure completed.	

TABLE NO. 8-2-3-11. ENG OIL PRESS 1 And/Or ENG OIL PRESS 2 Display Inoperative.

**TEST OR INSPECTION
CORRECTIVE ACTION**

1. Are both displays inoperative?

Step 1. If both displays are inoperative, replace CDU (PARA 8-4-10). Go to 5.

Step 2. If only one display is inoperative, go to 2.

2. Is ENG OIL PRESS 1 inoperative?

Step 1. If ENG OIL PRESS 1 is inoperative, go to 3.

Step 2. If ENG OIL PRESS 1 is operative, go to 4.

3. Check continuity between P151-45 and P154-45.

Step 1. If continuity is present, replace No. 1 SDC (PARA 8-4-15). Go to 5.

Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 5.

Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 5.

4. Check continuity between P152-45 and P155-45.

Step 1. If continuity is present, replace No. 2 SDC (PARA 8-4-15). Go to 5.

Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 5.

Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 5.

5. Procedure completed.

TABLE NO. 8-2-3-12. ENG OIL TEMP 1 And/Or ENG OIL TEMP 2 Display Inoperative.

**TEST OR INSPECTION
CORRECTIVE ACTION**

1. Are both displays inoperative?

Step 1. If both displays are inoperative, replace CDU (PARA 8-4-10). Go to 5.

Step 2. If only one display is inoperative, go to 2.

2. Is ENG OIL TEMP 1 inoperative?

Step 1. If ENG OIL TEMP 1 is inoperative, go to 3.

Step 2. If ENG OIL TEMP 1 is operative, go to 4.

TABLE NO. 8-2-3-12. ENG OIL TEMP 1 And/Or ENG OIL TEMP 2 Display Inoperative. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
3. Check continuity between P151-44 and P154-44.	Step 1. If continuity is present, replace No. 1 SDC (PARA 8-4-15). Go to 5. Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 5. Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 5.
4. Check continuity between P152-44 and P155-44.	Step 1. If continuity is present, replace No. 2 SDC (PARA 8-4-15). Go to 5. Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 5. Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 5.
5. Procedure completed.	

TABLE NO. 8-2-3-13. NO. 1 HIGH OIL TEMP And/Or NO. 2 HIGH OIL TEMP Light Is Off.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Are both lights off?	Step 1. If both lights are off, replace CDU (PARA 8-4-10). Go to 7. Step 2. If only one light is off, go to 2.
2. Is NO. 1 HIGH OIL TEMP light off?	Step 1. If NO. 1 HIGH OIL TEMP light is off, go to 3. Step 2. If NO. 1 HIGH OIL TEMP light is on, go to 5.
3. Check if OIL TEMP 1 fuse is good.	Step 1. If fuse is good, go to 4. Step 2. If fuse is not good, replace fuse (PARA 8-4-13). Go to 7.
4. Check continuity between P151-32 and P154-32.	Step 1. If continuity is present, replace No. 1 SDC (PARA 8-4-15). Go to 7.

TABLE NO. 8-2-3-13. NO. 1 HIGH OIL TEMP And/Or NO. 2 HIGH OIL TEMP Light Is Off. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
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Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 7.
 Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 7.

5. Check if OIL TEMP 2 fuse is good.

Step 1. If fuse is good, go to 6.
 Step 2. If fuse is not good, replace fuse (PARA 8-4-13). Go to 7.

6. Check continuity between P152-32 and P155-32.

Step 1. If continuity is present, replace No. 2 SDC (PARA 8-4-15). Go to 7.
 Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 7.
 Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 7.

7. Procedure completed.

TABLE NO. 8-2-3-14. Ng SPEED 1 And/Or Ng SPEED 2 Vertical Display Inoperative.

TEST OR INSPECTION	CORRECTIVE ACTION
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1. Are both displays inoperative?

Step 1. If both displays are inoperative, replace CDU (PARA 8-4-10). Go to 5.
 Step 2. If only one display is inoperative, go to 2.

2. Is Ng SPEED 1 vertical display inoperative?

Step 1. If Ng SPEED 1 vertical display is inoperative, go to 3.
 Step 2. If Ng SPEED 1 vertical display is operative, go to 4.

3. Check continuity between P151-47 and P154-47.

Step 1. If continuity is present, replace No. 1 SDC (PARA 8-4-15). Go to 5.
 Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 5.
 Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 5.

4. Check continuity between P152-47 and P155-47.

TABLE NO. 8-2-3-14. Ng SPEED 1 And/Or Ng SPEED 2 Vertical Display Inoperative. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If continuity is present, replace No. 2 SDC (PARA 8-4-15). Go to 5. Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 5. Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 5.
5. Procedure completed.	

TABLE NO. 8-2-3-15. Ng SPEED 1 And/Or Ng SPEED 2 Digital Display Inoperative.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Are both digital displays inoperative?	Step 1. If both displays are inoperative, replace CDU (PARA 8-4-10). Go to 5. Step 2. If only one display is inoperative, go to 2.
2. Is Ng SPEED 1 digital display inoperative?	Step 1. If Ng SPEED 1 display is inoperative, go to 3. Step 2. If Ng SPEED 1 display is operative, go to 4.
3. Check continuity between P151-49 and P154-49.	Step 1. If continuity is present, replace No. 1 SDC (PARA 8-4-15). Go to 5. Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 5. Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 5.
4. Check continuity between P152-49 and P155-49.	Step 1. If continuity is present, replace No. 2 SDC (PARA 8-4-15). Go to 5. Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 5. Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 5.
5. Procedure completed.	

TABLE NO. 8-2-3-16. % TRQ 1 And/Or % TRQ 2 Vertical Display Or % TRQ Digital Readout Inoperative.

TEST OR INSPECTION	CORRECTIVE ACTION
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1. Are both vertical displays inoperative?

Step 1. If both displays are inoperative, replace PDU (PARA 8-4-16). Go to **6**.

Step 2. If only one display is inoperative, go to **2**.

2. Is CHAN 1 light on?

Step 1. If CHAN 1 light is on, replace No. 1 SDC (PARA 8-4-15). Go to **6**.

Step 2. If CHAN 1 light is off, go to **3**.

3. Is CHAN 2 light on?

Step 1. If CHAN 2 light is on, replace No. 2 SDC (PARA 8-4-15). Go to **6**.

Step 2. If CHAN 2 light is off, go to **4**.

4. Interchange PDUs. Go to 5.

5. Is trouble on same PDU?

Step 1. If trouble is on same PDU, replace PDU (PARA 8-4-16). Go to **6**.

Step 2. If trouble is on different PDU, repair/replace wiring, as required, between the following (Appendix F), then go to **6**:

P149-20 and P157-20

P149-21 and P157-21

P149-22 and P157-22

P149-23 and P157-23

P150-20 and P158-20

P150-21 and P158-21

P150-22 and P158-22

P150-23 and P158-23

6. Procedure completed.

TABLE NO. 8-2-3-17. XMSN PRESS Vertical Scale Display Inoperative.

TEST OR INSPECTION CORRECTIVE ACTION
1. Check continuity between P152-28 and P155-28. Step 1. If continuity is present, replace No. 2 SDC (PARA 8-4-15). Go to 2. Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 2. Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 2. 2. Procedure completed.

TABLE NO. 8-2-3-18. XMSN TEMP Vertical Scale Display Inoperative.

TEST OR INSPECTION CORRECTIVE ACTION
1. Check continuity between: P152-22 and P155-22 P152-23 and P155-23 Step 1. If continuity is present, replace No. 2 SDC (PARA 8-4-15). Go to 2. Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 2. Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 2. 2. Procedure completed.

TABLE NO. 8-2-3-19. PDU's % RPM 1 Or % RPM 2 Vertical Display Inoperative.

TEST OR INSPECTION CORRECTIVE ACTION
1. Is CHAN 1 light on? Step 1. If CHAN 1 light is on, replace No. 1 SDC (PARA 8-4-15). Go to 5. Step 2. If CHAN 1 light is not on, go to 2. 2. Is CHAN 2 light on? Step 1. If CHAN 2 light is on, replace No. 2 SDC (PARA 8-4-15). Go to 5.

TABLE NO. 8-2-3-19. PDU's % RPM 1 Or % RPM 2 Vertical Display Inoperative. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If CHAN 2 light is off, go to 3.

3. Interchange PDUs. Go to 4.

4. Is trouble on same PDU?

Step 1. If trouble is on same PDU, replace malfunctioning PDU (PARA 8-4-16).
Go to 5.

Step 2. If trouble is on different PDU, repair/replace wiring, as required, between the following (Appendix F), then go to 5.

P149-17 and P157-17

P149-19 and P157-19

P150-17 and P158-17

P150-19 and P158-19

5. Procedure completed.

TABLE NO. 8-2-3-20. TGT TEMP 1 Or TGT TEMP 2 Vertical Display Inoperative.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Is CHAN 1 light on?

Step 1. If CHAN 1 light is on, replace No. 1 SDC (PARA 8-4-15). Go to 4.

Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 4.

Step 3. If CHAN 1 light is not on, go to 2.

2. Is CHAN 2 light on?

Step 1. If CHAN 2 light is on, replace No. 2 SDC (PARA 8-4-15). Go to 4.

Step 2. If CHAN 2 light is off, go to 3.

3. Check continuity between:

P151-46 and P154-46

P152-46 and P155-46

Step 1. If continuity is present, replace CDU (PARA 8-4-10). Go to 4.

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.

TABLE NO. 8-2-3-20. TGT TEMP 1 Or TGT TEMP 2 Vertical Display Inoperative. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
4. Procedure completed.	

TABLE NO. 8-2-3-21. TGT TEMP 1 And/Or TGT TEMP 2 Digital Display Inoperative.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Are both digital displays inoperative?	Step 1. If both displays are inoperative, replace CDU (PARA 8-4-10). Go to 7. Step 2. If only one display is inoperative, go to 2.
2. Is CHAN 1 light on?	Step 1. If CHAN 1 light is on, replace No. 1 or No. 2 SDC as applicable (PARA 8-4-15). Go to 7. Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 7. Step 3. If CHAN 1 light is off, go to 3.
3. Is CHAN 2 light on?	Step 1. If CHAN 2 light is on, replace No. 2 SDC (PARA 8-4-15). Go to 7. Step 2. If trouble remains, replace CDU (PARA 8-4-10). Go to 7. Step 3. If CHAN 2 light is off, go to 4.
4. Is TGT TEMP 1 digital display inoperative?	Step 1. If display is inoperative, go to 5. Step 2. If display is operative, go to 6.
5. Check continuity between P151-48 and P154-48.	Step 1. If continuity is present, replace CDU (PARA 8-4-10). Go to 7. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 7.
6. Check continuity between P152-48 and P155-48.	Step 1. If continuity is present, replace CDU (PARA 8-4-10). Go to 7.

TABLE NO. 8-2-3-21. TGT TEMP 1 And/Or TGT TEMP 2 Digital Display Inoperative. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 7.
	7. Procedure completed.

TABLE NO. 8-2-3-22. CDU Faceplate Lighting Inoperative.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Turn upper console INST LT NON FLT control to BRT. Go to 2.
	2. Remove CDU faceplate and check for 115 VAC at faceplate lighting connector.
	Step 1. If voltage is as specified, replace CDU faceplate (PARA 8-4-11). Go to 9.
	Step 2. If voltage is not as specified, go to 3.
	3. Check for 115 VAC between P154-128 and gnd.
	Step 1. If voltage is as specified, replace CDU (PARA 8-4-10). Go to 9.
	Step 2. If voltage is not as specified, go to 4.
	4. Check continuity between P151-128 and P154-128.
	Step 1. If continuity is present, go to 5.
	Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 9.
	5. Check helicopter configuration.
	Step 1. EMEP, go to 6.
	Step 2. W/O EMEP, go to 8.
EMEP	6. Remove and check pin filter adapter (PFA) P138 (PARA 9-4-98).
	Step 1. If PFA is good, go to 7.
	Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 9.
EMEP	7. Install PFA (PARA 9-4-98). Go to 8.

TABLE NO. 8-2-3-22. CDU Faceplate Lighting Inoperative. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
8. Check for 5 VAC between P138-GG and gnd.	Step 1. If voltage is as specified, replace No. 1 SDC (PARA 8-4-15). Go to 9. Step 2. If voltage is not as specified, repair/replace wiring between P280-L and P138-GG (Appendix F). Go to 9. Step 3. If trouble remains, troubleshoot instrument panel lights (PARA 9-2-4). Go to 9.
9. Procedure completed.	

TABLE NO. 8-2-3-23. Pilot's PDU Faceplate Lighting Inoperative.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Interchange pilot's PDU and copilot's PDU and check that faceplate on pilot's PDU lights.	Step 1. If faceplate lights, replace malfunctioning PDU faceplate (PARA 8-4-17). Go to 10. Step 2. If faceplate does not light, go to 2.
2. Check for 115 VAC between P158-99 and P158-100.	Step 1. If voltage is as specified, replace PDU (PARA 8-4-16). Go to 10. Step 2. If voltage is not as specified, go to 3.
3. Check continuity between: P150-99 and P158-99 P150-100 and P158-100	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 10.
4. Check for 5 VAC between P386-A and gnd.	Step 1. If voltage is as specified, go to 5. Step 2. If voltage is not as specified, repair/replace wiring between P246-a and P386-A (Appendix F). Go to 10.

TABLE NO. 8-2-3-23. Pilot's PDU Faceplate Lighting Inoperative. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 3. If trouble remains, troubleshoot instrument panel lights (PARA 9-2-4). Go to 10. 5. Check helicopter configuration. Step 1. EMEP , go to 6. Step 2. W/O EMEP , go to 8. 6. Remove and check PFA P137 (PARA 9-4-98). Step 1. If PFA is good, go to 7. Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 10. 7. Install PFA (PARA 9-4-98). Go to 8. 8. Check for 115 VAC between P137-FF and gnd. Step 1. If voltage is as specified, replace No. 2 SDC (PARA 8-4-15). Go to 10. Step 2. If voltage is not as specified, go to 9. 9. Check continuity between P137-FF and P386-C. Step 1. If continuity is present, replace pilot's flight instruments 5v/115v transformer (PARA 9-4-55). Go to 10. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 10. 10. Procedure completed.

TABLE NO. 8-2-3-24. Copilot's PDU Faceplate Lighting Inoperative.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Interchange copilot's and pilot's PDUs and check that faceplate on copilot's PDU lights. Step 1. If faceplate lights, replace malfunctioning PDU faceplate (PARA 8-4-17). Go to 10. Step 2. If faceplate does not light, go to 2.

TABLE NO. 8-2-3-24. Copilot's PDU Faceplate Lighting Inoperative. (Cont)

	TEST OR INSPECTION CORRECTIVE ACTION	
	2. Check for 115 VAC between P157-99 and P157-100.	
	Step 1. If voltage is as specified, replace PDU (PARA 8-4-16). Go to 10. Step 2. If voltage is not as specified, go to 3.	
	3. Check continuity between: P149-99 and P157-99 P149-100 and P157-100	
	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 10.	
	4. Check helicopter configuration.	
	Step 1. EMEP , go to 5. Step 2. W/O EMEP , go to 7.	
EMEP	5. Remove and check PFA P138 (PARA 9-4-98).	
	Step 1. If PFA is good, go to 6. Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 10.	
EMEP	6. Install PFA (PARA 9-4-98). Go to 7.	
	7. Check for 115 VAC between P138-FF and gnd.	
	Step 1. If voltage is as specified, replace No. 1 SDC (PARA 8-4-15). Go to 10. Step 2. If voltage is not as specified, go to 8.	
	8. Check for 5 VAC between P387-A and gnd.	
	Step 1. If voltage is as specified, go to 9. Step 2. If voltage is not as specified, repair/replace wiring between P246-g and P387-A (Appendix F). Go to 10.	
	9. Check continuity between P138-FF and P387-C.	
	Step 1. If continuity is present, replace copilot's flight instrument 5v/115v transformer (PARA 9-4-55). Go to 10. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 10.	

TABLE NO. 8-2-3-24. Copilot's PDU Faceplate Lighting Inoperative. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
10. Procedure completed.	

TABLE NO. 8-2-3-25. Master Warning Panel LOW ROTOR RPM, #1 ENG OUT, Or #2 ENG OUT Capsule Is Off.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check helicopter configuration.	Step 1. EMEP , go to 2. Step 2. W/O EMEP , go to 4.
2. Remove and check PFAs P137 and P138 (PARA 9-4-98).	Step 1. If PFAs are good, go to 3. Step 2. If PFAs are not good, replace PFAs as required (PARA 9-4-98). Go to 11.
3. Install PFAs (PARA 9-4-98). Go to 4.	
4. Disconnect P130, P131, P137, P138, P242, and P902. Go to 5.	
5. Check continuity between:	P130-3 and P138-<u>n</u> P130-3 and P242-<u>a</u> P130-3 and P902-<u>u</u> P130-9 and P137-<u>u</u> P130-9 and P138-<u>u</u> P130-9 and P243-<u>h</u> P130-10 and P137-<u>n</u> P130-10 and P243-<u>a</u> P130-10 and P243-J Step 1. If continuity is present, go to 6. Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 11.
6. Check continuity between:	

TABLE NO. 8-2-3-25. Master Warning Panel LOW ROTOR RPM, #1 ENG OUT, Or #2 ENG OUT Capsule Is Off. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
P130-3 and P131-3 P130-9 and P131-9 P130-10 and P131-10
Step 1. If continuity is present, go to 7. Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 11.
7. Connect P130, P131, P137, P138, P242, and P902 to their respective units. Go to 8.
8. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 9.
9. Check that all master warning panel capsules go on.
Step 1. If capsules go on, repair/replace wiring, as required, between the following (Appendix F), then go to 11: P130-3 and P138-<u>n</u> P130-9 and P137-<u>u</u> P130-9 and P138-<u>u</u> P130-10 and P137-<u>n</u> P131-3 and P138-<u>n</u> P131-9 and P137-<u>u</u> P131-9 and P138-<u>u</u> P131-10 and P137-<u>n</u>
Step 2. If capsules do not go on, go to 10.
10. Check capsule lamps.
Step 1. If lamps are good, replace master warning panel (PARA 8-4-24). Go to 11. Step 2. If lamps are not good, replace lamps as required (PARA 8-4-25). Go to 11.
11. Procedure completed.

TABLE NO. 8-2-3-26. Caution/Advisory Panel #1 ENGINE OIL PRESS Or #2 ENGINE OIL PRESS Capsules Do Not Go On.

**TEST OR INSPECTION
CORRECTIVE ACTION**

1. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 2.

2. Check that capsules go on.

Step 1. If capsules go on, go to 3.

Step 2. If capsules do not go on, go to 7.

3. Check helicopter configuration.

Step 1. **EMEP** , go to 4.

Step 2. **W/O EMEP** , go to 6.

4. Remove and check PFAs P117, P137, and P138 (PARA 9-4-98).

Step 1. If PFAs are good, go to 5.

Step 2. If PFAs are not good, replace PFAs (PARA 9-4-98). Go to 8.

5. Install PFAs (PARA 9-4-98). Go to 6.

6. Check continuity between:

P137-X and P117-51

P138-X and P117-3

Step 1. If continuity is present, system is operational. Go to 8.

Step 2. If continuity is not present, repair/replace wiring as required (Appendix F).
Go to 8.

7. Check capsule for burnt out lamps.

Step 1. If lamps are burnt out, replace lamps (PARA 8-4-20). Go to 8.

Step 2. If lamps are not burnt out, replace caution/advisory panel (PARA 8-4-19).
Go to 8.

8. Procedure completed.

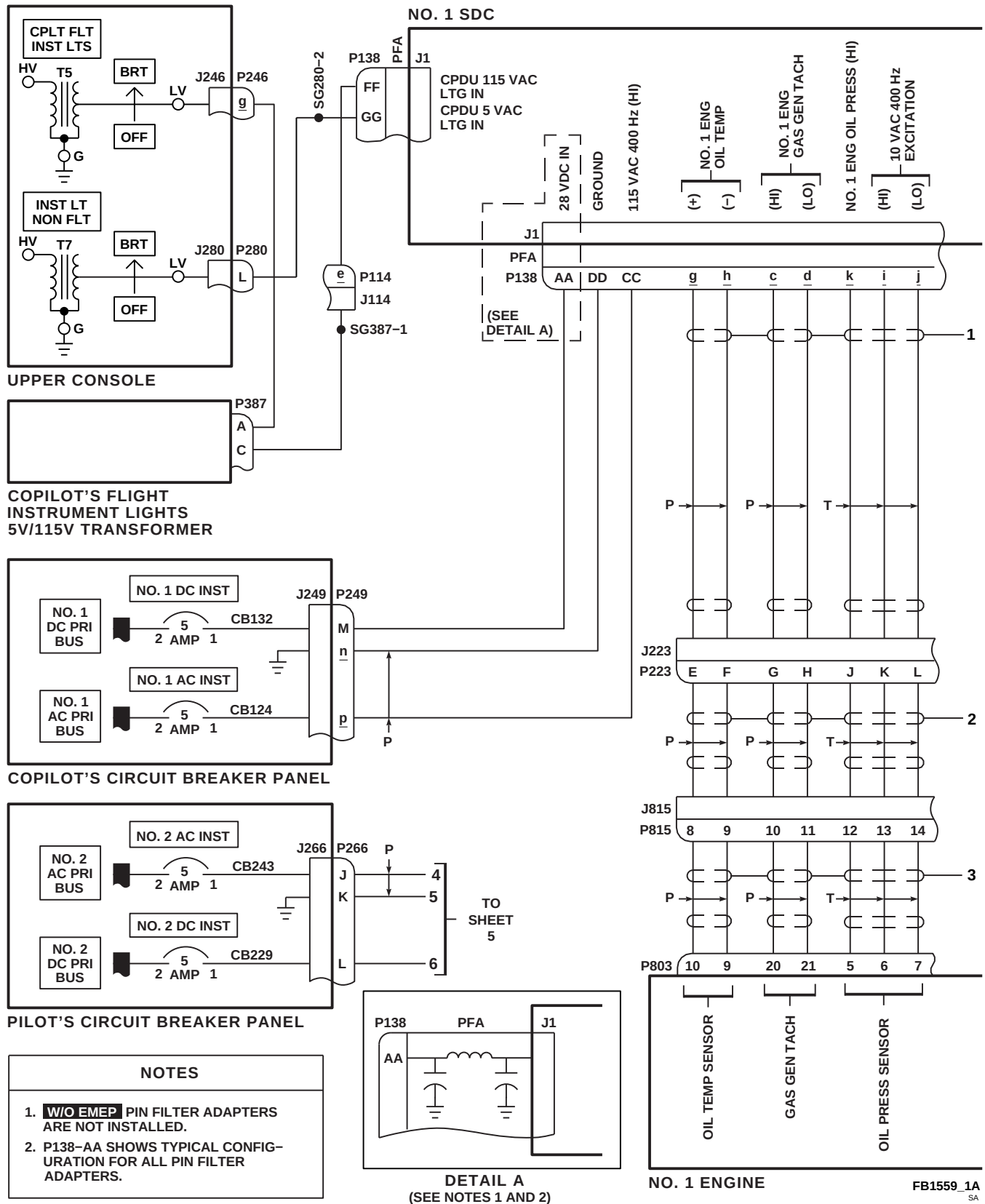
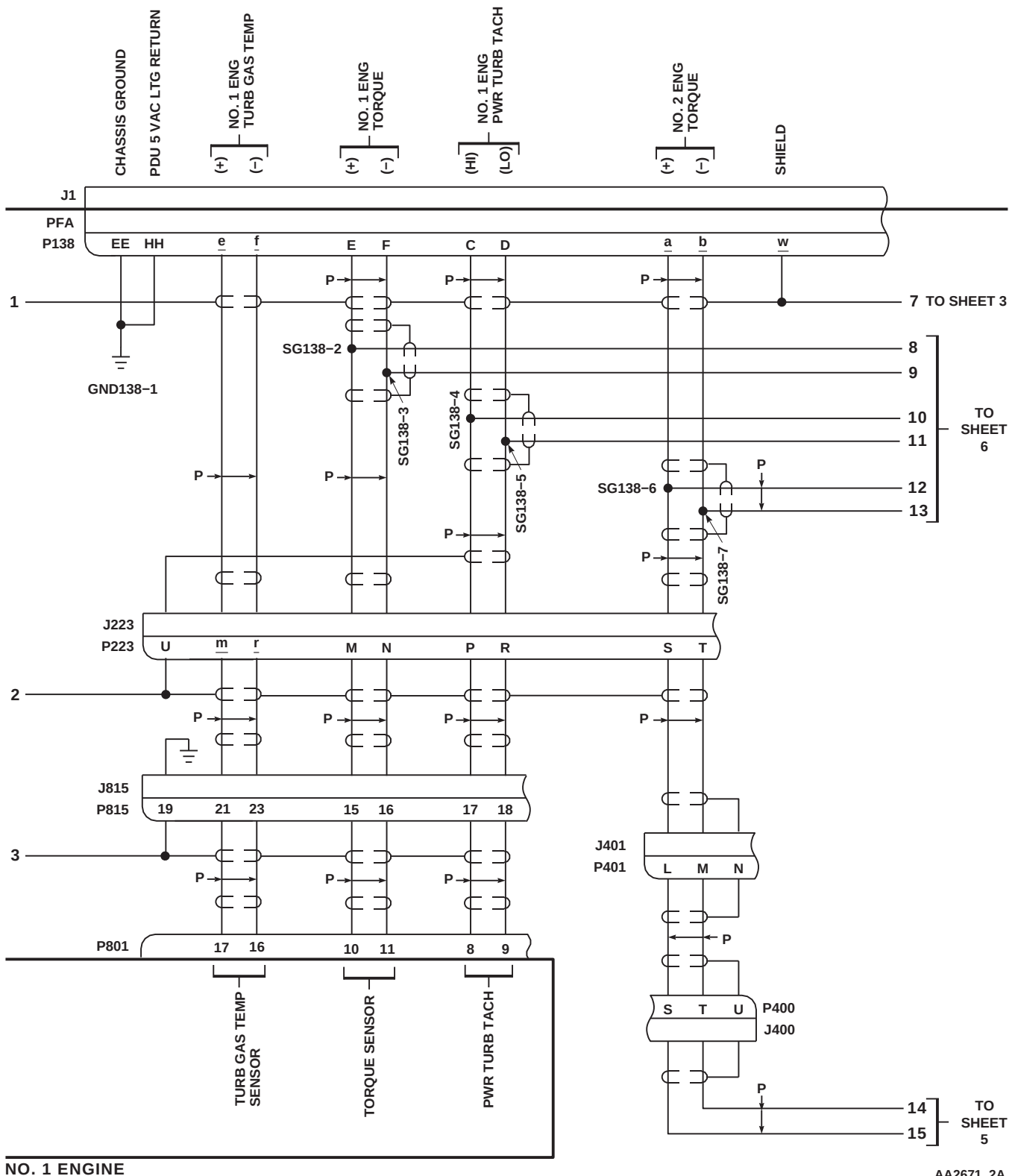


FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 21)

NO. 1 SDC



AA2671_2A
SA

FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 21)

NO. 1 SDC

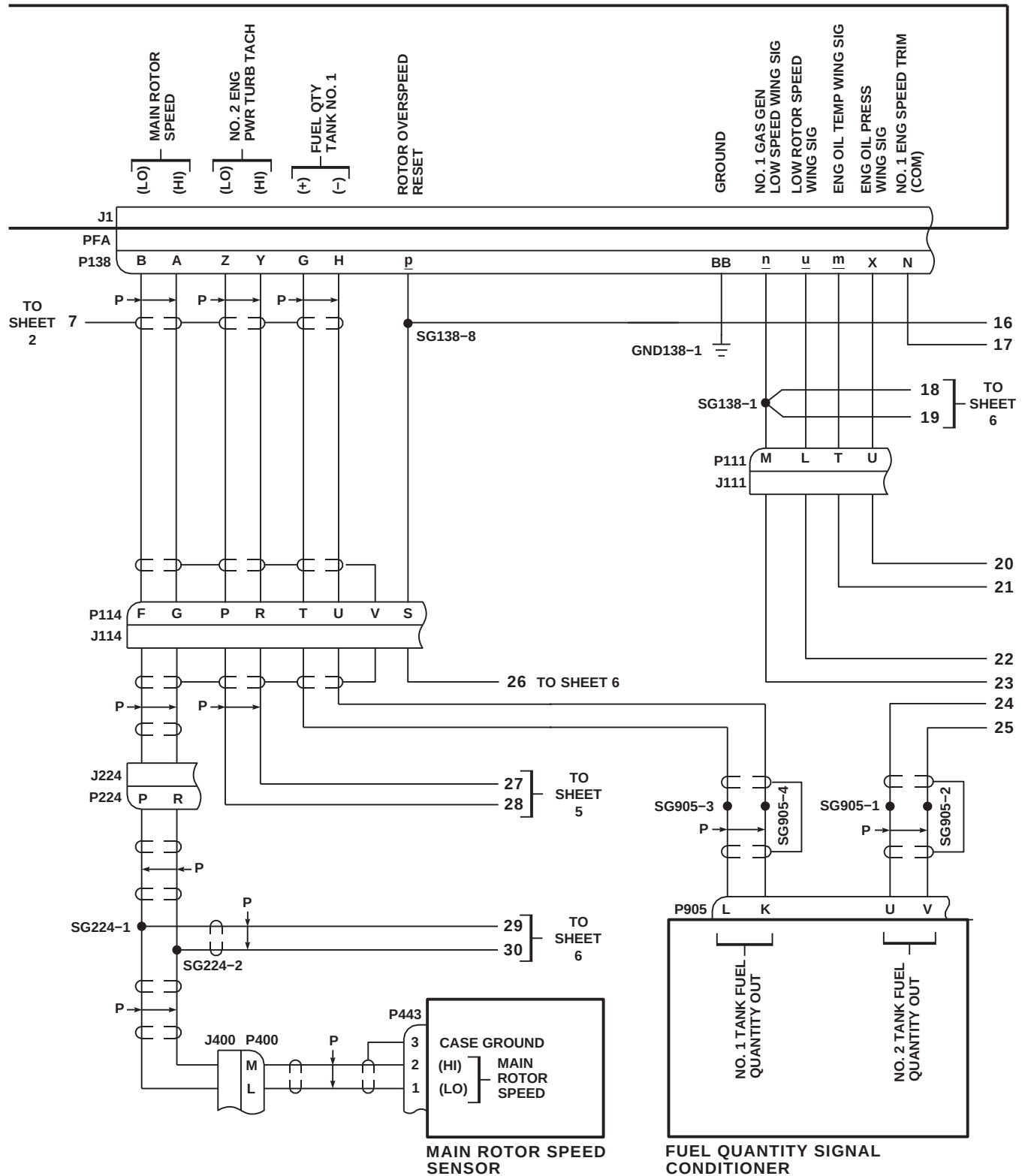
AA2671_3A
SA

FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 21)

Change 8

8-2-3-45

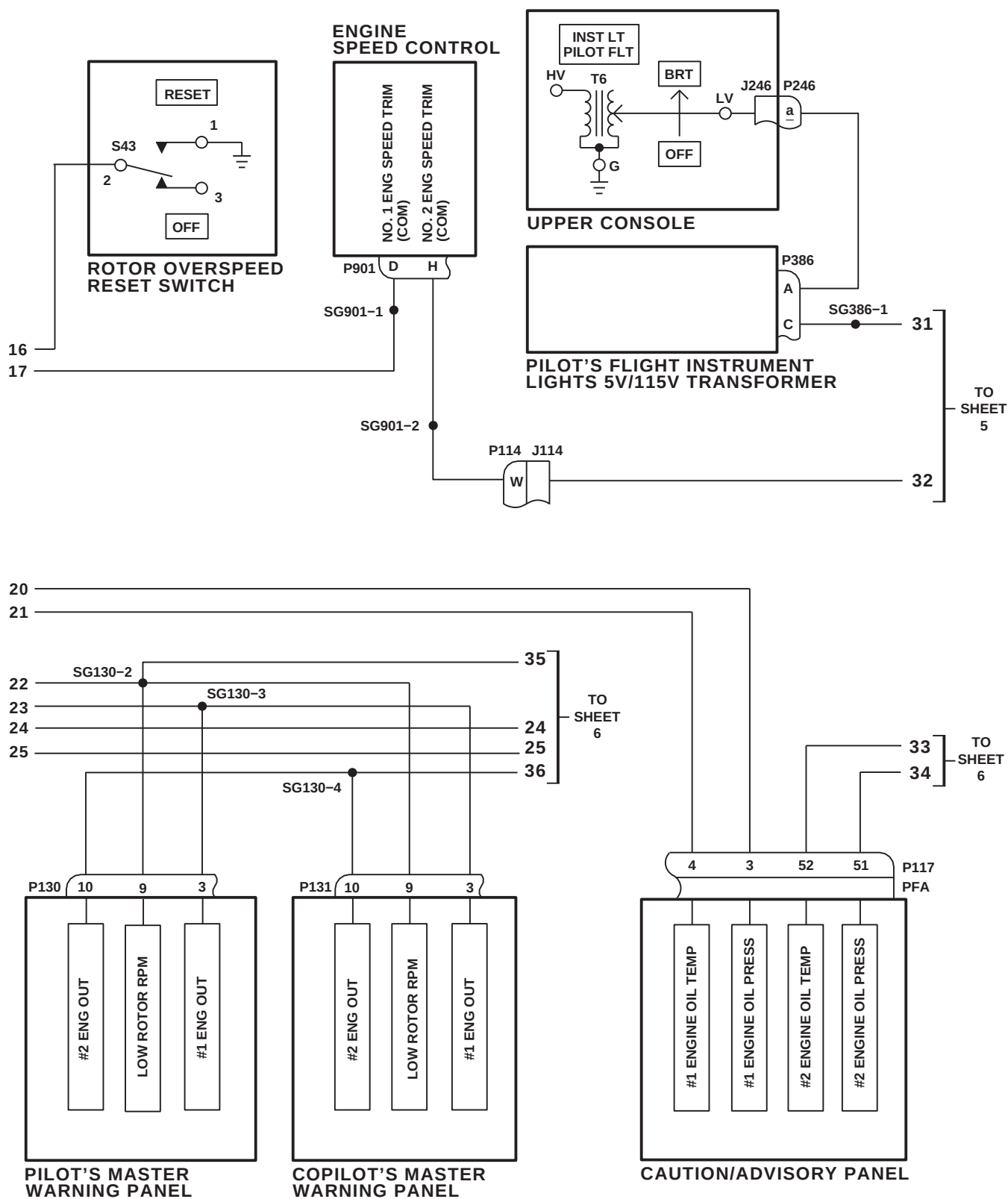
FK1636_2B
SA

FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 21)

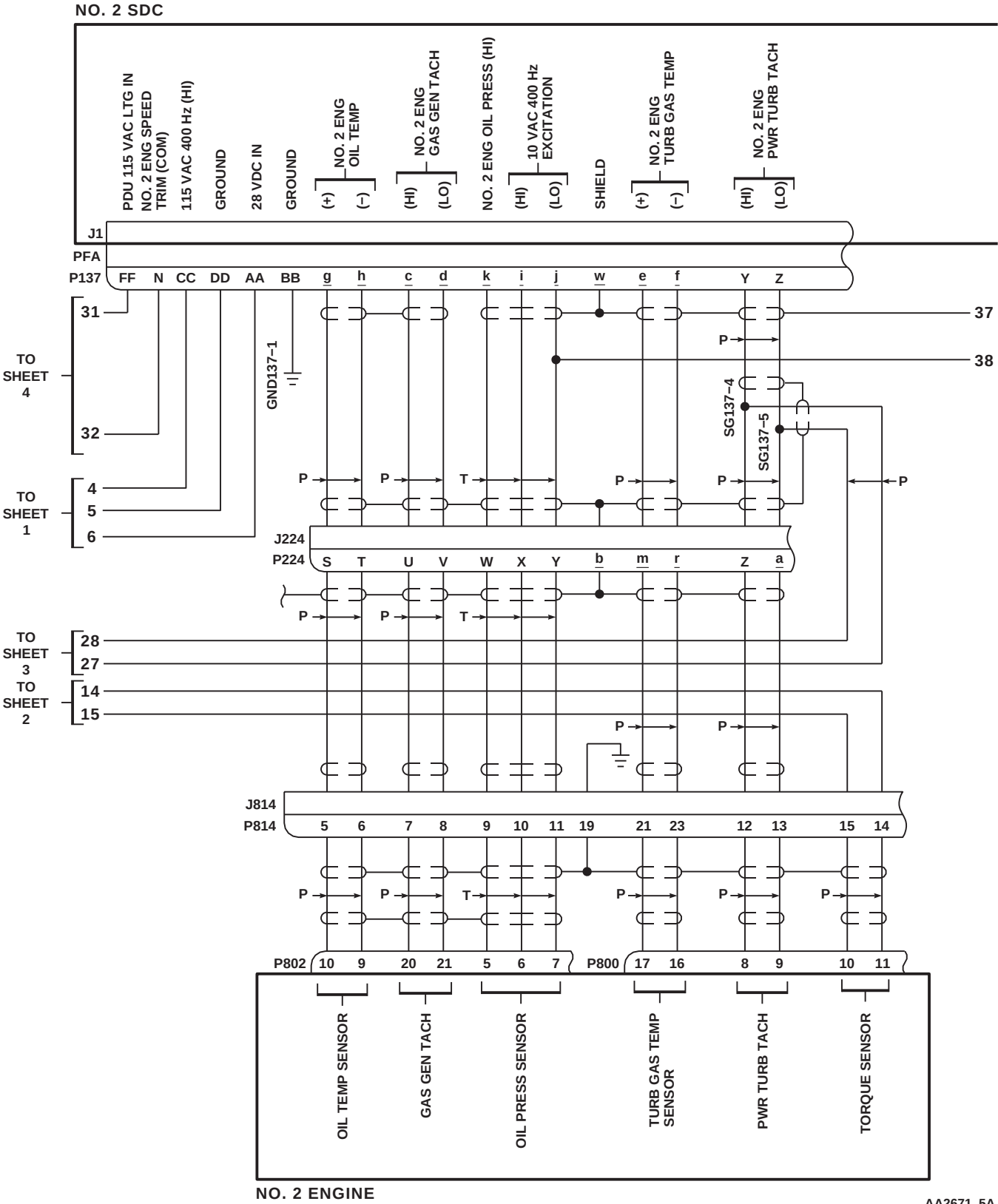


FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 5 OF 21)

NO. 2 SDC

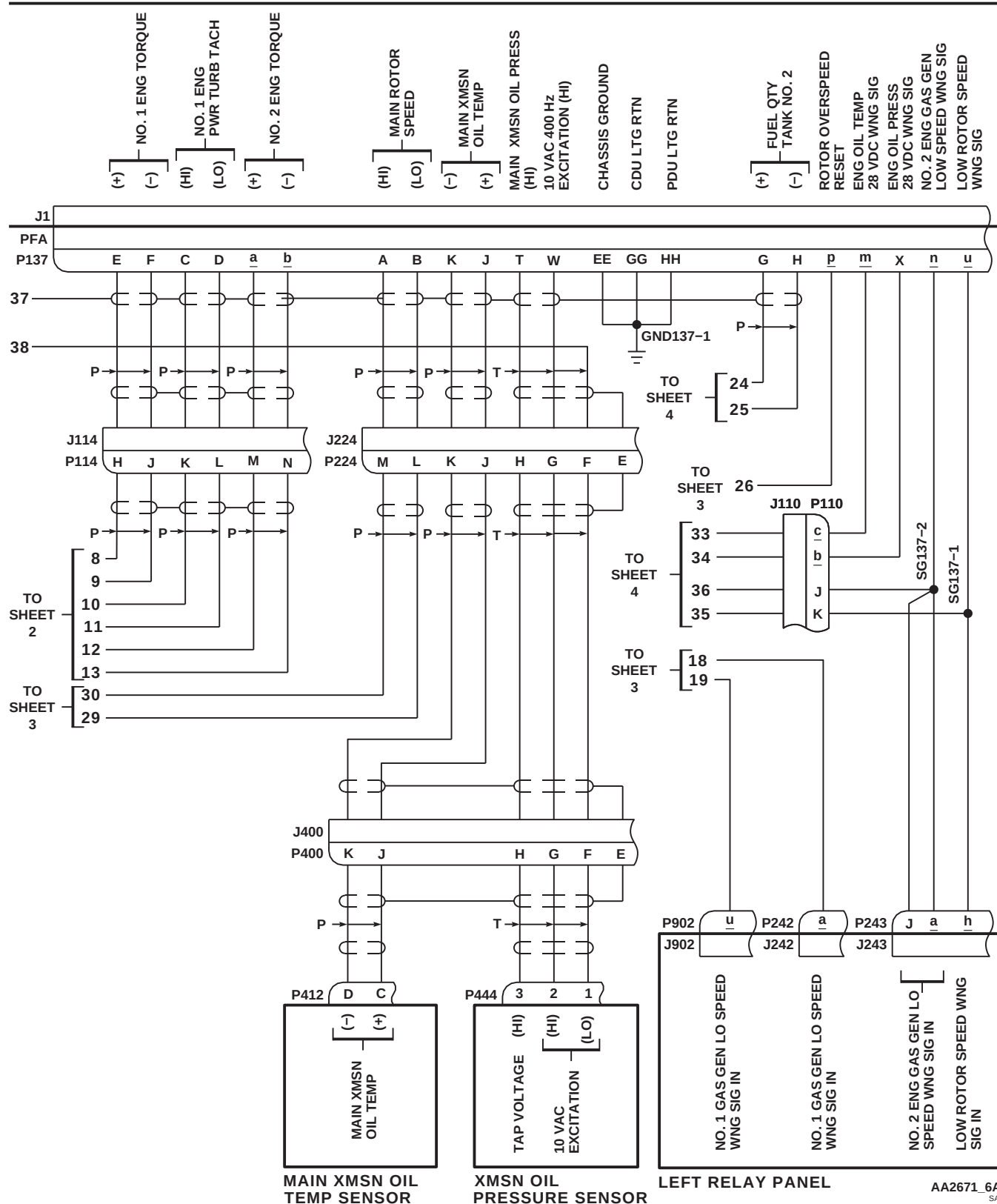
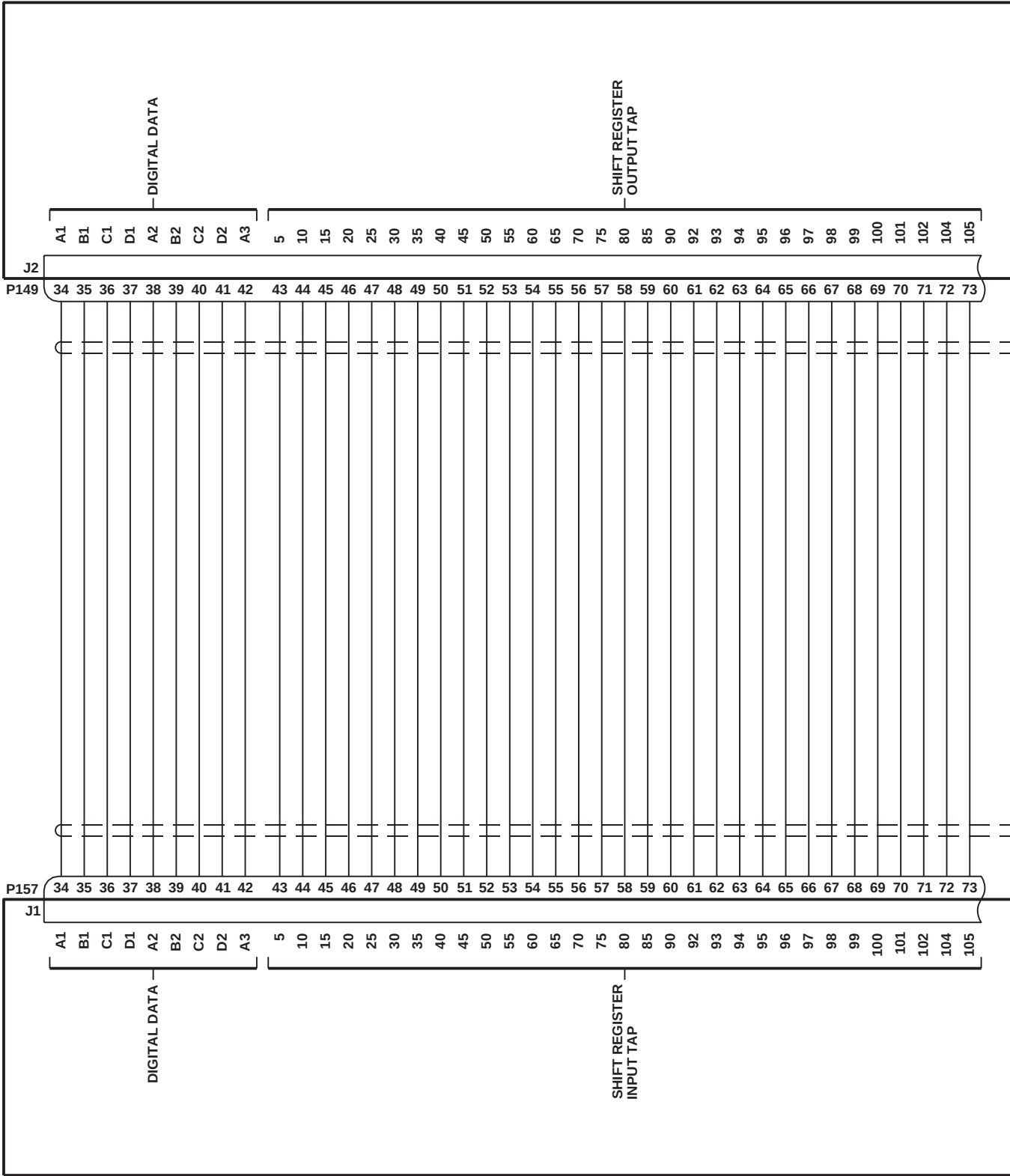
AA2671_6A
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FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 6 OF 21)

NO. 1 SDC

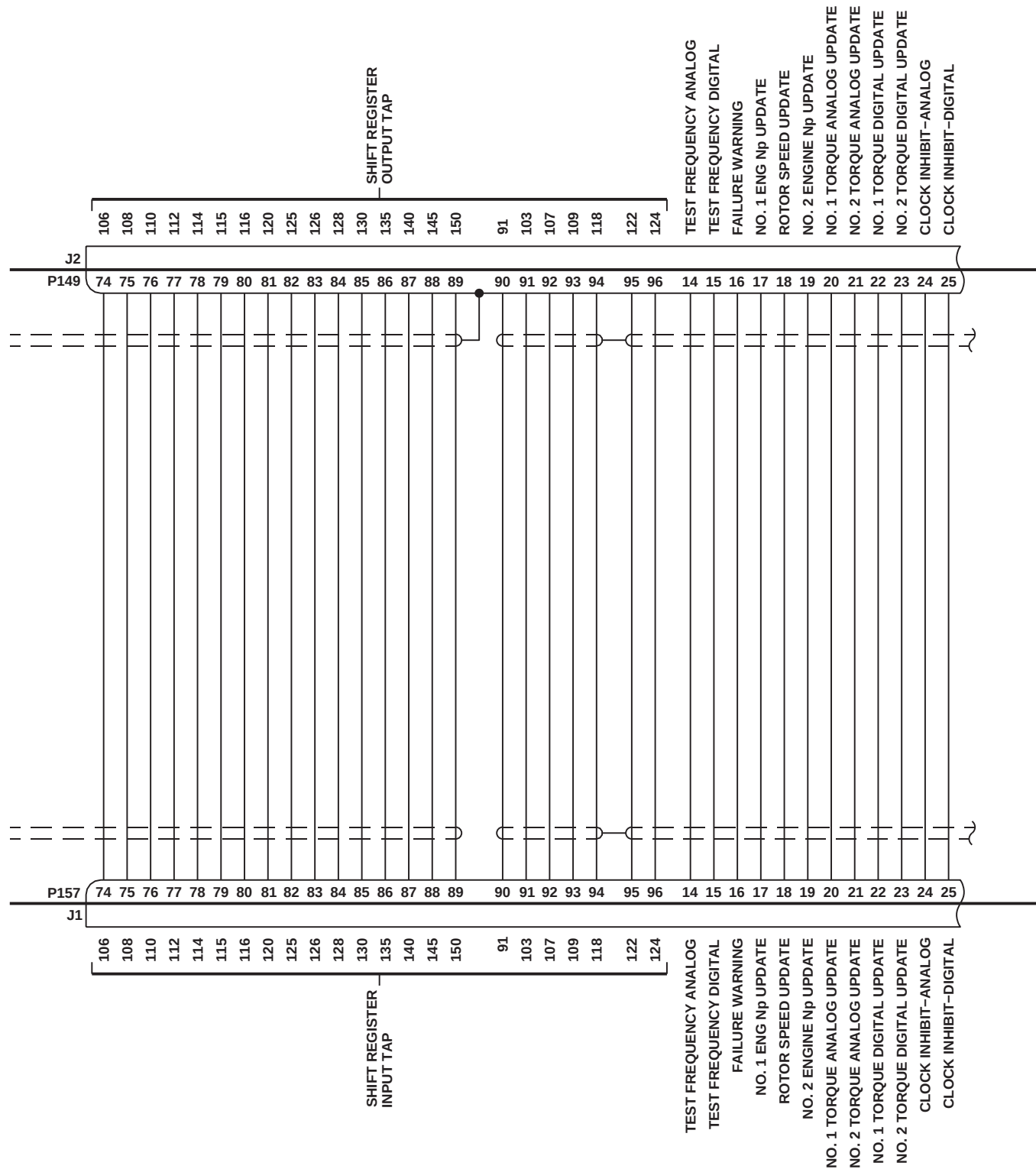


COPILOT DISPLAY UNIT

AA2671_7A
SA

FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 7 OF 21)

NO. 1 SDC



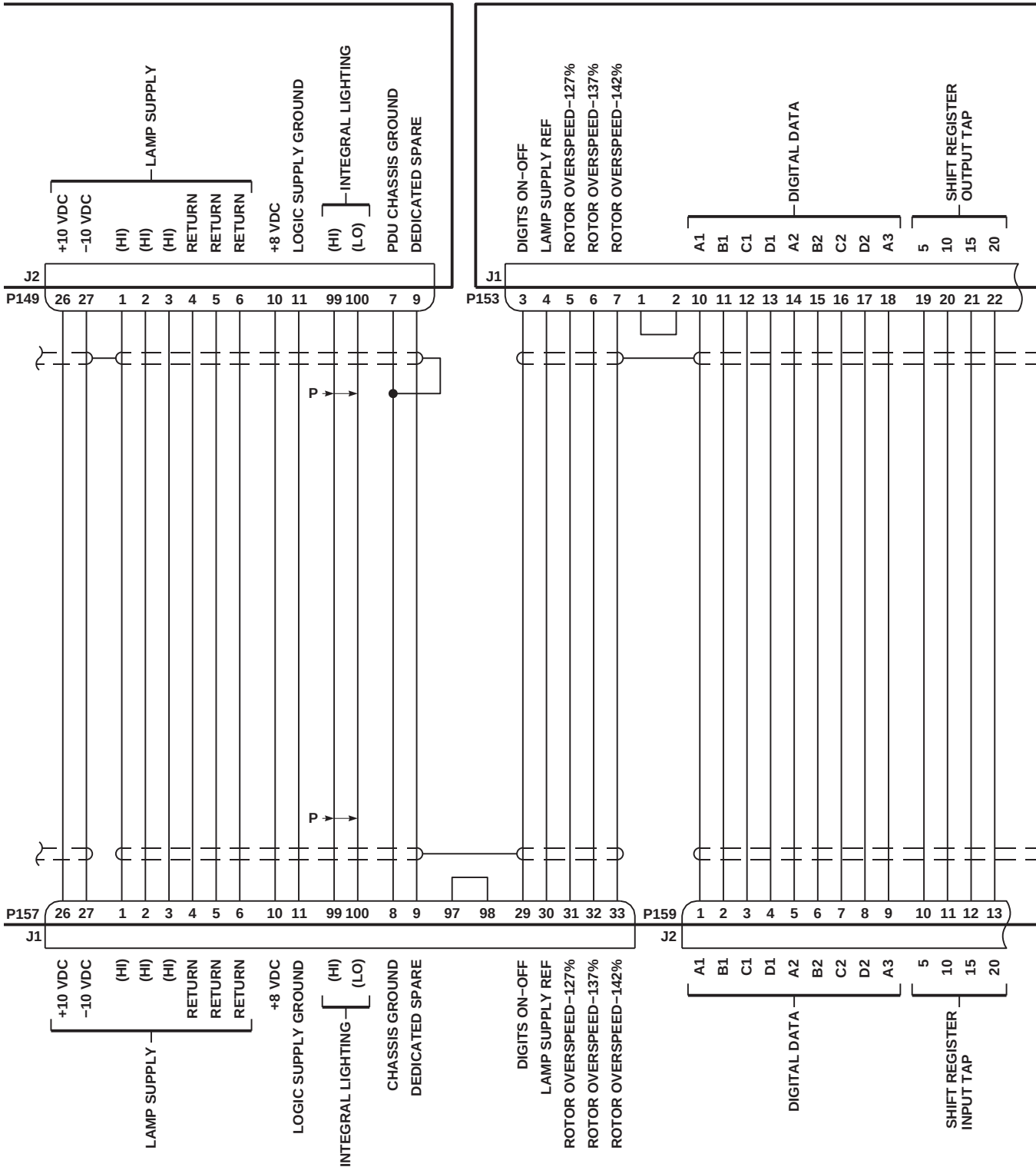
COPILOT DISPLAY UNIT

AA2671_8A
SA

FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 8 OF 21)

NO. 1 SDC

CDU

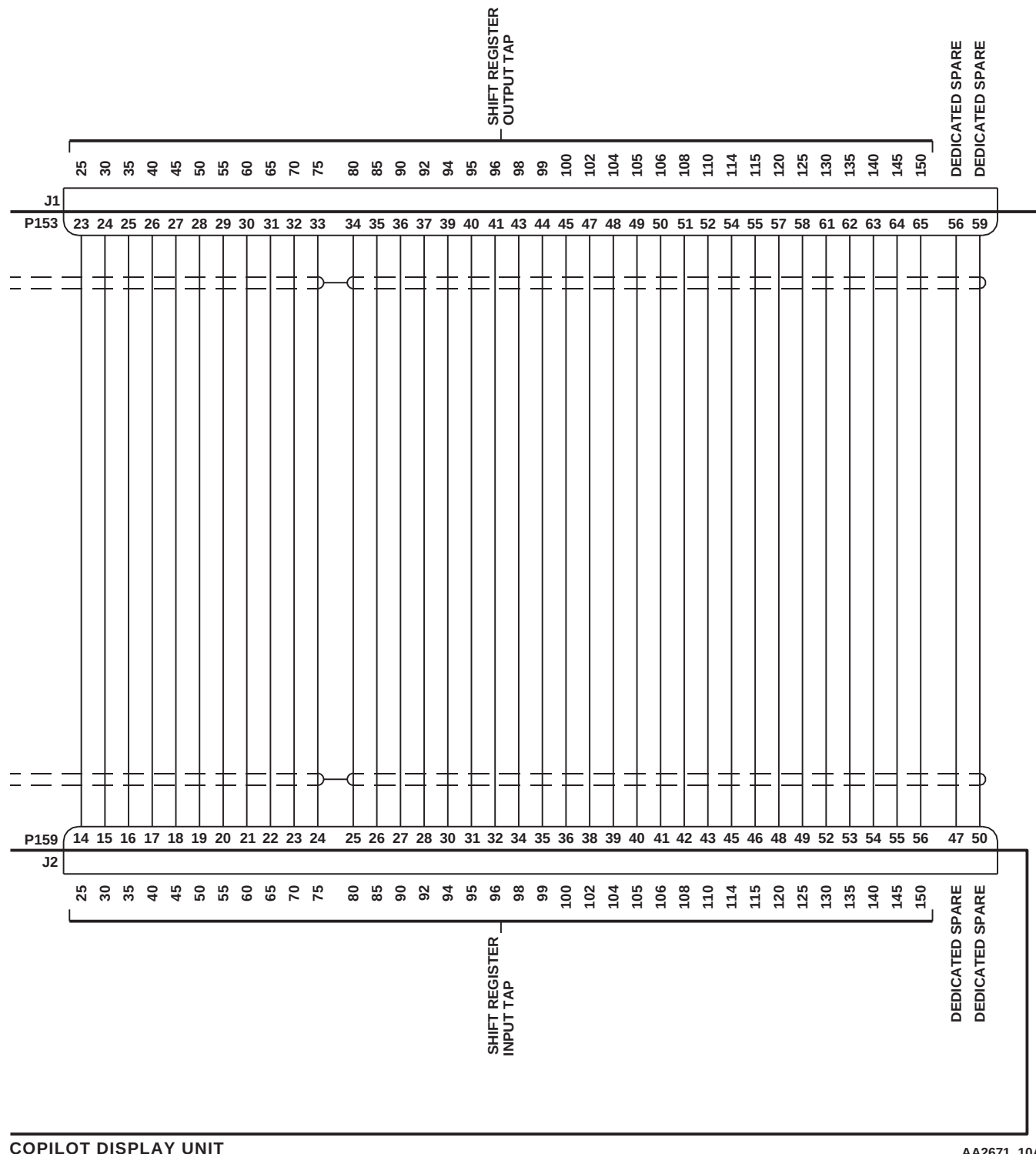


COPILOT DISPLAY UNIT

AA2671_9A
SA

FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 9 OF 21)

CDU



COPILOT DISPLAY UNIT

AA2671_10A
SA

FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 10 OF 21)

8-2-3-52 Change 8

CDU

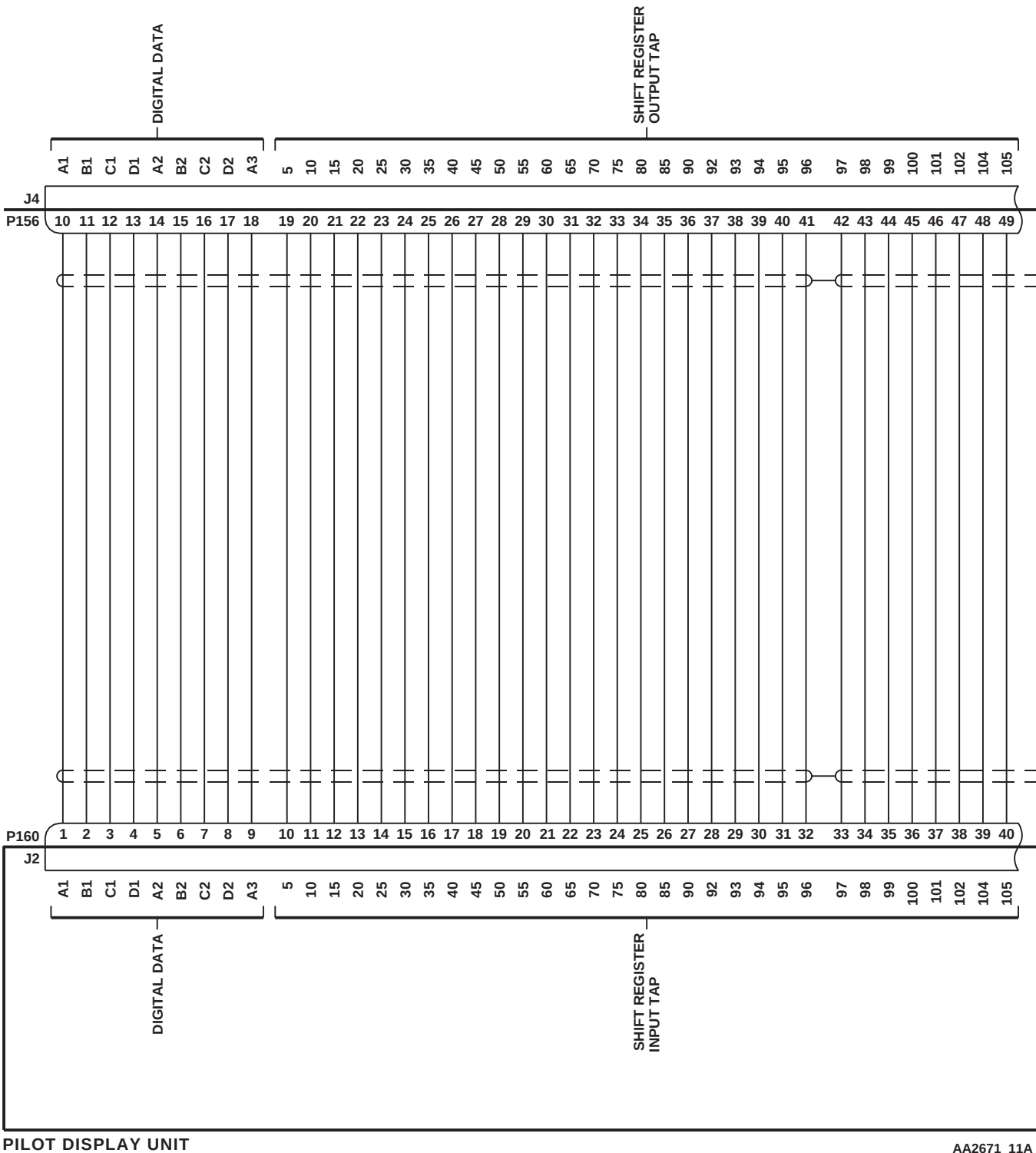
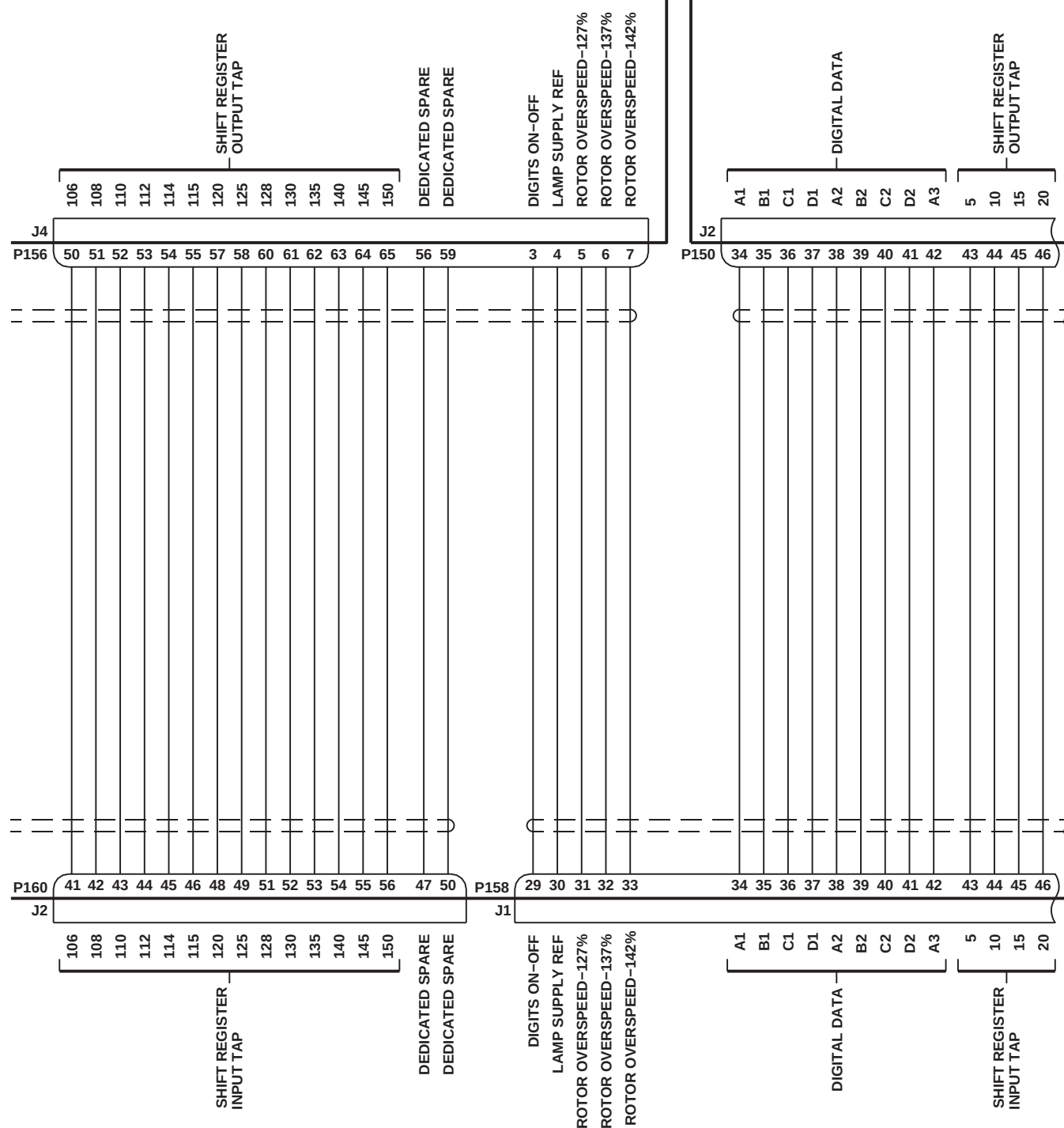


FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 11 OF 21)

CDU

NO. 2 SDC



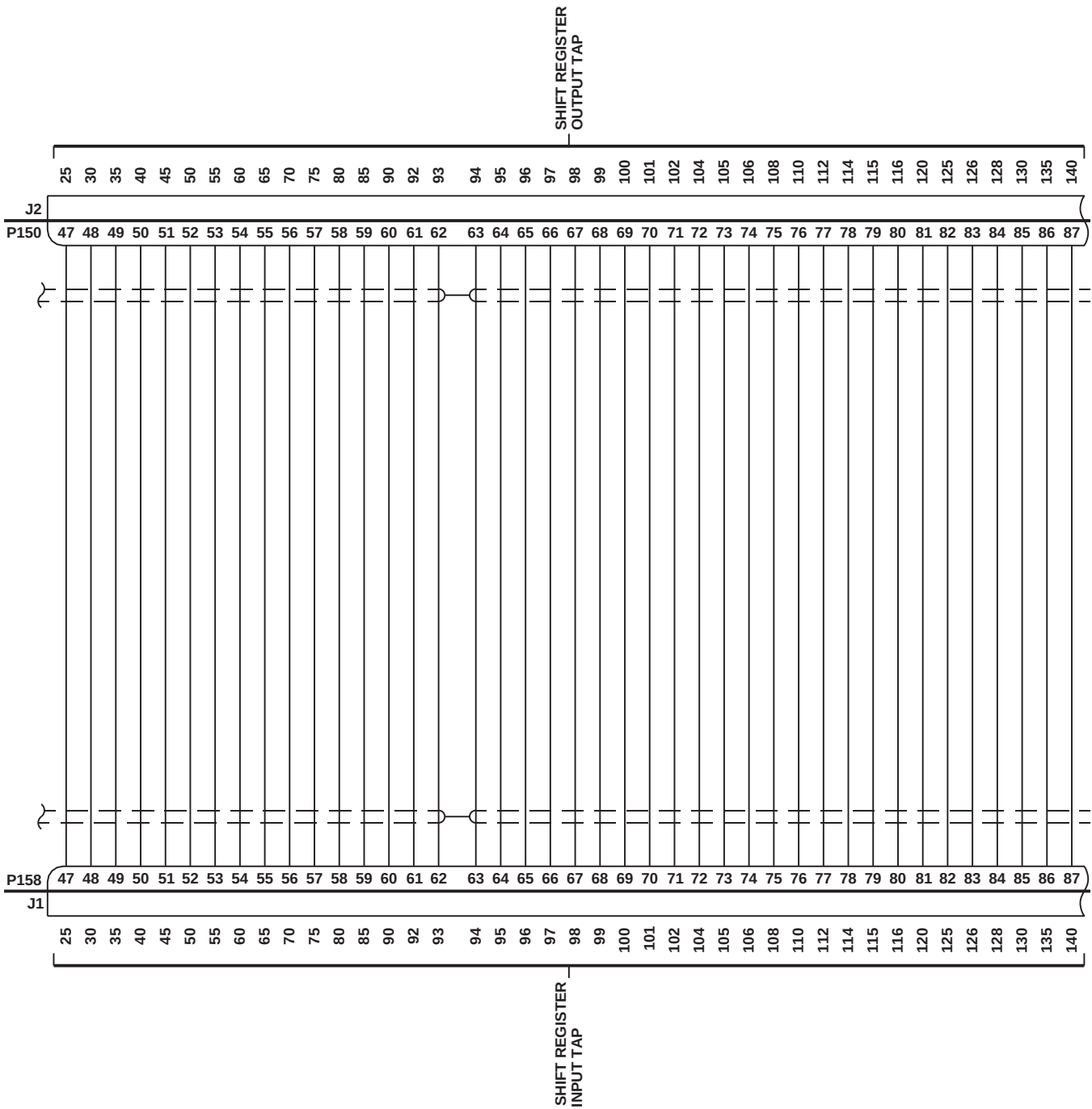
PILOT DISPLAY UNIT

AA2671_12A
SA

FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 12 OF 21)

8-2-3-54 Change 8

NO. 2 SDC



PILOT DISPLAY UNIT

AA2671_13A
SA

FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 13 OF 21)

NO. 2 SDC

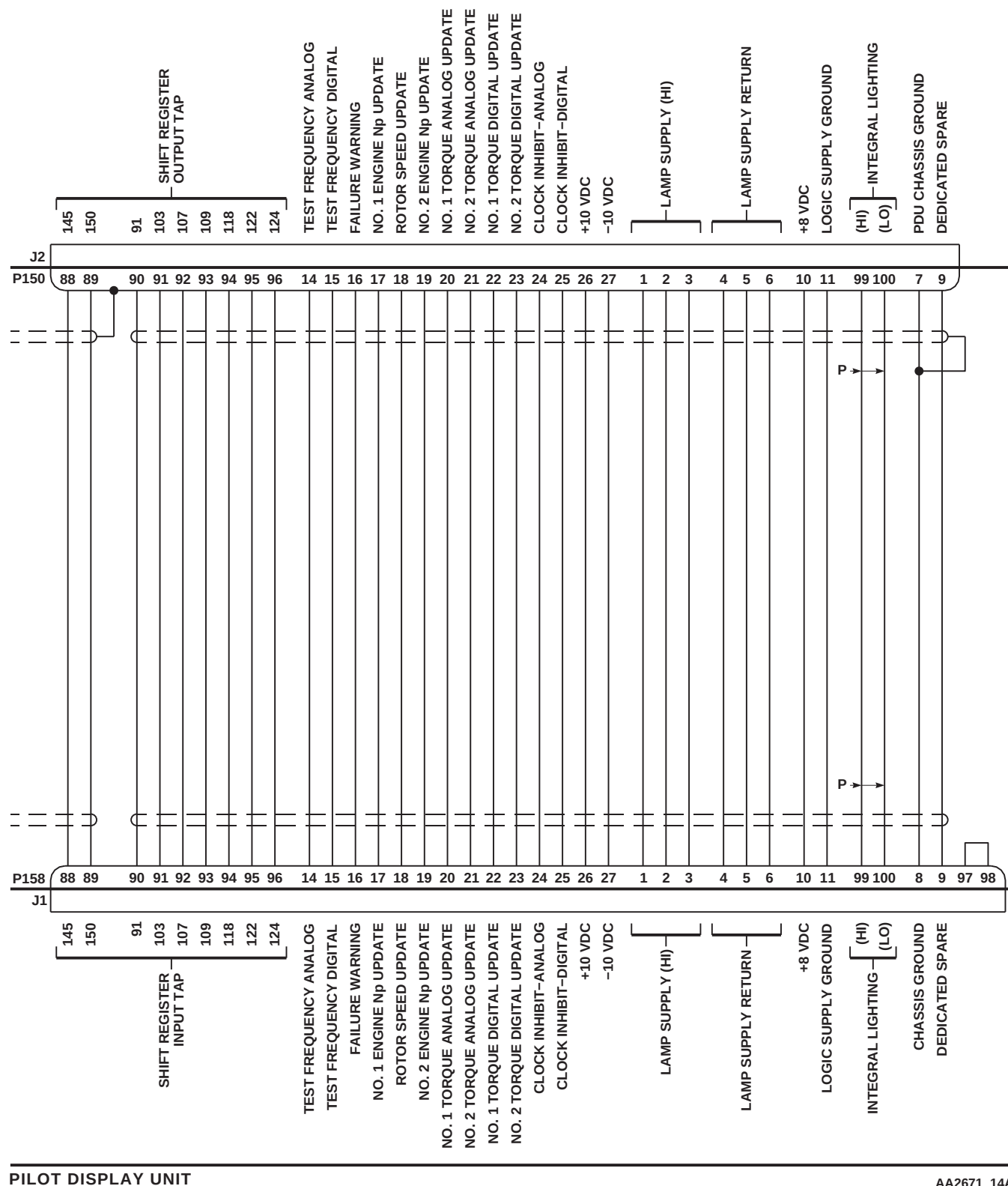
AA2671_14A
SA

FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 14 OF 21)

NO. 2 SDC

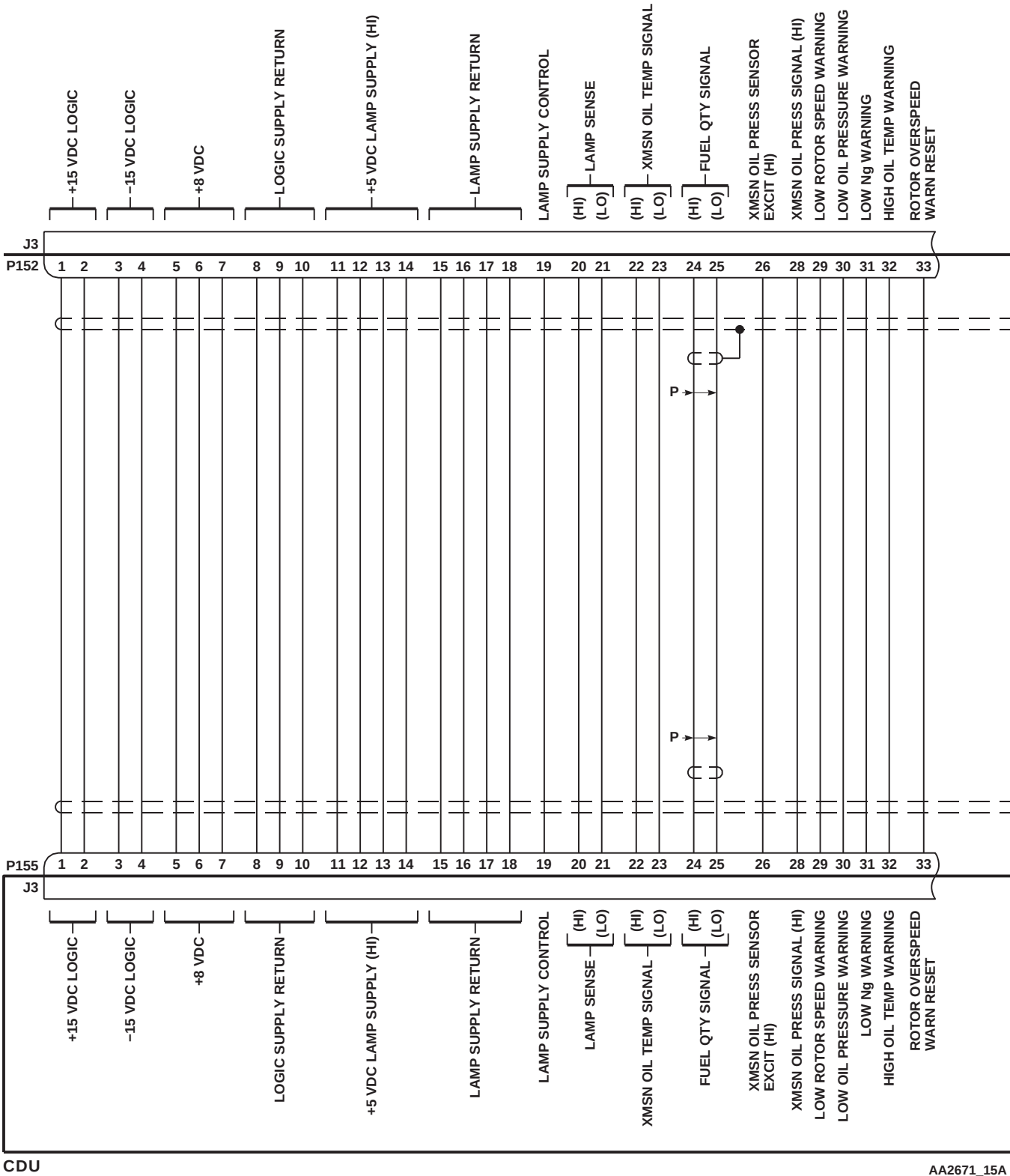
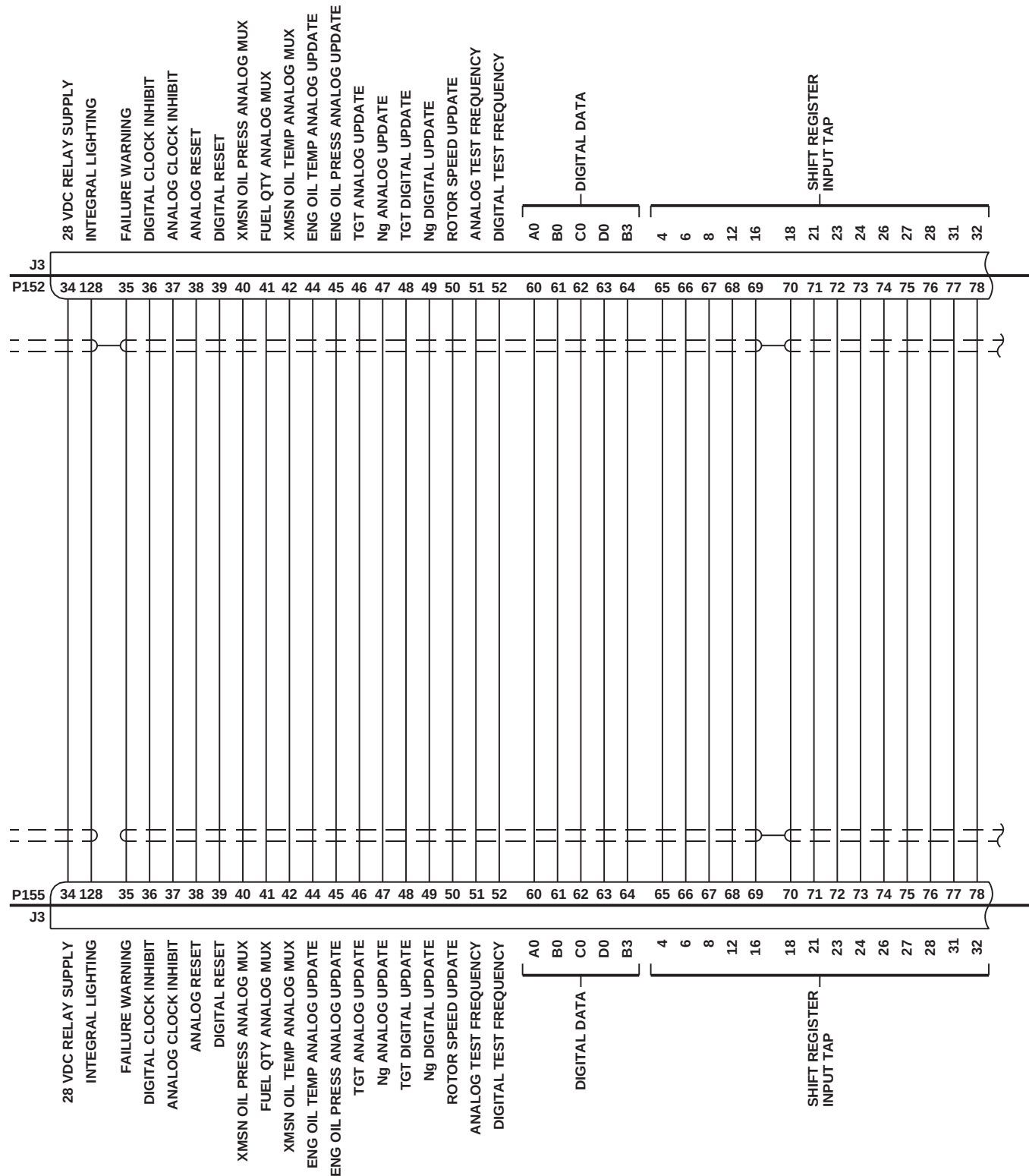


FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 15 OF 21)

NO. 2 SDC



CDU

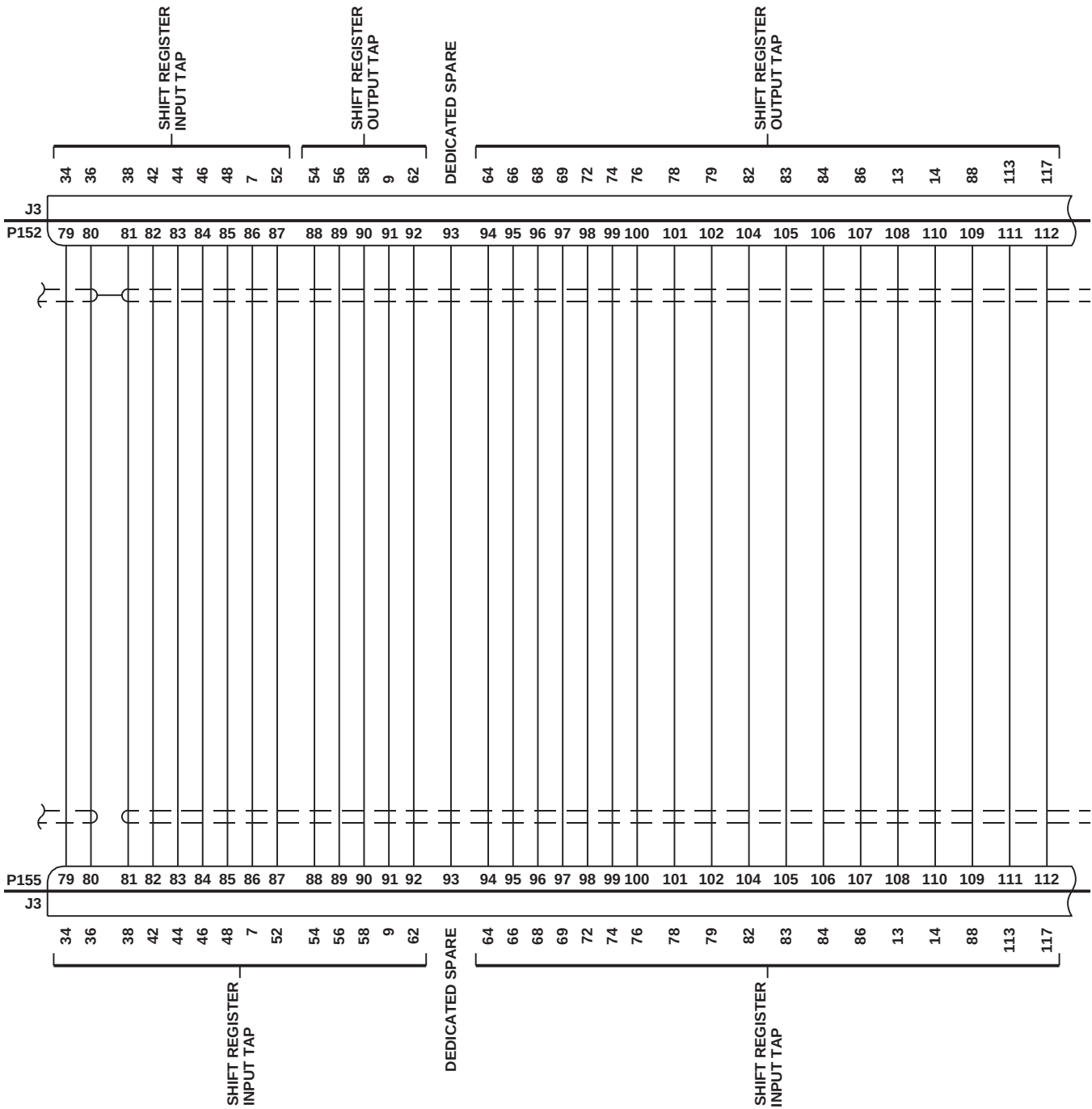
AA2671_16A
SA

FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 16 OF 21)

8-2-3-58

Change 8

NO. 2 SDC



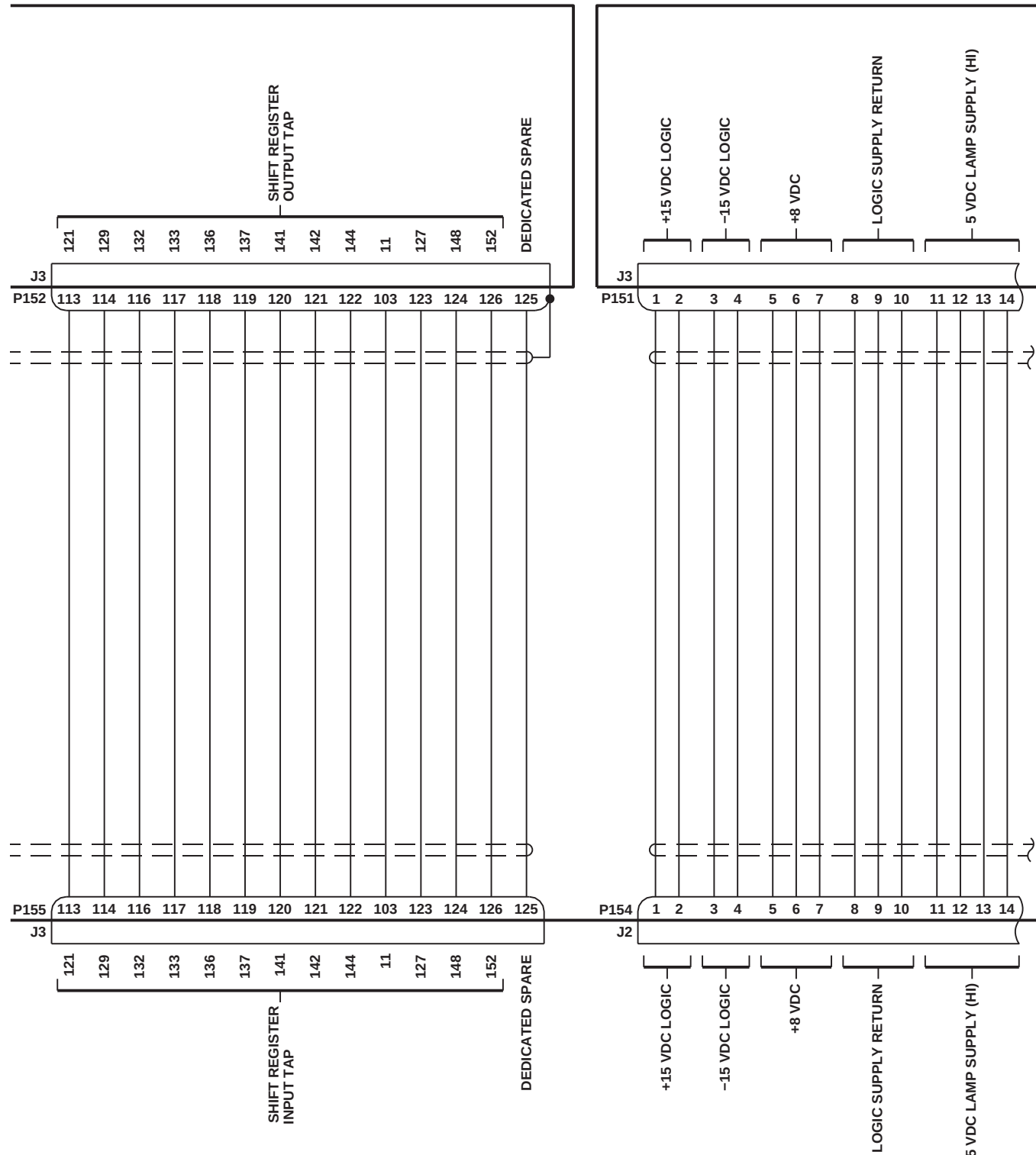
CDU

AA2671 17A
SA

FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 17 OF 21)

NO. 2 SDC

NO. 1 SDC



CDU

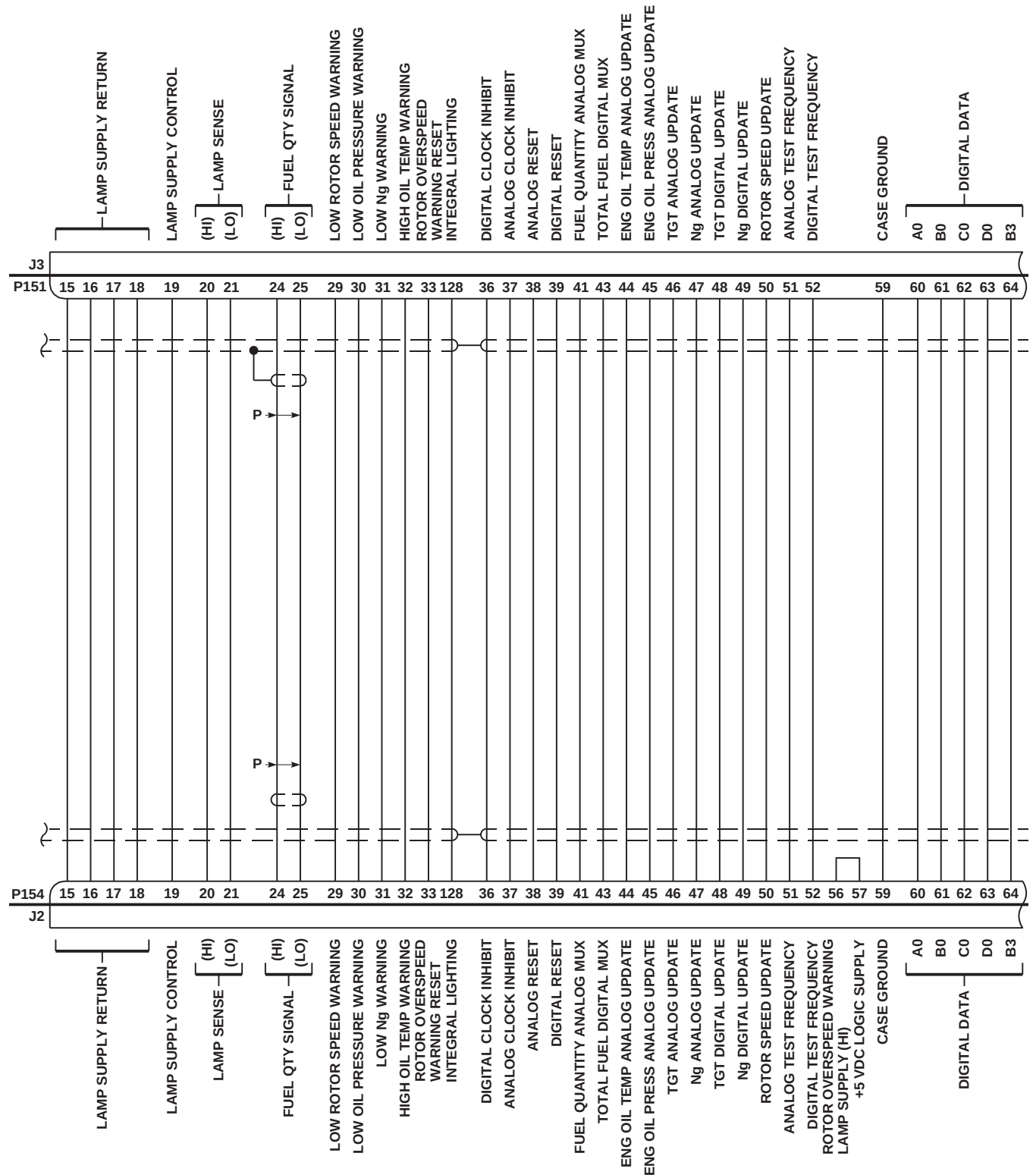
AA2671_18A
SA

FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 18 OF 21)

8-2-3-60 Change 8

This Document contains technical data subject to ITAR. See WARNING and classifications on first page.

NO. 1 SDC

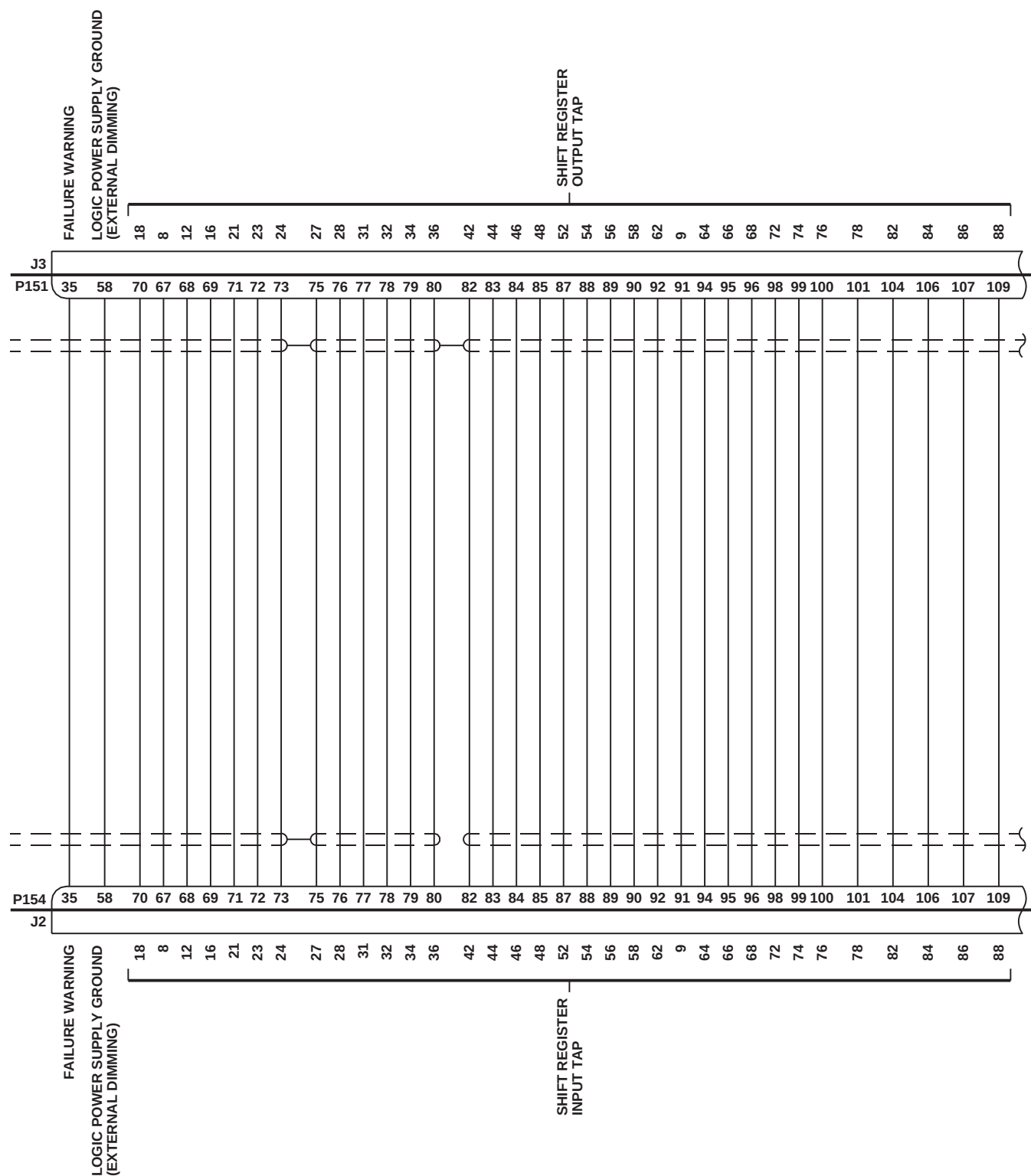


CDU

AA2671 19A
SA

FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 19 OF 21)

NO. 1 SDC



CDU

AA2671_20A
SA

FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 20 OF 21)

NO. 1 SDC

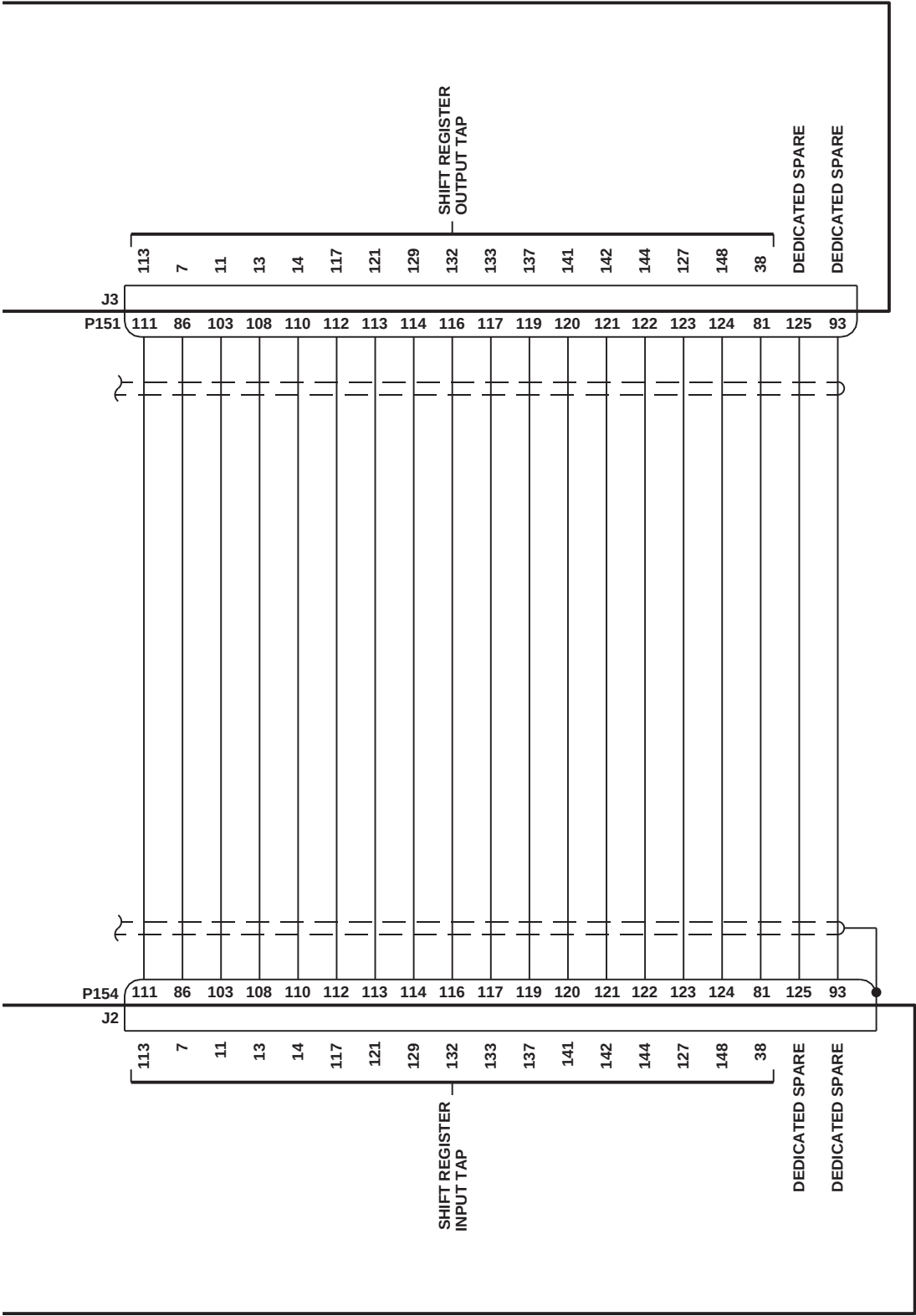


FIGURE 8-2-3-1. INSTRUMENT DISPLAY SYSTEM SCHEMATIC DIAGRAM (SHEET 21 OF 21)

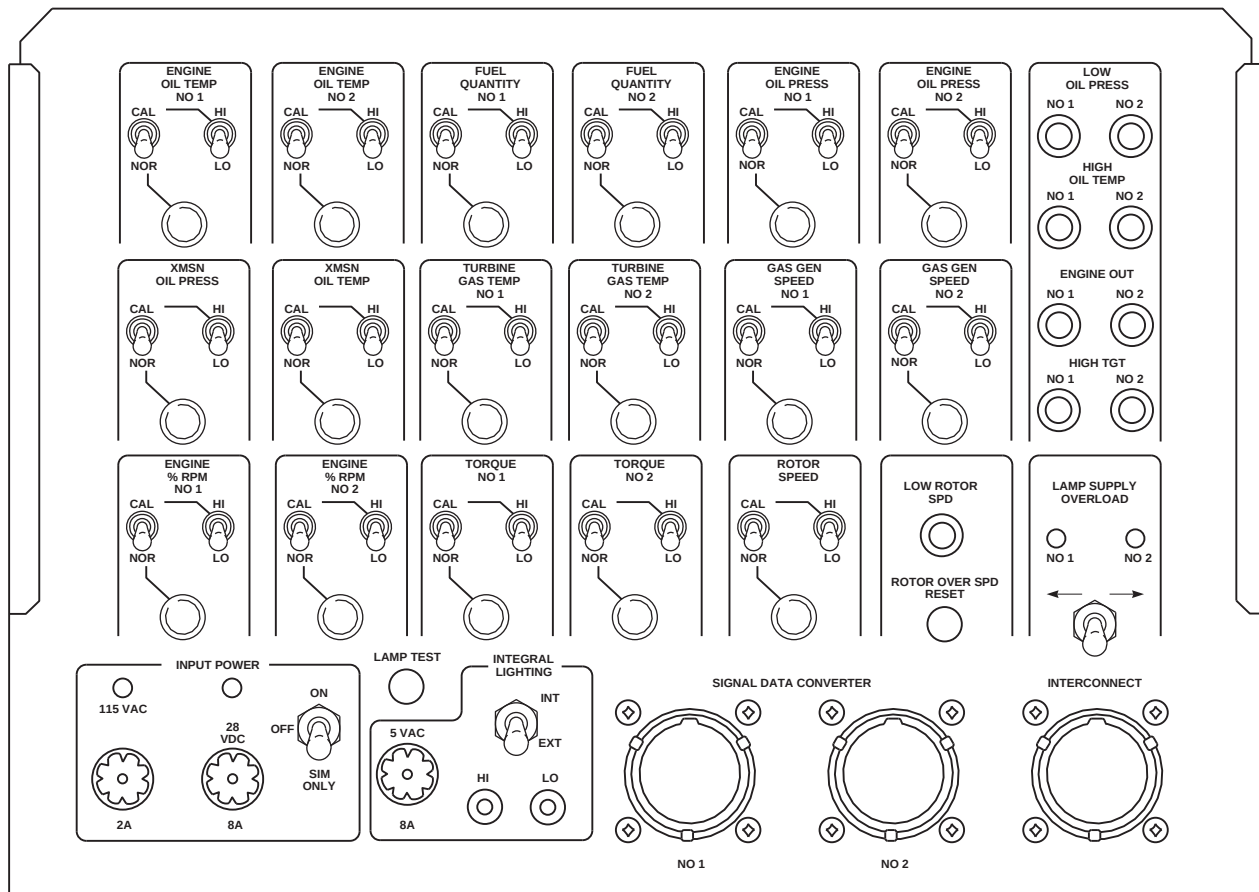
AA1737
SA

FIGURE 8-2-3-2. LINE TEST SET FRONT PANEL

8-2-4 CAUTION/ADVISORY WARNING SYSTEM.

INITIAL SETUP

Test Equipment

- Headset - Microphone
- Multimeter, Fluke 45

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer Toolkit
- Locally-Made Infrared Light Test Set, Figure H-149, Appendix H
- Locally-Made Drag Beam Wedge, Figure H-167, Appendix H

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available Or APU Operational

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

NOTE

A momentary flicker of some caution/advisory panel capsules is normal and requires no corrective action.

8-2-4.1 Operational/Troubleshooting Procedure.

NOTE

Step a. must be performed when a system malfunction has been reported; otherwise, continue to step b.

- EMEP** Perform pin filtered connector and adapter checks (TABLE NO. 8-2-4-1).
- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Pull out the following circuit breakers:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
BACKUP HYD CONTR	Upper console

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
-----------------	-----------------------

- | | |
|----------------|---------------|
| ICS PILOT | Upper console |
| NO. 1 GEN WARN | Copilot’s |
| NO. 2 GEN WARN | Pilot’s |
| NO. 2 AC INST | Pilot’s |
| NO. 2 DC INST | Pilot’s |
- Turn on electrical power (PARA 1-6-16).
 - Turn upper console INST LT PILOT FLT control to OFF, then turn slightly clockwise to get out of OFF detent.
 - Momentarily press copilot’s master warning panel MASTER CAUTION PRESS TO RESET capsule.

RESULT: If pilot's and copilot's MASTER CAUTION PRESS TO RESET capsules were on, they shall go off.

- If result is not as specified, go to TABLE NO. 8-2-4-2.

NOTE

If caution/advisory capsules do not go on, press and release copilot's cyclic stick PNL LTS switch. The dimming mode of the caution/advisory panel may get out of synchronization with the other panel lights; that is, when the PNL LTS button is pressed, the caution/advisory panel capsules may go on bright when the other lights go off. To correct this condition, place the caution/advisory panel capsules at high intensity, using the PNL LTS switch. Remove power from the helicopter and then reapply power.

- g. Place and hold caution/advisory panel BRT/DIM-TEST switch to BRT/DIM.

NOTE

In step h. start with CAUT/ADVSY NVG DIMMING control in fully counterclockwise position.

- h. Turn instrument panel CAUT/ADVSY NVG DIMMING control fully clockwise, then fully counterclockwise.

RESULT: Caution/advisory capsules shall go on and shall increase and then decrease in brightness.

- If result is not as specified, replace CAUT/ADVSY NVG DIMMING control (PARA 8-4-22).
 - If no caution/advisory capsules are on or dim, go to TABLE NO. 8-2-4-3.
- i. Place power switch of locally-made infrared light test set to ON.

RESULT: POWER ON lamp (green) on locally-made infrared light test set shall go on.

- If result is not as specified, replace infrared light test set, test set lamp, or test set battery, as required.

- j. Turn CAUT/ADVSY NVG DIMMING and MA WRN NVG DIMMING controls fully clockwise.

- k. While holding down the caution/advisory panel BRT/DIM-TEST switch, place locally-made infrared light test set IR detector lead over each caution/advisory panel and master warning capsule that is not flashing.

RESULT:

- (1) Locally-made infrared light test set IR LIGHT ON lamp shall go on each time detector lead is placed over capsule that is not flashing.
- (2) Master warning capsules will not illuminate if soft lighting system (SLS) is installed.

- If results are not as specified, go to TABLE NO. 8-2-4-4.

NOTE

Make sure instrument panel MA WRN NVG DIMMING and CAUT/ADVSY NVG DIMMING controls are rotated fully clockwise.

- l. Push in NO. 1 GEN WARN circuit breaker (copilot's circuit breaker panel).

RESULT:

- (1) Both MASTER CAUTION PRESS TO RESET capsules shall go on.
- (2) #1 GEN caution capsule shall go on.

8-2-4-2

- If results are not as specified, troubleshoot AC Electrical System (PARA 9-2-1).
- m. Momentarily press copilot's master warning panel MASTER CAUTION PRESS TO RESET capsule.

RESULT: Both MASTER CAUTION PRESS TO RESET capsules shall go off.

- If result is not as specified, go to TABLE NO. 8-2-4-2.
- n. Push in NO. 2 GEN WARN circuit breaker (pilot's circuit breaker panel).

RESULT:

- (1) Both MASTER CAUTION PRESS TO RESET capsules shall go on.
 - (2) #2 GEN caution capsule shall go on.
- If result (1) is not as specified, check if lamps are good.
 - (a) If lamps are good, replace caution/advisory panel or master warning panel (PARA 8-4-19 or PARA 8-4-24).
 - (b) If lamps are not good, replace lamps (PARA 8-4-20 or PARA 8-4-25).
 - If result (2) is not as specified, troubleshoot AC Electrical System (PARA 9-2-1).
 - o. Turn instrument panel MA WRN NVG DIMMING control counterclockwise.

RESULT: All master warning panel capsules that are on shall dim.

- If result is not as specified, replace MA WRN NVG DIMMING control (PARA 8-4-27).
- p. Momentarily press pilot's master warning panel MASTER CAUTION PRESS TO RESET capsule.

RESULT: Both MASTER CAUTION PRESS TO RESET capsules shall go off.

- If result is not as specified, go to TABLE NO. 8-2-4-2.
- q. Connect headset to pilot's ICS control panel.
- r. Push in ICS PILOT circuit breaker (upper console circuit breaker panel).

WARNING

To prevent injury to personnel, make sure area around the helicopter's stabilator is clear of personnel and equipment.

NOTE

Make sure left landing gear drag beam switch is in ground position (weight on wheels).

- s. Place stabilator control/auto flight control panel MAN SLEW switch down or up depending on the stabilator's position.

RESULT: Beeping tone shall be heard in headset, and stabilator control/auto flight control panel AUTO CONTROL ON legend shall be on.

- If result is not as specified, go to TABLE NO. 8-2-4-5.
- t. Press pilot's master warning panel MASTER CAUTION PRESS TO RESET capsule.
- u. Push in NO. 2 DC INST and NO. 2 AC INST circuit breaker (pilot's circuit breaker panel).

RESULT:

- (1) # 1 ENG OUT and # 2 ENG OUT master warning capsules shall go on.
- (2) LOW ROTOR RPM capsules shall flash (master warning panels).

- (3) MASTER CAUTION PRESS TO RESET capsules shall go on.

- (4) A steady tone shall be heard in headset.

- If results (1), (2), and (3) are not as specified, troubleshoot Instrument Display System (IDS) (PARA 8-2-3).
- If result (4) is not as specified, go to TABLE NO. 8-2-4-6.

- v. Press pilot's master warning panel MASTER CAUTION PRESS TO RESET capsule and check for tone in headset.

RESULT: No tone shall be heard.

- If result is not as specified, replace left relay panel (PARA 9-4-2).

- w. Push in BACKUP HYD CONTR circuit breaker (upper console circuit breaker panel).

- x. Insert locally-made drag beam wedge in left drag beam switch to simulate in-flight (weight-off-wheels) condition.

RESULT: Steady tone shall be heard in headset.

- If result is not as specified, go to TABLE NO. 8-2-4-7.

- y. Press pilot's master warning panel MASTER CAUTION PRESS TO RESET capsule and check for tone in headset.

RESULT: Steady tone shall be heard in headset.

- If result is not as specified, go to TABLE NO. 8-2-4-7.

- z. Remove locally-made drag beam wedge from left drag beam switch.

RESULT: No tone shall be heard.

- If result is not as specified, troubleshoot digital AFCS (TM 11-70-23).

WARNING

To prevent injury to personnel, make sure area around the helicopter's stabilator is clear of personnel and equipment.

NOTE

In the following step, disregard vertical instrument display indications.

- aa. Reset aircraft power by using the EXT PWR RESET switch (center of the upper console), and then turn on.

RESULT:

- (1) # 1 ENG OUT and # 2 ENG OUT caution capsules shall go on.

- (2) LOW ROTOR RPM capsules shall flash and MASTER CAUTION PRESS TO RESET capsules shall go on.

- (3) A steady tone shall be heard in headset.

- If results are not as specified, troubleshoot IDS (PARA 8-2-3).

- ab. Make sure stabilator control/auto flight control panel AUTO CONTROL ON legend is on first, and then place stabilator control/auto flight control panel MAN SLEW switch to the DOWN position.

RESULT:

- (1) Caution/advisory panel STABILATOR capsule shall go on.

- (2) Tone shall change from steady to beeping.

- (3) Stabilator control/auto flight control panel AUTO CONTROL ON legend shall go off.

8-2-4-4

- If result (1) is not as specified, troubleshoot Stabilator Unit (TM 11-70-23).
 - If result (2) is not as specified, replace audible warning unit (PARA 8-4-23).
 - If result (3) is not as specified, troubleshoot digital AFCS (TM 11-70-23).
 - If trouble remains, replace left relay panel (PARA 9-4-2).
- ac. Momentarily press either master warning panel MASTER CAUTION PRESS TO RESET capsule.

RESULT:

- (1) Tone shall change from beeping to steady.
 - (2) MASTER CAUTION PRESS TO RESET capsules shall go off.
- If result (1) is not as specified, troubleshoot digital AFCS (TM 11-70-23).
 - If result (2) is not as specified, go to TABLE NO. 8-2-4-2.
- ad. Momentarily press either master warning panel MASTER CAUTION PRESS TO RESET capsule.

RESULT: Tone shall not be heard in headset.

- If result is not as specified, troubleshoot digital AFCS (TM 11-70-23).
- ae. Insert locally-made drag beam wedge in left drag beam switch to simulate in-flight (weight-off-wheels) condition.
- RESULT: Steady tone shall be heard in headset.
- If result is not as specified, go to TABLE NO. 8-2-4-7.
- af. Momentarily press stabilator control/auto flight control panel AUTO CONTROL RESET switch.

RESULT:

- (1) Caution/advisory panel STABILATOR capsule shall go off.
 - (2) Stabilator control/auto flight control panel AUTO CONTROL switch ON legend shall go on.
- If results are not as specified, troubleshoot stabilator unit (TM 11-70-23).
- ag. Press and hold stabilator control/auto flight control panel TEST switch.

RESULT:

- (1) Caution/advisory panel STABILATOR capsule shall go on.
 - (2) MASTER CAUTION PRESS TO RESET capsules shall go on.
 - (3) Stabilator control/auto flight control panel AUTO CONTROL switch ON legend shall go off.
 - (4) Beeping tone shall be heard in headset.
- If result (1) is not as specified, troubleshoot stabilator unit (TM 11-70-23).
 - If results (2), (3), or (4) are not as specified, troubleshoot digital AFCS (TM 11-70-23).
- ah. Momentarily press either master warning panel MASTER CAUTION PRESS TO RESET capsule.

RESULT:

- (1) Tone shall change from beeping to steady.
 - (2) MASTER CAUTION PRESS TO RESET capsules shall go off.
- If result (1) is not as specified, troubleshoot digital AFCS (TM 11-70-23).
 - If result (2) is not as specified, go to TABLE NO. 8-2-4-2.

- ai. Remove locally-made drag beam wedge from left drag beam switch.
- aj. Turn off electrical power (PARA 1-6-16).

RESULT: No tone shall be heard in headset.

- If result is not as specified, troubleshoot digital AFCS (TM 11-70-23).

TABLE NO. 8-2-4-1. Pin Filtered Connector And Adapter Check.

EMEP	TEST OR INSPECTION CORRECTIVE ACTION
	1. Disconnect P117, P118, P130, P131, P137, P243, and P902. Go to 2. 2. Make sure INST LT PILOT FLT switch is OFF. Go to 3. 3. Remove and check pin filtered adapters from P117 and P118 (PARA 9-4-98). - Step 1. If pin filtered adapters are good, go to 4. - Step 2. If pin filtered adapters are not good, replace pin filtered adapters as required (PARA 9-4-98). Go to 9. 4. Install pin filtered adapters P117 and P118 (PARA 9-4-98). Go to 5. 5. Check continuity between each pin of P130 and connector shell. - Step 1. If continuity is present, replace pin filtered connector (PARA 9-4-98). Go to 9. - Step 2. If continuity is not present, go to 6. 6. Check continuity between: P130-7 and terminal A (INST LT PILOT FLT switch) P130-2 and P118-e P130-3 and P138-n P130-8 and P118-i P130-9 and P138-u P130-9 and P243-h P130-10 and P137-n P130-11 and ground P130-10 and P243-a P130-1 and P118-j P130-3 and P242-a P130-4 and P243-G P130-5 and P118-X P130-6 and P118-a P130-6 and P243-U P130-7 and P118-W

TABLE NO. 8-2-4-1. Pin Filtered Connector And Adapter Check. (Cont)

EMEP	TEST OR INSPECTION CORRECTIVE ACTION
	Step 1. If continuity is present, go to 7. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 9. 7. Check continuity between each pin of P131 and connector shell. Step 1. If continuity is present, replace pin filtered connector (PARA 9-4-98). Go to 9. Step 2. If continuity is not present, go to 8. 8. Check continuity between: P130-1 and P131-1 P130-2 and P131-2 P130-3 and P131-3 P130-5 and P131-5 P130-6 and P131-6 P130-7 and P131-7 P130-8 and P131-8 P130-9 and P131-9 P130-10 and P131-10 P902-H and P131-4 P131-11 and ground Step 1. If continuity is present, go to 9. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 9. 9. Connect P117, P118, P130, P131, P137, P243, and P902. Go to 10. 10. Procedure completed.

TABLE NO. 8-2-4-2. MASTER CAUTION PRESS TO RESET Capsules Do Not Go Off When Reset.

TEST OR INSPECTION CORRECTIVE ACTION
1. Check helicopter configuration. Step 1. EMEP , go to 2. Step 2. W/O EMEP , go to 4.

**TABLE NO. 8-2-4-2. MASTER CAUTION PRESS TO RESET Capsules Do Not Go Off When Reset.
(Cont)**

	TEST OR INSPECTION CORRECTIVE ACTION
EMEP	2. Remove and check pin filtered adapter from P118 (PARA 9-4-98). Step 1. If pin filtered adapter is good, go to 3. Step 2. If pin filtered adapter is not good, replace pin filtered adapter (PARA 9-4-98). Go to 12.
EMEP	3. Install pin filtered adapter (PARA 9-4-98). Go to 4. 4. Press pilot's capsule and check that both capsules go off. Step 1. If both capsules are off, go to 5. Step 2. If both capsules are not off, go to 6. 5. Check continuity between: P118-W and P130-7 P118-a and P130-6 Step 1. If continuity is present, replace copilot's master warning panel (PARA 8-4-24). Go to 12. Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 12. 6. Press copilot's capsule and check that both capsules go off. Step 1. If both capsules go off, go to 7. Step 2. If both capsules do not go off, go to 8. 7. Check continuity between: P118-W and P131-7 P118-a and P131-6 Step 1. If continuity is present, replace pilot's master warning panel (PARA 8-4-24). Go to 12. Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 12. 8. Check for 28 vdc between: P130-7 and P130-11

**TABLE NO. 8-2-4-2. MASTER CAUTION PRESS TO RESET Capsules Do Not Go Off When Reset.
(Cont)**

TEST OR INSPECTION	CORRECTIVE ACTION
P131-7 and P131-11	
Step 1. If voltage is as specified, go to 9 .	
Step 2. If voltage is not as specified, go to 10 .	
9. Check for 28 vdc between P118-a and ground while pressing MASTER CAUTION PRESS TO RESET capsule (pilot's).	
Step 1. If voltage is as specified, replace caution/advisory panel (PARA 8-4-19). Go to 12 .	
Step 2. If voltage is not as specified, repair/replace wiring as required between (Appendix F), then go to 12 . P118-a and P130-6 P118-a and P131-6	
10. Check continuity between: P130-11 and ground P131-11 and ground	
Step 1. If continuity is present, go to 11 .	
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 12 .	
11. Check for 28 vdc between terminal 1 (CAUT/ADVSY PNL circuit breaker) and ground.	
Step 1. If voltage is as specified, repair/replace wiring as require between (Appendix F). Go to 12 .	
Terminal 1 (CAUT/ADVSY PNL circuit breaker) and P130-7 Terminal 1 (CAUT/ADVSY PNL circuit breaker) and P131-7	
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 12 .	
12. Procedure completed.	

TABLE NO. 8-2-4-3. No Caution/Advisory Capsules Are On.

TEST OR INSPECTION CORRECTIVE ACTION	
1. Check helicopter configuration.	
Step 1. EMEP , go to 2.	
Step 2. W/O EMEP , go to 4.	
EMEP	2. Remove and check pin filtered adapter from P118 (PARA 9-4-98).
	Step 1. If pin filtered adapter is good, go to 3.
	Step 2. If pin filtered adapter is not good, replace pin filtered adapter (PARA 9-4-98). Go to 6.
EMEP	3. Install pin filtered adapter (PARA 9-4-98). Go to 4.
	4. Check for 28 vdc between terminal 1 (LIGHTS CAUT ADVSY circuit breaker) and ground.
	Step 1. If voltage is as specified, go to 5.
	Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 6.
	5. Check continuity between: P118-U and terminal 1 (LIGHTS CAUT ADVSY circuit breaker) P118-V and terminal 1 (LIGHTS CAUT ADVSY circuit breaker)
	Step 1. If continuity is present, replace caution/advisory panel (PARA 8-4-19). Go to 6.
	Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 6.
	6. Procedure completed.

TABLE NO. 8-2-4-4. Master Warning Capsules Do Not Go On.

TEST OR INSPECTION CORRECTIVE ACTION	
1. Check helicopter configuration.	
Step 1. EMEP , go to 2.	
Step 2. W/O EMEP , go to 4.	
EMEP	2. Remove and check pin filtered adapter from P118 (PARA 9-4-98).

TABLE NO. 8-2-4-4. Master Warning Capsules Do Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If pin filtered adapter is good, go to 3. Step 2. If pin filtered adapter is not good, replace pin filtered adapter (PARA 9-4-98). Go to 6.
EMEP	3. Install pin filtered adapter (PARA 9-4-98). Go to 4.
	4. Check for resistance of 0 to 1.5 K ohms between P118-k and P118-m. Step 1. If resistance is as specified, replace caution/advisory panel (PARA 8-4-19). Go to 6. Step 2. If resistance is not as specified, go to 5.
	5. Check continuity between: R2-1 and P118-k R2-2 and P118-m Step 1. If continuity is present, replace MA WRN NVG DIMMING control (PARA 8-4-22). Go to 6. Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 6. 6. Procedure completed.

TABLE NO. 8-2-4-5. No Stabilator Warning Tone Heard In Headset.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Is steady tone heard in headset? Step 1. If tone is as specified, go to 2. Step 2. If tone is not as specified, go to 7.
	2. Check for 28 vdc between P243-S and ground. Step 1. If voltage is as specified, replace left relay panel (PARA 9-4-2). Go to 12. Step 2. If voltage is not as specified, go to 3.
	3. Check helicopter configuration.

TABLE NO. 8-2-4-5. No Stabilator Warning Tone Heard In Headset. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. EMEP , go to 4. Step 2. W/O EMEP , go to 6.
EMEP	4. Remove and check pin filtered adapter from P117 (PARA 9-4-98). Step 1. If pin filtered adapter is good, go to 5. Step 2. If pin filtered adapter is not good, replace pin filtered adapter (PARA 9-4-98). Go to 12.
EMEP	5. Install pin filtered adapter (PARA 9-4-98). Go to 6. 6. Repair/replace wiring between P243-S and P117-27 (Appendix F). Go to 12. 7. Press master warning panel MASTER CAUTION PRESS TO RESET capsule (pilot's) and listen for a steady tone in headset. Step 1. If steady tone is heard in headset, go to 8. Step 2. If steady tone is not heard in headset, go to 9. 8. Check continuity between P242-G and P961-D. Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to 12. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 12. 9. Check for 28 vdc between P961-A and P961-B. Step 1. If voltage is as specified, go to 10. Step 2. If voltage is not as specified, go to 11. 10. Check continuity between: P242-E and P656R-M P961-B and ground Step 1. If continuity is present, replace audible warning unit (PARA 8-4-23). Go to 12. Step 2. If trouble remains, troubleshoot Intercommunication Set (TM 11-70-23). Go to 12. Step 3. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 12.

TABLE NO. 8-2-4-5. No Stabilator Warning Tone Heard In Headset. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
11. Check continuity between: P961-A and P242-H P961-B and ground	Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to 12. Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 12.
12. Procedure completed.	

TABLE NO. 8-2-4-6. No Engine Out Warning Steady Tone Heard In Headset.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Is beeping tone heard in headset?	Step 1. If beeping tone is heard, go to 2. Step 2. If beeping tone is not heard, go to 3.
2. While pressing MASTER CAUTION PRESS TO RESET capsule (pilot's master warning panel), check for 28 vdc between P243-U and ground.	Step 1. If voltage is as specified, replace left relay panel (PARA 9-4-2). Go to 5. Step 2. If voltage is not as specified, repair/replace wiring between P243-U and P130-6 (Appendix F). Go to 5.
3. Check for 28 vdc between P961-A and P961-B.	Step 1. If voltage is as specified, go to 4. Step 2. If voltage is not as specified, replace left relay panel (PARA 9-4-2). Go to 5.
4. Check continuity between P242-F and P961-F.	Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to 5. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 5.
5. Procedure completed.	

TABLE NO. 8-2-4-7. No Low Rotor RPM Warning Tone Heard In Headset.

TEST OR INSPECTION CORRECTIVE ACTION	
1. Check for 28 vdc between P243-h and ground.	
Step 1. If voltage is as specified, go to 2 . Step 2. If voltage is not as specified, repair/replace wiring between P243-h and P130-9 or P131-9 (Appendix F). Go to 6 .	
2. Check for 28 vdc between P902-CC and ground.	
Step 1. If voltage is as specified, go to 3 . Step 2. If voltage is not as specified, go to 4 .	
3. Press landing gear drag beam switch and check continuity between P242-L and ground.	
Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to 6 . Step 2. If continuity is not present, go to 5 .	
4. Check for 28 vdc between terminal 1 (BACKUP HYD CONTR circuit breaker) and ground.	
Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P902-CC (Appendix F). Go to 6 . Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 6 .	
5. Check continuity between: J163-A and ground J163-C and P242-L	
Step 1. If continuity is present, replace landing gear drag beam switch (PARA 3-4-6). Go to 6 . Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 6 .	
6. Procedure completed.	

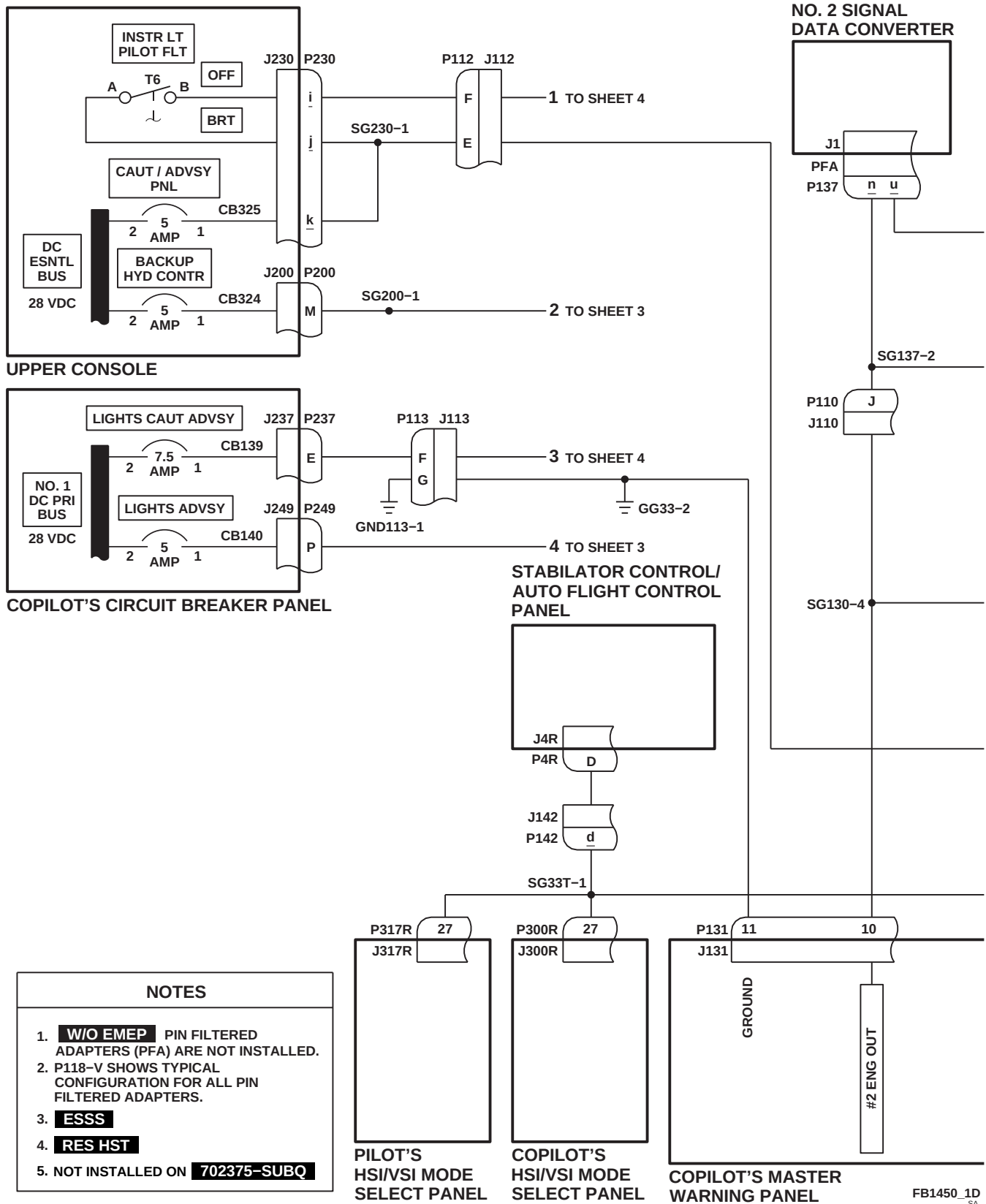


FIGURE 8-2-4-1. CAUTION/ADVISORY WARNING SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 10)

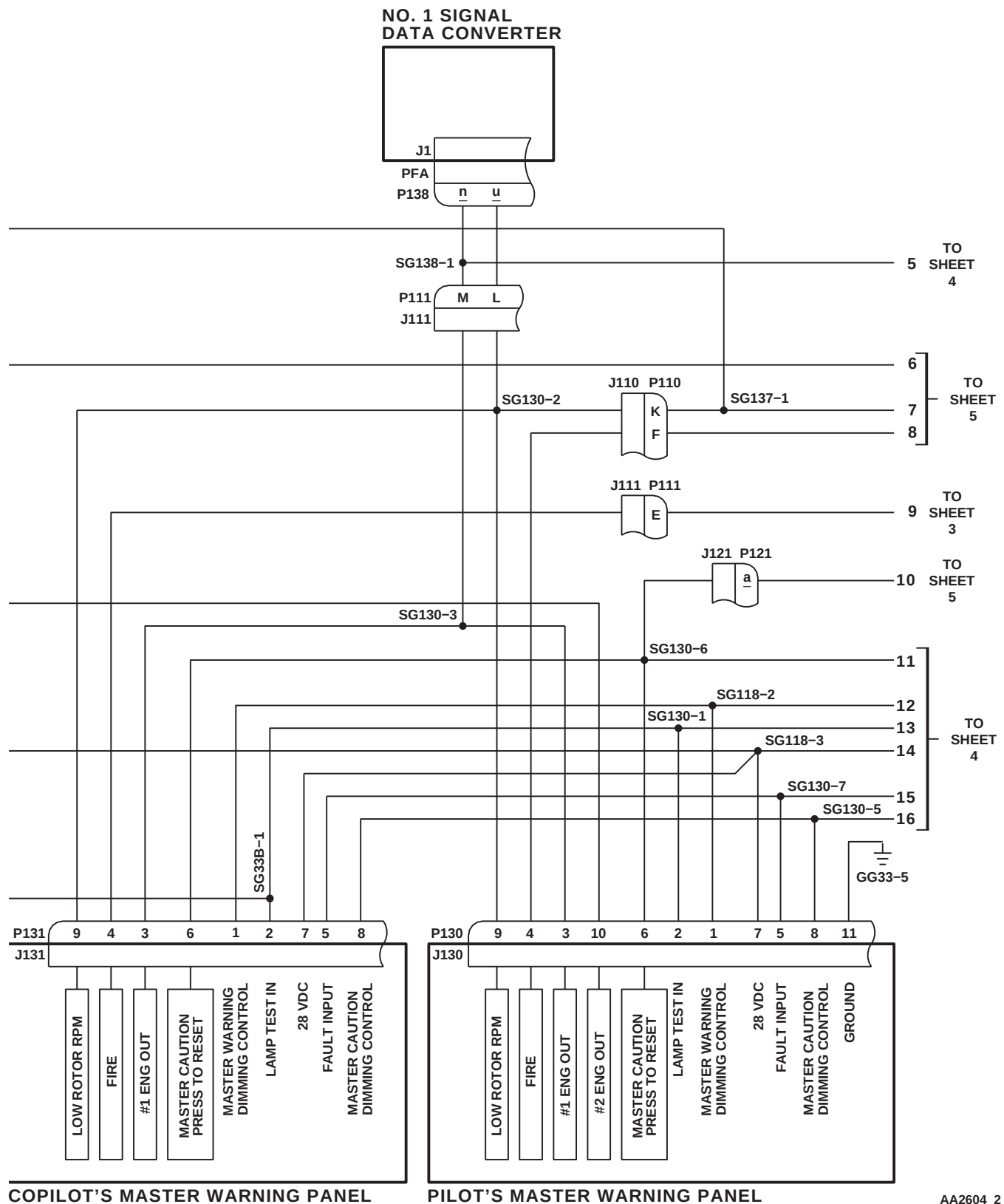
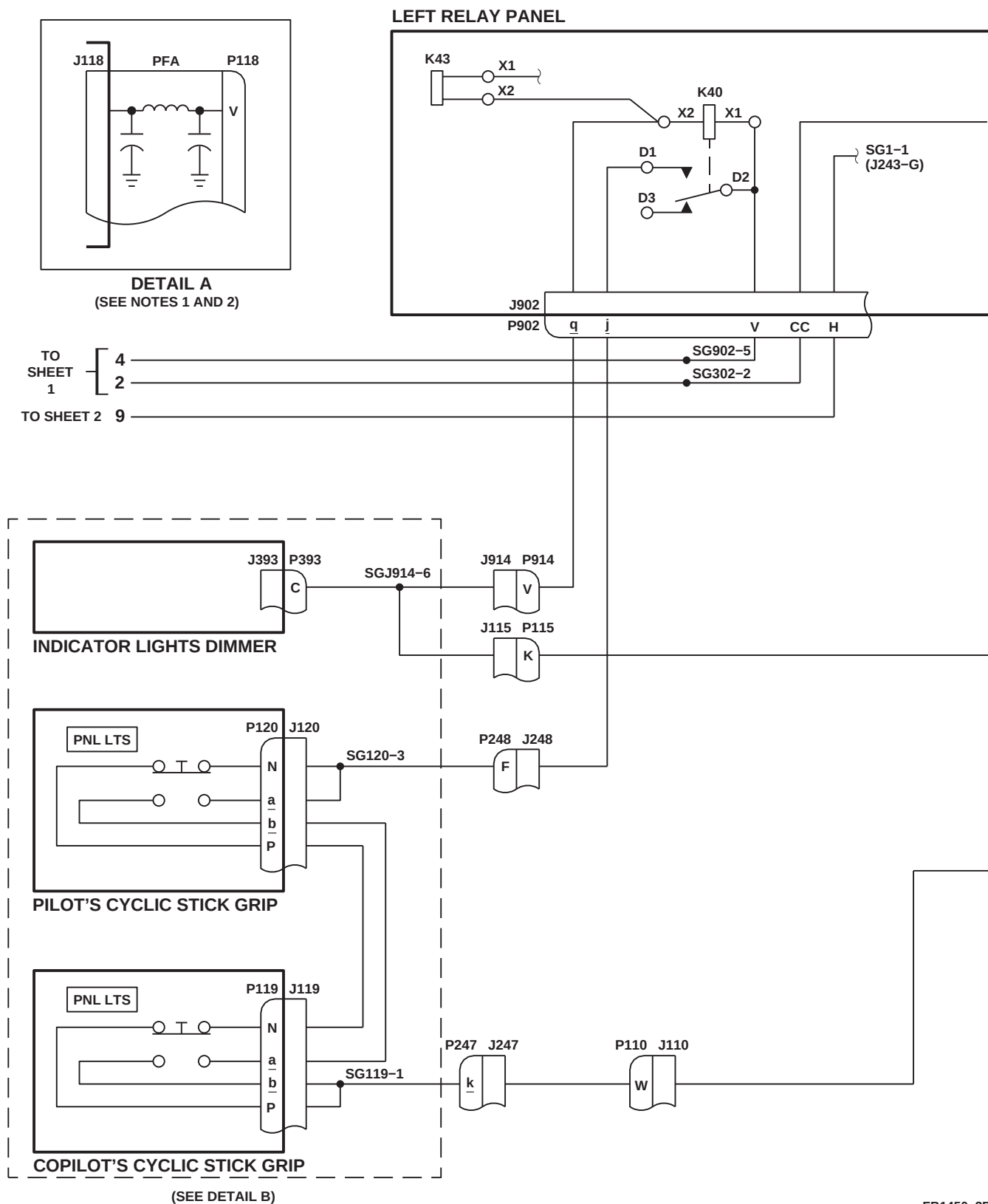


FIGURE 8-2-4-1. CAUTION/ADVISORY WARNING SYSTEM SCHEMATIC DIAGRAM
(SHEET 2 OF 10)



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FIGURE 8-2-4-1. CAUTION/ADVISORY WARNING SYSTEM SCHEMATIC DIAGRAM
(SHEET 3 OF 10)

LEFT RELAY PANEL

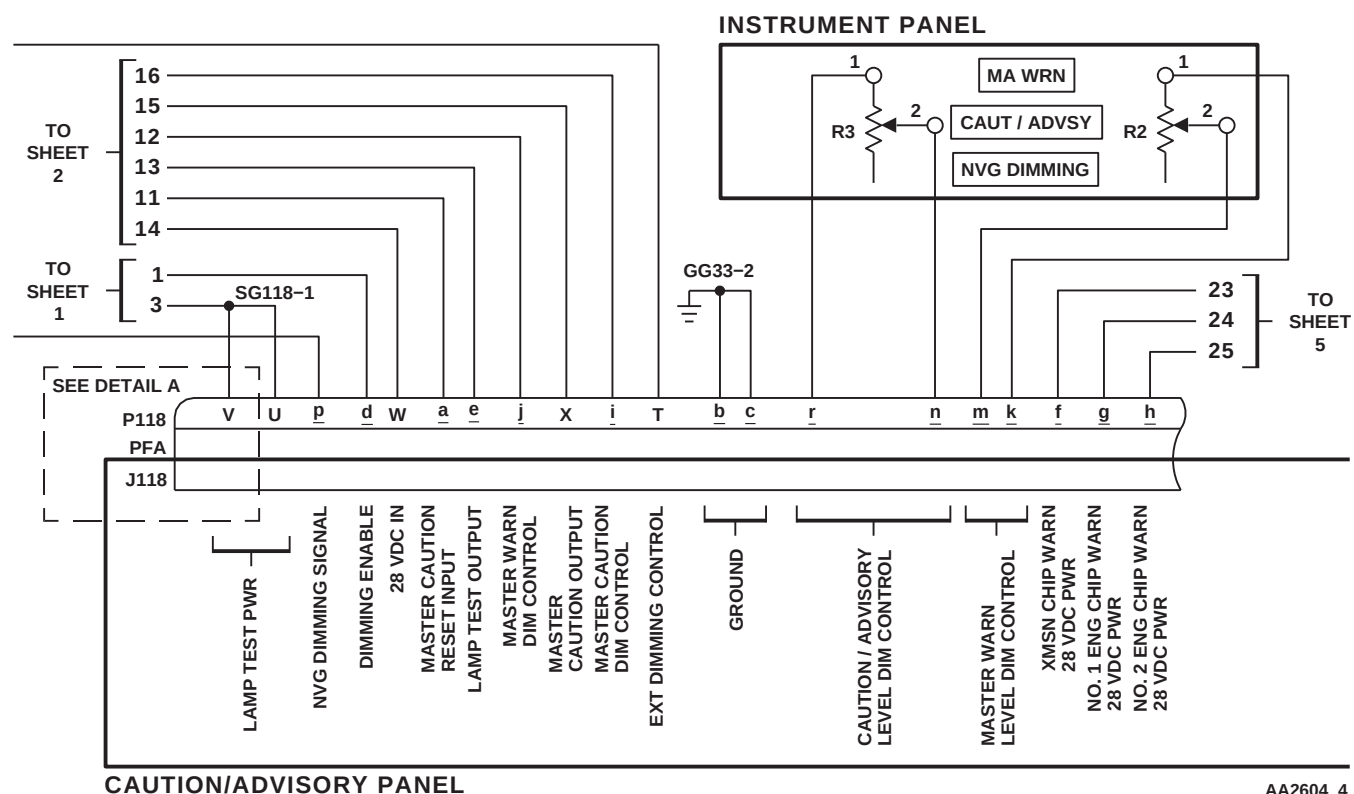
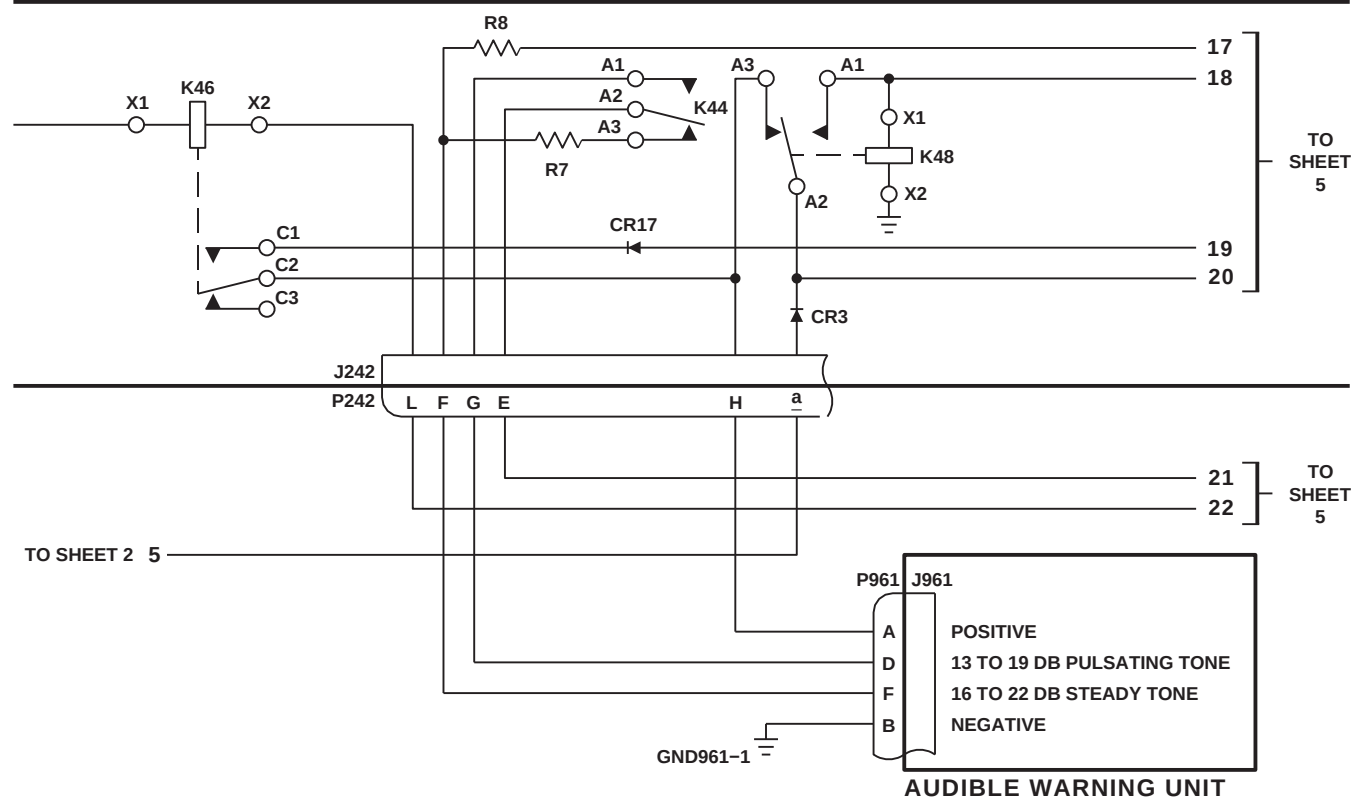


FIGURE 8-2-4-1. CAUTION/ADVISORY WARNING SYSTEM SCHEMATIC DIAGRAM
(SHEET 4 OF 10)

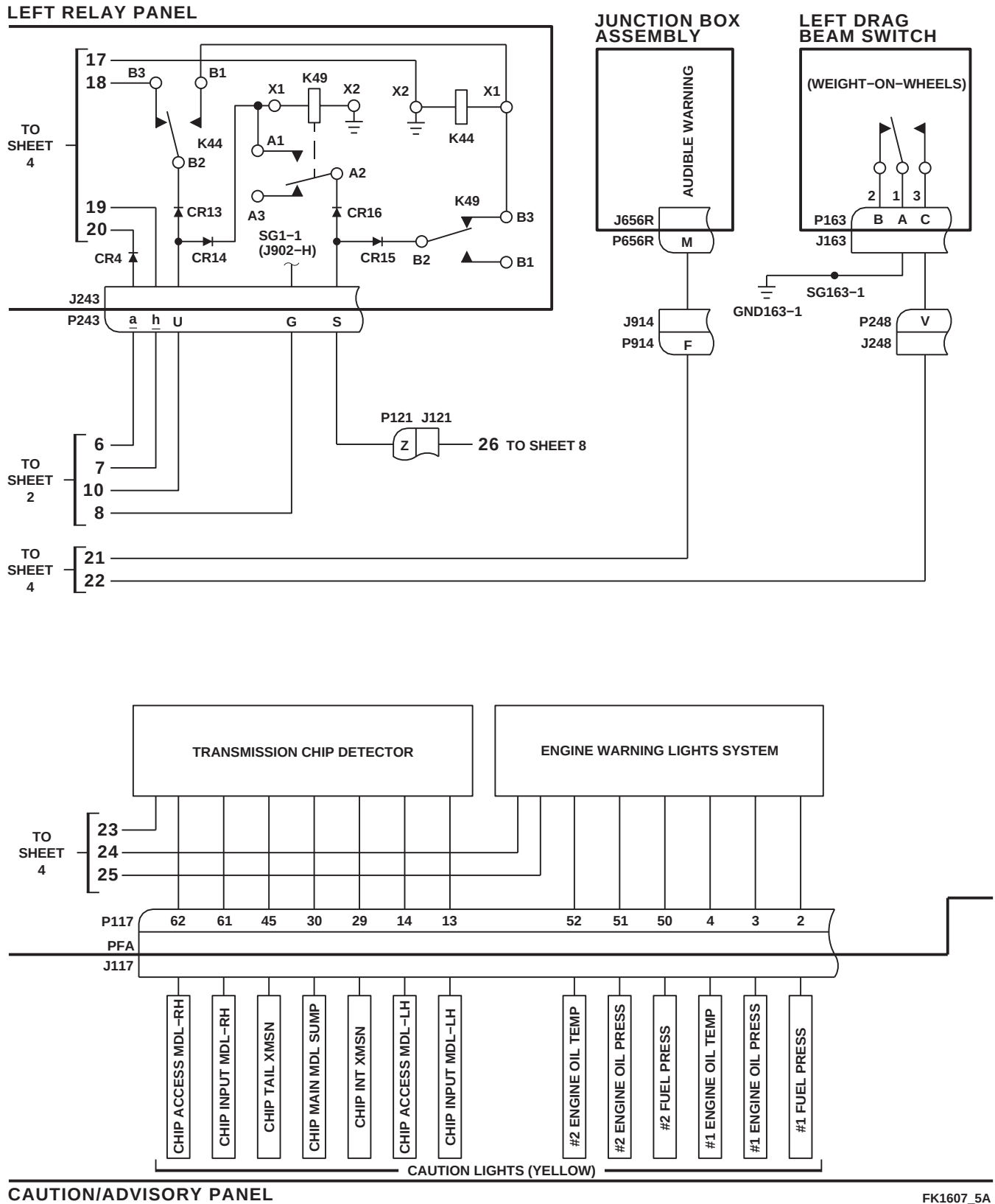


FIGURE 8-2-4-1. CAUTION/ADVISORY WARNING SYSTEM SCHEMATIC DIAGRAM (SHEET 5 OF 10)

CAUTION/ADVISORY PANEL

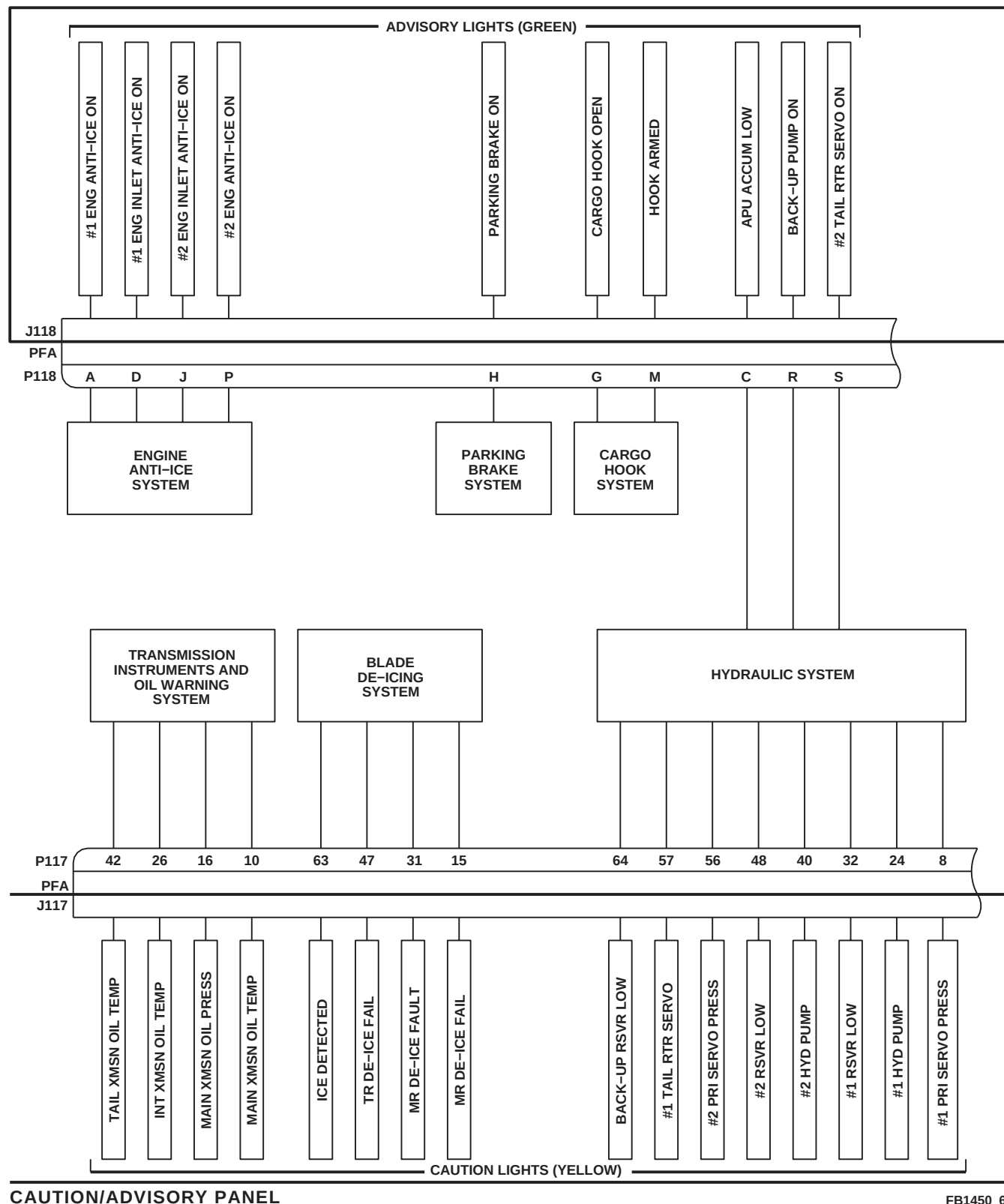


FIGURE 8-2-4-1. CAUTION/ADVISORY WARNING SYSTEM SCHEMATIC DIAGRAM
(SHEET 6 OF 10)

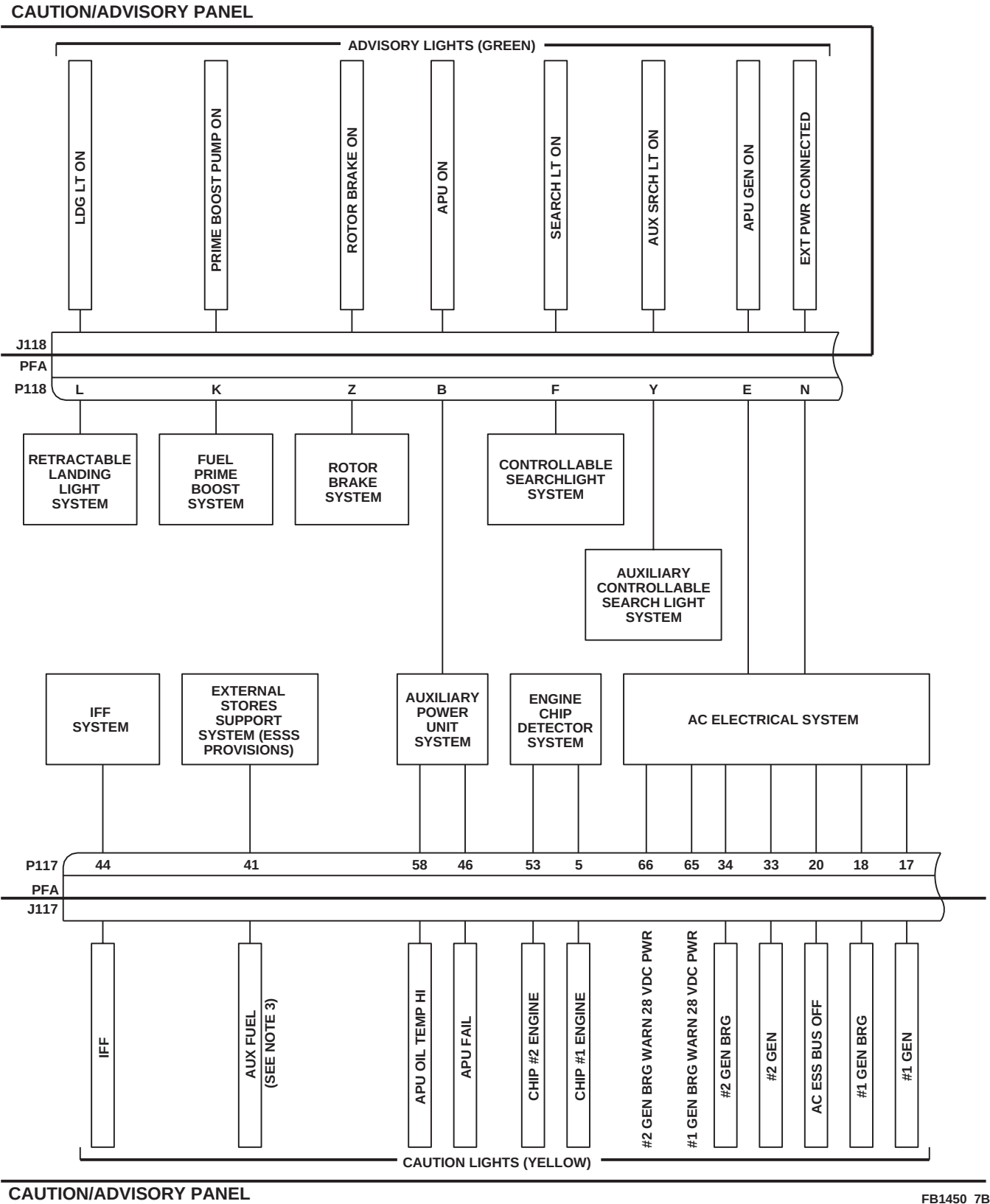


FIGURE 8-2-4-1. CAUTION/ADVISORY WARNING SYSTEM SCHEMATIC DIAGRAM (SHEET 7 OF 10)

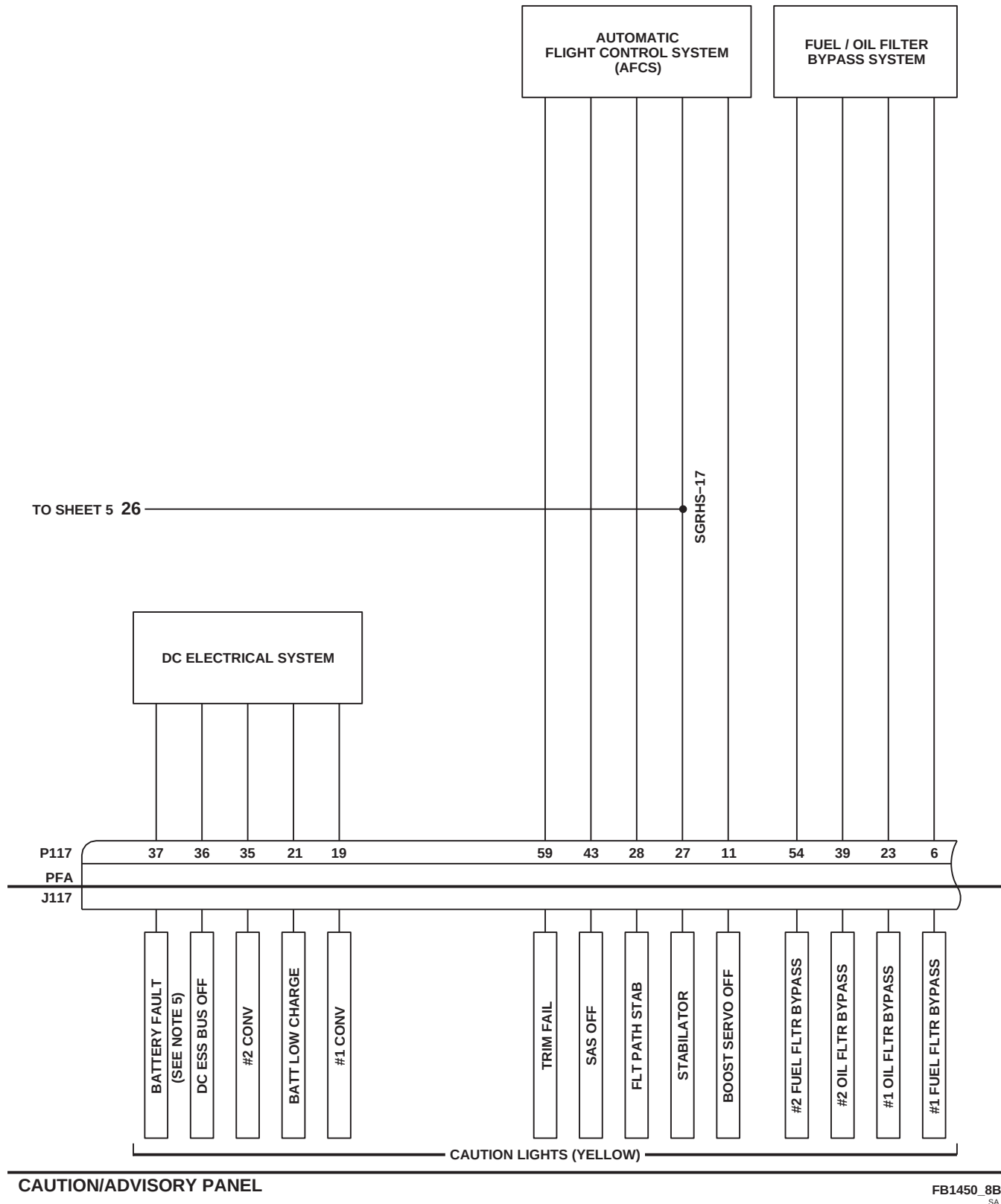


FIGURE 8-2-4-1. CAUTION/ADVISORY WARNING SYSTEM SCHEMATIC DIAGRAM
(SHEET 8 OF 10)

8-2-4-22

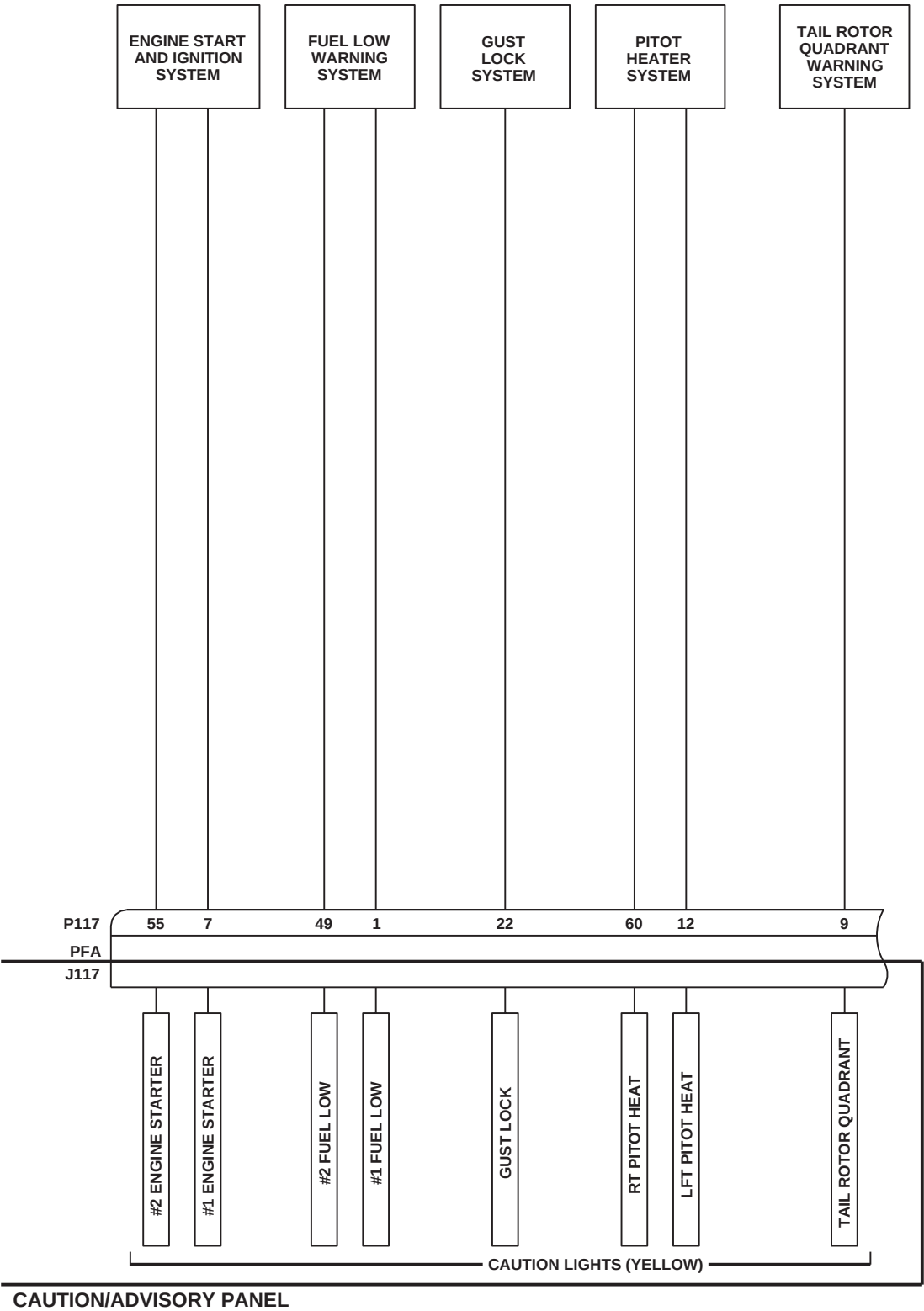


FIGURE 8-2-4-1. CAUTION/ADVISORY WARNING SYSTEM SCHEMATIC DIAGRAM (SHEET 9 OF 10)

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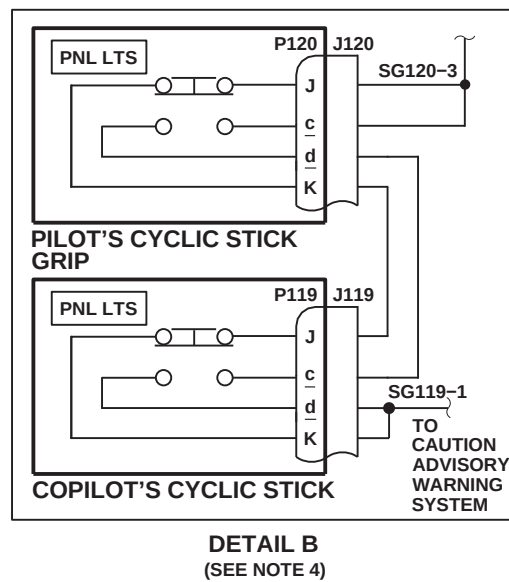


FIGURE 8-2-4-1. CAUTION/ADVISORY WARNING SYSTEM SCHEMATIC DIAGRAM
(SHEET 10 OF 10)

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8-2-5 DIGITAL CLOCK.

INITIAL SETUP

Tools

- Electronic Equipment Toolkit
- Multimeter, Fluke 27/AN

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or APU Operational

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

8-2-5.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).

RESULT:

- Pilot's and copilot's digital clock shall go into self-test mode and display all digits during the first 5 seconds of operation.
- At the end of the self-test cycle, pilot's and copilot's clock shall display a "C" on the clock face, indicating clock time mode.
 - If only some segments illuminate, replace affected clock (PARA 8-4-28).
 - If results (1) and (2) are not as specified, go to TABLE NO. 8-2-5-1 for pilot's clock or TABLE NO. 8-2-5-2 for copilot's clock.
- Press SEL and CTRL buttons simultaneously. When both hour digits flash, hold down CTRL button until desired hour is reached.
- Press SEL button. When both minute hands are flashing, hold down CTRL button until desired minutes are reached.

- Press SEL button. When both second digits are flashing, hold down CTRL button until desired seconds are reached.
- Press SEL button.

RESULT: Selected time shall be displayed and clock timer shall begin.

- If result is not as specified, replace digital clock (PARA 8-4-28).
- With digital clock in clock mode, press SEL to initiate elapsed time mode.
 - Press CTRL button.

RESULT: Elapsed timer shall begin.

- If result is not as specified, replace digital clock (PARA 8-4-28).
- Press CTRL button.

RESULT: Elapsed timer shall stop.

- If result is not as specified, replace digital clock (PARA 8-4-28).
- Press CTRL button again.

RESULT: Elapsed timer shall reset.

- If result is not as specified, replace digital clock (PARA 8-4-28).

k. Press CTRL button again.

RESULT: Elapsed timer shall begin at 0.0 seconds.

- If result is not as specified, replace digital clock (PARA 8-4-28).

l. Push circuit breakers back in.

RESULT: Clocks shall display correct time after approximately 5 seconds.

- If results are not as specified, replace clock battery (PARA 8-4-29).
- m. Turn off electrical power (PARA 1-6-16).

TABLE NO. 8-2-5-1. Pilot's Clock Does Not Function.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 vdc between P186-1 and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3.
2. Check continuity between: P186-2 and ground P186-5 and ground	Step 1. If continuity is present, replace digital clock (PARA 8-4-28). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
3. Check for 28 vdc between terminal 1 (PILOT ALTM circuit breaker) and ground.	Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P186-1 (Appendix F). Go to 4. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 4.
4. Procedure completed.	

TABLE NO. 8-2-5-2. Copilot's Clock Does Not Function.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 vdc between P195-1 and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3.

TABLE NO. 8-2-5-2. Copilot's Clock Does Not Function. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
2. Check continuity between:	P195-2 and ground
	P195-5 and ground
	Step 1. If continuity is present, replace digital clock (PARA 8-4-28). Go to 4.
	Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
3. Check for 28 vdc between terminal 1 (CPLT ALTM circuit breaker) and ground.	
	Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and
	P195-1 (Appendix F). Go to 4.
	Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 4.
4. Procedure completed.	

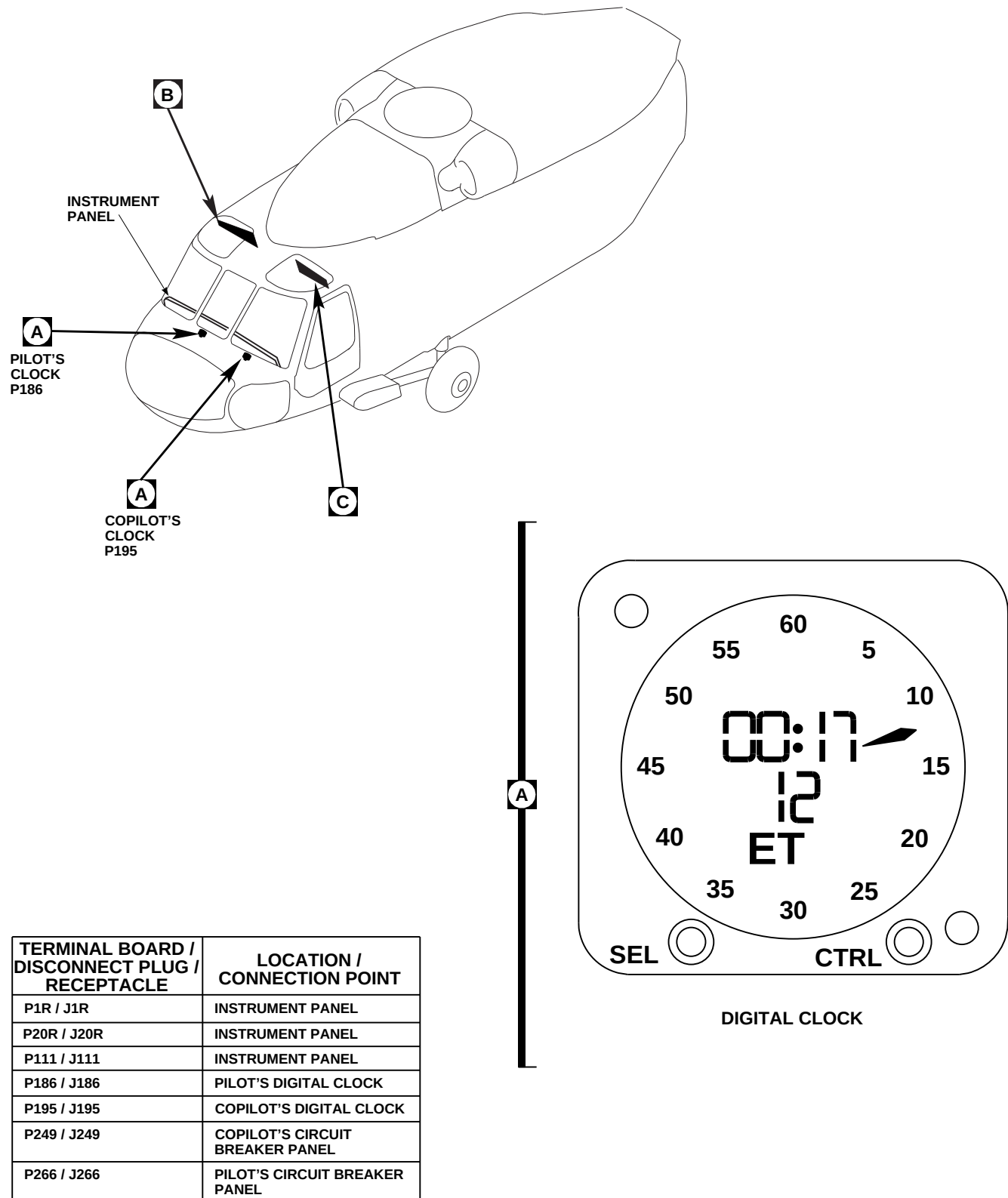
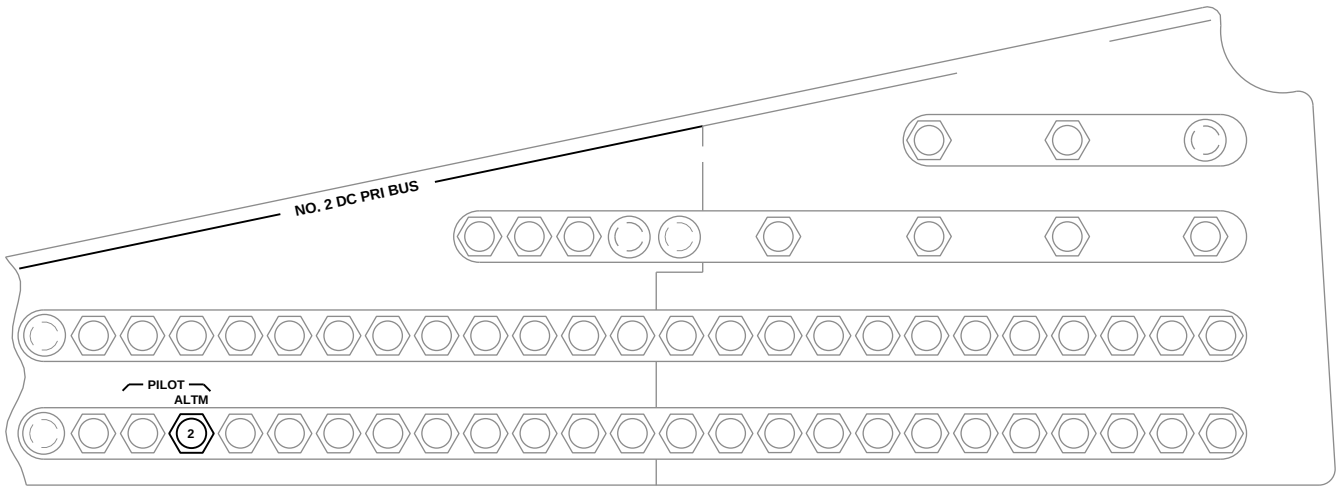


FIGURE 8-2-5-1. DIGITAL CLOCK LOCATION DIAGRAM (SHEET 1 OF 2)

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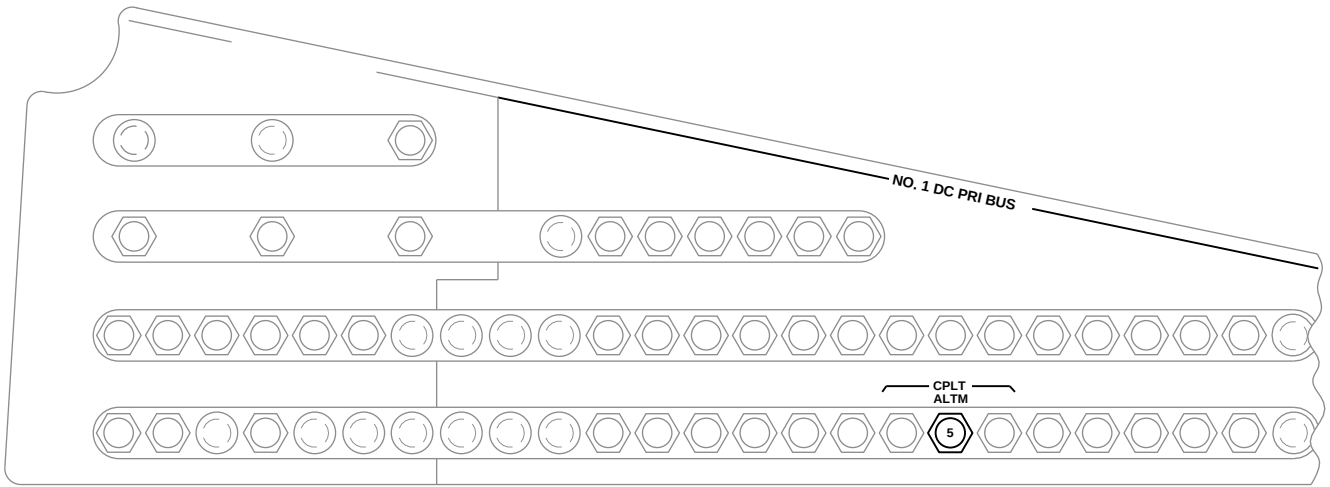
8-2-5-4



PILOT CIRCUIT BREAKER PANEL

B

C



COPILOT CIRCUIT BREAKER PANEL

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FIGURE 8-2-5-1. DIGITAL CLOCK LOCATION DIAGRAM (SHEET 2 OF 2)

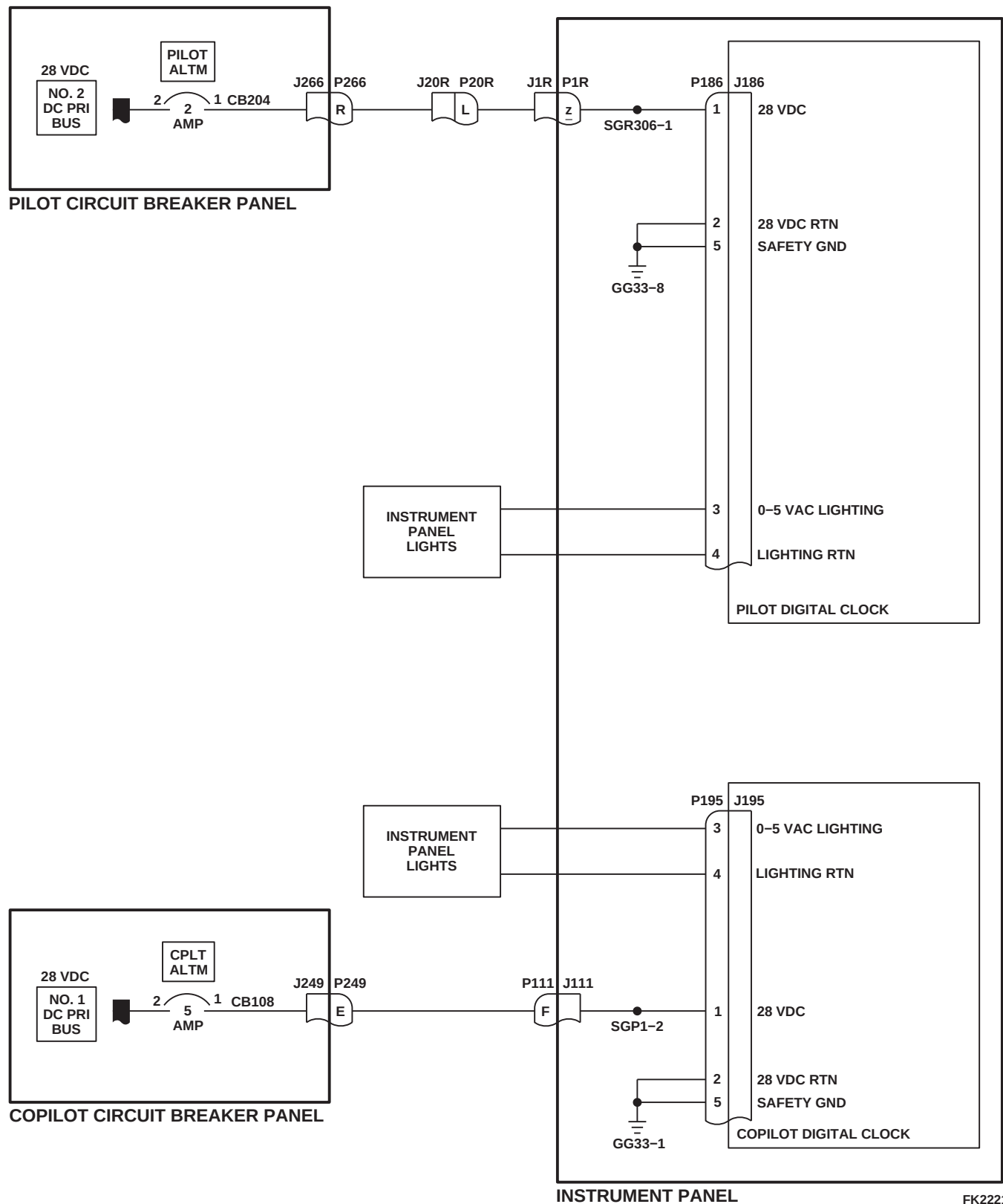
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FIGURE 8-2-5-2. DIGITAL CLOCK SCHEMATIC DIAGRAM

8-2-6 CENTRAL DISPLAY UNIT (70450-01043-122) (BENCH).

INITIAL SETUP

Applicable Configuration

- Central Display Unit 70450-01043-122

Test Equipment

- Bench Test Set, 100-601165-000
- Multimeter, Fluke 27/AN

Tools

- Electrical Repairer Toolkit

Equipment Conditions

- Bench Power Available, 28 vdc, 115 vac, 400 Hz, Single-Phase

Special Instructions

The central display unit (CDU) operational/troubleshooting procedure is subdivided as follows:

- Setup/Power On

- Digital and Analog Update
- CDU Push To Test
- Display Auto-Dim
- Logic Power Supply Redundancy
- Regulator Voltage And Single SDC
- Analog Parameter Indication
- Digital Parameter Indication
- No. 1 Fuel Quantity (Analog)
- No. 2 Fuel Quantity (Analog)
- Main Transmission Oil Temperature (Analog)
- Main Transmission Oil Pressure (Analog)
- Parameter Measurement (Digital)
- Failure Mode
- Shutdown

8-2-6.1 Setup/Power On.

- Make sure simulator tester INPUT POWER switch is OFF.
- Connect cable assembly, 217-419742-000, between simulator tester INTERCONNECT connector and unit tester J5.
- Connect simulator tester dc power connector to 28 vdc bench power.
- Connect simulator tester ac power connector to 115 vac, 400 Hz, single-phase bench power.
- Place simulator tester INPUT POWER switch to ON.

- RESULT: Simulator tester INPUT POWER 115 VAC and 28 VDC lamps shall go on.
- If results are not as specified, troubleshoot bench test set (TM 55-4920-412-13&P).
- Momentarily press simulator tester LAMP TEST switch.

RESULT:

- All simulator tester lamps shall go on when LAMP TEST switch is pressed.
- All unit tester lamps shall go on when LAMP TEST switch is pressed.

- If results are not as specified, troubleshoot bench test set (TM 55-4920-412-13&P).

- g. Place simulator tester INPUT POWER switch to OFF.
- h. Connect cable assembly, 217-419744-000, between unit tester J1 and CDU J1.
- i. Connect cable assembly, 217-419748-000, between unit tester J2 and CDU J2.
- j. Connect cable assembly, 217-419748-000, between unit tester J3 and CDU J3.
- k. Connect cable assembly, 217-419744-000, between unit tester J4 and CDU J4.

- l. Place simulator tester switches as follows:

All CAL-NOR switches	NOR
All HI-LO switches	LO
All NOR controls	Fully counterclockwise
INTEGRAL LIGHTING	INT

- m. Place unit tester DISPLAY CONTROLS section switches and controls as follows:

DIGITAL SET	0
DIGITS	ON
DISPLAY UPDATE	ON
ANALOG SET	STOP
ROTOR OVERSPEED 127%	Off (down)
ROTOR OVERSPEED 137%	Off (down)
ROTOR OVERSPEED 142%	Off (down)

- n. Place unit tester SIGNAL DATA CONVERTER CONVERSION switch to 1.
- o. Place unit tester CENTRAL DISPLAY UNIT section switches and controls as follows:

PILOTS	Fully counterclockwise
COPILOTS	Fully counterclockwise
P/S CONT	NORMAL
SDC FAIL	Off (center)

- p. Position unit tester INTER CAL SELECT switch to OFF.
- q. Place CDU - DIGITS switch to ON and turn DIM control fully clockwise.
- r. Place simulator tester INPUT POWER switch to ON.

RESULT:

- (1) Simulator tester INPUT POWER 115 VAC and 28 VDC lamps shall go on.
 - (2) After approximately 3 seconds, CENTRAL DISPLAY UNIT SENSING A and B lamps on unit tester shall be on.
 - (3) CDU integral panel lighting shall go on without any dark spots.
- If result (2) is not as specified, CDU chassis assembly is malfunctioning.
 - If result (3) is not as specified, replace CDU faceplate (PARA 8-4-11).

If trouble remains, the CDU chassis assembly is malfunctioning.

8-2-6.2 Digital And Analog Update.

- a. Do Setup/Power On, if not already done.
- b. Place unit tester switches as follows:

INTER CAL SELECT	21
DISPLAY UPDATE	ON
DIGITAL SET	5
DIGITS	ON
ANALOG SET	FAST

8-2-6-2

RESULT:

- (1) All CDU vertical scales shall continually cycle from minimum to maximum.
 - (2) TGT 1, TGT 2, Ng SPEED 1, and Ng SPEED 2 digital readouts shall indicate 555.
 - (3) TOTAL FUEL digital readout shall indicate 1550.
- If results are not as specified, record result obtained and continue with operational/troubleshooting procedure.
- c. If no other check is to be done, go to Shutdown.

8-2-6.3 CDU Push To Test.

- a. Do Setup/Power On, if not already done.
- b. Press, then release CDU - PUSH TO TEST switch.

RESULT: With PUSH TO TEST switch pressed:

- (1) All vertical scale lamps shall go on.
 - (2) TGT 1, TGT 2, Ng SPEED 1, and Ng SPEED 2 digital readouts shall indicate 888.
 - (3) TOTAL FUEL digital readout shall indicate 8888.
 - (4) CHAN 1 and CHAN 2 lamps shall go on.
 - (5) Unit tester 127%, 137%, and 142% ROTOR OVERSPEED lamps shall be on.
- If none of the results are as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (1) is not as specified, replace display unit lamp assembly (PARA 8-4-14)

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (2) or (3) is not as specified, replace digital readout module A18 (PARA 8-5-3).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (4) is not as specified, replace digital readout module A18 (PARA 8-5-3).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (5) is not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- c. If no other check is to be done, go to Shutdown.

8-2-6.4 Display Auto-Dim.

- a. Do Setup/Power On, if not already done.

NOTE

Make sure unit tester INTER CAL SELECT switch is placed to 21.

- b. Turn CDU - DIM control fully counterclockwise.

RESULT: CDU display lighting shall go off.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

NOTE

Opaque object must remain in place for steps c. through l.

- c. Cover light sensor, located between CHAN lamps on CDU, with an opaque object.

- d. Set multimeter to indicate dc voltage and connect leads between PDU AD/O and SDC COM test jacks on unit tester.

RESULT: Multimeter shall indicate less than 0.5 vdc.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- e. Turn CDU - DIM control clockwise, but not enough to activate the auto-dim override switch.

RESULT: Multimeter shall indicate less than 1.75 vdc.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- f. Connect voltmeter between CDU AD/P and SDC COM test jacks on unit tester.

- g. Adjust unit tester PILOTS control to obtain between 1.95 to 2.05 vdc on voltmeter.

RESULT: CDU display lighting shall go on.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- h. Connect multimeter between PDU AD/O and SDC COM test jacks on unit tester.

RESULT: Multimeter shall indicate between 3.3 and 4.1 vdc.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- i. Turn unit tester COPILOTS control fully clockwise.

RESULT: Multimeter shall indicate between 4.9 and 5.9 vdc.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- j. Slowly turn CDU - DIM control fully counterclockwise.

RESULT: CDU display lighting intensity shall gradually dim.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- k. Turn unit tester PILOTS and COPILOTS controls fully counterclockwise.

- l. Position CDU - DIM control to mid-position.

- m. Remove opaque object from light sensor.

- n. If no other check is to be done, go to Shutdown.

8-2-6.5 Logic Power Supply Redundancy.

- a. Do Setup/Power On, if not already done.

- b. Place unit tester LOGIC P/S switch to 1.

- c. Connect multimeter between SDC +15.5V and COM test jacks on unit tester.

RESULT: Multimeter shall indicate between 14 and 21 vdc.

- If result is not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

8-2-6-4

- d. Connect multimeter between SDC -15.5V and COM test jacks on unit tester.

RESULT: Multimeter shall indicate between -21 and -14 vdc.

- If result is not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- e. Connect multimeter between SDC 8V and COM test jacks on unit tester.

RESULT: Multimeter shall indicate between 6.5 and 10.5 vdc.

- If result is not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- f. Place unit tester LOGIC P/S switch to 2.

- g. Repeat steps c. through e.

- h. If no other check is to be done, go to Shutdown.

8-2-6.6 Regulator Voltage And Single SDC.

- a. Do Setup/Power On, if not already done.

- b. Place unit tester LOGIC P/S switch to 1.

- c. Connect multimeter between CDU 5V and SDC COM test jacks on unit tester.

RESULT: Multimeter shall indicate between 4.75 and 5.25 vdc.

- If result is not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- d. Connect multimeter between CDU +10 and SDC COM test jacks on unit tester.

RESULT: Multimeter shall indicate between 9.5 and 10.5 vdc.

- If result is not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- e. Connect multimeter between CDU -10 and SDC COM test jacks on unit tester.

RESULT: Multimeter shall indicate between -10.5 and -9.5 vdc.

- If result is not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- f. Disconnect multimeter.

- g. If no other check is to be done, go to Shutdown.

8-2-6.7 Analog Parameter Indication.

- a. Do Setup/Power On, if not already done.

- b. Place simulator tester INPUT POWER switch to ON.

NOTE

Cycling rate of vertical scales is controlled by unit tester ANALOG SET switch.

- c. Place unit tester ANALOG SET switch to SLOW.

RESULT:

- (1) FUEL QTY 1 vertical scale on CDU shall continually cycle from 0 to full scale.
- (2) At about 250 lbs indication, FUEL QTY 1 lower amber scale segment on CDU shall go off.

- (3) FUEL QTY 2 vertical scale on CDU shall continually cycle from 0 to full scale.
 - (4) At about 250 lbs indication, FUEL QTY 2 lower amber scale segment on CDU shall go off.
 - (5) XMSN TEMP vertical scale on CDU shall continually cycle from -50°C (-58°F) to full scale.
 - (6) XMSN PRESS vertical scale on CDU shall continually cycle from 0 to full scale.
 - (7) At about 37.5 psi, XMSN PRESS lower red and amber scale segments on CDU shall go off.
 - (8) ENG OIL TEMP 1 vertical scale on CDU shall continually cycle from -50°C (-58°F) to full scale.
 - (9) Unit tester HIGH OIL 1 TEMP lamp shall go on whenever scale indication is greater than 150°C (302°F).
 - (10) ENG OIL TEMP 2 vertical scale on CDU shall continually cycle from -50°C (-58°F) to full scale.
 - (11) Unit tester HIGH OIL 2 TEMP lamp shall go on when scale indication is greater than 150°C (302°F).
 - (12) ENG OIL PRESS 1 vertical scale on CDU shall continually cycle from 10 or 12 psi to full scale.
 - (13) At about 47.5 psi indication, ENG OIL PRESS 1 lower red and amber scale segments on CDU shall go off.
 - (14) Unit tester LOW OIL 1 PRESS lamp shall be on when scale indication is less than 25 psi.
 - (15) ENG OIL PRESS 2 vertical scale on CDU shall continually cycle from 10 or 12 psi to full scale.
 - (16) At about 47.5 psi indication, ENG OIL PRESS 2 lower red and amber scale segments on CDU shall go off.
 - (17) Unit tester LOW OIL 2 PRESS lamp shall be on when scale indication is less than 25 psi.
 - (18) TGT TEMP 1 vertical scale on CDU shall continually cycle from 0 to full scale.
 - (19) TGT TEMP 2 vertical scale on CDU shall continually cycle from 0 to full scale.
 - (20) Ng SPEED 1 vertical scale on CDU shall continually cycle from 0 to full scale.
 - (21) Unit tester LOW ENG 1 OUT lamp shall be on when scale indication is less than 55%.
 - (22) Ng SPEED 2 vertical scale on CDU shall continually cycle from 0 to full scale.
 - (23) Unit tester LOW ENG 2 OUT lamp shall be on when scale indication is less than 55%.
- If result (1) is not as specified, replace display driver module A3 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.
 - If result (2) is not as specified, replace display driver module A3 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.
 - If result (3) is not as specified, replace display driver module A4 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.
 - If result (4) is not as specified, replace display driver module A4 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (5) is not as specified, replace display driver module A5 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (6) is not as specified, replace display driver module A6 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (7) is not as specified, replace display driver module A6 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (8) is not as specified, replace display driver module A7 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (9) is not as specified, check CDU OIL TEMP 1 fuse.

(a) If fuse is not good, replace fuse (PARA 8-4-13).

(b) If fuse is good, replace voltage regulator module A17 (PARA 8-5-2).

(c) If trouble remains, the CDU chassis assembly is malfunctioning.

- If HIGH OIL 1 TEMP lamp does not go off when scale indication is less than 150°C (302°F), replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (10) is not as specified, replace display driver module A8 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (11) is not as specified, check CDU OIL TEMP 2 fuse.

(a) If fuse is not good, replace fuse (PARA 8-4-13).

(b) If fuse is good, replace voltage regulator module A17 (PARA 8-5-2).

(c) If trouble remains, the CDU chassis assembly is malfunctioning.

- If HIGH OIL 2 TEMP lamp does not go off when scale indication is less than 150°C (302°F), replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (12) is not as specified, replace display driver module A9 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (13) is not as specified, replace display driver module A9 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (14) is not as specified, check CDU OIL PRESS 1 fuse.

(a) If fuse is not good, replace fuse (PARA 8-4-13).

(b) If fuse is good, replace voltage regulator module A17 (PARA 8-5-2).

(c) If trouble remains, the CDU chassis assembly is malfunctioning.

- If LOW OIL 1 PRESS lamp does not go off when scale indication is more than 25 psi, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (15) is not as specified, replace display driver module A10 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (16) is not as specified, replace display driver module A10 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (17) is not as specified, check CDU OIL PRESS 2 fuse.

(a) If fuse is not good, replace fuse (PARA 8-4-13).

(b) If fuse is good, replace voltage regulator module A17 (PARA 8-5-2).

(c) If trouble remains, the CDU chassis assembly is malfunctioning.

- If LOW OIL 2 PRESS lamp does not go off when scale indication is more than 25 psi, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (18) is not as specified, replace display driver module A11 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (19) is not as specified, replace display driver module A12 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (20) is not as specified, replace display driver module A13 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (21) is not as specified, check CDU Ng 1 fuse.

(a) If fuse is not good, replace fuse (PARA 8-4-13).

(b) If fuse is good, replace voltage regulator module A17 (PARA 8-5-2).

(c) If trouble remains, the CDU chassis assembly is malfunctioning.

- If LOW ENG 1 OUT lamp does not go off when scale indication is more than 55%, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (22) is not as specified, replace display driver module A14 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (23) is not as specified, check CDU Ng 2 fuse.

(a) If fuse is not good, replace fuse (PARA 8-4-13).

(b) If fuse is good, replace voltage regulator module A17 (PARA 8-5-2).

(c) If trouble remains, the CDU chassis assembly is malfunctioning.

- If LOW ENG 2 OUT lamp does not go off when scale indication is more than 55%, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- d. Place unit tester ANALOG SET switch to STOP when unit tester INTERFACE MONITOR indicates less than 700.
- e. Momentarily press unit tester RESET switch.

RESULT:

- (1) Unit tester LOW ROTOR SPEED lamp shall be on.
- (2) Unit tester 127%, 137%, and 142% ROTOR OVERSPEED lamps shall go off.
- If result (1) is not as specified, check CDU RTR SPD fuse.
 - (a) If fuse is not good, replace fuse (PARA 8-4-13).
 - (b) If fuse is good, replace voltage regulator module A17 (PARA 8-5-2).
 - (c) If trouble remains, the CDU chassis assembly is malfunctioning.
- If result (2) is not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- f. Place unit tester ANALOG SET switch to SLOW.

RESULT:

- (1) Unit tester LOW ROTOR SPEED lamp shall go off when unit tester INTERFACE MONITOR indicates more than 950.
- (2) Unit tester 127% ROTOR OVERSPEED lamp shall go on when INTERFACE MONITOR initially indicates between 1269 and 1271, and shall remain on during subsequent INTERFACE MONITOR cycles.
- (3) Unit tester 137% ROTOR OVERSPEED lamp shall go on when INTERFACE

MONITOR initially indicates between 1369 and 1371, and shall remain on during subsequent INTERFACE MONITOR cycles.

- (4) Unit tester 142% ROTOR OVERSPEED lamp shall go on when INTERFACE MONITOR initially indicates between 1419 and 1421, and shall remain on during subsequent INTERFACE MONITOR cycles.

- If result (1) is not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If results (2) through (4) are not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- g. Place unit tester ANALOG SET switch to STOP when unit tester INTERFACE MONITOR indicates less than 1200.

- h. Momentarily press unit tester RESET switch.

RESULT: Unit tester 127%, 137%, and 142% ROTOR OVERSPEED lamps shall go off.

- If result is not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- i. If no other check is to be done, go to Shutdown.

8-2-6.8 Digital Parameter Indication.

- a. Do Setup/Power On, if not already done.
- b. Set unit tester DIGITAL SET thumbwheel to the following positions:

DIGITAL SET	TOTAL FUEL	1 TGT 2 1 Ng SPEED 2
0	0	0
1	1110	111
2	2220	222
3	3330	333
4	440	444
5	1550	555
6	2660	666
7	3770	777
8	880	888
9	1990	999

RESULT: CDU TOTAL FUEL, TGT 1, TGT 2, Ng SPEED 1, and Ng SPEED 2 digital readouts shall indicate corresponding numbers.

- If result is not as specified, replace digital readout module A18 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- c. Place CDU - DIGITS switch to OFF.

RESULT: CDU digital readouts shall go off.

- If result is not as specified, replace digital readout module A18 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- d. If no other check is to be done, go to Shutdown.

8-2-6.9 No. 1 Fuel Quantity (Analog).

- Do Setup/Power On, if not already done.
- Place CDU - DIGITS switch to ON.

- Place simulator tester NO. 1 FUEL QUANTITY CAL-NOR switch to CAL and HI-LO switch to HI.

- Position unit tester INTER CAL SELECT switch to 15 and DISPLAY UPDATE SWITCH to OFF.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 1216 and 1234.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- Place simulator tester NO. 1 FUEL QUANTITY HI-LO SWITCH to LO.

RESULT: Unit tester INTERFACE MONITOR shall indicate 0 and 9.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- Place simulator tester NO. 1 FUEL QUANTITY CAL-NOR switch to NOR.

- Slowly turn simulator tester NO. 1 FUEL QUANTITY variable control fully clockwise.

RESULT: Unit tester INTERFACE MONITOR count shall increase smoothly.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- Slowly turn simulator tester NO. 1 FUEL QUANTITY variable control fully counterclockwise.

RESULT: Unit tester INTERFACE MONITOR count shall decrease smoothly.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

8-2-6-10

If trouble remains, the CDU chassis assembly is malfunctioning.

- i. If no other check is to be done, go to Shutdown.

8-2-6.10 No. 2 Fuel Quantity (Analog).

- a. Do Setup/Power On, if not already done.
- b. Place simulator tester NO. 2 FUEL QUANTITY CAL-NOR switch to CAL and HI-LO switch to HI.
- c. Position unit tester INTER CAL SELECT switch to 16.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 1216 and 1234.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- d. Place simulator tester NO. 2 FUEL QUANTITY HI-LO switch to LO.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 0 and 9.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- e. Place simulator tester NO. 2 FUEL QUANTITY CAL-NOR switch to NOR.
- f. Slowly turn simulator tester NO. 2 FUEL QUANTITY variable control fully clockwise.

RESULT: Unit tester INTERFACE MONITOR count shall increase smoothly.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- g. Slowly turn simulator tester No. 2 FUEL QUANTITY variable control fully counterclockwise.

RESULT: Unit tester INTERFACE MONITOR count shall decrease smoothly.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- h. If no other check is to be done, go to Shutdown.

8-2-6.11 Main Transmission Oil Temperature (Analog).

- a. Do Setup/Power On, if not already done.
- b. Place simulator tester MAIN XMSN OIL TEMP CAL-NOR switch to CAL and HI-LO switch to HI.
- c. Position unit tester INTER CAL SELECT switch to 17.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 1216 and 1264.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- d. Place simulator tester MAIN XMSN OIL TEMP HI-LO switch to LO.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 0 and 18.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

e. Place simulator tester MAIN XMSN OIL TEMP CAL-NOR switch to NOR.

f. Slowly turn simulator tester MAIN XMSN OIL TEMP variable control fully clockwise.

RESULT: Unit tester INTERFACE MONITOR count shall increase smoothly.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

g. Slowly turn simulator tester MAIN XMSN OIL TEMP variable control fully counterclockwise.

RESULT: Unit tester INTERFACE MONITOR count shall decrease smoothly.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

h. If no other check is to be done, go to Shutdown.

8-2-6.12 Main Transmission Oil Pressure (Analog).

a. Do Setup/Power On, if not already done.

b. Place simulator tester XMSN OIL PRESS CAL-NOR switch to CAL and HI-LO switch to HI.

c. Position unit tester INTER CAL SELECT switch to 18.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 1108 and 1132.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

d. Place simulator tester XMSN OIL PRESS HI-LO switch to LO.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 0 and 12.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

e. Place simulator tester XMSN OIL PRESS CAL-NOR switch to NOR.

f. Slowly turn simulator tester XMSN OIL PRESS variable control fully clockwise.

RESULT: Unit tester INTERFACE MONITOR count shall increase smoothly.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

g. Slowly turn XMSN OIL PRESS variable control fully counterclockwise.

RESULT: Unit tester INTERFACE MONITOR count shall decrease smoothly.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

h. If no other check is to be done, go to Shutdown.

8-2-6.13 Parameter Measurement (Digital).

a. Do Setup/Power On, if not already done.

b. Place simulator tester NO. 1 FUEL QUANTITY and NO. 2 FUEL QUANTITY CAL-NOR switches to CAL.

- c. Position unit tester INTER CAL SELECT switch to 19.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 0 and 7.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- d. Place simulator tester NO. 1 FUEL QUANTITY HI-LO switch to HI.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 1218 and 1232.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- e. Place simulator tester NO. 2 FUEL QUANTITY HI-LO switch to HI.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 2443 and 2457.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- f. Place simulator tester NO. 1 FUEL QUANTITY CAL-NOR switch to NOR.

- g. Slowly turn simulator tester NO. 1 FUEL QUANTITY variable control fully clockwise.

RESULT: Unit tester INTERFACE MONITOR count shall increase smoothly.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- h. Slowly turn simulator tester NO. 1 FUEL QUANTITY variable control fully counterclockwise.

RESULT: Unit tester INTERFACE MONITOR count shall decrease smoothly.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- i. Position unit tester INTER CAL SELECT switch to OFF.

- j. Place unit tester DISPLAY UPDATE switch to ON.

- k. If no other check is to be done, go to Shutdown.

8-2-6.14 Failure Mode.

- a. Do Setup/Power On, if not already done.

- b. Place unit tester SDC FAILURE switch to 1 and release.

RESULT: With SDC FAILURE switch to 1, CDU CHAN 1 lamp shall go on.

- If result is not as specified, the display unit chassis assembly is malfunctioning.

- c. Place unit tester SDC FAILURE switch to 2 and release.

RESULT: With SDC FAILURE switch to 2, CDU CHAN 2 lamp shall go on.

- If result is not as specified, the display unit chassis assembly is malfunctioning.

- d. Place unit tester DISPLAY UPDATE switch to OFF.

RESULT:

- (1) All CDU vertical displays shall blank out.

- (2) All CDU digital displays shall blank out.

(3) CHAN 1 and CHAN 2 lamps shall go on.

- If results are not as specified, CDU chassis assembly is malfunctioning.

e. Place unit tester ANALOG SET switch to STOP when INTERFACE MONITOR indicates less than 930.

f. Press, then release unit tester RESET switch.

RESULT: Unit tester 127%, 137%, and 142% lamps shall go off.

- If result is not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

g. If no other check is to be done, go to Shutdown.

8-2-6.15 Shutdown.

a. Place simulator tester INPUT POWER switch to OFF.

b. Disconnect cable assemblies.

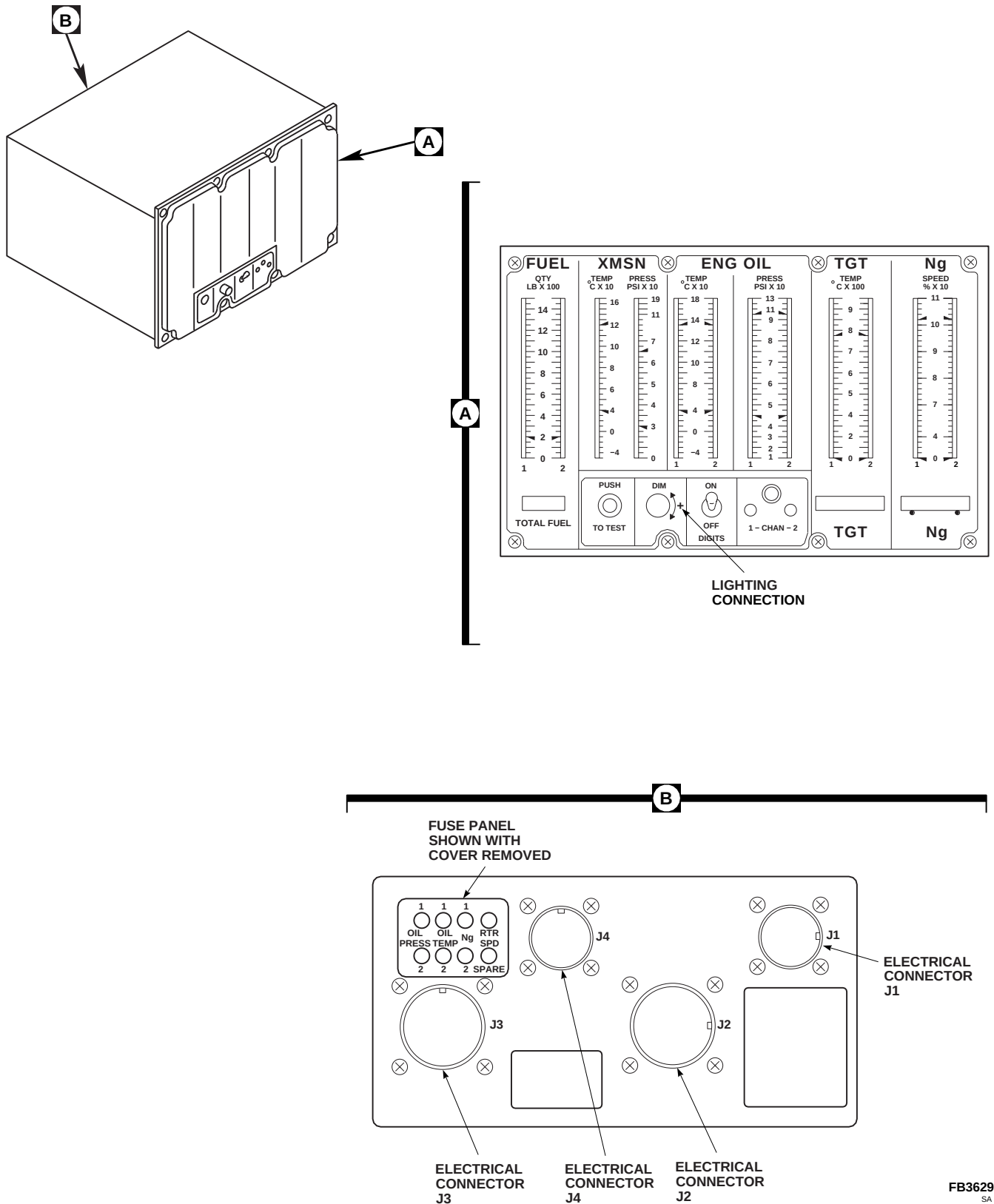


FIGURE 8-2-6-1. CENTRAL DISPLAY UNIT (CDU)

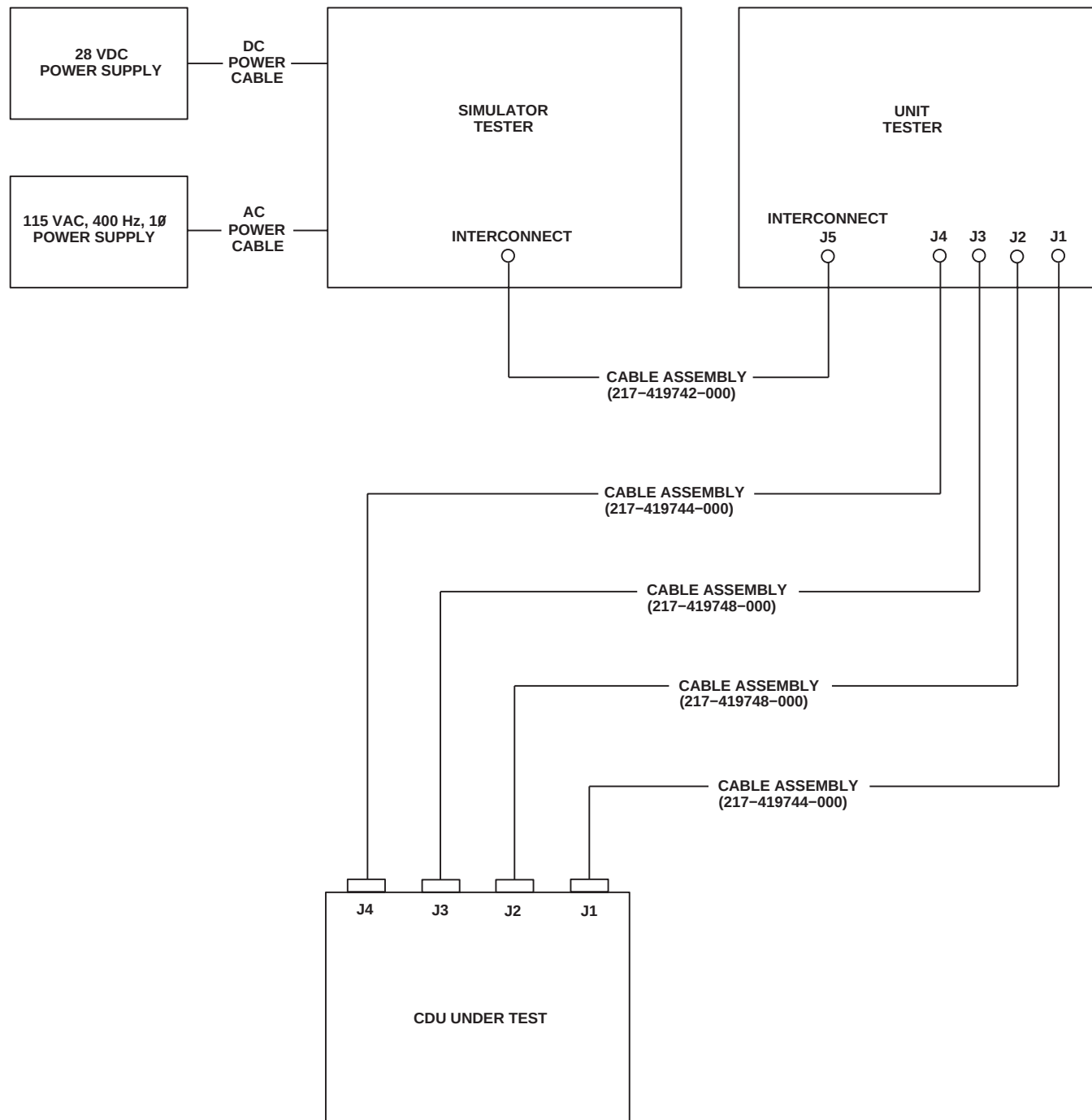
FI0972
SA

FIGURE 8-2-6-2. CENTRAL DISPLAY UNIT TEST SETUP

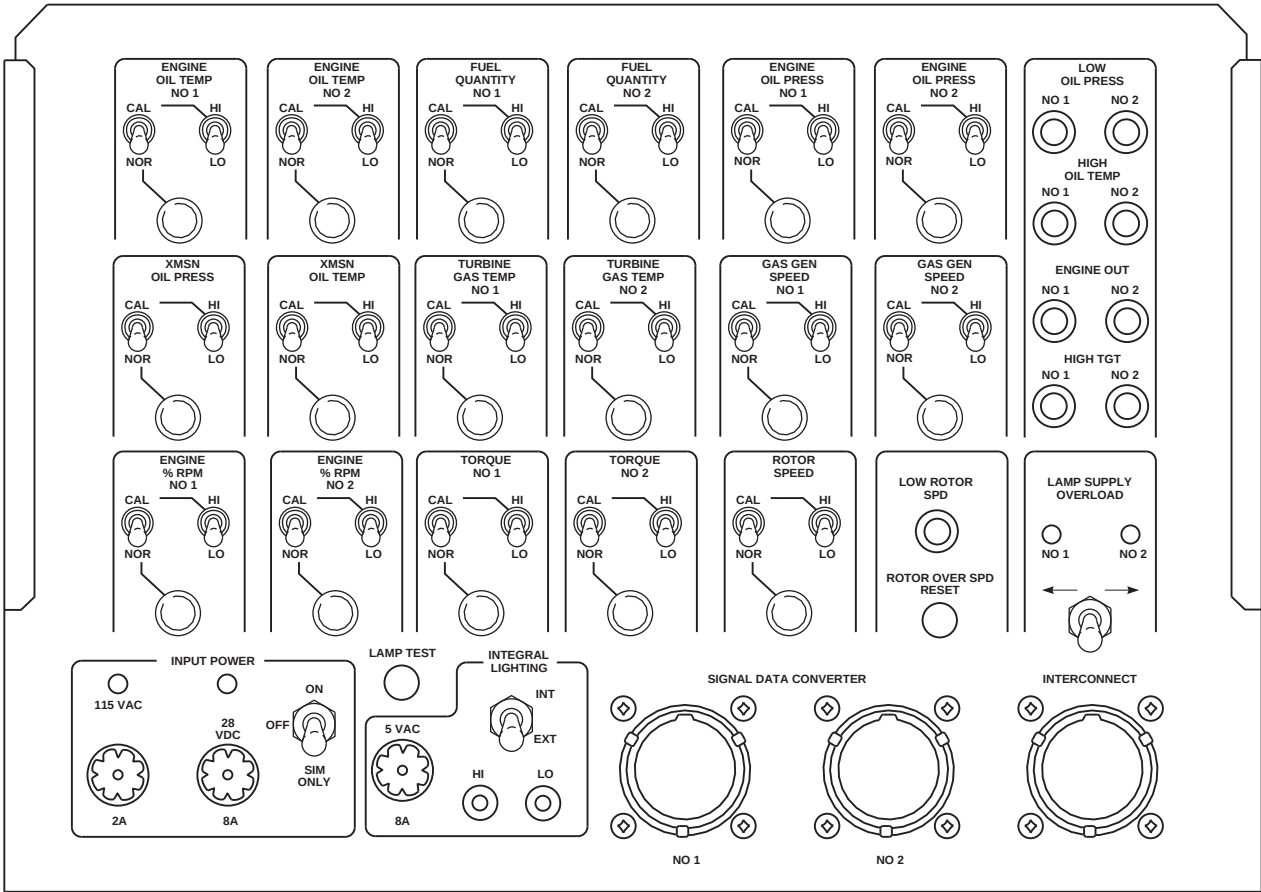


FIGURE 8-2-6-3. BENCH TEST SET (SHEET 1 OF 2)

AA1737
SA

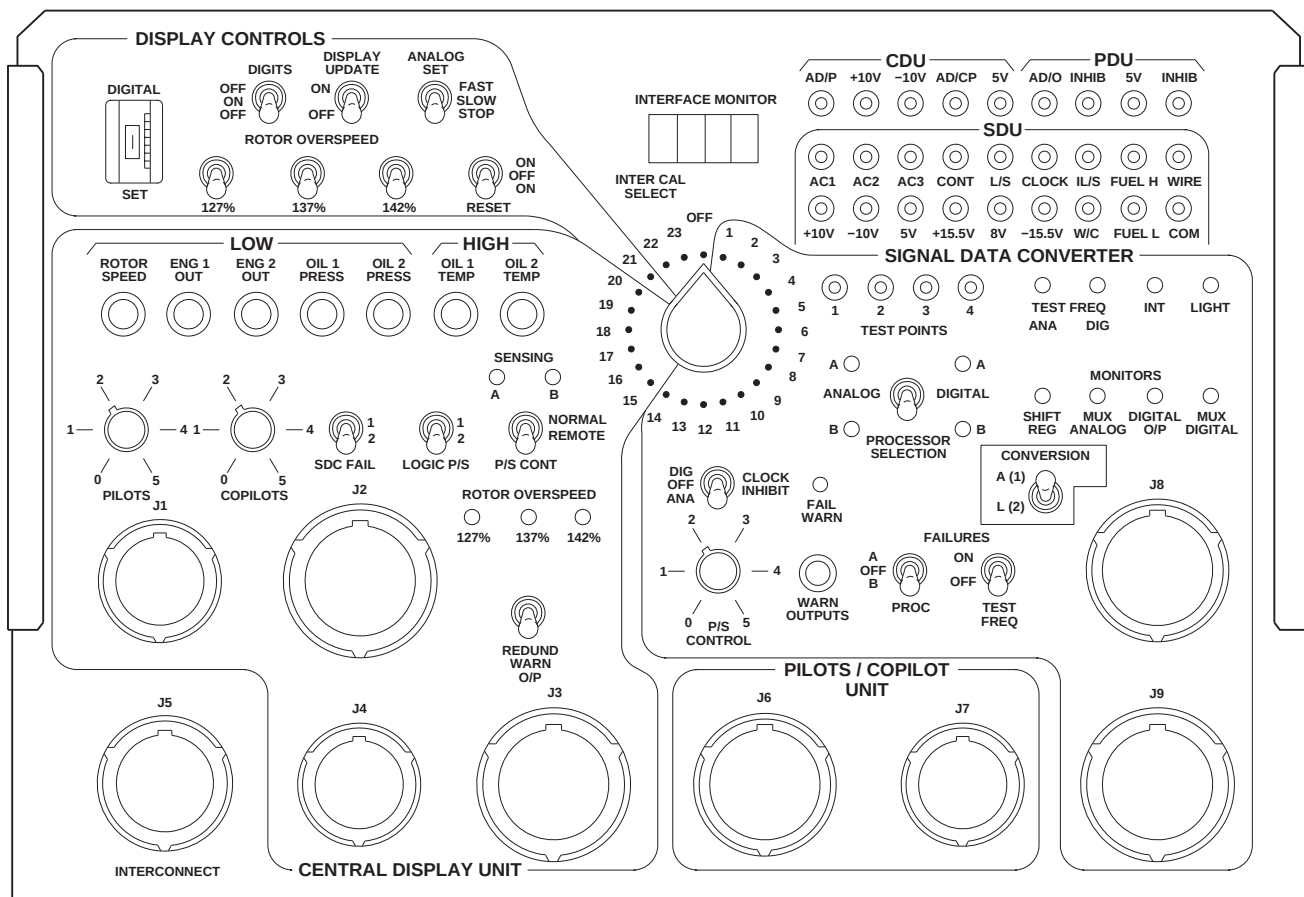


FIGURE 8-2-6-3. BENCH TEST SET (SHEET 2 OF 2)

AA1738
SA

8-2-7 CENTRAL DISPLAY UNIT (70450-01916-105) (BENCH).

INITIAL SETUP

Applicable Configuration

- Central Display Unit 70450-01916-105

Test Equipment

- Bench Test Set, 100-601165-000
- Multimeter, Fluke 27/AN

Tools

- Electrical Repairer Toolkit

Equipment Conditions

- Bench Power Available, 28 vdc, 115 vac, 400 Hz, Single-Phase

Special Instructions

The central display unit (CDU) operational/troubleshooting procedure is subdivided as follows:

- (1) Setup/Power On
- (2) Digital and Analog Update
- (3) CDU Lamp Test
- (4) Display Auto-Dim
- (5) Regulator Voltage And Single SDC
- (6) Analog Parameter Indication
- (7) Digital Parameter Indication
- (8) No. 1 Fuel Quantity (analog)
- (9) No. 2 Fuel Quantity (analog)
- (10) Main Transmission Oil Temperature (Analog)
- (11) Main Transmission Oil Pressure (Analog)
- (12) Parameter Measurement (Digital)
- (13) Failure Mode
- (14) Shutdown

8-2-7.1 Setup/Power On.

- Make sure simulator tester INPUT POWER switch is OFF.
- Connect cable assembly, 217-419742-000, between simulator tester INTERCONNECT connector and unit tester J5.
- Connect simulator tester dc power connector to 28 vdc bench power.
- Connect simulator tester ac power connector to 115 vac, 400 Hz, single-phase bench power.
- Place simulator tester INPUT POWER switch to ON.

- RESULT: Simulator tester INPUT POWER 115 VAC and 28 VDC lamps shall go on.
- If result is not as specified, troubleshoot bench test set (TM 55-4920-412-13&P).
- Press simulator tester LAMP TEST switch, then release.
- RESULT:
- All simulator tester lamps shall go on when LAMP TEST switch is pressed.
 - All unit tester lamps shall go on when LAMP TEST switch is pressed.
- If results are not as specified, troubleshoot bench test set (TM 55-4920-412-13&P).

- g. Place simulator tester INPUT POWER switch to OFF.
- h. Connect cable assembly, 217-419744-000, between unit tester J1 and CDU J1.
- i. Connect cable assembly, 217-419748-000, between unit tester J2 and CDU J2.
- j. Connect cable assembly, 217-419748-000, between unit tester J3 and CDU J3.
- k. Connect cable assembly, 217-419744-000, between unit tester J4 and CDU J4.
- l. Place simulator tester switches as follows:

All CAL-NOR switches	NOR
All HI-LO switches	LO
All NOR controls	Fully counterclockwise
INTEGRAL LIGHTING	INT

- m. Place unit tester DISPLAY CONTROLS section switches and controls as follows:

DIGITAL SET	0
DIGITS	ON
DISPLAY UPDATE	ON
ANALOG SET	STOP
ROTOR OVERSPEED 127%	Off (down)
ROTOR OVERSPEED 137%	Off (down)
ROTOR OVERSPEED 142%	Off (down)

- n. Place unit tester SIGNAL DATA CONVERTER CONVERSION switch to 2.
- o. Place unit tester CENTRAL DISPLAY UNIT section switches and controls as follows:

PILOTS	Fully counterclockwise
COPILOTS	Fully counterclockwise

P/S CONT	NORMAL
SDC FAIL	Center position

- p. Position unit tester INTER CAL SELECT switch OFF.
- q. Place CDU - DIGITS switch to ON and turn DIM control fully clockwise.
- r. Place simulator tester INPUT POWER switch to ON.

RESULT:

- (1) Simulator tester INPUT POWER 115 VAC and 28 VDC lamps shall go on.
 - (2) After approximately 3 seconds, CENTRAL DISPLAY UNIT SENSING A and B lamps on unit tester shall go on.
 - (3) CDU integral panel lighting shall go on without any dark spots.
- If result (2) is not as specified, CDU chassis assembly is malfunctioning.
 - If result (3) is not as specified, replace CDU faceplate (PARA 8-4-11).

If trouble remains, the CDU chassis assembly is malfunctioning.

8-2-7.2 Digital And Analog Update.

- Do Setup/Power On, if not already done.
- Place unit tester switches as follows:

INTER CAL SELECT	21
DISPLAY UPDATE	ON
DIGITAL SET	5
DIGITS	ON
ANALOG SET	FAST

RESULT:

- (1) All CDU vertical scales shall continually cycle from minimum to maximum.

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- (2) TGT 1, TGT 2, Ng SPEED 1, and Ng SPEED 2 digital readouts shall indicate 555.
- (3) MAIN FUEL digital readout shall indicate 1550.
- If results are not as specified, record result obtained and continue with operational/troubleshooting procedure.
- c. If no other check is to be done, go to Shutdown.

8-2-7.3 CDU Lamp Test.

- a. Do Setup/Power On, if not already done.
- b. Momentarily press CDU - LAMP TEST switch.

RESULT: With LAMP TEST switch pressed:

- (1) All vertical scale lamps shall go on.
- (2) TGT 1, TGT 2, Ng SPEED 1, and Ng SPEED 2 digital readouts shall indicate 888.
- (3) MAIN FUEL digital readout shall indicate 8888.
- (4) CHAN 1 and CHAN 2 lamps shall go on.
- (5) Unit tester 127%, 137%, and 142% ROTOR OVERSPEED lamps shall be on.
- If all results are not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (1) is not as specified, replace display unit lamp assembly (PARA 8-4-14).
- If trouble remains, the CDU chassis assembly is malfunctioning.
- If result (2) or (3) is not as specified, replace digital readout module A18 (PARA 8-5-3).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (4) is not as specified, replace digital readout module A18 (PARA 8-5-3).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (5) is not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- c. If no other check is to be done, go to Shutdown.

8-2-7.4 Display Auto-Dim.

- a. Do Setup/Power On, if not already done.

NOTE

Make sure unit tester INTER CAL SELECT switch is placed to 21.

- b. Turn CDU - DIM control fully counterclockwise.

RESULT: CDU display lighting shall go off.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

NOTE

Opaque object must remain in place for steps c. through l.

- c. Cover light sensor, located between CHAN lamps on CDU, with an opaque object.
- d. Set multimeter to indicate dc voltage and connect leads between PDU AD/O and SDC COM test jacks on unit tester.

RESULT: Multimeter shall indicate less than 0.5 vdc.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- Turn CDU - DIM control clockwise, but not enough to activate the auto-dim override switch.

RESULT: Multimeter shall indicate less than 1.75 vdc.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- Connect voltmeter between CDU AD/P and SDC COM test jacks on unit tester.
- Adjust unit tester PILOTS control to obtain between 1.95 to 2.05 vdc on voltmeter.

RESULT: CDU display lighting shall go on.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- Connect multimeter between PDU AD/O and SDC COM test jacks on unit tester.

RESULT: Multimeter shall indicate between 3.3 and 4.1 vdc.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- Turn unit tester COPILOTS control fully clockwise.

RESULT: Multimeter shall indicate between 4.9 and 5.9 vdc.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- Slowly turn CDU - DIM control fully counterclockwise.

RESULT: CDU display lighting intensity shall gradually dim.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

- Turn unit tester PILOTS and COPILOTS controls fully counterclockwise.

- Position CDU - DIM control to mid-position.

- Remove opaque object from light sensor.

- If no other check is to be done, go to Shutdown.

8-2-7.5 Regulator Voltage And Single SDC.

- Do Setup/Power On, if not already done.

- Place unit tester LOGIC P/S switch to 1.

- Connect multimeter between CDU 5V and SDC COM test jacks on unit tester.

RESULT: Multimeter shall indicate between 4.75 and 5.25 vdc.

- If result is not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- Connect multimeter between CDU +10 and SDC COM test jacks on unit tester.

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RESULT: Multimeter shall indicate between 9.5 and 10.5 vdc.

- If result is not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- e. Connect multimeter between CDU -10 and SDC COM test jacks on unit tester.

RESULT: Multimeter shall indicate between -10.5 and -9.5 vdc.

- If result is not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- f. Disconnect multimeter.

- g. If no other check is to be done, go to Shutdown.

8-2-7.6 Analog Parameter Indication.

- a. Do Setup/Power On, if not already done.
- b. Place simulator tester INPUT POWER switch to ON.

NOTE

Cycling rate of vertical scales is controlled by unit tester ANALOG SET switch.

- c. Place unit tester ANALOG SET switch to SLOW.

RESULT:

- (1) FUEL QTY 1 vertical scale shall continually cycle from 0 to full scale.
- (2) At approximately 250 lbs indication, FUEL QTY 1 lower amber scale segment shall go off.

- (3) FUEL QTY 2 vertical scale shall continually cycle from 0 to full scale.
- (4) At approximately 250 lbs indication, FUEL QTY 2 lower amber scale segment shall go off.
- (5) XMSN TEMP vertical scale shall continually cycle from -50°C (-58°F) to full scale.
- (6) XMSN PRESS vertical scale shall continually cycle from 0 to full scale.
- (7) At approximately 37.5 psi, XMSN PRESS lower red and amber scale segments shall go off.
- (8) ENG OIL TEMP 1 vertical scale shall continually cycle from -50°C (-58°F) to full scale.
- (9) Unit tester HIGH OIL 1 TEMP lamp shall go on whenever scale indication is greater than 150°C (302°F).
- (10) ENG OIL TEMP 2 vertical scale shall continually cycle from -50°C (-58°F) to full scale.
- (11) Unit tester HIGH OIL 2 TEMP lamp shall go on when scale indication is greater than 150°C (302°F).
- (12) ENG OIL PRESS 1 vertical scale shall continually cycle from 10 psi to full scale.
- (13) At approximately 47.5 psi indication, ENG OIL PRESS 1 lower red and amber scale segments shall go off.
- (14) Unit tester LOW OIL 1 PRESS lamp shall be on when scale indication is less than 24 psi.
- (15) ENG OIL PRESS 2 vertical scale shall continually cycle from 10 psi to full scale.

- (16) At approximately 47.5 psi indication, ENG OIL PRESS 2 lower red and amber scale segments shall go off.
- (17) Unit tester LOW OIL 2 PRESS lamp shall be on when scale indication is less than 24 psi.
- (18) TGT TEMP 1 vertical scale shall continually cycle from 0 to full scale.
- (19) TGT TEMP 2 vertical scale shall continually cycle from 0 to full scale.
- (20) Ng SPEED 1 vertical scale shall continually cycle from 0 to full scale.
- (21) Unit tester LOW ENG 1 OUT lamp shall be on when scale indication is less than 55%.
- (22) Ng SPEED 2 vertical scale shall continually cycle from 0 to full scale.
- (23) Unit tester LOW ENG 2 OUT lamp shall be on when scale indication is less than 55%.

- If result (1) is not as specified, replace display driver module A3 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (2) is not as specified, replace display driver module A3 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (3) is not as specified, replace display driver module A4 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (4) is not as specified, replace display driver module A4 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (5) is not as specified, replace display driver module A5 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (6) is not as specified, replace display driver module A6 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (7) is not as specified, replace display driver module A6 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (8) is not as specified, replace display driver module A7 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (9) is not as specified, check CDU OIL TEMP 1 fuse.

(a) If fuse is good, replace voltage regulator module A17 (PARA 8-5-2).

(b) If fuse is not good, replace fuse (PARA 8-4-13).

(c) If trouble remains, the CDU chassis assembly is malfunctioning.

- If HIGH OIL 1 TEMP lamp does not go off when scale indication is less than 150°C (302°F), replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (10) is not as specified, replace display driver module A8 (PARA 8-5-1).

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If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (11) is not as specified, check CDU OIL TEMP 2 fuse.
 - (a) If fuse is good, replace voltage regulator module A17 (PARA 8-5-2).
 - (b) If fuse is not good, replace fuse (PARA 8-4-13).
 - (c) If trouble remains, the CDU chassis assembly is malfunctioning.
- If HIGH OIL 2 TEMP lamp does not go off when scale indication is less than 150°C (302°F), replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (12) is not as specified, replace display driver module A9 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (13) is not as specified, replace display driver module A9 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (14) is not as specified, check CDU OIL PRESS 1 fuse.

(a) If fuse is good, replace voltage regulator module A17 (PARA 8-5-2).

(b) If fuse is not good, replace fuse (PARA 8-4-13).

(c) If trouble remains, the CDU chassis assembly is malfunctioning.

- If LOW OIL 1 PRESS lamp does not go off when scale indication is more than 25 psi, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (15) is not as specified, replace display driver module A10 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (16) is not as specified, replace display driver module A10 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (17) is not as specified, check CDU OIL PRESS 2 fuse.

(a) If fuse is good, replace voltage regulator module A17 (PARA 8-5-2).

(b) If fuse is not good, replace fuse (PARA 8-4-13).

(c) If trouble remains, the CDU chassis assembly is malfunctioning.

- If LOW OIL 2 PRESS lamp does not go off when scale indication is more than 25 psi, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (18) is not as specified, replace display driver module A11 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (19) is not as specified, replace display driver module A12 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (20) is not as specified, replace display driver module A13 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (21) is not as specified, check CDU Ng 1 fuse.
 - (a) If fuse is good, replace voltage regulator module A17 (PARA 8-5-2).
 - (b) If fuse is not good, replace fuse (PARA 8-4-13).
 - (c) If trouble remains, the CDU chassis assembly is malfunctioning.
- If LOW ENG 1 OUT lamp does not go off when scale indication is more than 55%, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (22) is not as specified, replace display driver module A14 (PARA 8-5-1).

If trouble remains, the CDU chassis assembly is malfunctioning.

- If result (23) is not as specified, check CDU Ng 2 fuse.
 - (a) If fuse is good, replace voltage regulator module A17 (PARA 8-5-2).
 - (b) If fuse is not good, replace fuse (PARA 8-4-13).
 - (c) If trouble remains, the CDU chassis assembly is malfunctioning.
- If LOW ENG 2 OUT lamp does not go off when scale indication is more than 55%, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- d. Place unit tester ANALOG SET switch to STOP when unit tester INTERFACE MONITOR indicates less than 700.
- e. Momentarily press unit tester RESET switch.

RESULT:

- (1) Unit tester LOW ROTOR SPEED lamp shall be on.
- (2) Unit tester 127%, 137%, and 142% ROTOR OVERSPEED lamps shall go off.
- If result (1) is not as specified, check CDU RTR SPD fuse.
 - (a) If fuse is good, replace voltage regulator module A17 (PARA 8-5-2).
 - (b) If fuse is not good, replace fuse (PARA 8-4-13).
 - (c) If trouble remains, the CDU chassis assembly is malfunctioning.
- If result (2) is not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- f. Place unit tester ANALOG SET switch to SLOW.

RESULT:

- (1) Unit tester LOW ROTOR SPEED lamp shall go off when unit tester INTERFACE MONITOR indicates more than 950.
- (2) Unit tester 127% ROTOR OVERSPEED lamp shall go on when INTERFACE MONITOR initially indicates between 1269 and 1271 and shall remain on during subsequent INTERFACE MONITOR cycles.
- (3) Unit tester 137% ROTOR OVERSPEED lamp shall go on when INTERFACE

- MONITOR initially indicates between 1369 and 1371 and shall remain on during subsequent INTERFACE MONITOR cycles.
- (4) Unit tester 142% ROTOR OVERSPEED lamp shall go on when INTERFACE MONITOR initially indicates between 1419 and 1421 and shall remain on during subsequent INTERFACE MONITOR cycles.
- If result (1) is not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.
- If results (2) through (4) are not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.
- g. Place unit tester ANALOG SET switch to STOP when unit tester INTERFACE MONITOR indicates less than 1200.
- h. Momentarily press unit tester RESET switch.

RESULT: Unit tester 127%, 137%, and 142% ROTOR OVERSPEED lamps shall go off.

• If result is not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.
- i. If no other check is to be done, go to Shutdown.
- 8-2-7.7 Digital Parameter Indication.

a. Do Setup/Power On, if not already done.

b. Set unit tester DIGITAL SET thumbwheel to the following positions:

DIGITAL SET	MAIN FUEL	1 TGT 2 1 Ng SPEED 2
0	0	0
1	1110	111
2	2220	222
3	3330	333
4	440	444
5	1550	555
6	2660	666
7	3770	777
8	880	888
9	1990	999

- RESULT: CDU MAIN FUEL , TGT 1, TGT 2, Ng SPEED 1, and Ng SPEED 2 digital readouts shall indicate corresponding numbers.

• If result is not as specified, replace digital readout module A18 (PARA 8-5-3).

If trouble remains, the CDU chassis assembly is malfunctioning.
- c. Place CDU - DIGITS switch to OFF.

RESULT: CDU digital readouts shall go off.

• If result is not as specified, replace digital readout module A18 (PARA 8-5-3).

If trouble remains, the CDU chassis assembly is malfunctioning.
- d. If no other check is to be done, go to Shutdown.

8-2-7.8 No. 1 Fuel Quantity (Analog).

- a. Do Setup/Power On, if not already done.

b. Place CDU - DIGITS switch to ON.

- c. Place simulator tester NO. 1 FUEL QUANTITY CAL-NOR switch to CAL and HI-LO switch to HI.
- d. Position unit tester INTER CAL SELECT switch to 15 and DISPLAY UPDATE SWITCH to OFF.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 1216 and 1234.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- e. Place simulator tester NO. 1 FUEL QUANTITY HI-LO SWITCH to LO.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 0 and 9.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- f. Place simulator tester NO. 1 FUEL QUANTITY CAL-NOR switch to NOR.

- g. Slowly turn simulator tester NO. 1 FUEL QUANTITY variable control fully clockwise.

RESULT: Unit tester INTERFACE MONITOR count shall increase smoothly.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- h. Slowly turn simulator tester NO. 1 FUEL QUANTITY variable control fully counterclockwise.

RESULT: Unit tester INTERFACE MONITOR count shall decrease smoothly.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- i. If no other check is to be done, go to Shutdown.

8-2-7.9 No. 2 Fuel Quantity (Analog).

- a. Do Setup/Power On, if not already done.

- b. Place simulator tester NO. 2 FUEL QUANTITY CAL-NOR switch to CAL and HI-LO switch to HI.

- c. Position unit tester INTER CAL SELECT switch to 16.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 1216 and 1234.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- d. Place simulator tester NO. 2 FUEL QUANTITY HI-LO switch to LO.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 0 and 9.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- e. Place simulator tester NO. 2 FUEL QUANTITY CAL-NOR switch to NOR.

- f. Slowly turn simulator tester NO. 2 FUEL QUANTITY variable control fully clockwise.

RESULT: Unit tester INTERFACE MONITOR count shall increase smoothly.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

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If trouble remains, the CDU chassis assembly is malfunctioning.

- g. Slowly turn simulator tester NO. 2 FUEL QUANTITY variable control fully counterclockwise.

RESULT: Unit tester INTERFACE MONITOR count shall decrease smoothly.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- h. If no other check is to be done, go to Shutdown.

8-2-7.10 Main Transmission Oil Temperature (Analog).

- a. Do Setup/Power On, if not already done.
- b. Place simulator tester MAIN XMSN OIL TEMP CAL-NOR switch to CAL and HI-LO switch to HI.
- c. Position unit tester INTER CAL SELECT switch to 17.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 1216 and 1264.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- d. Place simulator tester MAIN XMSN OIL TEMP HI-LO switch to LO.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 0 and 18.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- e. Place simulator tester MAIN XMSN OIL TEMP CAL-NOR switch to NOR.

- f. Slowly turn simulator tester MAIN XMSN OIL TEMP variable control fully clockwise.

RESULT: Unit tester INTERFACE MONITOR count shall increase smoothly.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- g. Slowly turn simulator tester MAIN XMSN OIL TEMP variable control fully counterclockwise.

RESULT: Unit tester INTERFACE MONITOR count shall decrease smoothly.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- h. If no other check is to be done, go to Shutdown.

8-2-7.11 Main Transmission Oil Pressure (Analog).

- a. Do Setup/Power On, if not already done.
- b. Place simulator tester XMSN OIL PRESS CAL-NOR switch to CAL and HI-LO switch to HI.
- c. Position unit tester INTER CAL SELECT switch to 18.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 1108 and 1132.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- d. Place simulator tester XMSN OIL PRESS HI-LO switch to LO.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 0 and 12.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- e. Place simulator tester XMSN OIL PRESS CAL-NOR switch to NOR.

- f. Slowly turn simulator tester XMSN OIL PRESS variable control fully clockwise.

RESULT: Unit tester INTERFACE MONITOR count shall increase smoothly.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- g. Slowly turn XMSN OIL PRESS variable control fully counterclockwise.

RESULT: Unit tester INTERFACE MONITOR count shall decrease smoothly.

- If result is not as specified, replace interface No. 1 module A15 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- h. If no other check is to be done, go to Shutdown.

8-2-7.12 Parameter Measurement (Digital).

- a. Do Setup/Power On, if not already done.

- b. Place simulator tester NO. 1 FUEL QUANTITY and NO. 2 FUEL QUANTITY CAL-NOR switches to CAL.

- c. Position unit tester INTER CAL SELECT switch to 19.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 0 and 7.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- d. Place simulator tester NO. 1 FUEL QUANTITY HI-LO switch to HI.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 1218 and 1232.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- e. Place simulator tester NO. 2 FUEL QUANTITY HI-LO switch to HI.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 2443 and 2457.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- f. Place simulator tester NO. 1 FUEL QUANTITY CAL-NOR switch to NOR.

- g. Slowly turn simulator tester NO. 1 FUEL QUANTITY variable control fully clockwise.

RESULT: Unit tester INTERFACE MONITOR count shall increase smoothly.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

8-2-7-12

- h. Slowly turn simulator tester NO. 1 FUEL QUANTITY variable control full counterclockwise.

RESULT: Unit tester INTERFACE MONITOR count shall decrease smoothly.

- If result is not as specified, replace interface No. 2 module A16 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- i. Place unit tester INTER CAL SELECT switch to OFF.
- j. Place unit tester DISPLAY UPDATE switch to ON.
- k. If no other check is to be done, go to Shutdown.

8-2-7.13 Failure Mode.

- a. Do Setup/Power On, if not already done.
- b. Place unit tester SDC FAILURE switch to 1 and release.

RESULT: With SDC FAILURE switch to 1, CDU CHAN 1 lamp shall go on.

- If result is not as specified, the CDU chassis assembly is malfunctioning.
- c. Place unit tester SDC FAILURE switch to 2 and release.

RESULT: With SDC FAILURE switch to 2, CDU CHAN 2 lamp shall go on.

- If result is not as specified, the CDU chassis assembly is malfunctioning.

- d. Place unit tester DISPLAY UPDATE switch to OFF.

RESULT:

- (1) All CDU vertical displays shall blank out.
- (2) All CDU digital displays shall blank out.
- (3) CHAN 1 and CHAN 2 lamps shall go on.

- If results are not as specified, CDU chassis assembly is malfunctioning.

- e. Momentarily press unit tester RESET switch.

RESULT: Unit tester 127%, 137%, and 142% lamps shall go off.

- If result is not as specified, replace voltage regulator module A17 (PARA 8-5-2).

If trouble remains, the CDU chassis assembly is malfunctioning.

- f. If no other check is to be done, go to Shutdown.

8-2-7.14 Shutdown.

- a. Place simulator tester INPUT POWER switch to OFF.
- b. Disconnect cable assemblies.

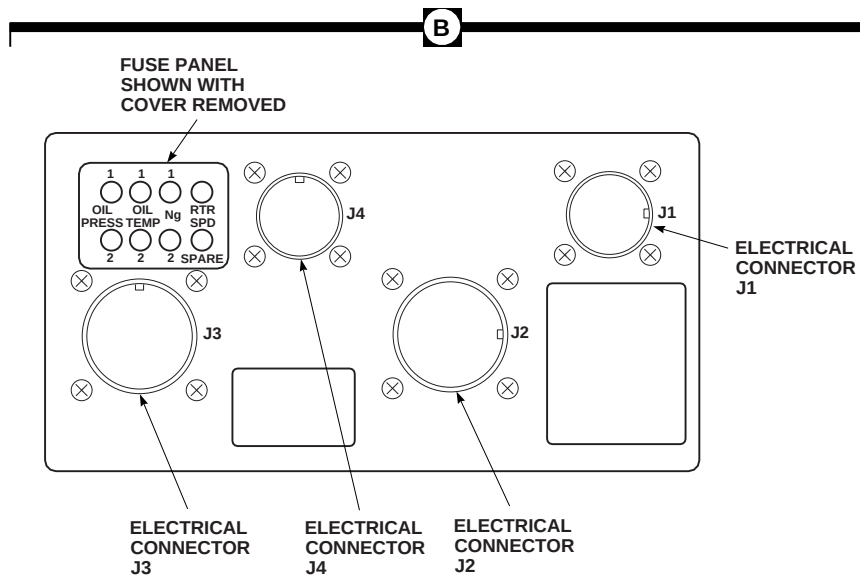
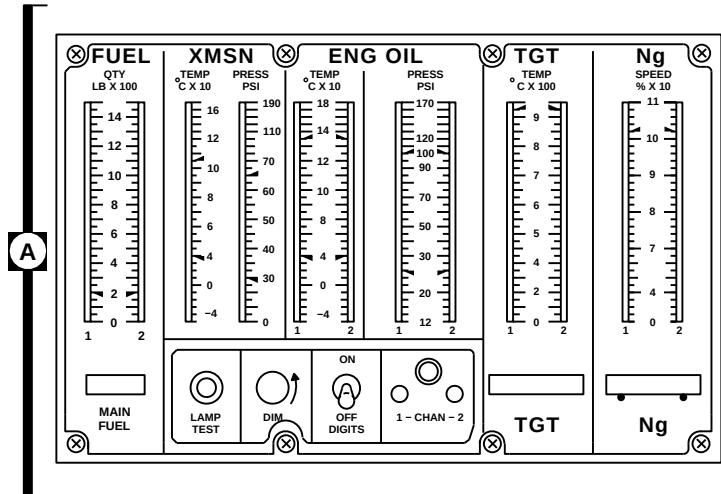
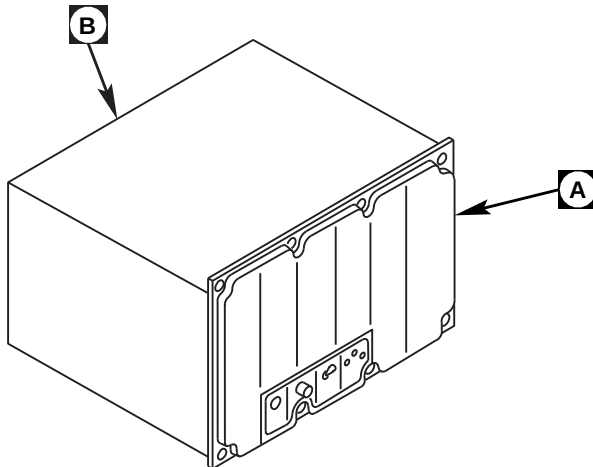


FIGURE 8-2-7-1. CENTRAL DISPLAY UNIT (CDU)

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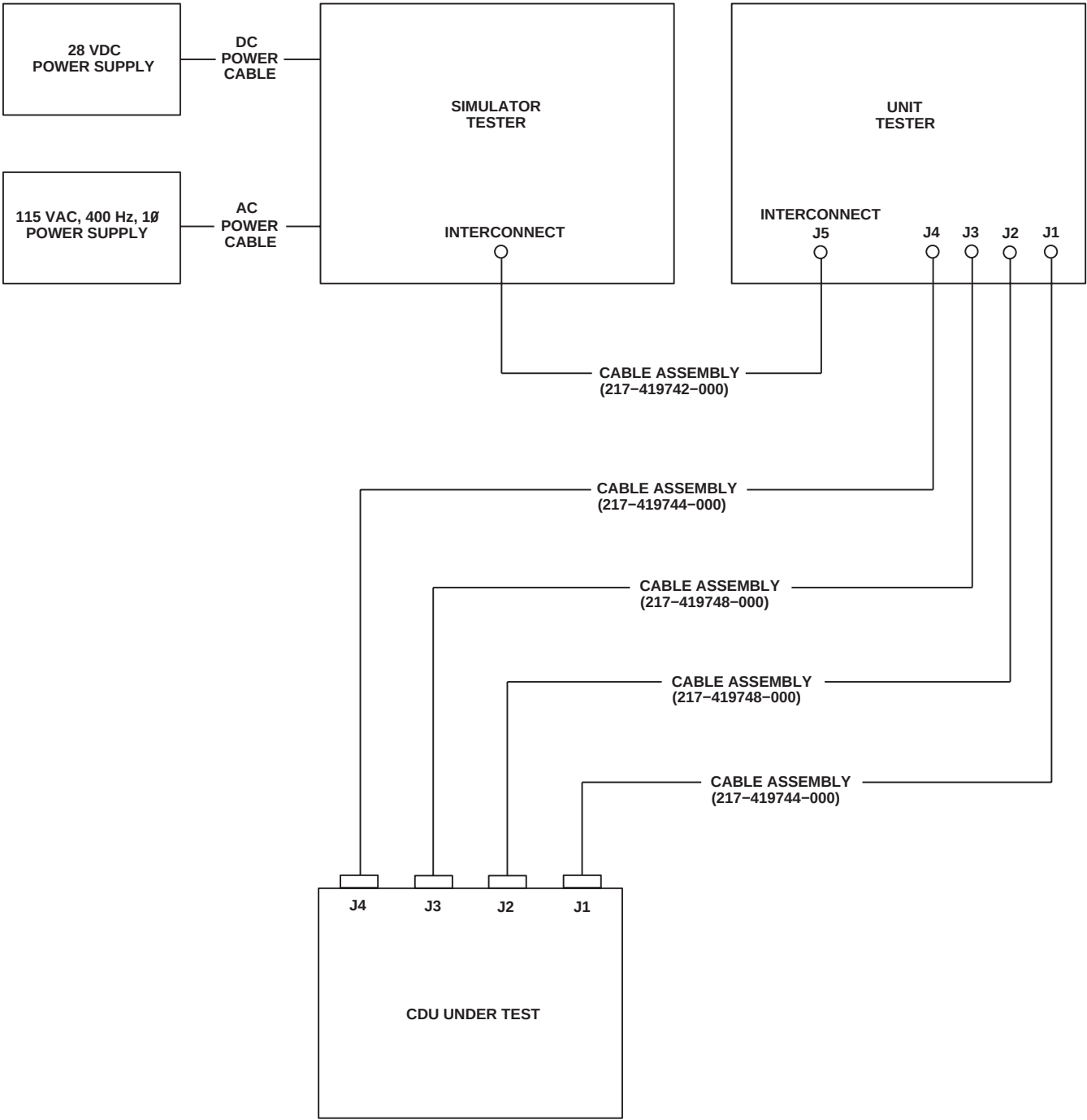


FIGURE 8-2-7-2. CENTRAL DISPLAY UNIT TEST SETUP

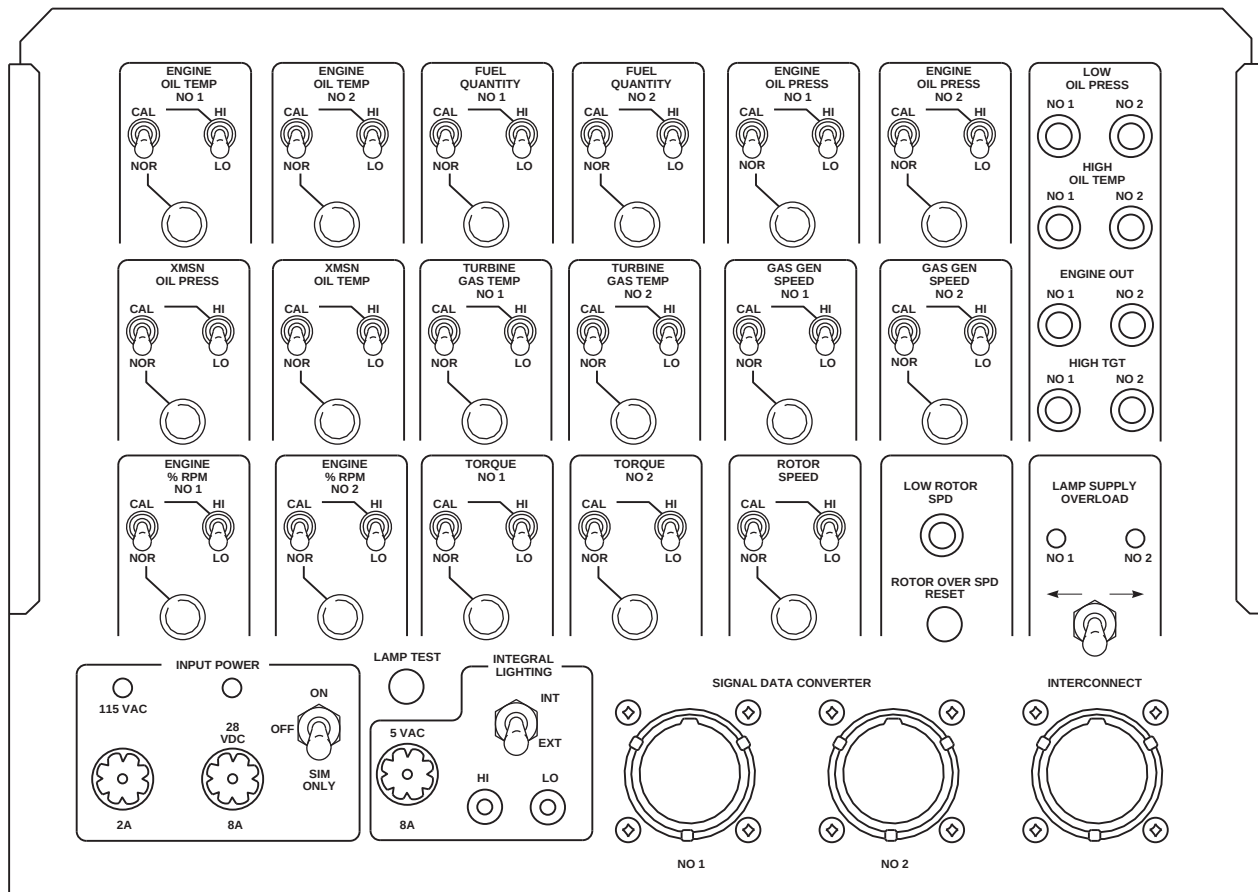
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FIGURE 8-2-7-3. BENCH TEST SET (SHEET 1 OF 2)

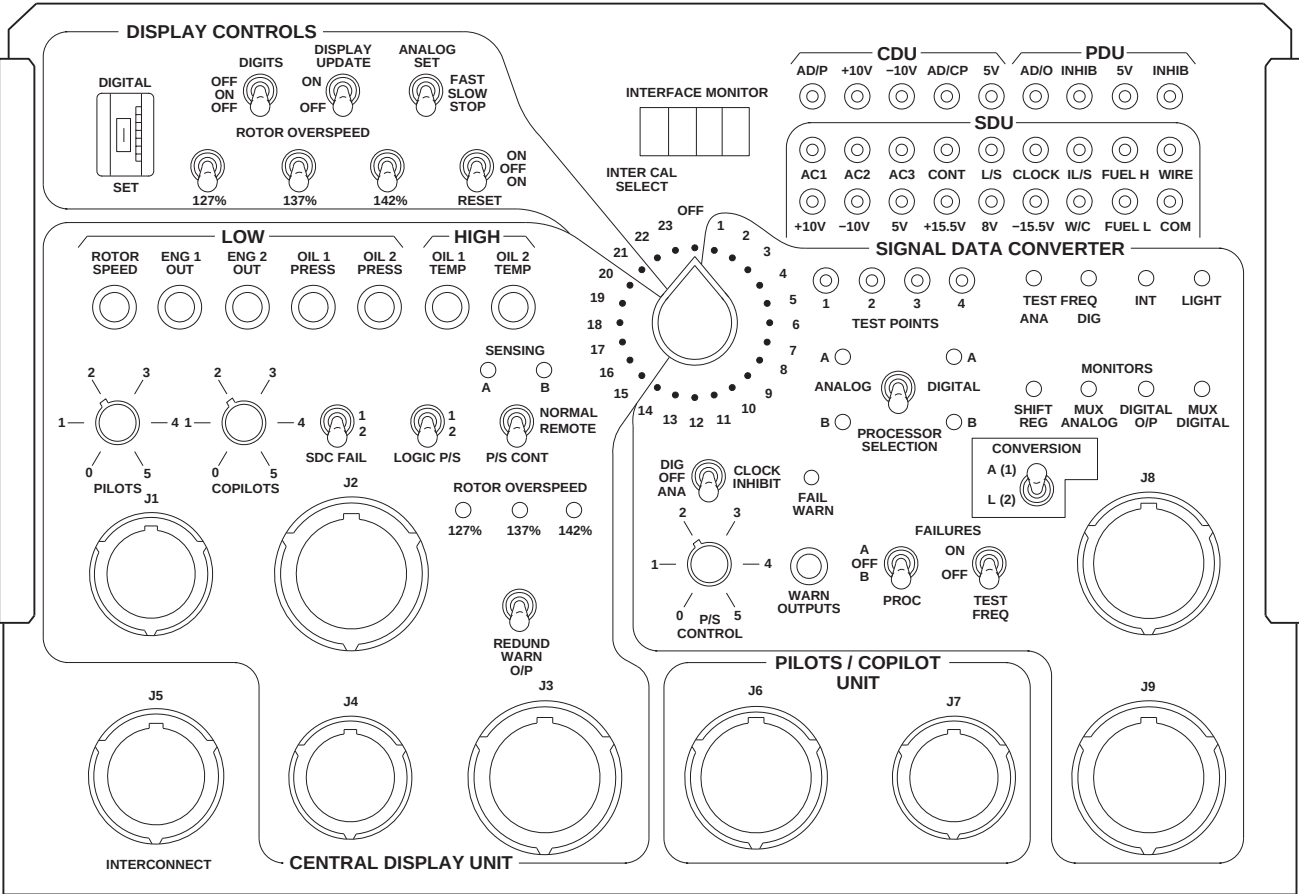


FIGURE 8-2-7-3. BENCH TEST SET (SHEET 2 OF 2)

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8-2-8 PILOTS DISPLAY UNIT (BENCH).

INITIAL SETUP

Test Equipment

- Bench Test Set, 100-601165-000
- Multimeter, Fluke 27/AN
- Oscilloscope, 7603

Tools

- Electrical Repairer Toolkit

Equipment Conditions

- Bench Power Available, 28 vdc, 115 vac, 400 Hz, Single-Phase

Special Instructions

The pilots display unit (PDU) operational/troubleshooting procedure is subdivided as follows:

- (1) Setup/Power On
- (2) 5 VDC Regulator
- (3) Clock Inhibit
- (4) Lamp/Digit
- (5) Parameter Indication (Analog)
- (6) Parameter Indication (Digital)
- (7) Failure Mode
- (8) Rotor Overspeed Lamp
- (9) Autodim
- (10) Shutdown

8-2-8.1 Setup/Power On.

- Make sure simulator tester INPUT POWER switch is OFF.
- Connect cable assembly, 217-419742-000, between simulator tester INTERCONNECT connector and unit tester J5 connector.
- Connect simulator tester dc power connector to 28 vdc bench power.
- Connect simulator tester ac power connector to 115 vac, 400 Hz, single-phase bench power.
- Place simulator tester INPUT POWER switch to ON.

RESULT: Simulator tester INPUT POWER 115 VAC and 28 VDC lamps shall go on.

- If result is not as specified, troubleshoot bench test set.

- Press simulator tester LAMP TEST switch, then release.

RESULT: All simulator tester lamps shall go on when LAMP TEST switch is pressed.

- If result is not as specified, troubleshoot bench test set.

- Place simulator tester INPUT POWER switch to OFF.
- Connect cable assembly, 217-419745-000, between display unit connector J1 and unit tester connectors J6 and J7.

NOTE

Apply 115 vac 400hz to simulator front panel integral lighting section HI and LO test points.

- Place simulator tester INTEGRAL LIGHTING switch to EXT.

- j. Place unit tester DISPLAY CONTROLS section switches as follows:

DIGITAL SET	1
DIGITS	ON
ROTOR OVERSPEED 127%, 137%, and 142%	Off (Down)
ANALOG SET	STOP
DISPLAY UPDATE	ON

- k. Position unit tester INTER CAL SELECT switch to 21.
- l. Place unit tester CENTRAL DISPLAY UNIT P/S CONT switch to REMOTE.
- m. Place simulator tester INPUT POWER switch to ON.

RESULT: Simulator tester INPUT POWER 115 VAC and 28 VDC lamps shall go on.

- If result is not as specified, troubleshoot bench test set.

8-2-8.2 5 VDC Regulator.

- a. Do Setup/Power On, if not already done.
- b. Set multimeter to indicate dc voltage and connect leads between PDU 5V and SDC COM test jacks on unit tester.

RESULT: Multimeter shall indicate between 4.75 and 5.25 vdc.

- If result is not as specified, replace pilot display autodim module A9 (PARA 8-5-12).

If trouble remains, the display unit chassis assembly is malfunctioning.

- c. Disconnect multimeter.

- d. If no other check is to be done, go to Shutdown.

8-2-8.3 Clock Inhibit.

- a. Do Setup/Power On, if not already done.
- b. Connect oscilloscope between PDU left IN-HIB and SDC COM test jacks on unit tester.

RESULT: Oscilloscope shall display 4 to 6 volt peak-to-peak square waves.

- If result is not as specified, replace pilot display autodim module A9 (PARA 8-5-12).

If trouble remains, the display unit chassis assembly is malfunctioning.

- c. Connect oscilloscope between PDU right IN-HIB and SDC COM test jacks on unit tester.

RESULT: Oscilloscope shall display 4 to 6 volt peak-to-peak square waves.

- If result is not as specified, replace pilot display autodim module A9 (PARA 8-5-12).

If trouble remains, the display unit chassis assembly is malfunctioning.

- d. Disconnect oscilloscope.

- e. If no other check is to be done, go to Shutdown.

8-2-8.4 Lamp/Digit.

- a. Do Setup/Power On, if not already done.
- b. Press display unit LT switch and turn unit tester P/S CONTROL fully counterclockwise, then fully clockwise. Next, adjust so you can see display normally. Release TEST or LT switch.

RESULT: With LT switch pressed:

- (1) All display unit vertical scale lamps shall go on.

8-2-8-2

- (2) Both display unit digital readouts shall indicate 188.
- (3) Display unit lighting shall vary with P/S CONTROL setting.
 - If results (1) and (2) are not as specified, display unit chassis assembly is malfunctioning.
 - If result (1) is not as specified, replace pilot's display driver module (A3, A4, A5, A6, and/or A7) as required (PARA 8-5-7 or PARA 8-5-8).

If trouble remains, the display unit chassis assembly is malfunctioning.

- If result (2) is not as specified, replace torque digital readout module A8 (PARA 8-5-11).

If trouble remains, the display unit chassis assembly is malfunctioning.

- If result (3) is not as specified, replace pilot display autodim module A9 (PARA 8-5-12).

If trouble remains, the display unit chassis assembly is malfunctioning.

- c. If no other check is to be done, go to Shutdown.

8-2-8.5 Parameter Indication (Analog).

- a. Do Setup/Power On, if not already done.

NOTE

Cycling rate of vertical scales is controlled by unit tester ANALOG SET switch.

- b. Place unit tester ANALOG SET switch to FAST.

RESULT:

- (1) Display unit % RPM 1 vertical scale shall continually cycle from 0 to full scale.

- (2) At 96%, % RPM 1 low red and yellow scale segments shall go off.
- (3) Display unit % RPM 2 vertical scale shall continually cycle from 0 to full scale.
- (4) At 96%, % RPM 2 low red and yellow scale segments shall go off.
- (5) Display unit % RPM R vertical scale shall continually cycle from 0 to full scale.
- (6) At 96%, % RPM R low red and yellow scale segments shall go off.
- (7) Display unit % TRQ 1 vertical scale shall continually cycle from 0 to full scale.
- (8) Display unit % TRQ 2 vertical scale shall continually cycle from 0 to full scale.
 - If results (1), (3), (5), (7), and (8) are not as specified, replace pilot's display driver module (A3, A4, A5, A6 and/or A7) as required (PARA 8-5-7 or PARA 8-5-8).

If trouble remains, the display unit chassis assembly is malfunctioning.

- If results (2), (4), and (6) are not as specified, replace pilot's display driver module (A3, A4 and/or A5) as required (PARA 8-5-7 or PARA 8-5-8).

- (a) If trouble remains, replace pilot display autodim module A9 (PARA 8-5-12).

- (b) If trouble still remains, the display unit chassis assembly is malfunctioning.

- c. If no other check is to be done, go to Shutdown.

8-2-8.6 Parameter Indication (Digital).

- a. Do Setup/Power On, if not already done.
- b. Set unit tester to the following positions:

DIGITAL SET	%	TORQUE DIGITAL
0		0
1		111
2		22
3		133
4		44
5		155
6		66
7		177
8		88
9		199

0	0
1	111
2	22
3	133
4	44
5	155
6	66
7	177
8	88
9	199

RESULT: Display unit % TRQ digital readouts shall indicate corresponding number.

- If specified digital readouts are not as specified, replace torque digital readout module A8 (PARA 8-5-11).

If trouble remains, the display unit chassis assembly is malfunctioning.

- c. If no other check is to be done, go to Shutdown.

8-2-8.7 Failure Mode.

- a. Do Setup/Power On, if not already done.
- b. Place unit tester DISPLAY UPDATE switch to OFF.

RESULT:

- (1) All display unit vertical displays shall blank out.
 - (2) Display unit digital displays shall blank out.
 - (3) Unit tester FAIL WARN lamp shall go on.
- If result (1) is not as specified, replace pilot's display driver (A3, A4, A5, A6 and/or A7) as required (PARA 8-5-7 or PARA 8-5-8).

- If result (2) is not as specified, replace torque digital readout module A8 (PARA 8-5-11).

- If result (3) is not as specified, display unit chassis assembly is malfunctioning.

- c. Place unit tester DISPLAY UPDATE switch to ON.

RESULT:

- (1) All display unit vertical displays shall continually cycle from 0 to full scale.
- (2) Display unit digital displays shall indicate 199.
- (3) Unit tester FAIL WARN lamp shall go off.

- If results are not as specified, the display unit chassis assembly is malfunctioning.

- d. If no other check is to be done, go to Shutdown.

8-2-8.8 Rotor Overspeed Lamp.

- a. Do Setup/Power On, if not already done.
- b. Place unit tester 127% ROTOR OVERSPEED switch up.

RESULT: Left RTR OVERSPEED lamp on display unit shall go on.

- If result is not as specified, display unit chassis assembly is malfunctioning.

- c. Place unit tester 137% ROTOR OVERSPEED switch up.

RESULT: Center RTR OVERSPEED lamp on display unit shall go on.

- If result is not as specified, display unit chassis assembly is malfunctioning.

- d. Place unit tester 142% ROTOR OVERSPEED switch up.

8-2-8-4

RESULT: Right RTR OVERSPEED lamp on display unit shall go on.

- If result is not as specified, display unit chassis assembly is malfunctioning.
- e. Place unit tester 127%, 137%, and 142% ROTOR OVERSPEED switches down.

RESULT: Three RTR OVERSPEED lamps on display unit shall go off.

- If result is not as specified, display unit chassis assembly is malfunctioning.
- f. If no other check is to be done, go to Shutdown.

8-2-8.9 Autodim.

- a. Do Setup/Power On, if not already done.
- b. Place unit tester P/S CONT switch to NORMAL.
- c. Cover light sensor, in lower right corner of display unit, with opaque object.

RESULT: All display unit vertical scale and digital readout lighting shall go off or decrease in brightness.

- If result is not as specified, replace pilot display autodim module A9 (PARA 8-5-12).

If trouble remains, the display unit chassis assembly is malfunctioning.

- d. Remove opaque object from light sensor.

RESULT: All display unit vertical scale and digital readout lighting shall increase in brightness.

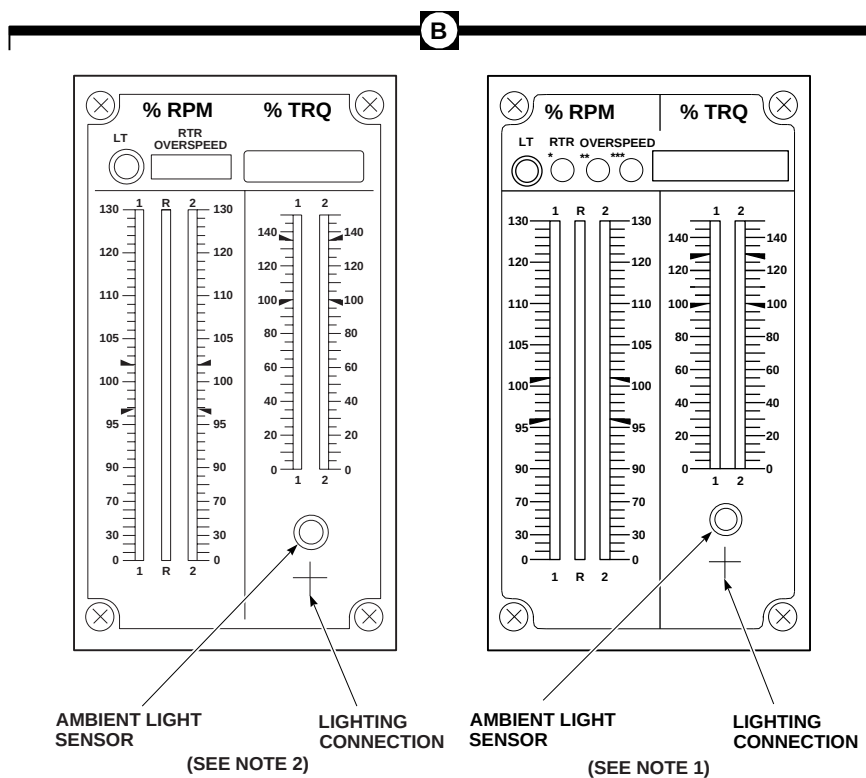
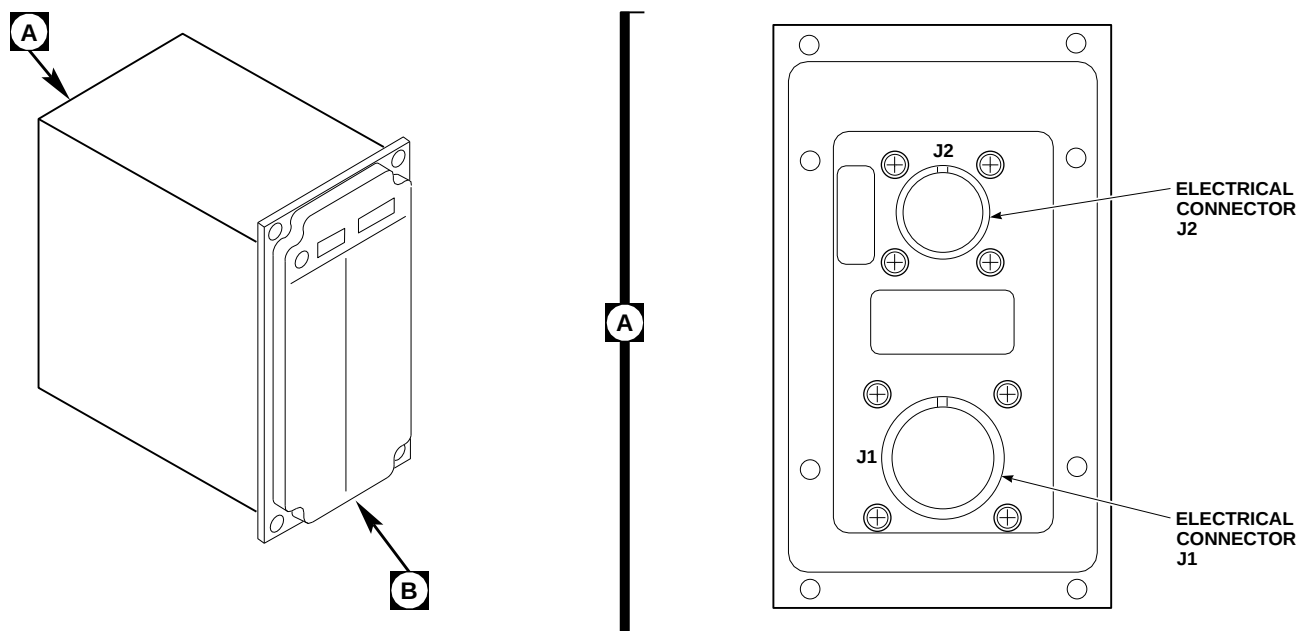
- If result is not as specified, replace pilot display autodim module A9 (PARA 8-5-12).

If trouble remains, the display unit chassis assembly is malfunctioning.

- e. If no other check is to be done, go to Shutdown.

8-2-8.10 Shutdown.

- a. Place simulator tester INPUT POWER switch to OFF.
- b. Disconnect cable assemblies.



NOTES
1. PDU 70450-01916-104.
2. PDU 70450-01043-121.

PILOTS AND COPILOTS DISPLAY UNIT

FB1561B
SA

FIGURE 8-2-8-1. PILOT'S/COPILOT'S DISPLAY UNIT

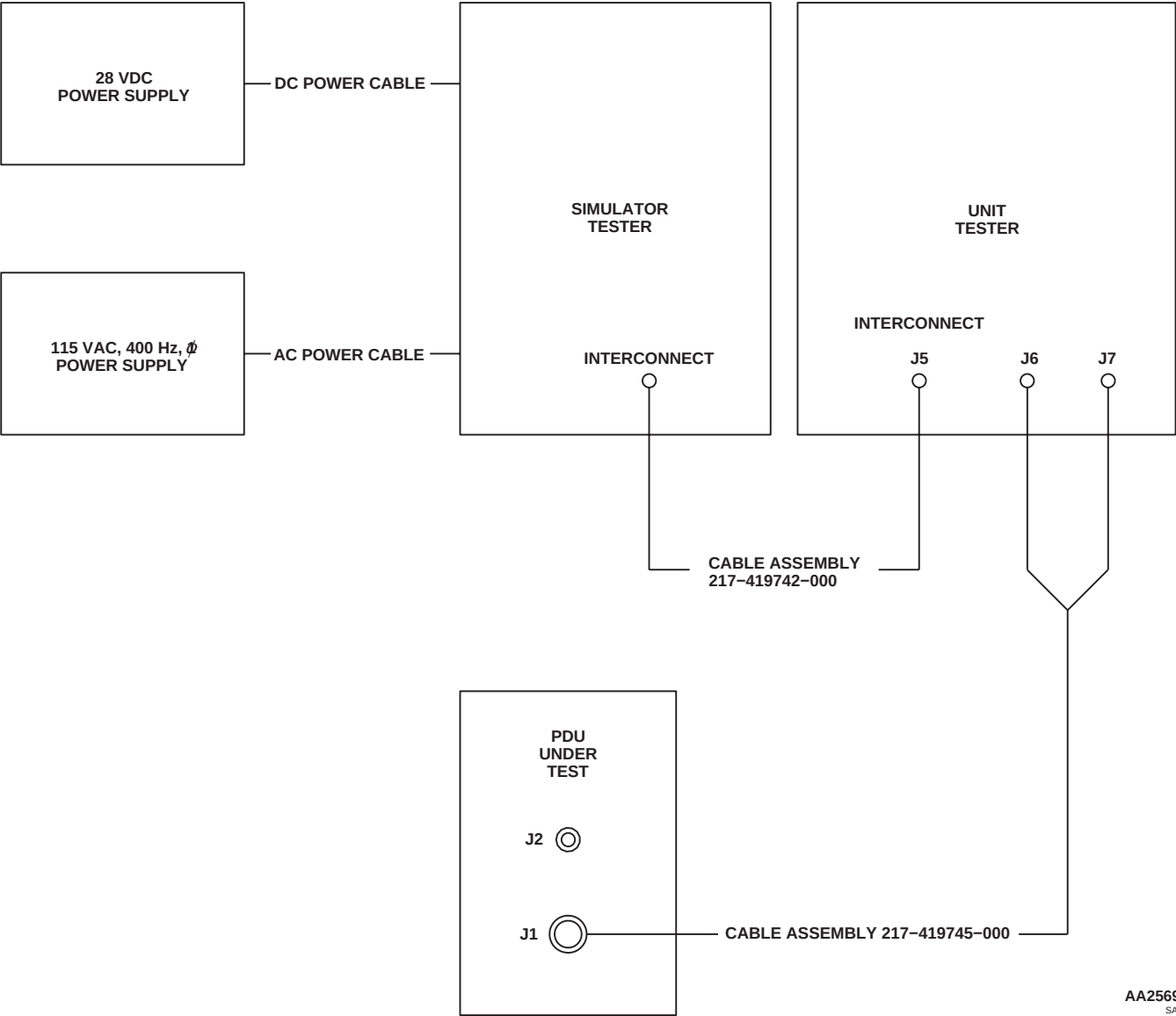


FIGURE 8-2-8-2. PILOTS DISPLAY UNIT TEST SETUP

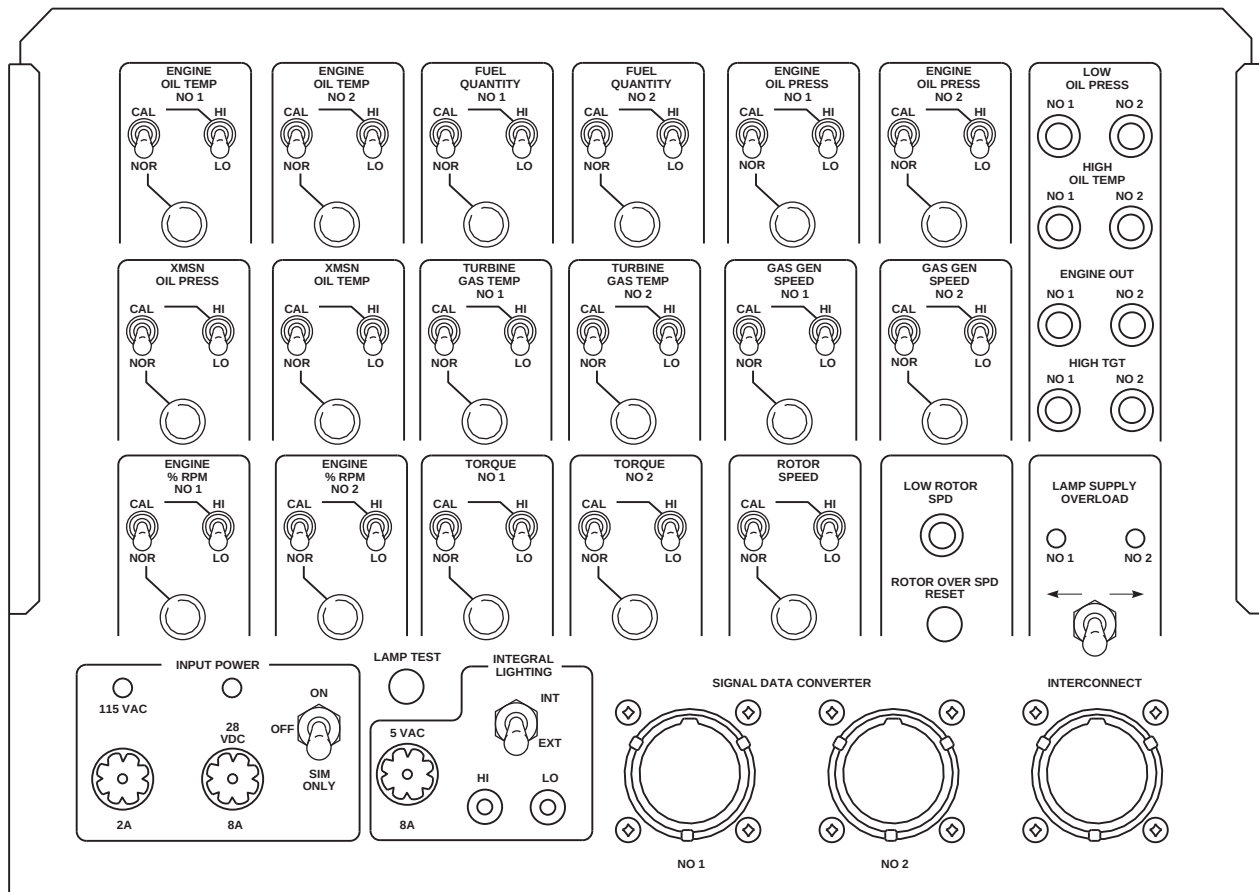
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FIGURE 8-2-8-3. BENCH TEST SET (SHEET 1 OF 2)

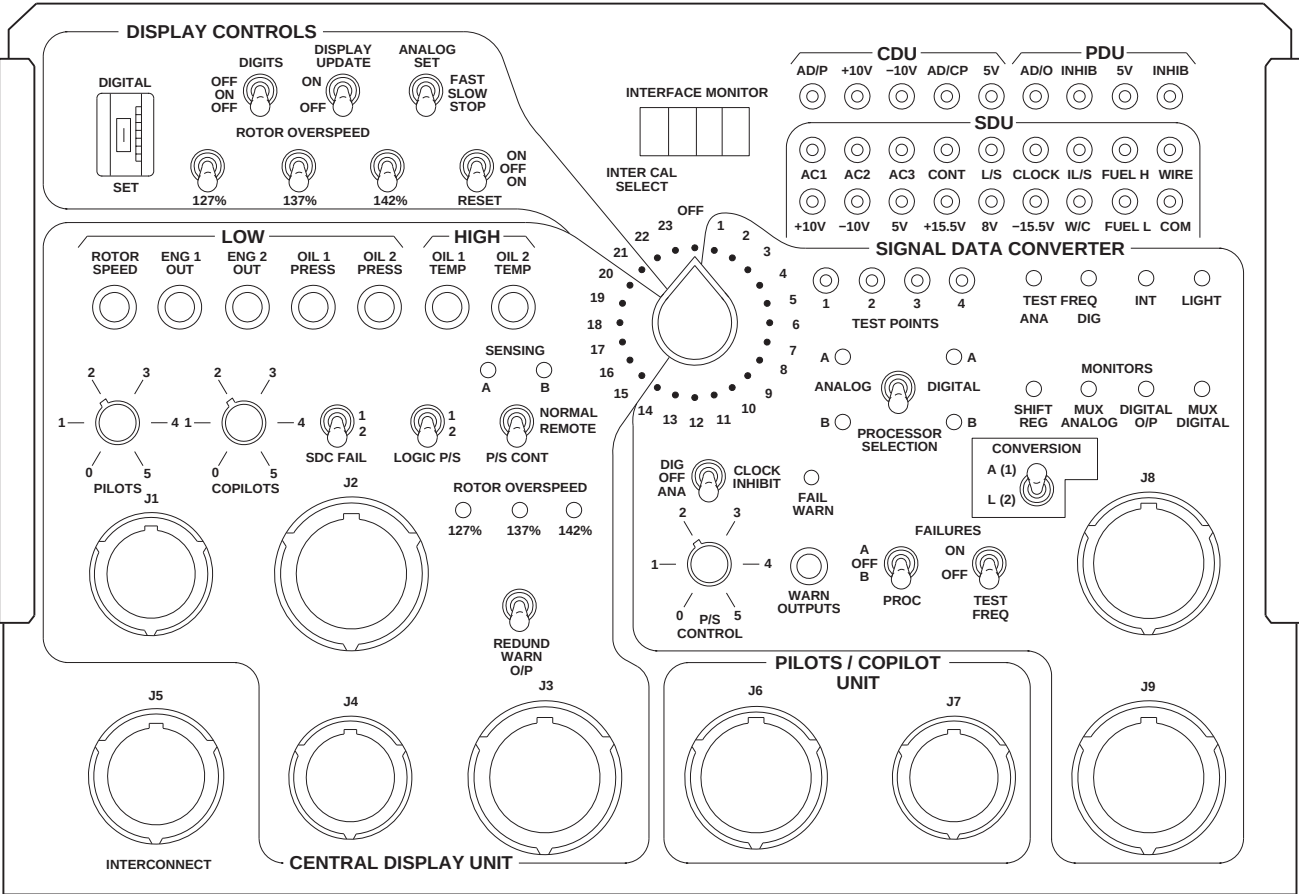


FIGURE 8-2-8-3. BENCH TEST SET (SHEET 2 OF 2)

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8-2-9 SIGNAL DATA CONVERTER (BENCH).

INITIAL SETUP

Test Equipment

- Bench Test Set, 100-601165-000
- Multimeter, Fluke 27/AN
- Oscilloscope, 7603

Tools

- Electrical Repairer Toolkit

Equipment Conditions

- Bench Power Available, 28 vdc, 115 vac, 400 Hz, Single-Phase

Special Instructions

The signal data converter operational/troubleshooting procedure is subdivided as follows:

- | | |
|---|---|
| <ul style="list-style-type: none"> (1) Setup/Power On (2) Logic Power Supply/Voltage Regulator (3) Lamp Power Supply (4) Processor Failure Switch-Over (5) Lamp Power Supply Failure (6) Sensor Excitation Voltage (7) Clock Monitor | <ul style="list-style-type: none"> (8) Clock Inhibit (Digital) (9) Clock Inhibit (Analog) (10) Sensor Interwiring/Warning Output (11) No. 1 Engine % RPM (Analog) (12) No. 2 Engine % RPM (Analog) (13) Rotor Speed (Analog) (14) No. 1 Torque (Analog) (15) No. 2 Torque (Analog) (16) Engine Oil Temperature (Analog) (17) Engine Oil Pressure (Analog) (18) Turbine Gas Temperature (Analog) (19) Gas Generator Speed (Analog) (20) No. 1 Torque (Digital) (21) No. 2 Torque (Digital) (22) Turbine Gas Temperature (Digital) (23) Gas Generator Speed (Digital) (24) Rotor Speed (Analog Relay) (25) Integral Lighting (26) Shutdown |
|---|---|

8-2-9.1 Setup/Power On.

- | | |
|--|--|
| <ul style="list-style-type: none"> a. Make sure simulator tester INPUT POWER switch is OFF. b. Connect cable assembly, 217-419742-000, between simulator tester INTERCONNECT connector and unit tester J5 connector. | <ul style="list-style-type: none"> c. Connect simulator tester dc power connector to 28 vdc bench power. d. Connect simulator tester ac power connector to 115 vac, 400 Hz, single-phase bench power. e. Place simulator tester INPUT POWER switch to ON. |
|--|--|

RESULT: Simulator tester INPUT POWER 115 VAC and 28 VDC lamps shall go on.

- If result is not as specified, troubleshoot bench test set.

f. Press simulator tester LAMP TEST switch, then release.

RESULT:

- (1) All simulator tester lamps shall go on when LAMP TEST switch is pressed.
 - (2) All unit tester lamps shall go on when LAMP TEST switch is pressed.
- If results are not as specified, troubleshoot bench test set.

g. Place simulator tester INPUT POWER switch to OFF.

h. Connect cable assembly, 217-419741-000, between simulator tester No. 1 SIGNAL DATA CONVERTER connector and signal data converter (SDC) J1.

i. Connect cable assembly, 217-419747-000, between unit tester J8 and SDC J2.

j. Connect cable assembly, 217-419748-000, between unit tester J9 and SDC J3.

k. Place simulator tester switches and controls as follows:

All CAL-NOR switches	NOR
All HI-LO switches	LO
All NOR controls	Fully counterclockwise
INTEGRAL LIGHTING	EXT

l. Place unit tester SIGNAL DATA CONVERTER section switches and controls as follows:

CLOCK INHIBIT	OFF
FAILURES PROC	OFF
P/S CONTROL	Midposition
CONVERSION	1
FAILURES TEST	OFF
FREQ	

m. Place unit tester CENTRAL DISPLAY UNIT section P/S CONT switch to REMOTE.

n. Place unit tester DISPLAY CONTROLS section DISPLAY UPDATE switch to ON.

o. Place unit tester INTER CAL SELECT switch to OFF.

p. Place simulator tester INPUT POWER switch to ON.

RESULT:

(1) Simulator tester INPUT POWER 115 VAC and 28 VDC and LAMP SUPPLY OVERLOAD NO. 1 lamps shall go on.

(2) The following unit tester lamps shall go on:

MONITORS SHIFT REG
MONITORS MUX ANALOG
MONITORS DIGITAL O/P
MONITORS MUX DIGITAL
TEST FREQ ANA (flash)
TEST FREQ DIG (flash)
PROCESSOR SELECTION ANALOG A or B
PROCESSOR SELECTION DIGITAL A or B

- If simulator tester LAMP SUPPLY OVERLOAD NO. 1 lamp does not go on, replace lamp power supply (PARA 8-5-5).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- If none of the unit tester lamps go on, do Logic Power Supply/Voltage Regulator check.

8-2-9-2

- If MONITORS SHIFT REG lamp does not go on, replace analog processor No. 1, module A9 (PARA 8-5-4).

- (a) If trouble remains, replace analog processor No. 2 module A8 (PARA 8-5-4).

- (b) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

- If MONITORS MUX ANALOG lamp does not go on, replace analog processor No. 2, module A8 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- If MONITORS DIGITAL O/P lamp does not go on, replace digital processor module A10 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- If MONITORS MUX DIGITAL lamp does not go on, replace digital processor module A10 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- If TEST FREQ ANA lamp does not flash, replace analog processor No. 2, module A8 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- If TEST FREQ DIG lamp does not flash, replace digital processor module A10 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- q. Place unit tester SIGNAL DATA CONVERTER FAILURES TEST FREQ switch to ON.

RESULT:

- (1) Unit tester SIGNAL DATA CONVERTER TEST FREQ ANA lamp shall flash.

- (2) Unit tester SIGNAL DATA CONVERTER TEST FREQ DIG lamp shall flash.

- If result (1) is not as specified, replace analog processor No. 2, module A8 (PARA 8-5-4).

- (a) If trouble remains, replace analog processor No. 1, module A9 (PARA 8-5-4).

- (b) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

- If result (2) is not as specified, replace digital processor module A10 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

8-2-9.2 Logic Power Supply/Voltage Regulator.

- a. Do Setup/Power On, if not already done.
- b. Set multimeter to indicate dc voltage and connect leads between SDC +15.5 and COM test jacks on unit tester.

RESULT: Multimeter shall indicate between 14 and 21 vdc.

- If result is not as specified, replace logic power supply (PARA 8-5-6).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- c. Connect multimeter between SDC -15.5V and COM test jacks on unit tester.

RESULT: Multimeter shall indicate between -21 and -14 vdc.

- If result is not as specified, replace logic power supply (PARA 8-5-6).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- d. Connect multimeter between SDC 8V and COM test jacks on unit tester.

RESULT: Multimeter shall indicate between 6.5 and 10.5 vdc.

- If result is not as specified, replace logic power supply (PARA 8-5-6).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- e. Place unit tester SIGNAL DATA CONVERTER FAILURES TEST FREQ switch to OFF.

RESULT: Multimeter shall indicate between 6.5 and 10.5 vdc.

- If result is not as specified, replace logic power supply (PARA 8-5-6).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- f. Place unit tester SIGNAL DATA CONVERTER FAILURES TEST FREQ switch to ON.

- g. Connect multimeter between SDC +10V and COM test jacks on unit tester.

RESULT: Multimeter shall indicate between 9.5 and 10.5 vdc.

- If result is not as specified, replace voltage regulator module A11 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- h. Connect multimeter between SDC -10V and COM test jacks on unit tester.

RESULT: Multimeter shall indicate between -10.5 and -9.5 vdc.

- If result is not as specified, replace voltage regulator module A11 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- i. Connect multimeter between SDC 5V and COM test jacks on unit tester.

RESULT: Multimeter shall indicate between 4.75 and 5.25 vdc.

- If result is not as specified, replace voltage regulator module A11 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- j. If no other check is to be done, go to Shutdown.

8-2-9.3 Lamp Power Supply.

- a. Do Setup/Power On, if not already done.

- b. Connect multimeter between SDC L/S and COM test jacks on unit tester.

- c. Adjust unit tester SIGNAL DATA CONVERTER P/S CONTROL to obtain 4.9 to 5.1 vdc indication on multimeter.

RESULT: Simulator tester LAMP SUPPLY OVERLOAD NO. 1 lamp shall be on.

- If result is not as specified, replace lamp power supply (PARA 8-5-5).

8-2-9-4

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- d. Connect multimeter between SDC CONT and COM test jacks on unit tester.

RESULT: Multimeter shall indicate between 4.5 and 4.9 vdc.

- If result is not as specified, replace voltage regulator module A11 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- e. Adjust unit tester SIGNAL DATA CONVERTER P/S CONTROL to obtain 1.70 to 1.90 vdc indication on multimeter.

RESULT: LAMP SUPPLY OVERLOAD NO. 1 lamp shall be dim.

- If result is not as specified, replace voltage regulator module A11 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- f. Connect multimeter between SDC L/S and COM test jacks on unit tester.

RESULT: Multimeter shall indicate between 2.0 and 2.2 vdc.

- If result is not as specified, replace lamp power supply (PARA 8-5-5).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- g. If no other check is to be done, go to Shutdown.

8-2-9.4 Processor Failure Switch-Over.

- a. Do Setup/Power On, if not already done.

- b. Momentarily place unit tester SIGNAL DATA CONVERTER PROCESSOR SELECTION switch to up position.

RESULT:

- (1) If not already on, SIGNAL DATA CONVERTER PROCESSOR SELECTION ANALOG A lamp on the unit tester shall go on and remain on.

- (2) If not already on, SIGNAL DATA CONVERTER PROCESSOR DIGITAL A lamp on the unit tester shall go on and remain on.

- If result (1) is not as specified, replace analog processor No. 1, module A9 (PARA 8-5-4).

- (a) If trouble remains, replace analog processor No. 2, module A8 (PARA 8-5-4).

- (b) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

- If result (2) is not as specified, replace digital processor module A10 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- c. Place and hold unit tester SIGNAL DATA CONVERTER PROC FAILURES switch to A.

RESULT: After approximately 5 seconds, the following shall occur:

- (1) PROCESSOR SELECTION ANALOG A lamp shall go off.

- (2) PROCESSOR SELECTION ANALOG B lamp shall go on.

- (3) Unit tester SIGNAL DATA CONVERTER FAIL WARN lamp shall go on.

- If results are not as specified, replace analog processor No. 1, module A9 (PARA 8-5-4).

- If trouble remains, replace analog processor No. 2, module A8 (PARA 8-5-4).
- If trouble still remains, the signal data converter chassis assembly is malfunctioning.

d. Release unit tester PROC FAILURES switch.

RESULT: FAIL WARN lamp shall go off.

- If result is not as specified, replace analog processor No. 1, module A9 (PARA 8-5-4).

- If trouble remains, replace analog processor No. 2, module A8 (PARA 8-5-4).
- If trouble still remains, the signal data converter chassis assembly is malfunctioning.

e. Place and hold unit tester PROC FAILURES switch to B.

RESULT: After approximately 5 seconds, the following shall occur:

- PROCESSOR SELECTION ANALOG B lamp shall go off.
- PROCESSOR SELECTION ANALOG A lamp shall go on.
- FAIL WARN lamp shall go on.

- If results are not as specified, replace analog processor No. 1, module A9 (PARA 8-5-4).

- If trouble remains, replace analog processor No. 2, module A8 (PARA 8-5-4).
- If trouble still remains, the signal data converter chassis assembly is malfunctioning.

f. Release unit tester PROC FAILURES switch.

RESULT: FAIL WARN lamp shall go off.

- If result is not as specified, replace analog processor No. 1, module A9 (PARA 8-5-4).

- If trouble remains, replace analog processor No. 2, module A8 (PARA 8-5-4).

- If trouble still remains, the signal data converter chassis assembly is malfunctioning.

g. If no other check is to be done, go to Shutdown.

8-2-9.5 Lamp Power Supply Failure.

a. Do Setup/Power On, if not already done.

b. If not already connected, connect multimeter between SDC L/S and COM test jacks on unit tester.

c. Turn unit tester SIGNAL DATA CONVERTER P/S CONTROL fully clockwise.

RESULT: Simulator tester LAMP SUPPLY OVERLOAD NO. 1 lamp shall be on.

- If result is not as specified, replace lamp power supply (PARA 8-5-5).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

d. Place and hold simulator tester LAMP SUPPLY OVERLOAD switch to NO. 1.

RESULT:

(1) Multimeter shall indicate less than 1.0 vdc.

(2) Simulator tester LAMP SUPPLY OVERLOAD NO. 1 lamp shall go off.

8-2-9-6

- If result is not as specified, replace lamp power supply (PARA 8-5-5).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- e. Release simulator tester LAMP SUPPLY OVERLOAD NO. 1 switch.

RESULT:

- (1) Multimeter shall indicate between 4.8 and 5.8 vdc.
- (2) Simulator tester LAMP SUPPLY OVERLOAD NO. 1 lamp shall go on.

- If results are not as specified, replace lamp power supply (PARA 8-5-5).

- f. Disconnect multimeter.
- g. If no other check is to be done, go to Shutdown.

8-2-9.6 Sensor Excitation Voltage.

- a. Do Setup/Power On, if not already done.
- b. Set multimeter to indicate ac voltage and connect leads between SDC AC1 and COM test jacks on unit tester.

RESULT: Multimeter shall indicate between 7.5 and 11.0 vac.

- If result is not as specified, replace logic power supply (PARA 8-5-6).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- c. Connect multimeter between SDC AC2 and COM test jacks on unit tester.

RESULT: Multimeter shall indicate between 7.5 and 11.0 vac.

- If result is not as specified, replace logic power supply (PARA 8-5-6).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- d. Connect multimeter between SDC AC3 and COM test jacks on unit tester.

RESULT: Multimeter shall indicate between 7.5 and 11.0 vac.

- If result is not as specified, replace logic power supply (PARA 8-5-6).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- e. Disconnect multimeter.
- f. If no other check is to be done, go to Shutdown.

8-2-9.7 Clock Monitor.

- a. Do Setup/Power On, if not already done.
- b. Connect oscilloscope vertical input leads between SDC CLOCK and COM test jacks on unit tester.

RESULT: Oscilloscope shall display square wave with these requirements:

AMPLITUDE	4 to 6 Volts
PERIOD	6.8 to 8.8 Microseconds

- If result is not as specified, replace digital processor module A10 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- c. Disconnect oscilloscope.
- d. If no other check is to be done, go to Shutdown.

8-2-9.8 Clock Inhibit (Digital).

- a. Do Setup/Power On, if not already done.
- b. Place and hold unit tester SIGNAL DATA CONVERTER CLOCK INHIBIT switch to DIG.

RESULT:

- (1) Both SIGNAL DATA CONVERTER PROCESSOR SELECTION DIGITAL A and B lamps on unit tester shall be off.
- (2) The following unit tester SIGNAL DATA CONVERTER lamps shall go off:

TEST FREQ DIG
MONITORS DIGITAL O/P
MONITORS MUX DIGITAL

- (3) Unit tester SIGNAL DATA CONVERTER FAIL WARN lamp shall go on.
- If results are not as specified, replace digital processor module A10 (PARA 8-5-4).

- c. Release unit tester CLOCK INHIBIT switch.

RESULT:

- (1) The following lamps shall go on:

TEST FREQ DIG (flash)
MONITORS DIGITAL O/P
MONITORS MUX DIGITAL
Either PROCESSOR SELECTION DIGITAL A or B

- (2) FAIL WARN lamp shall go off.
- If results are not as specified, replace digital processor module A10 (PARA 8-5-4).

- d. If no other check is to be done, go to Shutdown.

8-2-9.9 Clock Inhibit (Analog).

- a. Do Setup/Power On, if not already done.
- b. Place and hold unit tester SIGNAL DATA CONVERTER CLOCK INHIBIT switch to ANA.

RESULT:

- (1) Both SIGNAL DATA CONVERTER PROCESSOR SELECTION ANALOG A and B lamps on unit tester shall be off.
- (2) The following unit tester SIGNAL DATA CONVERTER lamps shall go off:

TEST FREQ ANA
MONITORS SHIFT REG
MONITORS MUX ANALOG

- (3) Unit tester SIGNAL DATA CONVERTER FAIL WARN lamp shall go on.
- If results are not as specified, replace analog processor No. 1, module A9 (PARA 8-5-4).

- (a) If trouble remains, replace analog processor No. 2, module A8 (PARA 8-5-4).

- (b) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

- c. Release unit tester CLOCK INHIBIT switch.

RESULT:

- (1) The following lamps shall go on:

TEST FREQ ANA (flash)
MONITORS SHIFT REG
MONITORS MUX ANALOG
Either PROCESSOR SELECTION ANALOG A or B

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- (2) FAIL WARN lamp shall go off.
- If results are not as specified, replace analog processor No. 1, module A9 (PARA 8-5-4).
 - (a) If trouble remains, replace analog processor No. 2, module A8 (PARA 8-5-4).
 - (b) If trouble still remains, the signal data converter chassis assembly is malfunctioning.
- d. If no other check is to be done, go to Shutdown.

8-2-9.10 Sensor Interwiring/Warning Output.

- a. Do Setup/Power On, if not already done.
- b. Make sure FUEL QUANTITY NO. 1 CAL-NOR switch on simulator tester is placed to NOR.
- c. Set multimeter to indicate dc voltage and connect multimeter between SDC FUEL H and FUEL L test jacks on unit tester.
- d. Turn simulator tester FUEL QUANTITY NO. 1 variable control fully clockwise, then fully counterclockwise.

RESULT: Multimeter shall indicate between -0.5 and 0.5 vdc when the FUEL QUANTITY NO. 1 control is fully counterclockwise, and shall indicate between 6.5 and 8.5 vdc when the FUEL QUANTITY NO. 1 control is fully clockwise.

- If result is not as specified, the signal data converter chassis assembly is malfunctioning.
- e. Connect multimeter between SDC WIRE and COM test jacks on unit tester.

RESULT: Multimeter shall indicate between 4.5 and 5.5 vdc.

- If result is not as specified, the signal data converter chassis assembly is malfunctioning.

- f. Momentarily press unit tester SIGNAL DATA CONVERTER WARN OUTPUTS button.

RESULT: With WARNING OUTPUT button pressed, the following simulator tester lamps shall go on:

LOW ROTOR SPD
NO. 1 LOW OIL PRESS
NO. 1 HIGH OIL TEMP
NO. 1 ENGINE OUT

- If result is not as specified, the signal data converter chassis assembly is malfunctioning.

- g. Connect multimeter between SDC W/C and COM test jacks on unit tester.

RESULT: Multimeter shall indicate between 4.5 and 5.5 vdc.

- If result is not as specified, the signal data converter chassis assembly is malfunctioning.

- h. Momentarily press simulator tester ROTOR OVER SPD RESET button switch.

RESULT: With ROTOR OVER SPD RESET button pressed, multimeter shall indicate between -0.5 and 0.5 vdc.

- If result is not as specified, the signal data converter chassis assembly is malfunctioning.

- i. Disconnect multimeter.

- j. If no other check is to be done, go to Shutdown.

8-2-9.11 No. 1 Engine % RPM (Analog).

- a. Do Setup/Power On, if not already done.
- b. Place simulator tester ENGINE % RPM NO. 1 CAL-NOR switch to CAL and ENGINE % RPM NO. 1 HI-LO switch to HI.
- c. Place unit tester INTER CAL SELECT switch to 1.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 952 and 962.

- If result is not as specified, replace interface No. 2, module A6 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- d. Place simulator tester ENGINE % RPM NO. 1 HI-LO switch to LO.

RESULT: INTERFACE MONITOR shall indicate between 0 and 5.

- If result is not as specified, replace interface No. 2, module A6 (PARA 8-5-4).
- e. Place simulator tester ENGINE % RPM NO. 1 CAL-NOR switch to NOR.
- f. Slowly turn simulator tester ENGINE % RPM NO. 1 variable control fully clockwise.

RESULT: INTERFACE MONITOR indication shall increase smoothly.

- If result is not as specified, replace interface No. 2, module A6 (PARA 8-5-4).
- g. Slowly turn simulator tester ENGINE % RPM NO. 1 variable control fully counterclockwise.

RESULT: INTERFACE MONITOR indication shall decrease smoothly.

- If result is not as specified, replace interface No. 2, module A6 (PARA 8-5-4).
- h. If no other check is to be done, go to Shutdown.

8-2-9.12 No. 2 Engine % RPM (Analog).

- a. Do Setup/Power On, if not already done.
- b. Place simulator tester ENGINE % RPM NO. 2 CAL-NOR switch to CAL and ENGINE % RPM NO. 2 HI-LO switch to HI.

- c. Place unit tester INTER CAL SELECT switch to 2.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 952 and 962.

- If result is not as specified, replace interface No. 2, module A6 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- d. Place simulator tester ENGINE % RPM NO. 2 HI-LO switch to LO.

RESULT: INTERFACE MONITOR shall indicate between 0 and 5.

- If result is not as specified, replace interface No. 2, module A6 (PARA 8-5-4).
- e. Place simulator tester ENGINE % RPM NO. 2 CAL-NOR switch to NOR.
- f. Slowly turn simulator tester ENGINE % RPM NO. 2 variable control fully clockwise.

RESULT: INTERFACE MONITOR indication shall increase smoothly.

- If result is not as specified, replace interface No. 2, module A6 (PARA 8-5-4).
- g. Slowly turn simulator tester ENGINE % RPM NO. 2 variable control fully counterclockwise.

RESULT: INTERFACE MONITOR indication shall decrease smoothly.

- If result is not as specified, replace interface No. 2, module A6 (PARA 8-5-4).

- h. If no other check is to be done, go to Shutdown.

8-2-9.13 Rotor Speed (Analog).

- a. Do Setup/Power On, if not already done.

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- b. Place simulator tester ROTOR SPEED CAL-NOR switch to CAL and ROTOR SPEED HI-LO switch to HI.

- c. Place unit tester INTER CAL SELECT switch to 3.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 1016 and 1026.

- If result is not as specified, replace interface No. 1, module A7 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- d. Place simulator tester ROTOR SPEED HI-LO switch to LO.

RESULT: INTERFACE MONITOR shall indicate between 0 and 5.

- If result is not as specified, replace interface No. 1, module A7 (PARA 8-5-4).

- e. Place simulator tester ROTOR CAL-NOR switch to NOR.

- f. Slowly turn simulator tester ROTOR SPEED variable control fully clockwise.

RESULT: INTERFACE MONITOR indication shall increase smoothly.

- If result is not as specified, replace interface No. 1, module A7 (PARA 8-5-4).

- g. Slowly turn simulator tester ROTOR SPEED variable control fully counterclockwise.

RESULT: INTERFACE MONITOR indication shall decrease smoothly.

- If result is not as specified, replace interface No. 1, module A7 (PARA 8-5-4).

- h. If no other check is to be done, go to Shutdown.

8-2-9.14 No. 1 Torque (Analog).

- a. Do Setup/Power On, if not already done.

- b. Place simulator tester TORQUE NO. 1 CAL-NOR switch to CAL and TORQUE NO. 1 HI-LO switch to HI.

- c. Place unit tester INTER CAL SELECT switch to 4.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 1042 and 1062.

- If result is not as specified, replace interface No. 3, module A5 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- d. Place simulator tester TORQUE NO. 1 HI-LO switch to LO.

RESULT: INTERFACE MONITOR shall indicate between 0 and 10.

- If result is not as specified, replace interface No. 3, module A5 (PARA 8-5-4).

- e. Place simulator tester TORQUE NO. 1 CAL-NOR switch to NOR.

- f. Slowly turn simulator tester TORQUE NO. 1 variable control fully clockwise.

RESULT: INTERFACE MONITOR indication shall increase smoothly.

- If result is not as specified, replace interface No. 3, module A5 (PARA 8-5-4).

- g. Slowly turn simulator tester TORQUE NO. 1 variable control fully counterclockwise.

RESULT: INTERFACE MONITOR indication shall decrease smoothly.

- If result is not as specified, replace interface No. 3, module A5 (PARA 8-5-4).

- h. If no other check is to be done, go to Shutdown.

8-2-9.15 No. 2 Torque (Analog).

- Do Setup/Power On, if not already done.
- Place simulator tester TORQUE NO. 2 CAL-NOR switch to CAL and TORQUE NO. 2 HI-LO switch to HI.
- Place unit tester INTER CAL SELECT switch to 5.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 1042 and 1062.

- If result is not as specified, replace interface No. 3, module A5 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- Place simulator tester TORQUE NO. 2 HI-LO switch to LO.

RESULT: INTERFACE MONITOR shall indicate between 0 and 10.

- If result is not as specified, replace interface No. 3, module A5 (PARA 8-5-4).

- Place simulator tester TORQUE NO. 2 CAL-NOR switch to NOR.
- Slowly turn simulator tester TORQUE NO. 2 variable control fully clockwise.

RESULT: INTERFACE MONITOR indication shall increase smoothly.

- If result is not as specified, replace interface No. 3, module A5 (PARA 8-5-4).

- Slowly turn simulator tester TORQUE NO. 2 variable control fully counterclockwise.

RESULT: INTERFACE MONITOR indication shall decrease smoothly.

- If result is not as specified, replace interface No. 3, module A5 (PARA 8-5-4).

- h. If no other check is to be done, go to Shutdown.

8-2-9.16 Engine Oil Temperature (Analog).

- Do Setup/Power On, if not already done.
- Place simulator tester ENGINE OIL TEMP NO. 1 CAL-NOR switch to CAL and ENGINE OIL TEMP NO. 1 HI-LO switch to HI.

- Place unit tester INTER CAL SELECT switch to 6.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 1177 and 1223.

- If result is not as specified, replace interface No. 2, module A6 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- Place simulator tester ENGINE OIL TEMP NO. 1 HI-LO switch to LO.

RESULT: INTERFACE MONITOR shall indicate between 0 and 16.

- If result is not as specified, replace interface No. 2, module A6 (PARA 8-5-4).

- Place simulator tester ENGINE OIL TEMP NO. 1 CAL-NOR switch to NOR.
- Slowly turn simulator tester ENGINE OIL TEMP NO. 1 variable control fully clockwise.

RESULT: INTERFACE MONITOR indication shall increase smoothly.

- If result is not as specified, replace interface No. 2, module A6 (PARA 8-5-4).

- Slowly turn simulator tester ENGINE OIL TEMP NO. 1 variable control fully counterclockwise.

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RESULT: INTERFACE MONITOR indication shall decrease smoothly.

- If result is not as specified, replace interface No. 2, module A6 (PARA 8-5-4).
- h. If no other check is to be done, go to Shutdown.

8-2-9.17 Engine Oil Pressure (Analog).

- a. Do Setup/Power On, if not already done.
- b. Place simulator tester ENGINE OIL PRESS NO. 1 CAL-NOR switch to CAL and ENGINE OIL PRESS NO. 1 HI-LO switch to HI.
- c. Place unit tester INTER CAL SELECT switch to 7.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 900 and 940.

- If result is not as specified, replace interface No. 4, module A4 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.
- d. Place simulator tester ENGINE OIL PRESS NO. 1 HI-LO switch to LO.

RESULT: INTERFACE MONITOR shall indicate between 0 and 20.

- If result is not as specified, replace interface No. 4, module A4 (PARA 8-5-4).
- e. Place simulator tester ENGINE OIL PRESS NO. 1 CAL-NOR switch to NOR.
- f. Slowly turn simulator tester ENGINE OIL PRESS NO. 1 variable control fully clockwise.

RESULT: INTERFACE MONITOR indication shall increase smoothly.

- If result is not as specified, replace interface No. 4, module A4 (PARA 8-5-4).

- g. Slowly turn simulator tester ENGINE OIL PRESS NO. 1 variable control fully counterclockwise.

RESULT: INTERFACE MONITOR indication shall decrease smoothly.

- If result is not as specified, replace interface No. 4, module A4 (PARA 8-5-4).
- h. If no other check is to be done, go to Shutdown.

8-2-9.18 Turbine Gas Temperature (Analog).

- a. Do Setup/Power On, if not already done.
- b. Place simulator tester TURBINE GAS TEMP NO. 1 CAL-NOR switch to CAL and TURBINE GAS TEMP NO. 1 HI-LO switch to HI.
- c. Place unit tester INTER CAL SELECT switch to 8.

RESULT: Unit tester INTERFACE MONITOR shall indicate 1267 and 1283.

- If result is not as specified, replace interface No. 4, module A4 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.
- d. Place simulator tester TURBINE GAS TEMP NO. 1 HI-LO switch to LO.

RESULT: INTERFACE MONITOR shall indicate between 0 and 8.

- If result is not as specified, replace interface No. 4, module A4 (PARA 8-5-4).
- e. Place simulator tester TURBINE GAS TEMP NO. 1 CAL-NOR switch to NOR.
- f. Slowly turn simulator tester TURBINE GAS TEMP NO. 1 variable control fully clockwise.

RESULT: INTERFACE MONITOR indication shall increase smoothly.

- If result is not as specified, replace interface No. 4, module A4 (PARA 8-5-4).

g. Slowly turn simulator tester TURBINE GAS TEMP NO. 1 variable control fully counterclockwise.

RESULT: INTERFACE MONITOR indication shall decrease smoothly.

- If result is not as specified, replace interface No. 4, module A4 (PARA 8-5-4).

h. If no other check is to be done, go to Shutdown.

8-2-9.19 Gas Generator Speed (Analog).

a. Do Setup/Power On, if not already done.

b. Place simulator tester GAS GEN SPEED NO. 1 CAL-NOR switch to CAL and GAS GEN SPEED NO. 1 HI-LO switch to HI.

c. Place unit tester INTER CAL SELECT switch to 9.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 995 and 1005.

- If result is not as specified, replace interface No. 1, module A7 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

d. Place simulator tester GAS GEN SPEED NO. 1 HI-LO switch to LO.

RESULT: INTERFACE MONITOR shall indicate between 0 and 5.

- If result is not as specified, replace interface No. 1, module A7 (PARA 8-5-4).

e. Place simulator tester GAS GEN SPEED NO. 1 CAL-NOR switch to NOR.

f. Slowly turn simulator tester GAS GEN SPEED NO. 1 variable control fully clockwise.

RESULT: INTERFACE MONITOR indication shall increase smoothly.

- If result is not as specified, replace interface No. 1, module A7 (PARA 8-5-4).

g. Slowly turn simulator tester GAS GEN SPEED NO. 1 variable control counterclockwise.

RESULT: INTERFACE MONITOR indication shall decrease smoothly.

- If result is not as specified, replace interface No. 1, module A7 (PARA 8-5-4).

h. If no other check is to be done, go to Shutdown.

8-2-9.20 No. 1 Torque (Digital).

a. Do Setup/Power On, if not already done.

b. Place simulator tester TORQUE NO. 1 CAL-NOR switch to CAL and TORQUE NO. 1 HI-LO switch to HI.

c. Place unit tester INTER CAL SELECT switch to 10.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 1031 and 1051.

- If result is not as specified, replace interface No. 3, module A5 (PARA 8-5-4).

(a) If trouble remains, replace digital processor, module A10 (PARA 8-5-4).

(b) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

d. Place simulator tester TORQUE NO. 1 HI-LO switch to LO.

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RESULT: INTERFACE MONITOR shall indicate between 0 and 10.

- If result is not as specified, replace interface No. 3, module A5 (PARA 8-5-4).
- e. Place simulator tester TORQUE NO. 1 CAL-NOR switch to NOR.
- f. Slowly turn simulator tester TORQUE NO. 1 variable control fully clockwise.

RESULT: INTERFACE MONITOR indication shall increase smoothly.

- If result is not as specified, replace interface No. 3, module A5 (PARA 8-5-4).
- g. Slowly turn simulator tester TORQUE NO. 1 variable control fully counterclockwise.

RESULT: INTERFACE MONITOR indication shall decrease smoothly.

- If result is not as specified, replace interface No. 3, module A5 (PARA 8-5-4).
- h. If no other check is to be done, go to Shutdown.

8-2-9.21 No. 2 Torque (Digital).

- a. Do Setup/Power On, if not already done.
- b. Place simulator tester TORQUE NO. 2 CAL-NOR switch to CAL and TORQUE NO. 2 HI-LO switch to HI.
- c. Place unit tester INTER CAL SELECT switch to 11.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 1031 and 1051.

- If result is not as specified, replace interface No. 3, module A5 (PARA 8-5-4).
- (a) If trouble remains, replace digital processor, module A10 (PARA 8-5-4).

- (b) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

- d. Place simulator tester TORQUE NO. 2 HI-LO switch to LO.

RESULT: INTERFACE MONITOR shall indicate between 0 and 10.

- If result is not as specified, replace interface No. 3, module A5 (PARA 8-5-4).
- e. Place simulator tester TORQUE NO. 2 CAL-NOR switch to NOR.
- f. Slowly turn simulator tester TORQUE NO. 2 variable control fully clockwise.

RESULT: INTERFACE MONITOR indication shall increase smoothly.

- If result is not as specified, replace interface No. 3, module A5 (PARA 8-5-4).
- g. Slowly turn simulator tester TORQUE NO. 2 variable control fully counterclockwise.

RESULT: INTERFACE MONITOR indication shall decrease smoothly.

- If result is not as specified, replace interface No. 3, module A5 (PARA 8-5-4).
- h. If no other check is to be done, go to Shutdown.

8-2-9.22 Turbine Gas Temperature (Digital).

- a. Do Setup/Power On, if not already done.
- b. Place simulator tester TURBINE GAS TEMP NO. 1 CAL-NOR switch to CAL and TURBINE GAS TEMP NO. 1 HI-LO switch to HI.
- c. Place unit tester INTER CAL SELECT switch to 12.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 807 and 817.

- If result is not as specified, replace interface No. 4, module A4 (PARA 8-5-4).
 - (a) If trouble remains, replace digital processor module A10 (PARA 8-5-4).
 - (b) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

d. Place simulator tester TURBINE GAS TEMP NO. 1 HI-LO switch to LO.

RESULT: INTERFACE MONITOR shall indicate between 0 and 8.

- If result is not as specified, replace interface No. 4, module A4 (PARA 8-5-4).
- e. Place simulator tester TURBINE GAS TEMP NO. 1 CAL-NOR switch to NOR.
- f. Slowly turn simulator tester TURBINE GAS TEMP NO. 1 variable control fully clockwise.

RESULT: INTERFACE MONITOR indication shall increase smoothly.

- If result is not as specified, replace interface No. 4, module A4 (PARA 8-5-4).
- g. Slowly turn simulator tester TURBINE GAS TEMP NO. 1 variable control fully counterclockwise.

RESULT: INTERFACE MONITOR indication shall decrease smoothly.

- If result is not as specified, replace interface No. 4, module A4 (PARA 8-5-4).
- h. If no other check is to be done, go to Shutdown.

8-2-9.23 Gas Generator Speed (Digital).

- a. Do Setup/Power On, if not already done.

b. Place simulator tester GAS GEN SPEED NO. 1 CAL-NOR switch to CAL and GAS GEN SPEED NO. 1 HI-LO switch to HI.

c. Place unit tester INTER CAL SELECT switch to 13.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 995 and 1005.

- If result is not as specified, replace interface No. 1, module A7 (PARA 8-5-4).

(a) If trouble remains, replace digital processor module A10 (PARA 8-5-4).

(b) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

d. Place simulator tester GAS GEN SPEED NO. 1 HI-LO switch to LO.

RESULT: INTERFACE MONITOR shall indicate between 0 and 5.

- If result is not as specified, replace interface No. 1, module A7 (PARA 8-5-4).

e. Place simulator tester GAS GEN SPEED NO. 1 CAL-NOR switch to NOR.

f. Slowly turn simulator test GAS GEN SPEED NO. 1 variable control fully clockwise.

RESULT: INTERFACE MONITOR indication shall increase smoothly.

- If result is not as specified, replace interface No. 1, module A7 (PARA 8-5-4).

g. Slowly turn simulator tester GAS GEN SPEED NO. 1 variable control fully counterclockwise.

RESULT: INTERFACE MONITOR indication shall decrease smoothly.

- If result is not as specified, replace interface No. 1, module A7 (PARA 8-5-4).

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- h. If no other check is to be done, go to Shutdown.

8-2-9.24 Rotor Speed (Analog Relay).

- a. Do Setup/Power On, if not already done.
- b. Place simulator tester ROTOR SPEED CAL-NOR switch to CAL and ROTOR SPEED HI-LO switch to HI.
- c. Place unit tester INTER CAL SELECT switch to 14.

RESULT: Unit tester INTERFACE MONITOR shall indicate between 1016 and 1026.

- If result is not as specified, replace interface No. 1, module A7 (PARA 8-5-4).

If trouble remains, the signal data converter chassis assembly is malfunctioning.

- d. Place simulator tester ROTOR SPEED HI-LO switch to LO.

RESULT: INTERFACE MONITOR shall indicate between 0 and 5.

- If result is not as specified, replace interface No. 1, module A7 (PARA 8-5-4).
- e. Place simulator tester ROTOR CAL-NOR switch to NOR.
- f. Slowly turn simulator tester ROTOR SPEED variable control fully clockwise.

RESULT: INTERFACE MONITOR indication shall increase smoothly.

- If result is not as specified, replace interface No. 1, module A7 (PARA 8-5-4).

- g. Slowly turn simulator tester ROTOR SPEED variable control fully counterclockwise.

RESULT: INTERFACE MONITOR indication shall decrease smoothly.

- If result is not as specified, replace interface No. 1, module A7 (PARA 8-5-4).

- h. If no other check is to be done, go to Shutdown.

8-2-9.25 Integral Lighting.

- a. Do Setup/Power On, if not already done.
- b. Place simulator tester INTEGRAL LIGHTING switch to INT.

RESULT: Both SIGNAL DATA CONVERTER INT LIGHT lamps on unit tester shall go on.

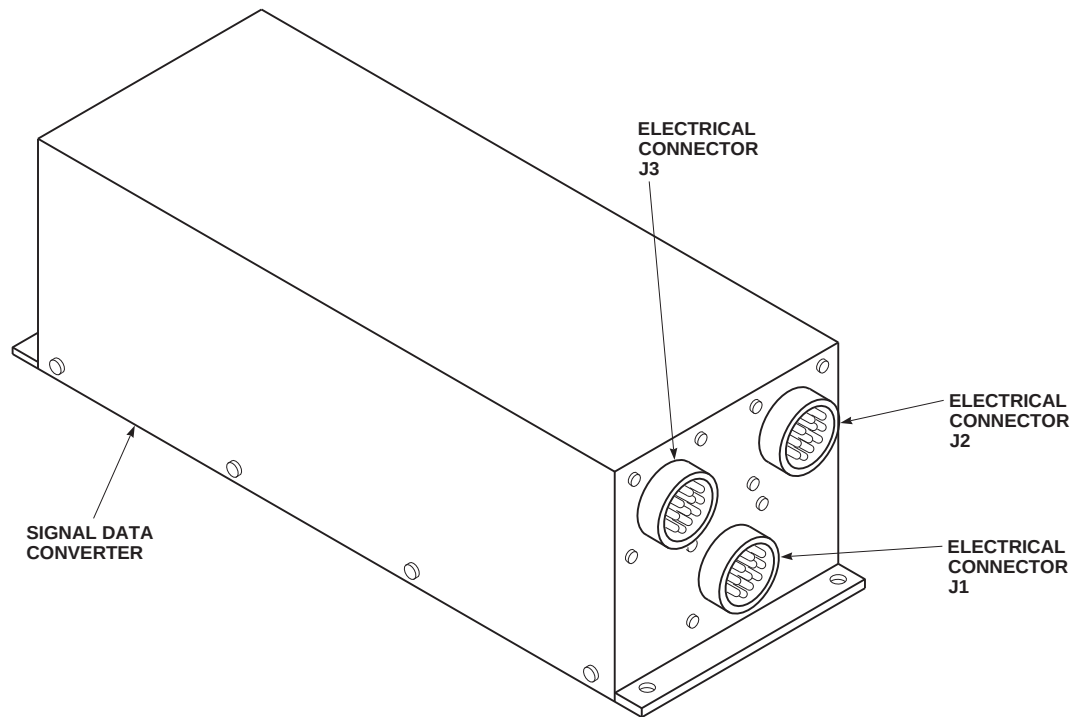
- If result is not as specified, the signal data converter chassis assembly is malfunctioning.

- c. Place simulator tester INTEGRAL LIGHTING switch to EXT.

- d. If no other check is to be done, go to Shutdown.

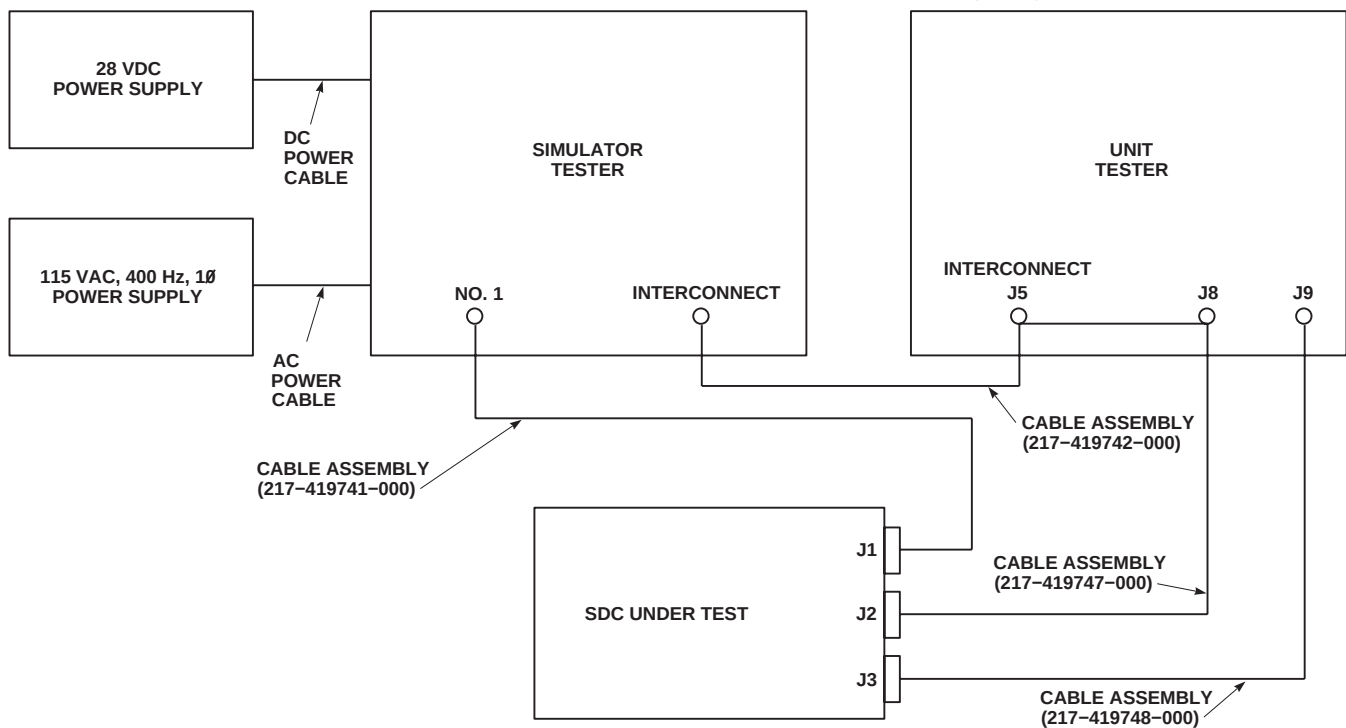
8-2-9.26 Shutdown.

- a. Place simulator tester INPUT POWER switch to OFF.
- b. Disconnect cable assemblies.



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FIGURE 8-2-9-1. SIGNAL DATA CONVERTER (SDC)



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FIGURE 8-2-9-2. SIGNAL DATA CONVERTER TEST SETUP

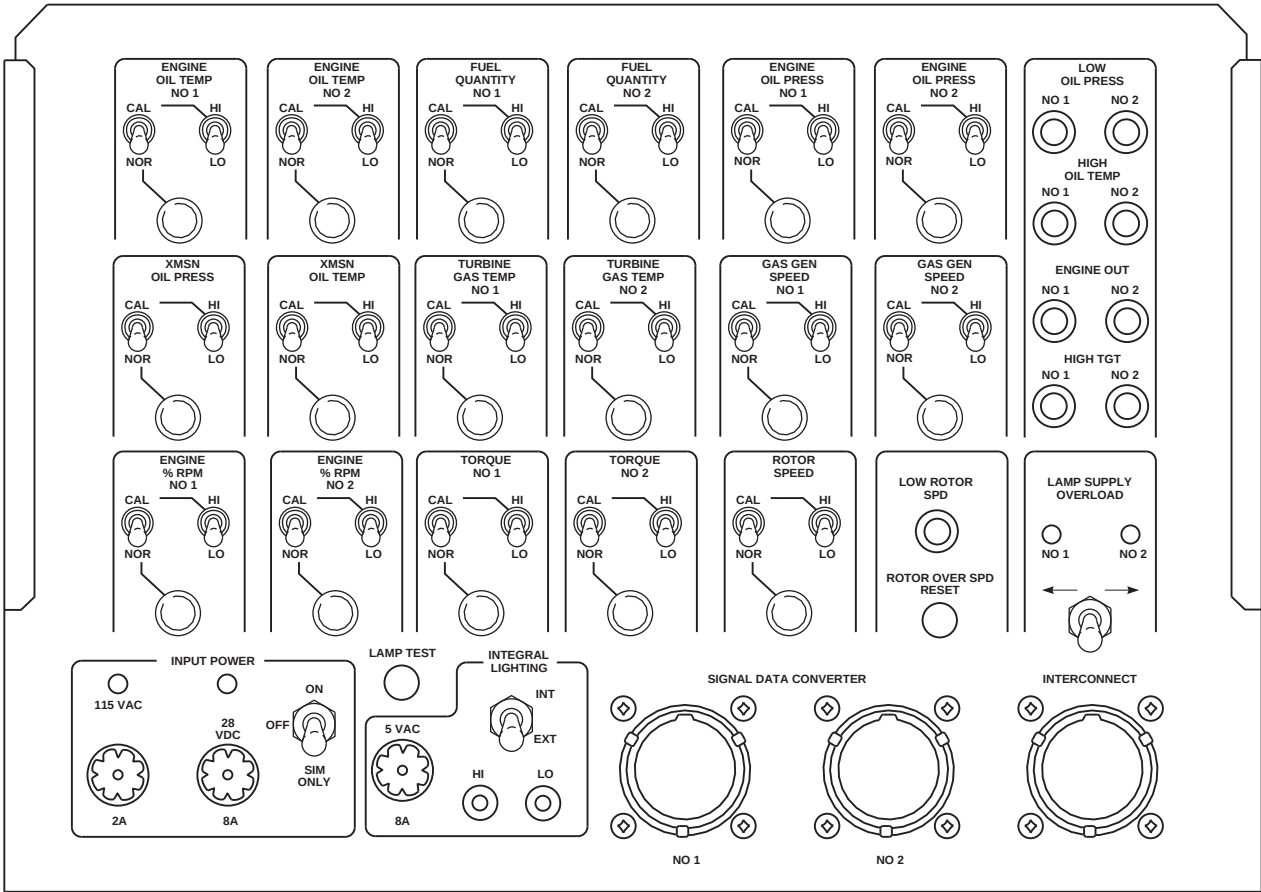


FIGURE 8-2-9-3. BENCH TEST SET (SHEET 1 OF 2)

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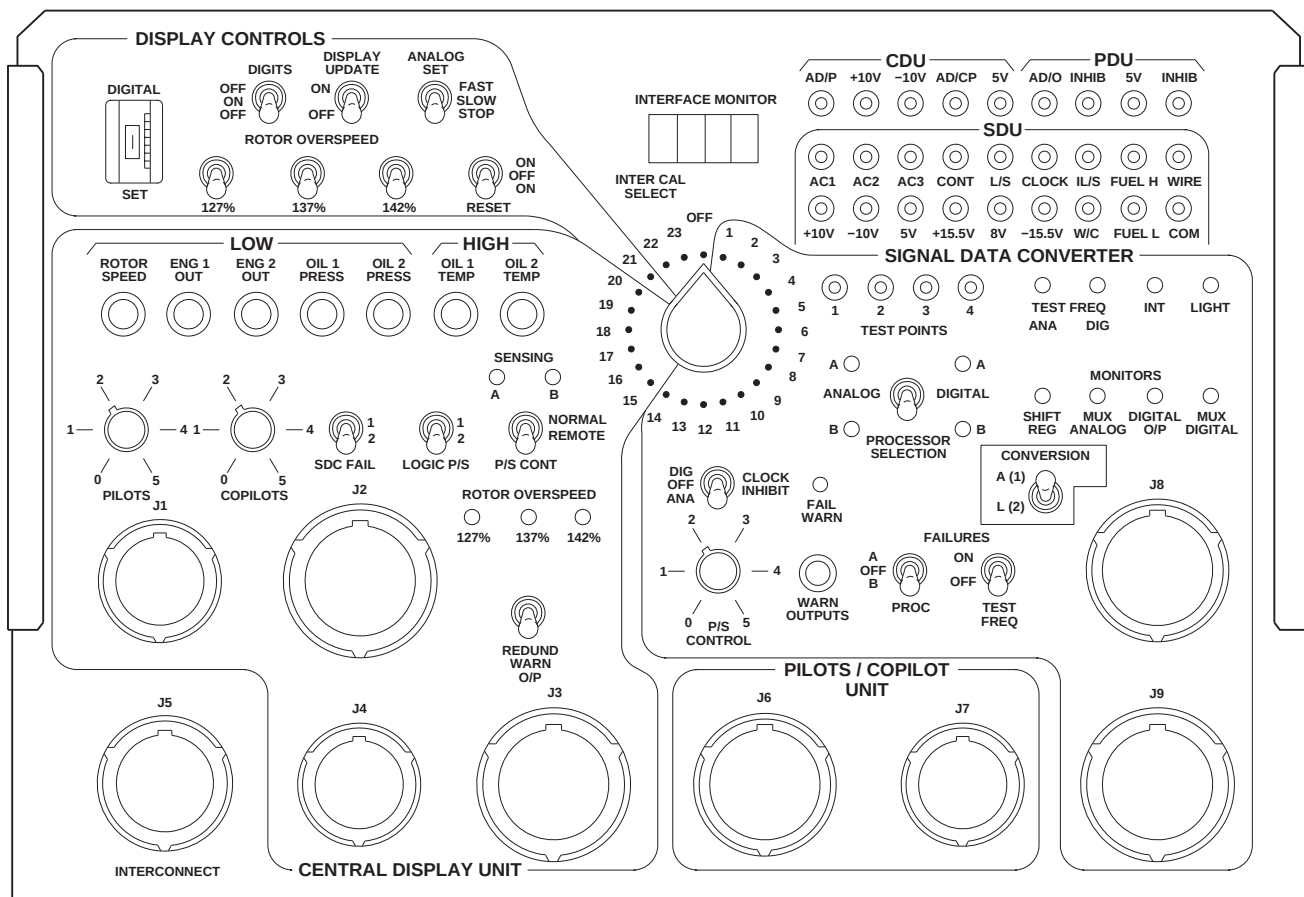


FIGURE 8-2-9-3. BENCH TEST SET (SHEET 2 OF 2)

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8-2-10 CAUTION/ADVISORY PANEL (BENCH).

INITIAL SETUP

Test Equipment

- Decade Resistor, ZM57U
- Multimeter, Fluke 45 (2 required)

Tools

- Electrical Repairer Toolkit
- Locally-Made Caution/Advisory Panel Test Set Adapter, Figure H-53, Appendix H
- Locally-Made Insulated Jumper Wire (6 required)

Equipment Conditions

- 27 to 28 vdc Bench Power Available

NOTE

In the following operational/troubleshooting procedure, all voltage indications are dependent upon the input voltage of 27 to 28 vdc.

NOTE

In the following operational/troubleshooting procedure, reference to the caution/advisory panel is abbreviated as panel.

8-2-10.1 Operational/Troubleshooting Procedure.

- Connect test adapter CB1 and CB2 input terminals to bench power.
- Make sure test adapter circuit breakers (circuit breakers) CB1 and CB2 are pulled out.
- Make sure test adapter switches S2 and S3 are off.
- Adjust test adapter resistors R1 and R2 fully counterclockwise.
- Push in circuit breakers CB1 and CB2.
- Connect No. 1 multimeter between test points TP8 and TP9.

RESULT: No. 1 multimeter shall indicate between 27 and 28 vdc.

- If result is not as specified, verify bench power is between 27 and 28 vdc.
- Connect test adapter test point (test point) TP9 to power ground.

- Pull out circuit breakers CB1 and CB2.
- Install an insulated jumper wire between test points TP8 and J117-65.
- Install an insulated jumper wire between test points TP8 and J117-66.
- Connect test adapter P118 to J118 at rear of panel.
- Connect test adapter P117 to J117 at rear of panel.
- Push in circuit breakers CB1 and CB2.
- Make sure bench power is between 27 and 28 vdc.
- Connect No. 1 multimeter between test points TP4 and TP5.
- Connect No. 2 multimeter between test points TP1 and TP2.
- Place and hold caution/advisory panel BRT/DIM-TEST switch at TEST.

RESULT:

- (1) All caution/advisory panel capsules (capsules) on front of panel shall go on brightly.
 - (2) Multimeter No. 1 shall indicate between 22 and 26 vdc.
 - (3) Multimeter No. 2 shall indicate between 22 and 26 vdc.
- If result (1) is not as specified, go to TABLE NO. 8-2-10-1.
 - If any single capsule or group of capsules listed by channel card in TABLE NO. 8-2-10-2 do not go on, replace associated channel card (PARA 8-5-14).
 - If results (2) and (3) are not as specified, replace control card A5 (PARA 8-5-14).

If trouble remains, replace chassis-mounted Q1 (PARA 8-5-16).

- If result (2) is not as specified, repair/replace wiring, as required, between:

J118-V and J118-e

J118-U and J118-e

- If result (3) is not as specified, replace channel card A4 (PARA 8-5-14).
- If only caution capsules do not go on, replace control card A5 (PARA 8-5-14).

If trouble remains, replace chassis-mounted Q3 (PARA 8-5-16).

- If only advisory capsules do not go on, replace control card A5 (PARA 8-5-14).

If trouble remains, replace chassis-mounted Q2 (PARA 8-5-16).

- r. Release caution/advisory panel BRT/DIM-TEST switch.

RESULT: Switch handle shall return to center, spring-loaded, neutral position.

- If result is not as specified, replace switch (PARA 8-5-15).
- s. Install an insulated jumper wire between test points TP8 and TB1-A.
 - t. Connect No. 1 multimeter between test points TP8 and TP6.
 - u. Connect No. 2 multimeter between test points TP1 and TP2.

RESULT:

- (1) No. 1 multimeter shall indicate greater than 25 vdc.
- (2) No. 2 multimeter shall indicate 0 vdc.
- (3) Capsule position 17 shall go on.
- (4) Test adapter lamp DS1 shall remain off.

- If result (1) is not as specified, replace control card A5 (PARA 8-5-14).

If trouble remains, replace chassis-mounted Q2 (PARA 8-5-16).

- If result (2) is not as specified, replace channel card A4 (PARA 8-5-14).
- If result (3) is not as specified, replace channel card A2 (PARA 8-5-14).
- If result (4) is not as specified, replace channel card A4 (PARA 8-5-14).

- v. Place test adapter switch S2 on.
- w. Place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM and release.

RESULT: No. 1 multimeter shall indicate between 3.0 and 4.5 vdc.

- If result is not as specified, replace control card A5 (PARA 8-5-14).

8-2-10-2

If trouble remains, replace chassis-mounted Q2 (PARA 8-5-16).

- x. Slowly adjust test adapter resistor R1 from fully counterclockwise to fully clockwise.

RESULT: No. 1 multimeter shall indicate from between 3.0 and 4.5 vdc (fully counterclockwise) to between 12 and 13 vdc (fully clockwise).

- If result is not as specified, replace control card A5 (PARA 8-5-14).

If trouble remains, replace chassis-mounted Q2 (PARA 8-5-16).

- y. Adjust test adapter resistor R1 fully counterclockwise.
- z. Place test adapter switch S3 on.
- aa. Adjust test adapter resistor R1 from fully counterclockwise to fully clockwise.

RESULT: No. 1 multimeter shall indicate between 3.0 and 4.5 vdc and not change.

- If result is not as specified, replace control card A5 (PARA 8-5-14).

If trouble remains, replace chassis-mounted Q2 (PARA 8-5-16).

- ab. Adjust test adapter resistor R1 fully counterclockwise.
- ac. Momentarily pull out then reset circuit breaker CB1.
- ad. Place test adapter switches S2 and S3 off.

RESULT: Multimeter shall indicate between 3.0 and 4.5 vdc.

- If result is not as specified, replace control card A5 (PARA 8-5-14).
- ae. Remove insulated jumper wire from between test points TP8 and TB1-A.

- af. Momentarily pull out then reset circuit breaker CB1.

- ag. Install an insulated jumper wire between test points TP9 and J117-34.

- ah. Connect No. 1 multimeter between test points TP8 and TP7.

RESULT:

- (1) Multimeter shall indicate greater than 25 vdc.
- (2) Capsule position 42 shall go on.
- (3) Test adapter lamp DS1 shall go on.
- If result (1) is not as specified, replace control card A5 (PARA 8-5-14).

If trouble remains, replace chassis-mounted Q3 (PARA 8-5-16).

- If result (2) or (3) is not as specified, replace channel card A3 (PARA 8-5-14).
- ai. Connect No. 1 multimeter between test points TP1 and TP2.
- aj. Momentarily press test adapter switch S1.

RESULT:

- (1) Multimeter shall indicate 0 vdc.
- (2) Test adapter lamp DS1 shall go off.
- If results are not as specified, replace channel card A4 (PARA 8-5-14).

- ak. Make sure test adapter switches S2 and S3 are off.
- al. Momentarily pull out then reset circuit breaker CB1.
- am. Set decade resistor to 85 ohms.
- an. Connect decade resistor, set to 85 ohms, between test points TP8 and TP3.

- ao. Connect No. 1 multimeter between test points TP8 and TP3.

RESULT:

- (1) Multimeter shall indicate less than 0.3 vdc.
- (2) Capsule position 42 shall go on.

- If results are not as specified, replace channel card A3 (PARA 8-5-14).

- ap. Place test adapter switch S2 on.

- aq. Place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM and release.

RESULT: Multimeter shall indicate greater than 25 vdc.

- If result is not as specified, replace channel card A4 (PARA 8-5-14).

- ar. Disconnect decade resistor connections from TP8 and TP3.

- as. Connect No. 1 multimeter between TP8 and TP7.

RESULT: Multimeter shall indicate between 3.0 and 4.5 vdc.

- If result is not as specified, replace control card A5 (PARA 8-5-14).

If trouble remains, replace chassis-mounted Q3 (PARA 8-5-16).

- at. Slowly adjust test adapter resistor R1 from fully counterclockwise to fully clockwise.

RESULT: No. 1 multimeter shall indicate from between 3.0 and 4.5 vdc (fully counterclockwise) to between 12 and 13 vdc (fully clockwise).

- If result is not as specified, replace control card A5 (PARA 8-5-14).

If trouble remains, replace chassis-mounted Q3 (PARA 8-5-16).

- au. Adjust test adapter resistor R1 fully counterclockwise.

- av. Place test adapter switch S3 on.

- aw. Adjust test adapter resistor R1 from fully counterclockwise to fully clockwise.

RESULT: No. 1 multimeter shall indicate between 3.0 and 4.5 vdc and not change.

- If result is not as specified, replace control card A5 (PARA 8-5-14).

If trouble remains, replace chassis-mounted Q3 (PARA 8-5-16).

- ax. Adjust test adapter resistor R1 fully counterclockwise.

- ay. Place test adapter switches S2 and S3 off.

- az. Momentarily pull out then reset circuit breaker CB1.

- ba. Connect No. 2 multimeter between test points TP1 and TP2.

RESULT: No. 2 multimeter shall indicate greater than 25 vdc.

- If result is not as specified, replace channel card A4 (PARA 8-5-14).

- bb. Place test adapter switch S2 on.

- bc. Place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM and release.

RESULT: No. 2 multimeter shall indicate between 3.0 and 4.5 vdc.

- If result is not as specified, replace channel card A4 (PARA 8-5-14).

- bd. Slowly adjust test adapter resistor R2 from fully counterclockwise to fully clockwise.

8-2-10-4

RESULT: No. 2 multimeter shall indicate from between 3.0 and 4.5 vdc (fully counterclockwise) to between 14 and 15 vdc (fully clockwise).

- If result is not as specified, replace channel card A4 (PARA 8-5-14).
- be. Adjust test adapter resistor R2 fully counterclockwise.
- bf. Connect No. 1 multimeter between test points TP8 and TP7.
- bg. Place test adapter switch S3 on.
- bh. Slowly adjust test adapter resistor R2 from fully counterclockwise to fully clockwise.

RESULT: No. 1 multimeter shall indicate between 3.0 and 4.5 vdc and not change.

- If result is not as specified, replace channel card A4 (PARA 8-5-14).
- bi. Adjust test adapter resistor R2 fully counterclockwise.
- bj. Place test adapter S2 and S3 off.
- bk. Momentarily pull out then reset circuit breaker CB1.
- bl. Connect No. 1 multimeter between test points TP8 and TP5.

RESULT: Multimeter shall indicate greater than 25 vdc.

- If result is not as specified, replace control card A5 (PARA 8-5-14).
- If trouble remains, replace chassis-mounted Q1 (PARA 8-5-16).
- bm. Place test adapter switch S2 on.
- bn. Place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM and release.

RESULT: Multimeter shall indicate between 3.0 and 4.5 vdc.

- If result is not as specified, replace channel card A4 (PARA 8-5-14).

If trouble remains, replace chassis-mounted Q1 (PARA 8-5-16).

- bo. Slowly adjust test adapter resistor R2 from fully counterclockwise to fully clockwise.

RESULT: No. 1 multimeter shall indicate from between 3.0 and 4.5 vdc (fully counterclockwise) to between 14.5 and 15.5 vdc (fully clockwise).

- If result is not as specified, replace channel card A4 (PARA 8-5-14).

If trouble remains, replace chassis-mounted Q1 (PARA 8-5-16).

- bp. Adjust test adapter resistor R2 fully counterclockwise.
- bq. Place test adapter switch S3 on.
- br. Adjust test adapter resistor R2 from fully counterclockwise to fully clockwise.

RESULT: No. 1 multimeter shall indicate from between 3.0 and 4.5 vdc (fully counterclockwise) to between 14.5 and 15.5 vdc (fully clockwise).

- If result is not as specified, replace control card A5 (PARA 8-5-14).

If trouble remains, replace chassis-mounted Q1 (PARA 8-5-16).

- bs. Adjust test adapter resistor R2 fully counterclockwise.
- bt. Place test adapter S2 and S3 off.
- bu. Momentarily pull out then reset circuit breaker CB1.

- bv. Remove the insulated jumper wire installed in step ag.
- bw. Momentarily install an insulated jumper between test point TP8 (28 vdc) and test points listed in TABLE NO. 8-2-10-3.

RESULT:

- (1) Each advisory capsule shall go on brightly when 28 vdc is applied to indicated test point and shall go off when 28 vdc is removed.
- (2) Test adapter lamp DS1 shall remain off.
 - If results are not as specified, replace indicated channel card (PARA 8-5-14).
 - If none of the capsules go on when 28 vdc is applied, replace overvoltage protector CR3 (PARA 8-5-20).
- bx. Momentarily install an insulated jumper wire between test point TP8 (28 vdc) and test points listed in TABLE NO. 8-2-10-4.

RESULT:

- (1) Each caution capsule and test adapter lamp DS1 shall go on brightly when 28 vdc is applied to indicated test point and shall go off when 28 vdc is removed.
- (2) Capsule positions 1 and 62 shall flash.
 - If results are not as specified, replace indicated channel card (PARA 8-5-14).
 - If none of the capsules go on when 28 vdc is applied, replace overvoltage protector CR3 (PARA 8-5-20).
- by. Momentarily connect decade resistor, set to 50 ohms, between test point TP9 and P117-5.

RESULT:

- (1) Capsule position 5 shall go on when signal is applied.

- (2) Test adapter lamp DS1 shall go on.
- (3) Capsule position 5 shall go off when signal is removed.
- (4) Test adapter lamp DS1 shall go off when signal is removed.

- If results (1) and (2) are not as specified, adjust decade resistor between 45 and 55 ohms and repeat step.
- If results (3) and (4) are not as specified, replace channel card A3 (PARA 8-5-14).

- bz. Momentarily connect decade resistor, set to 50 ohms, between test point TP9 and J117-53.

RESULT:

- (1) Capsule position 66 shall go on when signal is applied.
- (2) Test adapter lamp DS1 shall go on.
- (3) Capsule position 66 shall go off when signal is removed.
- (4) Test adapter lamp DS1 shall go off when signal is removed.
 - If results (1) and (2) are not as specified, adjust decade resistor between 45 and 55 ohms and repeat step.
 - If results (3) and (4) are not as specified, replace channel card A3 (PARA 8-5-14).
- ca. Momentarily install an insulated jumper wire between test point TP9 and J117-18.

RESULT:

- (1) Capsule position 21 shall go on when signal is applied and go off when signal is removed.
- (2) Test adapter lamp DS1 shall go on when signal is applied and go off when signal is removed.

8-2-10-6

- If results are not as specified, replace channel card A3 (PARA 8-5-14).
- cb. Momentarily connect decade resistor, set to 50 ohms, between test point TP9 and J117-30.

RESULT:

- (1) Capsule position 33 shall go on between 30 and 33 seconds after signal is applied and shall go off when signal is removed.
 - (2) Test adapter lamp DS1 shall go on when signal is applied and go off when signal is removed.
- If results are not as specified, replace channel card A3 (PARA 8-5-14).
- cc. Momentarily connect decade resistor, set to 50 ohms, between test point TP9 and J117-61.

RESULT:

- (1) Capsule position 74 shall go on when signal is applied and shall go off when signal is removed.
 - (2) Test adapter lamp DS1 shall go on when signal is applied and go off when signal is removed.
- If results are not as specified, replace channel card A3 (PARA 8-5-14).
- cd. Momentarily connect decade resistor, set to 50 ohms, between test point TP9 and J117-62.

RESULT:

- (1) Capsule position 75 shall go on when signal is applied and shall go off when signal is removed.
- (2) Test adapter lamp DS1 shall go on when signal is applied and go off when signal is removed.

- If results are not as specified, replace channel card A3 (PARA 8-5-14).
- ce. Momentarily install an insulated jumper wire between test point TP9 and J117-26.

RESULT:

- (1) Capsule position 29 shall go on when signal is applied, and shall go off when signal is removed.
 - (2) Test adapter lamp DS1 shall go on when signal is applied and go off when signal is removed.
- If results are not as specified, replace channel card A4 (PARA 8-5-14).
- cf. Momentarily install an insulated jumper wire between test point TP9 and J117-42.

RESULT:

- (1) Capsule position 50 shall go on when signal is applied and shall go off when signal is removed.
 - (2) Test adapter lamp DS1 shall go on when signal is applied and go off when signal is removed.
- If results are not as specified, replace channel card A4 (PARA 8-5-14).
- cg. Momentarily connect decade resistor, set to 50 ohms, between test point TP9 and J117-29.

RESULT:

- (1) Capsule position 32 shall go on when signal is applied and shall go off when signal is removed.
- (2) Test adapter lamp DS1 shall go on when signal is applied and go off when signal is removed.

- If results are not as specified, adjust decade resistor between 45 and 55 ohms and repeat step.

If trouble remains, replace channel card A4 (PARA 8-5-14).

- ch. Momentarily connect decade resistor, set to 50 ohms, between test point TP9 and J117-45.

RESULT:

- (1) Capsule position 53 shall go on when signal is applied and shall go off when signal is removed.
 - (2) Test adapter lamp DS1 shall go on when signal is applied and go off when signal is removed.
- If results are not as specified, adjust decade resistor between 45 and 55 ohms and repeat step.

If trouble remains, replace channel card A4 (PARA 8-5-14).

- ci. Momentarily connect decade resistor, set to 50 ohms, between test point TP9 and J117-13.

RESULT:

- (1) Capsule position 13 shall go on when signal is applied and shall go off when signal is removed.
- (2) Test adapter lamp DS1 shall go on when signal is applied and go off when signal is removed.

- If results are not as specified, replace channel card A4 (PARA 8-5-14).

- cj. Momentarily connect decade resistor, set to 50 ohms, between test point TP9 and J117-14.

RESULT:

- (1) Capsule position 14 shall go on when signal is applied and shall go off when signal is removed.
- (2) Test adapter lamp DS1 shall go on when signal is applied and go off when signal is removed.

- If results are not as specified, replace channel card A4 (PARA 8-5-14).

- ck. Remove bench power from test adapter.

- cl. Disconnect decade resistor.

- cm. Disconnect test adapter from caution/advisory panel.

TABLE NO. 8-2-10-1. No Caution/Advisory Panel Capsules Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. With switch S1 held at TEST, check for 28 vdc between K1-X1 and K1-X2.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3.
2. Check for 28 vdc between J118-e and test point TP9.	Step 1. If voltage is as specified repair/replace wiring, as required between, then go to 5:

TABLE NO. 8-2-10-1. No Caution/Advisory Panel Capsules Go On. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
	J118-e and XA1-27 J118-e and XA1-72 J118-e and XA2-27 J118-e and XA2-72 J118-e and XA3-25 J118-e and XA3-70 J118-e and XA4-15
	Step 2. If voltage is not as specified, replace relay K1 (PARA 8-5-17). Go to 5.
3. Check for 28 vdc between S1-1 and test point TP9.	
	Step 1. If voltage is as specified, go to 4. Step 2. If voltage is not as specified, repair/replace wiring as required between, then go to 5: S1-1 and J118-U S1-1 and J118-V
4. With switch S1 held at TEST, check for 28 vdc between S1-2 and test point TP9.	
	Step 1. If voltage is as specified, repair/replace wiring, as required, between, then go to 5: S1-2 and K1-X1 K1-X2 and ground Step 2. If voltage is not as specified, replace switch S1 (PARA 8-5-15). Go to 5.
5. Procedure completed.	

TABLE NO. 8-2-10-2. Capsules and Associated Channel Cards.

CARD TO BE REPLACED	CAPSULE POSITION NO.
A1	2, 3, 4, 6, 7, 8, 9, 10, 11, 12, 15, 16, 20, 22, 23, 24, 25, 26, 27, 28, 30, 31, 34, 35, 41, 43, 44
A2	17, 18, 19, 45, 46, 47, 48, 49, 51, 52, 54, 55, 56, 63, 64, 65, 67, 68, 69, 70, 71, 72, 73, 76, 77
A3	5, 21, 33, 36, 37, 38, 39, 40, 42, 57, 58, 59, 60, 61, 66, 74, 75, 78, 79, 80

TABLE NO. 8-2-10-2. Capsules and Associated Channel Cards. (Cont)

CARD TO BE REPLACED	CAPSULE POSITION NO.
A4	1, 13, 14, 29, 32, 50, 53, 62, 81, 82

TABLE NO. 8-2-10-3. Advisory Capsules and Associated Channel Cards.

TEST ADAPTER TEST POINT	CAPSULE POSITION NO.	CHANNEL CARD
TB1-A	17	A2
TB1-B	18	A2
TB1-C	19	A2
TB1-D	36	A3
TB1-E	37	A3
TB1-F	38	A3
TB1-G	39	A3
TB1-H	40	A3
TB1-J	57	A3
TB1-K	58	A3
TB1-L	59	A3
TB1-M	60	A3
TB1-N	61	A3
TB1-P	78	A3
TB1-R	79	A3
TB1-S	80	A3
TB1-Y	81	A4
TB1-Z	82	A4

TABLE NO. 8-2-10-4. Positive Input Caution Capsules.

TEST ADAPTER TEST POINT	CAPSULE POSITION NO.	CHANNEL CARD
J117-1	*1	A4
J117-2	2	A1
J117-3	3	A1
J117-4	4	A1
J117-6	6	A1
J117-7	7	A1
J117-8	8	A1
J117-9	9	A1
J117-10	10	A1
J117-11	11	A1
J117-12	12	A1
J117-15	15	A1
J117-16	16	A1
J117-17	20	A1
J117-19	22	A1
J117-20	23	A1
J117-21	24	A1
J117-22	25	A1
J117-23	26	A1
J117-24	27	A1
J117-25	28	A1
J117-27	30	A1
J117-28	31	A1
J117-31	34	A1
J117-32	35	A1
J117-33	41	A1
J117-35	43	A1
J117-36	44	A1
J117-37	45	A2
J117-38	46	A2
J117-39	47	A2
J117-40	48	A2
J117-41	49	A2
J117-43	51	A2
J117-44	52	A2
J117-46	54	A2
J117-47	55	A2

**TABLE NO. 8-2-10-4. Positive Input Caution
Capsules. (Cont)**

J117-48	56	A2
J117-49	*62	A4
J117-50	63	A2
J117-51	64	A2
J117-52	65	A2
J117-54	67	A2
J117-55	68	A2
J117-56	69	A2
J117-57	70	A2
J117-58	71	A2
J117-59	72	A2
J117-60	73	A2
J117-63	76	A2
J117-64	77	A2

*Indicates capsule flashes

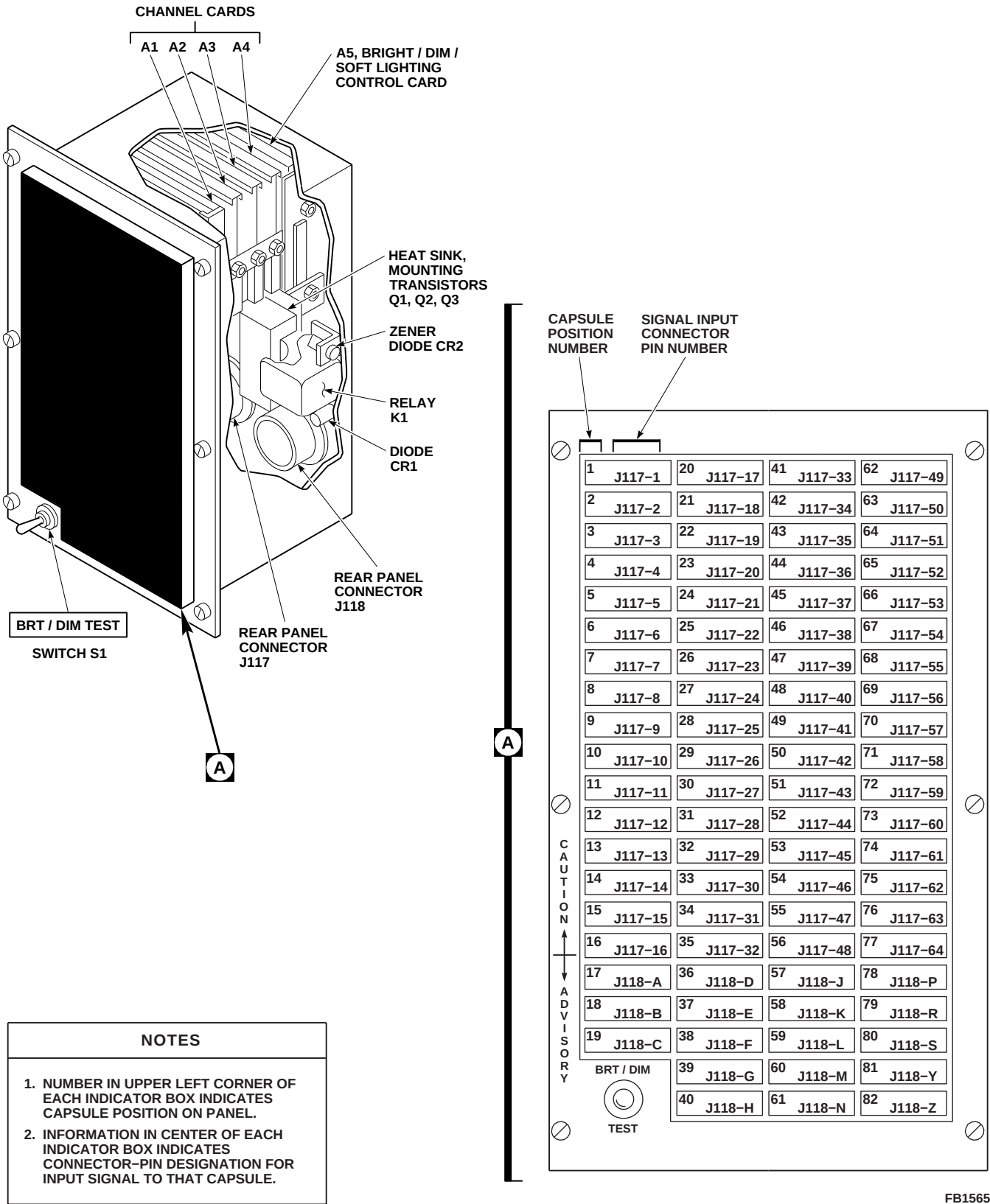


FIGURE 8-2-10-1. CAUTION/ADVISORY PANEL COMPONENTS LOCATION DIAGRAM

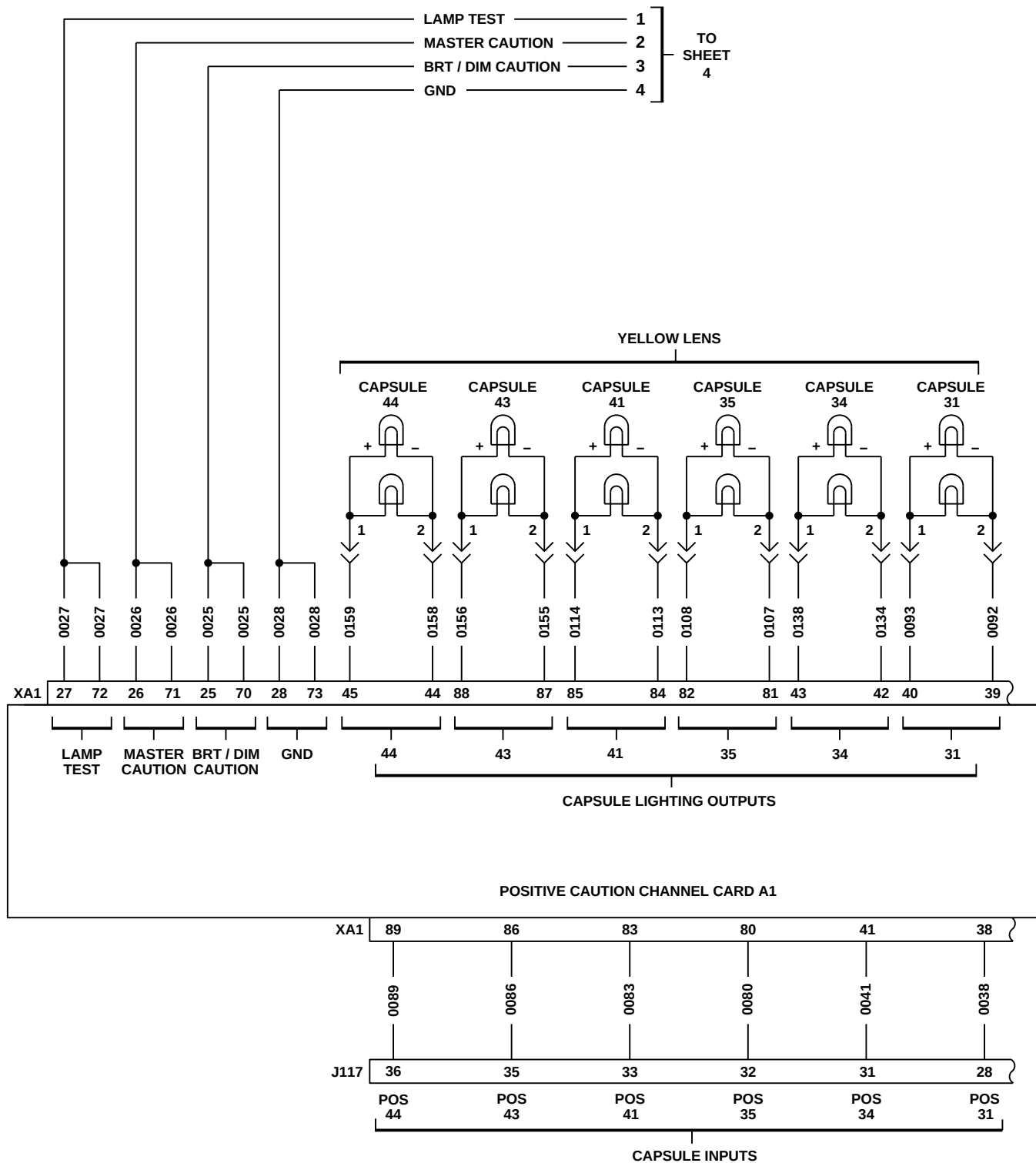
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FIGURE 8-2-10-2. CAUTION/ADVISORY PANEL SCHEMATIC DIAGRAM (SHEET 1 OF 14)

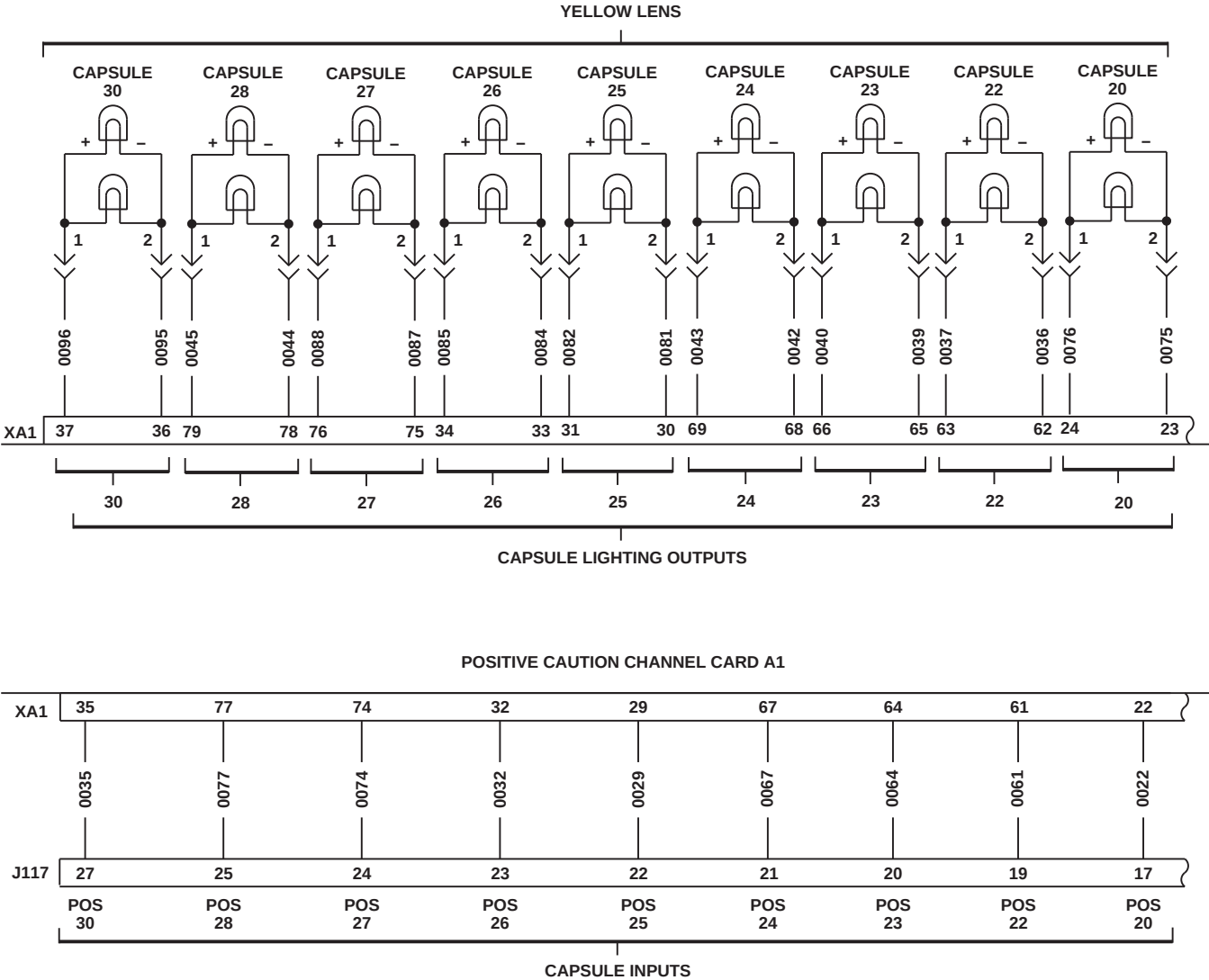


FIGURE 8-2-10-2. CAUTION/ADVISORY PANEL SCHEMATIC DIAGRAM (SHEET 2 OF 14)

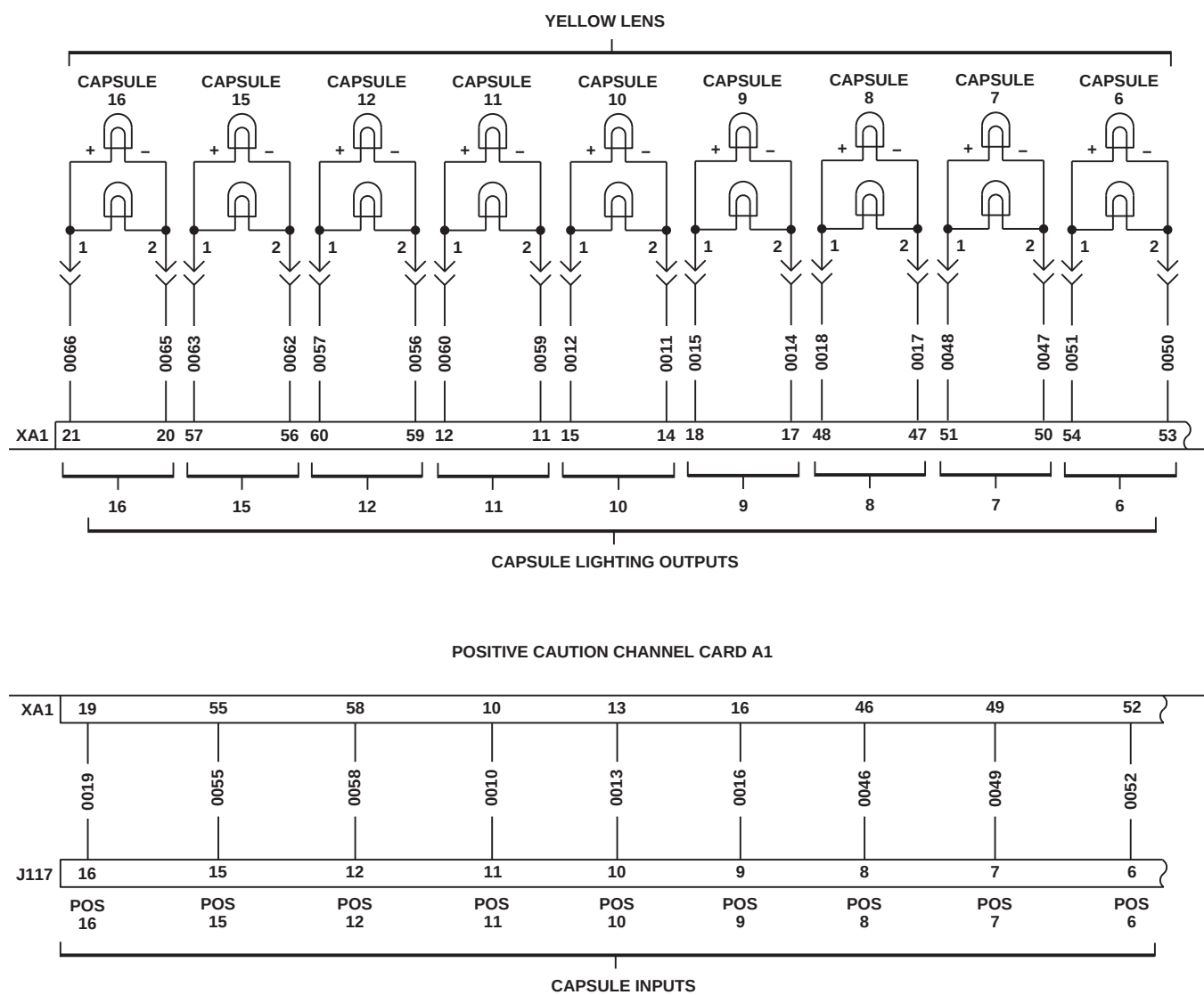


FIGURE 8-2-10-2. CAUTION/ADVISORY PANEL SCHEMATIC DIAGRAM (SHEET 3 OF 14)

8-2-10-16

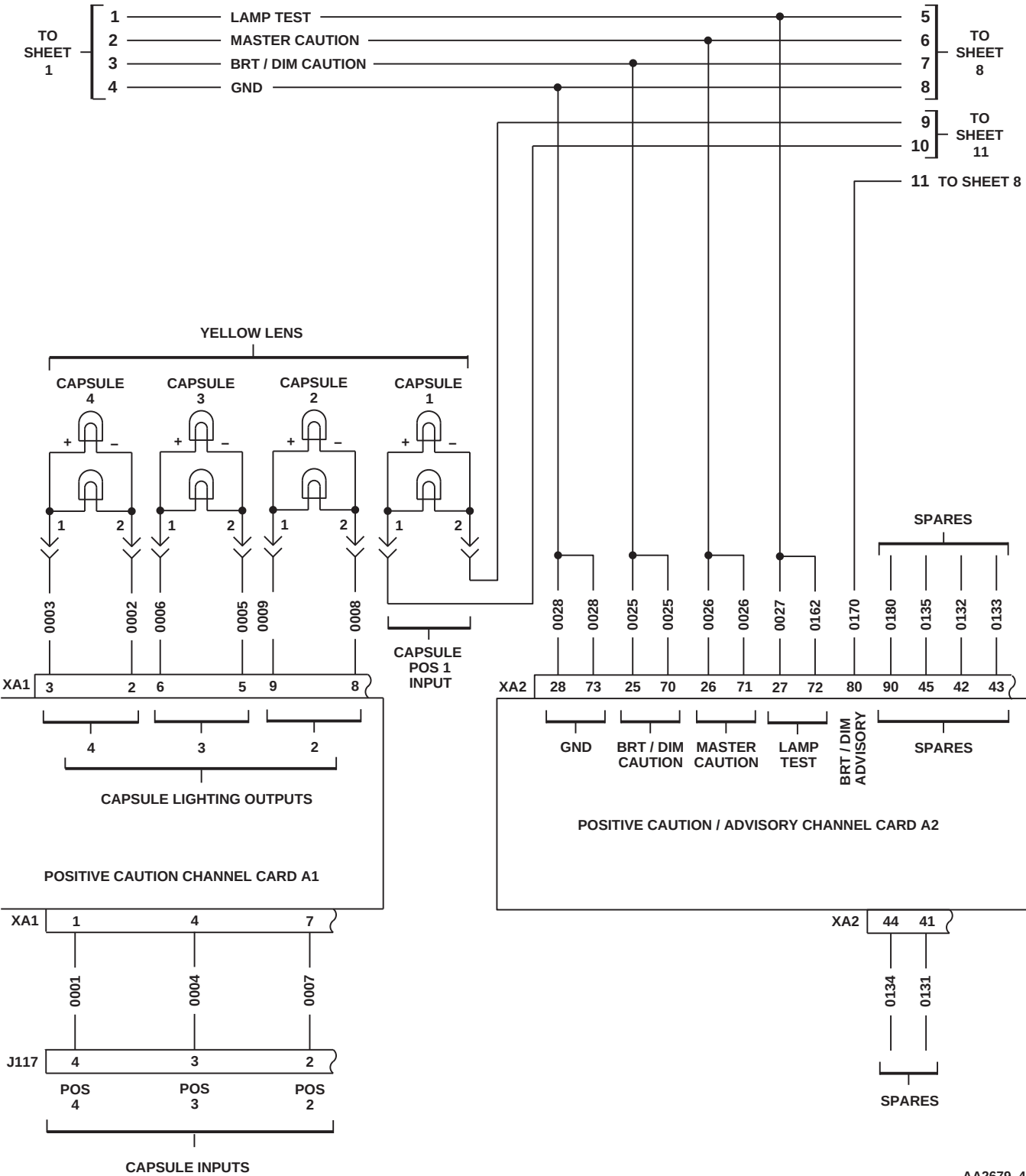


FIGURE 8-2-10-2. CAUTION/ADVISORY PANEL SCHEMATIC DIAGRAM (SHEET 4 OF 14)

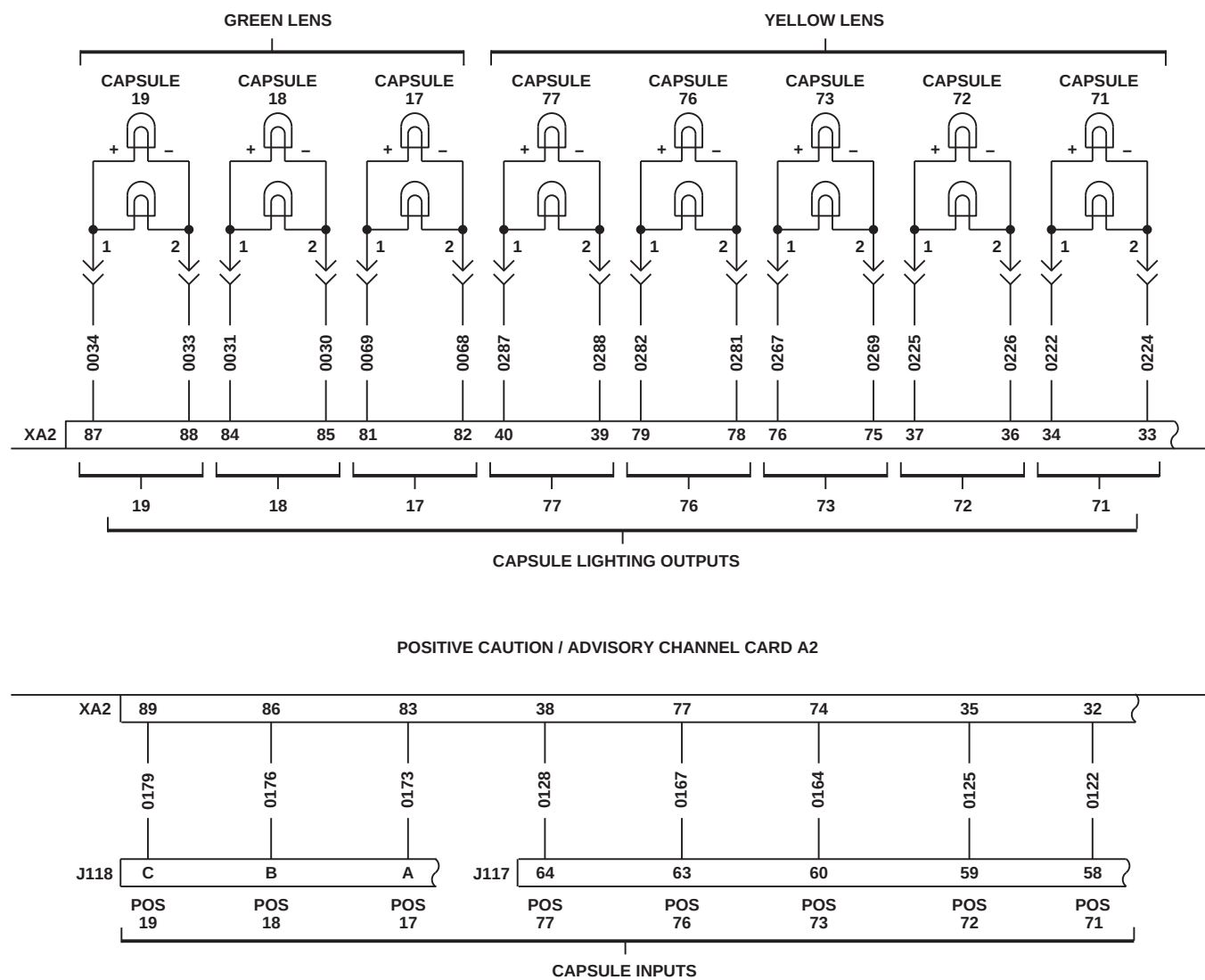
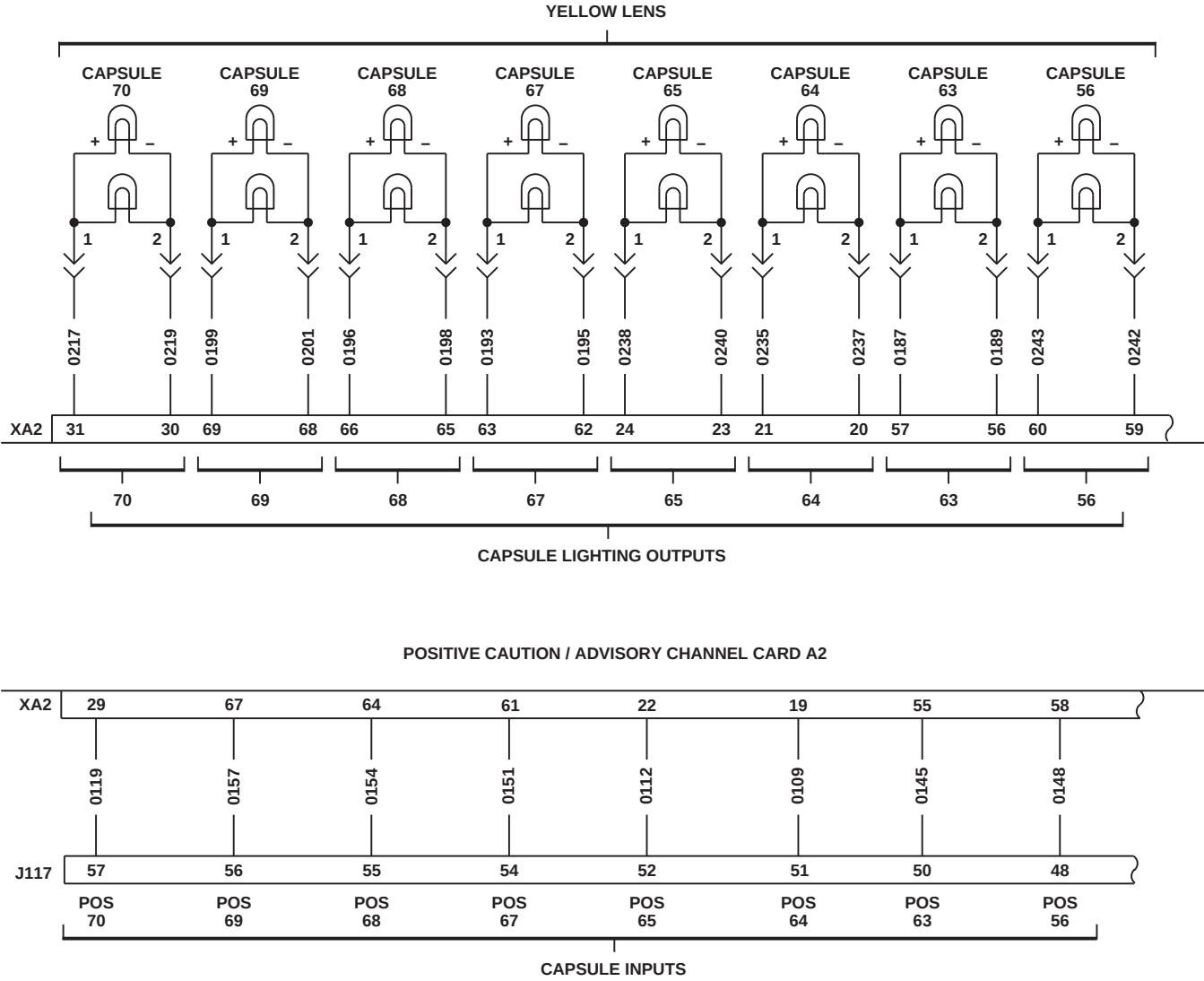
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FIGURE 8-2-10-2. CAUTION/ADVISORY PANEL SCHEMATIC DIAGRAM (SHEET 5 OF 14)

8-2-10-18

This Document contains technical data subject to ITAR. See WARNING and classifications on first page.



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FIGURE 8-2-10-2. CAUTION/ADVISORY PANEL SCHEMATIC DIAGRAM (SHEET 6 OF 14)

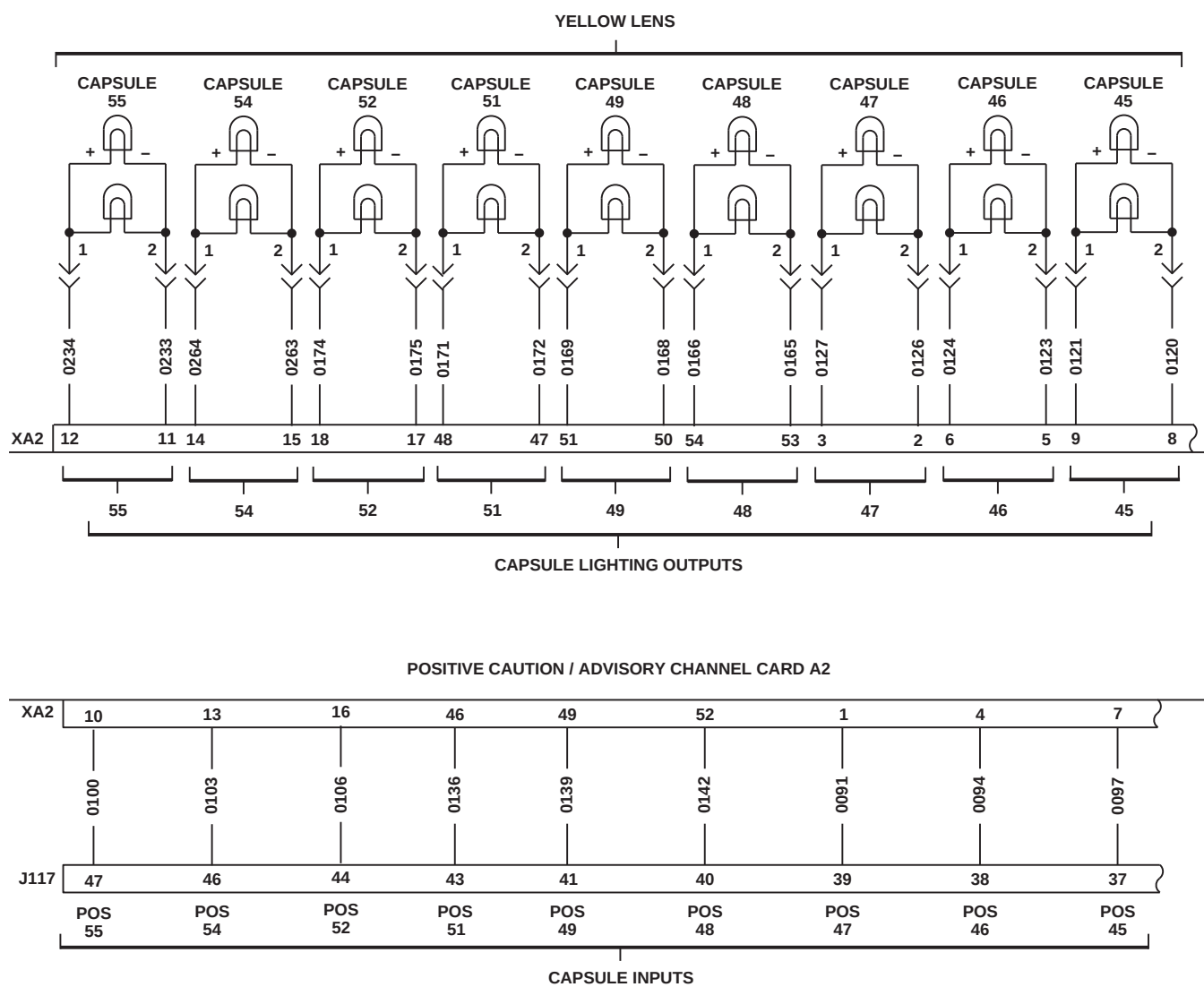
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FIGURE 8-2-10-2. CAUTION/ADVISORY PANEL SCHEMATIC DIAGRAM (SHEET 7 OF 14)

8-2-10-20

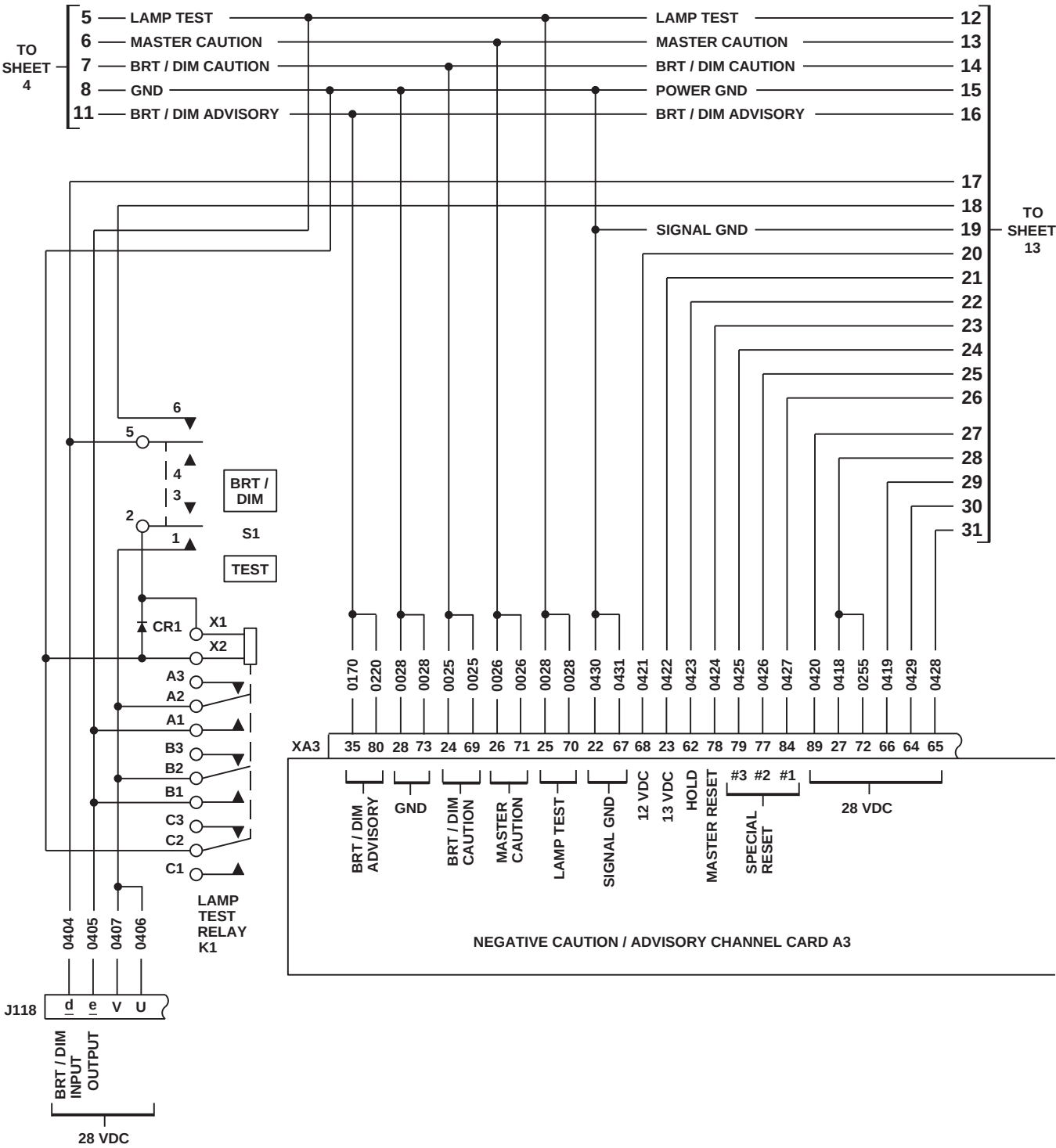


FIGURE 8-2-10-2. CAUTION/ADVISORY PANEL SCHEMATIC DIAGRAM (SHEET 8 OF 14)

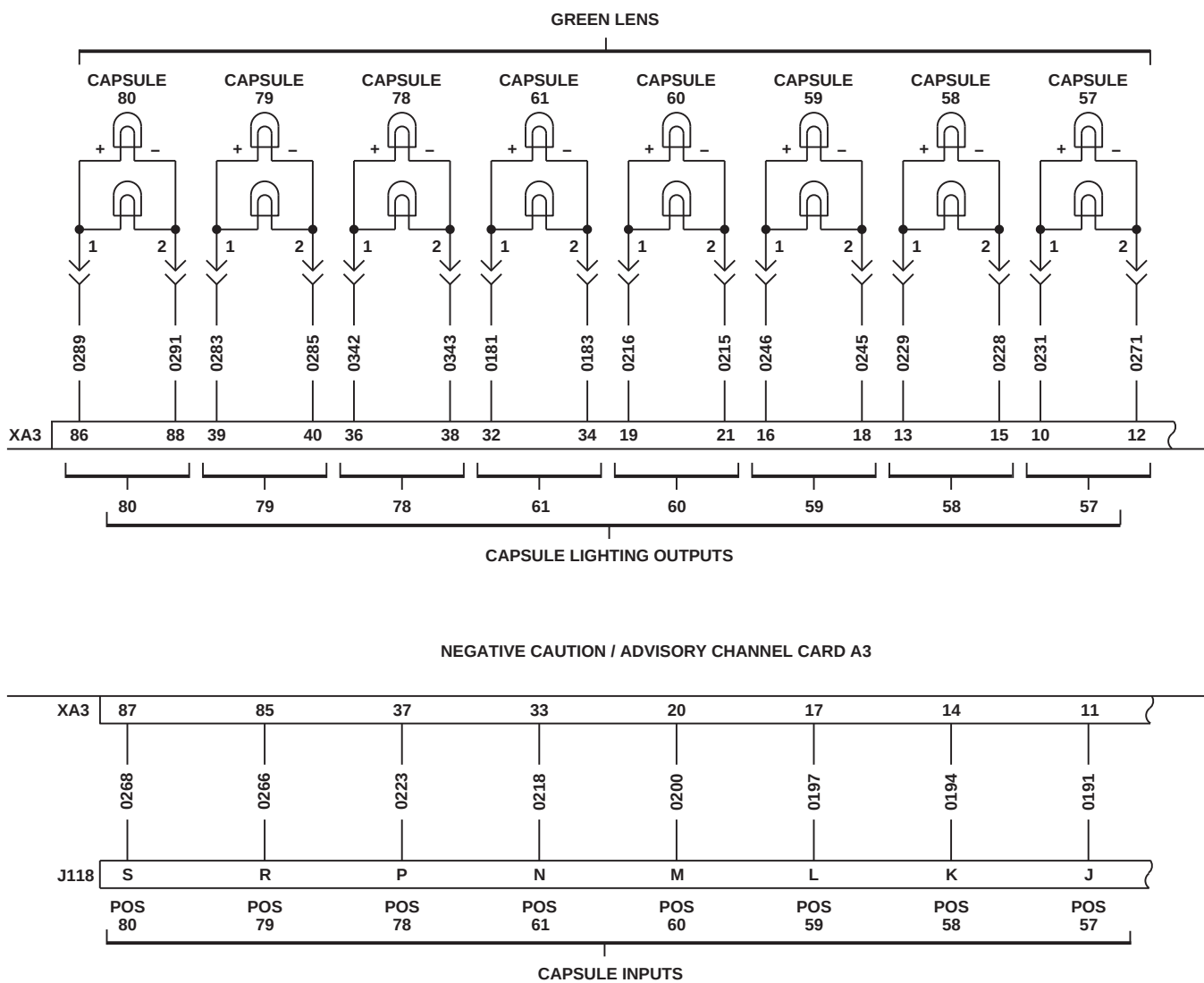
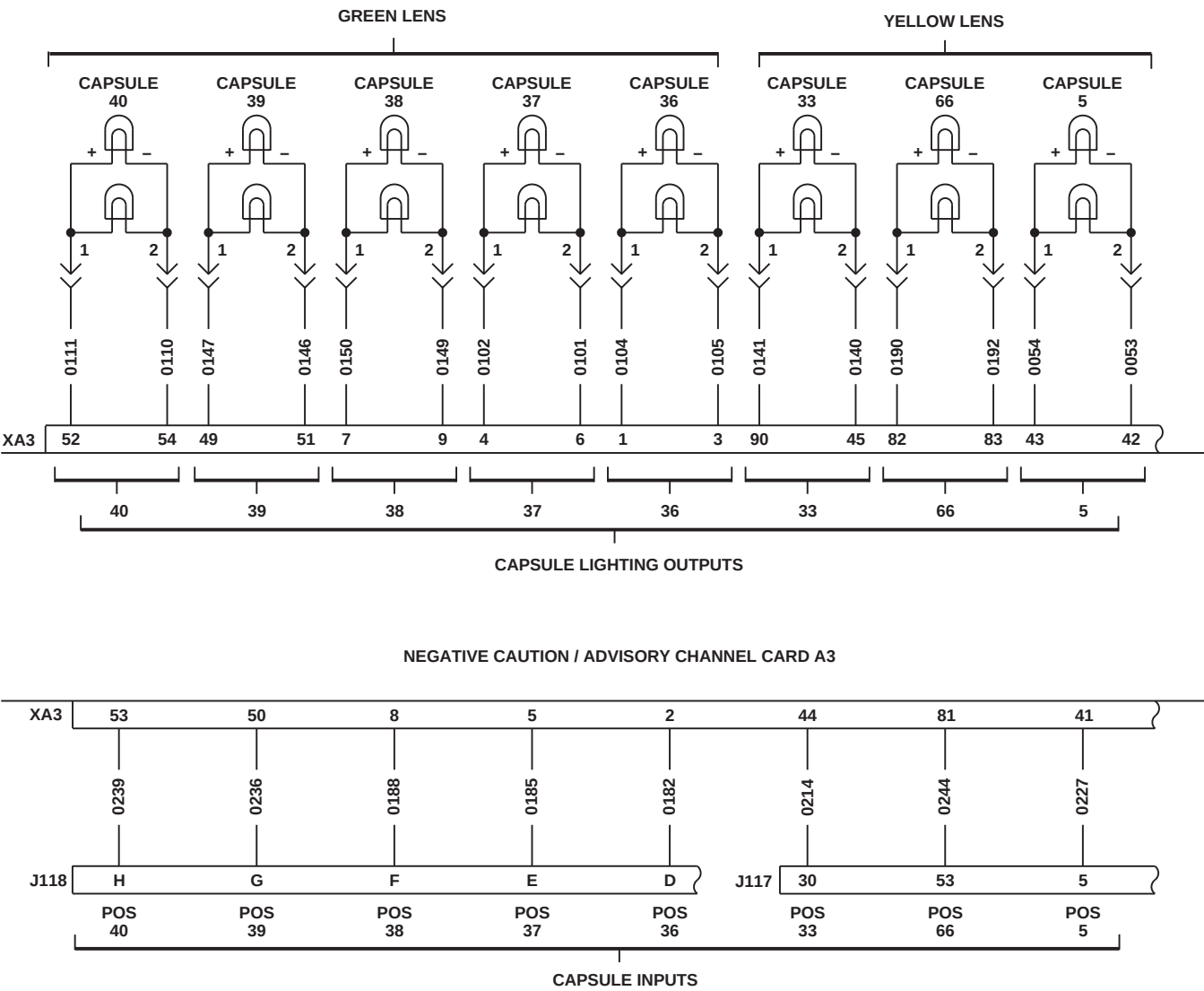
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FIGURE 8-2-10-2. CAUTION/ADVISORY PANEL SCHEMATIC DIAGRAM (SHEET 9 OF 14)

8-2-10-22



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FIGURE 8-2-10-2. CAUTION/ADVISORY PANEL SCHEMATIC DIAGRAM (SHEET 10 OF 14)

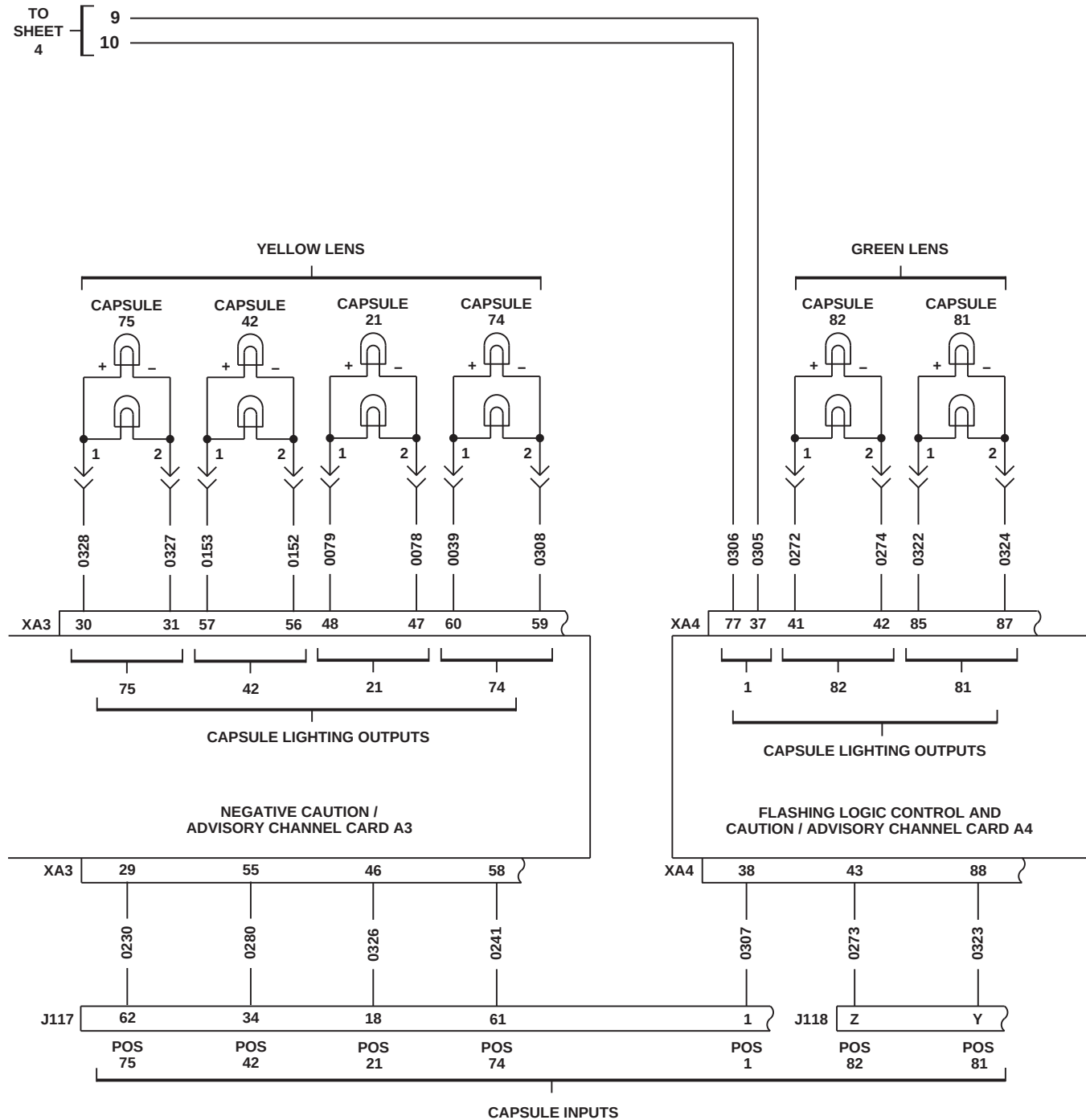
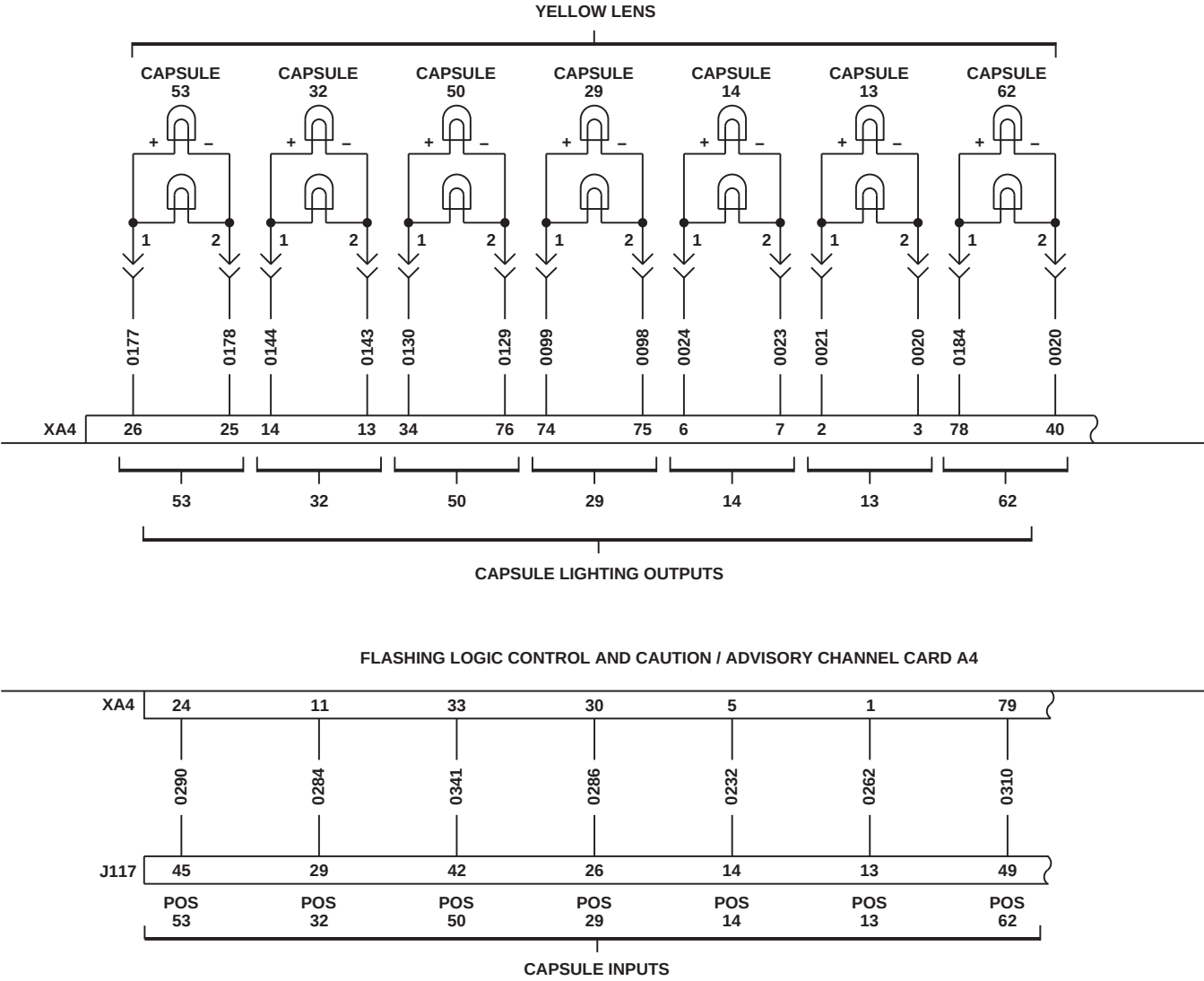
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FIGURE 8-2-10-2. CAUTION/ADVISORY PANEL SCHEMATIC DIAGRAM (SHEET 11 OF 14)

8-2-10-24



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FIGURE 8-2-10-2. CAUTION/ADVISORY PANEL SCHEMATIC DIAGRAM (SHEET 12 OF 14)

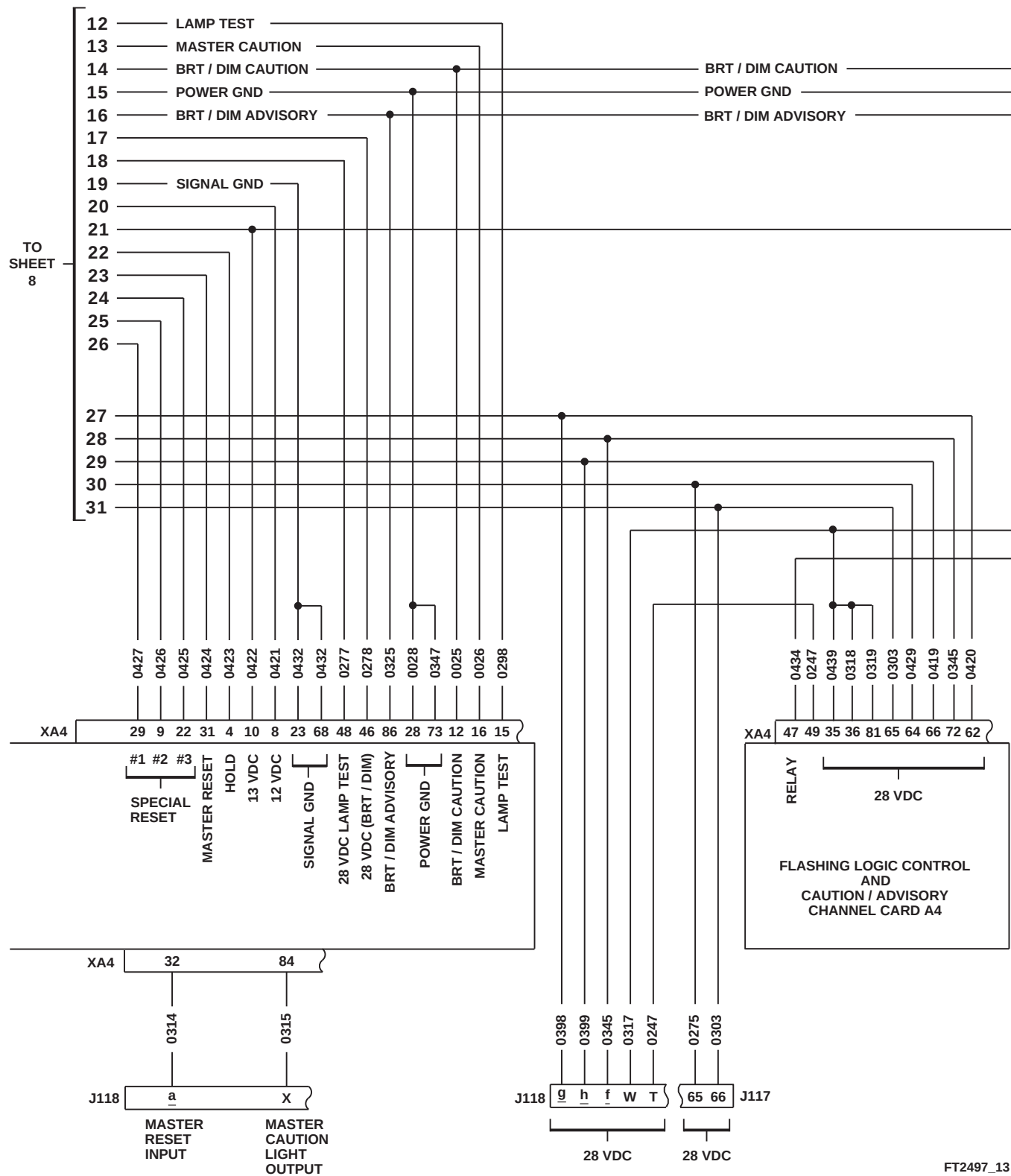
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FIGURE 8-2-10-2. CAUTION/ADVISORY PANEL SCHEMATIC DIAGRAM (SHEET 13 OF 14)

8-2-10-26

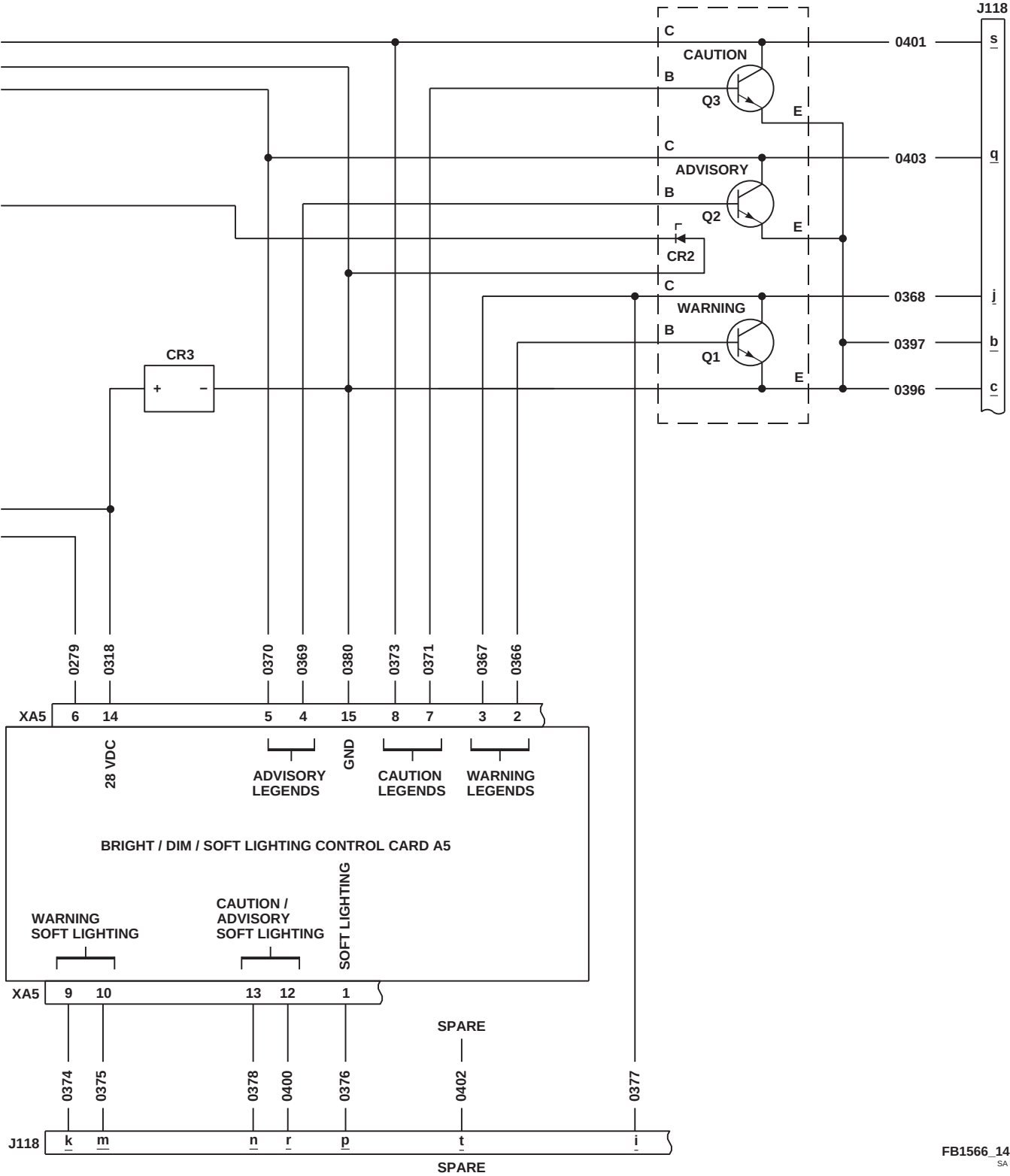


FIGURE 8-2-10-2. CAUTION/ADVISORY PANEL SCHEMATIC DIAGRAM (SHEET 14 OF 14)

8-2-10-27/(8-2-10-28 Blank)

8-2-11 MISCELLANEOUS SWITCH PANEL (BENCH).

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

- Locally-Made Miscellaneous Switch Panel Test Harness, Figure H-52, Appendix H

Equipment Conditions

- Bench Power Available, Between 27 and 28 vdc, and Between 114 and 116 vac, 395 to 405 Hz, 1ø

8-2-11.1 Operational/Troubleshooting Procedure.

NOTE

Perform steps a. through d. if GYRO ERECT switch installed. If not, begin at step e.

- Remove cover from panel and inspect for broken or burnt wires, security of components, and general condition.

RESULT: Wires shall not be broken or burnt and components shall be secure and in good condition.

- If result is not as specified, repair/replace as required.
- Connect locally-made test harness to miscellaneous switch panel as follows: P1 to J659R, and P2 to J135.
 - Check continuity between the following P1 test harness terminal block posts:

1	and	2
2	and	3

RESULT: Meter shall not indicate continuity.

- If result is not as specified, replace GYRO ERECT switch (PARA 9-5-13).

- Press and hold miscellaneous switch panel GYRO ERECT switch while checking continuity between the following P1 terminal block posts:

1	and	2
2	and	3

RESULT:

- Meter shall indicate continuity between terminal block posts 1 and 2.
- Meter shall indicate continuity between terminal block posts 2 and 3.

- If results (1) and (2) are not as specified, check continuity between J659R-E and GYRO ERECT switch, section A terminal C.
 - If continuity is present, replace GYRO ERECT switch (PARA 9-5-13).
 - If continuity is not present, repair/replace wiring.
- If result (1) is not as specified, check continuity between J659R-D and GYRO ERECT switch, section A terminal NO.
 - If continuity is present, replace GYRO ERECT switch (PARA 9-5-13).

(b) If continuity is not present, repair/replace wiring.

- If result (2) is not as specified, check continuity between J659R-F and GYRO ERECT switch, section C terminal NO, and between GYRO ERECT switch, section C terminal C and section A terminal C.

(a) If continuity is present, replace GYRO ERECT switch (PARA 9-5-13).

(b) If continuity is not present, repair/replace wiring as required.

- e. Press and hold miscellaneous switch panel FUEL IND TEST switch while checking continuity between the following P2 test harness terminal block posts:

11	and	12
13	and	14
15	and	16
17	and	18

RESULT: Meter shall not indicate continuity.

- If result is not as specified, replace FUEL IND TEST switch (PARA 9-5-13).

- f. Release miscellaneous switch panel FUEL IND TEST switch and check continuity between the following P2 test harness terminal block posts:

11	and	12
13	and	14
15	and	16
17	and	18

RESULT:

- (1) Meter shall indicate continuity between terminal block posts 11 and 12.

(2) Meter shall indicate continuity between terminal block posts 13 and 14.

(3) Meter shall indicate continuity between terminal block posts 15 and 16.

(4) Meter shall indicate continuity between terminal block posts 17 and 18.

- If more than one result is not as specified, replace FUEL IND TEST switch (PARA 9-5-13).

- If result (1) is not as specified, check continuity between J135-H and FUEL IND TEST switch, section A terminal C, and between J135-J and FUEL IND TEST switch, section A terminal NC.

(a) If continuity is present, replace FUEL IND TEST switch (PARA 9-5-13).

(b) If continuity is not present, repair/replace wiring as required.

- If result (2) is not as specified, check continuity between J135-K and FUEL IND TEST switch, section B terminal C, and between J135-L and FUEL IND TEST switch, section B terminal NC.

(a) If continuity is present, replace FUEL IND TEST switch (PARA 9-5-13).

(b) If continuity is not present, repair/replace wiring as required.

- If result (3) is not as specified, check continuity between J135-M and FUEL IND TEST switch, section C terminal C, and between J135-N and FUEL IND TEST switch, section C terminal NC.

(a) If continuity is present, replace FUEL IND TEST switch (PARA 9-5-13).

(b) If continuity is not present, repair/replace wiring as required.

8-2-11-2

- If result (4) is not as specified, check continuity between J135-P and FUEL IND TEST switch, section D terminal C, and between J135-R and FUEL IND TEST switch, section D terminal NC.
 - (a) If continuity is present, replace FUEL IND TEST switch (PARA 9-5-13).
 - (b) If continuity is not present, repair/replace wiring as required.
- g. Place miscellaneous switch panel TAIL SERVO switch to NORMAL.
- h. Check continuity between the following terminal block posts:

19	and	20
22	and	23

RESULT: Meter shall not indicate continuity.

- If result is not as specified, replace TAIL SERVO switch (PARA 9-5-14).
- i. Check continuity between the following terminal block posts:

20	and	21
23	and	24

RESULT:

- (1) Meter shall indicate continuity between terminal block posts 20 and 21.
- (2) Meter shall indicate continuity between terminal block posts 23 and 24.
- If result (1) is not as specified, check continuity between J135-S and TAIL SERVO switch terminal 2, and between J135-U and TAIL SERVO switch terminal 3.
 - (a) If continuity is present, replace TAIL SERVO switch (PARA 9-5-14).

- (b) If continuity is not present, repair/replace wiring as required.
- If result (2) is not as specified, check continuity between J135-V and TAIL SERVO switch terminal 5, and between J135-X and TAIL SERVO switch terminal 6.
 - (a) If continuity is present, replace TAIL SERVO switch (PARA 9-5-14).
 - (b) If continuity is not present, repair/replace wiring as required.
- j. Place miscellaneous switch panel TAIL SERVO switch to BACKUP.
- k. Check continuity between the following terminal block posts:

19	and	20
22	and	23

RESULT:

- (1) Meter shall indicate continuity between terminal block posts 19 and 20.
- (2) Meter shall indicate continuity between terminal block posts 22 and 23.
- If result (1) is not as specified, check continuity between J135-T and TAIL SERVO switch terminal 1.
 - (a) If continuity is present, replace TAIL SERVO switch (PARA 9-5-14).
 - (b) If continuity is not present, repair/replace wiring as required.
- If result (2) is not as specified, check continuity between J135-W and TAIL SERVO switch terminal 4.
 - (a) If continuity is present, replace TAIL SERVO switch (PARA 9-5-14).

- (b) If continuity is not present, repair/replace wiring as required.

1. Apply 115 vac to terminal block post 4 and ground to 5. Apply 28 vdc to terminal block post 7 and ground to 5.

RESULT:

- (1) LOCK legend shall light evenly on TAIL WHEEL pushbutton switch.
 - (2) UNLK legend shall light evenly on TAIL WHEEL pushbutton switch.
 - (3) Information plate legends shall light evenly.
- If results (1), (2), and (3) are not as specified, repair/replace wiring between J659R-H and J1-GND.
 - If results (1) and (2) are not as specified, check continuity between J1-GND and TAIL WHEEL switch, lamp terminal C.

- (a) If continuity is present, replace TAIL WHEEL switch (PARA 9-5-13).

- (b) If continuity is not present, repair/replace wiring.

- If result (1) is not as specified, replace lamp that is not lit (PARA 9-4-8).

If trouble remains, check continuity between J135-C and TAIL WHEEL switch lamp terminal 1, and between TAIL WHEEL switch lamp terminal 1 and terminal 4.

- 1 If continuity is present, replace TAIL WHEEL switch (PARA 9-5-13).

- 2 If continuity is not present, repair/replace wiring.

- If result (2) is not as specified, replace lamp that is not lit (PARA 9-4-8).

If trouble remains, check continuity between J135-G and TAIL WHEEL switch lamp terminal 3, and between TAIL WHEEL switch lamp terminal 2 and terminal 3.

- 1 If continuity is present, replace TAIL WHEEL switch (PARA 9-5-13).

- 2 If continuity is not present, repair/replace wiring.

- If result (3) is not as specified, replace information plate lamps as required (PARA 9-4-9).

If trouble remains, check continuity between J659R-G and J1-1.

- 1 If continuity is present, replace information panel (PARA 9-4-9).

- 2 If continuity is not present, repair/replace wiring.

- m. Remove 28 vdc and 115 vac from test harness.

- n. Press miscellaneous switch panel TAIL WHEEL switch once.

- o. Check continuity between terminal block posts 6 and 8.

RESULT: Meter shall not indicate continuity.

- If result is not as specified, press TAIL WHEEL switch, then check continuity between posts 6 and 8 in the UNLK position.

- (a) If continuity is present in UNLK position, continue with operational/troubleshooting procedure.

- (b) If continuity is not present in UNLK position, check continuity between J135-A and TAIL WHEEL switch, section A terminal NC, and between

J135-D and TAIL WHEEL switch, section A terminal C.

1 If continuity is present, replace TAIL WHEEL switch (PARA 9-5-13).

2 If continuity is not present, repair/replace wiring.

p. Check continuity between terminal block posts 9 and 8.

RESULT: Meter shall not indicate continuity.

- If result is not as specified, replace TAIL WHEEL switch (PARA 9-5-13).

q. Press miscellaneous switch panel TAIL WHEEL switch once.

r. Check continuity between terminal block posts 9 and 5.

RESULT: Meter shall indicate continuity.

- If result is not as specified, check continuity between J135-F and TAIL WHEEL switch, section C terminal C.

(a) If continuity is present, replace TAIL WHEEL switch (PARA 9-5-13).

(b) If continuity is not present, repair/replace wiring.

s. Press miscellaneous switch panel TAIL WHEEL switch once.

t. Check continuity between the following terminal block posts:

6	and	9
8	and	5

RESULT: Meter shall indicate continuity.

- If result is not as specified, check continuity between TAIL WHEEL switch, section A terminal NC and TAIL WHEEL switch, section C terminal NO.

(a) If continuity is present, replace TAIL WHEEL switch (PARA 9-5-13).

(b) If continuity is not present, repair/replace wiring.

u. Remove test harness from panel connectors J659R and J135.

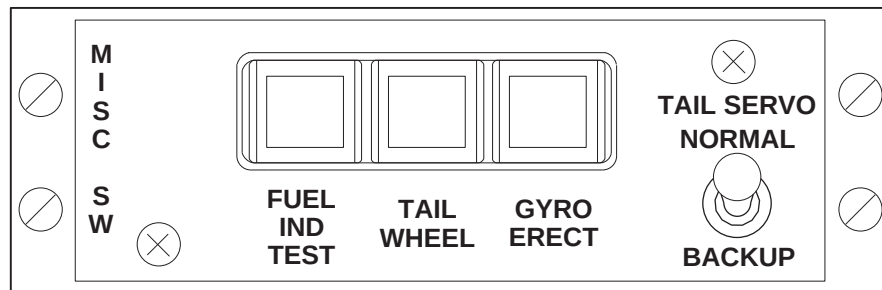
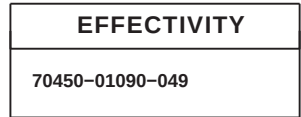
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FIGURE 8-2-11-1. TYPICAL MISCELLANEOUS SWITCH PANEL



8-2-11-7

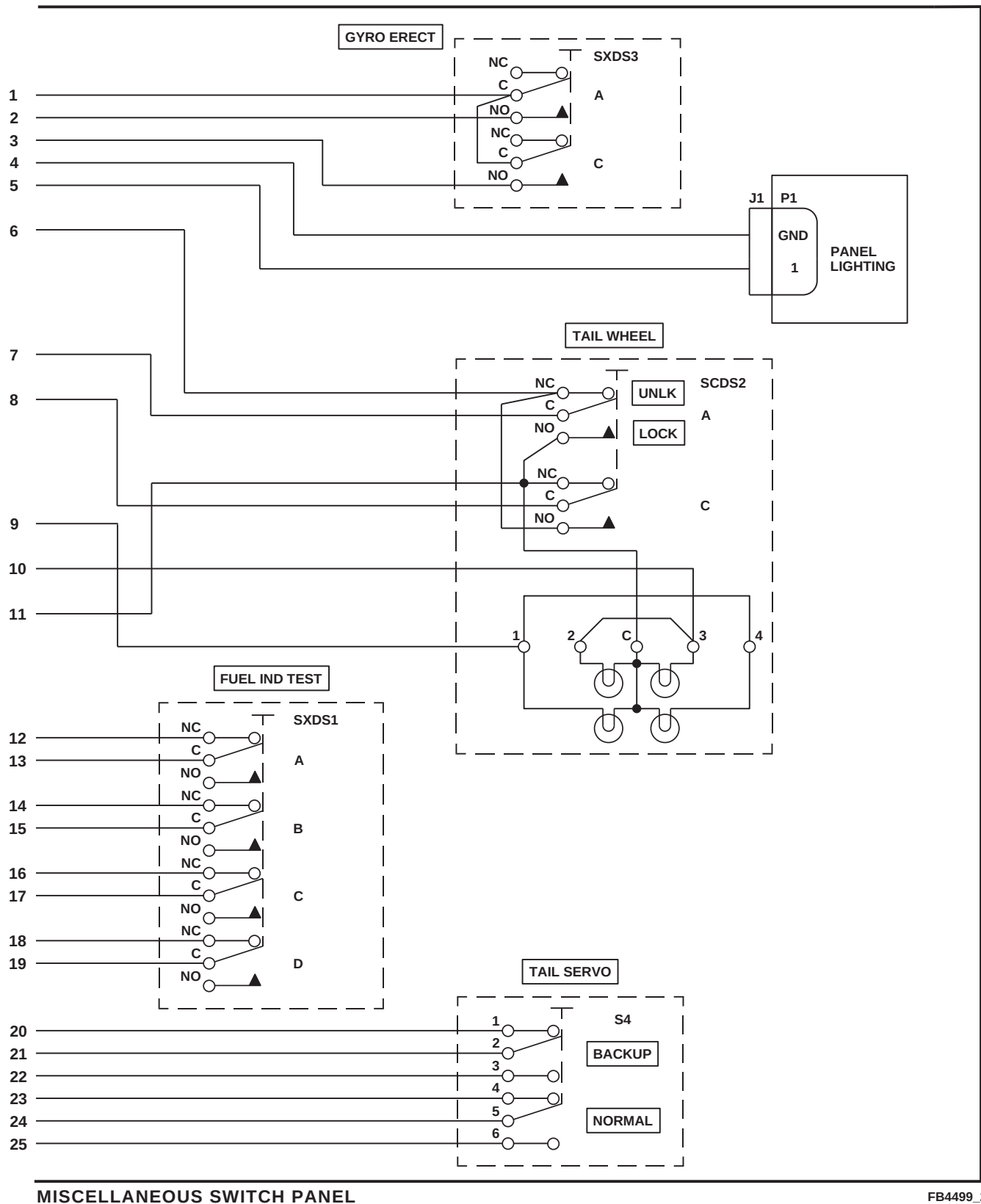


FIGURE 8-2-11-2. MISCELLANEOUS SWITCH PANEL SCHEMATIC DIAGRAM (SHEET 2 OF 2)

8-2-11-8

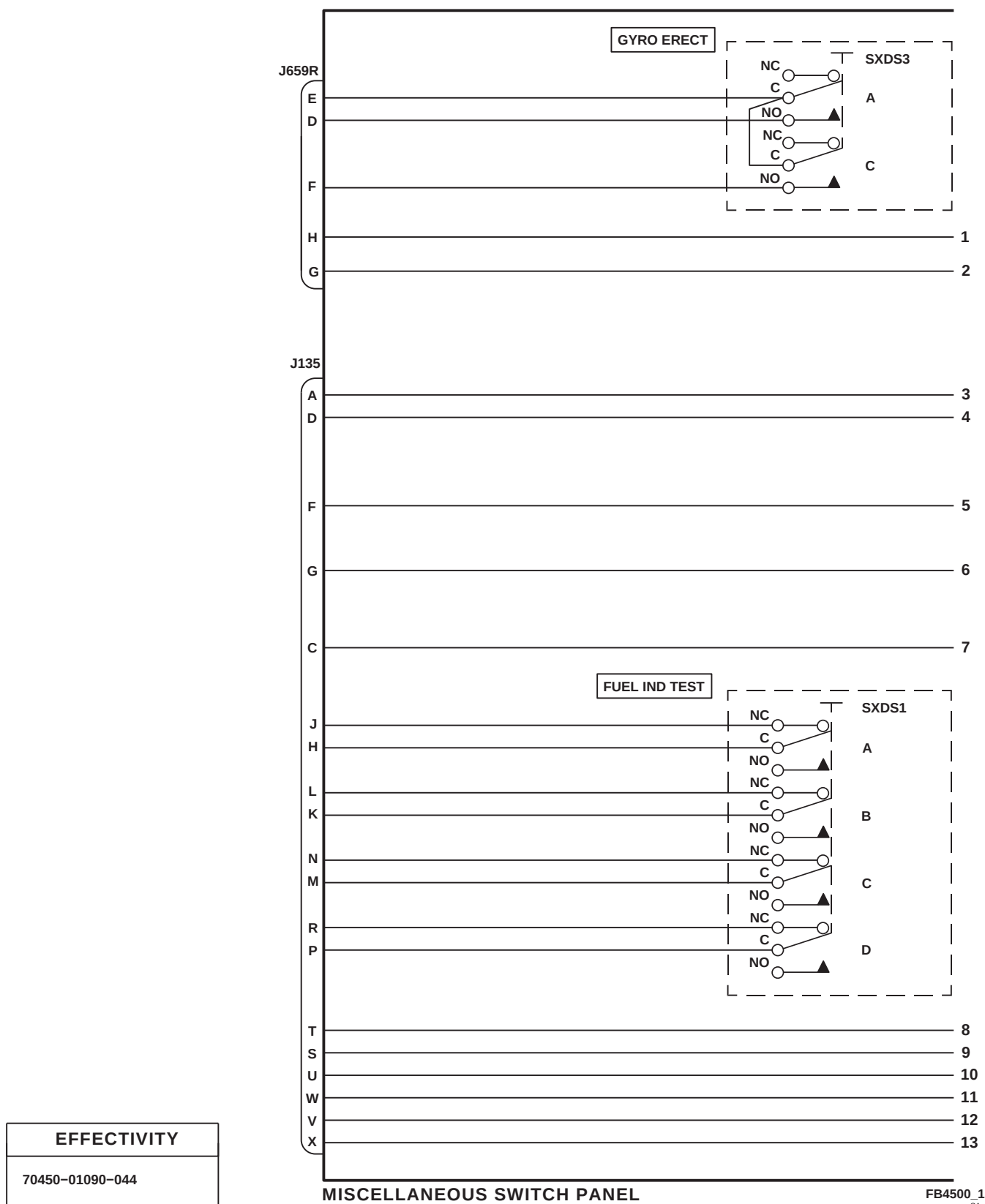
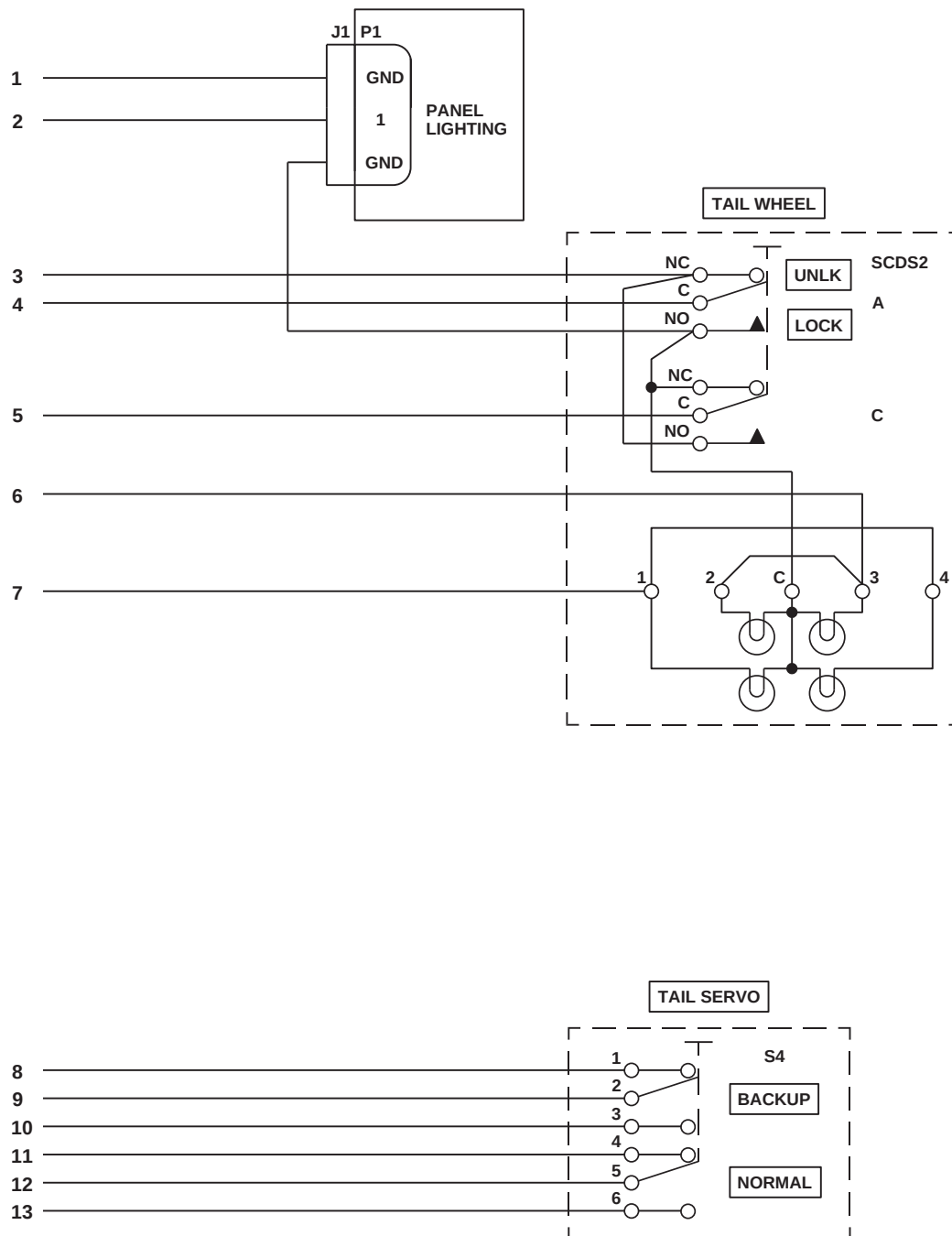


FIGURE 8-2-11-3. MISCELLANEOUS SWITCH PANEL SCHEMATIC DIAGRAM (SHEET 1 OF 2)



MISCELLANEOUS SWITCH PANEL

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FIGURE 8-2-11-3. MISCELLANEOUS SWITCH PANEL SCHEMATIC DIAGRAM (SHEET 2 OF 2)

8-2-11-10

SECTION 8-3
ON AIRCRAFT INSPECTIONS
NOT APPLICABLE

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SECTION 8-4

MAINTENANCE PROCEDURES

SECTION OVERVIEW

PARAGRAPH	TITLE
8-4-1	Pitot-Static Head
8-4-2	Deleted
8-4-3	Pitot-Static Lines and Components
8-4-4	Vertical Speed Indicator
8-4-5	Airspeed Indicator
8-4-6	Barometric Altimeter
8-4-7	Standby Compass
8-4-8	Standby Compass Lamp SLS
8-4-9	Standby Compass Light Switch
8-4-10	Central Display Unit
8-4-11	Central Display Unit Faceplate and EMI Shield
8-4-12	Central Display Unit Bar Graph Filament Lamps and Direct View Filament Modules EMEP
8-4-13	Central Display Unit Fuses
8-4-14	Central Display Unit Lamp Assembly W/O EMEP
8-4-15	Signal Data Converter, No. 1 and No. 2
8-4-16	Pilot Display Unit
8-4-17	Pilot Display Unit Faceplate and EMI Shield
8-4-18	Pilot Display Unit Bar Graph Filament Lamps and Direct View Filament Modules EMEP
8-4-19	Caution/Advisory Panel

PARAGRAPH

TITLE

8-4-20	Caution/Advisory Panel Lamps
8-4-21	Caution/Advisory Panel Capsule Parts
8-4-22	Caution/Advisory System and Master Warning Dimming Controls
8-4-23	Caution/Advisory Audible Warning Unit
8-4-24	Master Warning Panel
8-4-25	Master Warning Panel Lamps
8-4-26	Master Warning Panel Capsule
8-4-27	Master Warning Dimming Control
8-4-28	Clock
8-4-29	Clock Battery
8-4-30	Free-Air Thermometer
8-4-31	Rotor Overspeed Reset Switch
8-4-32	Chip Detector Resistor Unit 70006-70720

8-4-1 PITOT-STATIC HEAD.

PARAGRAPH BREAKDOWN

	PAGE
8-4-1.1 Replace Pitot-Static Head	8-4-1-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 297, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 8-2-1
- PARA 13-4-11

8-4-1.1 Replace Pitot-Static Head.

NOTE

Left and right pitot-static heads are not interchangeable. Replacement of right and left pitot-static heads is the same, except as noted.

8-4-1.1.1 Remove.

- Remove heater ducts (PARA 13-4-11).
- Remove soundproofing panels (PARA 2-4-69).

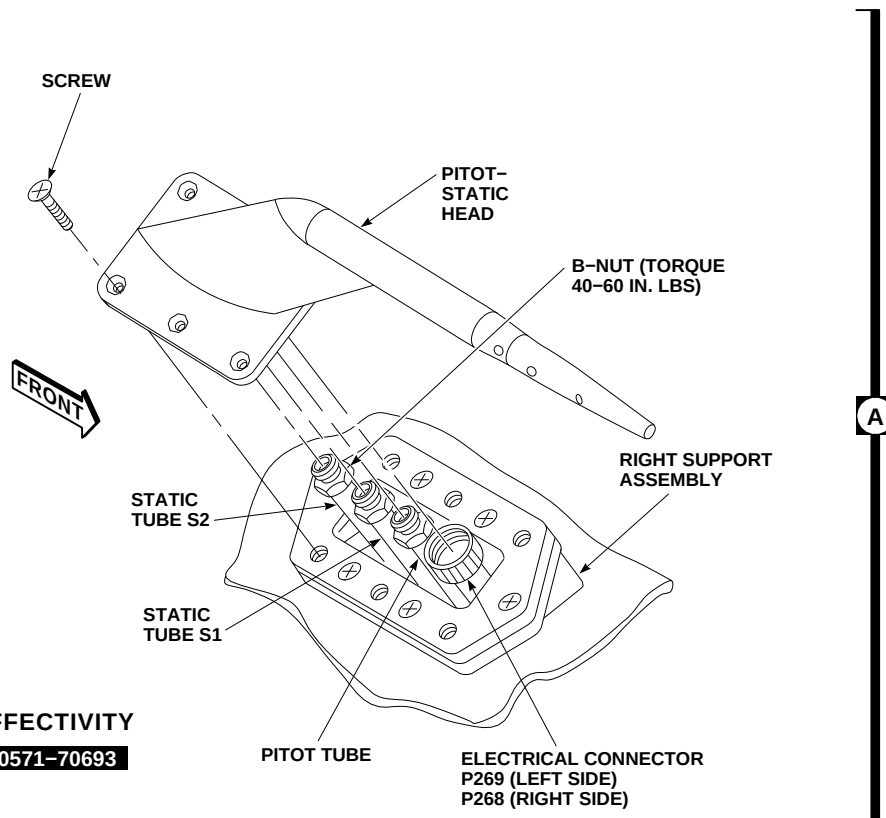
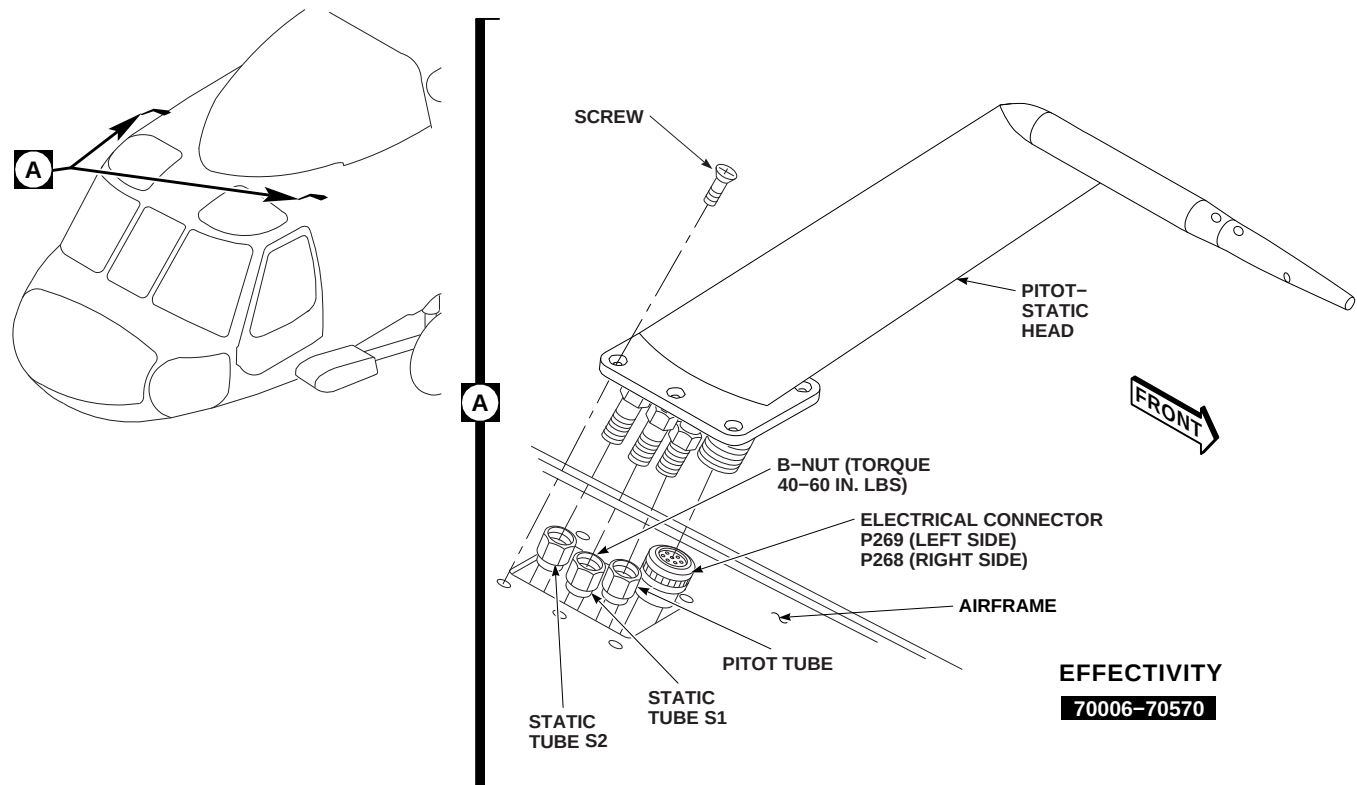
WARNING

Injury to personnel will result if pitot-static head is hot. Make sure pitot-static head is cool before attempting to remove.

- Disconnect pitot and static tubes, and electrical connector, P268 (right) or P269 (left), from pitot-static head (Figure 8-4-1-1, Sheet 1, Detail A or Sheet 2, Detail A).
- Install protective caps and plugs on electrical connectors.
- Cap pitot and static tubes to prevent foreign material from entering tubes.
- Remove screws and pitot-static head from 70006-70570 airframe 70571 - SUBQ support assembly .

8-4-1.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Prepare mounting surface for electrical bonding.



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FIGURE 8-4-1-1. REPLACE PITOT-STATIC HEAD (SHEET 1 OF 2)

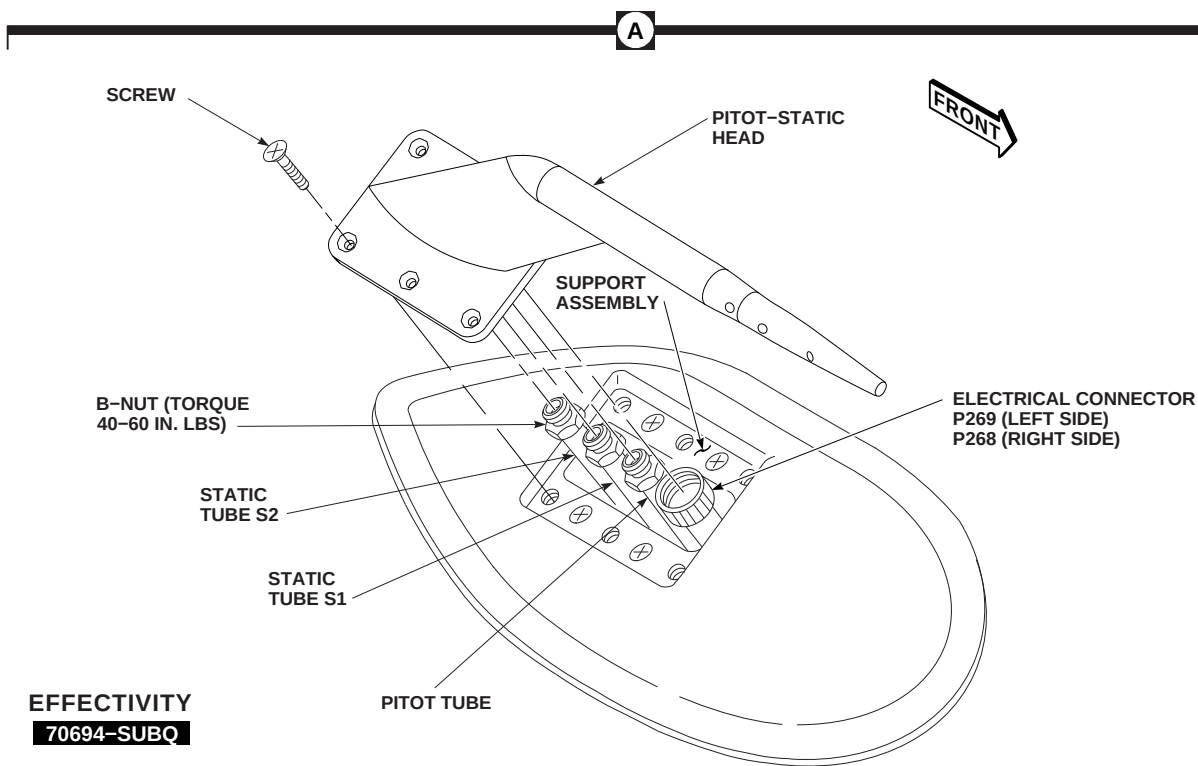


FIGURE 8-4-1-1. REPLACE PITOT-STATIC HEAD (SHEET 2 OF 2)

- c. Apply thin coat of sealing compound, Item 292, Appendix D, to bottom surface of pitot-static head.
- d. Secure pitot-static head to **70006-70570** airframe **70571-SUBQ** support assembly with screws (Figure 8-4-1-1, Sheet 1, Detail A or Sheet 2, Detail A).
- e. Seal joint between pitot-static head and **70006-70570** airframe **70571-SUBQ** support assembly with sealing compound, Item 292, Appendix D.
- f. Remove protective caps and plugs from electrical connectors.
- g. Inspect and treat electrical connectors (PARA 1-7-20).
- h. In cabin area at STA 247.0, connect electrical connector, P269 (left) or P268 (right), into receptacle on pitot-static head.
- i. Coat all mating threads on pitot and static tubes, and nuts with sealing compound, Item 297, Appendix D.
- j. Uncap and connect pitot and static tubes to pitot-static head. **TORQUE B-NUTS TO 40 - 60 INCH-POUNDS.**
- k. Make sure area is clean and free of foreign material.
- l. Install soundproofing panels (PARA 2-4-69).
- m. Install heater ducts (PARA 13-4-11).
- n. Perform operational check of pitot-static system (PARA 8-2-1).

8-4-2 PITOT-STATIC HEAD SUPPORT 70571-70693 .

PARAGRAPH BREAKDOWN

	PAGE
8-4-2.1 Replace Pitot-Static Head Support	8-4-2-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Sealing Compound, Item 292, Appendix D

- Trichloroethane, 1,1,1, Item 341, Appendix D
- Wooden Spatula, Item 349, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 8-4-1

8-4-2.1 Replace Pitot-Static Head Support.

NOTE

Left and right support assemblies are not interchangeable. Replacement of right and left pitot-static head supports is the same.

8-4-2.1.1 Remove.

- a. Remove pitot-static head (PARA 8-4-1).
- b. Remove screws and plate (Figure 8-4-2-1, Detail A).
- c. Remove sealing compound from screw wells in pitot-static head support using wooden spatula, Item 349, Appendix D, trichloroethane, 1,1,1, Item 341, Appendix D, and cheesecloth, Item 90, Appendix D.
- d. Remove screws and pitot-static head support.

8-4-2.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).

- b. Prepare mounting surface for electrical bonding.
- c. Apply thin coat of sealing compound, Item 292, Appendix D, to bottom surface of pitot-static head support.
- d. Install pitot-static head support and secure with screws (Figure 8-4-2-1, Detail A).
- e. Apply thin coat of sealing compound, Item 292, Appendix D, to top surface of pitot-static head support.
- f. Install plate and secure with screws.
- g. Completely fill screw wells in pitot-static head support with sealing compound, Item 292, Appendix D.
- h. Seal joint between support and fuselage with sealing compound, Item 292, Appendix D.
- i. Install pitot-static head (PARA 8-4-1).

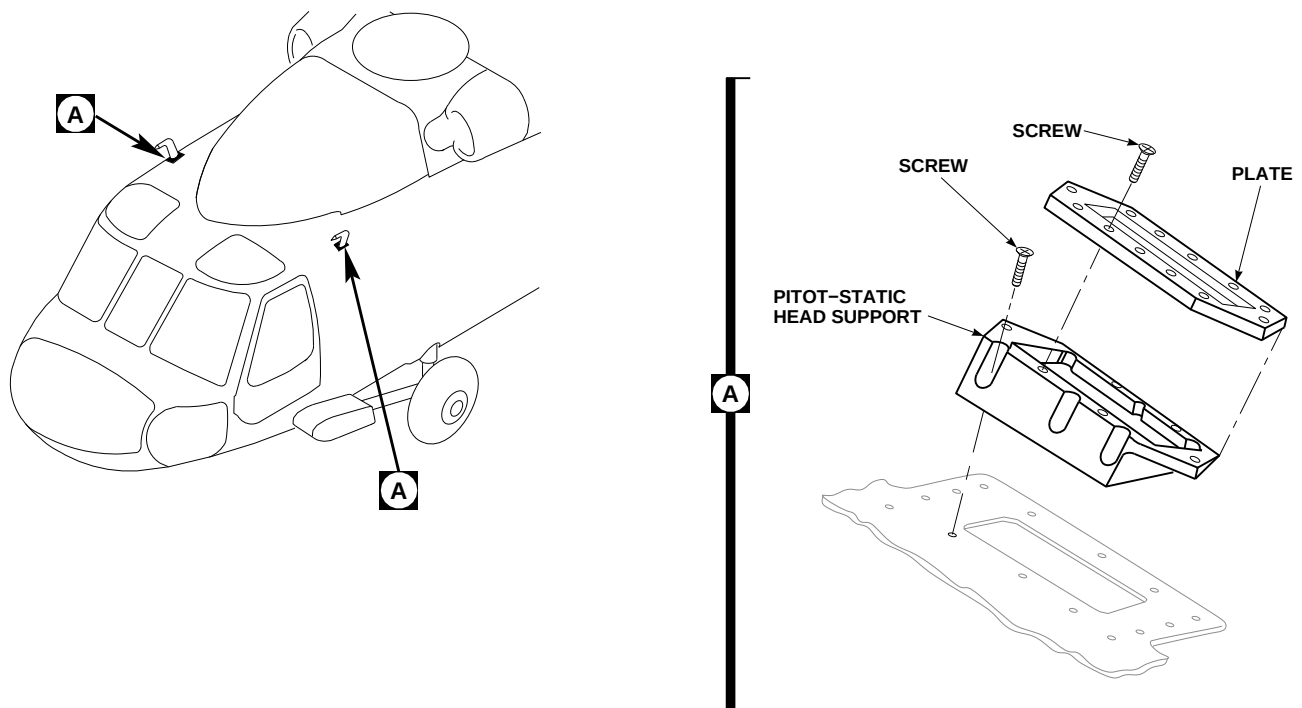
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FIGURE 8-4-2-1. REPLACE PITOT-STATIC HEAD SUPPORT

8-4-3 PITOT-STATIC LINES AND COMPONENTS.

PARAGRAPH BREAKDOWN

		PAGE
8-4-3.1	Replace Pitot-Static Lines and Components	8-4-3-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs

- Sealing Compound, Item 297, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-69
- PARA 8-2-1
- PARA 13-4-11

8-4-3.1 Replace Pitot-Static Lines and Components.

8-4-3.1.1 Remove.

- Remove heating and ventilation system heater ducts as required (PARA 13-4-11).
- Remove soundproofing panels (PARA 2-4-69).
- Remove clamps securing pitot or static line to helicopter as required.

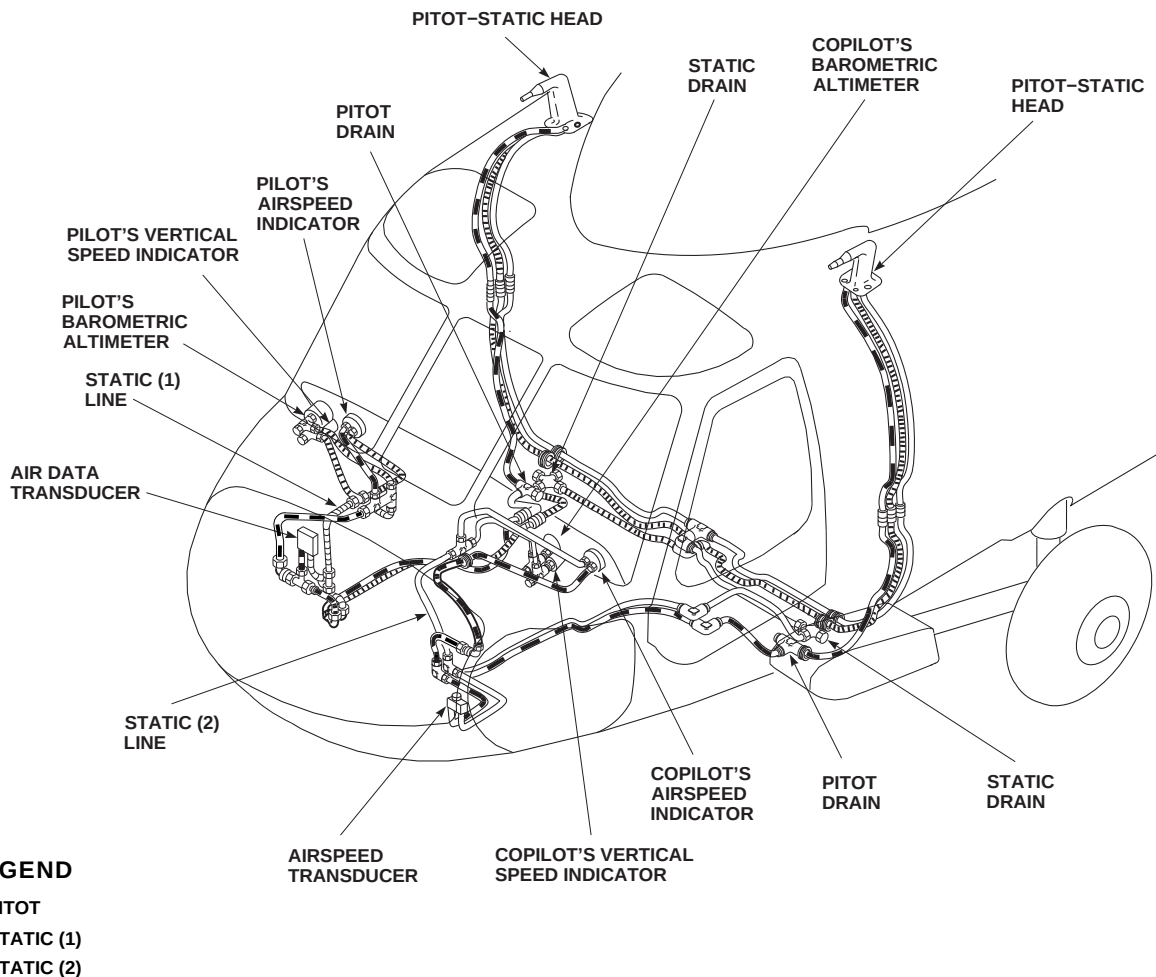
NOTE

70571 – SUBQ To make sure of proper installation, note orientation of restrictor unions prior to removal.

- Disconnect line at connections to fittings and components, and remove line, tee fittings, 70571 – SUBQ restrictor unions, balance chambers, packings, and elbows from helicopter as required (Figure 8-4-3-1, Sheet 1, and Sheet 2, Detail A, Detail B, and Detail C). Discard packings.
- Cap all lines on helicopter.

8-4-3.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).

**EFFECTIVITY****70006-70570**FB3711_2
SA**FIGURE 8-4-3-1. REPLACE PITOT-STATIC LINES AND COMPONENTS (SHEET 1 OF 2)****WARNING**

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure

all personnel are wearing safety glasses when compressed air is being used.

- 70571 - SUBQ** If restrictor unions were removed, clean internally using dry-cleaning solvent, Item 124, Appendix D, and dry with compressed air.
- Using sealing compound, Item 297, Appendix D, coat all threads in components.
- Remove caps from lines.

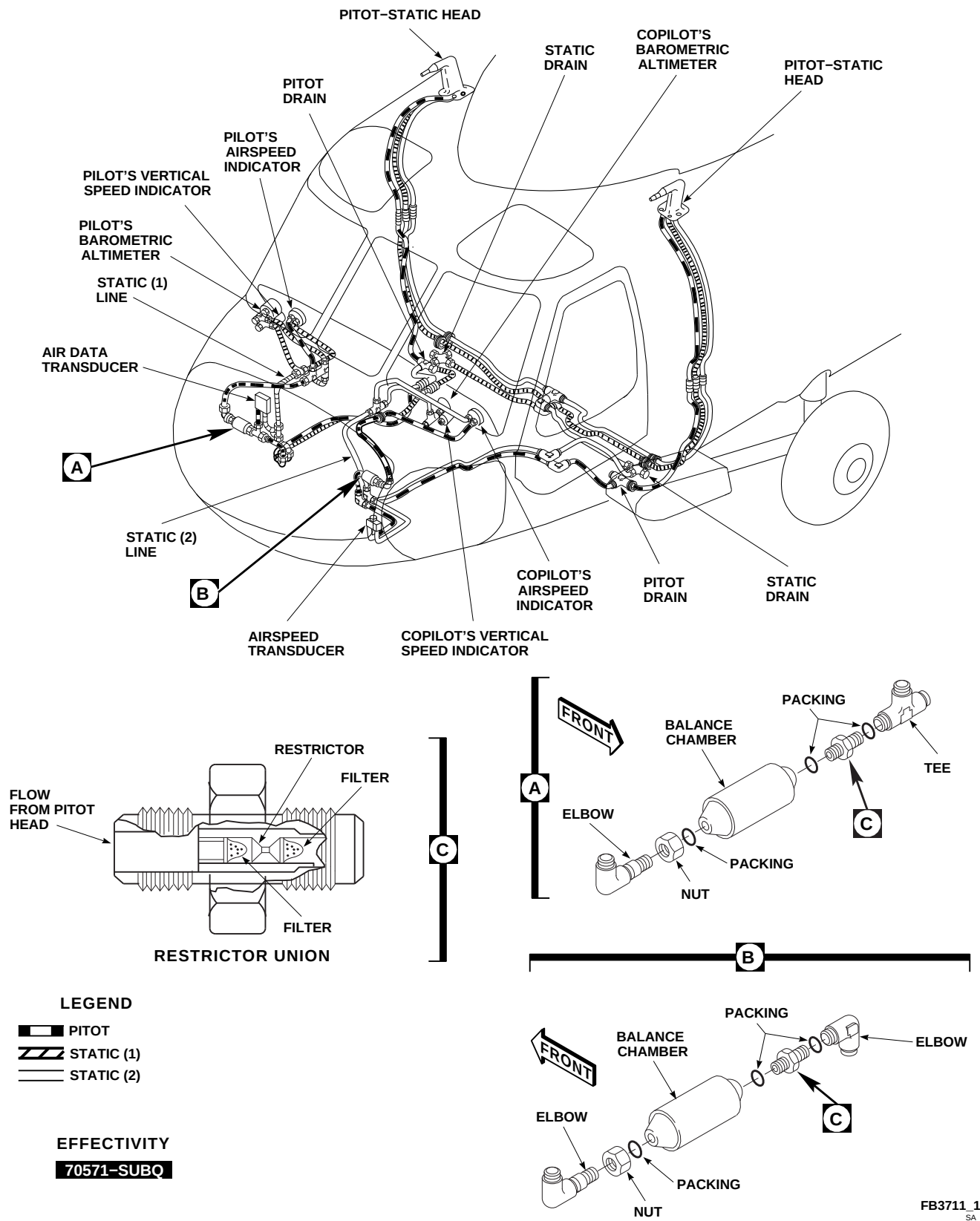


FIGURE 8-4-3-1. REPLACE PITOT-STATIC LINES AND COMPONENTS (SHEET 2 OF 2)

CAUTION

70571 - SUBQ Damage to pitot-static system will result if restrictor union is installed incorrectly and dirt is allowed to enter system. Make sure restrictor union is oriented the same as when removed.■

- e. Connect adjoining lines with fittings, elbows, unions, tees, **70571 - SUBQ** restrictor unions, balance chambers,■ and packings to components as required (Figure 8-4-3-1, Sheet 1, and Sheet 2, Detail A, Detail B, and Detail

C). TORQUE B-NUTS TO 40 - 60 INCH-POUNDS.

- f. Secure all lines to helicopter with clamps. ■
- g. Make sure area is clean and free of foreign material.
- h. Install soundproofing panels (PARA 2-4-69).
- i. Install heating and ventilation system heater ducts (PARA 13-4-11). ■
- j. Perform operational check of pitot-static system (PARA 8-2-1).

8-4-4 VERTICAL SPEED INDICATOR.

PARAGRAPH BREAKDOWN

		PAGE
8-4-4.1	Replace Vertical Speed Indicator	8-4-4-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 87, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 8-2-1

8-4-4.1 Replace Vertical Speed Indicator.

NOTE

Replacement of pilot’s and copilot’s vertical speed indicators is the same, except as noted.



Damage to lighted bezel assembly will result if not careful when removing from instrument panel. Make sure to use care when removing lighted bezel assembly from the instrument panel.

8-4-4.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- W/O IAS** Disconnect static tube, S2, from tee fitting behind vertical speed indicator and cap static tube (Figure 8-4-4-1, Detail A).◀
- SLS** Remove vertical speed indicator as follows:◀
 - Remove screws securing lighted bezel assembly to instrument panel.

- Remove lighted bezel assembly and vertical speed indicator from instrument panel to gain access to electrical connector. Disconnect electrical connector, P181 (pilot’s) or P194 (copilot’s), from lighted bezel assembly.
- W/O SLS** Remove vertical speed indicator as follows:◀
 - Remove screws securing vertical speed indicator to instrument panel.

- (2) Disconnect electrical connector, P8 (pilot's) or P3 (copilot's), from vertical speed indicator.
- (3) Remove vertical speed indicator from instrument panel.
- e. Install protective cap and plug on electrical connectors.
- f. **W/O IAS** Remove tee fitting and packing from vertical speed indicator. Discard packing. 
- g. Cap port on vertical speed indicator.

8-4-4.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Uncap port on vertical speed indicator.
- c. **W/O IAS** Install new tee fitting and packing on indicator (Figure 8-4-4-1, Detail A). 
- d. Remove protective cap and plug from electrical connectors.
- e. Inspect and treat electrical connectors (PARA 1-7-20).
- f. **SLS** Install vertical speed indicator as follows: 
 - (1) Connect electrical connector, P181 (pilot's) or P194 (copilot's) to lighted bezel assembly.
 - (2) Install vertical speed indicator and lighted bezel assembly into instrument panel with screws.
- g. **W/O SLS** Install vertical speed indicator as follows: 
 - (1) Connect electrical connector, P8 (pilot's) or P3 (copilot's), into electrical receptacle on vertical speed indicator.
- h. **W/O IAS** Connect tee fitting to static tube as follows: 
 - (1) Coat mating threads on nut and tee fitting with antiseize compound, Item 87, Appendix D, and fasten static tube, S2, on tee fitting. **TORQUE B-NUT TO 40 - 60 INCH-POUNDS.**
 - (2) Make sure tee fitting has cap installed.
- i. Lightly tap on front bezel assembly of vertical speed indicator.
- j. If indicator pointer is offset from zero, perform the following:
 - (1) Turn zero adjust clockwise until pointer indicates about 500 FT PER MIN.
 - (2) Slowly turn zero adjust counterclockwise until pointer indicates between 100 FT PER MIN UP and 100 FT PER MIN DOWN.
 - (3) Lightly tap on front bezel assembly of vertical speed indicator and check pointer indication.
 - (4) If pointer does not indicate between 100 FT PER MIN UP and 100 FT PER MIN DOWN, repeat above adjustment procedure.
- k. Make sure area is clean and free of foreign material.
- l. Perform operational check of pitot-static system (PARA 8-2-1).

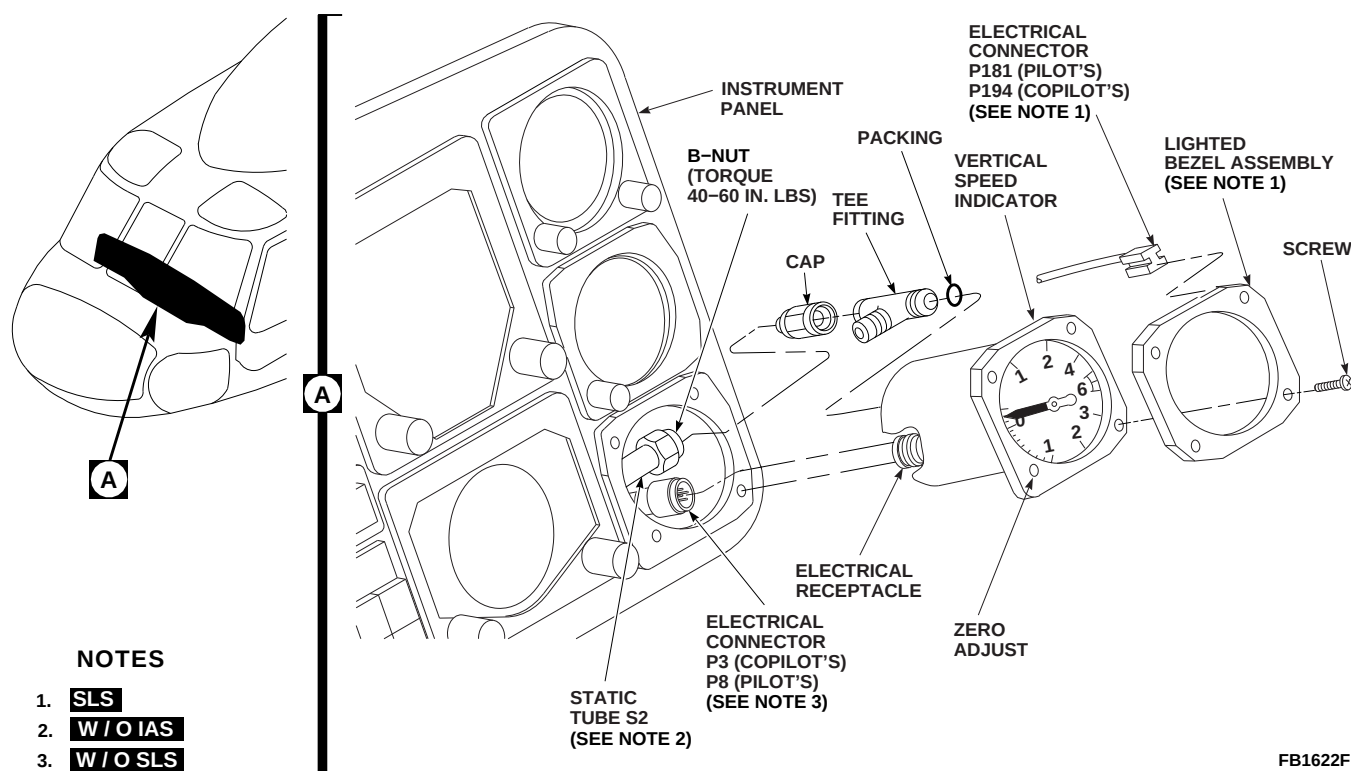


FIGURE 8-4-4-1. REPLACE VERTICAL SPEED INDICATOR

8-4-5 AIRSPEED INDICATOR.

PARAGRAPH BREAKDOWN

		PAGE
8-4-5.1	Replace Airspeed Indicator	8-4-5-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Protection, Eye

- Protective Caps and Plugs
- Sealing Compound, Item 297, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 8-2-1

8-4-5.1 Replace Airspeed Indicator.

NOTE

Replacement of pilot’s and copilot’s airspeed indicators is the same, except as noted.

8-4-5.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect pitot and static lines from union and restrictor union on airspeed indicator from under instrument panel (Figure 8-4-5-1, Detail A).
- Install protective plugs in pitot and static lines.
- SLS** Remove screws from lighted bezel assembly. ◀



SLS Damage to lighted bezel assembly will result if not careful when removing from instrument panel. Use care when removing lighted bezel assembly from the instrument panel. ◀

- SLS** Slowly lift lighted bezel assembly and airspeed indicator from instrument panel to gain access to electrical connector. Remove connector, P187 (pilot’s) or P189 (copilot’s), from lighted bezel assembly. ◀
- W/O SLS** Disconnect electrical connector, P7 (pilot’s) or P2 (copilot’s), from airspeed indicator. ◀
- Install protective cap and plug on electrical connectors.

- h. **W/O SLS** Remove screws and airspeed indicator from instrument panel. ◀

NOTE

If airspeed indicator is being replaced, retain union and restrictor union for installation on replacement indicator.

- i. If required, remove union, restrictor union, and packings from airspeed indicator. Discard packings.
- j. If unions are installed, install protective caps on unions, or install protective plugs in airspeed indicator ports.

8-4-5.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. If replacement airspeed indicator will be installed, perform the following:

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- (1) Using dry-cleaning solvent, Item 124, Appendix D, and dry compressed air, clean restrictor union internally.
- (2) Remove protective plugs from airspeed indicator ports.
- (3) Using sealing compound, Item 297, Appendix D, coat threads of union and restrictor union.
- (4) Install packing on union and restrictor union (Figure 8-4-5-1, Detail A).

CAUTION

Damage to pitot-static system will result if dirt is allowed to enter system. Make sure restrictor union is oriented the same as when removed.

- (5) Install union in static port of airspeed indicator and restrictor union in pitot port.
 - (6) Tighten restrictor union and union in airspeed indicator until they bottom tightly. **TORQUE UNIONS TO 40 - 60 INCH-POUNDS.**
- c. Remove protective plugs from pitot and static lines. If required, remove protective caps from unions.
- d. Remove protective cap and plug from electrical connectors.
- e. Inspect and treat electrical connectors (PARA 1-7-20).
- f. **SLS** Connect electrical connector, P187 (pilot's) or P189 (copilot's), to lighted bezel assembly. ◀
- g. **SLS** Position airspeed indicator and lighted bezel assembly in instrument panel and secure with screws. ◀
- h. **W/O SLS** Connect electrical connector, P7 (pilot's) or P2 (copilot's), to airspeed indicator. ◀
- i. **W/O SLS** Position airspeed indicator in instrument panel and secure with screws. ◀
- j. Connect pitot and static lines to union and restrictor union from under instrument panel. **TORQUE B-NUTS TO 40 - 60 INCH-POUNDS.**
- k. Make sure area is clean and free of foreign material.
- l. Perform operational check of pitot-static system (PARA 8-2-1).

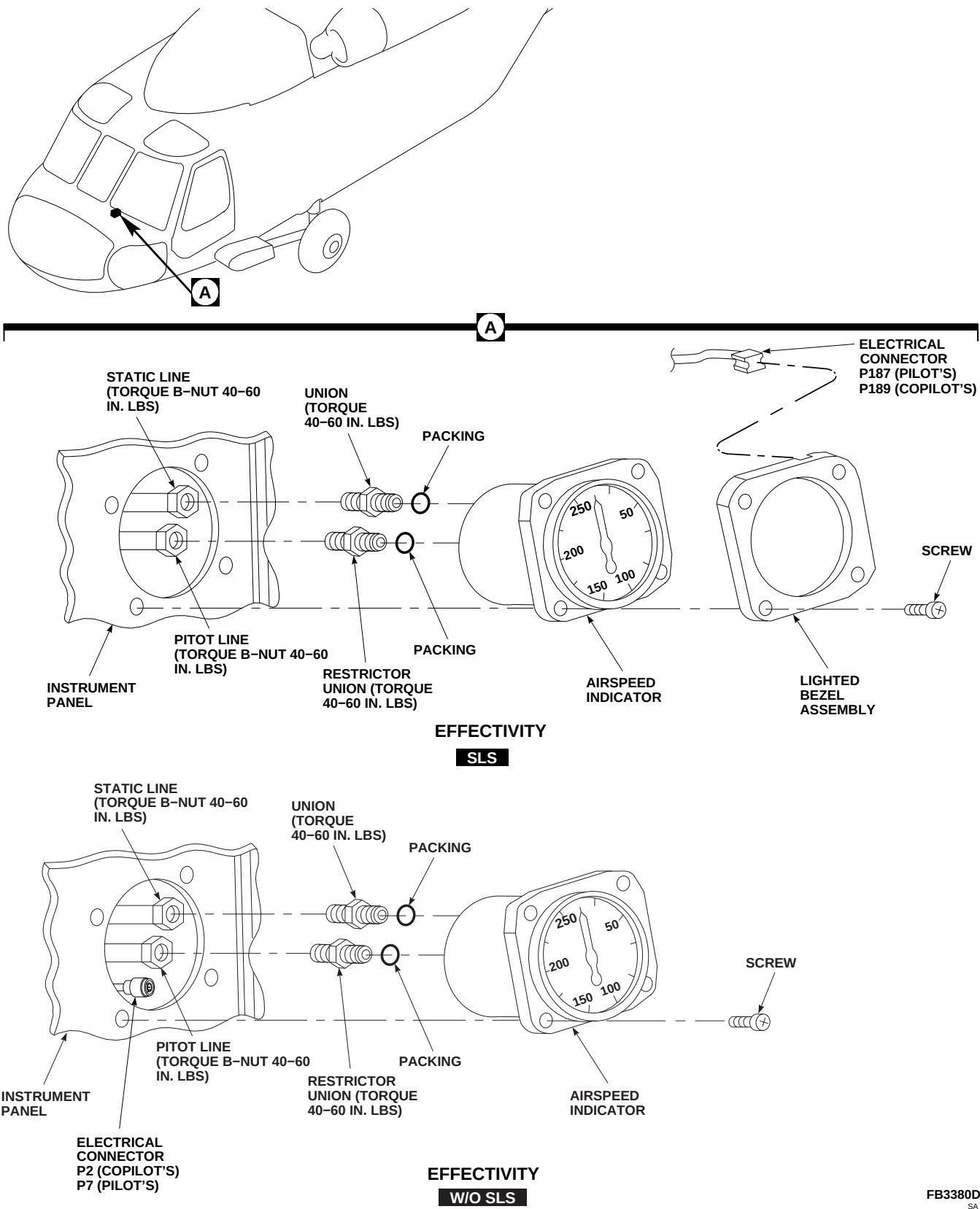


FIGURE 8-4-5-1. REPLACE AIRSPEED INDICATOR

8-4-6 BAROMETRIC ALTIMETER.

PARAGRAPH BREAKDOWN

		PAGE
8-4-6.1	Replace Barometric Altimeter	8-4-6-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Protective Caps and Plugs
- Sealing Compound, Item 297, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 8-2-1

8-4-6.1 Replace Barometric Altimeter.

NOTE

- Pilot’s and copilot’s barometric altimeters are not interchangeable.
- Replacement of pilot’s and copilot’s barometric altimeters is the same, except as noted.



Damage to lighted bezel assembly will result if care is not taken when removing from instrument panel. Use care when removing lighted bezel assembly from instrument panel.

8-4-6.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect static line from tee fitting behind barometric altimeter (Figure 8-4-6-1, Detail A).
- Install protective plug in static line.
- W/O SLS** Disconnect electrical connector, P306R (pilot’s) or P1 (copilot’s), from behind barometric altimeter.
- SLS** Remove screws from lighted bezel assembly.

- SLS** Slowly lift lighted bezel assembly and barometric altimeter from instrument panel to gain access to electrical connector.
- SLS** Disconnect electrical connector, P183 (pilot’s) or P193 (copilot’s), from lighted bezel assembly.
- W/O SLS** Remove screws and barometric altimeter from instrument panel.
- Install protective cap and plug on electrical connectors.

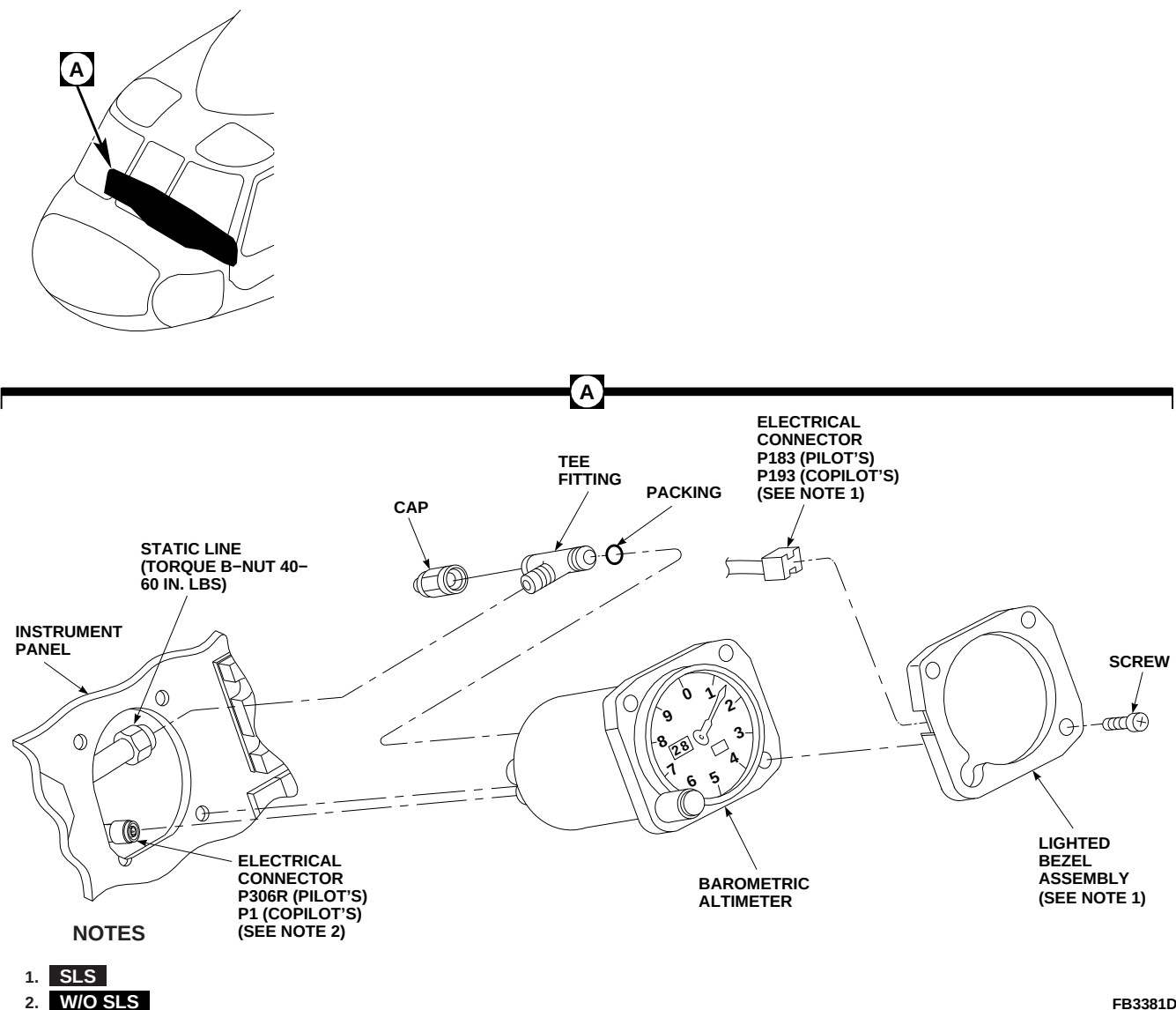


FIGURE 8-4-6-1. REPLACE BAROMETRIC ALTIMETER

NOTE

If new barometric altimeter will be installed, retain tee fitting for installation on replacement altimeter.

- j. If required, remove tee fitting and packing from barometric altimeter. Discard packing.
- k. If tee fitting is installed, install protective cap on tee fitting, or install protective plug in barometric altimeter port.

8-4-6.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).

- b. If new barometric altimeter will be installed, perform the following:

- (1) Remove protective plug from altimeter port.
- (2) Install packing on tee fitting (Figure 8-4-6-1, Detail A).
- (3) Using sealing compound, Item 297, Appendix D, coat threads of tee fitting.
- (4) Install tee fitting in port of barometric altimeter and cap on open end of tee fitting.

- c. Remove protective plug from static line. If required, remove protective cap from tee fitting.
- d. Remove protective cap and plug from electrical connectors.
- e. Inspect and treat electrical connectors (PARA 1-7-20).
- f. **W/O SLS** Connect electrical connector, P306R (pilot's) or P1 (copilot's), to rear of barometric altimeter.■
- g. **SLS** Connect electrical connector, P183 (pilot's) or P193 (copilot's), to lighted bezel assembly.■
- h. **SLS** Position barometric altimeter and lighted bezel assembly in instrument panel and secure with screws.■
- i. **W/O SLS** Position barometric altimeter in instrument panel and secure with screws.■
- j. Connect static line to tee fitting on barometric altimeter. **TORQUE B-NUT TO 40 - 60 INCH-POUNDS.**
- k. Make sure area is clean and free of foreign material.
- l. Perform operational check of pitot-static system (PARA 8-2-1).

8-4-7 STANDBY COMPASS.

PARAGRAPH BREAKDOWN

	PAGE
8-4-7.1	Replace Standby Compass 8-4-7-1

INITIAL SETUP

Tools

- Crimping Tool
- Electrical Repairer's Toolkit
- Magnet, Small Permanent

- Nonmagnetic Screwdriver

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20

8-4-7.1 Replace Standby Compass.

8-4-7.1.1 Remove.

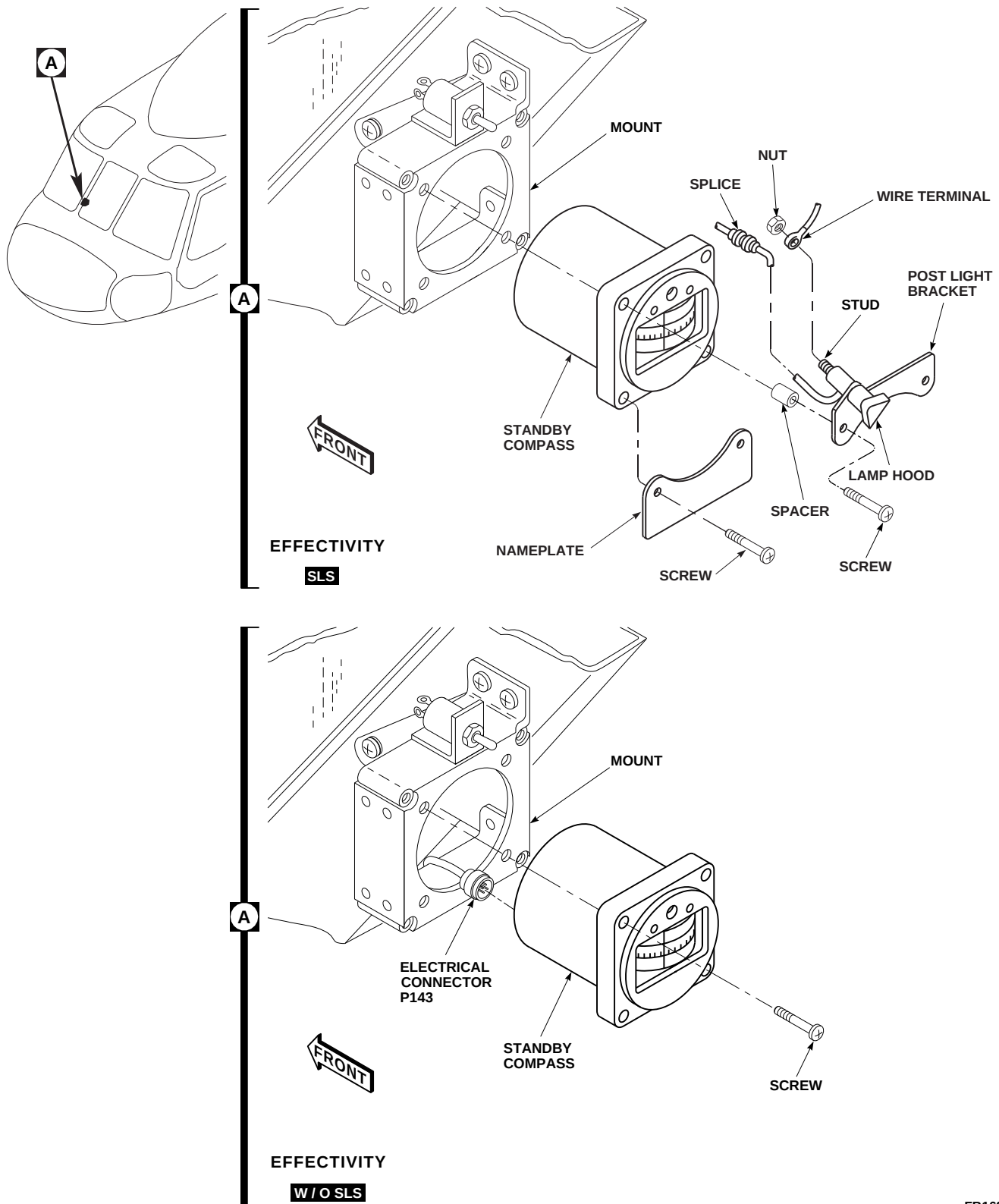
- Turn off all electrical power (PARA 1-6-16).
- SLS** On helicopters with external post light bracket installed, perform the following (Figure 8-4-7-1, Detail A):
 - Remove nut and wire terminal from post light stud.
 - Remove splice.
 - Remove screws and spacers securing post light bracket, standby compass, and nameplate to mount.
 - Remove standby compass from mount.
- W/O SLS** On helicopters without external post light bracket installed, perform the following (Figure 8-4-7-1, Detail A):
 - Remove screws securing standby compass to mount.

- Disconnect electrical connector, P143, from standby compass.

- Install protective caps and plugs on electrical connectors.

8-4-7.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- SLS** On helicopters with external post light bracket, perform the following (Figure 8-4-7-1, Detail A):
 - Place standby compass into mount.
 - Install nameplate, post light bracket, and standby compass to mount with screws and spacers.
 - Using crimping tool, splice other lead wire to wire from post light bracket.
 - Install wire terminal to post light stud with nut.
- W/O SLS** On helicopters without external post light bracket, perform the following (Figure 8-4-7-1, Detail A):



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FIGURE 8-4-7-1. REPLACE STANDBY COMPASS

- (1) Remove protective caps and plugs from electrical connectors.
 - (2) Inspect and treat electrical connector (PARA 1-7-20).
 - (3) Connect electrical connector, P143, to standby compass.
 - (4) Install standby compass to mount with screws.
- d. Compensate standby compass (PARA 8-4-7.1.3).

8-4-7.1.3 Compass Compensation.

NOTE

- Read compass directly in front of lubber line to avoid erroneous readings.
 - Inertial navigation system (INS) may be used as heading reference in place of compass rose if variation is known to be within 0.2°.
 - Example of Compensating Swing and Residual Swing is shown in Figure 8-4-7-2, Sheet 1.
 - Compensating Swing, Residual Swing, and Computation Tables shown in Figure 8-4-7-2, Sheet 2, must be locally reproduced and used during these procedures.
- a. Make sure helicopter is at least 100 yards away from any steel structure, underground cables, or other aircraft.
 - b. Make sure all items in helicopter containing ferrous materials are located in positions they will occupy during flight.
 - c. Have personnel remove all metallic objects and jewelry from their person.
 - d. Pass a small permanent magnet near compass to make compass card deflect through a small angle. **COMPASS CARD MUST ROTATE**

FREELY ON ITS PIVOT. PATH OF ROTATION MUST BE ALONG HORIZONTAL PLANE. If compass card does not rotate freely or stay on a horizontal plane, replace compass.

- e. Remove one compensator cover plate screw. Loosen other screw and rotate plate to allow access to compensating screws.
- f. With qualified pilot at controls, run engines and apply electrical power to all electrical equipment.
- g. Perform COMPENSATING SWING as follows (Figure 8-4-7-2, Sheet 1 and Sheet 2):
 - (1) Park helicopter on compass rose with nose pointing toward magnetic north. Record compass reading on Compensating Swing Table ACFT COMP column. Example shown assumes reading is 8°.
 - (2) Turn helicopter on compass rose with nose pointing toward east. Record compass reading on Compensating Swing Table ACFT COMP column. Example shown assumes reading is 86°.
 - (3) Turn helicopter on compass rose with nose pointing toward south. Record compass reading on Compensating Swing Table ACFT COMP column. Example shown assumes reading is 176°.
 - (4) Turn helicopter on compass rose with nose pointing toward west. Record compass reading on Compensating Swing Table ACFT COMP column. Example shown assumes reading is 278°.
 - (5) Subtract ACFT COMP value from ACTUAL HEAD (M) value for N, E, S, and W. Record result in DEVIATION column (ACTUAL HEAD (M) - ACFT COMP = DEVIATION).
 - (6) Turn helicopter on compass rose with nose pointing toward magnetic north. Using equation for Coefficient C, determine what compass should read when compensated. Adjust NS compensating

screw using nonmagnetic screwdriver until compass indicates compensated value.

- (7) Turn helicopter on compass rose with nose pointing toward east. Using equation for Coefficient B, determine what compass should read when compensated. Using nonmagnetic screwdriver, adjust EW compensating screw until compass indicates compensated value.

- (8) Using equation for Coefficient A, determine total deviation. **TOTAL MAXIMUM POSITIVE AND NEGATIVE DEVIATION AFTER COMPENSATION MUST NOT EXCEED 5°**. If limit is exceeded, replace compass.

h. Perform RESIDUAL SWING as follows:

- (1) With helicopter nose still pointing toward east, record compass reading in Residual Swing Table ACFT COMP column.
- (2) Turn helicopter on compass rose with nose pointing toward north east. Record compass reading in Residual Swing Table ACFT COMP column.
- (3) Repeat every 45° around compass rose until helicopter is turned in a complete circle.

- (4) Turn off engines.

- (5) Subtract ACFT COMP value from ACTUAL HEAD (M) value for N, NE, E, SE, S, SW, W, and NW. Record result in C to M column in Computation Table (ACTUAL HEAD (M) - ACFT COMP = C to M).

- (6) Subtract ACTUAL HEAD (M) value from ACFT COMP value for N, NE, E, SE, S, SW, W, and NW. Record result in M to C column in Computation Table (ACFT COMP - ACTUAL HEAD (M) = M to C).

- (7) Add values in C to M column. **TOTAL DEVIATION OF EIGHT HEADINGS MUST NOT EXCEED $\pm 8^\circ$** . If limit is exceeded, replace compass.

- (8) Add values in M to C column. **TOTAL DEVIATION OF EIGHT HEADINGS MUST NOT EXCEED $\pm 8^\circ$** . If limit is exceeded, replace compass.

- i. Annotate helicopter's Pilot's Correction Card located in cockpit with values shown in Compass Correction Table, ACFT COMP column.

- j. Retain a copy of Compass Correction and Computation Tables with helicopter historical records.

COMPENSATING SWING TABLE				RESIDUAL SWING TABLE	
DIRECTION / COMP ROSE	ACTUAL HEAD (M)	ACFT COMP	DEVIATION	ACTUAL HEAD (M)	ACFT COMP
N = 00	00	008	-8	00	002
NE = 045				045	046
E = 090		086	+4	090	092
SE = 135				135	135
S = 180	180	176	+4	180	182
SW = 225	270			225	224
W = 270		278	-8	270	272
NW = 315		(ACTUAL HEAD (M) - ACFT COMP)		315	315

COMPUTATION TABLE		
C TO M (ACTUAL HEAD (M) - ACFT COMP)	ACTUAL HEAD (M)	M TO C (ACFT COMP- ACTUAL HEAD (M))
-2	00	+2
-1	045	+1
-2	090	+2
0	135	0
-2	180	+2
+1	225	-1
-2	270	+2
0	315	0

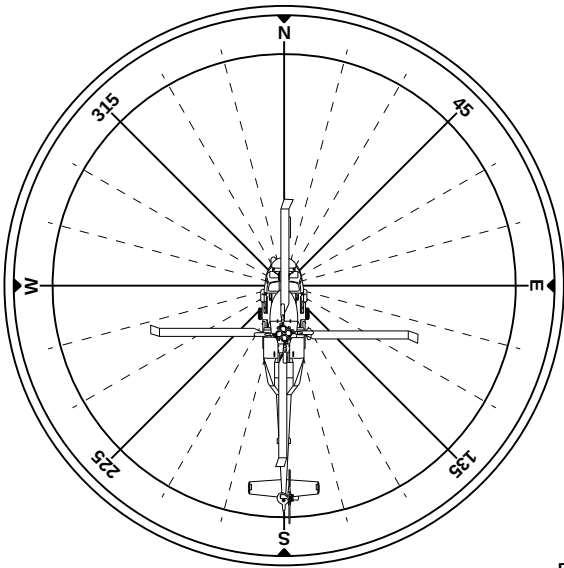
EQUATIONS

COEFFICIENT C = $\frac{N-S}{2} = \frac{(-8)-(+4)}{2} = \frac{-12}{2} = -6$

COEFFICIENT B = $\frac{E-W}{2} = \frac{(+4)-(-8)}{2} = \frac{12}{2} = +6$

COEFFICIENT A = $\frac{N+E+S+W}{4} = \frac{(-8)+(+4)+(+4)+(-8)}{4} = \frac{-8}{4} = -2$

SAMPLE



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FIGURE 8-4-7-2. COMPENSATING SWING, RESIDUAL SWING, AND COMPUTATION TABLES (SHEET 1 OF 2)

COMPASS CORRECTION TABLE

COMPENSATING SWING				RESIDUAL SWING	
DIRECTION / COMP ROSE	ACTUAL HEAD (M)	ACFT COMP	DEVIATION	ACTUAL HEAD (M)	ACFT COMP
N = 00	00			00	
NE = 045				045	
E = 090	090			090	
SE = 135				135	
S = 180	180			180	
SW = 225				225	
W = 270	270			270	
NW = 315				315	
			(ACTUAL HEAD (M) - ACFT COMP)		

COMPUTATION TABLE		
C TO M (ACTUAL HEAD (M) - ACFT COMP)	ACTUAL HEAD (M)	M TO C (ACFT COMP - ACTUAL HEAD (M))
	00	
	045	
	090	
	135	
	180	
	225	
	270	
	315	

EQUATIONS

$$\text{COEFFICIENT C} = \frac{N-S}{2}$$

$$\text{COEFFICIENT B} = \frac{E-W}{2}$$

$$\text{COEFFICIENT A} = \frac{N+E+S+W}{4}$$

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SAFIGURE 8-4-7-2. COMPENSATING SWING, RESIDUAL SWING, AND COMPUTATION TABLES
(SHEET 2 OF 2) (REPRODUCE LOCALLY)

8-4-8 **STANDBY COMPASS LAMP** **SLS** .

PARAGRAPH BREAKDOWN

		PAGE
8-4-8.1	Replace Standby Compass Lamp	8-4-8-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-6-16

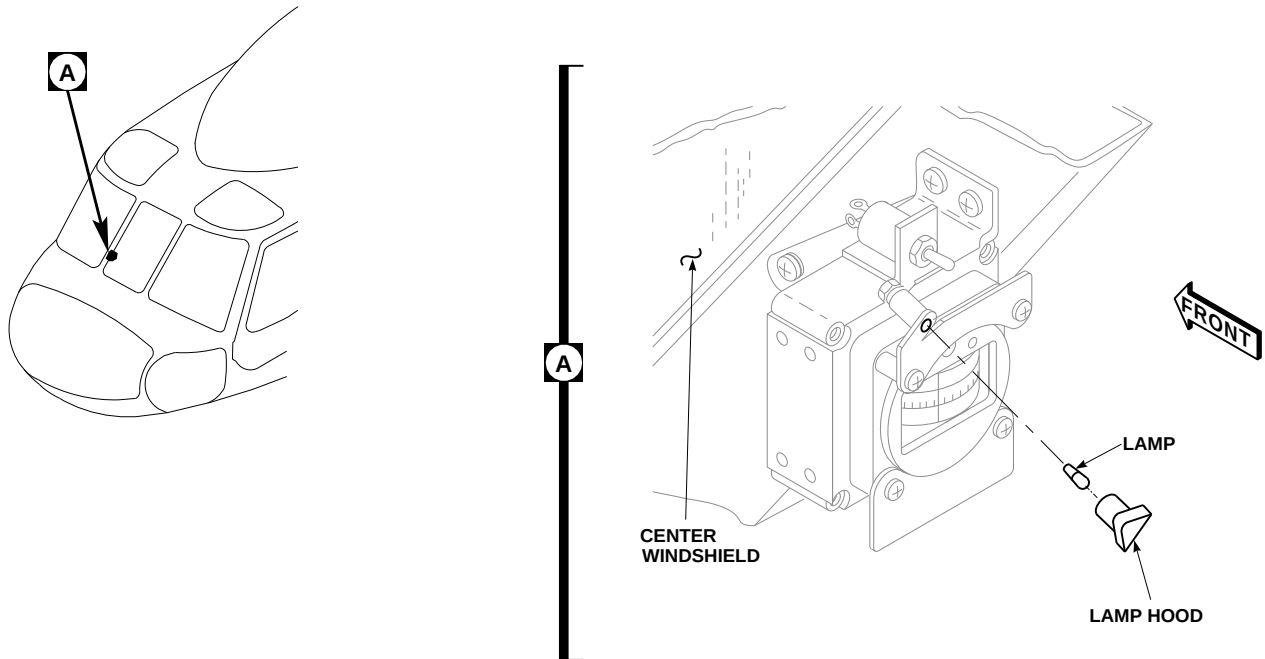
8-4-8.1 **Replace Standby Compass Lamp.**

8-4-8.1.1 **Remove.**

- Turn off all electrical power (PARA 1-6-16).
- Remove lamp hood (Figure 8-4-8-1, Detail A).
- Remove lamp from receptacle.

8-4-8.1.2 **Install.**

- Turn off all electrical power (PARA 1-6-16).
- Insert lamp into receptacle (Figure 8-4-8-1, Detail A).
- Install lamp hood.



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FIGURE 8-4-8-1. REPLACE STANDBY COMPASS LAMP

8-4-9 STANDBY COMPASS LIGHT SWITCH.

PARAGRAPH BREAKDOWN

	PAGE
8-4-9.1 Replace Standby Compass Light Switch	8-4-9-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Heat Gun
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Insulation Sleeving, Heat-Shrinkable, Item 180, Appendix D

- Solder, Item 313, Appendix D

- Tags

References

- Appendix D
- PARA 1-6-16

8-4-9.1 Replace Standby Compass Light Switch.

8-4-9.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove nut, lockwasher, and standby compass light switch from switch bracket (Figure 8-4-9-1, Detail A).
- Remove insulation sleeving from wires on standby compass light switch.
- Tag wires.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and

injury to personnel. Use protective clothing, including eye protection, while soldering. Make sure hot metal droppings do not come in contact with skin, eyes, or clothing.

- Using soldering iron, unsolder wires from standby compass light switch.

8-4-9.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install heat-shrinkable insulation sleeving, Item 180, Appendix D, over standby compass light switch wires.

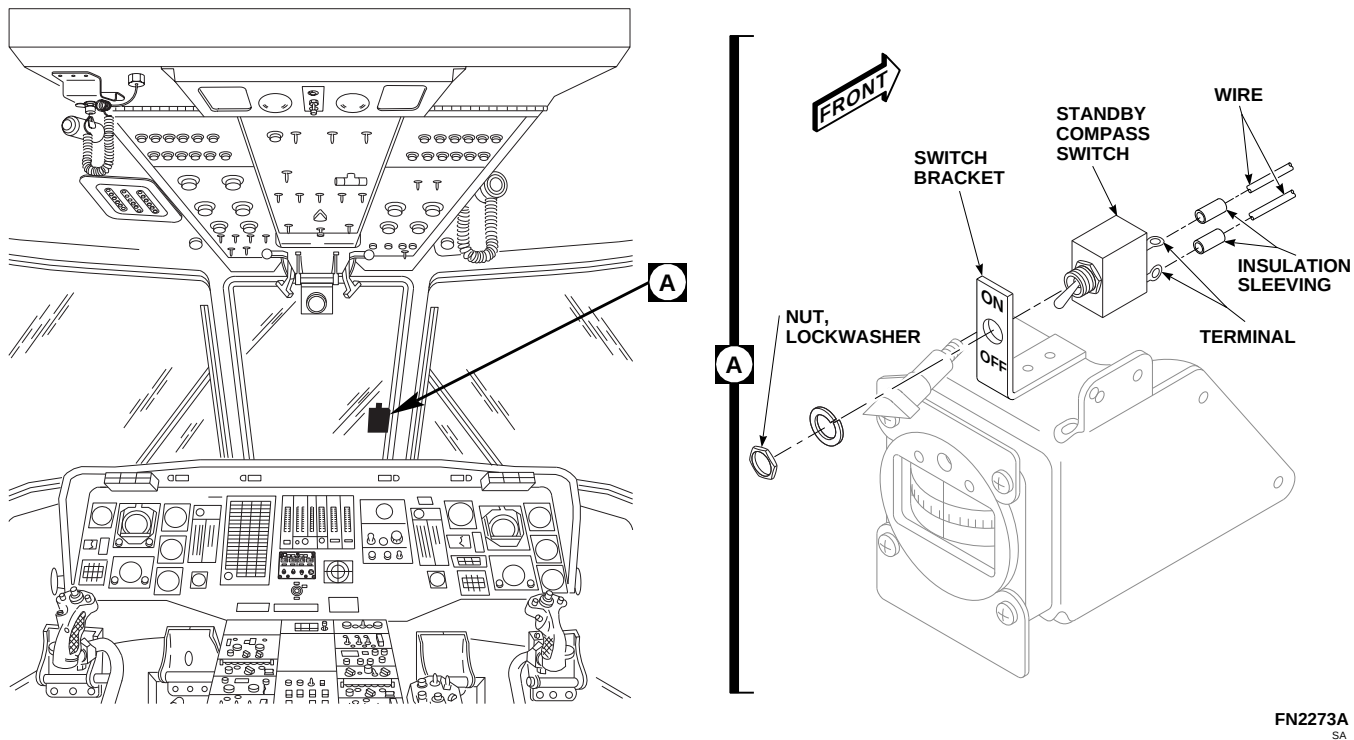


FIGURE 8-4-9-1. REPLACE STANDBY COMPASS LIGHT SWITCH

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection, while soldering. Make sure hot metal droppings do not come in contact with skin, eyes, or clothing.

- c. Using soldering iron and solder, Item 313, Appendix D, solder wires to switch lugs (Figure 8-4-9-1, Detail A).

- d. Remove tags.
- e. Slide insulation sleeving over terminals of standby compass light switch and using heat gun, shrink insulation sleeving around terminals.
- f. Install standby compass light switch on switch bracket with nut and lockwasher.
- g. Make sure area is clean and free of foreign material.

8-4-10 CENTRAL DISPLAY UNIT.

PARAGRAPH BREAKDOWN

	PAGE
8-4-10.1 Replace Central Display Unit	8-4-10-1

INITIAL SETUP

Tools

- Digital Low-Resistance Ohmmeter
- Electrical Repairer’s Toolkit

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 8-2-3

8-4-10.1 Replace Central Display Unit.

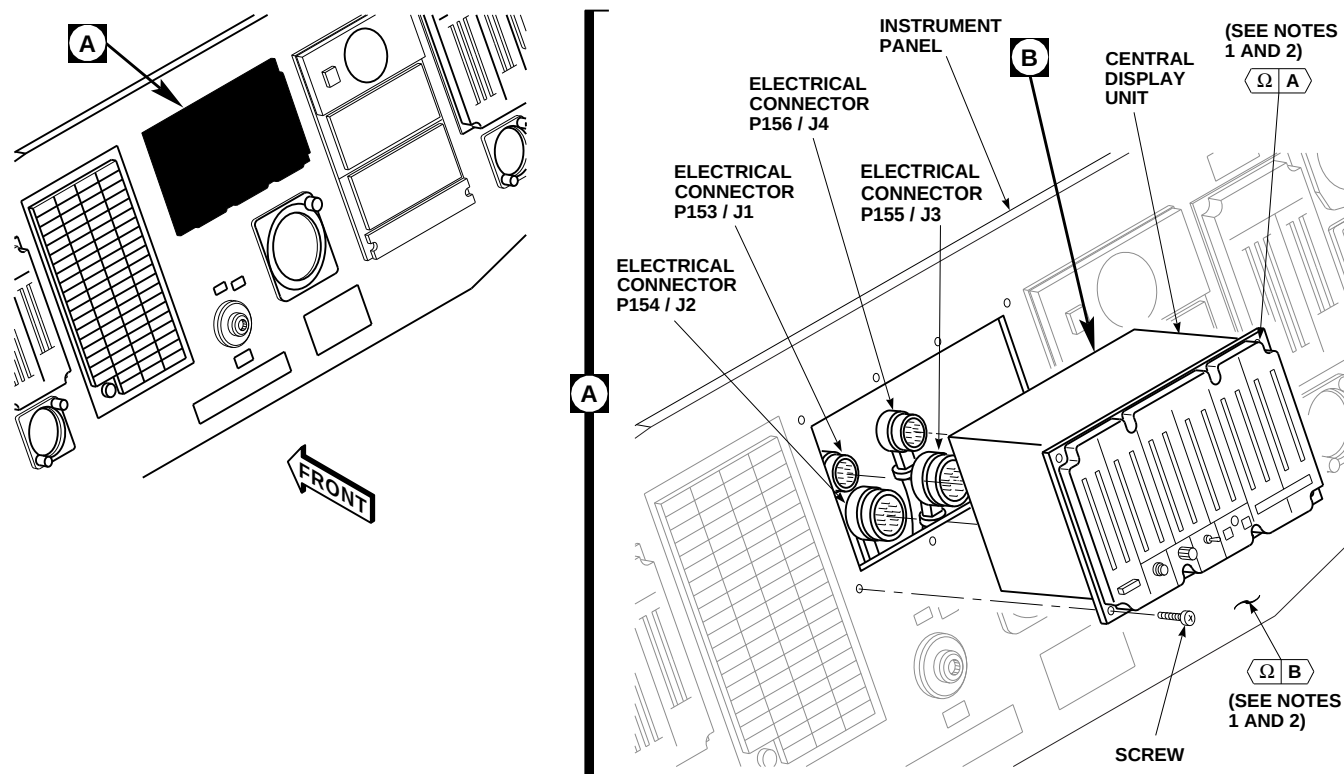
8-4-10.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove screws and lift central display unit (CDU) slowly from instrument panel to gain access to electrical connectors at rear (Figure 8-4-10-1, Detail A).
- c. Tag and disconnect electrical connectors, P153, P154, P155, and P156, from CDU. Remove CDU.
- d. Install protective caps and plugs on electrical connectors.

8-4-10.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).

- b. Remove protective caps and plugs from electrical connectors.
- c. Inspect and treat electrical connectors (PARA 1-7-20).
- d. Support central display unit (CDU) and connect electrical connectors, removing tags as follows (Figure 8-4-10-1, Detail A and Detail B):
 - (1) P153 to J1
 - (2) P154 to J2
 - (3) P155 to J3
 - (4) P156 to J4
- e. **EMEP** Inspect chemical conversion-coated surface as follows:■



NOTES

1. **EMEP**

POINTS FROM TO		RESISTANCE SHALL BE
A	B	2.5 MILLIOHMS OR LESS

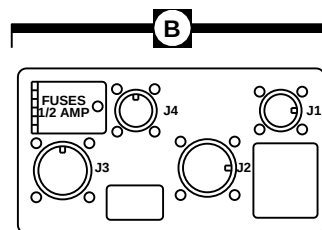
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FIGURE 8-4-10-1. REPLACE CENTRAL DISPLAY UNIT

- (1) Clean surface around all mounting holes with machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D. Dry surface with clean, dry machinery towel, Item 211, Appendix D.

- (2) Inspect chemical conversion-coated surfaces for breaks and scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches, apply chemical conversion coating (PARA 2-6-5).

- f. Make sure area is clean and free of foreign material.

- g. Carefully place CDU in instrument panel and secure with screws (Figure 8-4-10-1, Detail A).
- h. **EMEP** Perform EME resistance check of CDU as follows:

NOTE

Resistance measurements must be taken at least three times.

- (1) Using digital low-resistance ohmmeter, measure resistance reading from faceplate of CDU and instrument panel. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**

- (2) If resistance reading is greater than 2.5 milliohms, remove CDU and apply chemical conversion coating (PARA 2-6-5).
- i. Perform operational check of instrument display system (PARA 8-2-3).

8-4-11 CENTRAL DISPLAY UNIT FACEPLATE AND EMI SHIELD.

PARAGRAPH BREAKDOWN

	PAGE
8-4-11.1 Replace Central Display Unit Faceplate and EMI Shield	8-4-11-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 8-2-3

8-4-11.1 Replace Central Display Unit Faceplate and EMI Shield.

installing faceplate that screws are hand-tightened only until snug.

8-4-11.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove screws and washers from faceplate on central display unit (CDU) (Figure 8-4-11-1, Detail A).
- Starting at lower right corner, carefully pull faceplate straight outward to disconnect lighting jack at rear, and remove faceplate.
- On CDU, 70450-01043-125 or 70450-01916-105, carefully remove EMI shield from CDU.

NOTE

When installing faceplate to CDU, make sure lighting jack at rear of faceplate mates properly with chassis.

8-4-11.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- On central display unit (CDU), 70450-01043-125 or 70450-01916-105, position EMI shield on CDU (Figure 8-4-11-1, Detail A).

CAUTION

Damage to faceplate will result if screws are overtorqued. Make sure when

- Install faceplate on CDU with screws and washers. **HANDTIGHTEN SCREWS.**
- Make sure area is clean and free of foreign material.
- Turn on electrical power (PARA 1-6-16).
- Make sure LIGHTS NON FLT circuit breaker (NO. 2 AC PRI BUS - pilot’s circuit breaker panel) is pushed in.
- Turn upper console INSTR LT NON FLT control to BRT. CDU faceplate lights must go on bright. If result is not as specified, perform operational check (PARA 8-2-3).
- Turn off all electrical power (PARA 1-6-16).

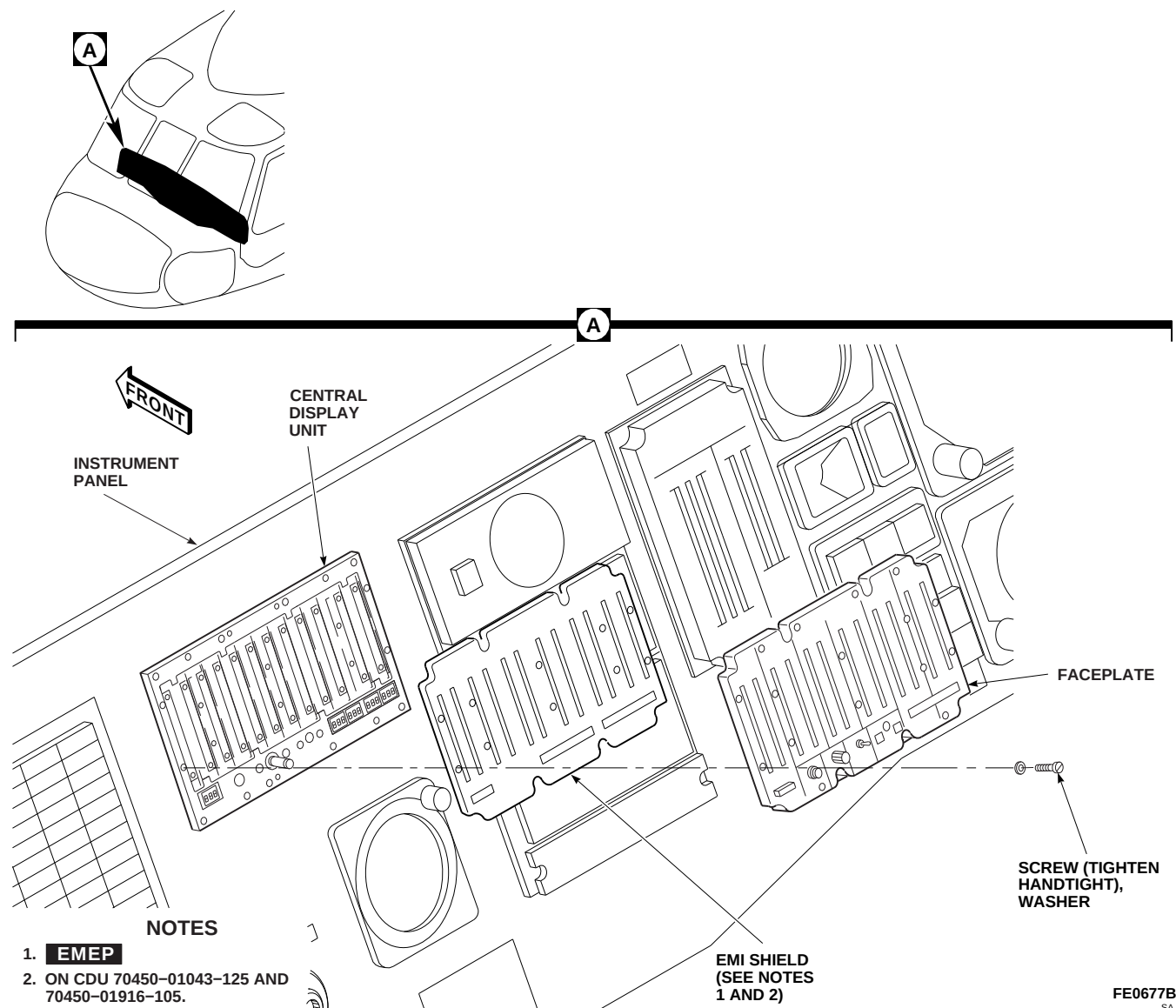


FIGURE 8-4-11-1. REPLACE CENTRAL DISPLAY UNIT FACEPLATE AND EMI SHIELD

8-4-12 CENTRAL DISPLAY UNIT BAR GRAPH FILAMENT LAMPS AND DIRECT VIEW FILAMENT MODULES **EMEP** .

PARAGRAPH BREAKDOWN

		PAGE
8-4-12.1	Replace Central Display Unit Bar Graph Filament Lamps and Direct View Filament Modules	8-4-12-1

INITIAL SETUP

- PARA 8-4-11

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 8-2-3

8-4-12.1 Replace Central Display Unit Bar Graph Filament Lamps and Direct View Filament Modules.

NOTE

Replacement is the same for all central display unit bar graph filament lamps and direct view filament modules.

8-4-12.1.1 Remove.

- Remove central display unit faceplate and EMI shield (PARA 8-4-11).
- Carefully remove bar graph filament lamp/direct view filament module from central display unit (Figure 8-4-12-1, Detail B).

8-4-12.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Position bar graph filament lamp/direct view filament module in required space and install (Figure 8-4-12-1, Detail B).
- Install central display unit faceplate and EMI shield (PARA 8-4-11).
- Perform operational check of instrument display system (PARA 8-2-3).

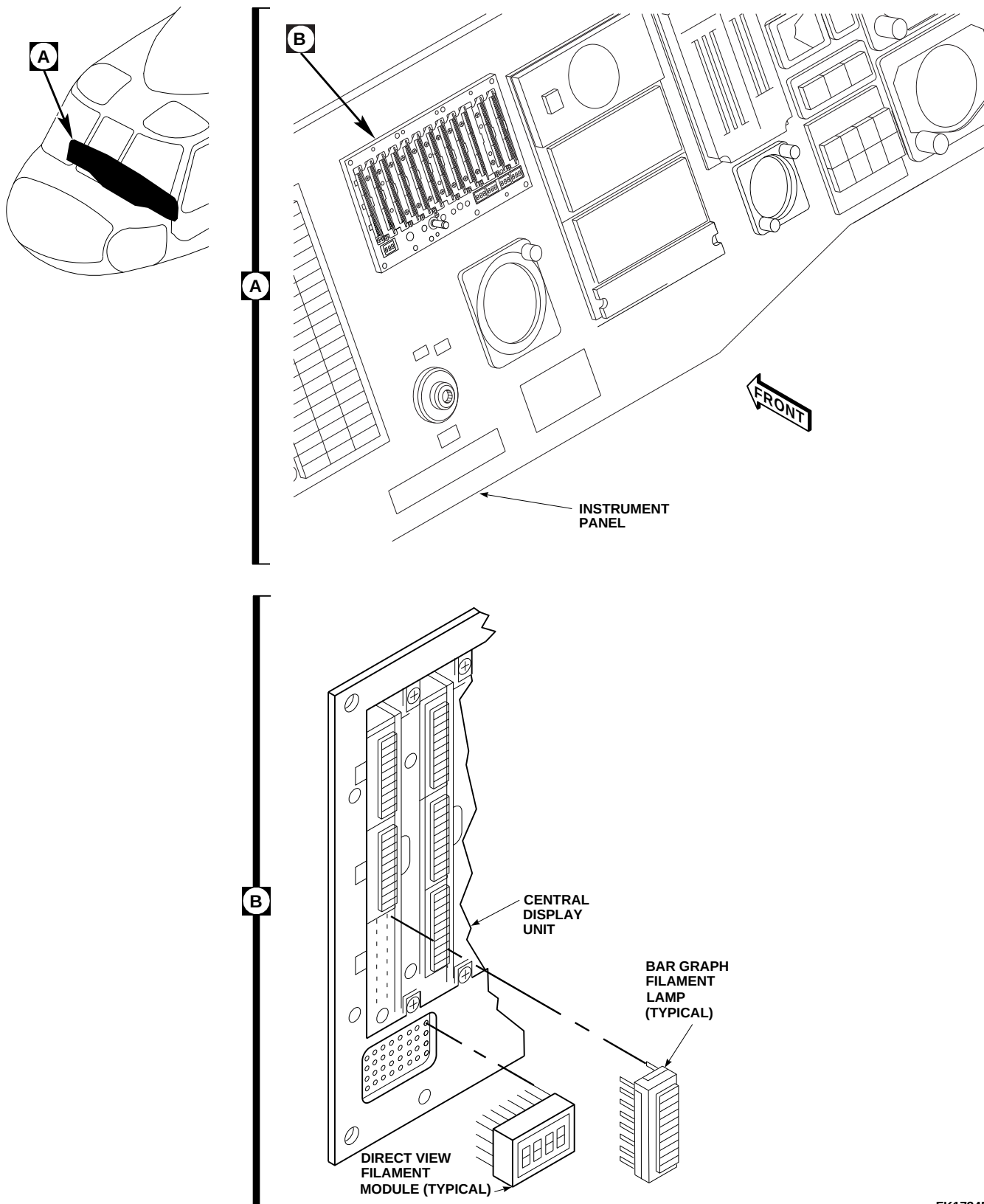
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FIGURE 8-4-12-1. REPLACE CENTRAL DISPLAY UNIT BAR GRAPH FILAMENT LAMPS AND DIRECT VIEW FILAMENT MODULES

8-4-13 CENTRAL DISPLAY UNIT FUSES.

PARAGRAPH BREAKDOWN

		PAGE
8-4-13.1	Replace Central Display Unit Fuses	8-4-13-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 8-4-10

8-4-13.1 Replace Central Display Unit Fuses.

NOTE

Replacement for all central display unit fuses is the same.

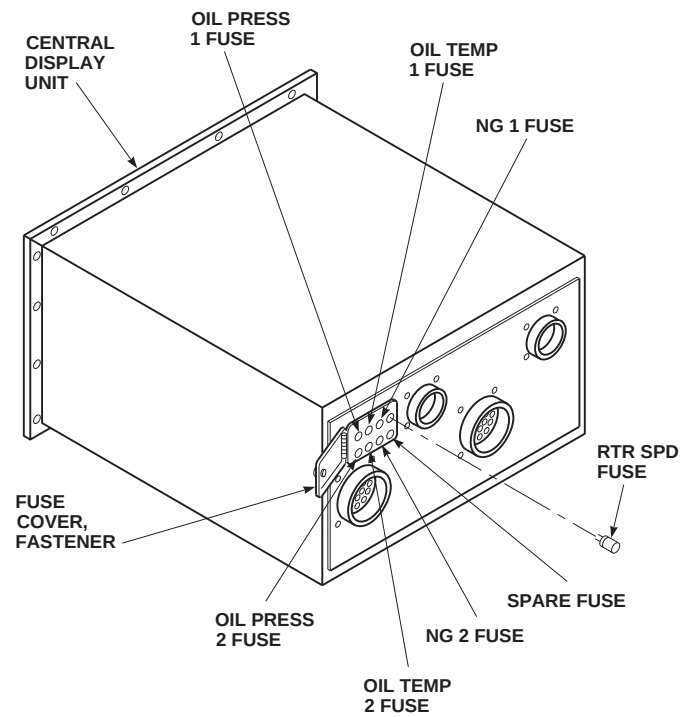
8-4-13.1.1 Remove.

- Remove central display unit (PARA 8-4-10).
- Loosen fastener and open fuse cover to gain access to fuse (Figure 8-4-13-1).

- Support central display unit and pull out fuse.

8-4-13.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Insert fuse (Figure 8-4-13-1).
- CLOSE FUSE COVER AND TIGHTEN FASTENER.**
- Install central display unit (PARA 8-4-10).

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8-4-14 CENTRAL DISPLAY UNIT LAMP ASSEMBLY **W/O EMEP** .

PARAGRAPH BREAKDOWN

		PAGE
8-4-14.1	Replace Central Display Unit Lamp Assembly	8-4-14-1

INITIAL SETUP

- PARA 8-4-11

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 8-2-3

8-4-14.1 Replace Central Display Unit Lamp Assembly.

NOTE

Replacement of all central display unit lamp assemblies is the same.

8-4-14.1.1 Remove.

- Remove faceplate (PARA 8-4-11).
- Loosen lockscrews securing lamp driver module to central display unit, and remove lamp driver module (Figure 8-4-14-1, Detail A).
- Remove screw securing display lamp assembly to lamp driver module, and carefully pull display lamp assembly away from lamp driver module (Figure 8-4-14-1, Detail B and Detail C).

8-4-14.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).

- Position and insert display lamp assembly into lamp driver module (Figure 8-4-14-1, Detail B).
- Install display lamp assembly with screw.

NOTE

These modules are interchangeable: transmission oil temperature (A5), No. 1 engine oil temperature (A7), No. 2 engine oil temperature (A8), No. 1 TGT (A11), No. 2 TGT (A12), No. 1 Ng speed (A13), No. 2 Ng speed (A14), No. 1 fuel quantity (A3) and No. 2 fuel quantity (A4), No. 1 engine oil pressure (A9) and No. 2 engine oil pressure (A10).

- Install lamp driver module into central display unit and secure with lockscrews (Figure 8-4-14-1, Detail A).
- Install faceplate (PARA 8-4-11).
- Make sure area is clean and free of foreign material.

- g. Perform operational check of instrument display system (PARA 8-2-3).

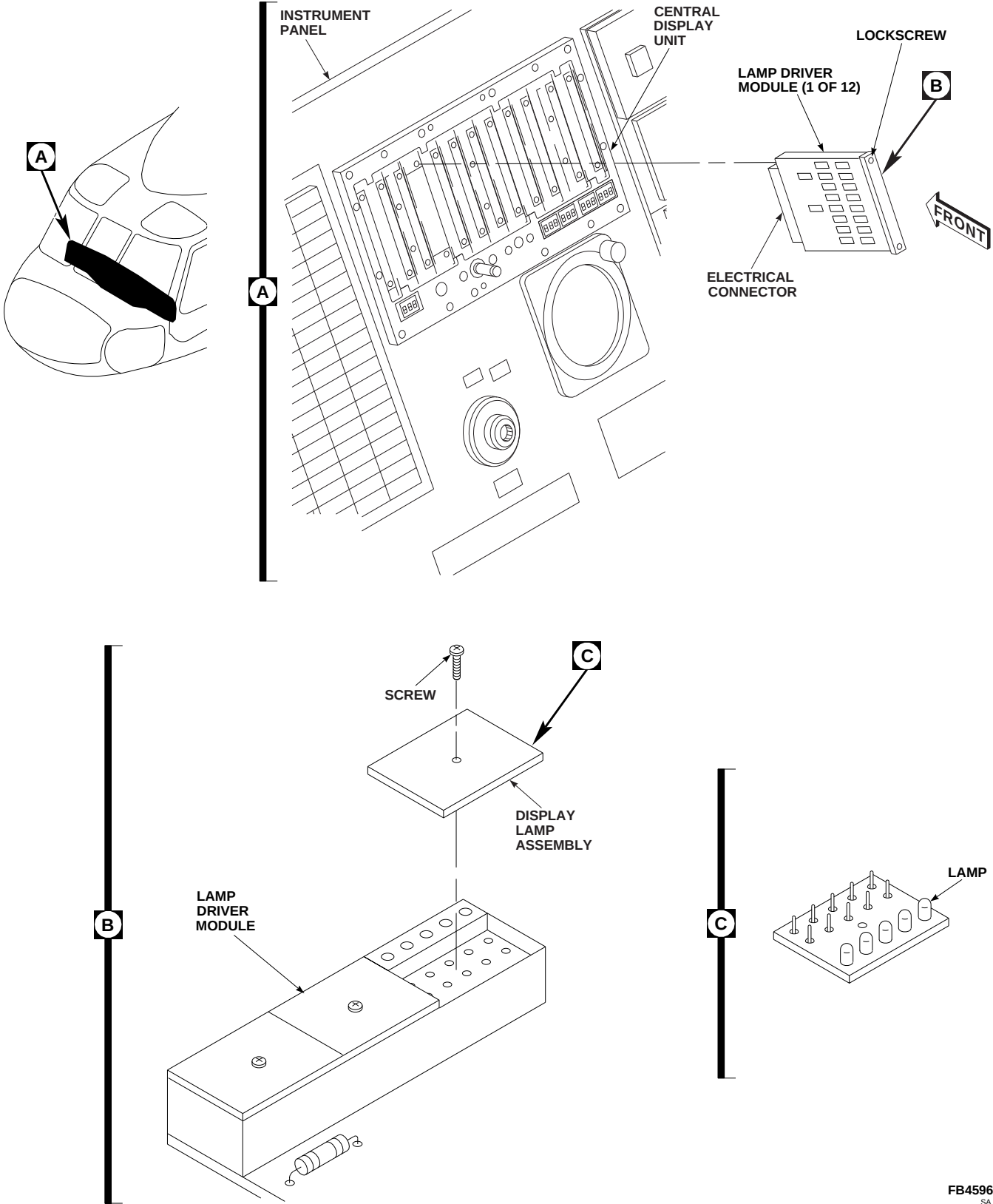


FIGURE 8-4-14-1. REPLACE CENTRAL DISPLAY UNIT LAMP ASSEMBLY

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8-4-15 SIGNAL DATA CONVERTER, NO. 1 AND NO. 2.

PARAGRAPH BREAKDOWN

		PAGE
8-4-15.1	Replace Signal Data Converter, No. 1 and No. 2	8-4-15-1

INITIAL SETUP

Tools

- Digital Low-Resistance Ohmmeter
- Electrical Repairer’s Toolkit

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 8-2-3

8-4-15.1 Replace Signal Data Converter, No. 1 and No. 2.

NOTE

- Replacement of No. 1 and No. 2 signal data converter is the same, except as noted.
- **EMEP** Pin filter adapters are considered part of harness assembly. Do not remove pin filter adapter from electrical connectors, P138 or P137.

8-4-15.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open nose door.
- Tag and disconnect electrical connectors as follows (Figure 8-4-15-1, Detail A):

No. 1 Signal Data Converter	No. 2 Signal Data Converter
-----------------------------	-----------------------------

- | | |
|--------------|--------------|
| P138 from J1 | P137 from J1 |
| P149 from J2 | P150 from J2 |
| P151 from J3 | P152 from J3 |

- Install protective caps and plugs on electrical connectors.
- Remove bolts, lockwashers, and washers securing signal data converter to nose shelf. Remove signal data converter.

8-4-15.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).

- b. **EMEP** Inspect chemical conversion-coated surfaces as follows:

- (1) Clean area around mounting surface using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D. Dry surface with machinery towel, Item 211, Appendix D.
- (2) Inspect chemical conversion-coated surfaces for breaks and scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches, apply chemical conversion coating (PARA 2-6-5).

NOTE

When installing signal data converter, make sure electrical connectors are positioned on left side of nose shelf for No. 1 signal data converter or right side for No. 2 signal data converter (Figure 8-4-15-1, Detail A).

- c. Install signal data converter to nose shelf and secure with bolts, lockwashers, and washers.
- d. Remove protective caps and plugs from electrical connectors.
- e. Inspect and treat electrical connectors (PARA 1-7-20).
- f. Remove tags, and connect electrical connectors as follows:

No. 1 Signal Data Converter	No. 2 Signal Data Converter
P138 to J1	P137 to J1
P149 to J2	P150 to J2
P151 to J3	P152 to J3

- g. **EMEP** Perform pin filter adapter EME resistance check as follows:

NOTE

Resistance measurements must be taken at least three times.

- (1) Using digital low-resistance ohmmeter, measure resistance between shell of each pin filter adapter and surface of signal data converter. **RESISTANCE SHALL BE 5.0 MILLIOHMS OR LESS.**
- (2) If resistance is greater than 5.0 milliohms, measure resistance between shell of pin filter adapter and connector on signal data converter. **RESISTANCE SHALL BE 5.0 MILLIOHMS OR LESS.**
- (3) If resistance is still beyond limits, clean surface between pin filter adapter and signal data converter (PARA 1-7-20).

- h. **EMEP** Perform EME resistance check as follows:

NOTE

Resistance measurements must be taken at least three times.

- (1) Using digital low-resistance ohmmeter, measure resistance between surface of signal data converter and nose shelf. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**
- (2) If resistance is greater than 2.5 milliohms, remove signal data converter and apply chemical conversion coating (PARA 2-6-5).

- i. Make sure area is clean and free of foreign material.
- j. Perform operational check of signal data converter (PARA 8-2-3).
- k. Close nose door and secure latches.

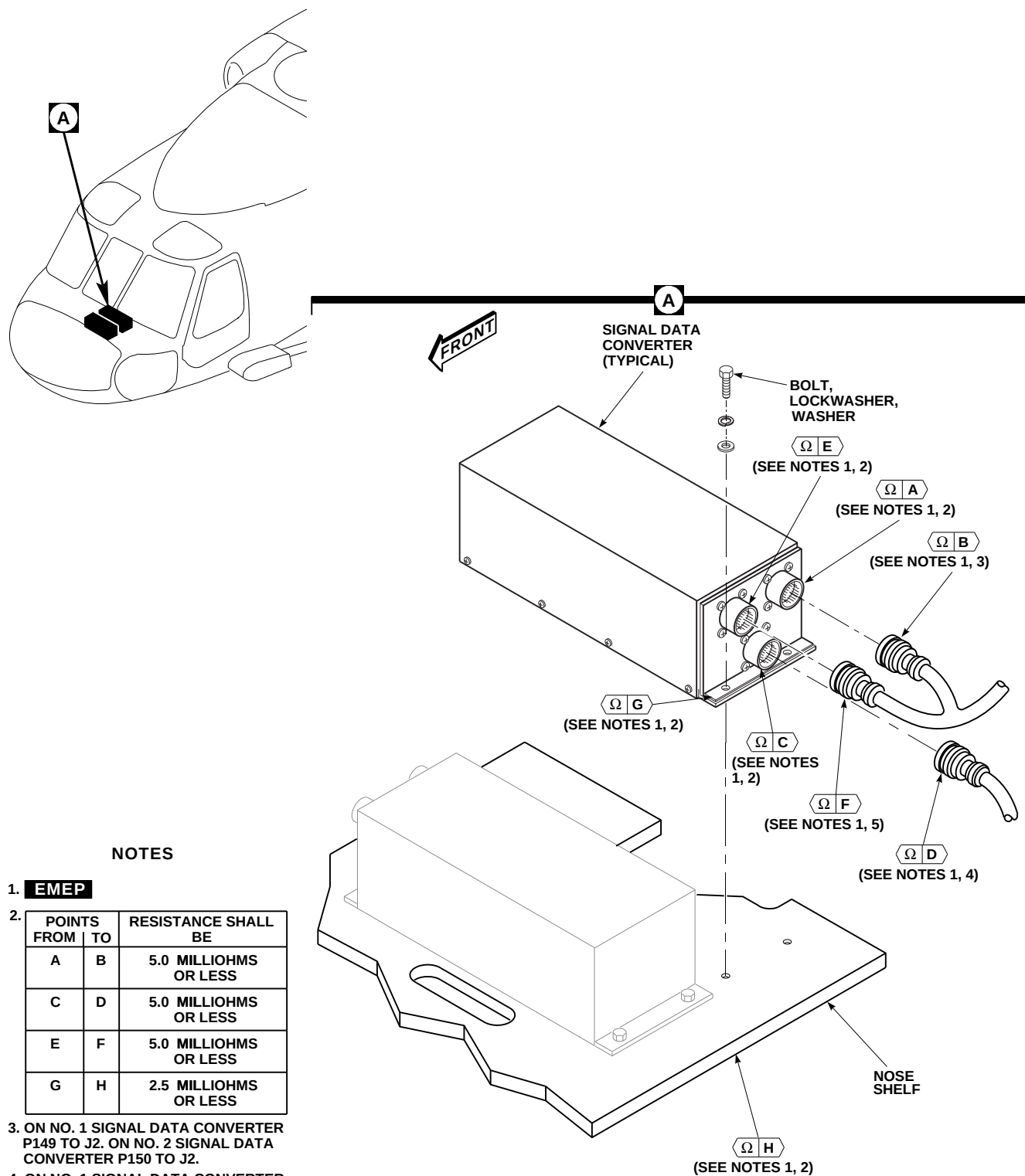


FIGURE 8-4-15-1. REPLACE SIGNAL DATA CONVERTER, NO. 1 AND NO. 2

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8-4-16 PILOT DISPLAY UNIT.

PARAGRAPH BREAKDOWN

	PAGE
8-4-16.1 Replace Pilot Display Unit	8-4-16-1

INITIAL SETUP

Tools

- Digital Low-Resistance Ohmmeter
- Electrical Repairer’s Toolkit

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 8-2-3

8-4-16.1 Replace Pilot Display Unit.

NOTE

Replacement of pilot’s and copilot’s pilot display unit (PDU) is the same, except as noted.

Copilot’s Pilot Display Unit Connectors	Pilot’s Pilot Display Unit Connectors
---	---------------------------------------

P157 from J1	P158 from J1
P159 from J2	P160 from J2

8-4-16.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove screws and lift PDU slowly from instrument panel until electrical connectors can be accessed (Figure 8-4-16-1, Detail A and Detail B).
- c. Support PDU. Tag and disconnect electrical connectors from unit as follows:

- d. Install protective caps and plugs on electrical connectors.
- e. Remove PDU.

8-4-16.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove protective caps and plugs from electrical connectors.

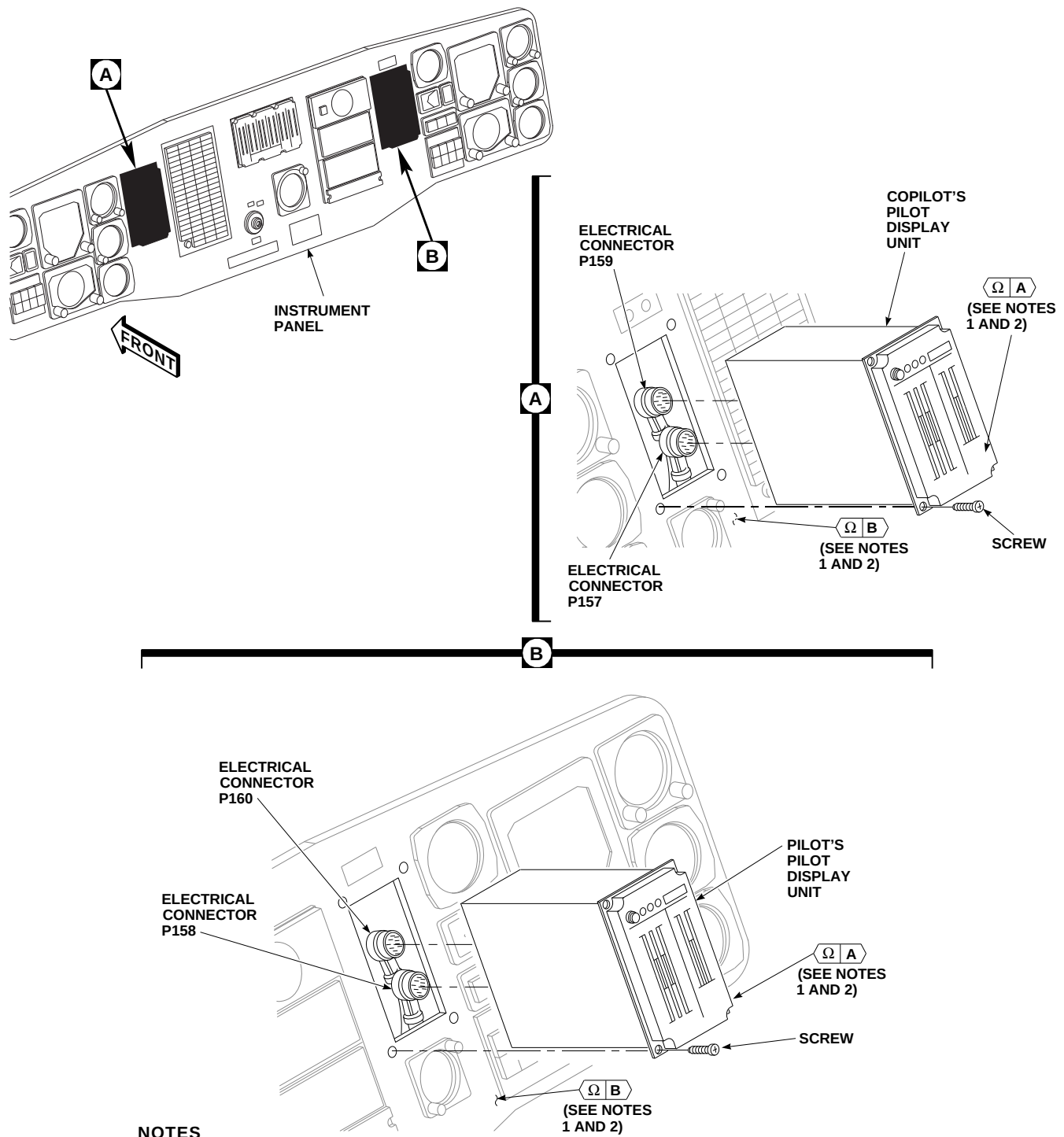


FIGURE 8-4-16-1. REPLACE PILOT DISPLAY UNIT

8-4-16-2 Change 4

- c. Inspect and treat electrical connectors (PARA 1-7-20).
- d. Support PDU. Remove tags and connect electrical connectors as follows (Figure 8-4-16-1, Detail A and Detail B):

Copilot's Display Unit Connectors	Pilot's Display Unit Connectors
P157 to J1	P158 to J1
P159 to J2	P160 to J2

- e. **EMEP** Inspect chemical conversion coated surfaces as follows:
 - (1) Clean area around mounting surface with machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D. Dry surface with machinery towel, Item 211, Appendix D.
 - (2) Inspect chemical conversion coated surfaces for breaks and scratches. **NONE ALLOWED.** If coated surfaces have

breaks or scratches apply chemical conversion coating (PARA 2-6-5).

- f. Make sure area is clean and free of foreign material.
- g. Carefully place PDU in instrument panel and fasten with screws.
- h. **EMEP** Perform EME resistance check as follows:

NOTE

Resistance measurements must be taken at least three times.

- (1) Using digital low-resistance ohmmeter, measure resistance from instrument panel to PDU. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**
- (2) If resistance is greater than 2.5 milliohms, remove PDU and apply chemical conversion coating (PARA 2-6-5).
- i. Perform operational check of instrument display system (PARA 8-2-3).

8-4-17 PILOT DISPLAY UNIT FACEPLATE AND EMI SHIELD.

PARAGRAPH BREAKDOWN

	PAGE
8-4-17.1 Replace Pilot Display Unit Faceplate and EMI Shield	8-4-17-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 8-2-3

8-4-17.1 Replace Pilot Display Unit Faceplate and EMI Shield.



NOTE

Replacement of pilot’s and copilot’s pilot display unit (PDU) faceplate and EMI shield is the same, except as noted.

Damage to faceplate will result if screws are overtorqued. Make sure screws are handtightened only until snug.

8-4-17.1.1. Remove.

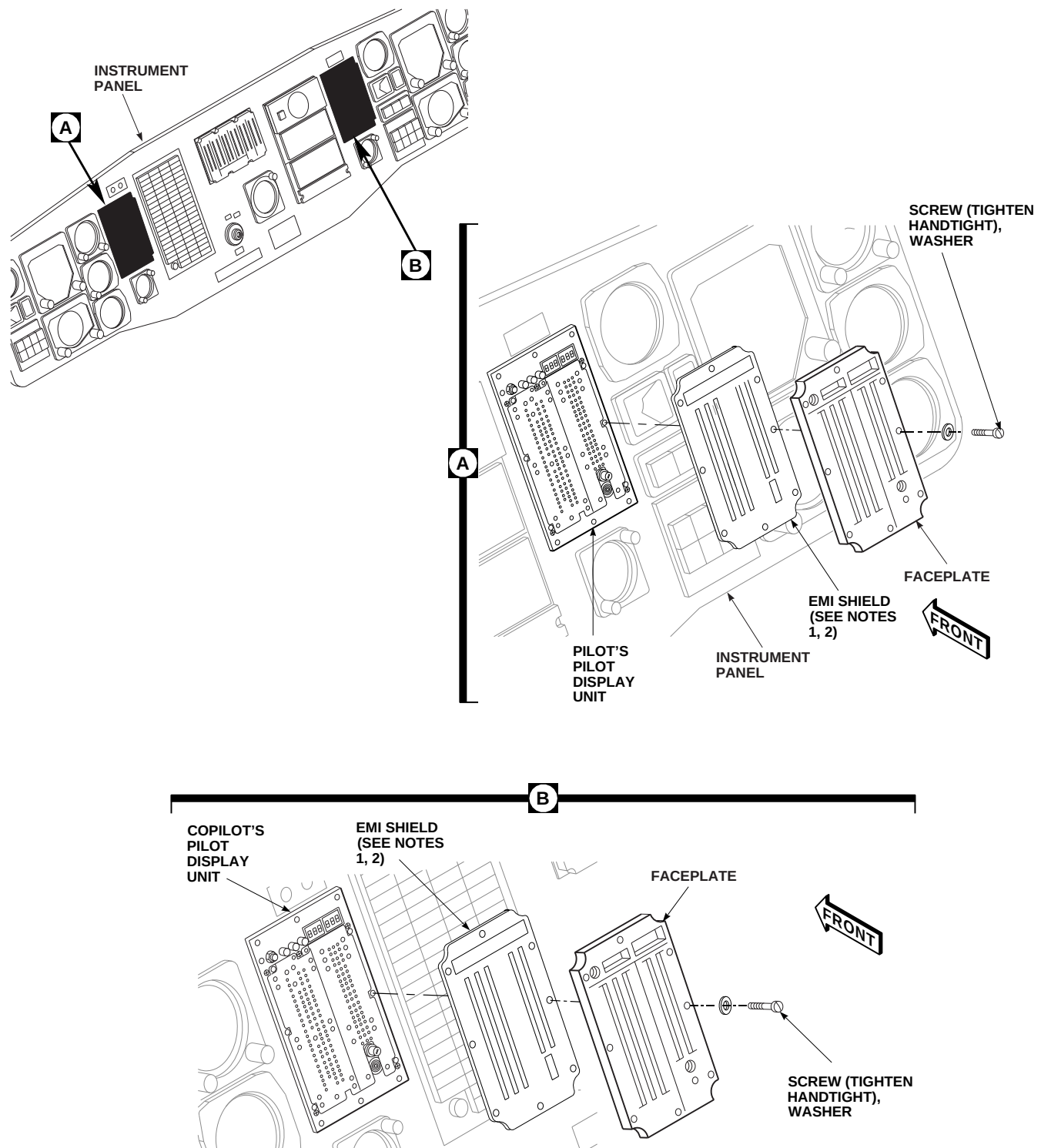
- Turn off all electrical power (PARA 1-6-16).
- Remove screws, washers, and faceplate from PDU (Figure 8-4-17-1, Detail A and Detail B).
- On PDU, 70450-01043-124 and 70450-01916-104, remove EMI shield from PDU.

- Install faceplate to PDU with screws and washers, making sure lighting jack at rear of faceplate mates properly with PDU. **TIGHTEN SCREWS HANDTIGHT.**
- Make sure area is clean and free of foreign material.
- Turn on electrical power (PARA 1-6-16).
- Make sure these circuit breakers are in: LIGHTS PLT FLT and LIGHTS CPLT FLT.

8-4-17.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- On PDU, 70450-01043-124 and 70450-01916-104, install EMI shield on PDU (Figure 8-4-17-1, Detail A and Detail B).

- Turn upper console INSTR LT PLT FLT control to BRT. Pilot’s PDU faceplate lights shall go on bright. If result is not as specified, perform operational check (PARA 8-2-3).
- Turn upper console CPLT FLT INST LTS control to BRT. Copilot’s PDU faceplate lights



NOTES

1. **EMEP**
2. ON PDU 70450-01043-124 AND 70450-01916-104.

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FIGURE 8-4-17-1. REPLACE PILOT DISPLAY UNIT FACEPLATE AND EMI SHIELD

8-4-17-2

shall go on bright. If result is not as specified, perform operational check (PARA 8-2-3).

- i. Turn off all electrical power (PARA 1-6-16).

8-4-18 PILOT DISPLAY UNIT BAR GRAPH FILAMENT LAMPS AND DIRECT VIEW FILAMENT MODULES **EMEP** .

PARAGRAPH BREAKDOWN

		PAGE
8-4-18.1	Replace Pilot Display Unit Bar Graph Filament Lamps and Direct View Filament Modules	8-4-18-1

INITIAL SETUP

- PARA 8-2-3

Tools

- PARA 8-4-17

- Digital Low-Resistance Ohmmeter
- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 2-6-5

8-4-18.1 Replace Pilot Display Unit Bar Graph Filament Lamps and Direct View Filament Modules.

board is gained by removing BGFs on either 1 or 2 scale.

- Carefully remove BGF lamp/direct view filament module from pilot display unit (Figure 8-4-18-1, Detail B).

NOTE

Replacement of pilot’s and copilot’s pilot display unit bar graph filament lamps and direct view filament modules is the same.

8-4-18.1.2 Install.

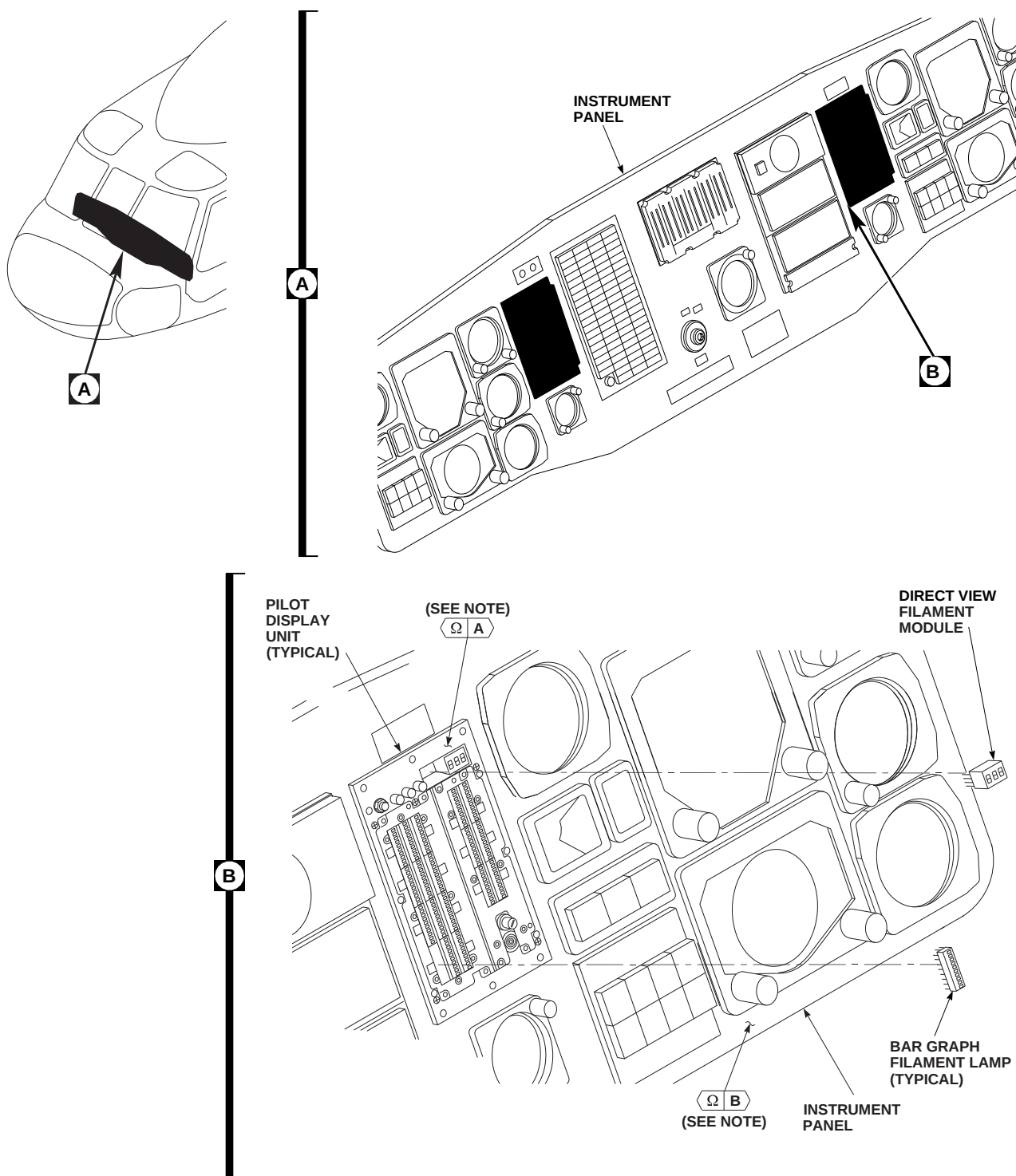
8-4-18.1.1 Remove.

- Remove pilot display unit faceplate and EMI shield (PARA 8-4-17).

- Turn off all electrical power (PARA 1-6-16).
- Position bar graph filament lamp/direct view filament module in required space and install (Figure 8-4-18-1, Detail B).
- Install pilot display unit faceplate and EMI shield (PARA 8-4-17).
- Perform EME resistance check of pilot display unit (PDU) as follows:

NOTE

Access to bar graph filament (BGF) lamps on R scale of RPM mounting



NOTE

POINTS FROM	TO	RESISTANCE SHALL BE
A	B	2.5 MILLIOHMS OR LESS

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FIGURE 8-4-18-1. REPLACE PILOT DISPLAY UNIT BAR GRAPH FILAMENT LAMPS AND DIRECT VIEW FILAMENT MODULES

NOTE

Resistance measurements must be taken at least three times.

- (1) Using digital low-resistance ohmmeter, measure resistance between PDU and instrument panel. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**

- (2) If resistance is greater than 2.5 milliohms, remove PDU and apply chemical conversion coating (PARA 2-6-5).

- e. Make sure area is clean and free of foreign material.
- f. Perform operational check of instrument display system (PARA 8-2-3).

8-4-19 CAUTION/ADVISORY PANEL.

PARAGRAPH BREAKDOWN

	PAGE
8-4-19.1 Replace Caution/Advisory Panel	8-4-19-1

INITIAL SETUP

Tools

- Digital Low-Resistance Ohmmeter
- Electrical Repairer’s Toolkit

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 8-2-4

8-4-19.1 Replace Caution/Advisory Panel.

NOTE

EMEP Pin filter adapters are considered part of harness assembly. Do not remove pin filter adapter from electrical connectors, P117 and P118 (Figure 8-4-19-1, Detail B).

8-4-19.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Unfasten and slowly lift caution/advisory panel from instrument panel to gain access to electrical connectors (Figure 8-4-19-1, Detail A).
- Tag and disconnect electrical connectors, P117 and P118, from caution/advisory panel and remove caution/advisory panel.

- Install protective caps and plugs on electrical connectors.

8-4-19.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- EMEP** Inspect chemical conversion coated surfaces as follows:
 - Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean area around mounting surface. Using machinery towel, Item 211, Appendix D, dry surface.
 - Inspect chemical conversion coated surfaces for breaks and scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches, apply chemical conversion coating (PARA 2-6-5).

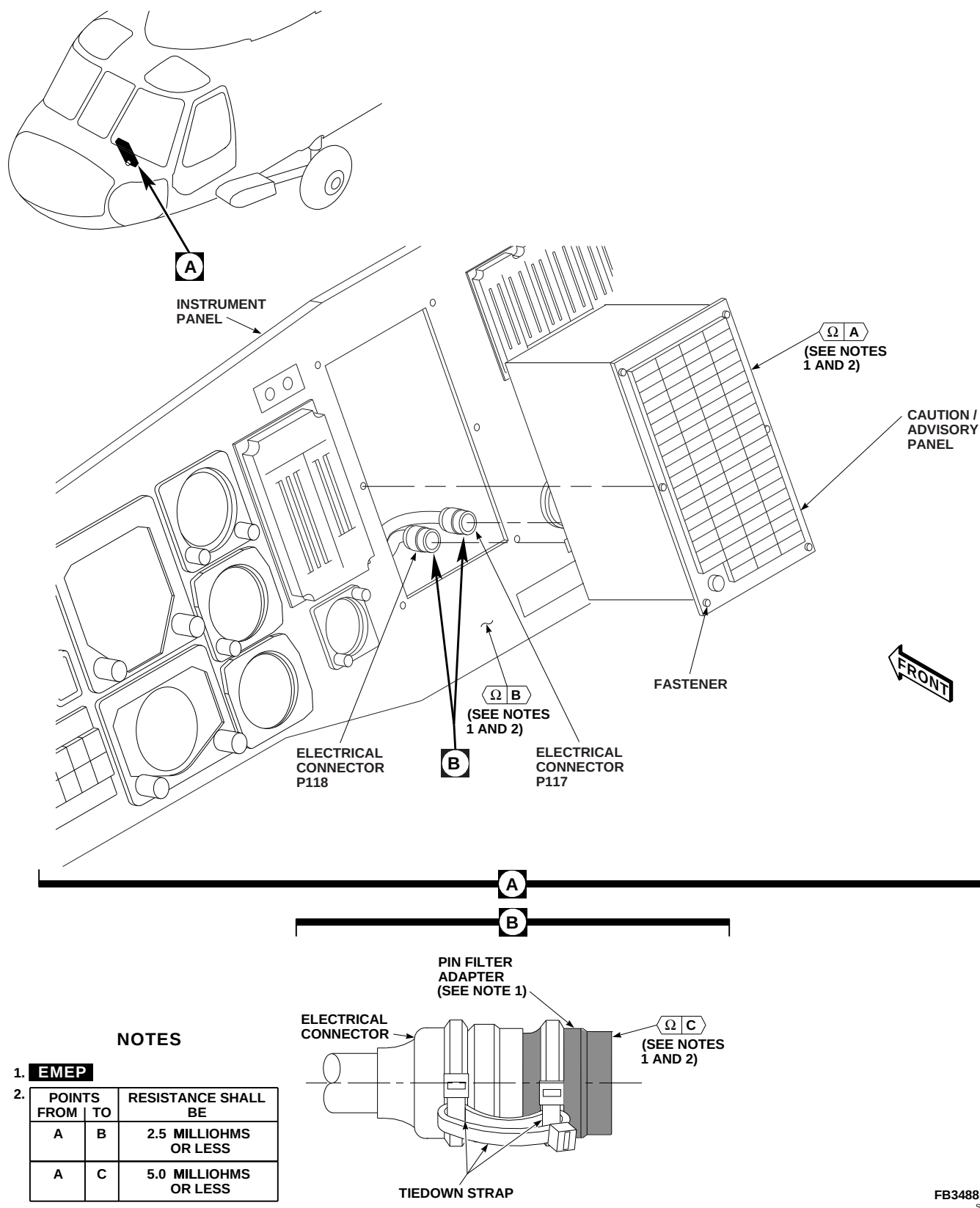


FIGURE 8-4-19-1. REPLACE CAUTION/ADVISORY PANEL

- c. Remove protective caps and plugs from electrical connectors.
- d. Inspect and treat electrical connectors (PARA 1-7-20).
- e. Support caution/advisory panel and connect electrical connectors, P117 to J117, and P118 to J118, to caution/advisory panel (Figure 8-4-19-1, Detail A). Remove tags.
- f. **EMEP** Perform EME resistance check of pin filter adapter as follows (Figure 8-4-19-1, Detail A and Detail B):

NOTE

Resistance measurements must be taken at least three times.

- (1) Using digital low-resistance ohmmeter, measure resistance between shell of each pin filter adapter and caution/advisory panel. **RESISTANCE SHALL BE 5.0 MILLIOHMS OR LESS.**
- (2) If resistance is greater than 5.0 milliohms, measure resistance between shell of pin filter adapter and connector on caution/advisory panel. **RESISTANCE SHALL BE 5.0 MILLIOHMS OR LESS.**

- (3) If resistance is still beyond limit, clean surface between pin filter adapter and generator control unit.
- g. Make sure area is clean and free of foreign material.
- h. Carefully place caution/advisory panel in instrument panel and tighten fasteners (Figure 8-4-19-1, Detail A).
- i. **EMEP** Perform EME resistance check of caution/advisory panel as follows:

NOTE

Resistance measurements must be taken at least three times.

- (1) Using digital low-resistance ohmmeter, measure resistance between caution/advisory panel and instrument panel. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**
- (2) If resistance is greater than 2.5 milliohms, remove caution/advisory panel and apply chemical conversion coating (PARA 2-6-5).
- j. Perform operational check of caution/advisory warning system (PARA 8-2-4).

8-4-20 CAUTION/ADVISORY PANEL LAMPS.

PARAGRAPH BREAKDOWN

	PAGE
8-4-20.1 Replace Caution/Advisory Panel Lamps	8-4-20-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 8-2-4

8-4-20.1 Replace Caution/Advisory Panel Lamps.

NOTE

Replacement of all caution/advisory panel lamps is the same.

8-4-20.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Push on right side of capsule to turn capsule from caution/advisory panel and expose lamps (Figure 8-4-20-1, Detail A).
- Remove lamp(s) from capsule.

8-4-20.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).

- Install lamp(s) in capsule (Figure 8-4-20-1, Detail A).
- Push capsule back into caution/advisory panel.
- Turn on electrical power (PARA 1-6-16).
- Make sure LIGHTS CAUT ADVSY, and LIGHTS ADVSY circuit breakers are in.
- Place caution/advisory panel BRT/DIM-TEST switch to TEST and check that capsule lights evenly.
- Perform operational check of caution/advisory system (PARA 8-2-4).
- Turn off all electrical power (PARA 1-6-16).

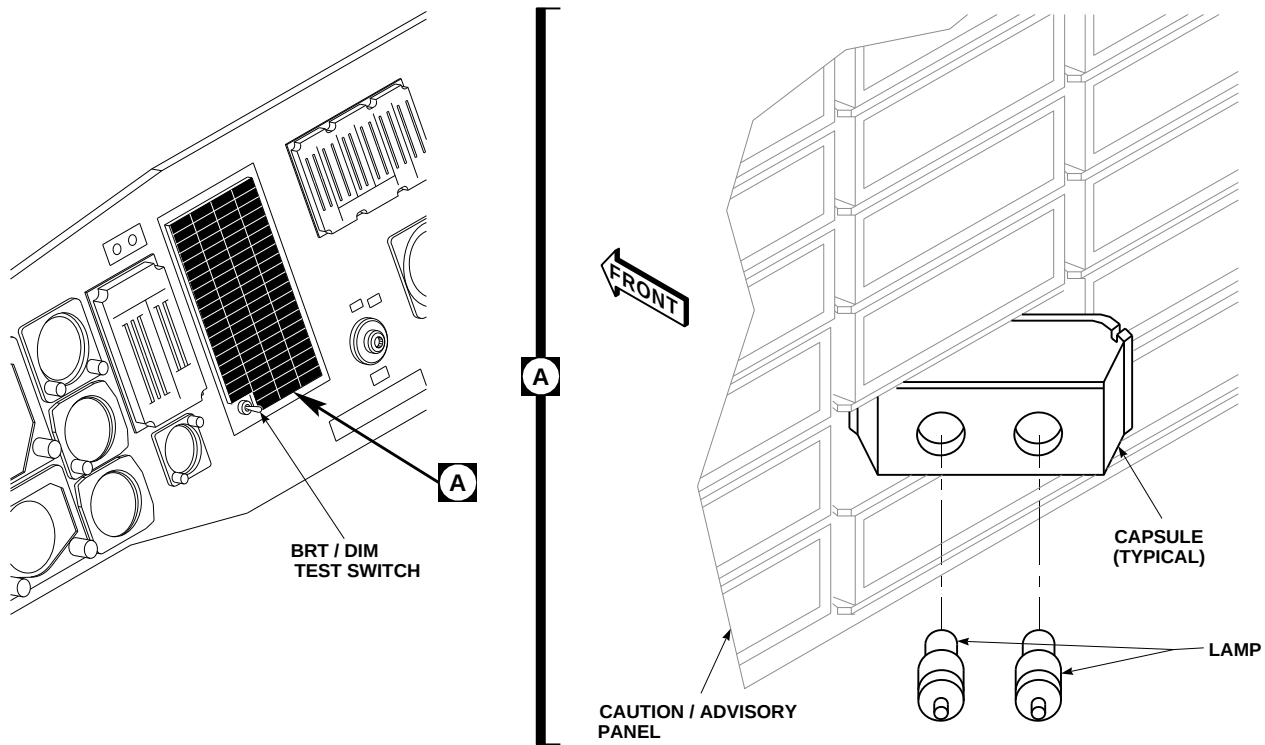
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FIGURE 8-4-20-1. REPLACE CAUTION/ADVISORY PANEL LAMPS

8-4-21 CAUTION/ADVISORY PANEL CAPSULE PARTS.

PARAGRAPH BREAKDOWN

	PAGE
8-4-21.1 Replace Caution/Advisory Panel Capsule Parts	8-4-21-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-6-16

8-4-21.1 Replace Caution/Advisory Panel Capsule Parts.

NOTE

Replacement of all caution/advisory panel capsule parts is the same.

8-4-21.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Push on right side of capsule to turn capsule from caution/advisory panel and expose end of lens retainer (Figure 8-4-21-1, Detail A).
- c. Lift exposed end of lens retainer away from capsule, then remove lens retainer, lenses, and legend plate.

8-4-21.1.2 Inspect.

Inspect lens retainer for damage. **NONE ALLOWED.** If lens retainer is damaged, replace lens retainer.

8-4-21.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Position lenses and legend plate on capsule (Figure 8-4-21-1, Detail A).
- c. Snap lens retainer on capsule.
- d. Push capsule into caution/advisory panel.
- e. Turn on electrical power (PARA 1-6-16).
- f. Place caution/advisory panel BRT/DIM-TEST switch to TEST and make sure capsule lights evenly.
- g. Turn off all electrical power (PARA 1-6-16).

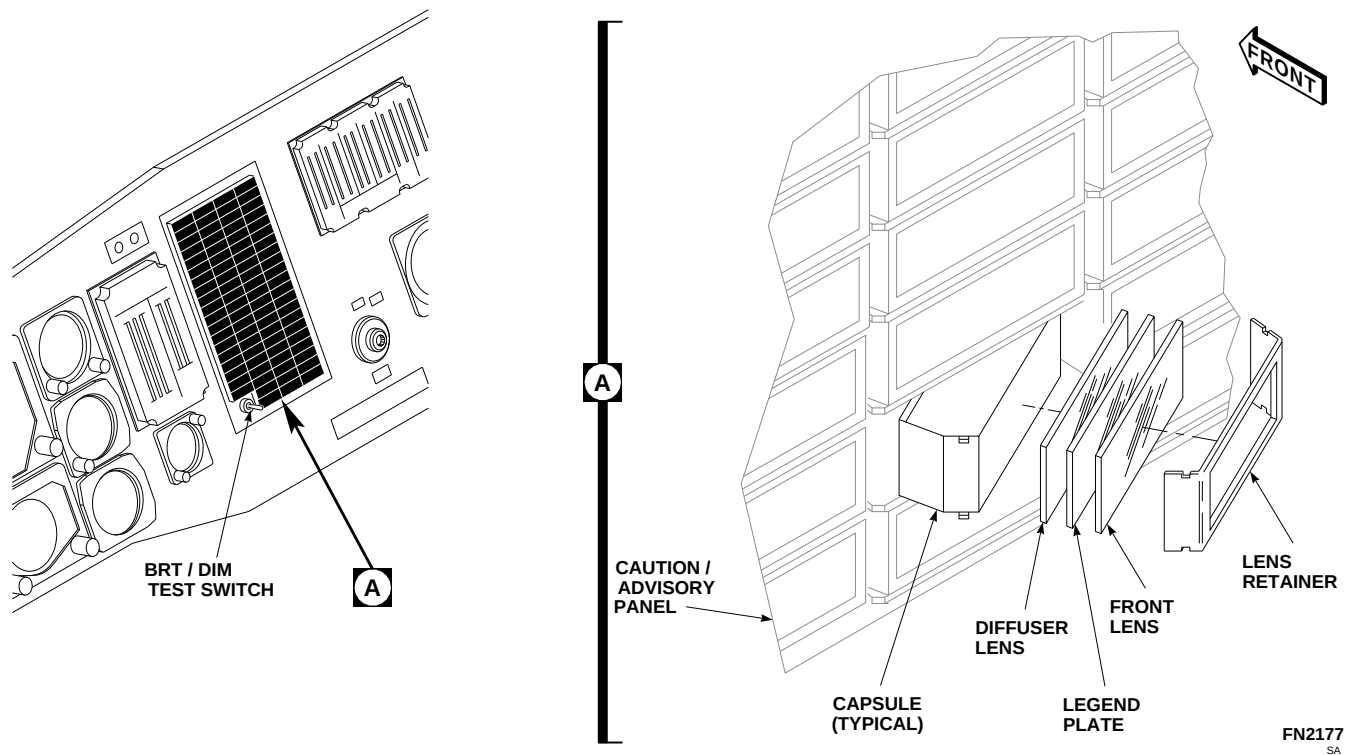


FIGURE 8-4-21-1. REPLACE CAUTION/ADVISORY PANEL CAPSULE PARTS

8-4-22 CAUTION/ADVISORY SYSTEM AND MASTER WARNING DIMMING CONTROLS.

PARAGRAPH BREAKDOWN

	PAGE
8-4-22.1 Replace Caution/Advisory System and Master Warning Dimming Controls	8-4-22-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 8-2-4
- PARA 8-4-16

8-4-22.1 Replace Caution/Advisory System and Master Warning Dimming Controls.

instrument panel, through copilot’s pilot display unit mounting hole.

NOTE

Replacement of master warning and caution/advisory dimming control is the same, except as noted.



8-4-22.1.1 Remove.

- Remove copilot’s pilot display unit (PARA 8-4-16).
- Loosen setscrews and remove knob from dimming control (Figure 8-4-22-1, Detail A).
- Remove nut and washer from dimming control.
- Push dimming control shaft through mounting hole and pull dimming control out from rear of

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection, while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires (Figure 8-4-22-1, Detail B). Using soldering iron, unsolder wires from dimming control.

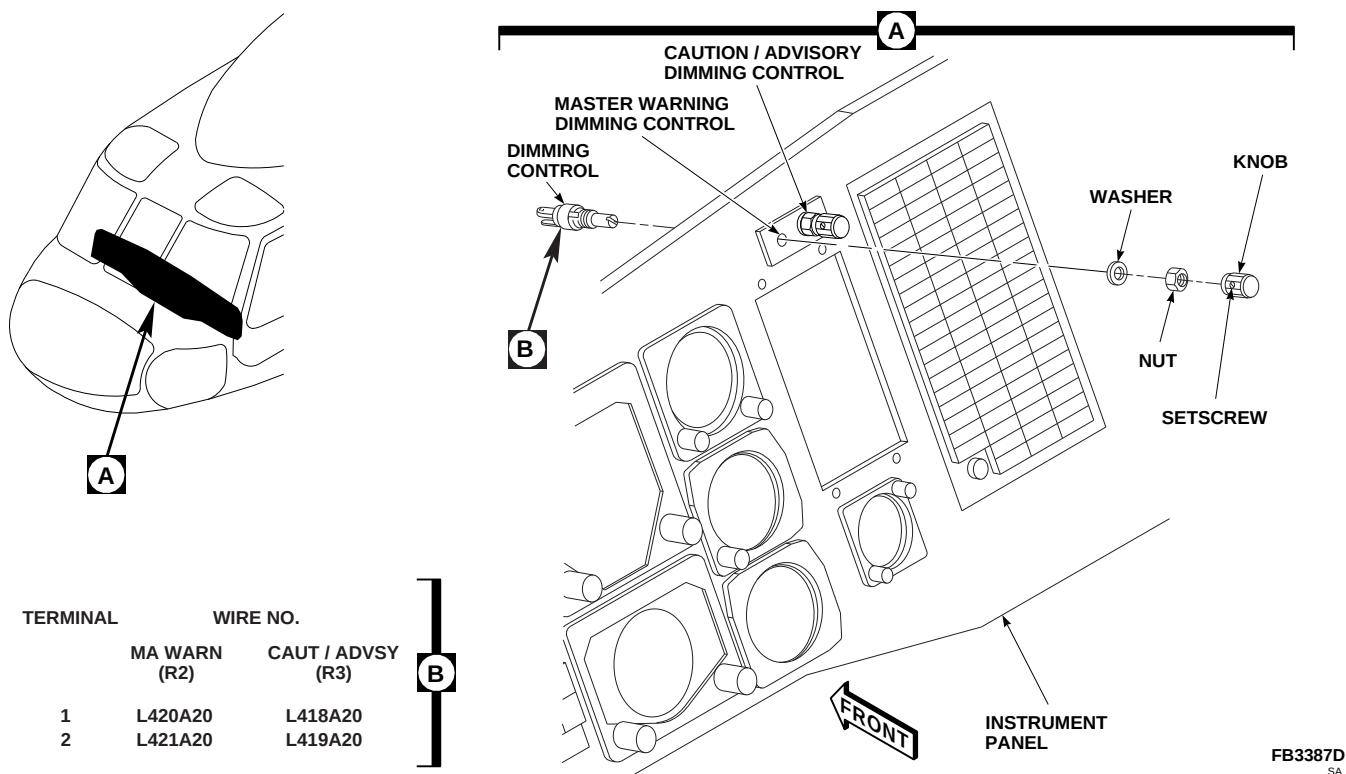


FIGURE 8-4-22-1. REPLACE CAUTION/ADVISORY SYSTEM AND MASTER WARNING DIMMING CONTROLS

8-4-22.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection, while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- b. Using soldering iron and solder, Item 313, Appendix D, solder wires to dimming control terminals (Figure 8-4-22-1, Detail A and Detail B). Remove tags from wires.
- c. Position dimming control shaft and tab in mounting holes and secure with nut and washer (Figure 8-4-22-1, Detail A).
- d. Position knob on shaft and tighten setscrew.
- e. Install copilot's pilot display unit (PARA 8-4-16).
- f. Perform operational check of caution/advisory warning system (PARA 8-2-4).

8-4-23 CAUTION/ADVISORY AUDIBLE WARNING UNIT.

PARAGRAPH BREAKDOWN

	PAGE
8-4-23.1 Replace Caution/Advisory Audible Warning Unit	8-4-23-1

INITIAL SETUP

Tools

- Digital Low-Resistance Ohmmeter
- Electrical Repairer’s Toolkit

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D

- Protective Cap and Plug

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 8-2-4

8-4-23.1 Replace Caution/Advisory Audible Warning Unit.

8-4-23.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open nose door.
- Disconnect electrical connector, P961, from caution/advisory audible warning unit (Figure 8-4-23-1, Detail A).
- Install protective cap and plug on electrical connectors.
- Remove screws, washers, and audible warning unit from canted bulkhead.

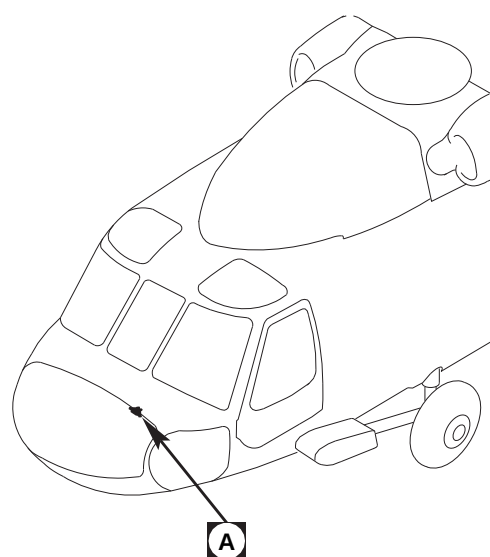
8-4-23.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- EMEP** Inspect chemical conversion coating as follows:

- Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean area around mounting surface. Using clean, dry machinery towel, Item 211, Appendix D, dry surface.

- Inspect chemical conversion-coated surfaces for breaks and scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches, apply chemical conversion coating (PARA 2-6-5).

- Position caution/advisory audible warning unit on canted bulkhead and secure with screws and washers (Figure 8-4-23-1, Detail A).
- Remove protective cap and plug from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).



NOTES

1. **EMEP**

POINTS FROM TO		RESISTANCE SHALL BE
A	B	2.5 MILLIOHMS OR LESS

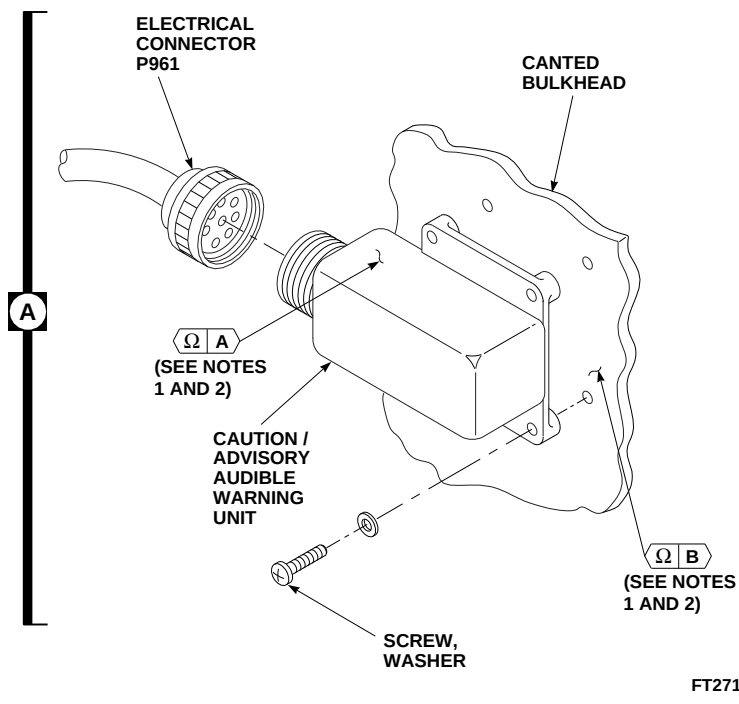
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FIGURE 8-4-23-1. REPLACE CAUTION/ADVISORY AUDIBLE WARNING UNIT

f. Connect electrical connector, P961, to caution/advisory audible warning unit.

chemical conversion coating (PARA 2-6-5).

g. **EMEP** Perform EME resistance check as follows:

h. Make sure area is clean and free of foreign material.

(1) Using digital low-resistance ohmmeter, measure resistance from canted bulkhead to audible warning unit. **RESISTANCE MUST BE 2.5 MILLIOHMS OR LESS.**

i. Perform operational check of caution/advisory system (PARA 8-2-4).

(2) If resistance is greater than 2.5 milliohms, remove audible warning unit and apply

j. Close nose door.

8-4-24 MASTER WARNING PANEL.

PARAGRAPH BREAKDOWN

		PAGE
8-4-24.1	Replace Master Warning Panel	8-4-24-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 8-2-4
- PARA 12-2-1

8-4-24.1 Replace Master Warning Panel.

NOTE

Replacement of pilot’s and copilot’s master warning panel is the same, except as noted.

8-4-24.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove screws and washers securing master warning panel to instrument panel glare shield. Lower master warning panel slowly until electrical connector can be accessed (Figure 8-4-24-1, Detail A).
- Support master warning panel and disconnect electrical connector, P130 (pilot’s) or P131 (copilot’s).
- Install protective caps and plugs on electrical connectors.

8-4-24.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).

- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Support master warning panel and connect electrical connector, P130 (pilot’s) or P131 (copilot’s), to master warning panel (Figure 8-4-24-1, Detail A).

NOTE

Short screws are installed on outboard sides of master warning panels.

- Carefully position master warning panel under instrument panel glare shield and fasten with screws and washers. **TORQUE SCREWS TO 2 INCH-POUNDS.**
- Make sure area is clean and free of foreign material.
- Perform operational check of caution/advisory warning system and fire detection system (PARA 8-2-4 and PARA 12-2-1).

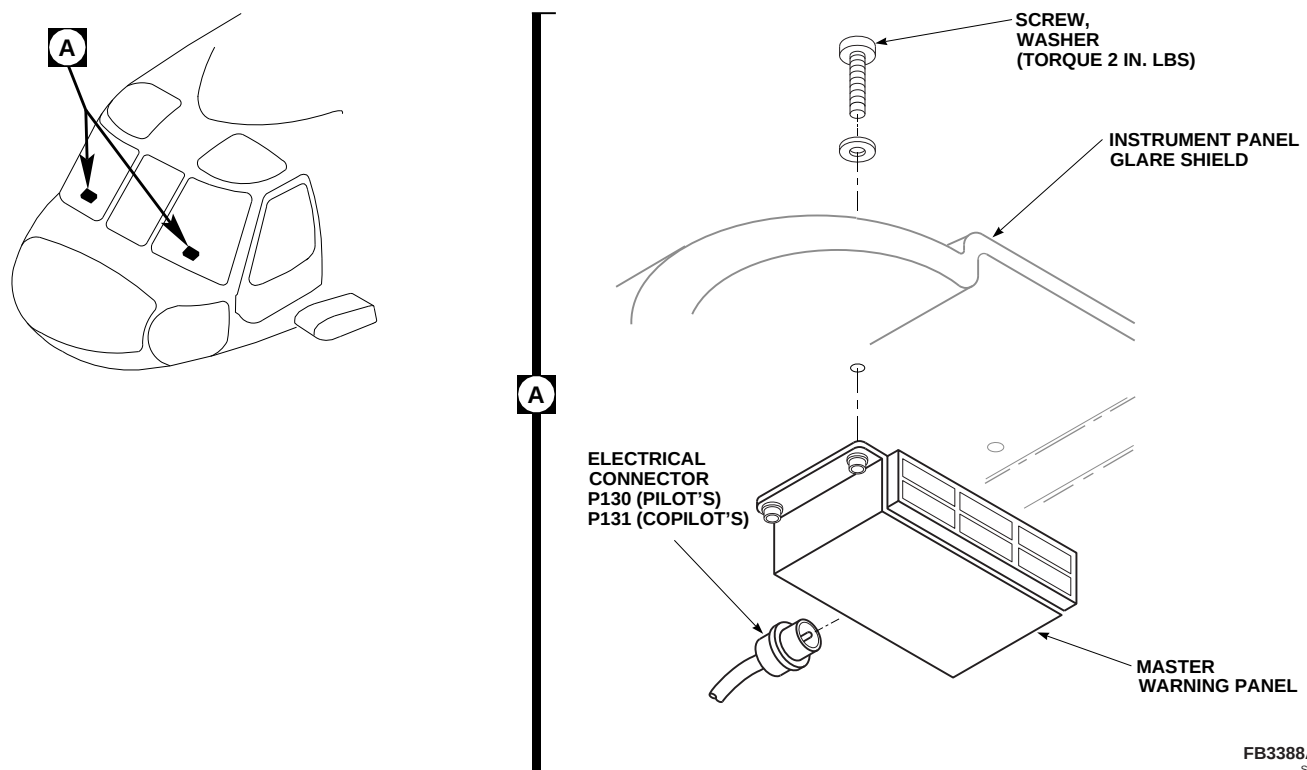


FIGURE 8-4-24-1. REPLACE MASTER WARNING PANEL

8-4-25 MASTER WARNING PANEL LAMPS.

PARAGRAPH BREAKDOWN

		PAGE
8-4-25.1	Replace Master Warning Panel Lamps	8-4-25-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 8-2-3

8-4-25.1 Replace Master Warning Panel Lamps.

NOTE

Replacement of all master warning panel lamps is the same.

8-4-25.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Push on right side of capsule to turn capsule on master warning panel and expose lamps (Figure 8-4-25-1, Detail A).

- Remove lamp(s) from capsule.

8-4-25.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install lamp(s) in capsule (Figure 8-4-25-1, Detail A).
- Push capsule back into master warning panel.
- Perform operational check of instrument display system (PARA 8-2-3).

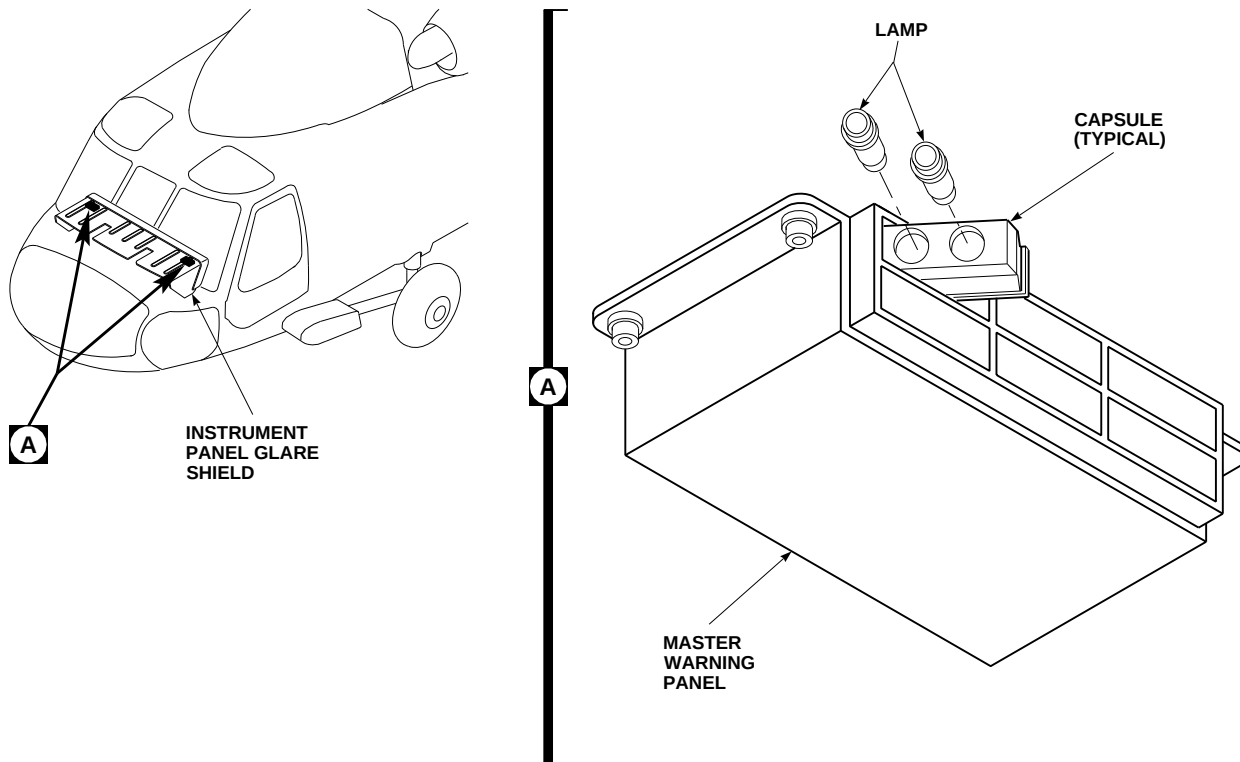
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FIGURE 8-4-25-1. REPLACE MASTER WARNING PANEL LAMPS

8-4-26 MASTER WARNING PANEL CAPSULE.

PARAGRAPH BREAKDOWN

		PAGE
8-4-26.1	Replace Master Warning Panel Capsule Parts	8-4-26-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 8-2-3

8-4-26.1 Replace Master Warning Panel Capsule Parts.

NOTE

Replacement of pilot’s and copilot’s master warning panel capsule parts is the same.

8-4-26.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Push on right side of capsule to turn capsule from master warning panel and expose end of lens retainer (Figure 8-4-26-1, Detail A).
- Lift exposed end of lens retainer away from capsule and remove lens retainer, lenses, and legend plate.

8-4-26.1.2 Inspect.

Inspect lens retainer for damage. **NONE ALLOWED.** If damaged, replace lens retainer.

8-4-26.1.3 Install.

- Turn off all electrical power (PARA 1-6-16).
- Position lenses and legend plate on capsule (Figure 8-4-26-1, Detail A).
- Snap lens retainer on capsule.
- Push capsule into master warning panel.
- Turn on electrical power (PARA 1-6-16).
- Place caution/advisory panel BRT/DIM-TEST switch to TEST and make sure capsule lights evenly.
- Turn off all electrical power (PARA 1-6-16).
- Perform operational check of instrument display system (PARA 8-2-3).

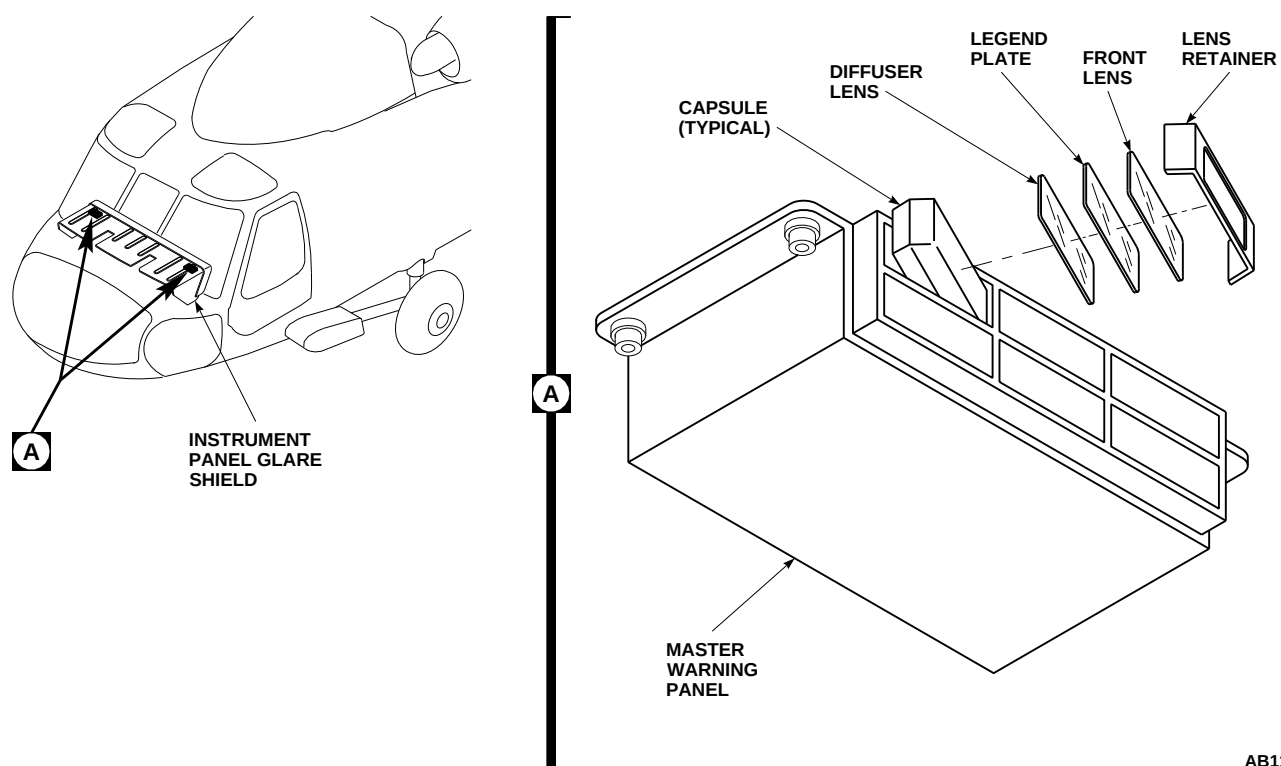
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FIGURE 8-4-26-1. REPLACE MASTER WARNING PANEL CAPSULE PARTS

8-4-27 MASTER WARNING DIMMING CONTROL.

PARAGRAPH BREAKDOWN

		PAGE
8-4-27.1	Replace Master Warning Dimming Control	8-4-27-1

INITIAL SETUP

References

- PARA 9-4-49

8-4-27.1 Replace Master Warning Dimming Control.

Replace master warning dimming control (PARA 9-4-49).

8-4-28 CLOCK.

PARAGRAPH BREAKDOWN

	PAGE
8-4-28.1 Replace Clock	8-4-28-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

Materials/Parts

- Gasket
- Protective Cap and Plug

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 8-2-5

8-4-28.1 Replace Clock.

NOTE

Replacement of pilot’s and copilot’s clock is the same, except as noted.

8-4-28.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- On clock, MC7545, remove as follows (Figure 8-4-28-1, Detail A):

- SLS** Remove screws from lighted bezel assembly.■

CAUTION

SLS Damage to lighted bezel assembly will result if not careful when removing from instrument panel. Make sure to use care when removing lighted bezel assembly from the instrument panel.■

- SLS** Slowly lift lighted bezel assembly and clock from instrument panel to gain access to electrical connector. Remove electrical connectors, P186 (pilot’s) or P195 (copilot’s), from lighted bezel assembly.■
- W/O SLS** Remove screws securing clock to instrument panel.■
- W/O SLS** Slowly lift clock from instrument panel to gain access to electrical connector.■
- W/O SLS** Disconnect electrical connector, P4 (pilot’s) or P6 (copilot’s), from clock.■
- W/O SLS** Remove clock from instrument panel.■

- On clock, PD89MME-637-3 remove as follows (Figure 8-4-28-1, Detail A):
 - Disconnect electrical connector, P186 (pilot’s) or P195 (copilot’s), from clock.
 - Remove screws, gasket, and clock from instrument panel.

- d. Install protective caps and plugs on electrical connectors.

8-4-28.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove protective caps and plugs from electrical connectors.
- c. Inspect and treat electrical connectors (PARA 1-7-20).
- d. On clock, MC7545, install as follows (Figure 8-4-28-1, Detail A):
 - (1) **SLS** Connect electrical connector, P186, (pilot's) or P195 (copilot's), to lighted bezel assembly.■
 - (2) **SLS** Install clock and lighted bezel assembly into instrument panel with screws.■
 - (3) **W/O SLS** Connect electrical connector, P4 (pilot's) or P6 (copilot's), to clock.■

- (4) **W/O SLS** Install clock into instrument panel with screws.■

- e. On clock, PD89MME-637-3, install as follows (Figure 8-4-28-1, Detail A):

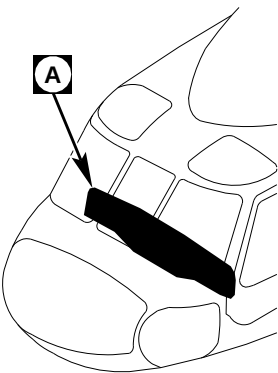
- (1) Install gasket onto back of clock.
- (2) Install clock into instrument panel and secure with screws.
- (3) Connect electrical connector, P186, (pilot's) or P195 (copilot's), to clock.

- f. Make sure area is clean and free of foreign material.

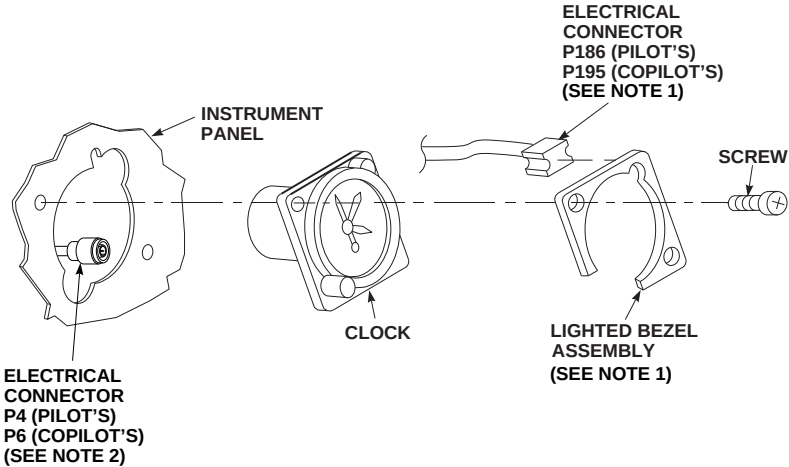
NOTE

Second clock may be used to obtain time setting.

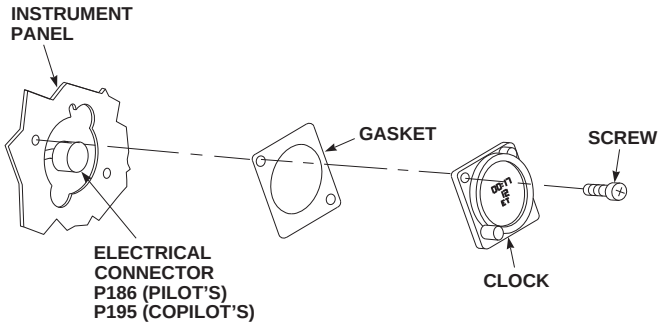
- g. Set proper time.
- h. On clock, PD89MME-637-3, perform operational check (PARA 8-2-5).



A



EFFECTIVITY
MC7545



EFFECTIVITY
PD89MME-637-3

NOTES

- 1. SLS
- 2. W/O SLS

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FIGURE 8-4-28-1. REPLACE CLOCK

8-4-29 **CLOCK BATTERY.**

PARAGRAPH BREAKDOWN

	PAGE
8-4-29.1 Replace Clock Battery	8-4-29-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

References

- PARA 8-4-28

8-4-29.1 **Replace Clock Battery.**

NOTE

This procedure applies only to helicopters with clock, PD89MME-637-3.

8-4-29.1.1 **Remove.**

NOTE

Note position of internal wiring harness. This will aid in installation.

- Remove clock (PARA 8-4-28).
- Remove battery board from clock (Figure 8-4-29-1).

CAUTION

Damage to battery board and bezel will result if excessive force is used when removing. Do not use excessive force.

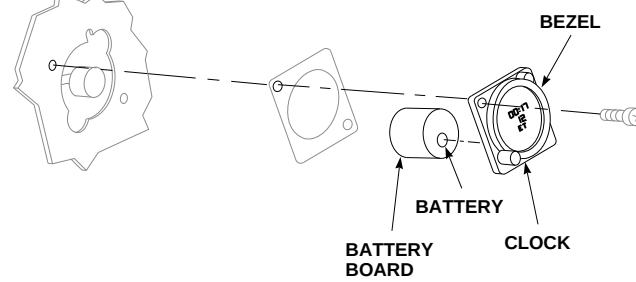
- Remove battery from battery board.

8-4-29.1.2 **Install.**

CAUTION

Damage to battery board and bezel will result if excessive force is used. Do not use excessive force.

- Install positive end of battery to battery board (Figure 8-4-29-1).
- Install battery board in clock.
- Install clock (PARA 8-4-28).

**EFFECTIVITY****PD89MME-637-3**FB3391B
SA**FIGURE 8-4-29-1. REPLACE CLOCK BATTERY**

8-4-30 FREE-AIR THERMOMETER.

PARAGRAPH BREAKDOWN

		PAGE
8-4-30.1	Replace Free-Air Thermo- meter	8-4-30-1
8-4-30.2	Replace Free-Air Thermo- meter Bonding Jumper	8-4-30-3
	HCW	

INITIAL SETUP

Tools

- Container, 1-Quart
- Drill
- Drill Bit, 1⁄8-inch diameter
- Electrical Repairer’s Toolkit
- Mercury Centigrade Thermometer
- Rivet Gun
- Torque Wrench, 0 - 30 in. lbs

- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Cleaning Compound, Solvent, Item 100A, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D
- Tiedown Strap, Item 337, Appendix D

References

- Appendix D

8-4-30.1 Replace Free-Air Thermometer.

NOTE

- **W/O HCW** One free-air thermometer (FAT) is mounted on the center windshield.◀
- **HCW** One free-air thermometer (FAT) is mounted on each upper window.◀

8-4-30.1.1 Remove.

- W/O HCW** Remove free-air thermometer (FAT) as follows (Figure 8-4-30-1, Detail B):◀
 - Remove screw, washers, and nut securing ends of bonding jumpers to canopy.
 - Unscrew and remove sunshield. Remove dishd washer and case washer.

NOTE

Keep reinforcement and flat washer for installation of new FAT.

- (3) Remove FAT, flat washer, bonding jumper, case washer, and reinforcement from center windshield.

- b. **HCW** Remove FAT as follows (Figure 8-4-30-1, Detail A):

- (1) Unscrew and remove sunshield. Remove dished washer and case washer.

NOTE

Keep reinforcement and flat washer for installation of new FAT.

- (2) Remove FAT, flat washer, case washer, and reinforcement from upper window.
- (3) Inspect bonding jumper for damage. Replace if damaged (PARA 8-4-30.2).

- c. **W/O HCW** Inspect center windshield bonding jumper for damage. If damaged, replace as follows (Figure 8-4-30-1, Detail B):

- (1) Remove screw, lockwasher, washers, bonding jumper, and nut at STA 223.0, WL 263.0.
- (2) Remove tiedown strap and bonding jumper from canopy.
- (3) Install replacement bonding jumper using screw, lockwasher, washers, and nut at STA 223.0, WL 263.0.
- (4) Tiedown bonding jumper on canopy using tiedown strap, Item 337, Appendix D.

8-4-30.1.2 Inspect.**NOTE**

Make sure sunshield is removed from free-air thermometer (FAT) prior to immersion into water.

- a. Perform ambient temperature test as follows:

- (1) Fill 1-quart container with fresh water. Allow water to stabilize at room temperature. If possible, allow water to set overnight.
- (2) Suspend FAT in water so stem is vertical and immersed 2 - 2 ½-inches into water. Simultaneously, suspend mercury centigrade thermometer into water at equal depth to FAT.
- (3) After 5 minutes, record temperature of mercury centigrade thermometer and FAT. Do not remove thermometer or FAT from water.
- (4) Allow additional 5 minutes to pass and record temperature of mercury centigrade thermometer and FAT. Repeat until temperatures stabilize.

- b. When temperatures stabilize, remove mercury centigrade thermometer and FAT from water.

- c. Perform cold temperature test as follows:

- (1) Fill 1-quart container with ice cubes or crushed ice and water.
- (2) Suspend FAT in ice water so stem is vertical and immersed 2 - 2 ½-inches into water. Simultaneously, suspend mercury centigrade thermometer into water at equal depth to FAT.
- (3) After 5 minutes, record temperature of mercury centigrade thermometer and FAT. Do not remove thermometer or FAT from ice water.
- (4) Allow additional 5 minutes to pass and record temperature of mercury centigrade thermometer and FAT. Repeat until temperatures stabilize.

- d. Compare stabilized readings. **FAT MUST BE WITHIN 2°C AT BOTH AMBIENT AND COLD TEST POINTS.** If temperature is not within limits, replace FAT.

- e. When temperatures stabilize, remove mercury centigrade thermometer and FAT from water.

8-4-30.1.3 Install.

- a. **W/O HCW** Install free-air thermometer (FAT) as follows (Figure 8-4-30-1, Detail B):
 - (1) Install flat washer, bonding jumper, case washer, and reinforcement on stem of FAT.
 - (2) Install FAT through hole in center windshield.
 - (3) Install case washer on stem of FAT on outside of center windshield.
 - (4) Coat threads inside sunshield and threads on FAT with corrosion preventive compound, Item 114, Appendix D.
 - (5) Install dished washer and sunshield. Do not tighten at this time.
 - (6) Position flat bonding jumper over top center screw hole of center windshield.
 - (7) Secure bonding jumper to center windshield with screw, washers, and nut. **TORQUE NUT TO 15 - 18 INCH-POUNDS.**
 - (8) Position FAT so words FREE AIR on dial are level.
 - (9) **TIGHTEN SUNSHIELD. TORQUE SUNSHIELD TO 38 - 42 INCH-POUNDS.**
- b. **HCW** Install FAT as follows (Figure 8-4-30-1, Detail A):
 - (1) Install flat washer on FAT.
 - (2) Position reinforcement and case washer between bonding jumper and upper window.
 - (3) Install FAT through hole in upper window.

- (4) Install case washer on stem of FAT on outside of upper window.
- (5) Coat threads inside sunshield and threads on FAT with corrosion preventive compound, Item 114, Appendix D.
- (6) Install dished washer and sunshield.
- (7) Position FAT so words FREE AIR on dial are level.
- (8) **TIGHTEN SUNSHIELD. TORQUE SUNSHIELD TO 38 - 42 INCH-POUNDS.**

8-4-30.2 Replace Free-Air Thermometer Bonding Jumper **HCW**.

NOTE

Replacement of left and right free-air thermometer (FAT) bonding jumper is the same.

8-4-30.2.1 Remove.

- a. Remove FAT (PARA 8-4-30.1.1).
- b. Carefully drill out rivets securing bonding jumper to canopy structure (Figure 8-4-30-2, Detail A).
- c. Remove bonding jumper and clean up rivet debris.

8-4-30.2.2 Inspect.

- a. Inspect bonding jumper for cracks and tears. **NONE ALLOWED.** If cracks or tears are found, replace jumper.
- b. Inspect area of bonding jumper between attachment rivets for buckling. **NONE ALLOWED.** If buckling is found, replace jumper.

8-4-30.2.3 Install.

- a. Position bonding jumper on canopy structure. Make sure jumper and structure are properly aligned (Figure 8-4-30-2, Detail A).

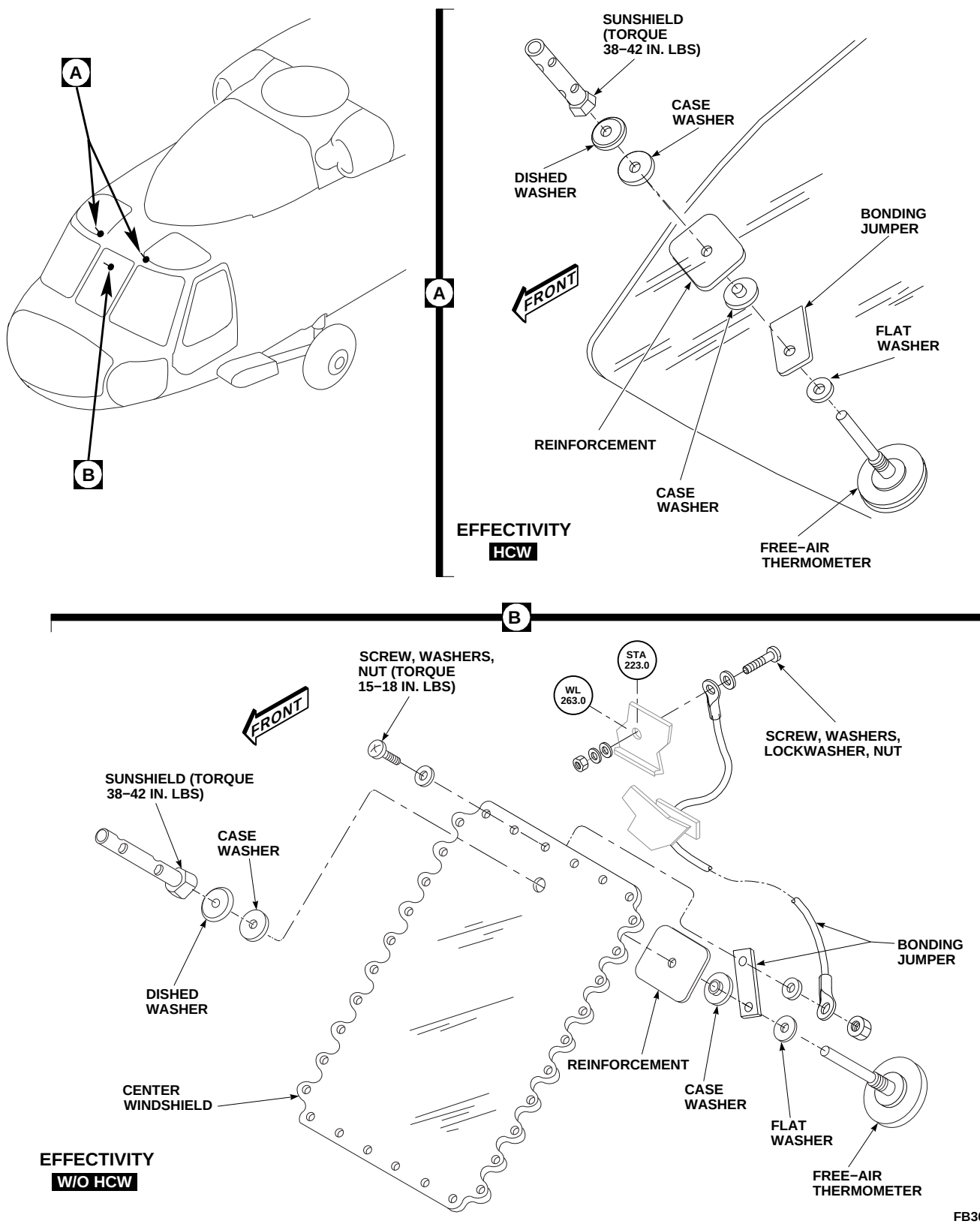
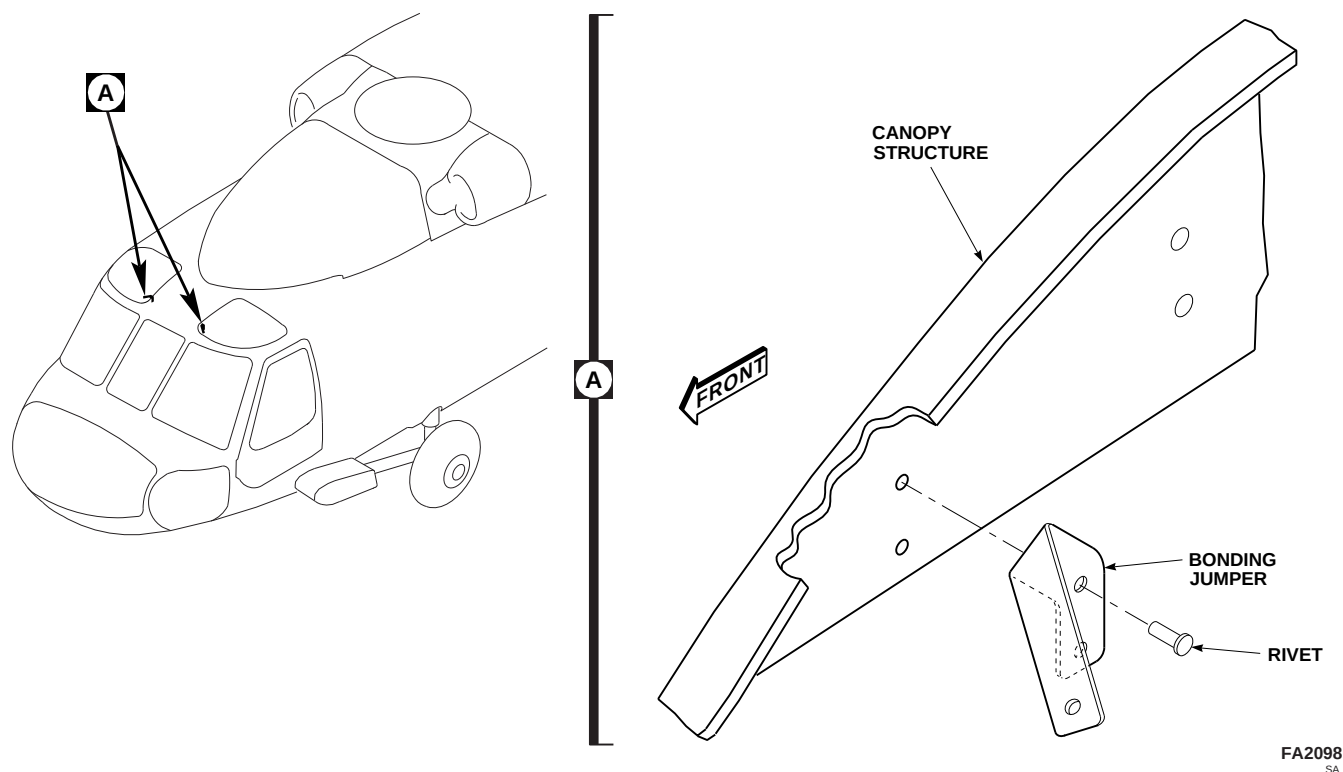


FIGURE 8-4-30-1. REPLACE FREE-AIR THERMOMETER



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FIGURE 8-4-30-2. REPLACE FREE-AIR THERMOMETER BONDING JUMPER HCW

- b. Using holes in canopy structure as a template, mark bonding jumper for drilling.
- c. Using drill with 1/8-inch diameter drill bit, drill through jumper where marked.
- d. Clean mating surfaces of jumper and upper canopy structure using cheesecloth, Item 90, Appendix D, dampened with cleaning compound solvent, Item 100A, Appendix D. Dry surface using clean, dry cheesecloth, Item 90, Appendix D.
- e. Position bonding jumper on canopy structure. Make sure mating surfaces of jumper and structure are bare metal and jumper is flat against structure.
- f. Using rivet gun, rivet bonding jumper to upper canopy structure.
- g. Install free-air thermometer (PARA 8-4-30.1.3).

8-4-31 ROTOR OVERSPEED RESET SWITCH.

PARAGRAPH BREAKDOWN

		PAGE
8-4-31.1	Replace Rotor Overspeed Reset Switch	8-4-31-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 8-2-3

8-4-31.1 Replace Rotor Overspeed Reset Switch.

8-4-31.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open nose door.
- In avionics compartment, remove nut holding rotor overspeed reset switch to canted bulkhead (Figure 8-4-31-1, Detail A). Remove switch.
- In cockpit, slide rubber boots from rotor over-speed reset switch terminals.
- Remove screws, lockwashers, and wiring from rotor overspeed reset switch terminals. Tag wires.

8-4-31.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Connect wiring to rotor overspeed reset switch terminals with screws and lockwashers (Figure 8-4-31-1, Detail A). Untag wires.
- Push rubber boots securely over terminals.
- In cockpit, install rotor overspeed reset switch in hole in canted bulkhead.
- In avionics compartment, fasten rotor over-speed reset switch to canted bulkhead with nut.
- Make sure area is clean and free of foreign material.
- Perform operational check of instrument display system (PARA 8-2-3).
- Close nose door.

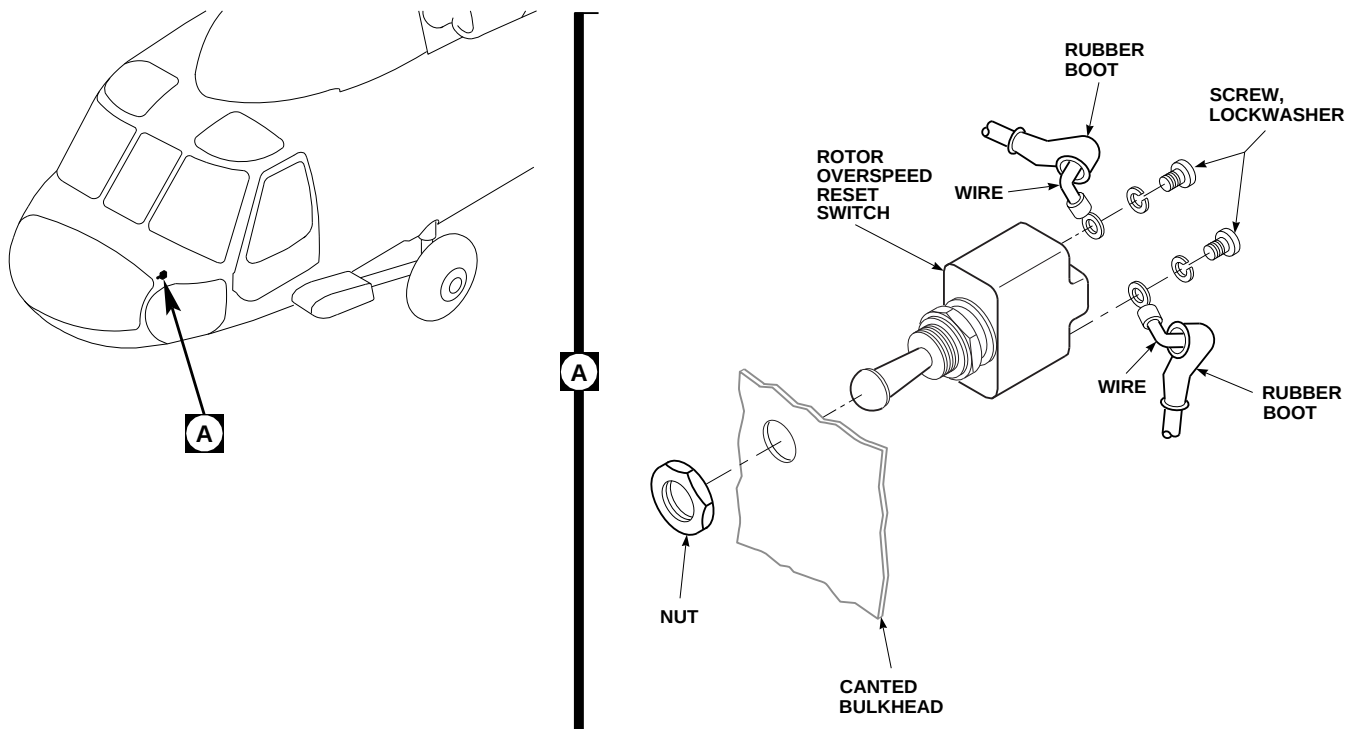
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FIGURE 8-4-31-1. REPLACE ROTOR OVERSPEED RESET SWITCH

8-4-32 CHIP DETECTOR RESISTOR UNIT 70006–70720

PARAGRAPH BREAKDOWN

		PAGE
8-4-32.1	Replace Chip Detector Resistor Unit	8-4-32-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D

- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 6-2-2

8-4-32.1 Replace Chip Detector Resistor Unit.

dampened with dry-cleaning solvent, Item 124, Appendix D.

8-4-32.1.1. Remove.

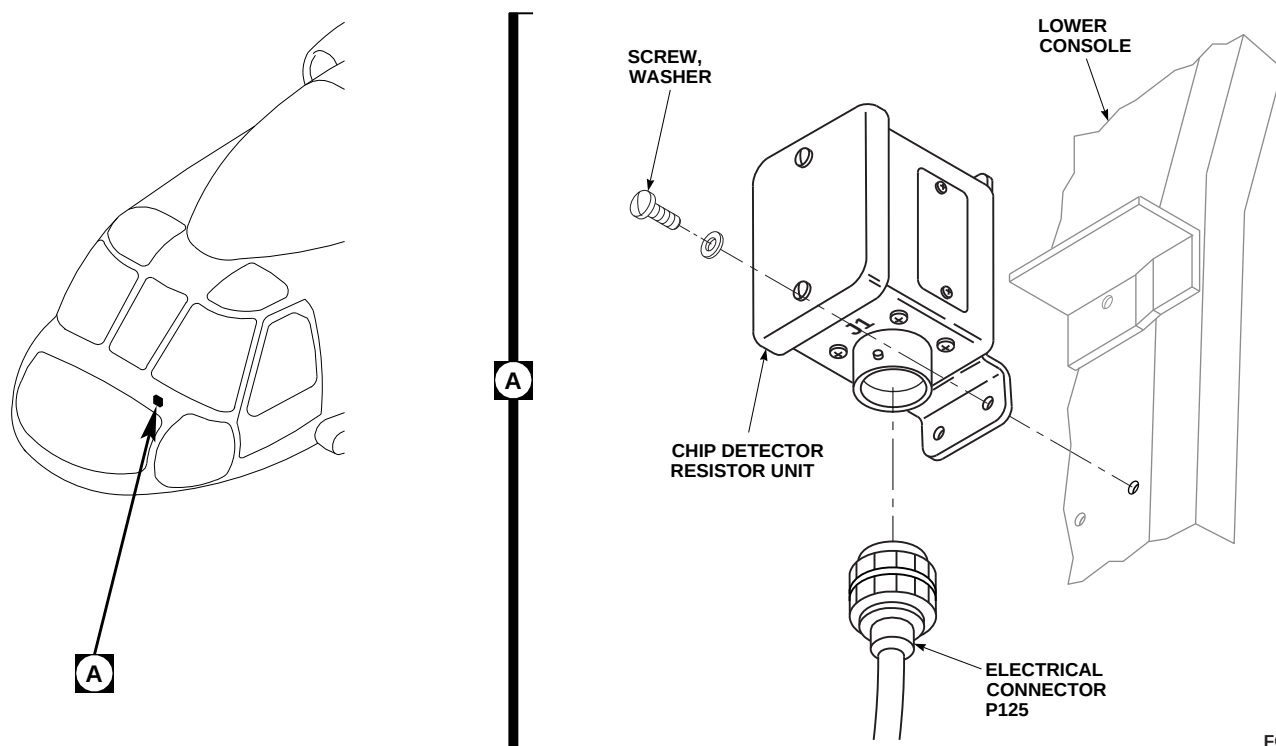
- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P125, from chip detector resistor unit (Figure 8-4-32-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Remove screws, washers, and chip detector resistor unit from lower console.

- Install chip detector resistor unit to lower console with screws and washers (Figure 8-4-32-1, Detail A).
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector, P125, to chip detector resistor unit.

8-4-32.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Clean surface around all mounting holes using machinery towel, Item 211, Appendix D,

- Make sure area is clean and free of foreign material.
- Perform operational check of transmission chip detector system (PARA 6-2-2).



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FIGURE 8-4-32-1. REPLACE CHIP DETECTOR RESISTOR UNIT

SECTION 8-5

MAINTENANCE PROCEDURES (BENCH)

SECTION OVERVIEW

PARAGRAPH	TITLE
8-5-1	Central Display Unit Display Lamp Driver Modules (BENCH)
8-5-2	Central Display Unit Voltage Regulator and Interface Modules (BENCH)
8-5-3	Central Display Unit Digital Readout Module (BENCH)
8-5-4	Signal Data Converter Modules (BENCH)
8-5-5	Signal Data Converter Lamp Power Supply (BENCH)
8-5-6	Signal Data Converter Logic Power Supply (BENCH)
8-5-7	Pilot Display Unit, 70450-01043-121 and 70450-01043-123, Driver Modules/Lamps (BENCH)
8-5-8	Pilot Display Unit, 70450-01043-124 and 70450-01916-104, Driver Modules (BENCH)
8-5-9	Pilot Display Unit, 70450-01043-121 and 70450-01043-123, Torque Display Analog Modules/Lamps (BENCH)
8-5-10	Pilot Display Unit, 70450-01043-124 and 70450-01916-104, Torque Display Analog Modules (BENCH)
8-5-11	Pilot Display Unit Torque Display/Digital Readout Module (BENCH)
8-5-12	Pilot Display Unit Autodim Module (BENCH)
8-5-13	Pilot Display Unit NVG Filters (BENCH)
8-5-14	Caution/Advisory Panel Channel/Control Cards (BENCH)
8-5-15	Caution/Advisory Panel Test Switch (BENCH)
8-5-16	Caution/Advisory Panel Heat Sink/Transistors (BENCH)
8-5-17	Caution/Advisory Panel Relay, K1 (BENCH)

PARAGRAPH	TITLE
8-5-18	Caution/Advisory Panel Diode, CR1 (BENCH)
8-5-19	Caution/Advisory Panel Diode, CR2 (BENCH)
8-5-20	Caution/Advisory Panel Diode, CR3 (BENCH)
8-5-21	Caution/Advisory Panel Connectors (BENCH)
8-5-22	Chip Detector Resistor Unit (BENCH)

8-5-1 CENTRAL DISPLAY UNIT DISPLAY LAMP DRIVER MODULES (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
8-5-1.1	Replace Central Display Unit Display Lamp Driver Modules (BENCH)	8-5-1-1

INITIAL SETUP

- PARA 8-4-11

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-7-20
- PARA 8-2-7

8-5-1.1 Replace Central Display Unit Display Lamp Driver Modules (BENCH).

NOTE

Replacement is the same for all lamp driver modules, except as noted.

8-5-1.1.1 Remove.

- Turn off all electrical power.
- Disconnect central display unit (CDU) from test set.
- Remove central display unit faceplate and EMI shield (PARA 8-4-11).
- On CDU, 70450-01043-125 or 70450-01916-105, remove EMI shield (PARA 8-4-11).

NOTE

Lamp driver modules are inserted into electrical connectors within CDU chas-

sis. Some resistance to removal is normal.

- On CDU, 70450-01043-122 or 70450-21943-118, loosen lockscrews at top and bottom of lamp driver module and pull outward to remove (Figure 8-5-1-1).
- On CDU, 70450-01043-125 or 70450-01916-105, remove lamp driver modules as follows (Figure 8-5-1-2, Detail A):
 - Loosen lockscrews at top and bottom of lamp driver module A3 and pull outward to remove.
 - Remove screws at top and bottom of lamp driver modules, A4 through A14, and pull outward to remove.

8-5-1.1.2 Install.

- Turn off all electrical power.
- Inspect and treat electrical connectors (PARA 1-7-20).

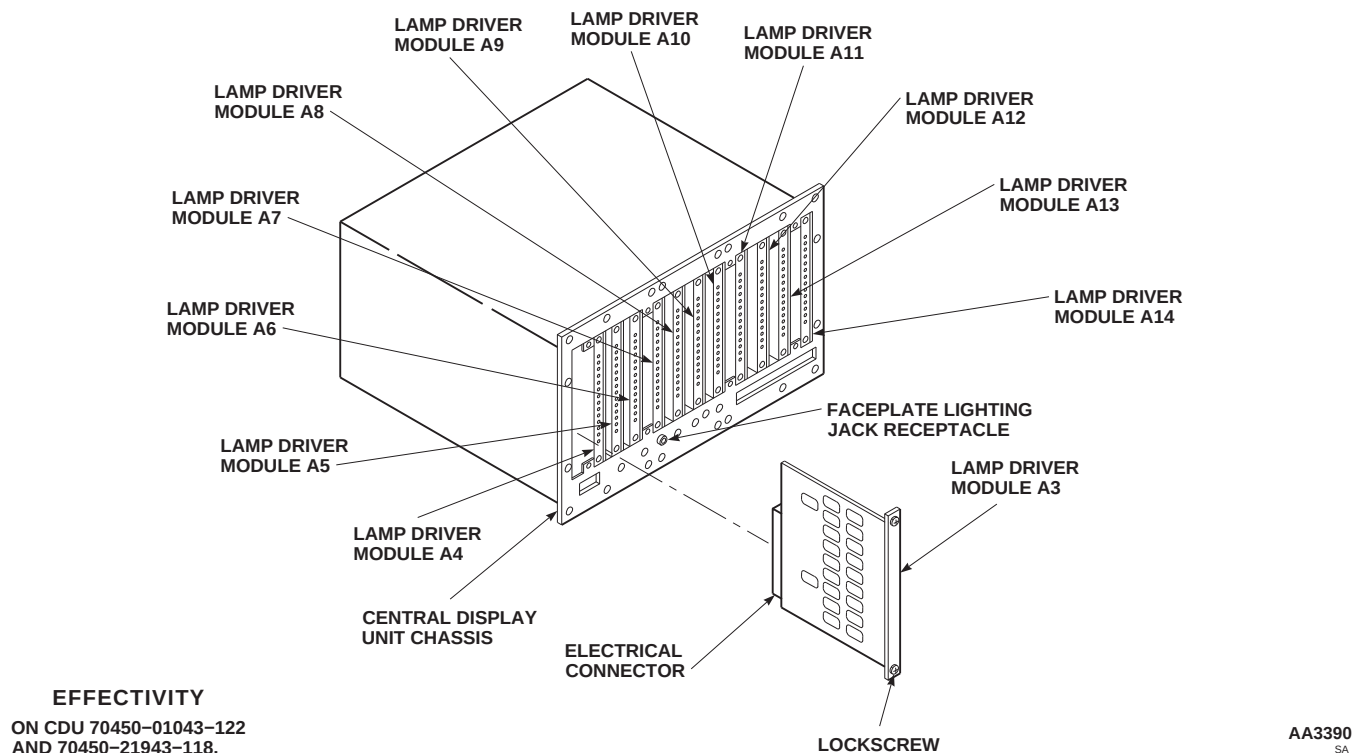


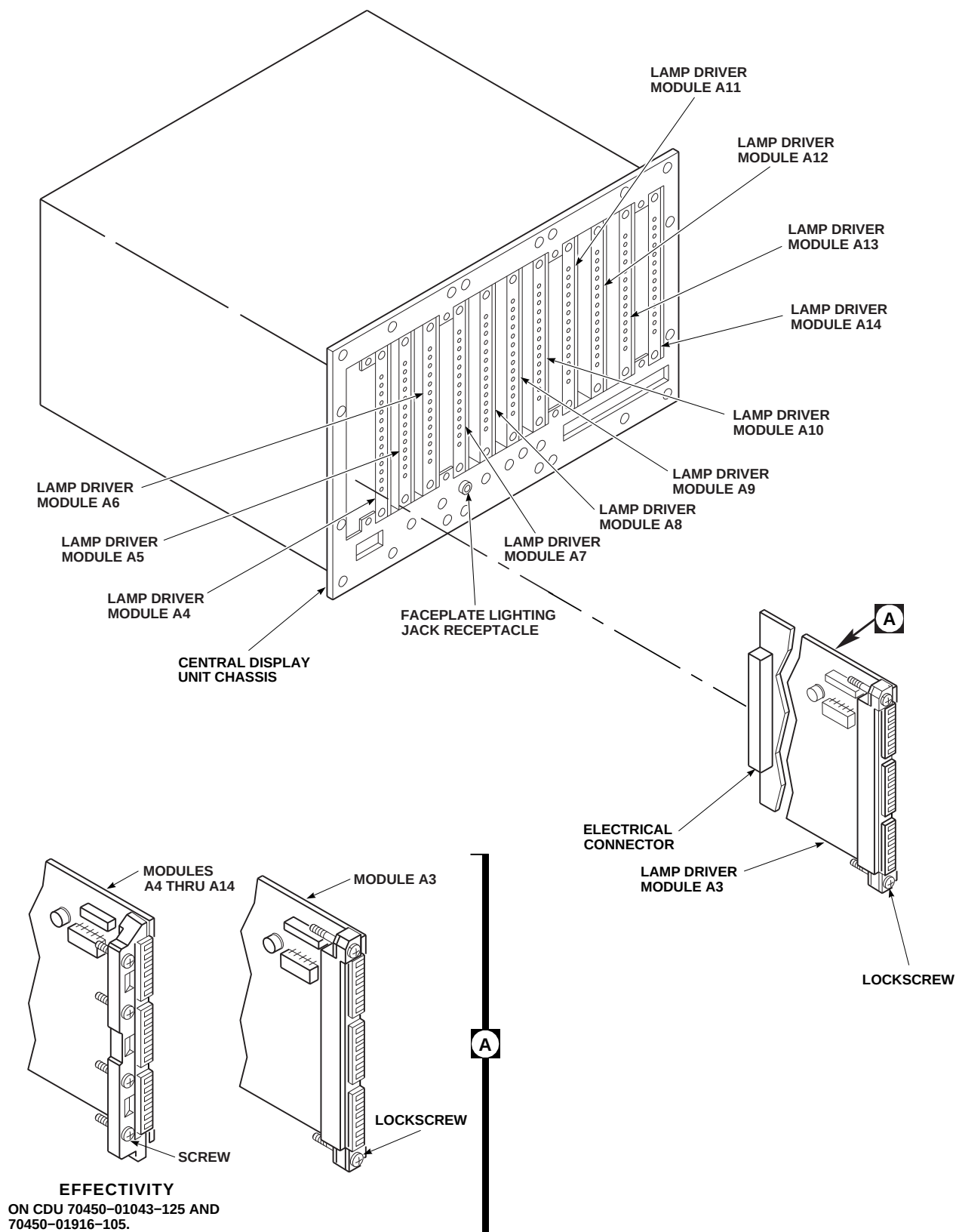
FIGURE 8-5-1-1. REPLACE CENTRAL DISPLAY UNIT DISPLAY LAMP DRIVER MODULES, 70450-01043-122 OR 70450-21943-118 (BENCH)

NOTE

On central display unit (CDU), 70450-01043-122 or 70450-21943-118, these lamp driver modules are interchangeable: A5, A7, A8, A11 through A14; A3 and A4; A9 and A10.

- c. On CDU, 70450-01043-122 or 70450-21943-118, insert lamp driver module into CDU and install with lock screws (Figure 8-5-1-1).
- d. On CDU, 70450-01043-125 or 70450-01916-105, install lamp drive modules as follows (Figure 8-5-1-2, Detail A):

- (1) Insert lamp driver module A3 into CDU and install with lock screws.
- (2) Insert lamp driver modules A4 through A14 into CDU and install with screws.
- e. On CDU, 70450-01043-125 or 70450-01916-105, install EMI shield (PARA 8-4-11).
- f. Install central display unit faceplate and EMI shield (PARA 8-4-11).
- g. Perform operational check of CDU (PARA 8-2-7).



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FIGURE 8-5-1-2. REPLACE CENTRAL DISPLAY UNIT DISPLAY LAMP DRIVER MODULES, 70450-01043-125 OR 70450-01916-105 (BENCH)

8-5-2 CENTRAL DISPLAY UNIT VOLTAGE REGULATOR AND INTERFACE MODULES (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
8-5-2.1	Replace Central Display Unit Voltage Regulator and Interface Modules (BENCH)	8-5-2-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-7-20
- PARA 8-2-7

8-5-2.1 Replace Central Display Unit Voltage Regulator and Interface Modules (BENCH).

NOTE

Module replacement is the same for voltage regulator, interface No. 1, and interface No. 2 modules.

8-5-2.1.1 Remove.

- Turn off all electrical power.
- Disconnect central display unit (CDU) from test set.
- Remove screws from rear of CDU, and remove dust cover (Figure 8-5-2-1).

NOTE

Modules are inserted into electrical connectors within CDU chassis. Some resistance to removal is normal.

- Loosen screws at each end of module and pull upward to remove.

8-5-2.1.2 Install.

- Turn off all electrical power.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Insert module into central display unit (CDU) chassis and apply pressure until module is completely engaged with chassis electrical connector and flush with other modules in chassis (Figure 8-5-2-1).
- TIGHTEN SCREWS AT EACH END OF MODULE TO SECURE TO CHASSIS.**
- Place dust cover on CDU chassis and fasten with screws.
- Perform operational check of CDU (PARA 8-2-7).

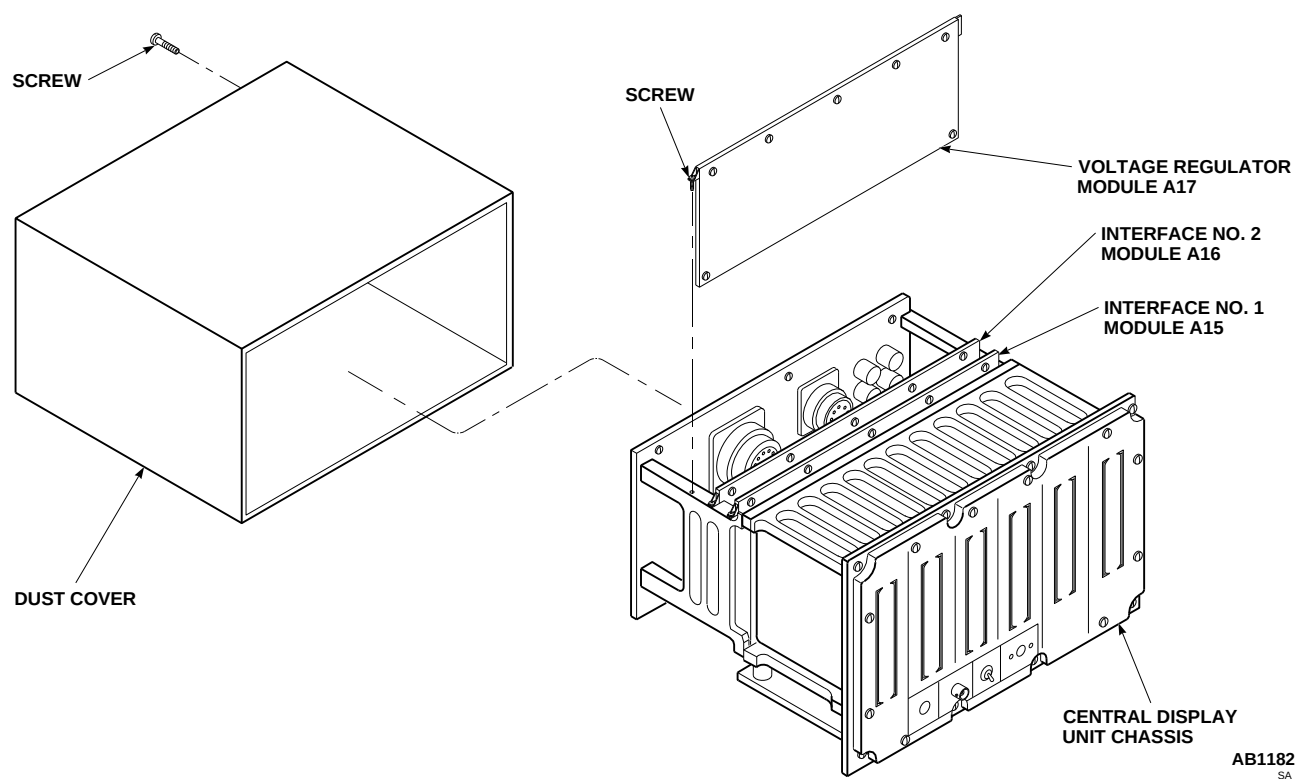


FIGURE 8-5-2-1. REPLACE CENTRAL DISPLAY UNIT VOLTAGE REGULATOR AND INTERFACE MODULES (BENCH)

8-5-3 CENTRAL DISPLAY UNIT DIGITAL READOUT MODULE (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
8-5-3.1	Replace Central Display Unit Digital Readout Module (BENCH)	8-5-3-1

INITIAL SETUP

- PARA 8-4-11

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-7-20
- PARA 8-2-7

8-5-3.1 Replace Central Display Unit Digital Readout Module (BENCH).

8-5-3.1.1 Remove.

- a. Turn off all electrical power.
- b. Disconnect central display unit (CDU) from test set.
- c. Remove faceplate (PARA 8-4-11).
- d. On CDU, 70450-01043-125 or 70450-01916-105, remove EMI shield (PARA 8-4-11).
- e. Remove screws securing CDU chassis to dust cover. Remove dust cover (Figure 8-5-3-1).
- f. Turn CDU chassis upside down and remove screws securing digital readout module.
- g. Lift at center of digital readout module releasing electrical connector from CDU chassis. Carefully remove module up at an angle and toward rear of CDU, making certain that digital display portion of module clears cutout in front of CDU chassis.

8-5-3.1.2 Install.

- a. Turn off all electrical power.
- b. Inspect and treat electrical connectors (PARA 1-7-20).



Mating electrical connectors of digital readout module and central display unit (CDU) chassis is done with gentle pressure on module. Any greater force exerted on electrical connectors can result in damage.

- c. Place digital readout module into CDU chassis at an angle with digital display portion directed forward and through cutout in front of CDU chassis (Figure 8-5-3-1). Carefully line up electrical connectors of digital readout module and CDU chassis and apply pressure until electrical connectors are engaged.

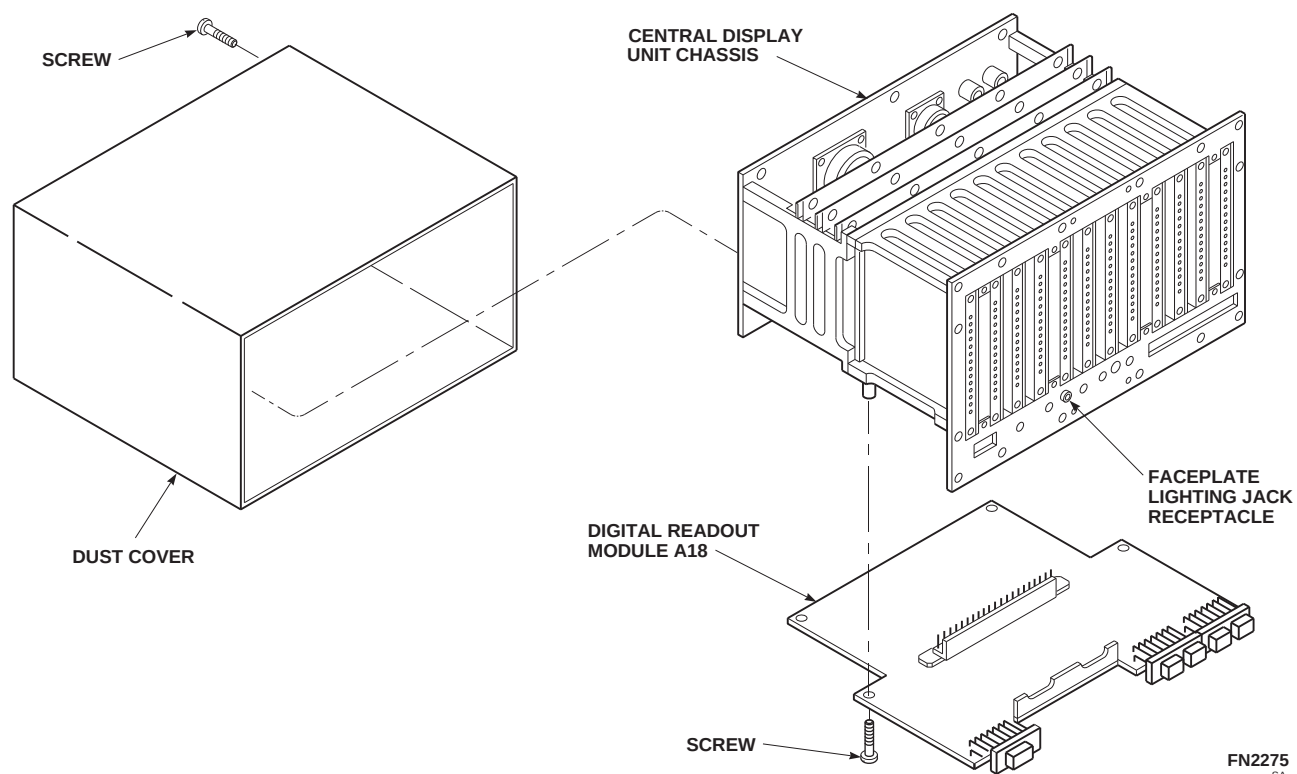


FIGURE 8-5-3-1. REPLACE CENTRAL DISPLAY UNIT DIGITAL READOUT MODULE (BENCH)

- d. Fasten digital readout module to CDU chassis with screws.
- e. Place dust cover on CDU chassis and fasten with screws.
- f. On CDU, 70450-01043-125 or 70450-01916-105, using locating pins for alignment, install EMI shield (PARA 8-4-11).
- g. Install faceplate (PARA 8-4-11).
- h. Perform operational check of CDU (PARA 8-2-7).

8-5-4 SIGNAL DATA CONVERTER MODULES (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
8-5-4.1	Replace Signal Data Converter Modules (BENCH)	8-5-4-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-7-20
- PARA 8-2-9

8-5-4.1 Replace Signal Data Converter Modules (BENCH).

- Slide module straight up and out of signal data converter.

NOTE

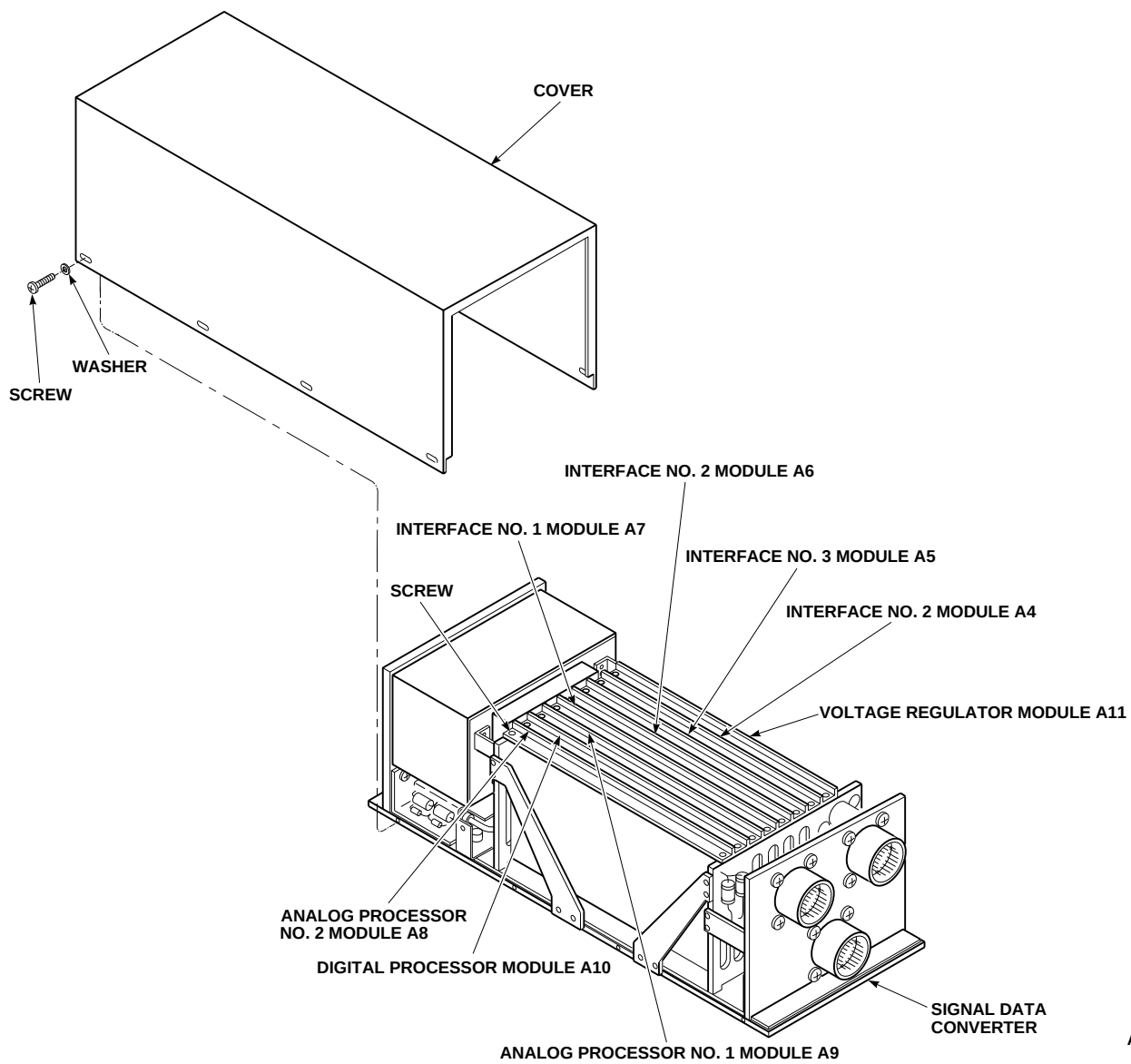
Replacement is the same for all modules.

8-5-4.1.2 Install.

8-5-4.1.1 Remove.

- Turn off all electrical power.
- Disconnect signal data converter from test set.
- Remove screws and washers from cover and remove cover (Figure 8-5-4-1).
- Remove screws from top of module to be replaced.

- Turn off all electrical power.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Slide replacement module into signal data converter (Figure 8-5-4-1).
- Fasten module to signal data converter with screws.
- Install cover on signal data converter and fasten with screws and washers.
- Perform operational check of signal data converter (PARA 8-2-9).



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FIGURE 8-5-4-1. REPLACE SIGNAL DATA CONVERTER MODULES (BENCH)

8-5-5 SIGNAL DATA CONVERTER LAMP POWER SUPPLY (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
8-5-5.1	Replace Signal Data Converter Lamp Power Supply (BENCH)	8-5-5-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-7-20
- PARA 8-2-9

8-5-5.1 Replace Signal Data Converter Lamp Power Supply (BENCH).

8-5-5.1.1 Remove.

- a. Turn off all electrical power.
- b. Disconnect signal data converter from test set.
- c. Remove screws and washers from cover and remove cover (Figure 8-5-5-1).
- d. Loosen screws securing connector of lamp power supply to receptacle and disconnect connector.
- e. Install protective caps and plugs on electrical connectors.
- f. Remove screws securing lamp power supply to signal data converter chassis and remove lamp power supply.

8-5-5.1.2 Install.

- a. Turn off all electrical power.
- b. Position lamp power supply in signal data converter chassis and secure with screws (Figure 8-5-5-1).
- c. Remove protective caps and plugs from electrical connectors.
- d. Inspect and treat electrical connectors (PARA 1-7-20).
- e. Connect lamp power supply connector to receptacle and secure by tightening captive screws on connector.
- f. Install cover on signal data converter and secure with screws and washers.
- g. Perform operational check of signal data converter (PARA 8-2-9).

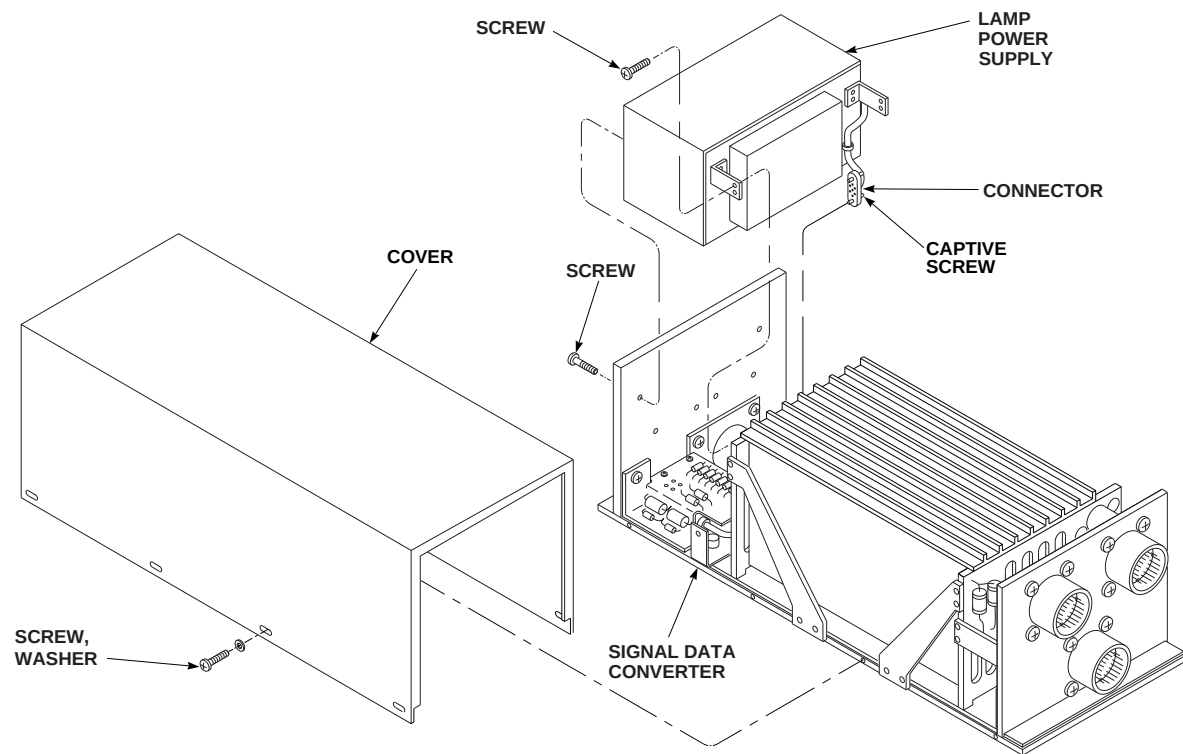
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FIGURE 8-5-5-1. REPLACE SIGNAL DATA CONVERTER LAMP POWER SUPPLY (BENCH)

8-5-6 SIGNAL DATA CONVERTER LOGIC POWER SUPPLY (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
8-5-6.1	Replace Signal Data Converter Logic Power Supply (BENCH)	8-5-6-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-7-20
- PARA 8-5-5

8-5-6.1 Replace Signal Data Converter Logic Power Supply (BENCH).

8-5-6.1.1 Remove.

- Remove signal data converter lamp power supply (PARA 8-5-5).
- Loosen screws securing connector of logic power supply to receptacle and disconnect connector (Figure 8-5-6-1).
- Install protective caps and plugs on electrical connectors.
- Remove screws securing logic power supply to signal data converter chassis and remove logic power supply.

8-5-6.1.2 Install.

- Turn off all electrical power.
- Place logic power supply into signal data converter chassis and secure with screws (Figure 8-5-6-1).
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect logic power supply connector to receptacle and secure by tightening connector-mounted screws.
- Install signal data converter lamp power supply (PARA 8-5-5).

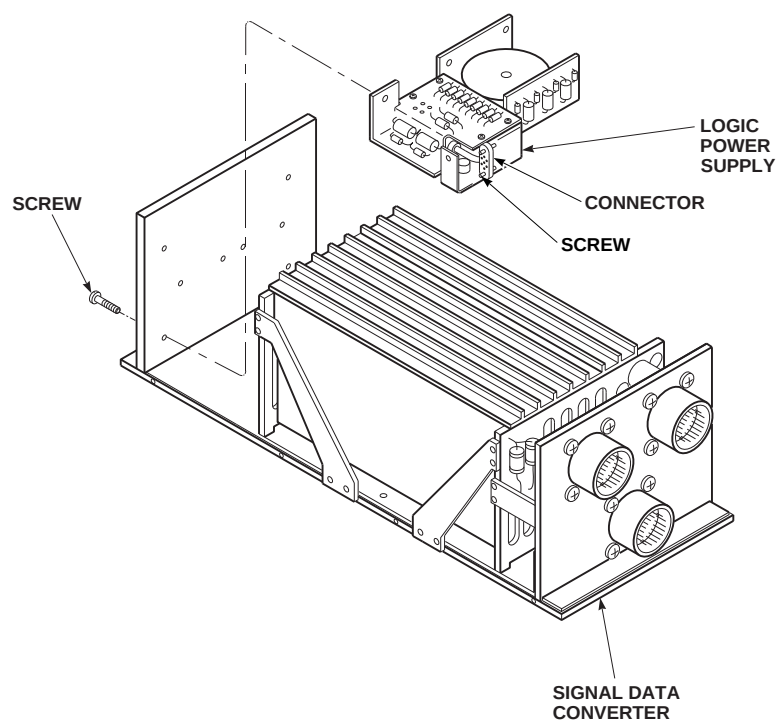
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FIGURE 8-5-6-1. REPLACE SIGNAL DATA CONVERTER LOGIC POWER SUPPLY (BENCH)

8-5-7 PILOT DISPLAY UNIT, 70450-01043-121 AND 70450-01043-123, DRIVER MODULES/LAMPS (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
8-5-7.1	Replace Pilot Display Unit, 70450-01043-121 and 70450-01043-123, Driver Modules/Lamps (BENCH)	8-5-7-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Screwdriver, Jeweler’s Cross-Tip

References

- PARA 1-7-20
- PARA 8-2-8
- PARA 8-4-17

8-5-7.1 Replace Pilot Display Unit, 70450-01043-121 and 70450-01043-123, Driver Modules/Lamps (BENCH).

NOTE

Module replacement is the same for all display driver modules.

NOTE

As screws become flush with aluminum block, aluminum block/modules move toward front, releasing electrical connector for each module at rear of pilot display unit housing.

8-5-7.1.1 Remove.

- Turn off all electrical power.
- Disconnect pilot display unit from test set.
- Remove pilot display unit faceplate (PARA 8-4-17).
- Remove screws at top and bottom of aluminum block (Figure 8-5-7-1, Detail A).
- Insert two of these screws into threaded holes (third from left) at top and bottom of aluminum block, and **ALTERNATELY TIGHTEN DOWN**.

- Carefully pull aluminum block/modules from housing.
- Remove two screws in aluminum block, to separate aluminum block/modules from housing.
- Using jeweler’s cross-tip screwdriver, remove small crosshead screws in line with each module to be replaced (Figure 8-5-7-1, Detail B).
- Remove large crosshead screws in line with module to be replaced and carefully separate lamp retainer/module from aluminum block.

8-5-7.1.2 Install.

- a. Turn off all electrical power.

NOTE

Before inserting module lamps into aluminum block, make sure they are lined up and uniform.

- b. Carefully insert module lamps into openings in aluminum block.

CAUTION

Damage to module lamps will result if not inserted into aluminum block properly. Make sure module lamps are properly inserted into aluminum block. Lamps can be viewed from front of block.

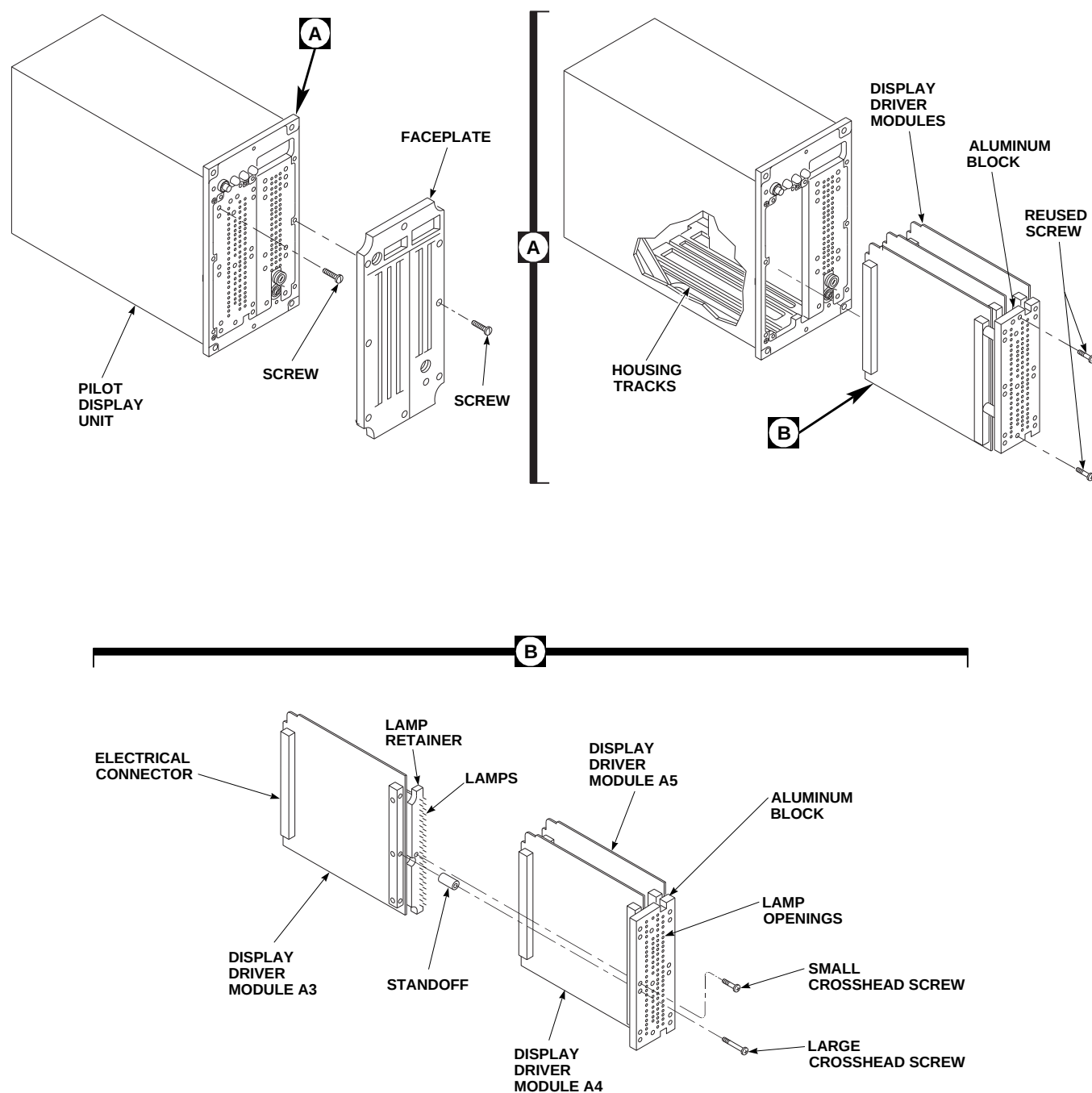
- c. Slide lamp retainer toward front to aluminum block (Figure 8-5-7-1, Detail B).
- d. Using jeweler's cross-tip screwdriver, secure lamp retainer to aluminum block with small crosshead screws.
- e. Position module standoffs to aluminum block and secure with large crosshead screws.

- f. Inspect and treat electrical connectors (PARA 1-7-20).
- g. Carefully slide modules into housing tracks and apply pressure until electrical connectors of modules reach rear of housing (Figure 8-5-7-1, Detail A).

CAUTION

Damage to pilot display unit may result if screws are threaded into third holes from left at top and bottom of aluminum block when installing aluminum block/modules. Make sure when installing aluminum block/modules not to install screws into third holes from left at top and bottom of aluminum block.

- h. Insert screws at top and bottom of aluminum block and **ALTERNATELY TIGHTEN UNTIL ALUMINUM BLOCK/MODULES ARE SECURED TO HOUSING.**
- i. Install pilot display unit faceplate (PARA 8-4-17).
- j. Perform operational check of pilot display unit (PARA 8-2-8).



EFFECTIVITY

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70450-01043-123

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FIGURE 8-5-7-1. REPLACE PILOT DISPLAY UNIT, 70450-01043-121 AND 70450-01043-123, DRIVER MODULES/LAMPS (BENCH)

8-5-8 PILOT DISPLAY UNIT, 70450-01043-124 AND 70450-01916-104, DRIVER MODULES (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
8-5-8.1	Replace Pilot Display Unit, 70450-01043-124 and 70450-01916-104, Driver Modules (BENCH)	8-5-8-1

INITIAL SETUP

- PARA 8-4-17

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-7-20
- PARA 8-2-8

8-5-8.1 Replace Pilot Display Unit, 70450-01043-124 and 70450-01916-104, Driver Modules (BENCH).

NOTE

Module replacement is the same for all display driver modules.

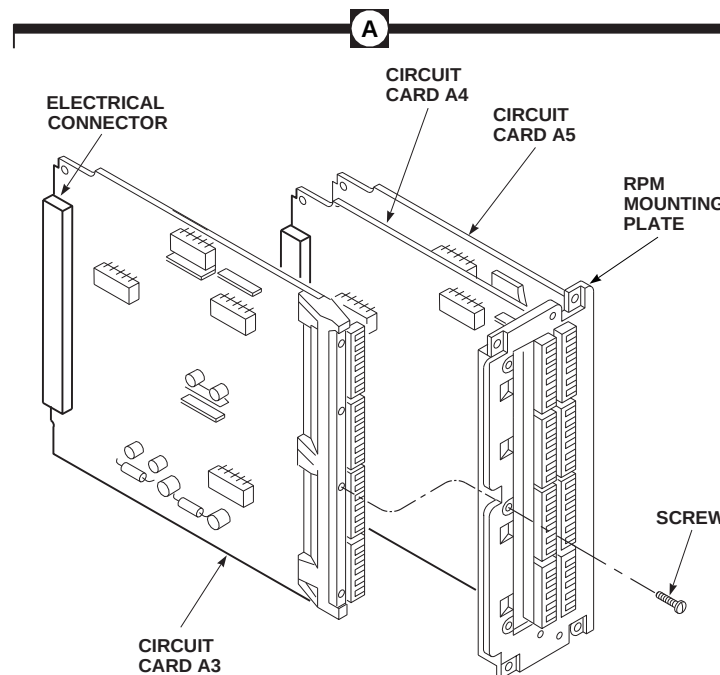
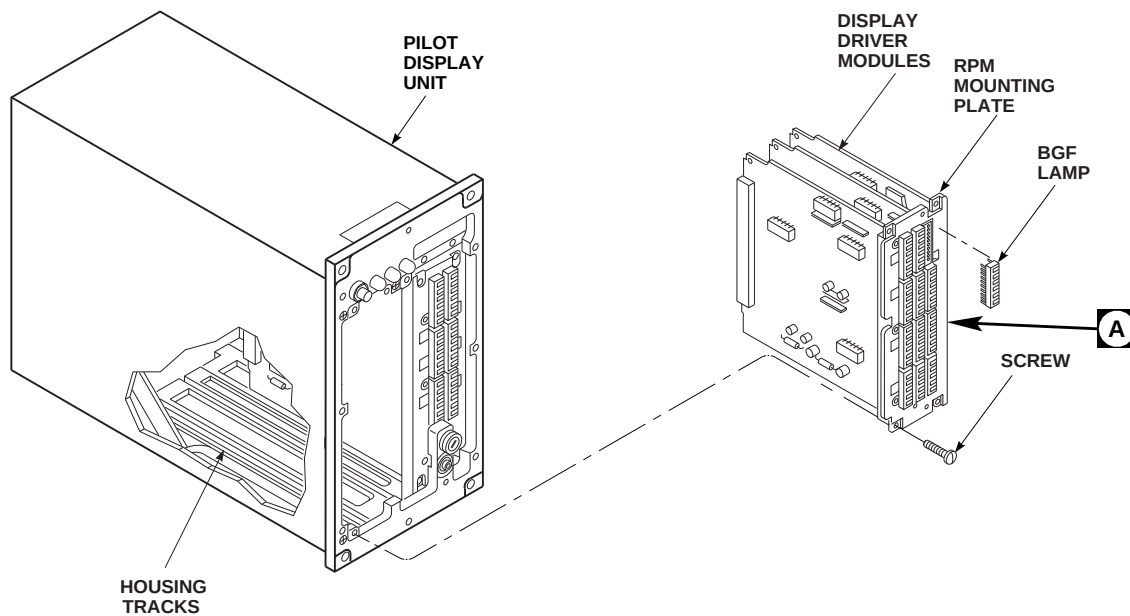
8-5-8.1.1 Remove.

- Turn off all electrical power.
- Disconnect pilot display unit (PDU) from test set.
- Remove faceplate and EMI shield (PARA 8-4-17).
- Remove screws and RPM mounting plate (Figure 8-5-8-1).
- Remove screws securing circuit cards A3, A4, A5 to RPM mounting plate and remove circuit cards (Figure 8-5-8-1, Detail A).

- Remove BGF lamps.

8-5-8.1.2 Install.

- Install BGF lamps on circuit cards A3, A4, and A5.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Install circuit cards on RPM mounting plate with screws in the sequence A5, A4, A3 (Figure 8-5-8-1, Detail A).
- Feed circuit card A3, A4, A5 into guide slots on pilot display unit (PDU) chassis, and push assembly until connectors are fully engaged to mating pins (Figure 8-5-8-1).
- Fasten assembly to chassis with screws.
- Install EMI shield and faceplate (PARA 8-4-17).



EFFECTIVITY
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 70450-01916-104

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FIGURE 8-5-8-1. REPLACE PILOT DISPLAY UNIT, 70450-01043-124 AND 70450-01916-104, DRIVER MODULES (BENCH)

8-5-8-2 Change 5

- g. Perform operational check of PDU (PARA 8-2-8).

8-5-9 PILOT DISPLAY UNIT, 70450-01043-121 AND 70450-01043-123, TORQUE DISPLAY ANALOG MODULES/LAMPS (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
8-5-9.1	Replace Pilot Display Unit, 70450-01043-121 and 70450-01043-123, Torque Display Analog Modules/Lamps (BENCH)	8-5-9-1

INITIAL SETUP

- PARA 8-4-17

Tools

- PARA 8-5-7

- Electrical Repairer’s Toolkit
- Screwdriver, Jeweler’s Cross-Tip

References

- PARA 1-7-20
- PARA 8-2-8

8-5-9.1 Replace Pilot Display Unit, 70450-01043-121 and 70450-01043-123, Torque Display Analog Modules/Lamps (BENCH).

NOTE

Module replacement is the same for both torque display analog modules.

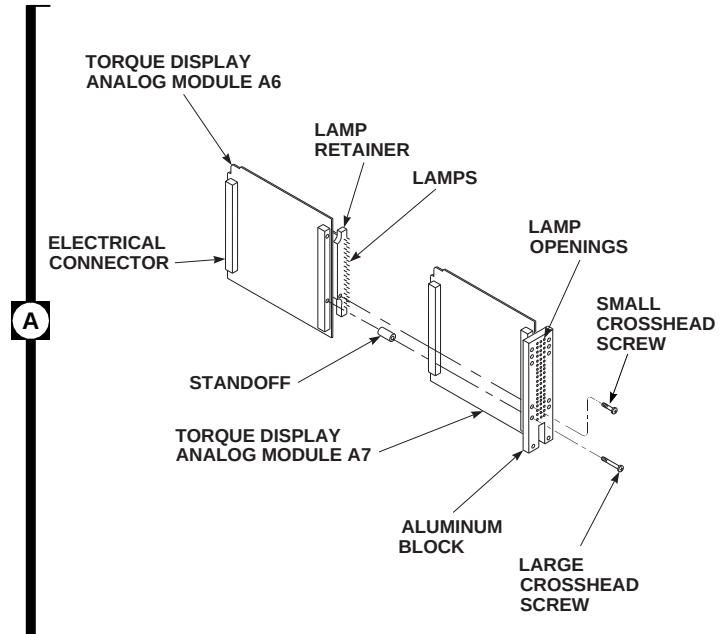
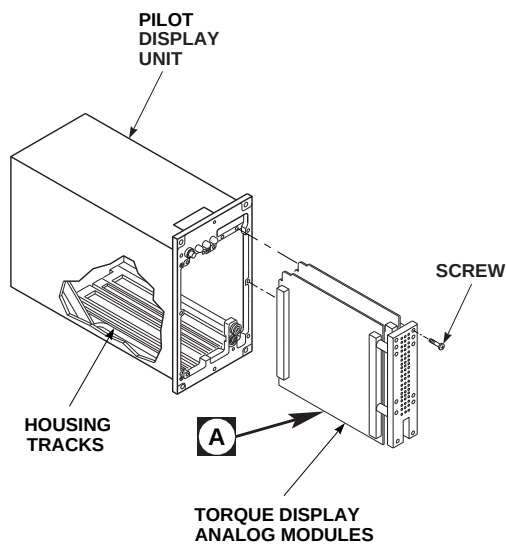
8-5-9.1.1 Remove.

- Turn off all electrical power.
- Disconnect pilot display unit (PDU) from test set.
- Remove pilot display unit faceplate (PARA 8-4-17).
- Remove modules A3, A4, and A5 (PARA 8-5-7).

- Remove screws at top and bottom of aluminum block (Figure 8-5-9-1).
- Carefully pull aluminum block/modules forward, releasing electrical connector for each module at rear of PDU housing. Remove aluminum block/modules.
- Using jeweler’s cross-tip screwdriver, remove small crosshead screws in line with each module to be replaced (Figure 8-5-9-1, Detail A).
- Remove large crosshead screws in line with module to be replaced and carefully separate lamp retainer/module from aluminum block.

8-5-9.1.2 Install.

- Turn off all electrical power.

**EFFECTIVITY**

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70450-01043-123

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FIGURE 8-5-9-1. REPLACE PILOT DISPLAY UNIT, 70450-01043-121 AND 70450-01043-123, TORQUE DISPLAY ANALOG MODULES/LAMPS (BENCH)

NOTE

Before inserting module lamps into aluminum block, make sure they are lined up and uniform.

- b. Carefully insert module lamps into openings in aluminum block (Figure 8-5-9-1, Detail A).

CAUTION

Damage to module lamps will result if not inserted into aluminum block properly. Make sure module lamps are properly inserted into aluminum block. Lamps can be viewed from front of block.

- c. Slide lamp retainer forward to aluminum block. Using jeweler's cross-tip screwdriver, secure lamp retainer to aluminum block with small cross-head screws.

- d. Position module standoffs to aluminum block and secure with large cross-head screws.
- e. Inspect and treat electrical connectors (PARA 1-7-20).
- f. Carefully slide modules into housing tracks and apply pressure until electrical connectors of modules reach rear of housing (Figure 8-5-9-1).
- g. Insert screws at top and bottom of aluminum block and **ALTERNATELY TIGHTEN UNTIL ALUMINUM BLOCK/MODULES ARE SECURED TO HOUSING.**
- h. Install modules A3, A4, and A5 (PARA 8-5-7).
- i. Install pilot display unit faceplate (PARA 8-4-17).
- j. Perform operational check of pilot display unit (PARA 8-2-8).

8-5-10 PILOT DISPLAY UNIT, 70450-01043-124 AND 70450-01916-104, TORQUE DISPLAY ANALOG MODULES (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
8-5-10.1	Replace Pilot Display Unit, 70450-01043-124 and 70450-01916-104, Torque Display Analog Modules (BENCH)	8-5-10-1

INITIAL SETUP

- PARA 8-4-17

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-7-20
- PARA 8-2-8

8-5-10.1 Replace Pilot Display Unit, 70450-01043-124 and 70450-01916-104, Torque Display Analog Modules (BENCH).

NOTE

Module replacement is the same for both torque display analog modules.

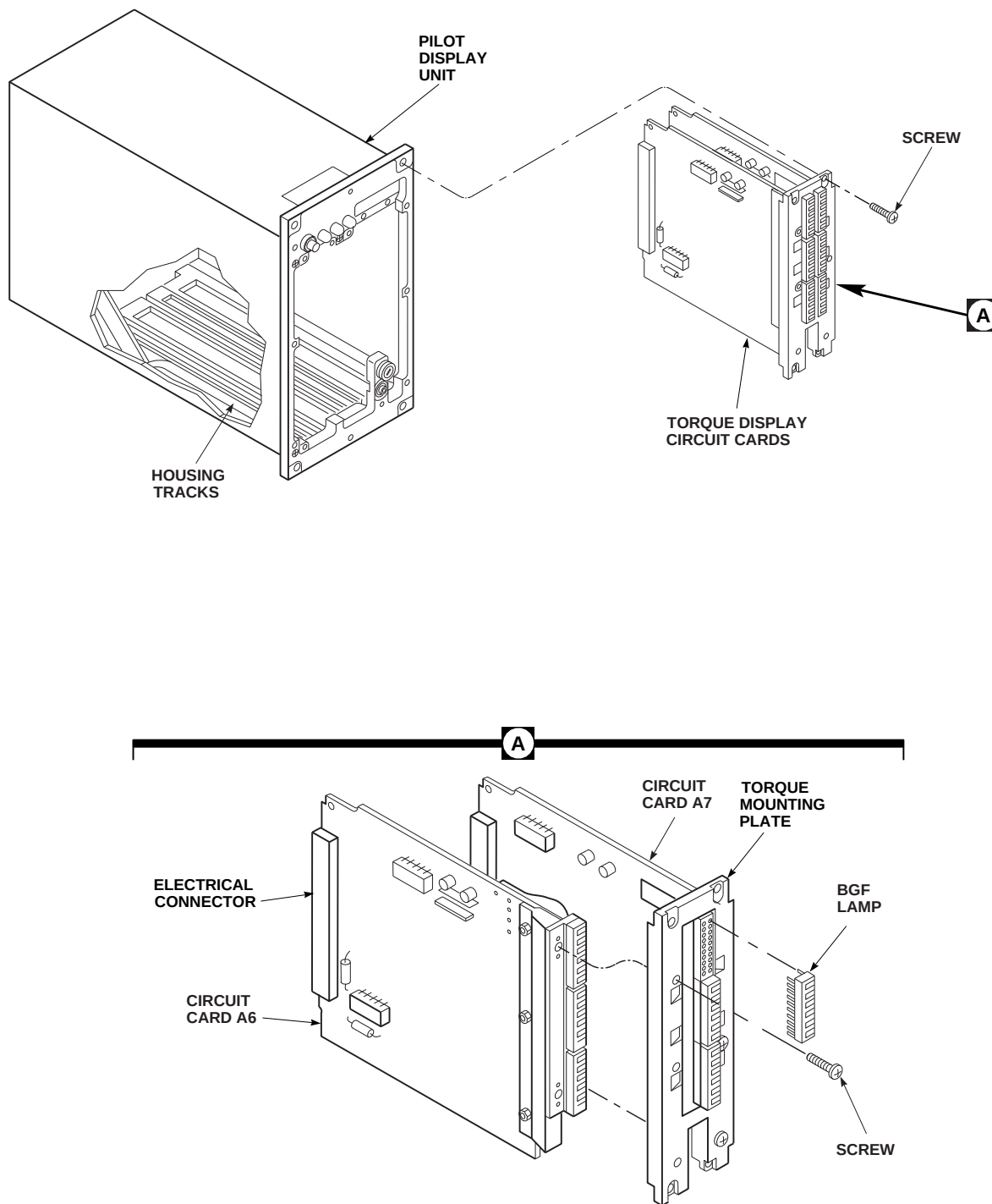
8-5-10.1.1 Remove.

- Turn off all electrical power.
- Disconnect pilot display unit (PDU) from test set.
- Remove faceplate and EMI shield (PARA 8-4-17).
- Remove screws and torque display circuit cards from PDU (Figure 8-5-10-1).

- Remove screws securing circuit cards, A6 and A7, to torque mounting plate and remove circuit cards (Figure 8-5-10-1, Detail A).
- Remove BGF lamps.

8-5-10.1.2 Install.

- Turn off all electrical power.
- Install BGF lamps.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Install circuit cards on torque mounting plate with screws (Figure 8-5-10-1, Detail A).
- Feed circuit cards, A6 and A7, into guide slots on PDU chassis, and push assembly until connectors are fully engaged to mating pins (Figure 8-5-10-1).

**EFFECTIVITY**

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70450-01916-104

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FIGURE 8-5-10-1. REPLACE PILOT DISPLAY UNIT, 70450-01043-124 AND 70450-01916-104, TORQUE DISPLAY ANALOG MODULES (BENCH)

- f. Fasten assembly to chassis with screws.
- g. Install EMI shield and faceplate (PARA 8-4-17).
- h. Perform operational check of PDU (PARA 8-2-8).

8-5-11 PILOT DISPLAY UNIT TORQUE DISPLAY/DIGITAL READOUT MODULE (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
8-5-11.1	Replace Pilot Display Unit Torque Display/Digital Readout Module (BENCH)	8-5-11-1

INITIAL SETUP

- PARA 8-4-17

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-7-20
- PARA 8-2-8

8-5-11.1 Replace Pilot Display Unit Torque Display/Digital Readout Module (BENCH).

NOTE

8-5-11.1.1 Remove.

- Turn off all electrical power.
- Disconnect pilot display unit (PDU) from test set.
- Remove screws from rear of PDU and remove dust cover (Figure 8-5-11-1).

NOTE

EMI shield installed on pilot display unit, 70450-01043-124 and 70450-01916-104.

- Remove pilot display unit faceplate and EMI shield (PARA 8-4-17).

Washers installed on pilot display unit, 70450-01043-124 and 70450-01916-104.

- Remove screws and washers securing torque display/digital readout module to PDU housing.
- Lift rear of torque display/digital readout module, releasing electrical connector from display unit housing. Carefully remove module out and toward rear of PDU. Make sure readout portion of module clears cutout in front of housing.
- On PDU, 70450-01043-124 and 70450-01916-104, remove direct view filament modules.

8-5-11.1.2 Install.

- a. On pilot display unit (PDU), 70450-01043-124 and 70450-01916-104, install direct view filament modules (Figure 8-5-11-1).
- b. Inspect and treat electrical connectors (PARA 1-7-20).
- c. Place torque display/digital readout module on PDU, with readout portion directed forward and through cutout in front of housing. Line up electrical connectors of module and housing and apply pressure until electrical connectors are engaged.

NOTE

Washers installed on pilot display unit, 70450-01043-124 and 70450-01916-104.

- d. Secure torque display/digital readout module to housing with screws and washers.

NOTE

EMI shield installed on pilot display unit, 70450-01043-124 and 70450-01916-104.

- e. Install pilot display unit faceplate and EMI shield (PARA 8-4-17).
- f. Secure dust cover on PDU with screws.
- g. Perform operational check of pilot display unit (PARA 8-2-8).

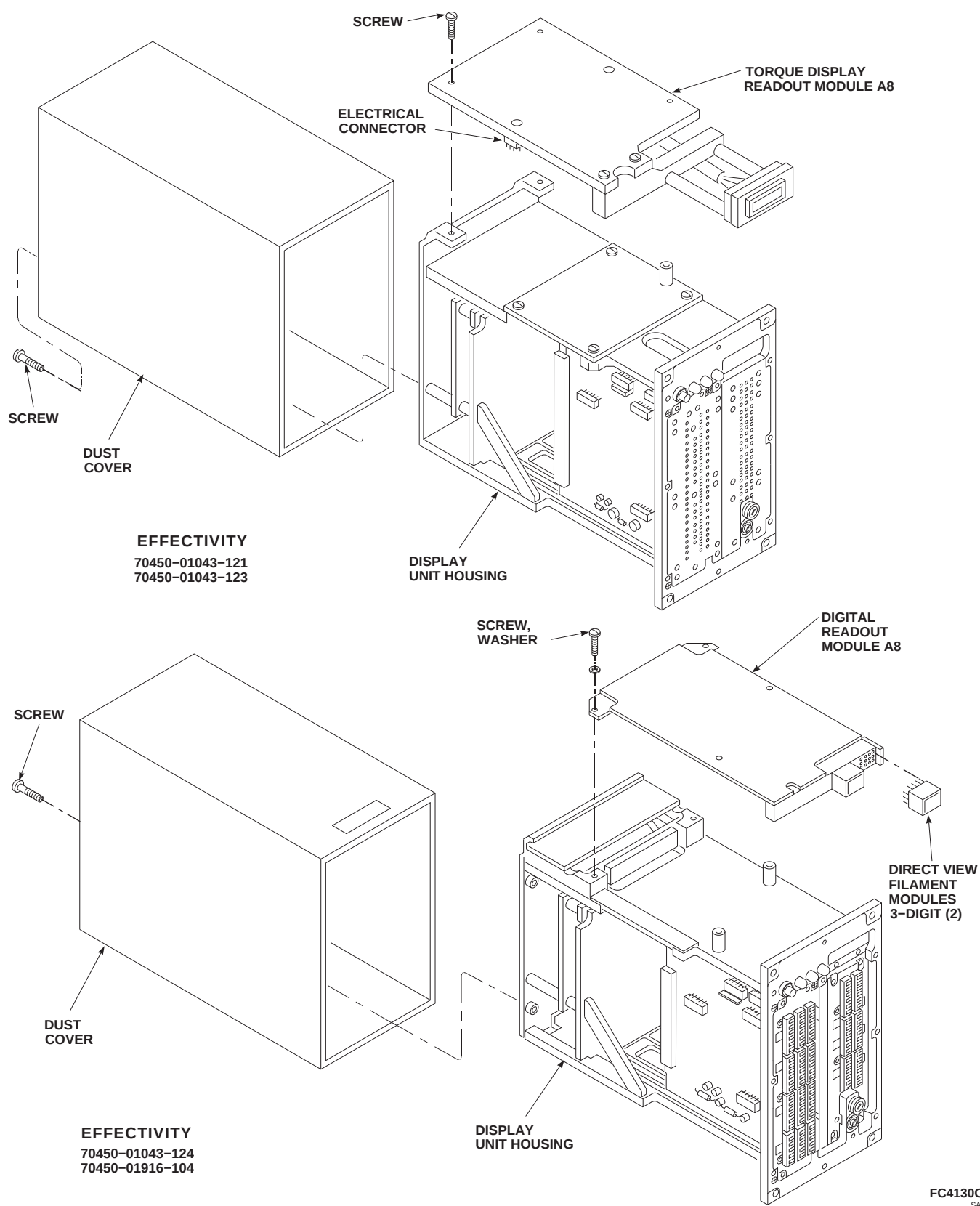


FIGURE 8-5-11-1. REPLACE PILOT DISPLAY UNIT TORQUE DISPLAY/DIGITAL READOUT MODULE (BENCH)

8-5-12 PILOT DISPLAY UNIT AUTODIM MODULE (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
8-5-12.1 Replace Pilot Display Unit Autodim Module (BENCH)	8-5-12-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-7-20
- PARA 8-2-8

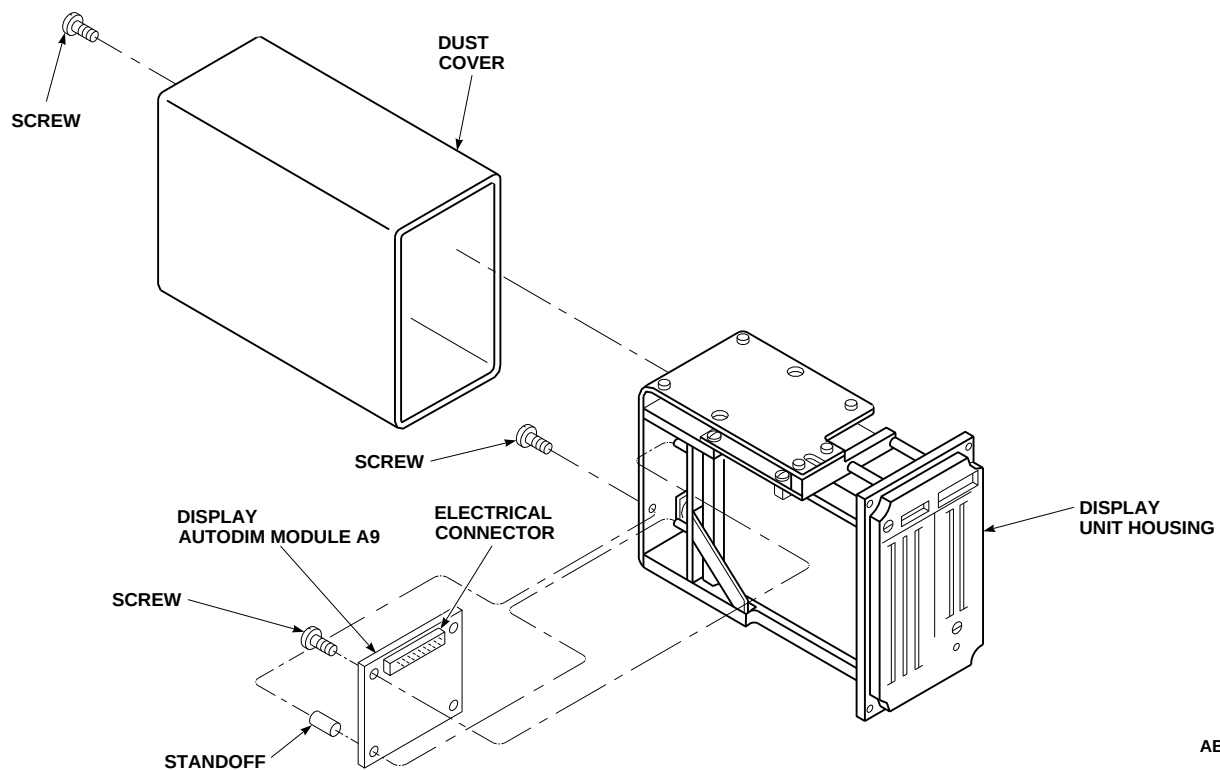
8-5-12.1 Replace Pilot Display Unit Autodim Module (BENCH).

8-5-12.1.1 Remove.

- Turn off all electrical power.
- Disconnect pilot display unit (PDU) from test set.
- Remove screws from rear of PDU, and remove dust cover (Figure 8-5-12-1).
- At rear of PDU, remove screws that pass through standoffs attaching bottom of module to housing. Remove standoffs from housing.
- Remove screws securing top of module to housing.
- Pull module toward rear of PDU, disconnecting electrical connector at top of module from housing.
- Carefully remove module from housing, making certain module clears wiring at bottom of PDU.

8-5-12.1.2 Install.

- Turn off all electrical power.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Carefully place module into housing, with electrical connector at top. Line up electrical connectors of module and housing and apply pressure until electrical connectors are engaged (Figure 8-5-12-1).
- Fasten top of module to housing with screws.
- Position standoffs between rear of housing and module. Insert screws through standoffs securing bottom of module to housing.
- Place dust cover on pilot display unit (PDU) and fasten screws.
- Perform operational check of PDU (PARA 8-2-8).

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SA**FIGURE 8-5-12-1. REPLACE PILOT DISPLAY UNIT AUTODIM MODULE (BENCH)**

8-5-13 PILOT DISPLAY UNIT NVG FILTERS (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
8-5-13.1	Replace Pilot Display Unit NVG Filters (BENCH)	8-5-13-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

Materials/Parts

- Paint, Polyurethane, Item 223, Appendix D

- Rubber Sheet, 0.188-inch thick, Item 269, Appendix D

References

- Appendix D
- PARA 1-6-16

8-5-13.1 Replace Pilot Display Unit NVG Filters (BENCH).

8-5-13.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Cut rubber sheet around NVG filter.
- Remove NVG filter (Figure 8-5-13-1, Detail A).

8-5-13.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Touch up paint chips on panel with polyurethane paint, Item 223, Appendix D.
- Install NVG filter (Figure 8-5-13-1, Detail A).
- Seal NVG filter with rubber sheet, 0.188-inch thick, Item 269, Appendix D.

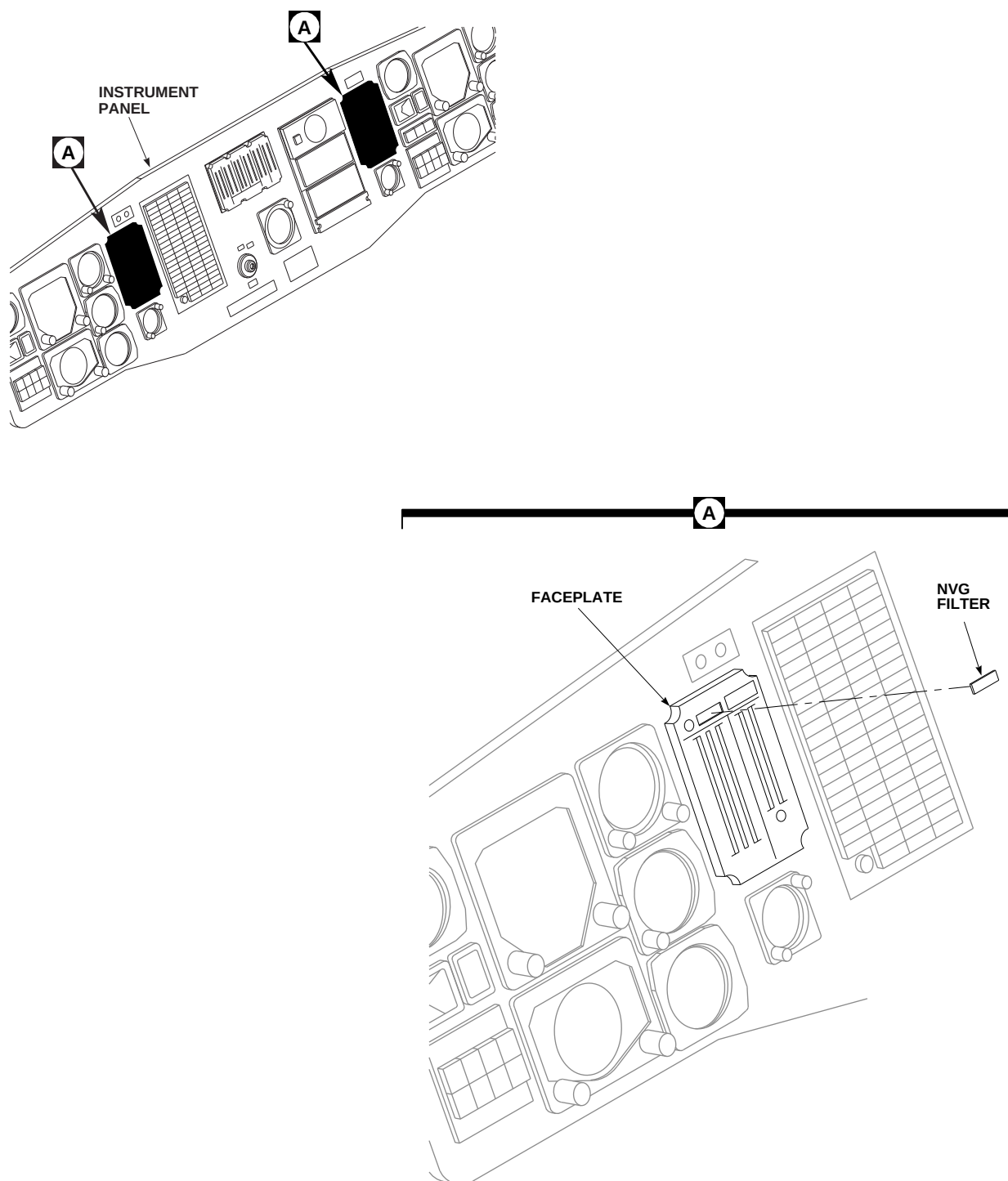


FIGURE 8-5-13-1. REPLACE PILOT DISPLAY UNIT NVG FILTERS (BENCH)

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8-5-13-2

8-5-14 CAUTION/ADVISORY PANEL CHANNEL/CONTROL CARDS (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
8-5-14.1 Replace Caution/Advisory Panel Channel/Control Cards (BENCH)	8-5-14-1

INITIAL SETUP

References

Tools

- Electrical Repairer’s Toolkit

- PARA 1-7-20
- PARA 8-2-10

8-5-14.1 Replace Caution/Advisory Panel Channel/Control Cards (BENCH).

NOTE

Channel card, A1, is positive caution, channel card, A2, is positive caution/advisory, channel card, A3, is negative caution/advisory, channel card, A4, is flashing logic control and caution/advisory, and control card, A5, is BRT/DIM/SLS. Replacement for all cards is the same.

- Locate card to be replaced; cards are in reference designation order with A1 nearest front panel.
- Grasp card at top and bottom of its stiffener and carefully pull out.

8-5-14.1.2 Install.

8-5-14.1.1 Remove.

- Turn off all electrical power.
- Disconnect caution/advisory panel from test set.
- Lay caution/advisory panel on its right side (viewed from front).
- Remove screws and left plate from caution/advisory panel (Figure 8-5-14-1).

- Turn off all electrical power.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Position top and bottom edges of card within retaining tracks (Figure 8-5-14-1).
- Grasp card at top and bottom of its stiffener and carefully push along track until card mates with connector. Make sure that card is securely mated with connector.
- Install left plate with screws.
- Perform operational check of caution/advisory panel (PARA 8-2-10).

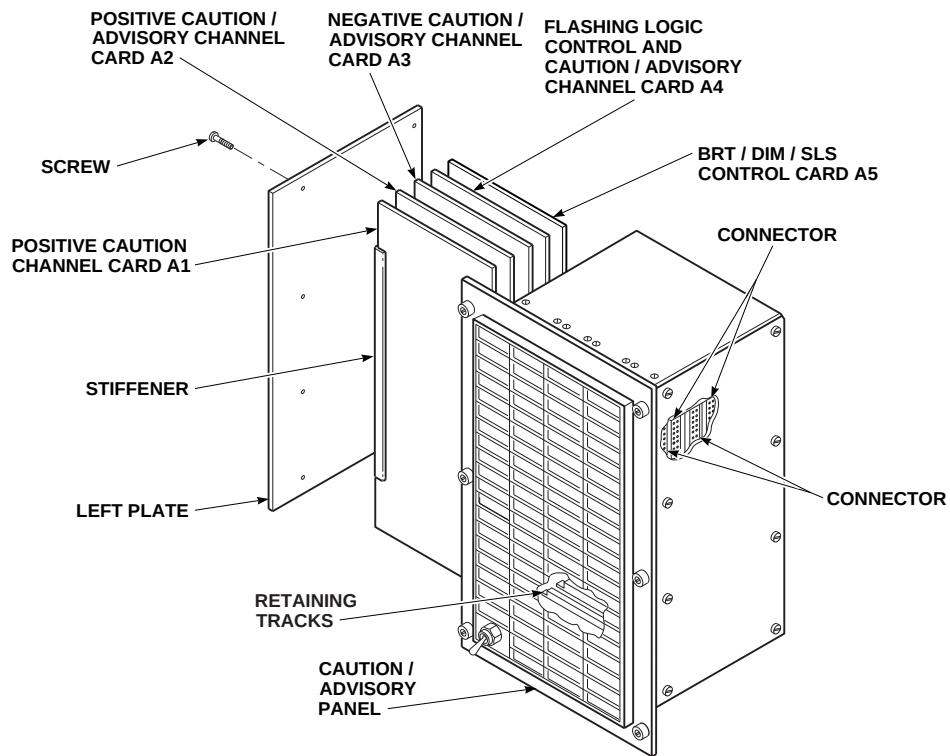
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FIGURE 8-5-14-1. REPLACE CAUTION/ADVISORY PANEL CHANNEL/CONTROL CARDS (BENCH)

8-5-15 CAUTION/ADVISORY PANEL TEST SWITCH (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
8-5-15.1 Replace Caution/Advisory Panel Test Switch (BENCH)	8-5-15-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 8-2-10
- PARA 8-5-14

8-5-15.1 Replace Caution/Advisory Panel Test Switch (BENCH).

8-5-15.1.1. Remove.

- Disconnect caution/advisory panel from test set.
- Remove channel cards, A1 through A4 (PARA 8-5-14).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and

injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder wires connected to test switch, S1, terminals (Figure 8-5-15-1).
- Remove nut and switch from rear of mounting bracket.

8-5-15.1.2. Install.

- Insert test switch, S1, into mounting bracket hole and fasten with nut (Figure 8-5-15-1).

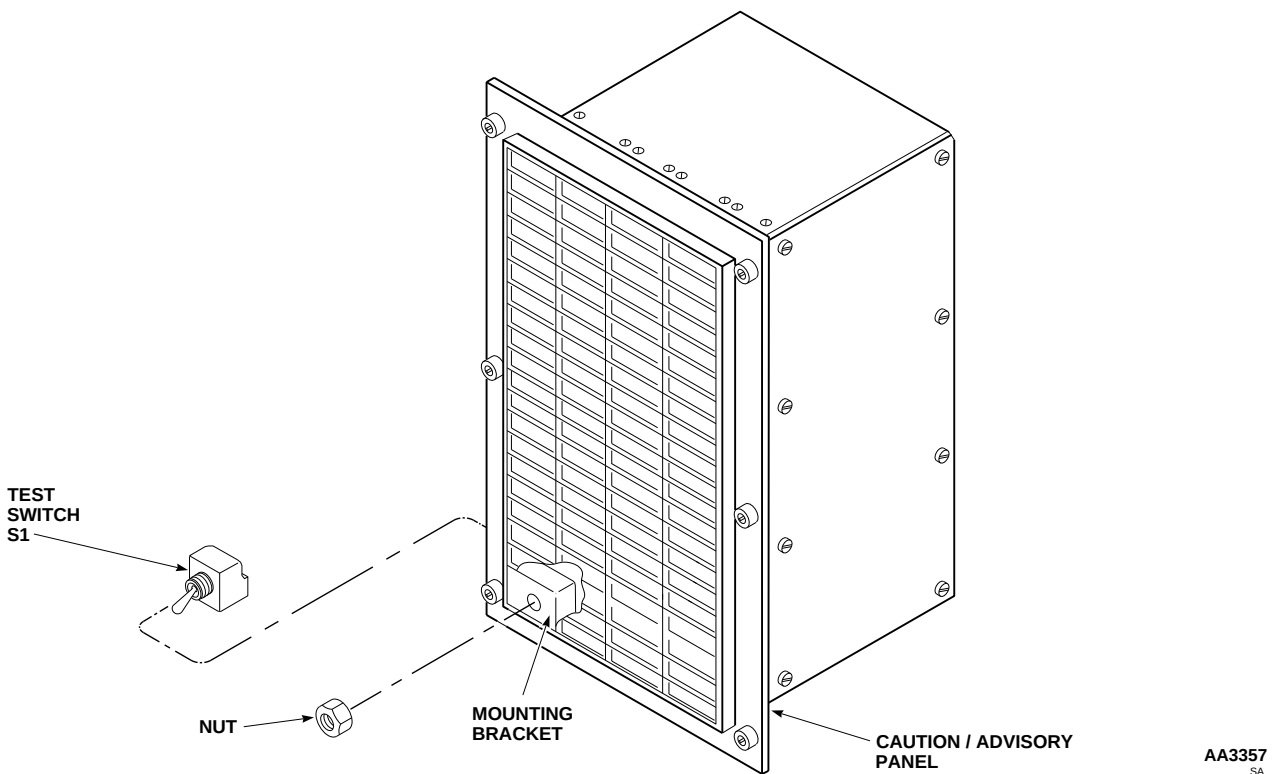


FIGURE 8-5-15-1. REPLACE CAUTION/ADVISORY PANEL TEST SWITCH (BENCH)

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- b. Using soldering iron and solder, Item 313, Appendix D, solder wiring to switch terminals. Untag wires.
- c. Install channel cards, A1 through A4 (PARA 8-5-14).
- d. Perform operational check of caution/advisory panel (PARA 8-2-10).

8-5-16 CAUTION/ADVISORY PANEL HEAT SINK/TRANSISTORS (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
8-5-16.1 Replace Caution/Advisory Panel Heat Sink/Transistors (BENCH)	8-5-16-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Silicone Compound, Item 309, Appendix D

- Solder, Item 313, Appendix D

- Tags

References

- Appendix D
- PARA 8-2-10
- PARA 8-5-14

8-5-16.1 Replace Caution/Advisory Panel Heat Sink/Transistors (BENCH).

NOTE

To replace heat sink, transistors Q1, Q2, and Q3, must be removed. Individual malfunctioning transistors may be removed from heat sink.

8-5-16.1.1. Remove.

- Disconnect caution/advisory panel from test set.
- Remove channel cards, A1 through A4 (PARA 8-5-14).



Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder all wires connected to heat sink terminal lugs and transistors (Figure 8-5-16-1, Detail A).

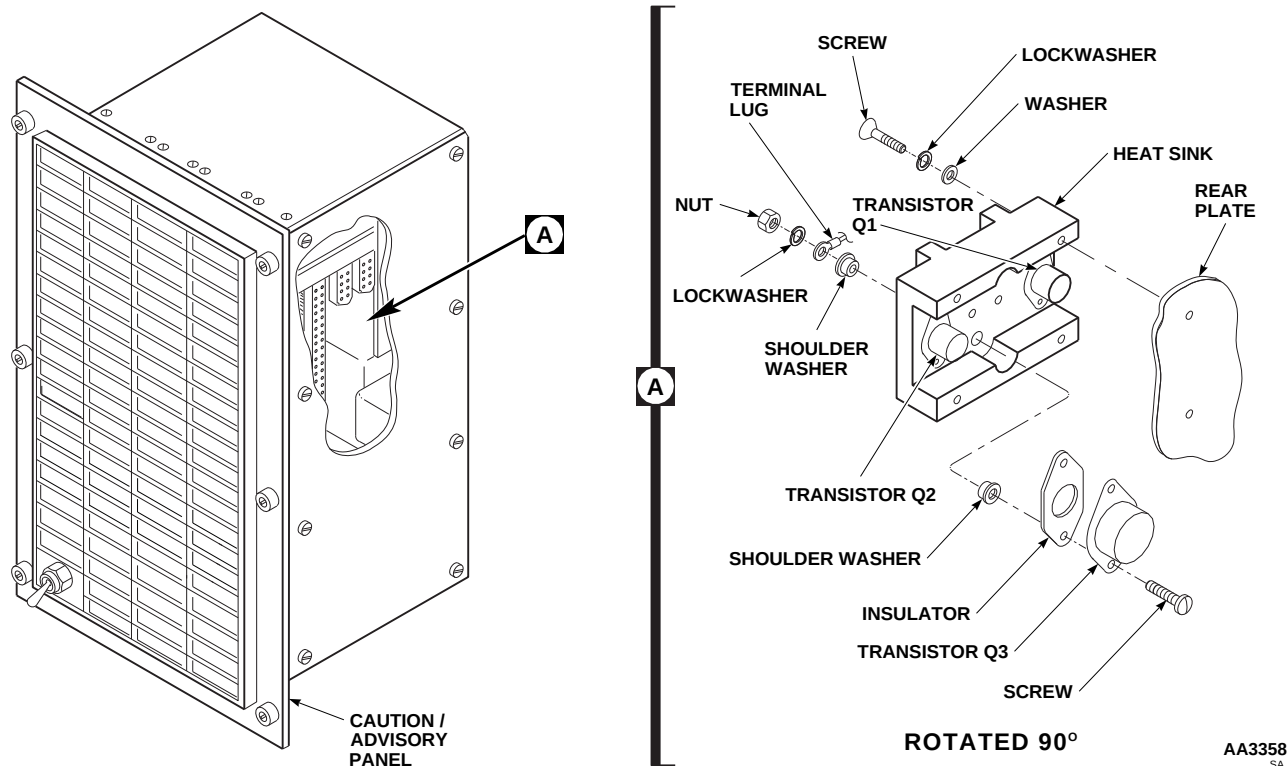


FIGURE 8-5-16-1. REPLACE CAUTION/ADVISORY PANEL HEAT SINK/TRANSISTORS (BENCH)

- d. Support heat sink and remove screws, lockwashers, and washers that hold it to rear plate of panel.
- e. Carefully remove heat sink.
- f. Remove screws, transistors, Q1, Q2, and Q3, insulators, shoulder washers, terminal lugs, lockwashers, and nuts from heat sink assembly. Do not crack mica insulators between transistors and heat sink.

8-5-16.1.2. Install.

- a. Apply silicone compound, Item 309, Appendix D, to both sides of insulators.
- b. Position insulators and transistors, Q1, Q2, and Q3, on heat sink and fasten with screws, shoulder washers, terminal lugs, lockwashers, and nuts (Figure 8-5-16-1, Detail A).
- c. Remove extra silicone compound from around transistors using cheesecloth, Item 90, Appendix D.

- d. Carefully position heat sink assembly to rear plate of panel.
- e. Fasten heat sink assembly to rear plate with screws, lockwashers, and washers.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- f. Using soldering iron and solder, Item 313, Appendix D, solder wiring to heat sink terminal lugs and transistors. Remove tags.
- g. Install channel cards, A1 through A4 (PARA 8-5-14).

- h. Perform operational check of caution/advisory panel (PARA 8-2-10).

8-5-17 CAUTION/ADVISORY PANEL RELAY, K1 (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
8-5-17.1 Replace Caution/Advisory Panel Relay, K1 (BENCH)	8-5-17-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 8-2-10
- PARA 8-5-14

8-5-17.1 Replace Caution/Advisory Panel Relay, K1 (BENCH).

8-5-17.1.1. Remove.

- Disconnect caution/advisory panel from test set.
- Remove channel cards, A1 through A4 (PARA 8-5-14).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and

injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder all wires connected to relay, K1, terminals.
- Remove screws, lockwashers, washers, nuts, and relay, K1, from rear plate of panel (Figure 8-5-17-1, Detail A).

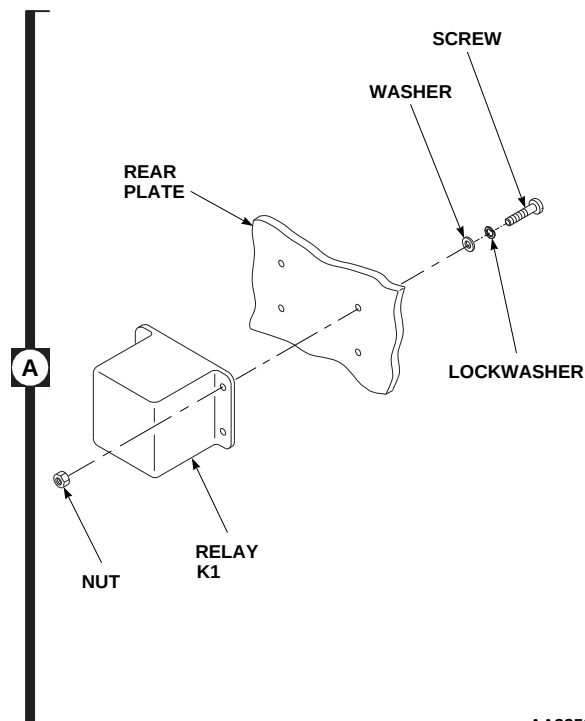
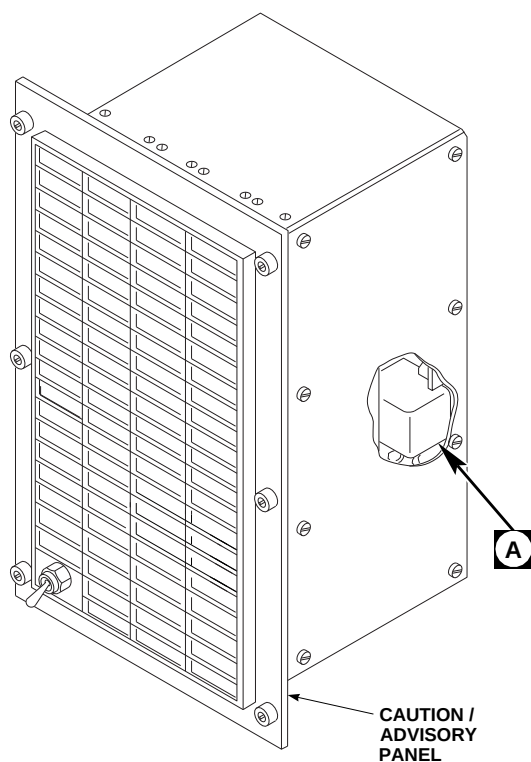
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FIGURE 8-5-17-1. REPLACE CAUTION/ADVISORY PANEL RELAY, K1 (BENCH)

8-5-17.1.2. Install.

- a. Install relay, K1, to rear plate of panel with screws, lockwashers, washers, and nuts (Figure 8-5-17-1, Detail A).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective

clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- b. Using soldering iron and solder, Item 313, Appendix D, solder wiring to relay terminals. Untag wires.
- c. Install channel cards, A1 through A4 (PARA 8-5-14).
- d. Perform operational check of caution/advisory panel (PARA 8-2-10).

8-5-18 CAUTION/ADVISORY PANEL DIODE, CR1 (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
8-5-18.1 Replace Caution/Advisory Panel Diode, CR1 (BENCH)	8-5-18-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 8-2-10
- PARA 8-5-14

8-5-18.1 Replace Caution/Advisory Panel Diode, CR1 (BENCH).

8-5-18.1.1. Remove.

- Disconnect caution/advisory panel from test set.
- Remove channel cards, A1 through A4 (PARA 8-5-14).
- Remove screws that hold right plate to panel (Figure 8-5-18-1). Remove plate.

injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder wires connected to diode, CR1.
- Remove diode from caution/advisory panel.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and

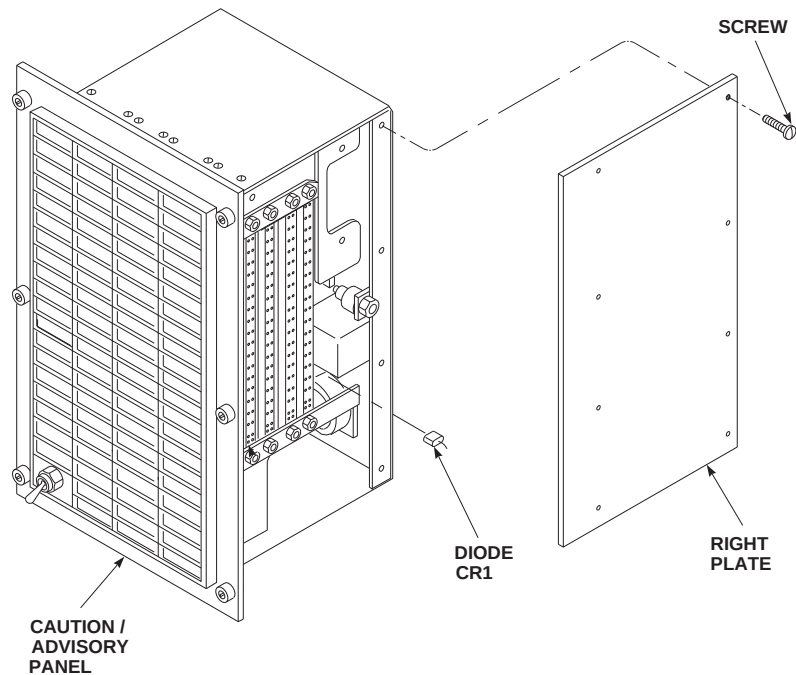
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FIGURE 8-5-18-1. REPLACE CAUTION/ADVISORY PANEL DIODE, CR1 (BENCH)

8-5-18.1.2. Install.**WARNING**

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- a. Using soldering iron and solder, Item 313, Appendix D, solder wiring to diode, CR1 (Figure 8-5-18-1). Untag wires.
- b. Install right plate with screws.
- c. Install channel cards, A1 through A4 (PARA 8-5-14).
- d. Perform operational check of caution/advisory panel (PARA 8-2-10).

8-5-19 CAUTION/ADVISORY PANEL DIODE, CR2 (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
8-5-19.1 Replace Caution/Advisory Panel Diode, CR2 (BENCH)	8-5-19-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 8-2-10
- PARA 8-5-14

8-5-19.1 Replace Caution/Advisory Panel Diode, CR2 (BENCH).

8-5-19.1.1. Remove.

- Disconnect caution/advisory panel from test set.
- Remove channel cards, A1 through A4 (PARA 8-5-14).
- Remove screws that hold right plate to panel (Figure 8-5-19-1). Remove plate.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and

injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder wires connected to diode, CR2.
- Remove nut that holds diode, CR2, to heat sink and remove diode.

8-5-19.1.2. Install.

- Install diode, CR2, to heat sink with nut (Figure 8-5-19-1).

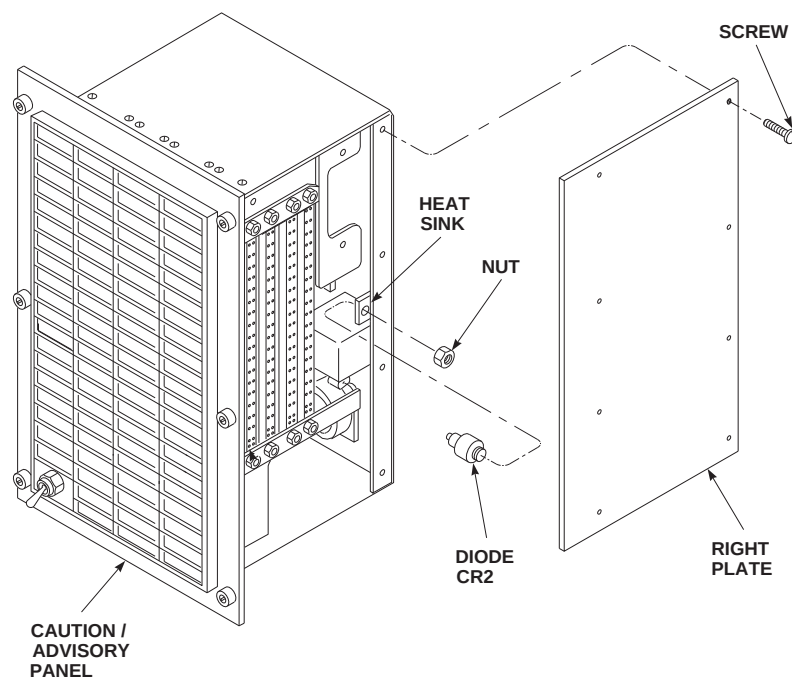
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FIGURE 8-5-19-1. REPLACE CAUTION/ADVISORY PANEL DIODE, CR2 (BENCH)

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- b. Using soldering iron and solder, Item 313, Appendix D, solder wiring to diode and remove tags.
- c. Install right plate with screws.
- d. Install channel cards, A1 through A4 (PARA 8-5-14).
- e. Perform operational check of caution/advisory panel (PARA 8-2-10).

8-5-20 CAUTION/ADVISORY PANEL DIODE, CR3 (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
8-5-20.1	Replace Caution/Advisory Panel Diode, CR3 (BENCH)	8-5-20-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 8-2-10
- PARA 8-5-14

8-5-20.1 Replace Caution/Advisory Panel Diode, CR3 (BENCH).

8-5-20.1.1 Remove.

- Remove channel cards, A1 through A4 (PARA 8-5-14).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and

injury to personnel. Use protective clothing, including eye protection, while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder wires connected to diode, CR3 (Figure 8-5-20-1).
- Remove screw, washer, lockwasher, and nut securing diode, CR3, to rear plate of caution/advisory panel. Remove diode, CR3.

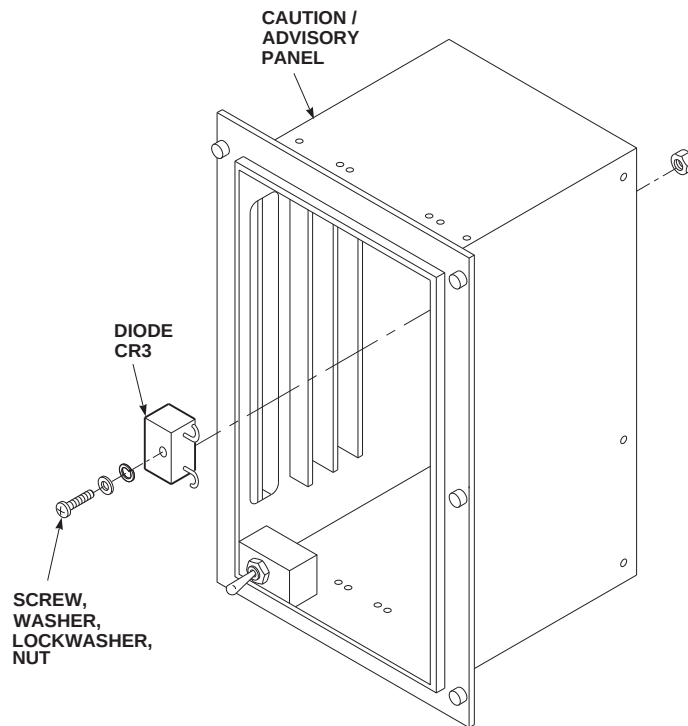
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FIGURE 8-5-20-1. REPLACE CAUTION/ADVISORY PANEL DIODE, CR3 (BENCH)

8-5-20.1.2 Install.

- a. Install diode, CR3, on rear plate of caution/advisory panel with screw, washer, lock-washer, and nut (Figure 8-5-20-1).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective

clothing, including eye protection, while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- b. Using soldering iron and solder, Item 313, Appendix D, solder wires to diode, CR3. Remove tags.
- c. Install channel cards, A1 through A4 (PARA 8-5-14).
- d. Perform operational check of caution/advisory panel (PARA 8-2-10).

8-5-21 CAUTION/ADVISORY PANEL CONNECTORS (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
8-5-21.1 Replace Caution/Advisory Panel Connectors (BENCH)	8-5-21-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-7-20
- PARA 8-2-10
- PARA 8-5-14

8-5-21.1 Replace Caution/Advisory Panel Connectors (BENCH).

NOTE

Replacement for connectors, XA1 through XA5, is the same. Replacement for connectors, J117 and J118, is the same.

8-5-21.1.1 Remove.

- Disconnect caution/advisory panel from test set.
- Remove channel cards, A1 through A4, and control card, A5 (PARA 8-5-14).
- Remove screws securing right plate to caution/advisory panel (Figure 8-5-21-1). Remove plate.



Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection, while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Using soldering iron, unsolder wires connected to connector terminals, XA1 through XA5. Tag connector terminals.
- Remove screws, lockwashers, washers, nuts, and connectors, XA1 through XA5, from caution/advisory panel.
- Using soldering iron, unsolder wires connected to electrical connector terminals, J117 and

J118. Tag connector terminals (Figure 8-5-21-1, Detail A).

- g. Remove screws, lockwashers, washers, nuts, and electrical connectors, J117 and J118, from rear plate of caution/advisory panel.

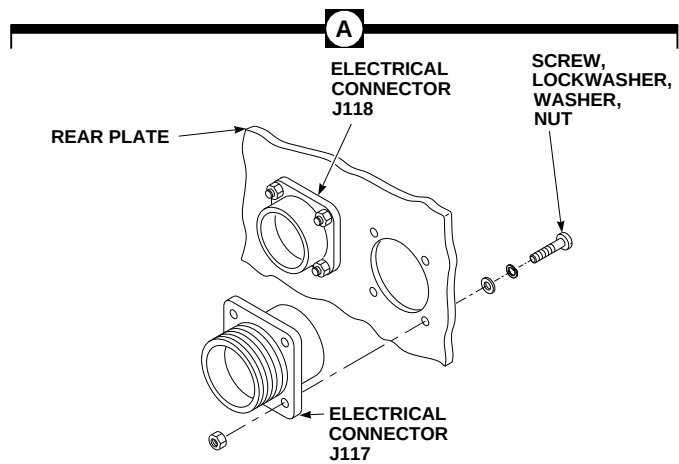
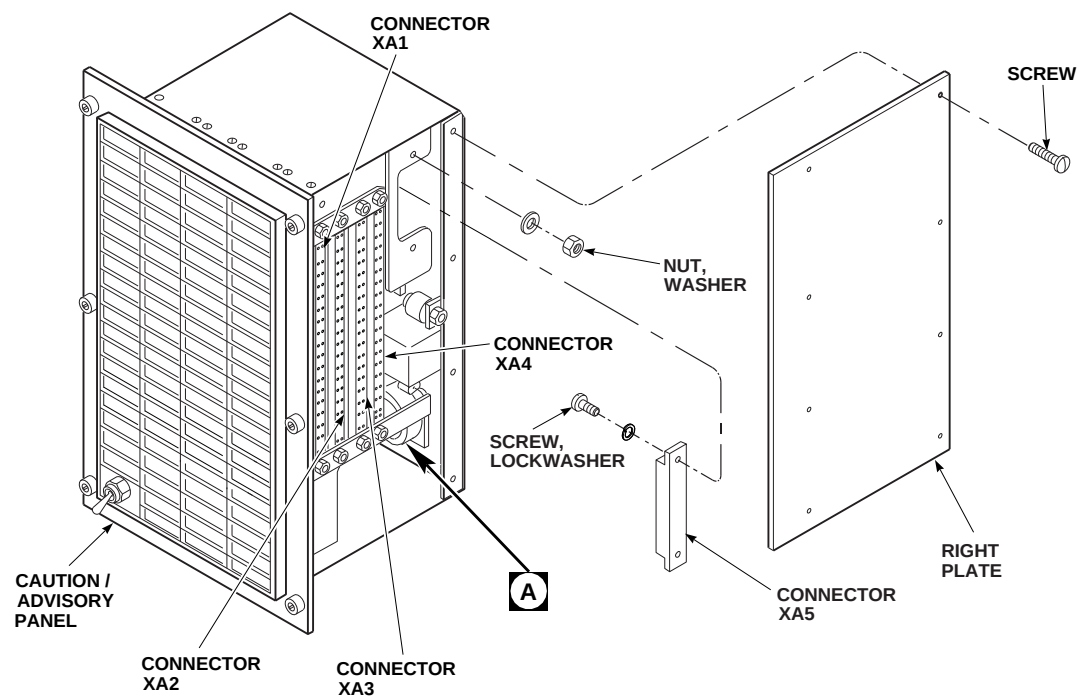
8-5-21.1.2 Install.

- a. Inspect and treat electrical connectors (PARA 1-7-20).
- b. Install electrical connectors, J117 and J118, in rear of caution/advisory panel using screws, lockwashers, washers, and nuts (Figure 8-5-21-1, Detail A).
- c. Using soldering iron and solder, Item 313, Appendix D, solder wiring to electrical connector terminals, J117 and J118. Untag wires.
- d. Inspect and treat electrical connectors (PARA 1-7-20).
- e. Install connectors, XA1 through XA5, in caution/advisory panel with screws, lockwashers, washers, and nuts (Figure 8-5-21-1).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection, while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- f. Using soldering iron and solder, Item 313, Appendix D, solder wiring to connector terminals, XA1 through XA5. Untag wires.
- g. Install right plate to caution/advisory panel with screws.
- h. Install channel cards, A1 through A4, and control card, A5 (PARA 8-5-14).
- i. Perform operational check of caution/advisory warning system (PARA 8-2-10).



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FIGURE 8-5-21-1. REPLACE CAUTION/ADVISORY PANEL CONNECTORS (BENCH)

8-5-22 **CHIP DETECTOR RESISTOR UNIT (BENCH).**

PARAGRAPH BREAKDOWN

		PAGE			PAGE
8-5-22.1	Replace Chip Detector Resistor Unit Resistors (BENCH)	8-5-22-1	8-5-22.3	Replace Chip Detector Resistor Unit Connector, J3 (BENCH)	8-5-22-4
8-5-22.2	Replace Chip Detector Resistor Unit Diode, CR1 (BENCH)	8-5-22-3			

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Insert/Extract Tool
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 8-2-3

8-5-22.1 **Replace Chip Detector Resistor Unit Resistors (BENCH).**

8-5-22.1.1 **Remove.**

- Turn off all electrical power (PARA 1-6-16).
- Loosen captive screws and cover from chip detector resistor unit (Figure 8-5-22-1).

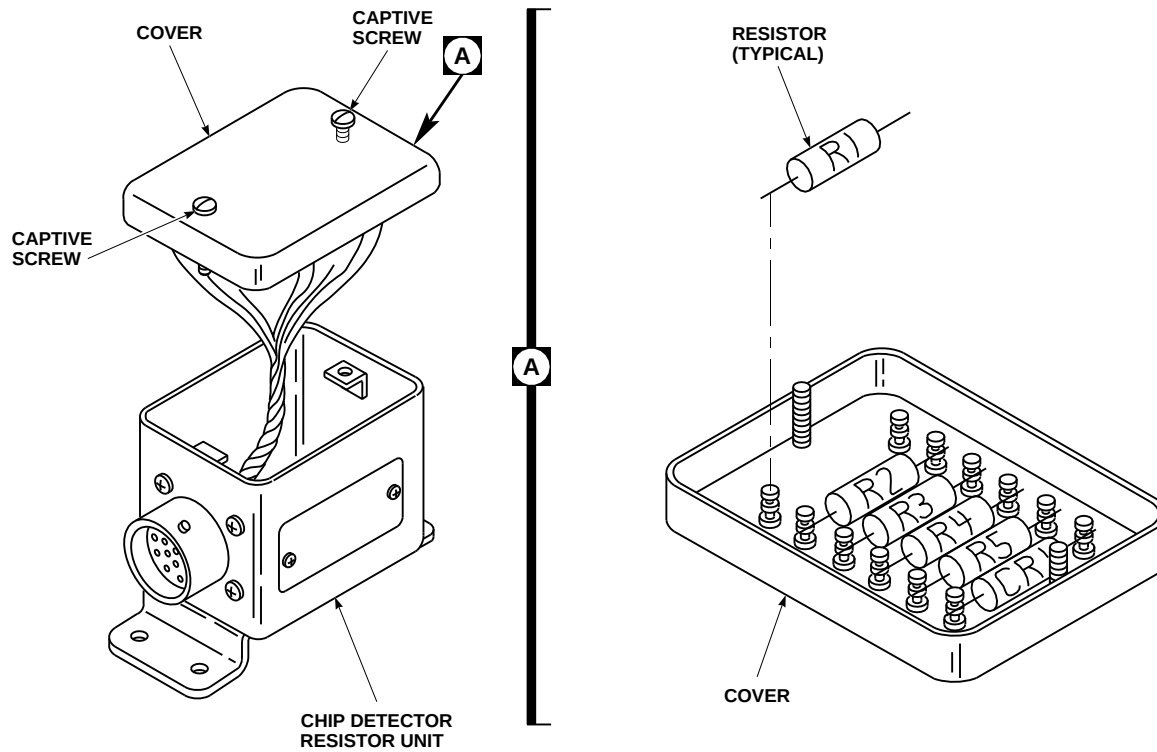
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FIGURE 8-5-22-1. REPLACE CHIP DETECTOR RESISTOR UNIT RESISTORS (BENCH)

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection, while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- c. Using soldering iron, unsolder and remove resistors as required (Figure 8-5-22-1, Detail A).

8-5-22.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).

WARNING

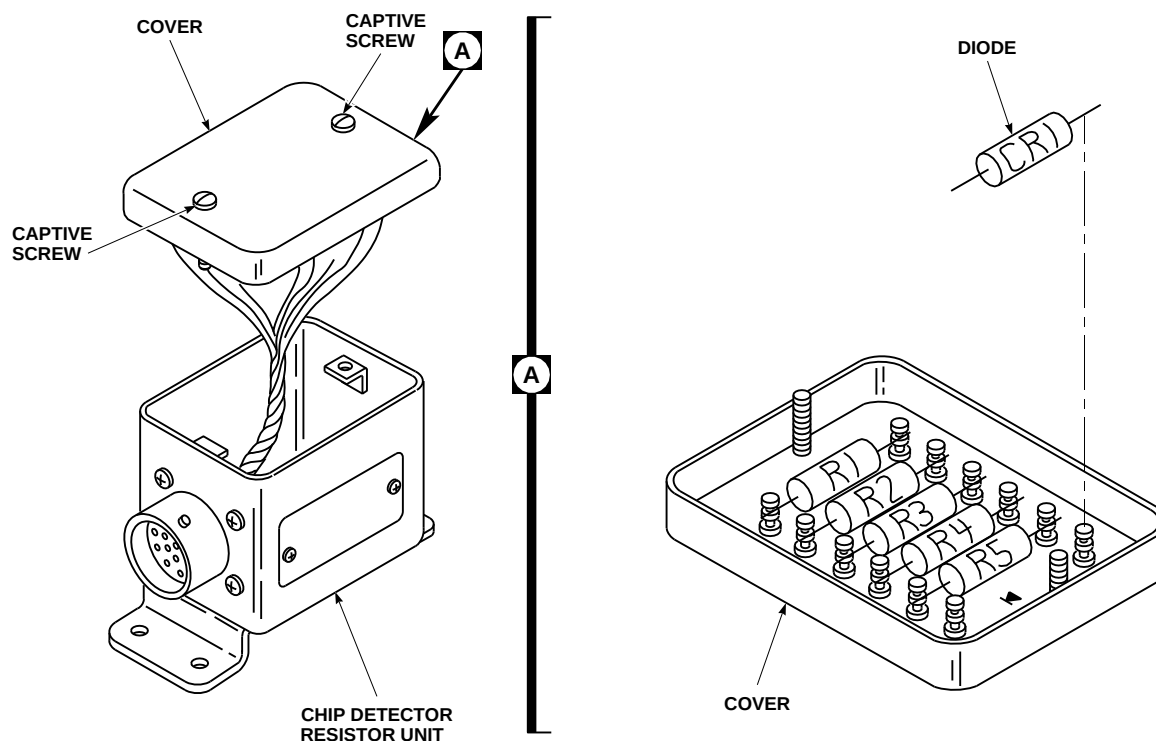
Hot metal droppings occur when soldering, causing both a fire hazard and

injury to personnel. Use protective clothing, including eye protection, while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- b. Using soldering iron, solder, Item 313, Appendix D, resistors to terminals per this chart (Figure 8-5-22-1, Detail A).

Resistor	Terminals
R1	1 and 7
R2	2 and 8
R3	3 and 9
R4	4 and 10
R5	5 and 11

- c. Install cover with captive screws (Figure 8-5-22-1).
- d. Do an operational check of chip detector resistor unit (PARA 8-2-3).



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FIGURE 8-5-22-2. REPLACE CHIP DETECTOR RESISTOR UNIT DIODE, CR1 (BENCH)

8-5-22.2 Replace Chip Detector Resistor Unit Diode, CR1 (BENCH).

8-5-22.2.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Loosen captive screws and cover from chip detector resistor unit (Figure 8-5-22-2).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection, while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Using soldering iron, unsolder diode, CR1, from terminals (Figure 8-5-22-2, Detail A).

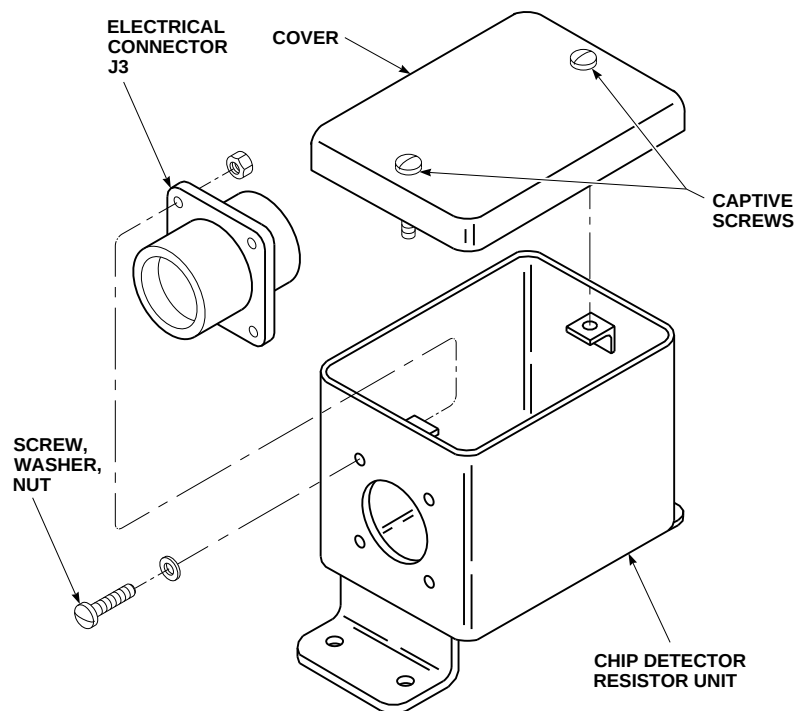
8-5-22.2.2 Install.

- Turn off all electrical power (PARA 1-6-16).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection, while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Using soldering iron, solder, Item 313, Appendix D, diode, CR1, anode to terminal 6 and cathode to terminal 12 (Figure 8-5-22-2, Detail A).
- Install cover with captive screws (Figure 8-5-22-2).

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SA**FIGURE 8-5-22-3. REPLACE CHIP DETECTOR RESISTOR UNIT CONNECTOR, J3 (BENCH)**

- d. Do an operational check of chip detector resistor unit (PARA 8-2-3).

8-5-22.3 Replace Chip Detector Resistor Unit Connector, J3 (BENCH).

8-5-22.3.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Loosen captive screws and cover from chip detector resistor unit (Figure 8-5-22-3).
- Using insert/extract tool, remove and tag wires from electrical connector, J3.
- Remove screws, washers, and nuts securing electrical connector, J3, to chip detector resistor unit and remove connector, J3.

8-5-22.3.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Using insert/extract tool, insert tagged wires to electrical connector, J3 (Figure 8-5-22-3). Remove tags.
- Inspect and treat electrical connector (PARA 1-7-20).
- Install electrical connector, J3, to chip detector resistor unit with screws, washers, and nuts.
- Install cover with captive screws.
- Do an operational check of chip detector resistor unit (PARA 8-2-3).